



EXTREME RISKS

VALUE IN A VAST AND UNCERTAIN WORLD



JEFFREY SANFORD RUSSELL



Last revised August 29, 2026

For my children, and any children they may have,
and so on.

Contents

Acknowledgments	1
1 Introduction	3
1.1 Big decisions	3
1.2 Some background ideas	9
1.2.1 Expectations	9
1.2.2 Pascal and St. Petersburg	12
1.2.3 Expected utility	14
1.2.4 Risk-weighting and probability discounting	19
1.3 Some foundational issues	25
1.3.1 What would be better	25
1.3.2 Information-sensitive value	26
1.3.3 Counterfactual value	29
1.3.4 States, outcomes, and random variables	33
1.3.5 Lotteries	36
1.4 Roadmap	39
I The Structure of Infinite Value	45
2 Pascal’s Principle	47
2.1 The bad argument	48
2.2 The good argument	57
2.3 Vector utilities	61
2.4 Hájek’s objection	70
2.5 “Wherever the infinite is”	72
2.6 Finite Fanaticism	77
2.7 The ugly argument	79

3	Against the Everything Bagel	83
3.1	Infinite futility	83
3.2	A toy model	85
3.3	The moral telescope	95
3.4	Morphing	106
3.5	By the numbers	115
3.6	Conclusion	124
4	Indeterminate Value	125
4.1	Incomparability and small improvements	125
4.2	Invariance	128
4.3	Non-constructiveness	136
4.4	The logic of comparatives	138
4.5	Opaque sweetening	142
4.6	Classical indeterminacy	147
4.7	Vague value	154
4.8	Precisifications and hyperreals	159
4.9	Conclusion	163
5	Infinitesimal Probabilities	165
5.1	Probability and weighing value	165
5.2	Return of the Everything Bagel	172
5.3	Tiny but powerful	179
5.4	Indeterminate probability	184
5.5	Williamson's paradox	186
5.6	Conditional probabilities	191
5.7	Conclusion	196
6	Paradoxes of Dominance	197
6.1	Cain's paradox	197
6.2	Subpopulation dominance	200
6.3	Paradoxes of risk	201
6.4	Dimensions of value	204
6.5	The path ahead	209
II	Weighing Small Probabilities	211
	Introduction to Part II	213
7	Individual Prospects	217

7.1	The <i>Ex Ante</i> Pareto principle	217
7.2	Ex Ante Pareto and infinity	223
7.3	Modes of presentation	227
7.4	From modes of presentation to permutations	235
7.5	The power of Pareto	241
7.6	From Pareto to paradox	245
7.7	What comes next	252
8	Distant Lives and Cosmic Value	253
8.1	Separability	253
8.2	The SSK paradox	262
8.3	A visit to Pasadena	269
8.4	Cosmic background value	273
8.5	Conclusion	281
9	Consistency and Coordination	283
9.1	The Sure Thing Principle	283
9.2	Redistributing risks	287
9.3	Symmetries of value	290
9.4	Eventwise Dominance	295
9.5	Is Fanaticism dynamically consistent?	305
9.6	Equilibrium strategies	310
9.7	Conclusion	321
10	Small Changes in Probability	323
10.1	Anti-Fanaticism	323
10.2	Small probabilities matter	327
10.3	Variable Fanaticism and infinite value	333
10.4	The direct argument for Pascalianism	335
10.5	Direct to St. Petersburg?	345
10.6	Conclusion	347
11	The Tyranny of the Infinite	349
11.1	Where we find ourselves	349
11.2	What should we do?	352
11.3	Many gods and infinite cluelessness	357
11.4	Epistemic responses	362
11.5	Deontic responses	369
11.6	Regrets	373

III Mathematical Appendices	375
Introduction to Part III	377
A Ordering Social Outcomes	379
A.1 Basics	379
A.2 Orders	380
A.3 Finitely symmetric strictly monotone orders	385
A.4 Social welfare orders	395
A.5 Symmetries	397
A.6 Invariant orders	404
B Aggregation	415
B.1 Quantities	415
B.2 Structures	422
B.3 Embeddings and representations	429
B.4 Finitary social aggregation	434
B.5 Filter powers	443
B.6 Quasi-finite sums	449
B.7 Representing social welfare	454
B.8 Symmetries and filter powers	459
C Frameworks and Foundations for Prospects	467
C.1 Possible worlds	468
C.2 Modes of presentation	473
C.3 The counterfactual framework	479
C.4 The Savage framework	483
C.5 The von Neumann–Morgenstern framework	492
C.6 The Sure Thing Principle and the Independence axiom	496
C.7 Game theory	496
D Discrete Probability	503
D.1 Simple probability distributions	503
D.2 Simple expectations	508
D.3 Simple stochastic dominance	513
D.4 Pascalian theorems	521
D.5 Ex ante Pareto and statewise dominance	526
D.6 Separability and additivity	528
D.7 Invariance and fanaticism	538
D.8 Infinite prospects	541

E	Standard Probability	549
E.1	Basic measure theory	549
E.2	Expectations and integration	557
E.3	Mixtures of probability measures	562
E.4	Stochastic dominance	566
E.5	Background randomness and infinite futility	575
E.6	The background value theorem	589
E.7	Conditional probability	593
F	Non-Archimedean Probability	599
F.1	Probability orders and measures	599
F.2	Limiting rings	604
F.3	Regular uniform probability measures and expectations . . .	608
F.4	Symmetric measures	612
F.5	Distributions	626
F.6	Stochastic dominance revisited	635
F.7	Valuing social prospects	642
	References	657

Acknowledgments

T-REX DID YOU KNOW THE OPINIONS OF YOUR FRIENDS ARE
OBJECTIVELY BETTER THAN YOURS

Ryan North, *Dinosaur Comics*

UNDER CONSTRUCTION.

Title page image: Detail from Magnus's *Carta Marina* of 1539. https://commons.wikimedia.org/wiki/File:Carta_Marina.jpeg

Chapter 1

Introduction

Maurice was more than a cat, he knew. He knew the world was big and complex and involved a lot more than wondering if the next meal was going to be beetles or chicken legs. The world was huge and difficult and full of amazing things ...

Terry Pratchett, *The Amazing Maurice and His Educated Rodents*

1.1 Big decisions

The global fight against disease is one of the great success stories of the past two centuries.¹ In 1800 no country in the world had an average life expectancy above 40 years; in 2023, no country in the world had an average life expectancy below 50 years. The global average has more than doubled, from 32 years in 1900 to 73 years in 2023. But there is massive inequality in how these health outcomes are distributed, with low-income countries much worse off than high-income countries. For example, in 2023 life expectancy in Nigeria was 54 years, while in Japan it was 85 years — reflecting, among other factors, a huge disparity in early childhood mortality rates. These gaps have been steadily closing over recent decades, though, and there are many relatively straightforward ways to close them faster. For example, in 2024 more children under five died of malaria worldwide than died in the United

¹All statistics are from *Our World in Data*: Ortiz-Ospina and Roser (2016); Dattani et al. (2023).

States of *any* cause — over 18 times more. Yet malaria can be treated and prevented by inexpensive and well-understood means, such as insecticide-treated bed nets. Global public health interventions in low-income countries are a highly effective way to improve the well-being of a huge number of people.

On the other hand, these steady improvements can be set back quickly by novel diseases. As of mid-2024, the COVID-19 pandemic was estimated to have resulted in around 27 million deaths.² Future pandemics might be even more catastrophic — especially as we have the newfound and poorly regulated capability to manufacture pathogens of our own design. In principle, a targeted pandemic could kill *billions* of people — maybe even all of humanity.

These facts present us with some hard choices. Consider an example.³

Medora has just completed a medical degree, and is considering two possible futures. She might pursue a career in public health, with a view toward advancing well-understood global health interventions against diseases like malaria. Or she might pursue a career in biosecurity, with the aim of improving defenses against as-yet-unimagined future pandemics. If Medora takes the first option, it is quite plausible that she will save thousands of lives. If she takes the second, given the immense uncertainty about future bioengineering, her work may well make *no* significant direct impact on actual diseases. But there is a *small* chance — it is hard to tell exactly how small — that her work will make a difference that saves billions of lives — and perhaps even secure a future for many, many more.

Choices like Medora's are hard and important. They require us to make trade-offs of two kinds. First, trade-offs between *people*. If Medora chooses to work toward sparing people from suffering from massive pandemics in the future, this has a real opportunity cost, giving up the chance to help many people who will suffer in the more immediate future. Promoting some people's good inevitably trades off against promoting other people's

²This estimate is highly uncertain. See Mathieu et al. (2020); see also G. Lewis (n.d.).

³This fictional example is loosely based on Dr. Cassidy Nelson's account of her own career choices (2020):

... [E]ven if you have low probability events, the scale of the impact that they'd have, both on near-term considerations as well as the long-term future really resonated with me and has really shaped my decisions .

good, and we must decide how to weigh these against one another. Second, trade-offs between *possibilities*. We do not know what will happen, and we must weigh the different ways our plans might turn out, and how good or bad they are, against each other. Medora is trading off a high probability of doing a certain amount of good against the low probability of doing much more good.

The central topic of this book is choices that have this structure, trading off outcomes of immense importance, but with minuscule probabilities — *big decisions*. To be more precise, though, our attention will be primarily directed not toward *choice*, but rather toward *value*. There are many questions Medora might ask that inform her choice. Which of these causes is closer to her heart? Does she have special duties based on her relationship to people who are suffering right now, rather than in the future? But among these, one important question she might ask is which of these options has better consequences for other people — taking into account that they have different chances of benefitting different people, and trading those off against one another.

The primary focus of this book is on the *good*, rather than the *should* — in the jargon, *axiology*, rather than *deontology*. This focus on the value of consequences is emphatically *not* supposed to come with a commitment to *consequentialism* — the view that the *only* thing that is relevant to which acts one ought to perform is the value of their consequences. Other considerations — duties, rights, demands of justice, personal prerogatives — may come apart from the value of consequences, and may matter a great deal.

Farid's child Chisom is suffering from a painful ear infection. Fortunately, Farid has one dose of an anti-biotic medication that will cure the infection. But Farid knows that a neighbor child, Neha, is also suffering from the same kind of infection — in fact, one that is even a bit *more* painful. There is no way for Farid or his neighbors to get any more of the medication.

It is not implausible that Farid morally ought to give the medication to his own child, rather than his neighbor's. But if so, this is because Farid stands in a special relationship of care toward his own child. It is not because it is *better* for Chisom to get the medicine than it is for Neha to get it. Presumably we will say that each child's suffering matters equally: and in fact, if Neha's suffering is worse than Chisom's, then giving Neha the medicine would do

more good overall.⁴

But even if considerations of what is better or worse for people do not *determine* what we should do, they are surely *relevant* to what we should do in many cases. And, for the sake of conceptual clarity, it is worth investigating what is better or worse for people in a way that substantially abstracts away from other non-axiological considerations such as special duties or prerogatives.

At least, that is our starting point. In the end, our investigations will put significant pressure on this idea — that there is sense to be made of an impartial, objective notion of betterness with respect to consequences for other people, which can be a useful guide for many of our choices. This will put pressure on consequentialist theories that require such a notion to even get off the ground, but it also puts pressure on any moral theory that gives some place to a kind of universal beneficence that aims to impartially promote the good of all. But let's not get ahead of ourselves: those are issues for the end of this book, rather than its beginning.

Another point of clarification. I will be talking a lot about *people*, what is better or worse for an individual person, and how we should weigh up value to individual people to make judgments of what is better or worse overall. This focus is because I believe that people and their well-being are morally important. But of course, it is not *only* people or their well-being that have moral value or disvalue. Dogs and chickens also matter morally, for their own sake, and so perhaps do jellies and nematodes, or artificial intelligences, or families, communities, ecosystems, or works of art. Again, my focus on people is not intended to imply that nothing else matters; but for the sake of conceptual clarity, it is helpful to narrow our focus. So, while I will sometimes talk about what is better or worse overall, often this should be understood to come with an implicit restriction, as what is better or worse overall *with respect to its consequences for individual people's well-being*: that is, *social welfare*.⁵

To sum up, our topic is narrowed to this: how to compare options that involve *extreme risks* — extreme with respect to the magnitude of their possible outcomes, and also with respect to the tininess of some of the probabilities involved — according to how *good* these options are for *individual people*, on

⁴Compare discussion in Parfit (1984, sec. 14).

⁵Note, though, that I will construe “people” broadly enough that it may include inhabitants of distant galaxies, even if they are not *humans*.

balance.

Even this is still a bit overambitious, and for the most part I will set aside some hard questions that fall within this subject matter.

- Is it worse for one person to permanently lose the use of their limbs than for many, many people to briefly suffer a slight headache? For the most part, I will set aside this question of *aggregation*,⁶ and instead focus on trade-offs involving the *same kind* of benefits or harms.
- Is it better for there to be many more happy people, rather than fewer people altogether? Again for the most part I will set aside this question of *population ethics*.⁷ Instead, our central question is, for a fixed (large) population of individuals, which option is *best for them*, on balance?

Plenty of difficult questions arise even when we have restricted our attention in these ways.

Pursuing this path is going to take us to some strange places. We will find ourselves thrown right away into considerations about not just *immense* value, but outright *infinite* value. And we will be considering prospects that involve not just outcomes with *small* probabilities, but *infinitely many* outcomes of *arbitrarily* small probabilities. And — let's be honest — it gets weird. *Here be dragons*.

When we start talking about infinitely long futures and infinitely large populations, you might start to back away slowly. You may not be inclined to take seriously the conclusions we bring back from these exotic climes. Honestly, *I'm* not sure how seriously to take them. Here is my general attitude toward infinity in ethics. It might be a completely irrelevant pointless distraction, a mere source of brainteasers. Alternatively, it might be literally the most important thing in the world. And I really have no idea which one it is. But I don't think we can find out which of these perspectives is right without taking infinity seriously — at least for a while. In fact, the decision to write (or read) this book may itself be a big decision: we risk throwing our efforts away on something unimportant, because of the small chance that it will be *very* important. Sadly, we will not have the general tools to evaluate whether this kind of decision is worth making until after we have already put in the time.

But I do not think the chance of any of these issues mattering is merely

⁶The classic discussion is Scanlon (1998, sec. 5.9).

⁷The classic discussion is Parfit (1984, part four).

minuscule. The world we live in might really be infinite. This is a hypothesis taken very seriously by cosmologists, as well as by distinguished and living traditions of both speculative philosophers and religious believers. We might even face choices that affect an infinite future. It may look like anything of value in our universe is on track to eventually fizzle out in some finite amount of time. Even so, many serious people profess belief in life everlasting or eternal cycles of rebirth; there is speculative physics that would permit us to rewind the ticking clock of entropy by creating our own “pocket universes”; there are also striking views from metaphysics, including liberal conceptions of personal identity compatible with immortality beyond physical death, and theories positing a plenitude of exotic person-like entities lurking in the penumbra of every person. (See section 11.3 for discussion.)

Appeals to religion, science fiction, or metaphysics may not impress you. But remember, our topic is *very small probabilities*. So even if you are unmoved to anything like *belief* in eternal life or eternal inflation, if it *might* be so, with non-zero probability, that is enough to drive the puzzles and arguments that follow.

Furthermore, even if infinity isn’t realistic, merely hypothetical situations are a useful testing ground for moral principles. I’ve never actually encountered a child drowning in a shallow pond, I’ve never had the opportunity to push anyone from a footbridge into the path of a runaway trolley, and I’ve never been surgically connected to a comatose violinist. You probably haven’t either. Even so, thinking about what I should do if I did can lead to moral insights. If there are truly general moral principles about what is better than what, then they should extend even to far-fetched realms. And if there are *no* such truly general principles, this would also be an important discovery.

Remember again, our topic is the *very large*. We surely do sometimes face options that can make a difference to vast numbers of people, or which might have effects that linger for vast spans of time. Many of the puzzles and arguments in this book arise even if we face not truly infinite consequences, but merely arbitrarily large finite consequences. Moreover, when we are studying the very large, a tool that has shown itself to be reliably useful since antiquity is thinking about the infinite limit.

So our journey will take us through some weird realms. But the destination we hope to reach eventually is not remote at all. We face big decisions right now, in the real world — decisions like Medora’s career planning, but also many others. Policy makers must decide how to allocate resources to prevent major catastrophes such as extreme climate change or nuclear winter.

They must decide when to prioritize proven interventions over “moon shot” projects. And even when we do not have responsibility for broader policies, individuals arguably face big decisions all the time, when we make our own tiny contributions to huge endeavors. One vote is very unlikely to swing a major election — but the results of close elections can have enormous consequences (for discussion see Barnett 2020, and references therein). Eating one chicken sandwich is very unlikely to result in the expansion of factory farming, but a single day on a single such farm can cause intense suffering for tens of thousands of birds (for discussion see Isaacs, Lerner, and Russell forthcoming, and references therein). We also face personal choices of what causes to support with our time and money — as well as how to cultivate our perhaps-immortal souls.

1.2 Some background ideas

1.2.1 Expectations

A tsunami is approaching two islands. You are driving a speedboat, and you only have time to rescue the people from one of them.

There is one person on *Sure Island*.

You don’t know how many people there are on *Unsure Island*. There might be three people, or there might be none, and both options are equally likely.

There are two options here, and it is uncertain which of them will turn out better. Taking the different possibilities into account, which option would be better on balance?

Probability:	1/2	1/2
Sure Island	1 life	1 life
Unsure Island	3 lives	0 lives

Since *The Port-Royal Logic*, the orthodox theory has been to calculate the value of a prospect as an *expectation*: harms and benefits are weighed in proportion to their probability.

... [I]n order to decide what we ought to do to obtain some good

or avoid some harm, it is necessary to consider not only the good or harm in itself, but also the probability that it will or will not occur, and to view geometrically the proportion all these things have when taken together. ([1683] 1996)

This way of putting things requires that we have some notion of the *size* of a good or harm. In the case at hand, there is an obvious way of doing this, according to the number of lives saved. So we may say that of these prospects, the better one is whichever maximizes the expected number of lives saved. *Sure* has probability one of saving one life, and thus its expectation is one. *Unsure* has probability 1/2 of saving three and probability 1/2 of saving zero lives, so its expectation is

$$\frac{1}{2} \cdot 3 + \frac{1}{2} \cdot 0 = \frac{3}{2} > 1.$$

So this expectational theory — specifically, call it the *Expected Number theory* — says that saving *Unsure* is better.

In general, the *expectation* of a quantity is the probability-weighted average of its various possible values. Say X is an unknown quantity which has various possible values $x_1, \dots, x_n$.⁸ Then the *expectation* of X , with respect to probabilities $\mathbb{P}$, is given by the formula

$$\mathbb{E}X = \mathbb{P}[X = x_1] \cdot x_1 + \dots + \mathbb{P}[X = x_n] \cdot x_n.$$

Multiply each possible value by its probability, then add up all of these re-weighted quantities.⁹

I have not given any positive argument for weighing prospects according to their expected number of lives saved, but this theory does have things going for it besides the weight of tradition. In the course of the book I'll discuss several strong arguments for the Expected Number theory — or in other words, for *risk neutrality* with respect to lives. But it has some troubling consequences. Consider a more extreme case.

A tsunami is again approaching two islands — in fact, this time they are two vast continents. Again you only have time to save one; everyone on the other will be destroyed.

⁸“Unknown quantities” are modeled as *random variables*; see section 1.3.4 and appendix D.

⁹See appendix D for explanation of the event notation $[X = x_1]$.

On *Big Sure*, there are certainly a billion people.

Very Unsure is almost certainly completely uninhabited. But with probability one in a billion, it is home to an unfathomably immense and prosperous civilization of a billion billion billion people.

Probability:	10^{-9}	$1 - 10^{-9}$
Big Sure	10^9 lives	10^9 lives
Very Unsure	10^{27} lives	0 lives

Which is better to save? Expected Number gives a straightforward answer. One in a billion times a billion billion billion lives is a billion billion lives, which is vastly greater than a billion lives.

$$10^{-9} \cdot 10^{27} = 10^{18} > 10^9.$$

So Expected Number says that it is better to save *Very Unsure* than *Big Sure*.

But this seems like a strange result. Saving *Very Unsure* is almost certain to do no good whatsoever, sacrificing a billion people's lives utterly in vain. If this isn't counterintuitive enough, consider *Very Very Unsure Island*, which has only probability one in 10^{1000} of a population of 10^{2000} — or make the odds as extreme as you like, while proportionally boosting the size the potential population. (Bostrom 2009 dramatizes this puzzle very effectively.)

We should be careful here. We shouldn't endorse the general principle that a prospect that is very likely to do no good is worse than one that is certain to do some substantial good. This principle would condemn ordinary good sense decisions, like buying insurance. Similarly, if you have only a small chance of carrying a very contagious and deadly disease, this can be sufficient reason to stay home rather than volunteering at the hospital. Sometimes small chances of very consequential outcomes are quite important. But even when we have taken this lesson on board, options like saving *Very Very Unsure Island* rather than *Big Sure* seem a bit ludicrous.

And it doesn't end there.

1.2.2 Pascal and St. Petersburg

An interesting thing about the history of decision theory is that some of its hardest problems are also some of its earliest. Two of these historic problems loom large in this book: *Pascal's wager*, and the *St. Petersburg paradox*. The original St. Petersburg paradox had to do with sums of money, and (as I'll discuss in chapter 2) Pascal originally offered his wager as an argument for cultivating religious faith. But both have at their core broader principles about how to weigh possibilities of infinite or unbounded goods. Both of these will receive more careful attention in later parts of the book, but here I'll briefly introduce versions of both ideas.

Pascal. Again, there are two islands in the path of the tsunami, and you can save the inhabitants of just one of them.

As before, it is certain that *Big Sure Island* has a billion inhabitants.

Pascal's Island is stranger. Almost certainly, it is uninhabited. But with probability one in a billion, it has *infinitely* many inhabitants.

Probability:	10^{-9}	$1 - 10^{-9}$
Big Sure	10^9 lives	10^9 lives
Pascal	Infinitely many lives	0 lives

It is not entirely clear how to apply the expectational theory to this case, since infinity is not a straightforward quantity like one or three. But it is very tempting to extend the theory by saying that the Pascalian prospect has *infinite* value:

$$\mathbb{E}(\text{Pascal}) \stackrel{?}{=} 10^{-9} \cdot \infty \stackrel{?}{=} \infty.$$

Indeed, Pascal ([1670] 1962, frag. 418, my translation) argued for the following principle:

This much is settled, wherever the infinite is and where there is not infinitely greater chance of loss than gain — there is no more weighing up to do: all must be given.

In short, it is better to sacrifice *arbitrarily great* finite goods for the sake of *arbitrarily tiny* chances at infinite goods. But can this be right? It is very

strange to think that we should consign a billion people to certain doom for the sake of such a ridiculous long shot. (If the long shot doesn't seem ridiculous enough, make the probability one in $10^{10^{10}}$, or whatever positive finite probability you like. You can also boost the population of Big Sure to whatever you like.) In chapter 2 I will take up the details of Pascal's argument — which I take to be widely misunderstood, and, consequently, underestimated.

Nicolaus Bernoulli (1713) introduced another famous case, which we will adapt.

St. Petersburg. Tsunami, two islands, etc.

You know that *Big Sure Island* has a billion inhabitants.

The other island is *St. Petersburg Island*, which has a peculiar history. An angel has flipped a fair coin repeatedly until it first came up heads. If it came up heads on the first flip, then two people were placed on St. Petersburg Island. If the first heads was on the second flip, then there were four people. If the third, then eight, and so on. (In the exceedingly unlikely event that the coin *never* came up heads, the island was left uninhabited.)

Probability:	1/2	1/4	1/8	...	$1/2^n$	...
Big Sure	10^9 lives	10^9 lives	10^9 lives	...	10^9 lives	...
St. Petersburg	2 lives	4 lives	8 lives	...	2^n lives	...

Which island would it be better to save? The expectational theory suggests that it is St. Petersburg island. We stated this theory in a way that only immediately makes sense for prospects with finitely many outcomes, but again it is tempting to extend it. Write out the formula:

$$\begin{aligned} \mathbb{E}(\text{St. Petersburg}) &\stackrel{?}{=} \frac{1}{2} \cdot 2 + \frac{1}{4} \cdot 4 + \frac{1}{8} \cdot 8 + \dots \\ &\stackrel{?}{=} 1 + 1 + 1 + \dots \\ &\stackrel{?}{=} \infty. \end{aligned}$$

This infinite sum diverges to infinity.¹⁰ So, we may conclude, saving St. Pe-

¹⁰Why the question marks — aren't these calculations just ordinary mathematics? In a

tersburg is a better prospect than saving Big Sure. But many people balk at this conclusion. The probability that St. Petersburg has more than a billion people is tiny — around one in a billion. It is nearly certain that saving Big Sure will turn out better than saving St. Petersburg. Furthermore, the same reasoning says that we should go for St. Petersburg no matter how many people are on Big Sure, as long as that number is finite.

Notice that unlike Pascal's wager, the St. Petersburg paradox does not involve the possibility of any infinitely good outcome — just the possibility of outcomes with arbitrarily great *finite* values. In a way, this is even more troubling than Pascal's case. Pascal said we should be willing to sacrifice arbitrarily great finite values for the sake of something that at least has a chance of being *better* than any finite value. St. Petersburg says we should be willing to sacrifice arbitrarily great finite values for the sake of something that is certain *not* to be better than every finite value. (I return to the St. Petersburg paradox and its variants in chapter 6 and sections D.8, 8.2 and 9.4.)

1.2.3 Expected utility

So far, the only reason I have offered for Very Unsure, Pascal, or St. Petersburg being better than Big Sure is that these seem to be consequences of the Expected Number theory (or natural-seeming generalizations of this theory) — and I have not given any argument at all for Expected Number. We will see in the chapters that follow that there are some powerful arguments to be made. But many people have proposed alternatives to Expected Number, which allow *risk aversion* with respect to happy lives. In this section and the next I'll introduce two such alternatives, and explain how each of them can be developed into replies to Pascal and St. Petersburg.

The first kind of theory — which is the prevailing orthodoxy in many quarters — retains the expectational idea from the *Port-Royal Logic*, but applies it differently. Instead of weighing prospects according to their expected *number of lives*, we can take the expectation of some other quantity instead — a quantity which is standardly called *utility*.¹¹ We call this *expected utility theory*. For a simple example, we might say that the utility of saving n

sense, yes, but the mathematical interpretation that would make these calculations correct is subtler than it may appear, and at this point I don't want to take for granted that this subtle interpretation is how symbols like $\mathbb{E}$ and ∞ are understood.

¹¹This idea comes from another Bernoulli, Nicolaus's cousin Daniel, in his discussion of the St. Petersburg paradox ([1738] 1954).

lives is given by the square root of n . Then these are the expected utilities corresponding to our original two islands Sure and Unsure:

$$\begin{aligned} \text{EU}(\text{Sure}) &= 1 \cdot \sqrt{1} &= 1 \\ \text{EU}(\text{Unsure}) &= \frac{1}{2} \cdot \sqrt{3} + \frac{1}{2} \cdot \sqrt{0} &\approx 0.866. \end{aligned}$$

According to this theory of the value of prospects, saving Sure is better than saving Unsure.

If we say this, we need not say that saving a second life is *less valuable* than saving a first. Rather, utilities encode how these values are to be weighed when we weigh risks.¹² The square root utility function encodes *risk aversion* with respect to happy lives: chance 1/2 at saving two is less valuable than the certainty of saving one. The same holds for any “concave down” utility function, which similarly grows more slowly as n increases. The Expected Number theory is a special case of expected utility theory, with the utility function $u(n) = n$. This represents *risk neutrality* with respect to happy lives: chance 1/2 at saving two is equally valuable as certainly saving one. There are also *risk-seeking* utility functions, which are “concave up” functions that grow more *quickly* as n increases, such as $u(n) = n^2$.

Risk-aversion is not enough to avoid the troubling consequences of the Expected Number theory. Consider again the square root utility function. This does say that saving Very Unsure is worse than saving Sure. But this conclusion is quite fragile. If the potential population of Very Unsure is a *billion* billion billion billion, rather than a mere billion billion billion, this tips the scales in its favor again. And we aren’t off the hook from Pascal or St. Petersburg, either. Consider, for example, *St. Petersburg Squared*: on this island, if heads first came up on the n th flip, then there are 2^{2^n} people. The basic trouble is that, even though the square root of n grows more slowly than n , it still gets arbitrarily large as n increases. It can be shown that for any *unbounded* utility function, expected utility theory still gives rise to both Pascalian prospects and St. Petersburg prospects that are more valuable than any merely finite benefit (Menger [1934] 1979; see sections 2.6, D.8 and 9.4).

The way to turn expected utility theory into a reply to Pascal and St. Petersburg is by deploying a utility function that has some *upper bound*. We will call this *bounded expected utility theory*. (To avoid analogous problems

¹²On the other hand, we might think that the *only* sensible notion of a cardinal scale of value — which would make sense of claims like “saving a second life is less valuable than saving the first” — arises from how we weigh risks (see Broome 1991, sec. 6.5 for discussion).

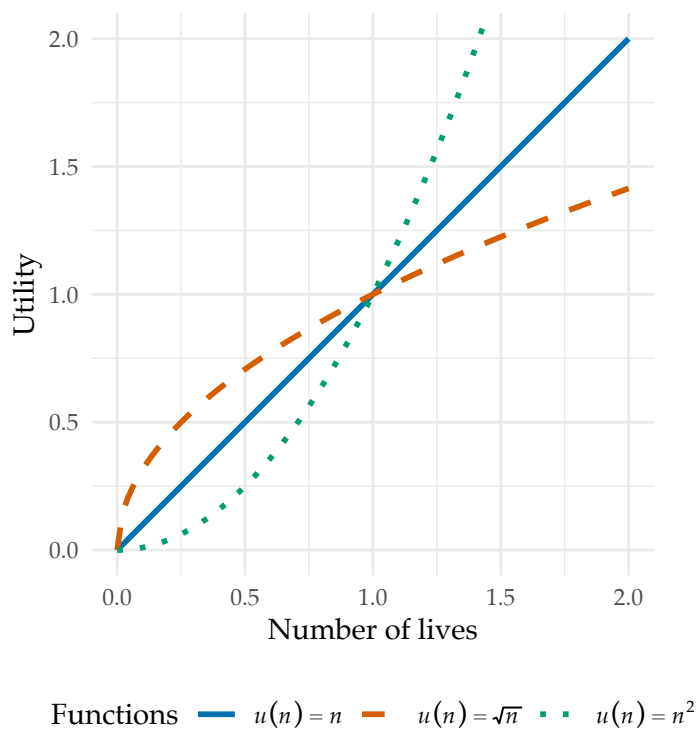


Figure 1.1: Three utility functions: linear utility encodes risk neutrality, square root encodes risk aversion, and squaring encodes risk-seeking values.

involving small chances of enormous *harms*, we will also want a *lower* bound. This has the upshot that for extremely bad outcomes, the theory will no longer be *risk-averse*, but rather *risk-seeking*.) But what could a reasonable bounded utility function look like?

One prevalent approach is *social discounting*. Suppose that we can measure individual well-being on a numerical scale from zero to one — the same scale for everyone. And suppose we can list out all of the people who will ever live in some order — say, by birth order. Then instead of weighing up each individual's well-being *equally*, we might assign earlier people greater weight. For example, we could use exponential discounting. Fix some discount factor β strictly between zero and one. Person 0 gets weight one, person 1 gets weight β , person 2 gets weight β^2 , and in general person n gets weight β^n . Then even if there are infinitely many people, the sum of weighted well-being only varies within a finite range (from 0 to $1/(1 - \beta)$). In general, social discounting will give rise to a bounded utility function over distributions of well-being as long as the discount function $\gamma(n)$ (the factor applied to individual n) approaches zero fast enough so that the sum $\sum_{n=0}^{\infty} \gamma(n)$ converges.¹³

From a technical perspective, social discounting is very attractive. It avoids nearly all of the difficulties of unbounded and infinite value that arise in this book. But taking social discounting as our theory of value amounts to giving up on the idea of *impartial* value of well-being. It is a highly *partial* theory of well-being, giving enormously greater weight to some individuals over others. Indeed, for any discounting function that gives rise to bounded social utilities, we end up treating an *arbitrarily small* increase in the well-being of one individual as outweighing an *arbitrarily large* decrease in the well-being of *infinitely many* other individuals.

I proposed that we might interpret a utility function not as telling us that saving additional lives is *less valuable* once we have already saved other lives, but rather as just telling us how to weigh risks. But the social discounting proposal cannot be interpreted that way: for in fact, social discounting revises how we rank *outcomes* even in a context of certainty. The socially discounted utility function is *not* an increasing function of the number of lives — indeed, it is not a function of the number of lives at all. If individual i 's weight $\gamma(i)$ is greater than individual j 's weight $\gamma(j)$, then the socially discounted utility of *certainly* saving i (say, taking their individual utility

¹³This account of social discounting differs in its details from the version most commonly used by economists; see below.

from 0 to 1) is greater than the socially discounted utility of *certainly* saving j .

Social discounting is more commonly applied to *times* — or sometimes “generations” — rather than ranked individuals.¹⁴ (This has a similar effect on total utility, if we suppose furthermore that the number of people at each time is bounded.) When economists are doing cost-benefit analyses, they often use what’s called a “positive rate of pure time preference”. A benefit next year might be counted for 1% less than a benefit today, and a benefit the year after that counts for 1% less again, and so on. Going on this way, the same benefit in a thousand years might be counted for more than 20,000 times less than the same benefit today. And a benefit in *two* thousand years is worth 20,000 times less than the same benefit in *one* thousand years.

There are some sensible things a discount rate could mean (see Parfit 1984, appendix F). It can make sense to discount things like wealth or food for people further in the future, if we think that the economy will grow and those people will be richer than us. It is better to give an extra sandwich to someone who doesn’t have one than to give it to someone already has several very nice sandwiches. But what we’re talking about is not discounting the value of *things* like wealth or money, but rather the value of well-being itself. And in any case, we are also concerned with planning for catastrophic futures where future people *aren’t* richer than us — for instance, because climate disasters lead to widespread agricultural collapse.

Alternatively, a discount rate might represent our greater *uncertainty* about the effects of our actions on the further future. This is generally sensible (see section 11.4 for discussion). But we shouldn’t double-count our uncertainty. Expected utility theory has two parts. First, we assign *utilities* to various possible outcomes. Second, we weigh these utilities according to their *probabilities*, representing how likely they are to come about. If we are less confident about how things will go for some people than others, this should be reflected in the probability distributions we consider.

But the idea that future people’s well-being simply *matters less*? Ramsey (1928) famously calls this view “ethically indefensible”. Parfit (1984) calls it “outrageous”. Broome (1994) calls it “reprehensible”. In any case, for now we can set it aside — though I will return to it in section 11.5, after we have

¹⁴Rather than discounting individual lifetime well-being, economists instead usually discount *consumption* during each period of time, according to when it occurs. You can think of an individual’s “consumption” as representing the contribution that some short period of time makes to that individual’s lifetime well-being.

explored the difficulties faced by impartial theories.

There is a different kind of bounded utility function that is more even-handed, and which does not appeal to social discounting. Suppose for the moment that there are only finitely many people, but we do not know how many, and once again individual well-being can be represented on a numerical scale. In this case, the sum of individual well-being can be represented by some finite number, though it may be arbitrarily large. We can then let the *utility* of an outcome be given by applying some bounded function to this total well-being (for example, as in figure 1.2).

In short, instead of calculating social utility by taking *discounting* individual well-being and then taking a *sum*, we take the *sum* of well-being, and then *discount* it. Unlike social discounting, this approach is compatible with impartiality. It turns out that it can also be extended to infinite outcomes — though we will find that it has odd features of its own, and does not escape every difficulty (sections 9.3 and 10.5).

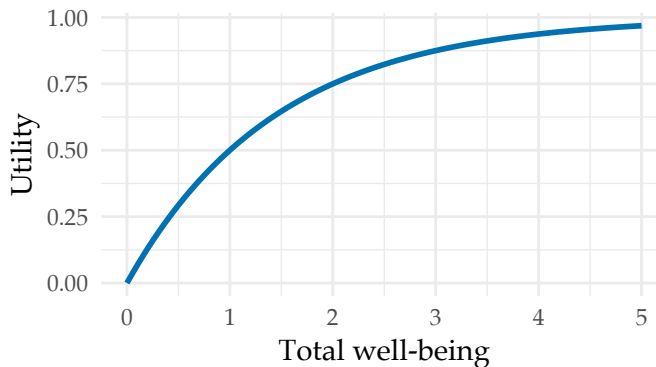


Figure 1.2: An example of a bounded utility function, $u(x) = 1 - 2^{-x}$, where x represents total well-being (assuming here that this is non-negative).

1.2.4 Risk-weighting and probability discounting

Now let's consider a second theory of risk aversion: *risk-weighting* (Quiggin 1993, under the label *rank dependence*; prominently defended by Buchak 2013). First, it's useful to take another more geometric perspective on ordinary expectations and expected utility theory. Consider for simplicity a prospect

with only finitely many possible numerical outcomes,

$$0 = x_0 < x_1 < x_2 < \cdots < x_n.$$

The expectation of this prospect is

$$\mathbb{E}X = \mathbb{P}[X = x_1] \cdot x_1 + \mathbb{P}[X = x_2] \cdot x_2 \cdots + \mathbb{P}[X = x_n] \cdot x_n.$$

This can be rewritten in terms of the *cumulative* probabilities — the probability of getting *at least* a certain value.¹⁵ The expectation of X is the same as the *area* under the graph of X 's (upward) cumulative probabilities, $\mathbb{P}[X \geq x]$ (figure 1.3). We can write this as an integral:

$$\mathbb{E}X = \int \mathbb{P}[X \geq x] dx.$$

We just considered one way of modifying this to incorporate risk-aversion, using a *utility* function. Mathematically, this amounts to first distorting the x -axis, and then finding the area under the stretched out plot (figure 1.4):

$$\mathbb{E}u(X) = \int \mathbb{P}[X \geq x] du(x) = \int \mathbb{P}[u(X) \geq t] dt.$$

But instead of distorting the x -axis, an alternative way to incorporate risk aversion is to distort the *probability* axis, by applying a *risk function* r . The result is a prospect's *rank-dependent* or *risk-weighted* expectation (figure 1.5):

$$\mathbb{E}^r X = \int r(\mathbb{P}[X \geq x]) dx.$$

We should let $r(0) = 0$ and $r(1) = 1$, to ensure that the risk-weighted expectation of a prospect with only one possible outcome agrees with that outcome. If we let $r(p) = p$ for all probabilities, then the risk-weighted

¹⁵Let $x_0 = 0$. Then, since $\mathbb{P}[X > x_{i-1}] = \mathbb{P}[X \geq x_i] = \mathbb{P}[X > x_i] + \mathbb{P}[X = x_i]$, we have the following, by “telescoping” the sum:

$$\begin{aligned} \mathbb{E}X &= \sum_{i=1}^n \mathbb{P}[X = x_i] \cdot x_i \\ &= \sum_{i=1}^n \mathbb{P}[X = x_i] \cdot x_i + (\mathbb{P}[X > x_i] \cdot x_i - \mathbb{P}[X > x_{i-1}] \cdot x_{i-1}) \\ &= \sum_{i=1}^n \mathbb{P}[X \geq x_i] \cdot (x_i - x_{i-1}). \end{aligned}$$

This is the same as the integral $\int \text{cdf } X$.

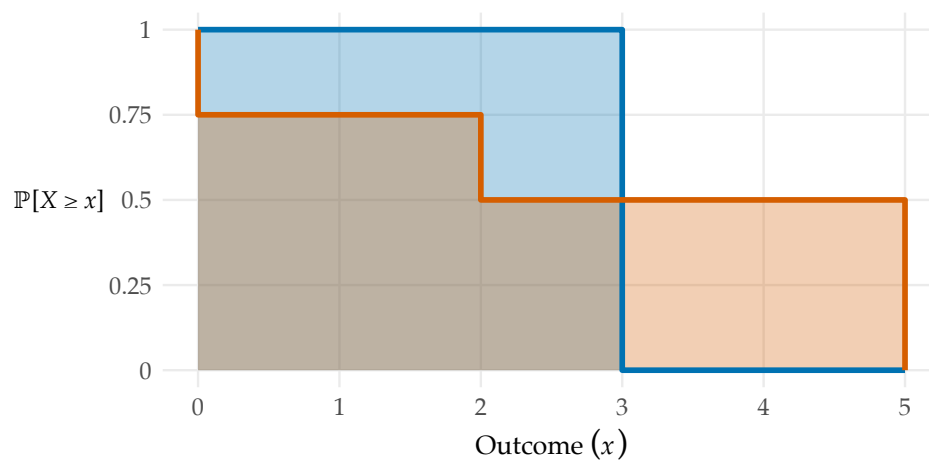


Figure 1.3: The expectation of X is the area under the graph of the (upward) cumulative probabilities $\mathbb{P}[X \geq x]$. In this case the two variables have the same expectation.

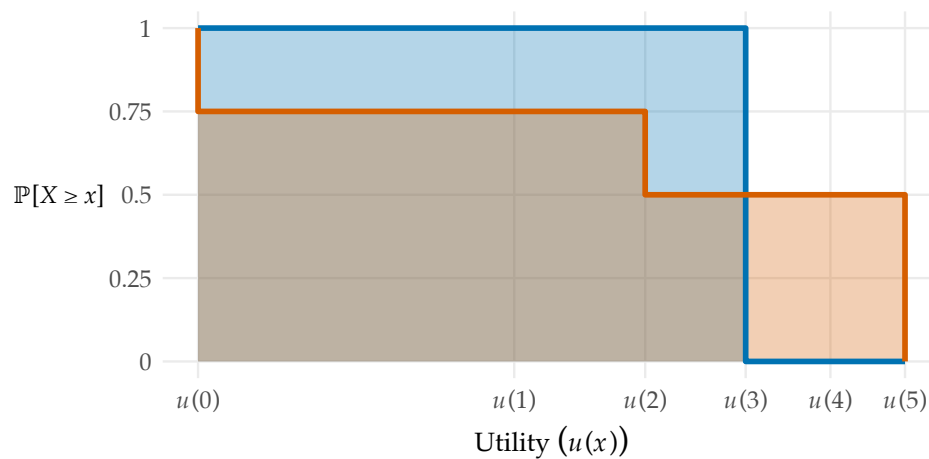


Figure 1.4: Expected utility distorts the x -axis. Applying a concave-down utility function (in this case $u(x) = \sqrt{x}$) favors the lower-variance distribution.

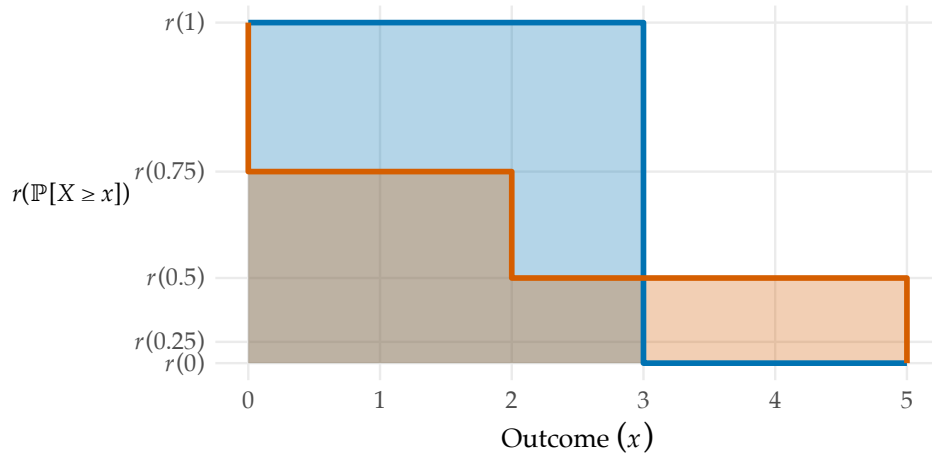


Figure 1.5: Risk-weighted expectation distorts the probability axis. Applying a concave-up risk function (in this case $r(p) = p^2$) also favors the lower-variance distribution.

expectation is exactly the same as the ordinary expectation. But if r is some other increasing function of probability, then we can get quite different results. For example, consider $r(p) = p^2$. Then we end up evaluating Unsure Island (from the earlier example: probability 1/2 of saving no lives, and probability 1/2 of saving three) as having value

$$1^2 \cdot 0 + \left(\frac{1}{2}\right)^2 \cdot (3 - 0) = \frac{3}{4} < 1.$$

So this theory recommends saving one life with certainty rather than the unsure prospect.

(The decision theory of Quiggin (1993) and Buchak (2013) actually incorporates *both* of these two ideas — both a utility function u and a risk function r . It is thus a theory of *risk-weighted* expected *utility*.)

As with non-linear utilities, risk-weighting by itself does not get us out of Pascal or St. Petersburg. For example, consider the risk function $r(p) = p^2$. At first this seems to help: the risk-weighted expectation of St. Petersburg Island no longer corresponds to a sum that diverges to infinity. But the relief is short-lived: once again, *St. Petersburg Squared* puts us right back where we were.

In the case of expected utility, only certain utility functions helped: namely,

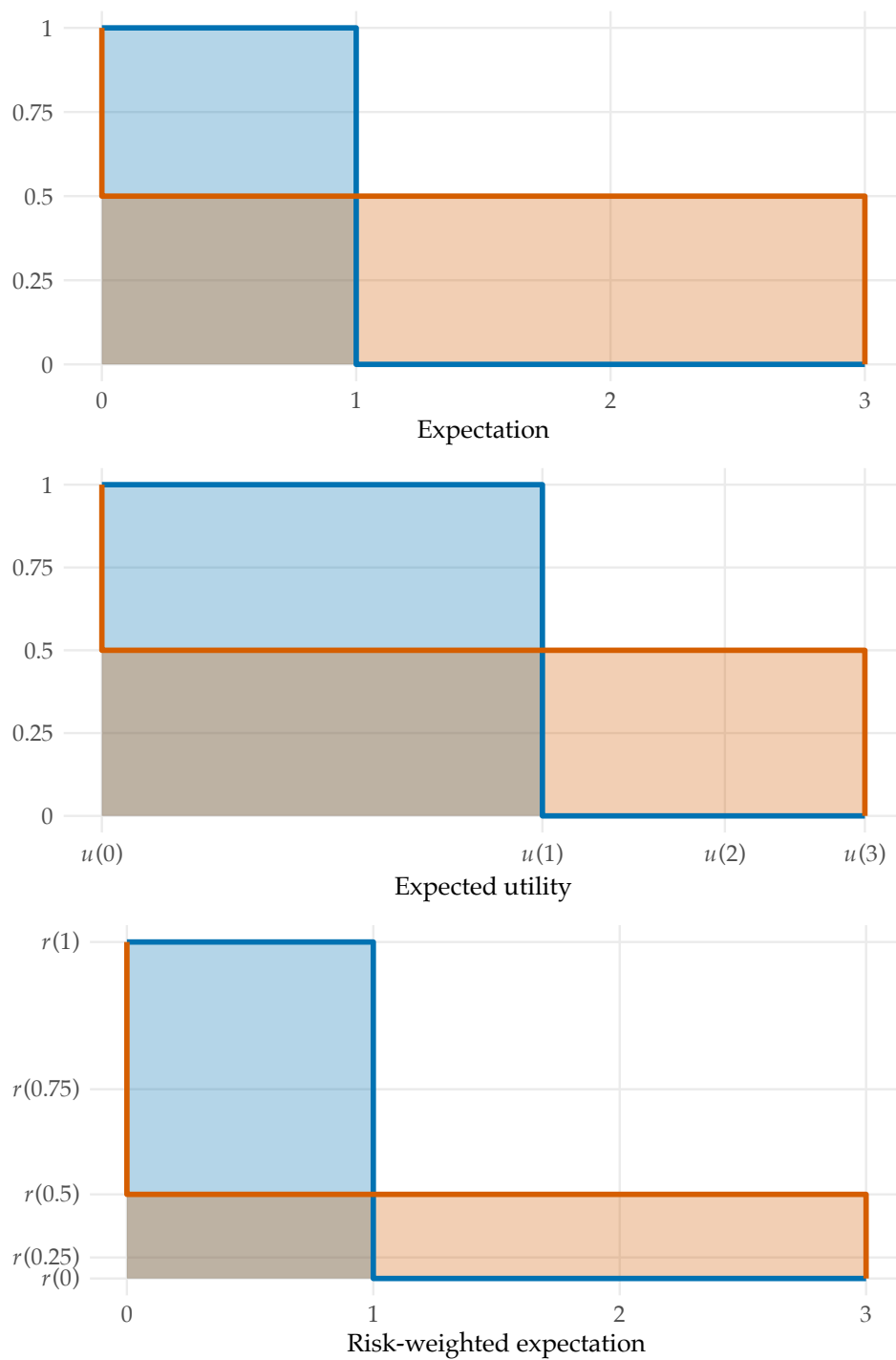


Figure 1.6: Even though Unsure saves more lives than Sure in expectation, Sure has a greater *expected utility* than Unsure (where $u(n) = \sqrt{n}$), and Sure also has a greater *risk-weighted* expectation than Unsure (where $r(p) = p^2$).

bounded utility functions. In the case of risk-weighting, once again only certain risk functions will help: we can let the risk function r take the value zero for sufficiently small probabilities (as suggested by Buchak 2013, 73–74). Then if the probability of saving a large number of lives is sufficiently small, it effectively gets ignored in the utility calculation. Call this theory *tail discounting*.

What we have arrived at is a version of an old idea, advocated by several 18th century authors as a reaction to the St. Petersburg paradox: *probability discounting*. Nicolaus Bernoulli asserts that “cases that have a very small probability must be neglected and assumed to be zero” (in a 1714 letter to Pierre Rémond de Montmort, quoted by Monton 2019, 6). Buffon ([1777] 2009, sec. VIII) argues more concretely, “Each fear or each expectation which is found below [probability one in] ten thousand, must neither affect us, or even occupy us a single instant the heart or the head.” Setting aside Buffon’s specific threshold of one in ten thousand, this idea has more recently been defended by Smith (2014) and Monton (2019). (For discussion see Kosonen 2022.) From this perspective (depending on the details), it can be better to save Big Sure Island rather than Pascal’s Island or St. Petersburg Island, since each of those have only a very, very small probability of saving more than a billion people.

In general, there is some difficulty in understanding exactly how probability discounting should work. After all, if we distinguish “cases” in a very fine-grained way, it can easily turn out that *every* outcome has a very small probability. Probability discounting requires some principled way of carving up “cases”.

Tail discounting is one way of providing a principled answer to this question. The idea is that the “cases” that are neglected are the *best* and *worst* outcomes. Effectively, tail discounting says to take a prospect’s cumulative distribution function and *truncate* it, lopping off the extreme ends of its tails. Then we evaluate the remaining truncated distribution according to its ordinary expectation. (Alternatively, we could combine this with a non-linear utility function, or a non-linear risk function for middling probabilities, or both.) Tail discounting has the upshot that Pascal’s Island and St. Petersburg Island are not so good after all.

Beckstead and Thomas (2024) propose a further modification of this theory, which allows us to treat small-probability extreme-utility events as tie-breakers: instead of letting $r(p) = 0$ for small p , we can let $r(p)$ be *infinitesimally small* for small p , but still positive and strictly increasing. This

yields a version of probability discounting that respects *stochastic dominance*: see section 1.3.5. (Again, we need to do a bit more to avoid analogous problems involving small chances of enormous *harms*. For probabilities p that are very close to *one*, we can also let $r(p) = 1$, or else let $r(p)$ be infinitesimally close to one.) I will use the label *tail discounting* for this theory as well.

I have discussed a few theories of the value of risky prospects, but these are by no means the only games in town. I will not try to survey the wide range of alternatives here, but others will come up in various parts of the book.

1.3 Some foundational issues

1.3.1 What would be better

What sort of thing is a risky prospect? I have represented them with descriptions like “saving Unsure Island”, and with tables showing probabilities and numbers of lives. For many purposes we don’t have to bear down further than this: prospects are just the sort of thing that have probabilities of turning out one way or another, and which can be better or worse than one another on balance. But for some purposes it will be useful to have a more official framework for describing prospects and making comparisons between them. These details do not make much difference to the book’s central story. But I think it is better to present something specific and concrete here, even if it is controversial, rather than keeping things more ecumenical but vague.

At the most basic level, I take a prospect to be a *way the world might be* — that is, a *proposition*. The central questions of this book are about whether it would be better for the world to be one way rather than another. Would it be better if the law were passed, or if it failed? If you lived a short happy life, or one that was longer but blander? If there were infinitely many happy people, or no one at all? The general form of the question is whether it would be better if A , or if instead B .

Note that this is not essentially a question about *choice*. Some of these propositions may describe acts we might consider taking, such as “I save Unsure Island”. We can ask whether it would be better to do one thing or another, but this is only a special case: the question of what would be better than what makes sense even for hypotheses that are practically unattainable. We can perfectly well ask whether it would be better for there to be fewer wildfires without supposing that someone faces that option in a practical choice.

I often speak about *value* (a certain quantity) or what is *good* (a certain property). Officially, most of the time what I will be talking about in the first instance is a certain *relation*: the comparative relation of being (*at least*) *as good*.¹⁶ We can abbreviate this $A \succeq B$. I will take for granted some logical principles about this relation, and some other closely related relations.

- A and B are *equally good* ($A \sim B$) iff each is as good as the other ($A \succeq B$ and $B \succeq A$).
- A is *strictly better* than B ($A \succ B$) if A is as good as B , while B is not as good as A ($A \succeq B$ and $B \not\succeq A$).
- I will also assume *Reflexivity*: any prospect is as good as itself ($A \succeq A$).
- A bit more controversially, I assume *Transitivity*: if A is as good as B , and B is as good as C , then A is as good as C . ($A \succeq B \succeq C$ implies $A \succeq C$.)

In general, whenever I call a relation an *order*, I will mean that it is reflexive and transitive.¹⁷

A further principle I will *not* take for granted is *Comparability*: for any prospects A and B , either $A \succeq B$ or $B \succeq A$. (Perhaps both, if the two are equally good.) I am inclined to believe this principle is true, but it is a complicated matter; I will take up the question of Comparability in chapter 4.

1.3.2 Information-sensitive value

As Aristotle told us, “good” is said in many ways. I have done some work already to narrow the sense of “good” that is of central interest for this project: what is good for individual people, impartially, on balance. But this still bears some further precisification.

We can sensibly ask, and sometimes answer, questions of what would be better in a context where we don’t *know* for sure exactly how things would go. Maybe if the law were passed it would achieve its intended effects, or perhaps not. We can consider whether it would be better for it to pass *on*

¹⁶Sometimes “as good” is naturally understood to mean “equally good”, rather than “at least as good”. I take this to be a case of scalar implicature, in the same way that “many students passed the exam” is often naturally understood to imply “not all students passed the exam”. (For further discussion see Dorr, Nebel, and Zuehl 2023, 429, and references therein.)

¹⁷In the more standard mathematical jargon, this is to say that it is a *preorder*. See section A.2 for a discussion of other order concepts.

balance, taking into account both the possibility of success and the possibility of failure.

Consider a standard example (slightly adapted from Parfit 1988).

Miners. There is been a terrible mining accident. One hundred miners are trapped underground, and they don't have much time before toxic air turns deadly. They are all together in one of two mineshafts, labeled A and B — but we don't know which one. We have three options.

We could put all our rescue efforts toward shaft A. If that is where they are, we can save them all. But if they are in shaft B, they will all perish.

Or we could put all of our rescue efforts toward shaft B, with the opposite results.

Finally, we could split our resources. In this case we could certainly save *most* of the miners — 99 of them — but we would not be able to save all of them in time.

	Miners are in Shaft A	Miners are in Shaft B
Save A	100	0
Save B	0	100
Split	99	99

There is a sense in which splitting our efforts is certainly not the best option. If the miners are in Shaft A, then saving A would be better than splitting resources. If they are in Shaft B, then saving B would be better. Either way, Split is worse than one of the other two options.

But there is another sense in which Split might well be the best option. This sense is connected to choice: given the uncertainty involved, it may seem like a wiser and better decision to choose Split. But it is not *only* connected to choice. Suppose we learn that the Split option is logistically impossible, after all; we might regret this, feel disappointment, feel sorry for the miners, say “that’s too bad”. These responses can be appropriate even though we know that this option is not the one that would turn out best for the miners.

One traditional label for this contrast is *objective* vs. *subjective* value (see F.

Jackson 1991, 465ff. for discussion). The miners really are in one shaft or the other. If they are in shaft A, then the option Save A is *objectively* best for them, even if we are not in a position to know this. If they are in shaft B, then instead Save B is objectively best. Either way, Split is not objectively best. In contrast, Split may be the best option in a more *subjective* sense. Another label for the contrast is “evidence-relative”, rather than “fact-relative” value.

Yet another label for the contrast is *ex ante* vs. *ex post* value. (This is the label I will generally favor in this book.) Those labels come from an idealized picture where we make *ex ante* evaluations *before* all of the relevant information about the world is revealed, while we make *ex post* evaluations *after* this. But of course we may never get such complete information; thus, so-called “*ex post*” evaluations only really come “after” something that may never occur.

In fact, there are not just *two* ways of understanding “better”, but very many. We can ask what is better than what with respect to *any* body of information. We might ask which option is better given whatever is actually true about where the miners are. We might ask which option is better in light of less information than this — say, just the information presented in the vignette, which says they are in one of two shafts but does not say which. But there are many other questions we could ask, though they do not stand out as so immediately important. Which is best supposing this coin comes up Heads? Given all the actual facts about the miners’ birthdays? If they are especially tall? In short, *value is information-sensitive*.

This information-sensitivity is a general feature of much of our moral language — not just “better” and its cognates, but also terms like “ought”, “should”, “permissible”, and “right”. Here is how Broome (1991, 128 and 130) puts the point.

For every set of probabilities, I suggest, there is an associated rightness. It is not really that ‘right’ has many senses. It has one sense, which is relative to particular probabilities. ‘It was right to go sailing’ and ‘It was not right to go sailing’ do not contradict each other, unless they are meant to be understood relative to the same probabilities.

... The [objective] rightness that Moore speaks of, which is determined by what would actually happen, may be thought of as the limit of probability-relative rightness. It is rightness relative to the set of probabilities that gives probability one to what would actually happen, and nought to everything else. It gives

probability one, that is, to the actual state of nature.

... [The view under discussion] should therefore take good as probability relative, like right. And everything I have been saying about rightness will apply to goodness too.

Broome talks about relativity to *probabilities*, where I prefer to talk about sensitivity to *information*. Rather than the relation *A is as good as B relative to probabilities P*, I instead focus on the relation *A is as good as B supposing E*, where *E* is a third proposition.¹⁸ I do not think this difference goes very deep.¹⁹ One reason I prefer to talk about information-sensitivity rather than probability-relativity is that it seems more intuitive: “It would be better if *A* than if *B*, supposing *E*” is a perfectly understandable expression in natural language (once the placeholders are filled in, of course), whereas “*A* is better than *B* relative to probabilities *P*” is a philosophical term of art. Another weightier reason is that we will encounter difficulties about how probabilities should be formally represented (chapter 5). I don’t want to build some opinionated take on probability representations right into our framework for asking value questions.

1.3.3 Counterfactual value

There is another choice point regarding what kind of value will be our focus. We can illustrate the contrast with an example.²⁰

¹⁸In appendix C I also sometimes have use for a *four-place* relation, *A, supposing E, is as good as B, supposing F*. For example: “It would be as good to put our resources to shaft A, supposing the miners are in shaft A, as it would be to put our resources to shaft B, supposing the miners are in shaft B.” The same four-place move could be made with probabilities rather than hypotheses.

¹⁹Fix some *prior probability function* P^* . This might represent very uninformative “ur-priors” (see Meacham 2020 for discussion), but this is not necessary; they just have to be uninformative enough to support whatever suppositions that are relevant in the context of discussion. Then, if you like probability-relative value, you can understand “*A* is as good as *B* supposing *E*” as “*A* is as good as *B* relative to the conditional probabilities $P^*(\cdot|E)$ ”. Conversely, if you like information-sensitive value, then you can understand “*A* is as good as *B* relative to probability function P ” as “*A* is as good as *B* supposing *E*” where *E* is some hypothesis such that $P^*(\cdot|E) = P$. (This direction requires that the prior is sufficiently rich so that such hypotheses exist; I defend this richness assumption in appendix C.)

²⁰This is a variant of the kind of “medical Newcomb problem” discussed by Gibbard and Harper (1978).

Miguel gets painful migraines, which are often triggered by low blood sugar. Low blood sugar also sometimes triggers strong food cravings. In these cases he often eats some chocolate. This doesn't prevent the migraine, but it's nice. Miguel loves chocolate, and he is not prone to overindulge, so the chocolate is not bad for him in itself — it is a net positive. But eating chocolate is bad news: if we find out that Miguel is eating chocolate, we can conclude that a migraine will probably follow soon.

Here are two different things we can say about this case.

- Suppose we find out that Miguel is eating chocolate. This is bad news. If he is eating chocolate, he is likely to have a migraine soon and be worse off.
- Eating chocolate does not *make* Miguel worse off. Suppose he is not in fact eating chocolate. If he *were* to eat a bit of chocolate now, he would be a bit better off than he is.

Both of these kinds of statement are perfectly sensible. The first is about *news value*: the new information that Miguel is eating chocolate is bad news. The second is about *counterfactual value*: Miguel is not worse off under the counterfactual supposition that he eats chocolate. (We should understand this counterfactual as holding his blood sugar fixed, which is causally “upstream” of the craving and eating. It is not a “backtracking” conditional (D. Lewis 1979).)

There is ongoing debate over which of these two kinds of value is more central to questions about choice: *evidential decision theorists* say it is news value, while *causal decision theorists* say it is counterfactual value. (For a nice overview, see Joyce 1999, Introduction.) I am broadly on the side of the causalists, and largely for this reason I will focus on counterfactual value (when it makes any difference). But we do not need to adjudicate this debate in decision theory in order to proceed the way I am going. Questions about news value and questions about counterfactual value are both perfectly sensible, and worthy of investigation. (In fact, my preferred framework has room for both of them.)

Note that we can sensibly ask whether things *would* be better a certain way, counterfactually, even in contexts where we know perfectly well that things are *not* actually that way. Here is Miguel before us, eating celery sticks. We can say he would be better off if he was eating chocolate instead. In

contrast, in this context where we know Miguel is not eating chocolate, it is hard to make any sense of the question, “Is it worse if he is actually eating chocolate?” (Trying to interpret this sends us searching for some other state of information for evaluating the question, different from the information we actually have.)

Another important feature of counterfactual value, in contrast with news value, is that it allows us to straightforwardly make sense of *dominance* relations between prospects. Dominance ideas are among this book’s central cast of characters. We say that a prospect *A* **(weakly) statewise dominates** a prospect *B* iff it is certain that *A* would turn out as well as *B*. This is a different idea from *A* being as good a *prospect* as *B*, *ex ante*; dominance is rather about the *ex post* value of *A* and *B*, how they would in fact turn out. There is a corresponding family of moral principles that link *ex post* and *ex ante* value.

Statewise Dominance. Let *A* and *B* be prospects.

If it is certain that *A* would turn out as well as *B*, then *A* is as good as *B* on balance.

If it is certain that *A* would turn out as well as *B*, and furthermore, *A* might turn out strictly *better* than *B*, then *A* is strictly better than *B* on balance.

(The word “statewise” in the label comes from the “states of nature” of Savage’s framework, discussed in section 1.3.4 below. But despite the terminology, neither the idea of statewise dominance nor this moral principle is essentially tied to Savage’s framework.)

In our journeys through dark and treacherous realms ahead, I will treat Statewise Dominance as a guiding light. I think that some version of this principle is central to the concept of *ex ante* value.

The whole reason for *caring* about which prospect is better than which given incomplete information is because we care about how well things *actually* would turn out. Sometimes we can’t be sure what would turn out best, and in these cases in order to compare prospects we must weigh up the various ways things might go better or worse against each other, in a way that hopefully points us in the right direction. But when we *are* sure what would turn out best, then there is nothing left worth worrying about — the right direction is obvious, and this is where our theory of value should point

us.²¹

That is not to say there is no room for debate. Indeed, I will consider challenges to these dominance principles in chapter 5 and chapter 6.

A terminological note before we move on. I will use “Statewise Dominance” as a name for the two-part principle. The first part is about what I am calling the relation of (*weak*) *statewise dominance* — being certain to turn out at least as well. The second part is about the “strict part” of that same relation (*A* dominates *B*, and *B* does not dominate *A*). But it is worth noting that this terminological choice comes into conflict with other common ways of using “weak dominance”. In my defense, I can plead that any choice I might make would run into similar problems, because the terminological conventions for structurally analogous ideas in different relevant fields (and sometimes within fields) are inconsistent (see table 1.6). Throughout the book, I will sometimes need to introduce names for various dominance-style relations and principles; I’ll do my best to avoid confusion, but the naming situation can never be really satisfying.

²¹Compare the arguments of Broome (1991, 129); Schoenfield (2014, 268); Hong and Russell (2025, sec. 3.1). See further discussion in section 4.5; section 5.1; section 9.4.

Tarsney, Lederman, and Spears (2025) oppose this kind of argument.

But this concern [with actual value] does not justify an insistence on statewise comparisons. Instead, this insistence represents (in our view) a mistaken concern with *counterfactuals* — with comparing the actual outcome of one’s choice to *what would have happened, if one had made a different choice in the same state of nature*. There is no particular reason, insofar as one is ultimately concerned with *how things actually turn out*, to privilege this counterfactual outcome over all the other possible unrealized outcomes of the gambles under consideration (which are all equally non-actual).

In my view, such concern with counterfactuals is no mistake at all (though others can have different preoccupations). We do need to be careful how we articulate the idea that actual value guides the value of prospects. Out of all the various epistemic possibilities concerning how well each option would turn out (if it obtained), one of these possibilities is the actual truth of the matter. The view I am expressing is that *information-sensitive* value comparisons of counterfactual value are “aiming at” this perfectly *informed* value comparison of counterfactuals. When there is no doubt what would turn out better, then our assessment of what would be better *on balance*, given our incomplete information, follows suit. (For further discussion see section C.1; see also Bacon 2022.)

Table 1.6: Standard terminology for different dominance relations from different fields.

	Social choice	Game theory	Statistics/ real analysis
Relations			
$\geq$ everywhere (R_1)	weak Pareto dominance	(very) weak dominance	(weak) (stochastic) dominance
$\geq$ everywhere and $>$ somewhere (R_2)	(strict) Pareto improvement (or dominance)	weak dominance	strict (stochastic) dominance
$>$ everywhere (R_3)	(strong) Pareto improvement (or dominance)	strict (or strong) dominance	strict dominance
Principles			
R_1 implies $\succeq$	monotonicity	–	(weak) monotonicity
R_2 implies $>$	strong Pareto principle	–	strict monotonicity
R_3 implies $>$	weak Pareto principle	–	–

1.3.4 States, outcomes, and random variables

As I have said, officially prospects are propositions; we make comparisons between them as to how good it would be if they were true, given some informational context. But while this is my preferred way of thinking about the foundations for our project, it is often useful to describe prospects in other terms, which abstract away from this framework. In this book we will often deploy two standard formalisms from decision theory: one associated with Savage (1954), and another associated with von Neumann and Morgenstern ([1944] 2007). These two frameworks are not in competition with one another. Rather, they each abstract away from different details. Let's take them one by

one. In this section I'll consider the Savage-style framework. (In section C.3 I also present a formal framework that corresponds more directly to my preferred foundational picture, and discuss in more detail how the three frameworks fit together.)

It is familiar in philosophy to represent propositions as sets of (conceptually) *possible worlds*: each possible world is a conceptually coherent and logically complete way the world might be. Rather than fully detailed possible worlds, it is helpful to consider more abstractly just their features that are immediately relevant to questions of value. The Savage framework deploys two different abstractions: *possible outcomes* and *states of nature*.

It is important to understand that, at least the way I will deploy this framework, there is not some single correct choice of what counts as an outcome, or what counts as a state of nature. Whether certain representational choices are appropriate will be sensitive to what we are interested in representing — in particular, which value questions we are asking.

Take *possible outcomes* first. One key question about a world is how good it is — that is, how good it is *ex post* (“objectively”), since a complete possible world leaves no questions open. An *outcome* represents this aspect of possible worlds. It really doesn't matter what kinds of things these outcomes are. If you like, you can think of them as propositions as well — perhaps identified with equivalence classes of equally good possible worlds. But it is often more convenient to use other abstract objects instead: for example, *numbers* of lives saved, or *sets* of happy people, or *functions* from individuals to their level of well-being. We can use anything we like as an outcome, as long as it encodes enough information to make value comparisons between the possible worlds we are immediately interested in, in a given context. More precisely, talking about outcomes is to posit some function O that takes each (relevant) possible world w to an outcome $O(w)$, with the property that if $O(w) = O(w')$, then w and w' are equally good.

The second key question arises only once we have selected certain prospects for evaluation. It is standard to call these prospects the *available options* (or *acts*). They might represent the available choices in some decision context, but that is not at all essential. They are just the propositions we are interested in evaluating, for whatever reason. Once we have chosen these, the second key question we can ask about a possible world w is how each available option would turn out, supposing that w is actual. A *state of nature* represents this aspect of possible worlds: the relevant *dependency hypotheses*. Just as with outcomes, we can choose any objects we like to play the role of states of

nature, as long as they encode enough information to distinguish between the relevant dependency hypotheses. When we talk about states, we posit some function S that takes each (relevant) possible world w to a state $S(w)$, with the property that if $S(w) = S(w')$, then every available option would result in the same outcome in w as it would in w' .²²

Using these two abstractions, we can represent each available option as a function from states to outcomes — a *Savage function*. In particular, if the proposition A is available, then it can be represented by the function that takes each state s to the unique outcome that *would* result from A , supposing that the actual state of nature is s . (This kind of function is often called a *Savage act*, but this terminology is over-restrictive, since we need not restrict our attention to prospects representing possible *actions*.)

We can conveniently represent Savage functions using *decision tables* — like the miners table from section 1.3.2, for example. (Again, though, the terminology is overrestrictive, since there need not be any *decision* involved at all — the propositions up for evaluation need not be potential objects of choice.) In this case, we can use numbers to represent outcomes, since we are supposing that the only thing relevant to how well things turn out is how many miners are saved. Our three available propositions are “We save A”, “We save B”, and “We split our resources”. How well each of these hypotheses would turn out is determined by where the miners are: for example, if the miners are in Shaft A (actually), then if we were to save A (counterfactually), 100 miners would be saved. So we can abstract from all of the details about the state of the actual world besides where the miners are, using these two propositions as our relevant “states of nature”.

	Miners are in Shaft A	Miners are in Shaft B
Save A	100	0
Save B	0	100
Split	99	99

In general, columns of the table represent states of nature, rows represent available propositions, and in each cell of the table we write the outcome that would obtain if its corresponding proposition were true, supposing its

²²There are some issues here concerning the principle of *counterfactual excluded middle*; see section C.3.

corresponding state of nature actually *does* obtain.

We can use this kind of representation for any propositions, as long as we appropriately abstract from possible worlds to states and outcomes. But when we do this, often we are making the tacit assumption that the table includes *enough information* to compare the value of prospects. This is a substantive assumption. The following principle suffices:

Statewise Equivalence. If it is certain that A and B would turn out equally well, then A and B are equally good on balance.

If this holds, then it follows that any prospects that are represented by the *same* Savage function are equally good. Thus abstracting from the prospect to its corresponding Savage function loses nothing as far as the value of prospects is concerned. Statewise Equivalence is a consequence of the Statewise Dominance principle stated in section 1.3.3. (For if A and B are certain to turn out equally well, then each of them weakly dominates the other.) I am happy to take it for granted in what follows.

One thing I have not mentioned in this section is *probability*. Savage left this out of his framework deliberately: it was part of his project to *reduce* probability to preferences between acts. But that is not part of my project, and normally we will have some other grip on the probabilities involved. Typically this will be represented by some probability measure over states of nature — though as I have said, there are some details about how to do this to be discussed later. When we include probabilities for sets of states, this gives the state space the structure of a *probability space*. In this setting, functions from states to outcomes are what statisticians call (outcome-valued) *random variables*.

1.3.5 Lotteries

We will also sometimes represent prospects using the formal framework of *lotteries* used by von Neumann and Morgenstern ([1944] 2007) — VNM, for short. This abstracts from possible worlds and counterfactuals in a different way.

The first abstraction is the same as in the Savage-style framework: once again, we will abstract from worlds to *outcomes*, which are certain features of the world that suffice to settle how good things are overall. In the VNM framework, though, we dispense with states of nature.

For now I'll focus on a simple case. A *simple prospect* is one that can only have

finitely many possible outcomes. Let us also suppose that we have some probability measure $\mathbb{P}$ over the space of possible worlds. Then any simple prospect A has a corresponding *finite probability distribution* over outcomes, $\mathbb{P}_A$: this assigns each outcome x the probability that A would result in x . Such a probability distribution over outcomes is called a *lottery*.

As we saw before in the case of Savage functions, making value comparisons between prospects based on their lottery representations involves an implicit assumption: namely, that lotteries includes enough information for such comparisons. Here is a principle that suffices:

Simple Stochastic Equivalence. For simple prospects A and B , if the probability that A would result in x is the same as the probability that B would result in x , for all outcomes x , then A and B are equally good. In short:

$$\text{if } \mathbb{P}_A = \mathbb{P}_B \text{ then } A \sim B.$$

Again, this is an assumption I am generally happy to make — though it is not entirely uncontroversial, and it will require some examination in what follows (see chapters 4 and 5). If it holds, then it makes sense for us to speak of the goodness relation between (at least some) *simple lotteries*.

The lottery framework leaves out information about how the possible outcomes of different prospects are *correlated* with one another. Does A turn out well in cases where B also would turn out well, or rather in cases where B would turn out badly? According to Simple Stochastic Equivalence, these correlations are irrelevant to the value of prospects. But without this information, *dominance* principles become more difficult to state. Consider an example.

Some time ago, three people — Amy, Bela, and Cleo — set out to populate some uninhabited islands. They made a peculiar arrangement. First, they flipped a fair coin.

If the coin came up Heads, then Amy set out for **Basic Island**. If it was Tails, she went home to safety on the mainland.

If that same coin flip came up Heads, Bela and Cleo went to **Correlated Island**. If it was Tails, then they went to **Anti-Correlated Island** instead.

Now all three islands are threatened by a tsunami.

Table 1.8: Correlated statewise dominates Basic, and Anti-Correlated is stochastically equivalent to Correlated.

	Heads	Tails
Basic	1	0
Correlated	2	0
Anti-Correlated	0	2

Saving Correlated Island is sure to save at least as many people as saving Basic Island, and it has probability 1/2 of saving strictly more. So Statewise Dominance tells us that Correlated is better than Basic.

Saving Anti-Correlated might turn out worse than saving Basic Island: if the coin came up Heads, then that would mean saving no one. So Anti-Correlated does *not* statewise dominate Basic. Nonetheless, it seems clear that Anti-Correlated is *better* than Basic. One way to argue this is by observing that Anti-Correlated is *stochastically equivalent* to Correlated, which in turn *statewise dominates* Basic.

Lotteries don't include enough information to capture statewise dominance relations. But there is a closely related relation that they *do* capture, which is called *stochastic dominance*.²³ We say that a lottery P **(weakly) stochastically dominates** a lottery Q iff there are *some* possible prospects A and B with distributions P and Q , respectively, such that A statewise dominates B .²⁴ Say that P **strongly stochastically dominates** Q iff there are some A and B with those distributions such that A statewise dominates B while B does not dominate A .

There is a corresponding moral principle, which will be another of our central

²³Throughout this book, I use "stochastic dominance" synonymously with "first-order stochastic dominance".

²⁴A bit more officially: P stochastically dominates Q iff there are random variables X and Y on some probability space (Ω, R) such that the distributions of X and Y are P and Q respectively, and X statewise dominates Y . There are other equivalent ways of formulating the relation of stochastic dominance — but we should be cautious, since some of these convenient equivalences break down in infinite contexts.

For example: for simple lotteries, strong stochastic dominance is equivalent to the *strict part* of weak stochastic dominance: P strongly stochastically dominates Q iff P weakly stochastically dominates Q and Q does not weakly stochastically dominate P . Unfortunately, this is one of the equivalences that breaks down more generally.

(See chapter 5 and sections D.3, E.4, F.6 and 8.4. See also J. S. Russell (forthcoming).)

cast of characters.

Simple Stochastic Dominance. Let P and Q be simple lotteries representing prospects.

If P weakly stochastically dominates Q , then P is as good as Q .

If P strongly stochastically dominates Q , then P is better than Q .

Each of these principles follows from the combination of Simple Stochastic Equivalence with its corresponding Statewise Dominance clause.

Since Simple Stochastic Equivalence is the principle that makes lotteries an appropriate formal representation for simple prospects in the first place, within this framework stochastic dominance has the same status as statewise dominance — which is to say, these moral principles are eminently plausible, and generally I am willing to take them on board. We will encounter some subtleties and complications, though (especially in chapter 5 and chapter 8).

Since the VNM lottery framework abstracts away from correlations between the outcomes of different prospects, in a way it is more ecumenical than the counterfactual or Savage formalisms. Lotteries are equally useful for determining the *news value* of a prospect, as opposed to their *counterfactual* value. Likewise, stochastic dominance is a more generally powerful tool than statewise dominance, since we can apply it even in cases where it is unclear how to “line up” the outcomes of different lotteries with a shared state space (see Bader 2018; Tarsney, Lederman, and Spears 2025).

I prefer to think of the lottery framework as an abstraction from a more basic story about counterfactual value. Occasionally this perspective will make some difference in the arguments that follow, but often it won’t matter. People who prefer to think about the significance of value comparisons between lotteries in some other terms, rather than the kind of counterfactual value that I am treating as central, can take away their own morals.

1.4 Roadmap

The central question of this book is what difference tiny probabilities of vast differences in value make to the value of prospects. There is a natural starting point for our investigation: Blaise Pascal’s notorious *wager* ([1670] 1962) — not only the earliest discussion of our topic I am aware of, but possibly the very earliest argument in decision theory. In the course of setting out his wager, Pascal advocates this principle: “wherever the infinite is, and

where there is not infinitely greater chance of loss than gain, there is no more weighing up to do: all must be given.” I call this *Pascal’s Principle*; one guiding question of this book is whether it is correct.

Pascal’s argument for his principle rests on a premise of *Risk Neutrality* with respect to happy lives — essentially, this is the Expected Number theory introduced in section 1.2.1. This premise can in turn be replaced by a weaker principle of (*Finite*) *Fanaticism*: that any certain benefit can be outweighed by an arbitrarily small chance at a sufficiently great *finite* benefit. So we will be investigating what grounds there are for all three of these interconnected principles.

Risk Neutrality $\Rightarrow$ Finite Fanaticism $\Rightarrow$ Pascal’s Principle

After this introduction, the book is divided into three parts.

I. The Structure of Infinite Value. In chapter 2 I set out Pascal’s argument for his principle. This argument is both powerful and — despite centuries of vigorous reactions to Pascal’s wager — surprisingly neglected. I also defend Pascal’s principle from charges of incoherence.

This discussion sets the stage for investigating a series of questions about what value in a possibly infinite world could possibly be like. In chapter 3, I defend the *Pareto* principle that making one life better makes the world better overall, even in an infinite random world. In chapter 4, I argue that value comparisons between possible worlds suffer either from rampant *value incomparability* or deep *arbitrariness*; I tentatively defend the second horn. In chapter 5, I argue that properly representing the value of uncertain prospects over infinite worlds goes beyond the resources of standard probability theory, and advocate a heterodox theory that admits *infinitesimal* probabilities. In chapter 6, we confront *paradoxes of dominance*: different structurally similar principles relating the value of prospects to value at worlds, times, events, or individuals conflict with one another. If we are to carry on the project of infinite ethics, we must distinguish correct dominance principles (if there are any, as I believe) from their *bad company*.

A running theme of part I is the rejection of *isomorphism* principles. There are structurally isomorphic possible worlds that are not equally good (*ex post*); there are structurally isomorphic propositions that are not equally probable; there are structurally isomorphic prospects that are not equally good (*ex ante*).

Another task throughout part I is to develop a working model for evaluating infinite prospects. My attitude toward this model is not that it ought to be our final theory, and there are many aspects of it that I will not defend. Rather, this model is a proof of concept, demonstrating the consistency of various desirable principles. It is meant to be a kind of simple baseline theory, against which future theories of value and risk can be compared.

II. Weighing Small Probabilities. Part I serves as a kind of prolegomenon, clarifying various issues and setting up a clean theoretical space in which we can approach the questions of Pascal's Principle and Finite Fanaticism. In part II, I evaluate several arguments for these principles. (These four chapters can each be read more or less independently.)

The central topic of chapter 7 is the *ex ante* Pareto principle, which says that a prospect which is better for each individual is also better overall. There is a powerful argument for Risk Neutrality based on this principle. But I will argue that this power is deceptive, and the argument is not dialectically stable; I argue that in fact we should reject the *ex ante* Pareto principle in general, based on considerations about its sensitivity to individual modes of presentation.

Chapter 8 is about *separability* principles, which say that value comparisons between prospects are not sensitive to the well-being of individuals who are not affected by the choice between them: we can just pay attention to the *difference* between prospects. Separability is a very powerful idea: in fact, with very modest side premises it implies Risk Neutrality, and thus both Fanaticism and Pascal's Principle as well. In the past, I have argued that it is *too* powerful to be true — indeed, it is inconsistent with other foundational principles. But I have changed my mind about this: in fact, in the framework of non-standard probability developed in part I, consistency can be restored. The separability arguments for Risk Neutrality, Fanaticism, and Pascal's principle are likewise restored, as coherent and dialectically stable after all.

But even if separability is *false*, this does not get us off the hook. In that case, comparisons of small-scale local interventions are sensitive to the probability distribution over the well-being of everyone else in the universe. Decisions about how best to do good impartially are thus hostage to speculative considerations about what I call *cosmic background value*. I consider and extend a line of argument from Christian Tarsney, to the effect that plausibly realistic conditions on cosmic background value also push us toward a restricted version of Risk Neutrality. I discuss the relationship between this restriction, Fanaticism, and Pascal.

Chapter 9 takes up a line of argument for Fanaticism that goes via a defense of Savage's *Sure Thing Principle*. I focus on the issue of *dynamic consistency*: how patterns of value judgments for separate choices cohere with the *global* value of an overall *plan*. Many philosophers have pushed back against dynamic consistency principles. In reply, I defend a version of dynamic consistency based on game-theoretic considerations, which I call the *Equilibrium Principle*. This principle survives objections based on "time-slice rationality", infinite sequences of decisions, and unbounded utility. Thus it can serve as the basis for a further line of argument in favor of Fanaticism and Pascal's Principle.

Again, though, even if we *reject* the Sure Thing Principle, we are not off the hook. If it fails, then we cannot simply pay attention to *changes* in probability of (say) heaven or hell. The *background* level of probability can make a difference to whether a small increase in probability outweighs some cost. This is the basis for yet another line of argument for Fanaticism and Pascal's Principle, which I take up in chapter 10. A guiding intuition against Fanaticism is that outcomes with sufficiently tiny probabilities shouldn't be decisive in comparisons between prospects, no matter how consequential they might be if they were sure to come to pass. Several philosophers have pointed out that this guiding intuition is unstable: putting it roughly, you can't *always* treat tiny changes in probability as negligible, because enough tiny changes can add up to a big change. The upshot is that a certain form of *Anti-Fanaticism* is untenable. I offer a related argument for another form of *Pascalianism*. Finite costs can't *always* outweigh tiny increases in the chance at infinite bliss: in uncountably many cases they do not.

The overall upshot of part II is that the case for Pascal's Principle is quite strong. In chapter 11 I take up the question of what it might imply for our moral lives if the principle is true. Spoiler: the situation is bleak. I argue that in this case we are profoundly ignorant — *clueless* — about which of our options are best, and it is not clear whether we have any path toward escaping this cluelessness. We begin the book from the idea that the impartial value of individual well-being plays some important role in our moral lives — whether this is the all-encompassing role assigned to it by certain consequentialists, or instead some more limited role. But when we take Pascal's Principle seriously, it turns out to be very difficult to see how impartial value can play any practical role at all.

Most of the book tells an encouraging story. The ethics of huge numbers is full of dangers and paradoxes — and yet, by and large, we are able to trace a course through this forbidding territory, piecing together as we go a

coherent picture of impartial value in a vast and uncertain world. But I'm afraid part II ends on a note more of despair than triumph. If impartial value really works the way that it seems it must, then even though we have made great theoretical progress, we are left with practical mysteries. If you were hoping for practical policy guidance from this book, I'm sorry to say that I have none. Perhaps, now that we have a clear framework in place, we can make further intellectual progress on the applied questions of how to assign probabilities and values to the vast and various possible futures before us. But that task remains incredibly daunting.

III. Mathematical Appendices. Despite the inevitability of some technicalities, I have attempted to keep the main text of this book as accessible as possible to readers without much mathematical background. But this involves writing quite a few mathematical checks that only get cashed in the supplementary chapters in part III. I have attempted to write these in a way that presupposes as little background as I can get away with. But the necessary substitute for that is a great deal of patience, as we work our way through the necessary math for understanding orders, symmetries and quantities (abstract algebra), structures and non-standard models (model theory), and three different flavors of probability theory (finite probability spaces, "classical" real-valued probability, and a new theory of non-standard "quasi-probability", which is part of a family of theories that includes hyperreal measure theory).

That's the plan. Let's go.

Part I

The Structure of Infinite Value

Of those countless universes, O Rāma, in some there
are only plants;
some have Brahmā, Visnu, Rudra and others as the
presiding deities,
and some have none at all;
in some there are only animals and birds;
in some there is only an ocean;
some are solid rocks;
some are inhabited only by worms;
some are pervaded by dense darkness;
in some gods dwell;
some are forever illumined.
Some seem to be heading towards dissolution;
some seem to be falling in space towards destruction.
Since consciousness exists everywhere for ever,
creation of these universes and their dissolution
also goes on everywhere for ever.

Yoga Vāsiṣṭha

Chapter 2

Pascal's Principle

Again, the kingdom of heaven is like a merchant seeking beautiful pearls, who, when he had found one pearl of great price, went and sold all that he had, and bought it.

Matthew 13.45–46

Many of us are in the dark about many important questions. These include questions about the long-term future of humanity, the ultimate destiny of the universe, and the fate of our immortal souls (if such there be). What are the practical consequences of such profound uncertainty?

A good place to start on such questions is a notorious argument from Pascal's posthumous *Pensées* ([1670] 1962). Pascal argues that if we are ignorant about religious truths — specifically, if we do not know whether God exists, or indeed even if we are justifiably very confident that God does *not* exist — we should endeavor to live a life of religious faith anyhow.

Here I will set out and tentatively defend one central part of Pascal's argument: in particular, the part of the argument in which Pascal defends this key principle:¹

This much is settled, wherever the infinite is and where there is not infinitely greater chance of loss than gain — there is no more weighing up to do: all must be given.²

¹All Pascal quotations come from Pascal ([1670] 1962), frag. 418, my translation.

²*Cela ôte tous parti partout où est l'infini et où il n'y a pas infinité de hasards de perte contre celui*

Pascal's argument is not the same as the one that has been much discussed under the heading of "Pascal's wager". *That* argument has been roundly rejected — and generally for good reasons. But the standard "Pascal's wager" is not *Pascal's* argument — his actual argument is much better, and it is one we should take very seriously.

I will not defend the specific further conclusion Pascal drew from this principle, concerning religious faith. As I will discuss, I think it is deeply unclear what the true practical upshot of Pascal's Principle is. But it surely tells us *something* important — and troubling.³

One aim of this chapter is historical: to understand Pascal's central argument for his key principle. But quite apart from anything Pascal believed, one of the central questions of this book is whether this principle is true. So a second aim of this book will be to frame the principle and some of the central ideas that support it in a way that serves the purposes of the rest of the book, anachronistically.

2.1 The bad argument

Pascal begins his central argument thus:

Let us examine this point, then. We say: "God exists, or he does not." But which way shall we lean? Reason can determine nothing. We are separated by infinite chaos. A game is played at the far side of this infinite distance, where heads or tails will land. What will you wager?

The standard interpretation of Pascal's wager centers on a diagram like table 2.1. (See for example Hájek 2008.) We face (it is supposed) two available choices: "wagering for God" or else "wagering against God". There are (it is supposed) two possibilities. Each of these can be assigned some probability. Pascal initially considers it to be a coin flip — setting the probability of God's existence to one half. But he dispenses with that simplifying assumption

de gain. Il n'y a point à balancer, il faut tout donner.

³The *Port Royal Logic* ([1683] 1996, ch. 16, p. 275) puts the principle like this:

Only infinite things such as eternity and salvation cannot be equalled by any temporal benefit. Thus we ought never balance them off against anything worldly. This is why the slightest bit of help for acquiring salvation is worth more than all the goods of the world taken together. And the least peril of being lost is more important than all temporal harms considered merely as harms.

almost immediately. It suffices for his argument that “there is not an infinity of chances of loss against the chance of winning” — that is, God’s existence might have any positive finite probability.

Table 2.1: “Pascal’s wager”

	God (p)	No God ($1 - p$)
Wager for God	∞	finite
Wager against God	finite	finite

We then consider the “pay-offs” for each available choice. It is worth taking a moment here to set aside a second argument that some have found in Pascal, besides the classic “wager”. According to Pascal, if you wager for God and win, “you win all”. He also says “if you lose, you lose nothing.” Hájek (2003a, sec. 2; following McClennen 1994) has read into this statement an “argument from superdominance”: the idea is that wagering for God is *certain* to turn out better than otherwise. If you are right, you gain infinitely, and if you are wrong, “you lose nothing”.

I think this is a misreading. Pascal is best understood here as deliberately hyperbolic. The context makes this clear. Consider what Pascal writes immediately before presenting the wager.

Unity joined to infinity does not increase it at all, no more than a foot increases an infinite measure; the finite is annihilated [*s’anéantit*] in the presence of the infinite, and it becomes a pure nothing [*néant*]. Thus our soul before God; thus also our justice before divine justice. There is not such a great disproportion between our justice and God’s as between unity and infinity. [...]

We know that the infinite exists, and we are ignorant of its nature, as we know that it is false that there are finitely many numbers. So it is true that there is infinity in number: but we do not know what that is. [...]

We know, therefore, the existence and the nature of what is finite because we are finite and finitely extended [*étendus comme lui*].

We know the existence of the infinite and are ignorant of its

nature because it is extended like us, but it does not have bounds like us.

But we know neither the existence nor the nature of God, because he has neither extension nor bounds.

The first two quoted sentences compactly evoke one of Zeno's paradoxes, which Simplicius reports thus (as quoted in Huggett 2019):

But if when it is subtracted, the other thing is no smaller, nor is it increased when it is added, clearly the thing being added or subtracted is nothing.

Consider an infinite ray, extending in one direction. If the ray is extended by one foot, what we have is another infinite ray exactly isomorphic to the first, still of infinite length — and, it would seem, the *same* infinite length. Since adding one foot adds no length at all, it seems that it must *be* nothing at all.⁴

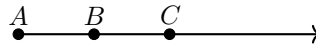


Figure 2.1: Zeno's paradox of addition: if the infinite ray $\overrightarrow{AB}$ is the same length as the infinite ray $\overrightarrow{BC}$, then "clearly the thing being added is nothing" — that is, the segment AB .

⁴Note that Huggett interprets Zeno as applying this remark to a continuous object of *finite* size. This makes some sense in the context of Zeno's other paradoxes about indefinitely dividing the continuum; but that version of the paradox is much more tendentious, and less relevant to our present purposes.

In the 9th century, al-Kindī carefully developed a version of this paradox of addition explicitly as an argument "that it is not possible that there be an infinite body" (Rescher and Khatchadourian 1965, 431; see also McGinnis 2007).

The 12th century Indian mathematician Bhāskarāchārya (Bhaskara II) offered another theological analogy, with a more Hindu inflection:

In this [*khahara*, infinity] there is no alteration, though many be inserted or extracted; as no change takes place in the infinite and immutable God, at the period of the destruction or creation of worlds, though numerous orders of beings are absorbed or put forth. (Brahmegupta and Bhascara 1817, 138)

(Colebrooke translates *khahara* as "quantity consisting of that which has cipher for its divisor", where "cipher" (*kha*) refers to either zero or an infinitesimal quantity. Sathaye (n.d.) provides helpful reconstruction of Bhāskarāchārya's system.) In contrast with Pascal, Bhāskarāchārya did not regard this property of infinity as paradoxical; instead, he developed a calculus of infinite and infinitesimal quantities that would allow statements like " $1 + \infty = \infty$ ", where ∞ represents a "zero-divided" quantity *khahara*. Sathaye (n.d.) remarks, "It is quite likely that the resulting algebraic systems turned out to be too far ahead of times and were abandoned by Bhāskarāchārya himself."

Pascal puts the point paradoxically for rhetorical effect. But he does not take the finite to be *literally* nothing. Indeed, he explicitly writes that “We know, therefore, the existence and the nature of what is finite because we are finite and finitely extended.” Rather the finite is as nothing *in comparison to the infinite* — because of the “great disproportion”. This geometric argument is a model for the practical argument that follows: finite gains and losses are not literally nothing, but they are to be considered as nothing in comparison to infinite gains and losses, so little weight do they carry.⁵ And as Pascal proceeds more carefully, he considers what is staked in the proposed wager not to be *nothing*, but rather “your life.”⁶

“You lose nothing” is a rhetorical gesture; the key point is just that what is lost, if one wagers for God while God does not exist, is merely *finite*, while what is to be gained is infinite. This is the claim that most discussion has rightly focused on.

It has become standard to articulate Pascal’s argument in the language of expected utility theory. If one wagers for God, and God exists, then the outcome is one that is assigned *infinite utility*. Meanwhile, if one wagers for God and God does not exist, or if one wagers against God, in each case the outcome is one that has *finite utility*. Then Pascal’s argument for wagering for God is interpreted as a matter of straightforward arithmetic. For any number $p > 0$, it is natural to calculate like this:

$$\begin{aligned} EU(\text{Wager for God}) &= p \cdot \infty + (1 - p) \cdot \text{finite} = \infty \\ EU(\text{Wager against God}) &= p \cdot \text{finite} + (1 - p) \cdot \text{finite} = \text{finite}. \end{aligned}$$

Thus, if one ought to choose an option that maximizes expected utility, then in this situation one ought to wager for God.

This argument has been widely rejected: for it faces decisive objections. I’ll quickly survey a few of the most influential (for overview see Hájek 2008, sec. 5.1).

Many Gods. This is an early and persistent objection to Pascal.⁷ If there is one epistemically live route to eternal happiness, then there are many. For

⁵Perhaps Pascal also intends to evoke St. Paul’s even stronger hyperbole in his letter to the Philippians: “More than that, I regard everything as loss because of the surpassing value of knowing Christ Jesus my Lord. For his sake I have suffered the loss of all things, and I regard them as rubbish, in order that I may gain Christ” (Philippians 3.8, NRSV).

⁶This is perhaps another Biblical allusion. John 12.25: “Those who love their life will lose it, and those who hate their life in this world will keep it for eternal life.”

⁷Diderot ([1746] 1950, *Addition*, LIX, my translation):

there are many competing religious claims trafficking in infinity. There are also further possibilities raised by secular science fiction or metaphysics. But if Pascal's reasoning is correct, how could we choose between alternative routes to infinite bliss? Indeed, if the naïve expected utility calculation is correct, then we can't.

Consider a simplified table with just two rival religions, the cults of God A and God B (table 2.2). Each religion promises eternal life to its followers. But God A (if there is such a being) demands fealty that excludes any worship of the false God B — and vice versa.

Table 2.2: The Many Gods objection.

	God A (p)	God B (q)	No God ($1 - p - q$)
Wager A	∞	finite	finite
Wager B	finite	∞	finite

If the two hypotheses have positive probabilities p and q , then it seems we must calculate thus:

$$EU(\text{Wager A}) = p \cdot \infty + q \cdot \text{finite} + (1 - p - q) \cdot \text{finite} = \infty$$

$$EU(\text{Wager B}) = p \cdot \text{finite} + q \cdot \infty + (1 - p - q) \cdot \text{finite} = \infty.$$

So both wagers have expected utility ∞ , and thus, it seems, both are equally good. Like the argument based on table 2.1. This result is insensitive to the probabilities involved, as long as they are finite and greater than zero.

The Many Gods objection raises an obstacle for concluding from Pascal's wager that one ought to wager on any *particular* religious faith. It is not immediately disastrous for Pascal's Principle itself, about the value of small chances of infinite goods. Rather, it suggests that if this principle is correct, then we may often face many-way ties between equally good options. But things get worse.

Evil God. Besides hypotheses involving eternal happiness, there are also hypotheses involving eternal torment. These, presumably, deserve *negative*

Pascal has said, "If your religion is false, you risk nothing in believing it true; if it is true, you risk everything in believing it false." An imam can say all the same things as Pascal.

infinite utility. One hypothesis that may be live is that God *infinitely punishes* precisely those people who wager for God (perhaps for being so mercenary with their allegiance). Call this the Evil God hypothesis, according to tradition. Then there is a missing column in Pascal's (purported) decision table.

Table 2.3: The Evil God objection.

	God (p)	Evil God (r)	No God ($1 - p - r$)
Wager	∞	$-\infty$	finite

Taking account of the extra column in table 2.3, we can write down the formula for the expected utility of wagering for God:

$$p \cdot \infty + r \cdot (-\infty) + (1 - p - r) \cdot \text{finite} = \infty - \infty = ?$$

But this calculation now requires us to *subtract* infinity from infinity, and this has no well-defined result. After all, if $1 + \infty = \infty$, then it would seem that $\infty - \infty = 1$. But by the same token, if $5 + \infty = \infty$, then $\infty - \infty = 5$, and if $\infty + \infty = \infty$, then $\infty - \infty = \infty$. So now the naïve calculus of infinite utility has collapsed, giving us no verdicts at all.

Mixed Strategies. Set aside the first two kinds of objections, and grant that only the original two hypotheses are live — a God who infinitely rewards those who earnestly seek God, or else no God, and no eternal rewards or punishments. Duff (1986) and Hájek (2003a) note that even this simple case is not so simple: for there are other options to consider besides simply wagering for God or declining. One might instead adopt a “mixed strategy”: flip a coin, and wager for God only if it comes up Heads (table 2.4).

Table 2.4: Mixed strategies

	God, Heads ($p/2$)	God, Tails ($p/2$)	No God, Heads ($(1-p)/2$)	No God, Tails ($(1-p)/2$)
Wager	∞	∞	finite	finite
Coin Flip	∞	finite	finite	finite

We calculate once again:

$$EU(\text{Coin Flip}) = \frac{p}{2} \cdot \infty + \frac{p}{2} \cdot \text{finite} + (1-p) \cdot \text{finite} = \infty.$$

This option once again has infinite expected utility. So, it seems, one might as well go for the Coin Flip rather than wagering. This still holds even if the coin is strongly biased toward Tails. For example, if it has just probability $1/100$ of coming up Heads, then the probability of eternal happiness is just $p/100$. But if p is positive and finite, then $p/100$ times infinity is still infinite! And thinking about it a bit more, it seems that *every* choice one might take has *some* non-zero probability of resulting in religious faith and eternal reward. Even if you set out resolutely to wager against God, you might be struck down on the road to Damascus and convert despite yourself. This is an even more extreme version of the Many Gods objection: not only do different religious faiths come out tied, but it seems that Pascal's wager does not favor *any* course of action over any other: it's infinite expected utilities all around. (Or if we allow negative infinities back into the decision table, then it's *undefined* expected utilities all around.)⁸

These objections are powerful. But in a way, they are tip-toeing around the central point. It is not just that this version of Pascal's argument is *unsound*: rather, the position attributed to Pascal is *incoherent*. The decision theory that assigns an infinite utility ∞ to some outcome, and then says to maximize expected utility according to the naïve rules that $p \cdot \infty = \infty$ and $\infty + \infty = \infty$, is a complete non-starter. We can show this by a more direct argument (closely related to Duff and Hájek's).

⁸This is a version of what is sometimes called the problem of *infinitarian paralysis* (for example Bostrom 2011, sec. 1.1). See also chapter 3 and section 5.2.

The voice of God resounds from the whirlwind: “Job, my faithful servant, I shall give you an eternal life of infinite bliss. Please simply raise your right hand to indicate that you accept this offer.”

Unimpressed, Job replies, “And what if I don’t?”

“Then I shall deliver you unto Satan, who shall submit you to a test. It’s not an especially fair test, I admit. Satan will toss a fair coin a thousand thousand times. If it ever comes up heads, then Satan will torment you for a thousand thousand lifetimes, after which you will be annihilated.”

“Ok, but what if it comes up tails every time?”

“In that case, I suppose I’ll give you eternal happiness anyway.”

Job thinks. “Well, let’s calculate the expected value of declining. One in two to the millionth power times *infinity* is still infinite. Subtract from that the finite disvalue of a thousand thousand lifetimes of torment (multiplied by almost one), and I still get infinity. So thanks but no thanks — I’ll take my chances with Satan.”

Table 2.5: Job’s wager.

	Lose Toss ($1 - \epsilon$)	Win Toss (ϵ)
Certain eternal happiness	∞	∞
Take your chances with Satan	finite (but very bad)	∞

The naïve calculus of infinity has the consequence that an option can be equally good as another option which is *certain* to be at least as good, and *nearly certain* to be infinitely better. This is absurd. In particular, it conflicts with a *dominance* principle: replacing an outcome in a gamble with a better outcome gives you a better gamble overall. (We might appeal either to the principle of *Statewise Dominance* introduced in section 1.3.4, or the principle of *Stochastic Dominance* introduced in section 1.3.5. But much weaker versions of these principles would suffice: for in this case, additionally, we have an outcome which is *nearly certain* to obtain, and an outcome that is *infinitely* better.) This absurdity arises even before the further absurdities that arise

from trying to extend this theory to *negative* infinite utilities.

These objections are powerful. The argument based on an expected utility calculation and simple arithmetic involving an infinite quantity ∞ leads only to confusion. But this was not Pascal's argument.

While Pascal was one of the discoverers of expectational decision theory, representing outcomes with *utilities* is completely anachronistic: this idea is usually credited to Daniel Bernoulli ([1738] 1954), over half a century after Pascal's death. Moreover, rather than clarifying Pascal's argument, utilities obscure it. The argument above in terms of "infinite utilities" does not make sense.[^other-problem-with-the-table]

I admit that Pascal himself muddies the waters here.⁹ His invocation of a Zeno-style paradox of infinite extension explicitly encourages the sort of theory of infinite quantities that would allow one to say things like " $\infty + 1 = \infty$ ", and more implicitly encourages one to go on with calculations like " $\frac{1}{2} \cdot \infty = \infty$ ". But Pascal's mystifier commentary on the infinite is raising a *paradox*, rather than a theory. What he is arguing there is that "We know that the infinite exists, but we are ignorant of its nature" — and that one should take a similar attitude toward God. This is very far from seriously advocating a calculus of infinite quantities.

There is a second way in which the decision table is anachronistic: it represents decisions in terms of the probabilities of *outcomes*. (This issue is less consequential for the argument.) This is not how Pascal or his contemporaries conceived of things. Rather, they speak of the probabilities of *gains* or *losses*. A gain or loss is not a matter of how a certain choice turns out in a certain state of nature. Rather, it is a *comparative* matter, of how much better or worse one choice would turn out than another. The *Port-Royal Logic* ([1683] 1996, 273–74) presents the central principle of decision theory more explicitly:

[I]n order to decide what we ought to do to obtain some good or avoid some harm, it is necessary to consider not only the good or harm in itself, but also the probability that it will or will not occur, and to view geometrically the proportion all these things

⁹Pascal is hardly to be blamed for this: what we know as Pascal's wager comes from extremely rough notes that he did not have a chance to edit. As Wetsel and Zoberman ([1670] 2021, 345, note 1) observe, "In the autograph, the text of the 'Wager' appears as an unreadable and total scramble of almost undecipherable and tentative notes, written in every direction on the page."

have when taken together.

In the case at hand, this formula says to consider the amount to be *gained* by wagering, if God exists, proportioned to its probability, and to compare this to the sacrifice required for wagering. Notice that this formula does not ask us to distinguish two questions: if God exists, how well off am I if I wager, and then how well off am I if I don't? We only ask one question: if God exists, how much *better off* am I if I wager than if I don't? (The relation between the two formulations of expected value is tied to the principle of *separability* discussed in chapter 8.)

There is some debate in more recent discussions of the wager about what value Pascal puts in the lower-left quadrant of table 2.1. (See overview in Hájek 2008, sec. 4). Since this question does not even arise for Pascal, it is not surprising that he does not clearly answer it.

Pascal's best argument for his principle — that “wherever the infinite is ... all must be given” — does not turn on any such calculation, nor on a decision table like table 2.1. Unfortunately, this argument has been neglected, and as a result the importance of Pascal's wager has been widely underrated.¹⁰ We turn there now.

2.2 The good argument

Pascal presents his argument in two steps. The first step is a special case, for illustration, and the second step generalizes it. I will take things in the same order, reconstructing Pascal's reasoning line by line. Here is the crucial passage in its entirety (still from [1670] 1962, frag. 418, my translation, numbers inserted).

Let us see. [1] Because there is equal chance of winning or losing, if you had to wager just two lives for one you might still wager [*vous pourriez encore gager*] — [2] but if there were three lives to win? You should play (since you are under the necessity of playing), and you would be imprudent, since you are forced to play, not to stake your life to to win three lives in a game where you have the same chance of losing and winning. [3] But there is an eternity of life and happiness. [4] And that being so, when there are infinitely many chances out of which just one is for you,

¹⁰That is, it has been neglected as a reading of Pascal. As I discuss in section 2.6 below, some philosophers have recently rediscovered it.

you still have reason to wager one to gain two, and you would act with bad sense (being obliged to play) to refuse to wager one life against three in a game with infinitely many chances and just one for you, if you had an infinity of infinitely happy life to gain. [5] But here there is an infinity of infinitely happy life to gain, one chance to win against a finite number of chances to lose, and what you stake is finite. [6] This much is settled, wherever the infinite is and where there is not infinitely greater chance of loss than gain — there is no more weighing up to do: all must be given.

Pascal's starting point [1] is a principle about risky prospects involving "lives". "Because there is equal chance of winning or losing, if you had to wager just two lives for one, you might still wager." And moreover, "you would be imprudent ... not to stake your life to to win three lives in a game where you have the same chance of losing and winning." It is not clear how to understand a "life" here, and as we will discuss, there is more than one philosophically interesting direction to take this (section 2.5). But it is natural to think of a "life" for Pascal as an ordinary lifespan filled with ordinary pleasures — for concreteness let's say 100 years of happiness. The abstract hypothetical gamble here involves a coin flip: on Tails, you lose out on an entire lifetime of happiness. On Heads, you win "two lives" — say 200 years of happiness. Pascal's starting place is that chance $1/2$ of two lives is exactly as good as one life for sure — so it is permissible to wager in that case. But chance $1/2$ of *three* lives is strictly better than this [2]. So given a choice between this chance, and one life for sure, one should take the gamble.

Let's use the notation $p * x$ for a gamble in which you have probability p of gaining good x , and otherwise get nothing. And let us write $>$ for the betterness relation between prospects: that is, $P > Q$ means that prospect P is better than Q . In Pascal's terms, this is to say that if you face a forced choice between P and Q , "you would be imprudent" not to choose P . (Incidentally, when Pascal reiterates that "you are under the necessity of playing", what he is emphasizing is that you do face a forced choice. There is no further option of just walking away from the table.) Then we can summarize Pascal's premise, in the simple case, thus:

$$\frac{1}{2} * 2 \text{ happy lives} \sim 1 \text{ happy life.}$$

Pascal continues: [3] "But there is an eternity of happy life." — or as he puts

it a few sentences later, “an infinity of infinitely happy life.” The implication is that what he has said for *three* lives should be all the more forceful for an *infinite* amount of happy life. That is, an infinity of infinitely happy life is better than two happy lives.

$$\infty \text{ happy lives} > 2 \text{ happy lives.}$$

This is also controversial, of course. Some have argued that immortality is undesirable — maybe it gets boring, or lacks meaning. I’ll return to this issue in section 2.5. But let’s continue.

That is the first step of Pascal’s argument. In order to complete the enthymeme, we must introduce an additional premise that I think Pascal thought sufficiently obvious as not to require stating: a *dominance* principle.¹¹

Dominance. If x^+ is better than x , then a gamble with chance $p > 0$ of x^+ is better than a gamble with chance p of x .

$$\text{If } p > 0 \text{ and } x^+ > x \text{ then } p * x^+ > p * x.$$

To be entirely explicit, we can also spell out a structural principle about betterness.

Better-Equal. For prospects P, Q, R , if $P > Q$ and $Q \sim R$ then $P > R$.

Now Pascal’s first conclusion follows.

First Conclusion. A gamble with chance $1/2$ of an infinity of infinitely happy life is better than the certainty of one happy life.

$$\frac{1}{2} * \infty \text{ happy lives} > 1 \text{ happy life.}$$

Pascal’s next step is to generalize this reasoning, dropping the simplifying assumption that God’s existence is a fair coin flip. His first move in this direction is a bit of a tangle; in the long sentence [4] Pascal considers gambles involving infinitely many chances — and thus, presumably, infinitesimal probabilities (as Hájek 2003a, 39 points out). I will skip over this. (I return to infinitesimal probabilities in chapter 5.) He reaches the point in [5]:

¹¹This is an instance of the Simple Stochastic Dominance principle introduced in section 1.3.5. Let E be an event with probability p . Let X be a random variable which results in the outcome x on E , and the “status quo” outcome otherwise. Let X^+ be a random variable which results in the outcome x^+ on E and the status quo otherwise. Then the distributions of X^+ and X are $p * x^+$ and $p * x$, respectively, and X^+ statewise dominates X , but not vice versa.

But here there is an infinity of infinitely happy life to gain, one chance to win against a finite number of chances to lose, and what you stake is finite.

Pascal does not go through the general argument step by step, and instead skips immediately to its conclusion [6]. But it is clear how to generalize his reasoning. First, Pascal evidently holds that one should be *risk-neutral* with respect to “lives”:

Risk Neutrality. For any $n > 0$, chance $1/n$ at n happy lives is equally good as one happy life for sure.

$$\frac{1}{n} * (n \text{ happy lives}) \sim 1 \text{ happy life.}$$

This starting point is very controversial. It may seem quite counterintuitive: why would it be “imprudent” to be *risk-averse* with respect to longevity? Pascal says nothing to support the premise. But there are strong things to be said for it; I will return to this issue in section 2.6 and throughout part II. We will also see there that a much weaker premise than risk neutrality would suffice for Pascal’s purposes. Even so, shockingly little of the controversy over Pascal’s wager has focused on this premise. Indeed, standard reconstructions of Pascal’s argument leave this premise out entirely.

The second premise is that an infinity of happiness is better than any finite amount of happiness.

More is Better. For any n , an infinity of happy life is better than n happy lives.

$$\infty \text{ happy lives} > n \text{ happy lives.}$$

The Dominance premise and and Better-Equal remain identical.

Second Conclusion. For any $n > 0$, a prospect with chance $1/n$ of an infinity of happy life is better than the certainty of one happy life.

$$\frac{1}{n} * \infty \text{ happy lives} > 1 \text{ happy life.}$$

This reconstruction of Pascal’s line of reasoning has several advantages over the more familiar interpretation. First, it does not involve any anachronistic appeal to expected utility theory. Second, it does not commit Pascal to a theory of the arithmetic of infinite quantities — something at odds with his stance that we are “ignorant” of the nature of the infinite, and we can know

only that the infinite exists. Third, it gives a logical place to a premise that Pascal presents as important for the argument — Risk Neutrality — rather than setting it aside. Finally, this reconstructed argument is valid, and (as I will discuss below) its conclusion is coherent. Indeed, I think it is quite powerful, and perhaps even sound.

I have never seen this argument attributed to Pascal. But in the 21st century it is no longer a new argument: Beckstead and Thomas (2024) give essentially the same argument. (More on the differences in section 2.6.)

I should emphasize again that I have only reconstructed *part* of Pascal’s argument: the part in which he argues for what I am calling Pascal’s Principle, concerning the value of small chances of infinite goods. This Principle is one of the central concerns of this book. Pascal, of course, goes on to apply the Principle to argue that we ought to pursue religious faith. This further step of the argument faces further difficulties, and I will return to some of them below (section 2.7 and chapter 11). But the first step to Pascal’s Principle is also important. It is the locus of many criticisms of Pascal’s wager, and — even if it turns out that Pascal’s own application was too hasty — it is still certainly *big if true*.

2.3 Vector utilities

At this point I will turn to evaluating Pascal’s Principle more anachronistically.

The challenge to the simplistic wager attributed to Pascal was not just that his argument was unsound, but that his position was incoherent: in particular, assigning infinite utility to an outcome violates eminently plausible dominance principles. So even once we have dispensed with the bad argument for Pascal’s Principle, and likewise with the naïve calculus of infinite utility, there may be lingering doubts about whether the improved argument also takes us to a similarly incoherent conclusion.

We might be even more concerned about this when we note that the standard axioms of expected utility are inconsistent with Pascal’s Principle. Let’s look at how this looks specifically in the influential framework of von Neumann and Morgenstern ([1944] 2007) (see section 1.3.5 and section D.2 for further details).

First, we need to generalize some of what we have been doing. We have been comparing the values of various prospects, and we have described

these prospects in terms of their probabilities of outcomes. So, in general, suppose we have some set of outcomes; a *simple lottery* is a probability distribution over finitely many outcomes. (Here we are only considering probabilities that are standard, finite real numbers between zero and one. I discuss *infinitesimal* probabilities in chapter 5.) There is a relation (*at least*) as good between these lotteries, written $\succeq$.

In this setting, standard expected utility theory can be characterized by certain axioms — *the VNM axioms*. Here are the first two. (Let variables P , Q , R , and P' be understood as generalized over all lotteries.)

Reflexivity. $P \succeq P$.

Transitivity. If $P \succeq Q$ and $Q \succeq R$, then $P \succeq R$.

We will understand these two conditions to be implied whenever we call $\succeq$ an *order*, and I won't repeat them in the statement of the representation theorems below. The next axiom is much more controversial:

Comparability. Either $P \succeq Q$ or $Q \succeq P$.

This says that no two lotteries are *incommensurable* in value; I will return to this idea in chapter 4. (This axiom is also commonly called *Completeness*.)

The final two axioms involve the idea of a *mixture* of lotteries. Say we have tickets for two lotteries P and Q . But instead of giving you either of these tickets outright, instead we will decide which one to give you by *another* random drawing. With probability p , we give you ticket P , and with probability $1 - p$ we give you ticket Q . Then the probability of getting any particular outcome x is a weighted average of the two probabilities $P(x)$ and $Q(x)$: namely, $p \cdot P(x) + (1 - p) \cdot Q(x)$. The probability distribution for this two-stage lottery is $p \cdot P + (1 - p) \cdot Q$.

Now, consider two different two-stage lotteries.

- With probability p , get lottery P ; otherwise, get lottery Q .
- With probability p , get lottery P' ; otherwise, get lottery Q .

With probability $1 - p$, both of these lotteries turn out exactly alike, giving you the outcome of Q . It is natural to suppose that we can “cancel out” the shared part of these two complex lotteries, and just pay attention to what is left. This is what the next characteristic axiom of expected utility theory says.

Independence. For any $p > 0$,

$$P \succeq P' \quad \text{iff} \quad p \cdot P + (1 - p) \cdot Q \succeq p \cdot P' + (1 - p) \cdot Q.$$

This axiom is also controversial. I will discuss it (and related ideas) in chapter 9.¹²

The final axiom specifically has to do with *infinite* value. The standard real numbers do not include any *infinite* quantities: no real number is greater than every number in the sequence

$$1, 2, 3, \dots$$

This is called the *Archimedean property*: for every real number r , there is some natural number $n > r$. Likewise (and equivalently), there are no *infinitesimal* real numbers. There is no standard real number which is greater than zero, and yet less than every number in the sequence

$$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

Likewise, no two real numbers are infinitesimally or infinitely far apart from one another. For any real numbers $y < z < x$, the proportion between $x - y$ and $z - y$ must be finite. So there is some natural number n such that

$$\frac{1}{n} \cdot x + \frac{n-1}{n} \cdot y < z.$$

The *Archimedean Axiom* says that an analogous property holds for the value of lotteries.

The Archimedean Axiom. If $p \cdot P + (1 - p) \cdot Q \succeq R$ for every finite probability $p > 0$, then $Q \succeq R$.

It may be easier to understand this in its contrapositive form (which is simplified if we also take Comparability for granted):

If $Q < R$, then there is some finite probability $p > 0$ such that $p \cdot P + (1 - p) \cdot Q < R$.

¹²Note that Independence generalizes the version of Dominance stated in section 2.2. Indeed, Independence implies Simple Stochastic Dominance. When we turn our attention to lotteries that are *not* simple, with infinitely many possible outcomes, this connection breaks down. In that more general setting, Independence and Stochastic Dominance are logically independent of one another. (See T. L. Fine 2008.)

The idea is that P cannot be so much more valuable than R that even an arbitrarily tiny chance at P is better than R .¹³

Von Neumann and Morgenstern's celebrated theorem says that an ordering of simple lotteries has a real-valued *expected utility representation* if and only if it satisfies Independence, Comparability, and the Archimedean Axiom (see section D.2). A real-valued expected utility representation is a function u that takes each outcome to a real number, with the property that for any simple lotteries P and Q , $P \succeq Q$ if and only if P has at least as great expected utility as Q .

The conflict between expected utility theory and Pascal's Principle arises from the Archimedean Axiom. Consider the case where P is the certainty of an infinity of happy life, Q is the certainty of getting nothing, and R is the certainty of one happy life. In this case, the axiom says:

If, for every finite probability $p > 0$,

$$p * \infty \text{ happy lives} \succeq 1 \text{ happy life,}$$

then

$$\text{nothing} \succeq 1 \text{ happy life.}$$

Since one happy life is better than nothing, the axiom implies that there must be *some* positive probability p — perhaps very small, but not infinitesimal — such that chance p of an infinity of happy life is not as good as getting one happy life for sure. This directly contradicts Pascal's Principle.

But this is not a very strong argument against Pascal's Principle. The Archimedean Axiom has received very little serious defense as a normative constraint. Rather, it is standardly defended just as a useful technical simplifying assumption. It allows us to use *real numbers* in our utility representation, without allowing any infinite or infinitesimal utilities. That makes our math simpler — but that is not a very strong reason to think it is true.¹⁴

¹³In the presence of the other axioms, the Archimedean Axiom is equivalent to the axiom of *Continuity*, which may be more familiar to some readers.

Continuity. If $P \succeq Q \succeq R$, then there is some $p \in [0, 1]$ such that $Q \sim p \cdot P + (1 - p) \cdot R$.

¹⁴See Luce and Raiffa (1989, 27); Hájek and Nover (2008, 649ff.); Isaacs (2014, 12); J. S. Russell and Isaacs (2021, 187). Gustafsson (2022, 65–67) considers the possibility of a money pump argument for Continuity, but he concludes that it is only “almost a money pump argument” (see also Hammond 1988, sec. 9.). In fact, “the money-pump argument for Continuity seems compelling [only] in case the agent has no (rationally allowed) infinite differences in their preferences” — precisely the point here at issue.

We should be especially skeptical because there is a quite strong argument *against* the Archimedean Axiom — namely, the argument Pascal gave for his principle.

While violations of the Archimedean Axiom are inconsistent with the letter of expected utility theory, they are perfectly consistent with its spirit. We just need to suitably expand our conception of *utilities* beyond the standard real numbers. A variety of ways of doing this have been defended. Hájek discussed several of these ways under the heading of “reformulating the wager” — that is, replacing the argument based on the utility table (table 2.1) with some other argument based on a different table of generalized utilities. As I have argued, Pascal’s argument was not really based on any table of utilities at all, and so it does not require any reformulation along these lines. But the resources of alternative theories of utility are helpful for Pascal’s project in a different way: they provide models that demonstrate the consistency of his principle and his argument’s premises, defending them against the everpresent threat of incoherence.

In fact, there is a general model of “infinite utilities” which has been known since the 1950’s to be adequate: *vector utilities* (Hausner and Wendel 1952; Hausner 1952; Chipman 1960).¹⁵ There are two different perspectives we can take on this: one more concrete, and one more abstract. Let’s start concretely.

Without the Archimedean Axiom, we may not be able to put the values of all outcomes on the same numerical scale. So instead we can put them on *different* numerical scales, corresponding to different dimensions of a vector space. A utility vector will have one coordinate corresponding to each of these dimensions of value, arranged from most significant to least significant. For example, we might represent a pesto sandwich with the vector $(0, 1)$, and represent an eternal life of infinite happiness with the vector $(1, 0)$. Say we also represent an outcome in which you get *both* a pesto sandwich *and* an eternal life of infinite happiness with the vector $(1, 1)$, and an outcome in which you just get two pesto sandwiches with $(0, 2)$. We order these utility vectors *lexicographically*: first we compare the first “heavenly” coordinate, and only in cases of ties do we consider the second “sandwichly” coordinate. So we get:

$$(1, 1) > (1, 0) > (0, 2) > (0, 1).$$

¹⁵Hájek (2003a, sec. 4.2) includes vector utilities in his laundry list, but he does not note its generality. The general vector utility representation theorems I discuss seem to be well-known among economists, but less so among philosophers.

Heaven and a sandwich is better than just heaven, which is in turn better than two sandwiches.

We can also straightforwardly calculate *expected* vector utilities of simple prospects. Consider a coin flip with chance $1/2$ of utility $(1,0)$, and chance $1/2$ of utility $(0,0)$. We calculate the expectation coordinate by coordinate, so this gives us expected utility $(1/2,0)$. This is still infinitely better than a pesto sandwich, since it has a non-zero heavenly component. But it is a *smaller* “infinite utility” than getting eternal life for sure:

$$(1,0) > \left(\frac{1}{2},0\right) > (0,1).$$

We do not have the kind of “infinity collapse” that led to all of the problems with the naïve argument attributed to Pascal.

Utilities might have more than just two dimensions. For example, perhaps true love is infinitely better than a pesto sandwich, while eternal happiness is infinitely better yet than true love. This is easy to capture in the framework: we just assign eternal happiness utility $(1,0,0)$, true love $(0,1,0)$, and the sandwich $(0,0,1)$. There might even be *infinitely* many dimensions of utility. (Indeed, in section 3.5 we will see how infinite-dimensional vector spaces arise naturally in ethics.)

That much is the concrete perspective on vector utilities: we can represent utilities by sequences of numbers, compared lexicographically, and with expectations calculated component by component. Now for the abstract perspective. Consider again the formula for calculating an expectation. If P is a probability distribution over finitely many numbers, then we can calculate its expectation thus:

$$\mathbb{E}P = P(x_1) \cdot x_1 + \cdots + P(x_n) \cdot x_n.$$

There are two things we have to be able to do to the values $x_1, \dots, x_n$ in order to make sense of this. First, we have to *rescale* them — that is, we multiply a value by a number (a probability). Second, we have to *add together* these results. These operations should be well-behaved in various ways: for instance, $\frac{1}{3} \cdot x_1 + \frac{2}{3} \cdot x_2$ is the same thing as $\frac{1}{3} \cdot x_2 + \frac{1}{3} \cdot x_1 + \frac{1}{3} \cdot x_2$. The name for an abstract mathematical structure with these two operations — addition and scalar multiplication, suitably well-behaved — is a *vector space*. A vector space has all the structure we need to make sense of the mathematical operation of taking an expectation. The real numbers are one example of

a vector space — the *one-dimensional* vector space. The multi-dimensional sequences of real numbers we just discussed are other examples of vector spaces.¹⁶

Accordingly, there is an elegant and more abstract version of von Neumann and Morgenstern’s representation theorem. Instead of the three axioms of Independence, Comparability, and the Archimedean Axiom, we only need one.

Theorem 2.1. An ordering of simple lotteries satisfies the Independence axiom iff it has an expectational representation in a partially ordered vector space.

This is stated more explicitly and proved in section D.2.

In order to go from a general vector representation to the particular case of the real numbers, we add on two things. Adding the axiom of Comparability takes us from a *partial* order to a *total* order. Adding the Archimedean Axiom ensures that we can represent outcomes with a *one-dimensional* vector space. Those properties characterize the real numbers: in fact, any one-dimensional vector space is isomorphic to the real numbers.

What’s more, the concrete case of lexicographically ordered sequences that we began with turns out to be essentially just as general as the more abstract version. Let’s focus here on *totally* ordered vector spaces — reinstating the Comparability axiom. It turns out that every totally ordered vector space can be isomorphically embedded in a lexicographically ordered set of sequences (Hausner and Wendel 1952).¹⁷ So as a corollary to theorem 2.1, we get another representation theorem.¹⁸

¹⁶To be precise, a vector space actually has a bit *more* structure than we really need for this: more generally, a *convex space* suffices. But these are less familiar, and we won’t have use for the additional level of abstraction. Furthermore, we are still assuming here that *probabilities* are represented by real numbers. Later we will relax this assumption (chapter 5). When we do, this will require us to generalize vector spaces in a different direction: namely, to *modules* (see section B.1).

¹⁷It takes a bit of subtlety to define a lexicographic order on an *infinite*-dimensional vector space. See section B.3 and section D.2 for the details.

¹⁸Fishburn (1971) provides additional axioms that restrict the representation to finite-dimensional vector spaces; but these are not especially well-motivated as normative constraints, and, as we will see in later chapters, they are incompatible with natural conditions on social welfare for infinite populations.

Theorem 2.2. An ordering of simple lotteries satisfies Independence and Comparability iff it has a lexicographic expected utility representation.

Restricting attention to simple lotteries, and insofar as we are sticking with the central orthodoxies of standard decision theory — the order axioms, Independence, and Comparability — there is no need to search any further than lexicographic vector spaces.

There is a generalization of this representation theorem that also drops the Comparability axiom, but I will postpone that to chapter 4. In later chapters, I will also consider theories that drop Independence. For now, though, let's focus on the minimal departure from orthodox expected utility theory that allows infinite utilities; this is completely characterized by lexicographic expected utility theory.

Let's check now how lexicographic vector utilities escape the objections to the naïve argument in section 2.1. For example, consider the Many Gods objection. We can represent the proposed situation as in table 2.6. (I've put in some particular numbers for concreteness, but they aren't important. We are treating the secular life as the baseline $(0, 0)$, and supposing that wagering comes at some finite cost.)

Table 2.6: The Many Gods objection in the setting of vector utilities.

	God A (p)	God B (q)	No God ($1 - p - q$)
Wager for A	$(1, -5)$	$(0, -5)$	$(0, -5)$
Wager for B	$(0, -5)$	$(1, -5)$	$(0, -5)$
Wager against both	$(0, 0)$	$(0, 0)$	$(0, 0)$

We can easily calculate the expected vector utilities:

$$EU(\text{Wager for A}) = (p, -5)$$

$$EU(\text{Wager for B}) = (q, -5)$$

$$EU(\text{Wager against both}) = (0, 0).$$

The result is that wagering for either god is better than wagering against both, as long as the probability of that god's existence is non-zero. And the best god to wager for is whichever has the higher probability: if $p > q$,

then you should wager for God A, and if $p < q$ you should wager for God B. (If it's a tie, either is fine.) There is no collapse here, where all options involving infinite outcomes end up looking exactly the same. It does mean that choosing your faith on Pascalian grounds may involve a bit more research in comparative religion than Pascal suggests, but the secular option is out. It is straightforward to check how things go for the Many Gods and Mixed Strategies objections as well.

Vector representations are also useful for demonstrating properties that *any* Pascalian value ordering must have — holding fixed the order axioms and Independence. For example, Schlesinger (1994, 97) proposes the principle that, in cases where multiple options have non-zero probability of achieving an infinite good, one should “try and increase the probability of obtaining the prospective prize.” Hájek (2003a, 33) characterizes this as “suggesting that decision theory should be supplemented with a new principle.” But in fact Schlesinger’s principle is already a consequence of the Independence axiom. We can see this by way of the vector representation theorem. Consider two options: P has probability p of obtaining the infinite good $(1, 0)$; Q has probability q of obtaining $(1, 0)$; and otherwise P and Q only involve finite (second-coordinate) goods. Then the vector expectation of P has the form (p, x) , and the vector expectation of Q has the form (q, y) . If $p > q$, it follows that $P \succ Q$.

Lexicographic utilities are a powerful tool for representing infinite value, but that doesn’t mean there is no place for other kinds of representation as well. Philosophers have deployed quite a few different approaches on Pascal’s behalf; I’ll very briefly mention a few of them here — and I will take up some of these ideas further in subsequent chapters.

One idea is *utility streams* (Hájek 2003a, sec. 4.3; E. Jackson and Rogers 2019): we represent each possible outcome by an infinite sequence of numbers, with each number intuitively representing the value of a particular period of time — a “life”, say. The same formalism can be used to represent the well-being of infinitely many different people. Economists standardly use utility streams to represent welfare distributions across future generations that may extend into the indefinite future; they call the task of ordering such streams the problem of *intergenerational equity*. I will also use the utility stream formalism frequently in this book (see section 3.5). But while it is a useful formalism, it is not much of a theory of infinite value taken on its own. There are many different ways of imposing orders on utility streams,

and none is obviously correct.¹⁹ This question is where the action is.

Another idea is to use the tools of *non-standard analysis*. The standard real numbers do not include any infinite or infinitesimal numbers, but there are plenty of alternative structures that do. These structures include various *hyperreal* structures, as well as Conway's *surreal* numbers (see Hájek 2003a, sec. 4.1; Chen and Rubio 2018). I will also deploy non-standard analysis later on in this book (see section 3.5 and chapter 5). (The particular approach I prefer is neither hyperreal nor surreal, but it is closely related.) Note, though, that this approach is not an *alternative* to vector utilities: a hyperreal structure is an ordered vector space, and so is a surreal structure.

Paul Bartha (2007, 2016) proposes a more radical approach that dispenses with utilities altogether, replacing them with a *relative utility function*. This is also a perfectly fine way of representing orders that satisfy Independence and Comparability without the Archimedean Axiom, as is shown by Bartha's representation theorem (2007, theorem 6, pp. 20–21). But there is no need for us to abandon utility representations to reckon with (relatively) infinite utilities.

2.4 Hájek's objection

Before moving on, I should briefly address a different objection that Hájek raises against the wager. Once we have a coherent model of infinite utility, we find that it is not true after all that “the finite is annihilated in the presence of the infinite.” If we represent an infinity of happy life by $(1, 0)$, and a pesto sandwich by $(0, 1)$, then adding a pesto sandwich to an infinity of happy life really does increase the value — $(1, 1) > (1, 0)$ in the lexicographic order. There are also other vectors like $(2, 0)$ which are better still. There are also higher-dimensional vector representations — so if we instead represent an

¹⁹In particular, I do not find the proposals of Hájek (2003a) or E. Jackson and Rogers (2019) to be especially promising as general theories of the value of well-being, though they may be helpful steps in that direction. Hájek (following Vallentyne 1993) suggests ordering utility streams according to their *long run average* utility, and ordering prospects according to their expected limiting average. Jackson and Roger's suggestion is similar. Instead of calculating each option's expected limiting average and then comparing them, we calculate the ratio of average expected utility over finite periods of time, and compare options according to the limit of this ratio. Note, though, that on either proposal, if we change a utility stream by simply changing its first zero to a one, this leaves all of the relevant limits unchanged. So neither theory says that the new utility stream is better than the original. But it is better for one life to be happy rather than sad, when everything else is held fixed. (See chapter 3 for further discussion.)

infinity of infinitely happy life by $(0, 1, 0)$, then we might worry about other outcomes that are infinitely better yet, say with utility $(1, 0, 0)$. "Salvation, then, is no longer the best possible thing after all" (Hájek 2003a, 45).

Why might this be a problem? I can see three reasons you might think so.

The first concern is about the original naïve argument, which relied on calculations like $1/2 \cdot \infty = \infty$. When we instead represent eternal bliss with a vector like $(1, 0)$, we find that $1/2 \cdot (1, 0) = (1/2, 0)$, which is not the same as $(1, 0)$. But it is untenable in any case to hold that chance $1/2$ at eternal happiness is just as good as getting it for sure. Good thing, then, that Pascal relied on no such argument. Nor should we.

The second reason is that Pascal himself writes that "unity joined to infinity does not increase it at all", and we have found this to be false, when read as a claim about the value of combining infinite and finite goods. But, first, this point is merely *ad hominem*. It might be embarrassing for Pascal, but it is not really relevant to the cogency of his argument, since that claim about infinite arithmetic is not one of its premises. Second, though, I have already defended Pascal from this potential embarrassment: as I argued in section 2.1, Pascal does not intend those claims to be taken literally. Those remarks are hyperbolic, used in the context of setting out a paradox that is intended to instill in us a suitable humility before the infinite (and before God) before we confront the wager itself.

The third reason — the one that I suspect Hájek really intends — is subtler. If "salvation is no longer the best possible thing", then the wager is vulnerable to "superwagers". Suppose, instead of a God who offers an infinity of infinitely happy life, there is a super-God who offers a super-infinity of super-infinitely happy life? The utility representation leaves room for a whole host of better and better infinite goods.

I don't think this problem is very serious in itself. Just because our abstract utility representation has a place for greater goods than salvation, it does not follow that there really *is* any conceivable good better than salvation which occupies that place. As Chen and Rubio (2018) write,

The point: where Hájek sees conceivable utilities, we see possible representations of utilities. But the fact that we could represent something's utility with a number that is infinite with respect to the number we have chosen to represent the utility of salvation does not imply that there is conceivably something with utility that is infinite with respect to that of salvation. For that, we

would need something in our space of lotteries that is infinitely preferable to salvation.

The same point arises even with respect to ordinary finite utilities. There are arbitrarily large finite real numbers. But the existence of the number 456,204 does not imply that there is anything that is exactly 456,204 times better for you than a pesto sandwich — that there is something so valuable that it is worth staking a pesto sandwich for chance $1/456,204$ of getting it (but no smaller chance). Perhaps there is such a possible good, but if so, we must argue for this on substantive grounds. The same goes for vector utilities. The mere existence of three dimensional vector spaces does not imply that there are three tiers of infinite goods, or that there is no *summum bonum*. Perhaps so, perhaps not, but either way we will need more argument than this.

Furthermore, even if there are super-infinite goods, while this raises complications for using the wager to push toward particular religious options, it is no objection at all to Pascal's central principle, which is the point I have been focusing on. To restate the principle: a positive finite probability of an infinite good is superior to any merely finite good. If there are multiple infinite goods (or evils) in play, then ranking prospects is more complicated, but it is true nevertheless that (adapting the words of St. Paul) all finite goods are as rubbish, in order that we may gain the infinite.

2.5 “Wherever the infinite is”

Pascal's Principle says that an arbitrarily small finite chance at an infinite benefit outweighs the certainty of an arbitrarily great finite benefit. But there is more than one way of understanding what we mean by “infinite” and “finite” benefits. It is worth taking a moment to make some distinctions.

One natural way to understand these ideas is in terms of *infinite cardinal value*. We have been taking for granted a notion of the *ordinal* value of prospects: this is a matter of whether one prospect is better than another. This kind of structure does not bring along with it any automatic way of understanding *how much* better one prospect is than another. Take three lotteries P, Q, R such that $P < Q < R$. These betterness comparisons exhaust what there is to say about their ordinal value. But there are other questions we might ask about how the value difference between P and Q compares to the value difference between Q and R . Are they the same? Is one much

greater than the other? Is one difference *infinitely* greater than the other? These are questions of *cardinal* value comparisons.

There is more than one way of finding cardinal value structure. One of the most prominent ties these questions directly to *risk* (for discussion see Broome 1991, sec. 6.5). Suppose there is some probability p such that Q is equally good as the mixed lottery $p \cdot P + (1 - p) \cdot R$. This provides us with one sense in which we can say that the value difference between P and Q is p times the value difference between P and R .

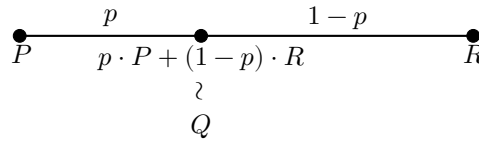


Figure 2.2: Cardinal value comparisons via equivalent mixtures.

The VNM axioms imply that such a probability p always exists.²⁰ But if the Archimedean Axiom fails, then there may not be any mixture of P and R which is just as good as Q . This could happen in either of two ways. (For simplicity I continue to take Comparability for granted.) It could be that any finite mixture of P and R is *better* than Q . In that case, intuitively the value difference between Q and R is *infinitely greater* than the value difference between P and Q . Let's stipulatively say that in this case R is *VNM-infinitely better* than Q (relative to the baseline P). (This is essentially how Bartha 2007, 18 defines *infinite relative utility*.) Alternatively, it could be that any finite mixture of P and R is *worse* than Q . Then the intuition is that the value difference between P and Q is infinitely greater than the value difference between Q and R .

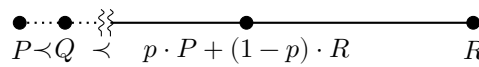


Figure 2.3: If Q is not as good as *any* finite mixture of P and R , we say R is *VNM-infinitely better* than Q (relative to P).

If we understand the "infinite benefit" and "finite benefit" in Pascal's Principle as prospects such that the first is *VNM-infinitely better* than the other, then Pascal's Principle becomes a tautology. To say that R is an "infinite benefit" and that Q is a "finite benefit" is just to say that an arbitrarily small chance at a prospect R is better than Q .

²⁰This follows directly from the Continuity axiom.

There are other ways of understanding cardinal scales of value, arising from other “extensive structures” besides that tied to risk. (For discussion see Nebel 2023; Cibinel, unpublished.) I’ll return to one such way of understanding cardinal quantitative value, in terms of *interpersonal* aggregation, in section 3.5. With those other cardinal structures come other ways of understanding “infinite value” and “finite value”.

But Pascal doesn’t do things this way at all. The “infinity” of the benefit he concerns himself with is infinite in a more concrete way, not directly tied to cardinal value structure at all. What is at issue is an *infinity of infinitely happy life*. His central claim is, in effect, that *this* prospect is VNM-infinitely better than the certainty of any ordinary finite happy life. This is substantive, not tautological.

This is a good place to return to Pascal’s premise that *More is Better*: an eternity of happiness is *better* than any finite duration of happiness. This claim is open to challenge. Williams (1973) influentially argues that “immortality would be, where conceivable at all, intolerable.” Eternal life must eventually degenerate into dreary repetition, or stultifying stasis, or loss of continuity of self. Williams’ critique is most naturally construed as the claim that eternal *happiness* is *impossible*. But it might instead be argued that eternal happiness would be *undesirable*. In his poem “Sunday Morning”, Wallace Stevens (1923) writes:

Death is the mother of beauty; hence from her,
Alone, shall come fulfilment to our dreams
And our desires.

And in the next stanza:

Is there no change of death in paradise?
Does ripe fruit never fall? Or do the boughs
Hang always heavy in that perfect sky,
Unchanging, yet so like our perishing earth,
With rivers like our own that seek for seas
They never find, the same receding shores
That never touch with inarticulate pang?
Why set the pear upon those river-banks
Or spice the shores with odors of the plum?
Alas, that they should wear our colors there,
The silken weavings of our afternoons,
And pick the strings of our insipid lutes!

Death is the mother of beauty, mystical,
 Within whose burning bosom we devise
 Our earthly mothers waiting, sleeplessly.

Stevens's picture is not that immortality would be "intolerable" (as Williams argues), but rather that — even if it is indeed a "paradise" — it would be "insipid". This is compatible with the idea that each *finite* increase in happy life is good for you. (See also May 2009.)

This is a serious challenge to one specific form of Pascal's wager. Consider a version of table 2.1, with a bit more descriptive detail filled in. Consider an "annihilationist" God, who rewards believers with eternal happiness, and consigns unbelievers to oblivion. Suppose that "wagering for God" comes at a heavy cost, sacrificing most earthly delights for a life of penitence. If Stevens is right, then this offer of a "reward" is not such a good thing at all. Due to its insipidity, eternal happiness is *worse* than a sufficiently long and pleasant mortal life. In this case, even assuming Risk Neutrality with respect to happy finite life, if the probability that God exists is low, then Pascal's wager will not go through.

Table 2.7: The "annihilationist" God.

	God	No God
Wager for God	eternal happiness	a dismal mortal life
Wager against God	a delightful mortal life	a delightful mortal life

But the power of this response to Pascal is very limited. Rather than making room for earthly pursuits, it rather points our attention to a different stock of infinite benefits. One salient alternative to this annihilationist picture is that God might consign unbelievers not to mere *finite* happiness, but rather to *eternal suffering*. In that case, we would be in the "intolerable" situation anyway: whatever meaninglessness comes with deathlessness is something we'd be stuck with. It would not count against eternal happiness in comparison to eternal suffering. So there does not seem to any good reason to say that the difference between eternal happiness and eternal suffering is not better than the difference between any finite duration of happiness and the same duration of suffering. If *this* "infernalist" hypothesis gets positive probability, then we can focus on it, and set the annihilationist hypothesis aside.

Table 2.8: The “infernalist” God.

	God	No God
Wager for God	eternal insipidity	a dismal mortal life
Wager against God	eternal suffering	a delightful mortal life

Another soteriological picture is that the infinite good we ought to pursue with all our hearts is to *escape* eternal life. According to both Hindus and Buddhists, our natural lot is *samsāra*, an eternal cycle of suffering and rebirth — and this is tragic, to be escaped if possible. One picture is that the result of such escape (*moksha*) is union with God.

Great souls make their lives perfect and discover me; they are freed from mortality and the suffering of this separate existence. Every creature in the universe is subject to rebirth, Arjuna, except the one who is united with me. (*The Bhagavad Gita* 8.15–16, E. Easwaran 2007, 167)

A different picture is that the result of escape is *nirvāṇa*, which literally means “blowing out” — extinguishing a flame. From the Buddhist Pāli Canon:

There being no passing away or arising, there is neither a here nor a there nor a between-the-two. This, just this, is the end of stress. (*Udāna* 8:4, Bhikkhu 2012, 113)

So we can consider yet another version of the wager: a tiny chance at escaping the eternal cycle of suffering to some negative state of “no passing away or arising” is worth the price of a certainty of a mundane lifetime of earthly sacrifice. This version of the wager is not threatened by the “Sunday Morning” objection, either.

For most of the rest of this book, I will shift attention to yet another alternative. While Pascal focuses on the *prudential* value of eternal life for an individual, the primary focus of this book is on the *impartial, moral* value of well-being for vast numbers of *different* individuals. We can directly reinterpret the “happy lives” of Pascal’s wager in this way: gaining n happy lives just means making n people happy. On this understanding of the wager, the premise More is Better says that it is better for infinitely many people to be happy than it is for only finitely many of them to be happy. The objection to Pascal based on the insipidity of immortality does not apply to this premise. It

really doesn't seem like much consolation to tell infinitely many sufferers that happiness would be meaningless if it were not finite.

The "Sunday Morning" objection may well make a difference to *which* outcomes really play the role of infinite goods. But it does not undermine the thrust of Pascal's wager, as far as how we should weigh small chances at such goods.

In the remainder of this book, I will primarily focus on the *interpersonal* interpretation of Pascal's Principle. "Infinite goods" are things like making infinitely many different people happy, rather than sad; "finite goods" are things like making only *finitely* many people happy rather than sad. Pascal's Principle says that such infinite goods are *VNM-infinitely better* than such finite goods. Crucially, though, this is not by the *definition* of "infinite good"; it is a substantive conclusion that requires argument — for instance, the argument presented in section 2.2.

2.6 Finite Fanaticism

As we saw, Pascal took the premise Risk Neutrality for granted: "Because there is equal chance of winning or losing, if you had to wager just two lives for one, you might still wager" — and similarly for numbers other than two. But on the face of it, Risk Neutrality may sound quite implausible. Who would really choose a one in a billion chance at a billion lives of happiness, and otherwise almost surely lose everything, rather than stick with the certainty of a single happy life?

Moreover, Risk Neutrality with respect to happy lives is not mandated by modern expectational decision theory, either. The orthodox theory says to maximize expected *utility* — but it permits you to assign any utilities you like to happy lives. Even preferring to have a short and miserable life rather than a glorious utopia is deemed "not contrary to reason" (*à la* Hume [1739] 2000, sec. 2.3.3.6). Risk Neutrality with respect to finite spans of happy life may be rationally *permitted* by standard decision theory, but how could it be required?

But here we are not narrowly concerned with normative constraints on choice that are requirements of "rationality" or "coherence", as this is normally construed in subjective expected utility theory. Pascal is quite clear that he is not making an appeal to *reason*, but rather to *prudence*. Prudence demands not just that you make choices that are *coherent*, but choices that are *good for*

you: in your own best interest. Furthermore, in this book we won't just be concerned with what is best for *you*, but with what is *morally* best, taking everyone's interests into account.

Even so, it is not obvious that Risk Neutrality is the correct theory of prudence (regarding your own life) or morality (regarding others'). We discussed some alternatives in section 1.2.3. Perhaps the value of a risky prospect is not determined by its expected *value*, but rather by its expected *utility*, where this is some increasing function of value (D. Bernoulli [1738] 1954). Or perhaps it is determined by some function other than an expectation, such as the *risk-weighted* expectation defended by Quiggin (1993) and Buchak (2013). These theories both allow for *risk aversion* with respect to happy lives, rather than risk neutrality. (Other prominent alternative theories also accommodate this possibility — for example, Bottomley and Williamson 2024; Tarsney 2025).

But many alternatives to Risk Neutrality offer no escape from Pascal's Principle. For example, consider expected utility theory, where the utility of n happy lives is given by $u(n) = \sqrt{n}$. This theory does not say that chance $1/n$ at n happy lives is as good as the certainty of one happy life — but it does say that chance $1/n$ at n^2 happy lives is good enough. The same goes for risk-weighting if, for example, we deploy the risk-weighting function $r(p) = p^2$. This theory also says that chance $\frac{1}{n}$ at n happy lives is not as good as one happy life for sure, but chance $\frac{1}{n}$ of n^2 happy lives will do. This quantitative adjustment makes no difference to Pascal's overall argument. Whether it takes a 1% chance of 100 lives or 10,000 lives to balance off the certainty of one happy life, an *infinite* amount of happy life is still better than either — so a 1% chance at an infinity of happy life is still better than one happy life. As long as there is *some* finite amount of happy life such that chance p of it is better than one happy life for sure — no matter how large it may be — we can still conclude that chance p of *infinite* happy life is likewise better.

Even though Pascal was happy to assume Risk Neutrality, his argument does not really require a premise that is so strong. It doesn't really matter for the argument that chance $1/n$ of n happy lives is as good as one happy life for sure: all that matters is that *some* amount of happy life suffices. We can replace Risk Neutrality with this:

- (F) For any finite probability $p > 0$, there is some finite number n of happy lives such that $p * (n \text{ happy lives}) > 1 \text{ happy life}$.

This thesis follows from Risk Neutrality, but it also follows from many other alternative theories. It follows from expected utility theory, with any *unbounded* utility function. It follows from risk-weighting, with any strictly increasing real-valued risk function. There are many other views that also have this consequence. Pascal's argument applies much more widely than just to risk-neutral theories of the value of happy lives.

Principle (F) is a version of the thesis often called *Fanaticism* (Wilkinson 2022a; J. S. Russell 2024; Beckstead and Thomas 2024 call a similar principle *Recklessness*). Let's consider a slightly more general formulation, and a corresponding generalization of Pascal's argument. Consider, abstractly, some set of *finite outcomes* and some set of *infinite outcomes*.

(Finite) Fanaticism. For any finite probability $p > 0$ and any finite outcome x , there is some finite outcome y such that $p * y > x$.

Infinity is Better. Any infinite outcome is better than any finite outcome.

Then, using the same Dominance premise as in Pascal's original argument, we can conclude:

Pascal's Principle. For any finite probability $p > 0$, chance p at any infinite outcome is better than the certainty of any finite outcome.

This more general version of the Pascalian argument has been put forward by Beckstead and Thomas (2024).

Finite Fanaticism is much weaker than Risk Neutrality, but it is still counter-intuitive and controversial. For now, I will leave it unargued for — but that is not because there are no strong arguments in its favor. In the course of part II of this book I will critically examine several such arguments.

2.7 The ugly argument

Pascal's Principle is coherent, and Pascal's argument for it is powerful. But I don't think this is the time for religious believers to pat themselves on the back for a job well done. For Pascal's Principle has troubling consequences, very different from those Pascal himself drew. I will return to this in chapter 11; but we can see some of the issues right away.

The first point is that the decision-theoretic case for religious faith is not nearly as open-and-shut as Pascal would have it. Rather, it is highly sensitive to one's probabilities. Chen and Rubio (2018) put it like this (see also E. Jackson and Rogers 2019, secs. 4.3–4.4):

But the argument does not deliver the result as advertised. Its partisans sell Pascal's Wager as a route to religion that does not depend on how the evidence falls between the theist and atheist. Our theory says otherwise. What one should do in a Pascalian decision problem depends on what one's credence function is like. It is not, after all, an end run around the evidence. For one's credence function should be sensitive to different kinds of evidence that support competing hypotheses to different degrees.

This is the true upshot of the problems of Many Gods, Evil Gods, and Mixed Strategies: not that the wager straightforwardly fails, nor that infinite utilities are incoherent, but that the probabilities matter. We saw this in the simple case we considered in section 2.3: if God A and God B both offer the same infinite afterlife, then you should wager for whichever has higher probability for you.

This might not seem like such a major concession for Pascal to make. It would fulfill quite a bit of his ambition if the wager still compels you to seek religious faith, wagering for whichever hypothesis involving supernatural reward is *most probable* for you (even if it is still extremely improbable). But things are even more delicate than that.²¹

Here is one simple illustration: suppose you assign positive probability to two different supernatural hypotheses. *Nice God* bestows eternal happiness on all those who earnestly seek God, and gives merely finite goods to the rest. *Nasty God* rewards those who earnestly seek God *and* live lives of abstinence and poverty and persecute unbelievers, and condemns all others to eternal torment. Even if you are much more confident in Nice God than Nasty God, you should still subject yourself to the Nasty God's strictures. The calculation isn't difficult. (See table 2.9. We arbitrarily set the life of secular pleasures as the baseline $(0,0)$. Suppose that earnest seeking comes at some modest finite cost $(0,-1)$, while abstinence, poverty, and persecution comes at much greater finite cost $(0,-10)$. Eternal happiness gets the lexicographically greater utility $(1,0)$, and eternal torment gets $(-1,0)$. But the precise numbers don't matter at all.)

²¹The following argument draws on unpublished work by Laura Siscoe. The argument in Duncan (2007) is related. Rota (2016, fn. 36, attributing the point to Johannes Grössl) raises the possibility of this kind of case in passing, but dismisses it. ("I raise this objection only to point out that it is extremely unlikely that one will find oneself in this sort of situation.")

Table 2.9: Nice God and Nasty God.

	Nice God (p)	Nasty God (q)	No God ($1 - p - q$)	Expected vector utility
Earnest seeking	$(1, -1)$	$(-1, -1)$	$(0, -1)$	$(p - q, -1)$
Earnest seeking + abstinence, poverty, and persecution	$(1, -10)$	$(1, -10)$	$(0, -10)$	$(p + q, -10)$
Secular pleasure	$(0, 0)$	$(-1, 0)$	$(0, 0)$	$(-q, 0)$

It is clear from the table that abstinence, poverty, and persecution ranks higher in the lexicographic order than either of the other two options, as long as the probability q of the Nasty God is positive. This is true even if, not only is No God overwhelmingly more likely than the religious hypotheses, but also Nice God is overwhelmingly more likely than Nasty God.

So even religious believers are not safe from Pascal's wager. The wager may require you not just to take counterintuitive leaps into religious faith, but leaps into the *most* counterintuitive and demanding of religious faiths.

But I think it is extremely unclear what Pascal's Principle really demands of us, if it is true. Supposing that outcomes of infinite value are really live possibilities, it is hard to tell which ones they are, how they compare, and how probable they are contingent on any of our actions. I will not pursue this line of thought further at this point. (See chapter 11.) For now, I only remark that while I think Pascal's argument is very strong, and widely underrated, I am far from comfortable with the principle it supports. I regard Pascal's wager as a deep and difficult problem.

Chapter 3

Against the Everything Bagel

JOBU TUPAKI: I got bored one day, and I put everything on a bagel. Everything. All my hopes and dreams, my old report cards, every breed of dog, every last personal ad on Craigslist, pepper, salt, sesame, poppy seed. It collapsed in on itself, because, you see, when you really put *everything* on a bagel, it becomes this: the truth.

EVELYN: What is the truth?

JOBU TUPAKI: Nothing matters.

Everything Everywhere All At Once

3.1 Infinite futility

The film *Everything Everywhere All At Once* takes place in a multiverse, full of an infinite amount of *stuff* — good, bad, beautiful, horrible, and absurd. The villain, Jobu Tupaki, creates the Everything Bagel as both a world-devouring superweapon and a totem representing her philosophical response to the multiverse that confronts her: *nothing matters*.

Jobu Tupaki's nihilism is a perfectly understandable reaction. Suppose the world is infinite, and it contains some of everything. There are infinitely many people at all levels of well-being. There are infinitely many children who drown in shallow ponds, and infinitely many who are rescued from



Figure 3.1: The Everything Bagel.

drowning. If that is the kind of world we live in, and *you* come across a child in a pond, then the overall pattern of the world looks the same whether or not you rescue them. Whatever you do, you are still in a world of infinitely many drownings and infinitely many savings. But then it is natural to conclude that your choice makes no difference to how good the world is overall. In this way too, we may fear, “the finite is annihilated in the presence of the infinite” (Pascal [1670] 1962, frag. 418).

Of course, the step from “no better overall” to “nothing matters” is tendentious. Finite interventions may still matter *to us*, and to the people we make better off. The specific issue I am investigating here is *impartial axiology* — whether, in the face of infinite plenitude, it is possible for a finite intervention to make the world better with respect to the *overall* value of individual well-being, taking an agent-neutral perspective on the whole of reality. Jobu Tupaki’s tempting and disturbing conclusion is that it is not.

The thing is, it is plausible that we really *do* live in an infinite world like this. It is a live, mainstream cosmological hypothesis that we really live in a world that is infinite in spatio-temporal extent. We should expect such a world to host infinitely many individuals at all possible levels of well-being, just like Jobu Tupaki’s world.¹ There are also many other routes to the

¹For now, suppose for simplicity that there are just finitely many “levels of well-being”

conclusion that our world is plenitudinous in this way.² Infinite worlds raise difficult problems and paradoxes, but they are not remote possibilities unworthy of serious examination. It is a pressing philosophical project to try to understand better how our own small, finite lives can fit into the moral landscape of a vast, infinite world. We too must confront the Everything Bagel.

3.2 A toy model

For the sake of conceptual clarity, let's start by considering some very simple possible worlds. In each of these worlds, there are infinitely many identical rooms labeled with the natural numbers and arranged in a row. Each room contains one person (and in each simple world, it's the *same* person in that room). Each person is at one of two levels of well-being: *happy* or *sad*. There is nothing else going on in these worlds that is relevant to how good they are overall. It isn't as if some people are more virtuous or deserving of happiness than others. No one will be jealous of another person's happiness. There are no families or societies or polities that group individuals together. And so on — you can fill in whatever further qualifications you think may be important. Call these *simple worlds*.

Let an *individual welfare level* be either “happy” (represented by 1 or 😊) or “sad” (0 or 😞). Let $\mathcal{X} = \{1, 0\}$ be the set of individual welfare levels. Any simple world w can be represented by a function $x : \mathcal{I} \rightarrow \mathcal{X}$ that takes each individual to their welfare level in w :

$$x(i) := \begin{cases} 1 & \text{if } i \text{ is happy in } w \\ 0 & \text{if } i \text{ is sad in } w. \end{cases}$$

Call any such function an **($\mathcal{X}$ -)social outcome (for $\mathcal{I}$)**. (These will play the role of *outcomes* discussed in section 1.3.4.) More generally, we can do this for any world where each individual is just as well off as they are in some simple world. Call these *happy-sad worlds*. (Later we will consider how to further generalize this to worlds with more variation in well-being.) If we conflate individuals with their numbers, and use the numerical labels for welfare-levels, then a social outcome amounts to a *utility stream*, as discussed in section 2.3.

for an individual — they are individuated somewhat coarsely.

²See section 1.1. See also J. S. Russell (2020b), and references therein.

Now consider two simple worlds, which we'll call Happy and Sad (figure 3.2). Each of them contains infinitely many happy people and infinitely many sad people. For concreteness, let's say that in both of them, all of the people in even-numbered rooms are happy, and all of the people in the odd-numbered rooms besides room one are sad. Call the person in room one Indy. In the Happy world, Indy is happy, and in the Sad world, Indy is sad. Otherwise, the two worlds are exactly the same. Everyone besides Indy is equally happy or sad in Happy or Sad.

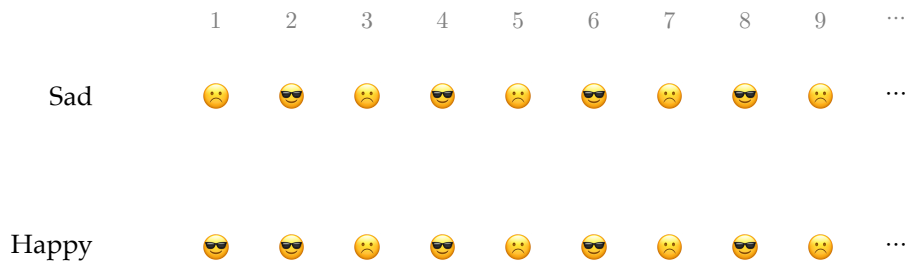


Figure 3.2: Two simple infinite worlds, Happy and Sad.

Here is a flat-footed argument that Happy and Sad are equally good. How good is a simple world overall? One standard thing some people like to say is that we can calculate the overall value of a population's well-being by representing each person's well-being with a number — called an *individual utility* — and then adding up all of those individual utilities. Let's give it a shot. Here are the utility streams representing these two worlds:

$$\begin{aligned}\text{Sad:} & \quad (0, 1, 0, 1, 0, 1, \dots) \\ \text{Happy:} & \quad (1, 1, 0, 1, 0, 1, \dots).\end{aligned}$$

Then we can try to calculate the sum of individual utilities in each case. For Sad, we calculate

$$0 + 1 + 0 + 1 + 0 + 1 + 0 + 1 + 0 + \dots = \infty.$$

For Happy, we get

$$1 + 1 + 0 + 1 + 0 + 1 + 0 + 1 + 0 + \dots = \infty.$$

Both sums are infinite, in the sense that as more and more terms are taken into account, both sums increase without bound. So (one might be tempted to say) the two worlds are equally good!

That's the most flat-footed argument for Everything-Bagelism.³ But it is not a good argument. It does show us something — but what it shows us is just that sideways eight is a bad way to represent infinite value, which we should consign to the flames.

This is the same lesson we learned from flat-footed discussions of Pascal's wager (in chapter 2). There the issue was that a single infinite utility was inadequate for distinguishing *certainty* of an infinite outcome from a smaller *probability* of an infinite outcome. Here we find that a parallel issue arises when it comes to distinguishing between *different* infinite outcomes — like Happy and Sad. As in Zeno's paradox of addition (section 2.1), it again seems that "unity joined to infinity does not increase it at all" (Pascal [1670] 1962, frag. 418). This time the issue has nothing to do with probabilities.

The *extended reals* consist of the standard real numbers together with two additional elements: ∞ and $-\infty$. The extended reals are inadequate for representing the value of infinite worlds. At this point we could turn to more sophisticated mathematical representations of infinite value, such as vector utilities or hyperreal utilities (as discussed in section 2.3). Those can be useful and appropriate, and ultimately I will want to deploy something along those lines (see section 3.5). But let's not get ahead of ourselves: as we already found in our discussion of Pascal's wager, once again utility representations are not the core issue.

When we talk about the overall value of the world, we shouldn't take this too literally, as if there was some cosmic scoreboard out there showing some *number*, and the goal of benevolence is to make that number bigger. We will have use for mathematical representations for the comparative value of

³Compare M. T. Nelson (1991, 343):

If maximal utility is to have any real sense at all, it has to denote a quantity, but in the case of [the two infinite worlds under discussion], the quantities of utility involved are actually the same.

Similarly, Hamkins and Montero (2000, 240):

We know that the intuition behind the principle that adding elements to an infinite set yields a strictly larger set is simply mistaken, and this is universally agreed. Why should we expect a difference when adding utility to an infinite world? We shouldn't...

Bostrom (2011, sec. 1.1) presents a similar argument, without endorsing it:

We can do only a finite amount of good or bad. Yet in cardinal arithmetic, adding or subtracting a finite quantity does not change an infinite quantity. Every possible act of ours therefore has the same net effect on the total amount of good and bad in a canonically infinite world: none whatsoever.

worlds, but we should choose representations that respect our judgments about what would be better or worse. We should not let our value judgments be pushed around by whatever mathematical representations happen to be ready to hand. For now, let's forget about utility representations entirely, so we can focus clearly on the question of what is better.

Furthermore, we should remember what *kind* of value we are talking about in this book: namely, the *overall value of individual well-being* (recall our discussion in section 1.1). In short, our focus is on the value of *social welfare*.⁴ (This restriction of our focus should not be understood as assuming *welfarism* — that well-being is all that matters.) When we ask whether one world is better than another, what we are asking is whether one is better than the other *for the individuals* in those worlds. We should not be misled into worrying about whether we might make the *world itself* happy or sad, over and above making *individuals* happy or sad.

Likewise, when we ask whether a world is *socially better*, or better for a group of individuals *overall*, we shouldn't be misled into thinking that we are talking about what benefits or harms some further entity — “society” or “individuals overall”. Rather, what we are asking about is what is better for those individuals *on balance*: it is a question of *trade-offs*. Making the world better on balance means making good trade-offs: helping some without harming others too much. Do the benefits to some outweigh the costs to others?

The best possible kind of “trade-off” is when we can make things better in some respect at no cost at all to anyone. If we can help someone without sacrificing anyone's well-being at all, then this unalloyed good has nothing to trade off against it, as far as the value of individual well-being is concerned. This should be the easiest kind of value comparison. We can sum up this idea with the following principle.

Pareto (*Ex Post*). Let $\mathcal{I}$ be a collection of individuals, and let v and w be any worlds.

If each individual in $\mathcal{I}$ is as well off in v as in w , then v is as good as w for the individuals in $\mathcal{I}$, on balance.

If each individual in $\mathcal{I}$ is as well off in v as in w , and furthermore, some individual in $\mathcal{I}$ is strictly better off in v than in w , then v is strictly better

⁴This use of “social welfare”, for the good of a group of individuals, should not be confused with a much broader use you also find in economics. Sometimes the label “social welfare function” is applied to a mathematical representation that corresponds to some “social ranking” of outcomes, regardless of what considerations that ranking is based on.

than w for the individuals in $\mathcal{I}$, on balance.

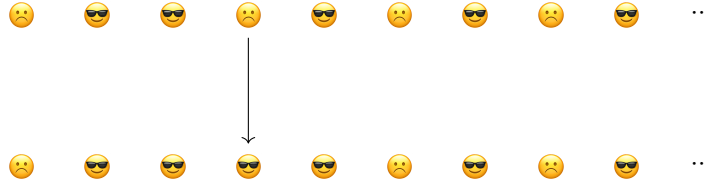


Figure 3.3: A Pareto improvement.

(The qualification “Ex Post” will be made clear in chapter 7. In what follows I will let the reference to a collection of individuals, and the qualification “for the individuals in $\mathcal{I}$ ” remain implicit. We also may wish to restrict the scope of the Pareto principle to worlds like those in our toy model, where there are not other confounding considerations like desert or community to worry about.)⁵

To accept Pareto is to reject the Everything Bagel. If you start with a world that already contains infinitely many happy people and infinitely many sad people, and you make just one sad person happy, this really does make the world better overall. It doesn’t matter that we haven’t changed the *number* of happy people. We don’t care about numbers — we care about people. (See also Vallentyne and Kagan 1997; Lauwers and Vallentyne 2004.)

One consequence of the Pareto principle is that there is no difference in *overall* well-being without any difference in *individual* well-being.

Pareto Equivalence (Ex Post). If every individual in $\mathcal{I}$ is equally well off in v as they are in w , then v and w are equally good for the individuals in $\mathcal{I}$, on balance.

(This follows by applying the weak clause of Pareto in both directions.) It could well be that two worlds with precisely the same distribution of well-being differ in some *other* morally important respect — say, in how just or diverse they are. But it is hard to see how they might differ with

⁵We may not be able to so easily set aside “pattern goods” like equality or diversity, if these are constitutively connected to individual well-being. But note that these other values (whether or not they are “pattern goods”) can only threaten the Pareto principle if they matter in a particular way: namely, by making a difference to how good things are *for these individuals*. A different view is that these other things matter *apart* from social welfare (narrowly construed): we shouldn’t just care about what is best for the individuals, but *also* about their desert, community, equality, or diversity. This version of the idea is no threat to the Pareto principle as stated above.

respect to how well-off individuals are, on balance, if there is nothing added or removed from the scales. (Again, though, we may wish to restrict the scope of Pareto: maybe it only applies to worlds without other confounding complications. That's fine, but I don't think it will make much difference in what follows, so I won't keep mentioning this.)

Here is another idea. As I have said, we are considering the value of worlds *impartially*. Jay's happiness may matter more to Jay than Kay's does, and Kay may have the opposite priorities. But here we are considering whether we can sensibly evaluate what is better or worse for a collection of individuals *overall*, giving priority to neither of these more parochial perspectives. Consider a status quo in which Jay and Kay are both sad, and we have the ability to either make Jay happy or else make Kay happy, with all other things being equal. From an impartial perspective, Jay's happiness and Kay's are equally weighty considerations. So, weighing everyone's well-being equally, we should say that these two outcomes are equally good. We can put the point like this.⁶

(Happy-Sad) Impartiality. For any individuals i and j (in a set of individuals $\mathcal{I}$) and any worlds v and w , if for each individual k (in $\mathcal{I}$) besides i and j , k is just as well off in v as in w , and in v , i is happy and j is sad, while in w , i is sad and j is happy, then v and w are equally good (for the individuals in $\mathcal{I}$).

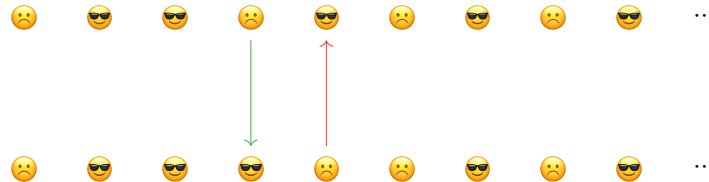


Figure 3.4: Impartiality.

⁶To be clear, this principle is not intended as a conceptual analysis of what it is for an axiology to be impartial in an intuitive sense. As Muñoz (2025) points out, theories can violate this principle even without being partial toward some individuals over others. Take, for instance, the *pure Paretian* order, which says that v is better than w if and only if v is a Pareto improvement over w . Crucially, this allows for *incomparability* between worlds, where neither Pareto-improves on the other. For a theory to be impartial, it must not say that i 's happiness is *weightier* than j 's; but it also need not say that i 's happiness is *equally* weighty as j 's: they may be incomparable. Nonetheless, I will be exploring axiologies that are impartial in this stronger sense, requiring *equivalence* rather than mere incomparability. (Indeed, in chapter 4 I will argue against value incomparability more generally. In particular, see discussion of *permutation invariance* in section 4.2.)

It can streamline discussion to put things at a slightly higher level of abstraction. We mentioned above that we can represent a world with a *social outcome*, a function from individuals to well-being levels. Pareto Equivalence is equivalent to the claim that any two worlds with the same social outcome (for $\mathcal{I}$) are equally good for the individuals in $\mathcal{I}$. Note, furthermore, that every social outcome represents some simple world. Thus, if Pareto Equivalence is true, we can stipulatively define a relation on social outcomes:

For any social outcomes x and y , x is **as good as** y (written $x \succeq y$)
iff for any world v that is represented by the social outcome x ,
and any world w that is represented by the social outcome y , v is
as good as w for the individuals in $\mathcal{I}$.

(We can also define $\sim$ and $>$ in terms of $\succeq$, in the usual way.) This definition lets us abstract from many details of worlds, and compare them just based on their individual well-being profiles. We can equivalently restate Pareto and Impartiality in terms of social outcomes.

Pareto. Let x and y be social outcomes.

If $x(i) \geq y(i)$ for all i in $\mathcal{I}$, then $x \succeq y$.

If, furthermore, $x(j) > y(j)$ for some j in $\mathcal{I}$, then $x > y$.

Impartiality. Let x and y be social outcomes, and let i and j be individuals in $\mathcal{I}$. Suppose:

$$\begin{aligned} x(k) &= y(k) \quad \text{for all } k \text{ in } \mathcal{I} \text{ besides } i \text{ and } j; \\ x(i) &= y(j); \\ x(j) &= y(i). \end{aligned}$$

Then $x \sim y$.

It is usually more convenient for technical purposes to work with social outcomes; but philosophically we often want to focus on more fully described worlds. I will move back and forth freely between these two ways of describing the issues.

There is a nice foundational technical result that formalizes the point about the inadequacy of the extended reals. (The point extends much more broadly than the basic observation about *adding up* individual utilities.)

Theorem 3.1 (Basu and Mitra (2003)). No ordering of social outcomes that satisfies Pareto and Impartiality can be represented by the extended reals. (That is, there is no function u from social outcomes to the extended reals

such that whenever $x \succ y$, $u(x) > u(y)$.)

(See section A.3 for details.)

As we have already discussed, if we are going to escape the Everything Bagel and maintain that improving a life can make the world a better place, we must leave the extended reals behind.

Even though orderings respecting Pareto and Impartiality are not real-representable, there certainly *are* such orderings. In the context of our toy model, the simplest example is the *counting order*. This says that switching from outcome y to outcome x makes things better iff only *finitely* many happy people in y are made sad in x , and *more* sad people in y are made happy in x .

The Counting Order. For $\{\text{😊}, \text{😞}\}$ -social outcomes x and y , $x \succeq^\# y$ iff there is some finite number of people n who are happy in y and sad in x , and there are at least n people who are sad in y and happy in x (perhaps infinitely many).

More concisely: for any $\{\text{😊}, \text{😞}\}$ -social outcome y , let $H(y)$ be the set of people who are happy in y . Then $x \succeq^\# y$ iff

$$\#(H(x) \setminus H(y)) \geq \#(H(y) \setminus H(x))$$

and the right-hand side is finite.⁷

The counting order satisfies Pareto and Impartiality.⁸

The counting order is very weak: it doesn't make lots of comparisons that we might like to make. (In the terms of section 2.3, it satisfies Reflexivity and

⁷The notation $\setminus$ stands for set difference: in general, $A \setminus B$ is the set of elements of A that are not in B .

Note that the requirement in the definition that $\#(H(y) \setminus H(x))$ is finite is crucial. Say we define $x \succeq^? y$ in the same way as $\succeq^\#$, except we also allow $x \succeq^? y$ in the case where $\#(H(x) \setminus H(y))$ and $\#(H(y) \setminus H(x))$ are both infinite. This relation $\succeq^?$ is not transitive. For example, let x, y, z be social outcomes with the following sets of happy people:

$$H(x) = \{1, 3, 5, 7, \dots\};$$

$$H(y) = \{2, 4, 6, 8, \dots\};$$

$$H(z) = \{3, 5, 7, 9, \dots\}.$$

You can check that $x \succeq^? y$ and $y \succeq^? z$, but $x \not\succeq^? z$.

⁸The counting order is the *Suppes–Sen grading principle*, specialized to the case of just two welfare levels (Suppes 1966; Sen [1970] 2017). It is also closely related to the *overtaking criterion* (see footnote 40 in section 3.5). For overview see Pivato and Fleurbaey (2024).

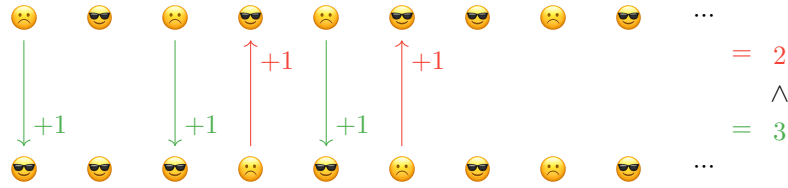


Figure 3.5: The counting order.

Transitivity, but not Comparability.) For example, it makes no comparison between the outcome where all and only the odd-numbered people are happy, versus the outcome where it is instead the even-numbered people who are happy.

For another example, consider the world *Twos*, each individual at an even numbered room is happy, and the rest are sad. In the world *Sevens*, each individual at a multiple of seven is happy, and the rest are sad. Many people have the intuition that *Twos* is better than *Sevens*, since it seems worse for happiness to be spread out so much more sparsely.⁹ The counting order does not deliver this verdict: it makes no comparison between *Twos* and *Sevens*.

There is some reason to doubt this intuition.¹⁰ We picked out the people in simple worlds according to their arrangement in rooms 1, 2, 3, But it is implausible that it is *essential* to these people to be arranged this way: they could have been arranged differently. Indeed, perhaps they could have been arranged so that the individuals we have labeled 2, 4, 6, ... were in fact in the *rooms* labeled 7, 14, 21, ..., while the individuals 1, 3, 5, ... were in rooms 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, Now consider two new worlds, *Twos*[†] and *Sevens*[†], in each of which people are arranged in this new way. (To be clear, *Twos*[†] and *Sevens*[†] do not count as *simple worlds* in our original sense; in all simple worlds, people were arranged the same way.) In *Twos*[†], just the people in even-numbered rooms are happy, while in *Sevens*[†], the people in multiple-of-seven-numbered rooms are happy. The same “happiness density” intuition that led us to say that *Twos* is better than *Sevens* would now lead us to say that *Twos*[†] is better than *Sevens*[†]. But note: *Twos* and *Sevens*[†] are *Pareto-equivalent*: exactly the same individuals are happy in each of these two worlds. Likewise, *Sevens* and *Twos*[†] are Pareto-equivalent.

⁹Lauwers and Vallentyne (2004); E. Jackson and Rogers (2019); see also discussion of the overtaking principle in footnote 40.

¹⁰Compare Askill (2018, 92ff.)’s discussion of “Clement” and “Stormy”.

So Pareto Equivalence is inconsistent with the combination of judgments $\text{Twos} > \text{Sevens}$ and $\text{Twos}^+ > \text{Sevens}^+$. We can have either one of these, but not both — and it might well seem unprincipled to accept one without the other.¹¹

So the rampant incomparability built into the counting order may be a feature rather than a bug. (I will return to this issue in section 4.2.) In any case, it certainly leaves much to say about how things go in more realistic worlds, where people have more going on than just being happy or sad. There is lively debate about these questions. But even if the counting order is not a good general theory of value, it is a good proof of concept. Since the counting order satisfies Pareto and Impartiality, it isn't as if the project of comparing worlds in a way that respects these principles is hopeless, even if we cannot represent such orders by real numbers. Furthermore, the counting order is the *most general* such order for simple worlds.

Proposition 3.2. Any order of social outcomes that satisfies Pareto and Impartiality extends the counting order. (See section A.3.)

Generally speaking, the project of extending the counting order looks to me like a fruitful research program. The deficiencies of *real-valued representations*

¹¹Some people might be driven by such thoughts away from Pareto altogether, and toward alternative views that give spatio-temporal arrangement a central role. This is a feature of many dominant views in infinite ethics (see for example Bostrom 2011; Arntzenius 2014; Wilkinson 2022b; Pivato 2023; see also Pivato and Fleurbaey 2024 for overview). Instead of *individuals*, these approaches often center *regions of space-time* as the basic “locations of value”. I will not pursue this in this book, but I want to note a couple of things.

First, even though I will not aspire to this much generality, much of this book's content could be straightforwardly adapted to alternative “locations” that are not individuals. So, people who like geometric approaches to ethics can still benefit from reading on.

Second, the reason switching from people to places might help is because it seems plausible enough that places could not have been arbitrarily rearranged. But if we really want to hang on to some kind of “density” idea, we might postulate some similar essentialism about individuals. Surely individuals could have been in different places — but that is not to say they could have been *arbitrarily* rearranged. Perhaps infinite rearrangements like Twos and Sevens⁺ are simply impossible. Concretely, we might propose that only rearrangements with *bounded displacement* are possible: for some distance d , each person is moved at most d . Bounded displacement permutations preserve limiting densities. (Or one might restrict further to *fixed step permutations*, which preserve the *overtaking order*: see Lauwers (1997), and footnote 40 below.) This kind of “arrangement essentialism” may be a bit surprising, but it doesn't seem wildly out of step with other more familiar ideas like origin essentialism. (Similar ideas can also be motivated by Leibniz shift style arguments; compare Maudlin (1988).) This would allow us to consistently maintain Pareto in conjunction with some kinds of “arrangements matter” intuitions. But I won't pursue this.

for orders should not make us fear the Everything Bagel.

3.3 The moral telescope

Another threat looms. Consider once again the two simple worlds Happy and Sad. Each of them contains infinitely many happy people and infinitely many sad people. So we can pair up people who are happy in the Happy outcome with people who are happy in the Sad outcome: the first person in Happy is paired up with the first happy person in Sad, the second happy person with the second happy person, and so on. We can also pair up the sad people from each outcome in the same way. This defines a one-to-one correspondence among individuals, such that each person i is associated with some other person j such that i 's well-being in the Happy outcome is precisely the same as j 's well-being in the Sad outcome. In short, these two worlds are *well-being isomorphic*.

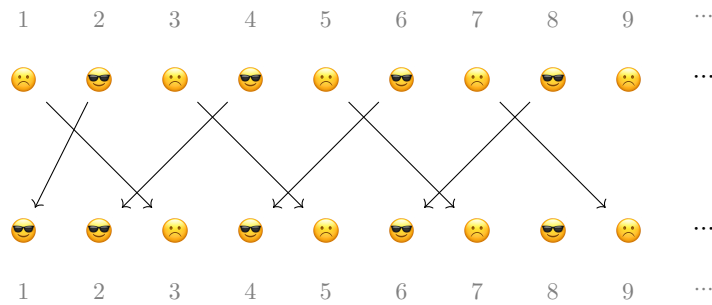


Figure 3.6: There is a well-being isomorphism π that maps Sad to Happy. (Namely, $(\pi(1), \pi(2), \pi(3), \dots) = (3, 1, 5, 2, 7, 4, \dots)$.)

You might think that this means the two worlds are equally good. They have the same overall distribution of happiness and sadness — the same infinite cardinal number of happy people, and the same infinite cardinal number of sad people. If that is all that matters, then Happy and Sad are equally good after all.¹² This is a different argument for Everything Bagel-ism, which is more powerful than the one from naïve extended-real addition.

¹²Compare Hamkins and Montero (2000, 233)'s discussion of a different case:

Those numbers are supposed to capture all that is important for a utilitarian to know in order to arrive at an abstract evaluation of the worth of the two [worlds]. For a utilitarian, it seems that any extra information over and above the spectrum of utility numbers just doesn't matter when comparing relative worth.

The situation is typical. Call a world *plenitudinous* iff it includes infinitely many individuals at every level of well-being. In a plenitudinous world, if you only change things for just finitely many individuals, then you get back something that's still well-being isomorphic to what you started with. If well-being isomorphic worlds are equally good, then no merely finite intervention can improve a plenitudinous world.

We can sum up the point like this. First, we can state the well-being isomorphism principle like this. Let π be a *permutation* of the relevant collection of individuals (a one-to-one and onto function from $\mathcal{I}$ to $\mathcal{I}$). Say two worlds v and w are *related by π* iff for every individual i , i in v is just as well off as $\pi(i)$ in w .

Strong Anonymity. Any two worlds that are related by a permutation are equally good.

Proposition 3.3. Strong Anonymity is inconsistent with Pareto.

Moreover, it may seem dialectically unstable to reject Strong Anonymity while affirming the Happy-Sad Impartiality principle we have already discussed. Let a *finite* permutation be one that leaves fixed all but finitely many individuals. Then Happy-Sad Impartiality can be equivalently restated like this.

Finite Anonymity. Any two happy-sad worlds that are related by a *finite* permutation are equally good.

If you put it this way, Finite Anonymity looks like just an artificial restriction of the more natural principle of Strong Anonymity.

I don't think that's right, though. Strong Anonymity is motivated by the idea that value resides in certain features of the *overall qualitative pattern* of individual values. That amounts to taking the cosmic scoreboard perspective: the *global pattern of the world as a whole* is the thing we are trying to improve. But — again — our topic is not whether one world is better than another with respect to its global pattern: rather, it is whether it is better *for certain individuals*, on balance. What we care about intrinsically is not the *world*, but *people*. Moreover, what it means for a world to be better than another for those people *overall, on balance*, is for its comparative benefits to individuals to outweigh its comparative costs. The ways in which people are better off in this world, rather than another, favorably trades off against the ways in which people are worse off.

The trade-off perspective motivates Impartiality: it is natural to think that making one sad person happy is the sort of thing that can balance off failing to make a different sad person happy. But this perspective does not motivate Strong Anonymity. If we recast Strong Anonymity in terms of trade-offs (in the setting of simple worlds), it says that making any (countably) *infinite* set of sad people happy is just as good overall as making any other (countably) infinite set of sad people happy. But this idea about improving different *infinite sets of lives* is in much worse standing than the idea about improving different *individual lives*. Indeed, we have good reason to think that different (countably) infinite sets of people are *not* equally weighty for social welfare — since one countably infinite set can be a proper subset of another. In general, we don't have any good reason to think that trade-offs between infinite collections of people are well-measured by their cardinalities. Infinite cardinalities are far too coarse-grained.¹³

There is another subtler disanalogy between Strong Anonymity and Impartiality. Strong Anonymity depends on the idea of two different individuals having the *same* level of well-being: this is built into the definition of “related by a permutation”. Finite Anonymity is also stated in these terms, but this is more of an accident. In fact, the core idea of the Impartiality principle stated above does not really rely on interpersonal comparisons of well-being in the same sense as Strong Anonymity. We said at the outset that in simple worlds each individual has one of two “levels of well-being”, “happy” or “sad”. But nothing actually turns on one individual's happiness or sadness being at the *same* level of well-being as any other individual's happiness or sadness. What matters, rather, is that the *change* from sadness to happiness for one individual is of equal moral weight as the corresponding change from sadness to happiness for another individual. Intuitively, what Impartiality relies on is just that there are certain *differences* or *intervals* in individual well-being

¹³A related question is whether infinite cardinal differences have any moral significance at all. Consider world with an *uncountable* infinity of individuals, each of which is either happy or sad. Should we think that a world with uncountably many happy people is better than a world with only countably many happy people? This might seem attractive, but I don't know of any good reason to accept it. Pareto and Impartiality do not settle whether this principle is true (see section A.3). In general, Cantorian cardinal numbers do not seem to be very morally important. In section 3.5 we will discuss a replacement notion of *quasi-cardinality* which is much more closely related to moral value. In general, though, these different cardinal notions can come apart dramatically. There is a family of different quasi-cardinal concepts. Some of them have this property: a set *A* can have greater *quasi-cardinality* than *B* even if *B* has greater *Cantorian* cardinality than *A*. (This requires making an appropriate choice of filter; see section B.5.)

that can be compared between individuals.¹⁴ The point here is not that such interval comparisons are in better standing than level comparisons. Rather, the point is that the difference in their presuppositions about the structure of well-being helps us see that Impartiality and Strong Anonymity are philosophically different ideas.

Sometimes Strong Anonymity is motivated by the idea that *renaming* individuals makes no moral difference.¹⁵ This idea is even built into the standard name of the principle; but it can easily lead to confusion. The difference

¹⁴In particular, what we really need for the statement of Impartiality is that there exist *some* individual outcomes — label them “happy_i” and “sad_i” for each of infinitely many individuals *i* — that are “interpersonally calibrated” in the right way: improving one individual from sad_i to happy_i is (socially) just as good as improving any other individual from sad_j to happy_j. For this to intuitively count as a principle of *impartiality*, the difference between happy_i and sad_i should be the “same” as the difference between happy_j and sad_j. (But we might think that the social value facts are what *make* the differences the same, rather than the other way around.)

Utilitarian aggregation is the classic example of a theory that requires interpersonal *interval* comparisons of well-being, without requiring interpersonal *level* comparisons. (For discussion, see Sen [1970] 2017; see also Nebel 2024, sec. 5.1.) The same holds for generalizations of utilitarianism to infinite settings (see sections B.7 and 3.5). In this context, there is a natural way to make sense of Impartiality (and indeed stronger principles like the one Nebel (2024) calls “Interval Anonymity”). But in this setting, Strong Anonymity is not obviously meaningful at all.

This example shows that level-comparisons are not necessary to make sense of Impartiality; but, to be clear, full *interval* comparisons are not necessary either. For example, we need not make sense of comparing the difference between happy_i and sad_i to other intervals elsewhere on an individual’s well-being scale, such as the difference between sad_i and miserable_i. Nebel (2024, 310–11 and appendix B) argues that a full-fledged *Interval Anonymity* principle is very strong, basically tantamount to a form of utilitarianism. (He says that this argument “requires [Finite] Anonymity to be *strengthened* to [Interval Anonymity]” (p. 311, my emphasis), but I think this is a slip: Finite Anonymity is not even meaningful for the kind of “hybrid difference structure” Nebel discusses.)

In short, *either one* of interpersonal level comparability or full interval comparability suffices for making sense of Impartiality, in our sense. But neither of these entails the other, so neither one of them is necessary.

¹⁵For example, D’Aspremont and Gevers (1977, 202):

... what really matters in each state is the list of individual utility levels, not the names attached to them. This idea is captured by the well known *anonymity* axiom.

Similarly, Gaertner (2009, 47):

Renumbering the individuals or changing their name-tags should not matter ...

(Though note that here Gaertner (2009) is discussing a principle more closely connected to *Permutation Invariance*, rather than Strong Anonymity; see section 4.2.)

between the worlds Happy and Sad has to do with an individual's *well-being*, not their *name*.

Here is one path to confusion: let's consider an alternative numbering scheme for the individuals. We originally numbered off the rooms and the individuals in them 1, 2, 3, 4, With respect to this ordering of individuals, we can describe the well-being distribution in Sad with the sequence (😞, 😞, 😞, 😞, ...), and the well-being distribution in Happy with the sequence (😎, 😎, 😞, 😞, ...). But, if we like, we could label the individuals in some other way — say, by the sequence of numbers

$$3, 1, 5, 2, 7, 4, 9, 6, \dots$$

(This is the sequence $(\pi(1), \pi(2), \pi(3), \dots)$ for the permutation π illustrated in figure 3.6.) Now if we list off the well-beings of each individual according to their place in the *new* ordering, we represent Sad with the sequence

$$(\text{😞}, \text{😞}, \text{😞}, \text{😞}, \text{😞}, \text{😞}, \dots).$$

(See figure 3.7.) This is just the same as our original representation of Happy! But surely we don't think that the choice of how to number off these individuals is morally important in itself.

New numbering	1	2	3	4	5	6	7	8	9	...
Old numbering	2	4	1	6	3	8	5	10	7	...
Sad	😞	😞	😞	😞	😞	😞	😞	😞	😞	...
Happy	😎	😎	😎	😎	😞	😎	😞	😎	😞	...

Figure 3.7: A different representation of the very same worlds Sad and Happy as before: the individuals are just numbered off in a different order. Note that if π is the permutation from figure 3.6, then the sequence of *old* numbers corresponding to the *new* numbers $(1, 2, 3, \dots)$ is $(\pi^{-1}(1), \pi^{-1}(2), \pi^{-1}(3), \dots) = (2, 4, 1, 6, \dots)$.

This is all correct so far — but it would be a mistake to conclude that Sad and Happy are equally good. We can use the same sequence to represent Sad as that we use to represent Happy — but that is to give the sequence two different *meanings*. By analogy, I could use the name “Joe” for a president, or for a cup of coffee; that doesn't mean that the president and the cup of

coffee have the same moral worth, or that they are alike in any other respect. Furthermore, note that when we introduce the new labels for individuals, this also changes what sequence we use to represent the world *Happy*. Its sequence is still better than Sad's sequence in one place — we have just relabeled the individual who is made better off with the number 3 rather than the number 1. This should be no surprise — we just redescribed the same pair of worlds!

We stated Strong Anonymity in terms of permutations of individuals. As mere mathematical functions, such permutations can be interpreted in two quite different ways. In the terminology from physics, we could interpret a permutation of individuals as a *passive* transformation, relating two different ways of *representing* a single world — merely exchanging the *labels* we use for individuals. Or we can understand a permutation as an *active* transformation, relating two different ways the world could be — holding fixed which individuals we are talking about, and reconfiguring what they are like.¹⁶ In the Strong Anonymity principle, permutations are understood as *active* transformations. It is not a trivial principle about alternative possible relabelings of the same situation; it is a very strong principle about the relationship between *different* possible ways of distributing well-being to individuals. (I have belabored this point a bit because similar issues arise throughout this book, in our discussion of other kinds of “symmetries of value”: see section 4.2; chapter 7; section 9.3.)

In short, it is reasonable to accept Impartiality without Strong Anonymity. Still, puzzles remain. Let's consider some new worlds. In *simple worlds*, we put people in rooms arranged along an infinite ray, in correspondence with the natural numbers. Now let's instead arrange people along an infinite line stretching out in both directions — call those directions East and West. We'll call these *East-West worlds*. We'll consider two particular East-West worlds: call them, Happy* and Sad*. Indy is an individual at a certain place in this line. In both worlds, everyone to the West of Indy is sad, and everyone to the East of Indy is happy; the two worlds only differ in how Indy fares: in

¹⁶For discussion see Earman (1989). Consider two models M and M' sharing the same domain, which are related by a non-trivial isomorphism h . For example, M and M' might each be some scalar field on space, represented by $\mathbb{R}^3$. To interpret h as an *active* transformation is to construe M and M' as representing different situations, holding fixed what the elements of the two models represent: for instance, we hold a coordinate system fixed, and “push around” the field in space. To interpret h as a *passive* transformation is to construe M and M' as representing the *same* situation in different ways: for instance, we hold fixed the field, and instead transform the *coordinate system*. In general, we can interpret each element x of the transformed model M' as representing the same thing that $h^{-1}(x)$ represents in M .

Happy*, Indy is happy, and in Sad*, Indy is sad. Everyone else is equally well off either way.

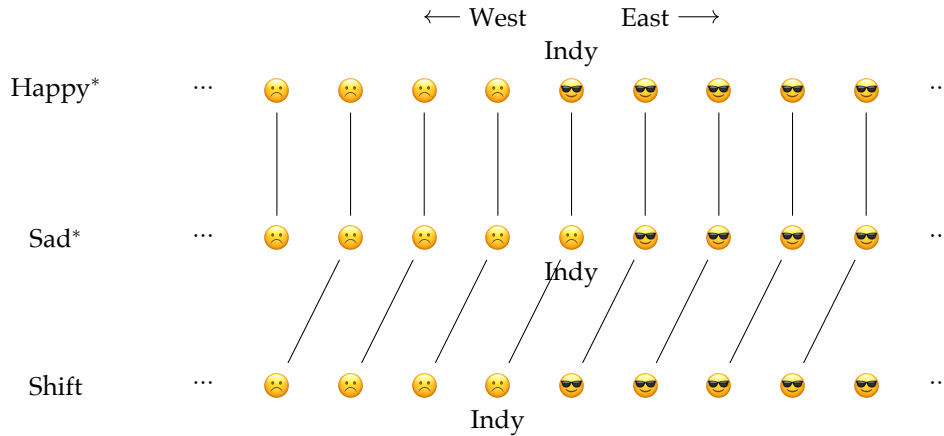


Figure 3.8: Happy* looks just like Sad* shifted one room to the left.

Again, the Pareto principle tells us that Happy* is better than Sad*. But these two worlds are not merely related by some arbitrary well-being-preserving permutation. In fact, suppose we start with Sad*, and then instruct each person to move to the position of their immediate neighbor to the West. Call this world Shift. But now we have a new puzzle. Surely uniformly moving everyone in space makes no moral difference to how good things are.¹⁷ So Sad* is just as good as Shift. But Shift is exactly like Happy*, except for the identities of the people involved. Surely, then, Shift is just as good as Happy*? And this contradicts the Paretian judgment that Happy* is strictly better than Sad*.

The function that maps each person to their immediate neighbor is not just a *well-being* isomorphism between Happy* and Sad*: it is an isomorphism

¹⁷We recall Leibniz's "shift" argument (Clarke and Leibniz [1717] 1956, 26):

... 'tis impossible there should be a reason, why God, preserving the same situations of bodies among themselves, should have placed them in space after one particular manner, and not otherwise ...

Leibniz concludes, based on the Principle of Sufficient Reason, that such a shift in space is impossible — and so absolute space is a mere "chimerical supposition". If we follow him to this conclusion, then this argument poses no threat to the Pareto principle — see the discussion of anti-haecceitism below. But we can affirm that the shifted possibility would be equally good, without affirming with Leibniz that it is absurd for there to be two such possibilities that are equally good.

in a stronger sense. Hamkins and Montero (2000, 235–36) argue:

It seems that what many take as the relevant information in evaluating a world is the natural order of locations, such as the order of time or the topological structure of space, and the amount of local goodness at each location. ... [I]t is natural to define a notion of isomorphism between worlds that respects this information. Specifically, let us call a counterpart function that exactly copies the local goodness and topological structure of the locations of one world to another world an *isomorphism* of the two worlds. ... Since isomorphisms preserve what we assumed is important in evaluating an individual world, we are led, almost tautologically, to the following principle:

Isomorphism Principle (IP): Any world is just as good as any isomorphic copy.

This reasoning suggests that Happy* and Sad* are equally good.

The Isomorphism Principle is tempting, but we should reject it. A non-trivial isomorphism does *not* preserve everything that matters. One of the things that matters is whether Indy is happy or sad — and while Indy is happy in Happy*, the shift isomorphism that relates Happy* to Sad* maps Indy to someone who is sad.

Hamkins and Montero consider isomorphisms of certain “relevant information”, including the structure of time and space and the distribution of local goodness across it. But we can go further, and consider isomorphisms in a stronger sense. Let us make another stipulation about Happy* and Sad*: all of the people in these worlds are very much alike. In particular, each of the happy people in these worlds is an exact duplicate of each of the others (including Indy in Happy*), and likewise each of the sad people is an exact duplicate of each of the others (including Indy in Sad*). Furthermore, the entire world they live in is spatially symmetric, in such a way as to guarantee that the function that maps each person in Happy* to their neighbor in Sad* does not just preserve their well-being and relative spatial arrangement, but in fact it preserves *every* qualitative feature of the world.¹⁸ Each individual in Happy* is mapped to an individual in Sad* who is *qualitatively indiscernible* (both intrinsically and extrinsically). Call such a function a *qualitative isomor-*

¹⁸Intuitively, qualitative features are *general* feature, that do not essentially say anything about particular individuals. See Adams (1979) for an influential discussion; see also D. Lewis (1986b), pp. 61–63; Cowling (2015); Plate (2022).

phism. But even though Happy* and Sad* are qualitatively isomorphic, I maintain that Happy* is better than Sad*. It is still better in one important respect — Indy’s happiness — and worse in none.¹⁹

This case is a simplified version of the situation that we may really face. Suppose we live in a *plenitudinous* world — an infinite world, teeming with people of every qualitative type (as we might).²⁰ Say I am sad in this world. Would it be better if I were happy, and otherwise everything was just the same? Yes! But the plenitudinous world already includes infinitely many sad people just like me, and infinitely many happy people just like I would be if I were happy. Merely making me happy makes no qualitative difference to the world as a whole, yet (I maintain) things would be better that way.

Recall a slogan from Kripke (1980, 44): “A possible world isn’t a distant country that we are coming across, or viewing through a telescope.” For example, I could have been a little bit taller. This is something we know. It’s a mistake to think that in order to establish this, we need to start by surveying the overall qualitative pattern of a possible world, and then somehow try to deduce from that which person is *me* (if any), and finally read off that person’s height. That’s not how modal epistemology works.

We shouldn’t be any more tempted by telescopic reasoning when it comes to value. Here you are again, considering whether to rescue a drowning child. It’s not as if, in order to determine whether things would be any better if you did, all you have to go on is the cosmic pattern of child-drownings and rescues. You could rescue *this child*. You don’t need to get out your modal telescope to establish this. Moreover, this child being rescued, rather than drowning, is a respect in which the world would be *better*. The world would not be *worse* in any countervailing respect — it is not as if you rescuing this child must be somehow offset by some other distant child being left to

¹⁹Thus, while I’m inclined to accept Leibniz’s value judgment in the case of the mere shift, I reject the general principle he invokes (Clarke and Leibniz [1717] 1956, 26):

But in truth the one would exactly be the same as the other, they being absolutely indiscernible; and consequently there is no room to enquire after a reason of the preference of the one to the other.

In fact, there can be such reasons to prefer one possibility from another qualitatively indiscernible possibility: Indy’s happiness provides just such a reason.

²⁰In order to fit them all, the infinite world must be sufficiently large, or else “qualitative types” must be sufficiently coarse-grained. Things are cleanest if we suppose we live in an infinite but *discrete* world, such that there are only finitely many different ways for physical objects to be configured within a bounded region of space-time. (See also J. S. Russell 2020b; J. S. Russell and Hawthorne 2018.)

drown. That is enough. We should not be distracted by isomorphisms.

One might be tempted by another line of reasoning. Here is Indy again, sad in the world Sad*. The world Happy* contains infinitely many qualitatively indiscernible individuals just like happy Indy, and infinitely many qualitatively indiscernible individuals very much like Indy except that they are happy. We might wonder: which of the people out of this multitude is Indy's *counterpart* — the one whose qualitative properties correspond to ways that Indy could have been? We may feel baffled. (Also, how can we say that one of the *happy* people is Indy's counterpart, when the sad people are even more similar to the way Indy actually is?) If we have trouble choosing a happy counterpart for Indy in the world Sad*, we might be led to doubt whether Sad* really presents the possibility of Indy being happy.

But we shouldn't let this "telescopic" picture of possible worlds push around our judgments of how things could have been. Indeed, while Kripke's injunction is often read as a criticism of *counterpart theory* (à la D. Lewis 1973, 1986a), in fact, even the counterpart theorist can and should accept this lesson.²¹

Let *anti-haecceitism* be the view that all facts supervene on qualitative facts. Let *haecceitism* be its negation. For instance, anti-haecceitism says that how happy *Indy* is supervenes on the qualitative pattern of the world. To be precise: Let *Q* be *qualitatively complete* iff for any qualitative proposition *Q'*, either *Q* entails *Q'* or else *Q* entails not-*Q'*. Let *Q* be *complete* iff for any proposition *P*, either *Q* entails *P* or else *Q* entails not-*P*. *Anti-haecceitism* says that any qualitatively complete proposition is complete. So anti-haecceitism implies that, necessarily, if things were *qualitatively* just as they are in Sad*, then *Indy* would be just as he is in Sad* — namely, sad.

But not even the counterpart theorist is committed to anti-haecceitism in this sense. D. Lewis (1986a, 221) confused matters by stipulating an alternative meaning for "anti-haecceitism" and planting his flag on that view (for discussion see Skow 2007; Kment 2012). We need not concern ourselves with the details of the view Lewis called "anti-haecceitism". But Lewis went on to endorse what he called "cheap haecceitism" — which is, in fact, a version of haecceitism in the sense I have stipulated.²²

²¹The following discussion draws on ideas from Hazen (1979, esp. p. 334); D. Lewis (1986a, 230ff.); Skow (2007); Kment (2012); J. S. Russell (2013).

²²I am bracketing some annoying details about how to understand non-qualitative *propositions* — this is also complicated within Lewis's system.

Here is how it works. Consider again Indy, and a possible world containing infinitely many people much like Indy, some happy and others sad.²³ We need not identify *one* of these people as Indy's counterpart. Indeed, they may *all* be Indy's counterparts — all of them witness ways that Indy could have been. Some are possibilities where Indy is happy, and others are possibilities where Indy is sad. These various possibilities are all exactly alike as far as the qualitative pattern of the world goes, and yet they differ in how Indy fares.

Properly understood, counterpart theory is perfectly compatible with haecceitism ("on the cheap", as Lewis put it).²⁴ Counterpart theorists do not need telescopes, either.

The counterpart theorist typically will put things a little differently than I have, but this difference is merely verbal. As D. Lewis (1986b, 230) puts it, using his own idiom, "*Possibilities are not always possible worlds.*" Where I have said we have qualitatively distinct possible worlds Happy* and Sad*, the Lewisian counterpart theorist will say that there is just *one* "world" here *w*. This world allows for two different ways of selecting counterparts: Indy has both a happy counterpart and a sad counterpart in *w*. These two counterparts give rise to two different *possibilities*, as Lewis puts it — and these possibilities are what I called Happy* and Sad*. Someone who prefers the Lewisian idiom will have to replace all of my references to possible worlds with talk of "maximally specific possibilities". The things Lewis calls "possible worlds" do not arise in my discussion at all.

Counterpart theorists, as such, are not committed to anti-haecceitism. Even so, some philosophers really are full-fledged anti-haecceitists — not in Lewis's idiosyncratic sense, but rather in the sense presently under discussion, concerning what is possible. The anti-haecceitist says that it is impossible for *me* to be different in any way without reality as a whole being different in some *qualitative* way. This view is independent of counterpart theory. The anti-haecceitist might be tempted to reason like this:

²³Note that these may be the very *same* possible world.

²⁴It must be admitted, though, that properly understanding counterpart theory is not a trivial matter. It is often tempting to go to D. Lewis (1968) as a source on this, because this formulation of the theory is especially precise. But D. Lewis (1986b, 232) recognized that his 1968 theory was inadequate — partly on the grounds I have just discussed. Unfortunately, while he gave programmatic suggestions (1986a, sec. 4.4), he never stated any theory nearly as precisely as in his original paper, leaving much work to be done. See Milford (2022) for one attempt to systematize things, as well as references therein.

Everything supervenes on the qualitative. In particular, the *value* of a world is determined by its qualitative features. Since Happy* and Sad* are qualitatively indiscernible, these two possibilities are equally good. This contradicts Pareto.

But this is not what the anti-haecceitist should say. Rather, they should say that one of Happy* and Sad* is *impossible*. For the anti-haecceitist, it is not just the *overall value* of the world that supervenes on its global qualitative features, but also *Indy's happiness*. In Happy*, Indy is happy. In Sad*, Indy is sad. Since these two situations are qualitatively indiscernible, by the anti-haecceitist's lights they are not both possible. There aren't really two possibilities here, but one.

The anti-haecceitist can affirm the principle that qualitatively isomorphic possibilities are equally good. But this is because they affirm the stronger claim that qualitatively isomorphic possibilities are alike in *every* respect, including how well off each individual is — indeed, there are no distinct isomorphic possibilities. This means, though, that for the anti-haecceitist there is no conflict between the isomorphism principle and the Pareto principle, which says that it would be better if some were better off while all are at least as well off. Anti-haecceitism implies that it is impossible for there to be a Pareto improvement without any qualitative difference.

Say I am in a plenitudinous world, and I am sad. I think “it would be better if I were happy, and everything else just the same”. The nihilist like Jobu Tupaki, daunted by infinite value, says *that possibility would be no better*. The anti-haecceitist does not endorse Jobu Tupaki's nihilism, but rather *fatalism*: *that could not have been*. Either I could not have been happy, or, if I could, then the whole world would have been different in some qualitative respect — and so the world would be different in infinitely many far-flung places as well. Either idea is very strange.

Fatalism is an idea with a long pedigree, and this is not the place to try to refute it. I will instead set it aside.

3.4 Morphing

Hare (2013, chs. 9, 11; also 2009) raises a related difficulty. Hare's argument is about *rational preferences of a benevolent person*. I am adapting it to construct an argument about what is *better for a group of individuals on balance*, but there will be places where the difference might matter. He presents two versions

of the argument: the first is aimed at counterpart theorists, and the second aimed at “identity theorists” (essentially, those who are not counterpart theorists). Let’s begin with the counterpart-theoretic formulation.

Let’s consider a possible world much like Happy*. There are infinitely many people arranged in a line from infinite West to infinite East. Indy is one of these people. Everyone to the West of Indy is sad, while Indy and everyone to Indy’s East is happy. This time, instead of being indiscernible duplicates, let’s suppose that these people are different from one another in various respects: they have various hair colors and skin tones and genders and heights and occupations. Call this world w_1 . Call the person immediately to Indy’s West Westley, and the person immediately to his East Esther.

Now consider a world w_2 , which has individuals each of whom is just *slightly different* from the individuals in w_1 . We are considering things as Lewisian counterpart theorists, so we will say that each of the people in w_2 is distinct from each of the people in w_1 ; but each w_2 -person is a *counterpart* of someone in w_1 . Indy has a counterpart in w_2 who is almost just like Indy is in w_1 , except “morphed” slightly so that the counterpart is a little more like Westley than Indy is (as they each are in w_1). In particular, since Indy is happy and Westley is sad, Indy’s counterpart in w_2 is slightly sadder than Indy is in w_1 . Indy’s neighbor Esther also has a counterpart in w_2 , who is almost just like Esther except “morphed” slightly to be a little more like Indy is in w_1 . In particular, Esther’s counterpart is just as happy as Esther is in w_1 . Similarly, Westley has a counterpart in w_2 who is a little more like the next person down the line (and equally sad). And so on for everyone. In short, each person i in w_1 has a counterpart $f(i)$ in w_2 who is nearly a duplicate of i (as they are in w_1) except a little more like the person to i ’s West in w_1 .

In world w_3 we take this transformation a step further. Each person $f(i)$ in w_2 has a counterpart in w_3 who is morphed to be a little *more* like the person to the West of i is in w_1 . And so on through $w_4, w_5, \dots$ — until finally the transformation is complete, culminating in a world w_N containing individuals *exactly* like the ones we originally found in w_1 — but shifted one place to the East. In fact, let us stipulate more than this: w_N is the very same possible world as w_1 (still speaking of “worlds” in Lewisian terms). Call the sequence $w_1, w_2, \dots, w_N$ a *morphing sequence* (figure 3.9).

We have described a counterpart function that associates each person in a world w_k with someone who is almost just like them in w_{k+1} . Furthermore, each person’s counterpart is either just as well off as they are, or else (in one case) slightly worse off. It is natural, then, to say that w_1 is better than w_2 ,

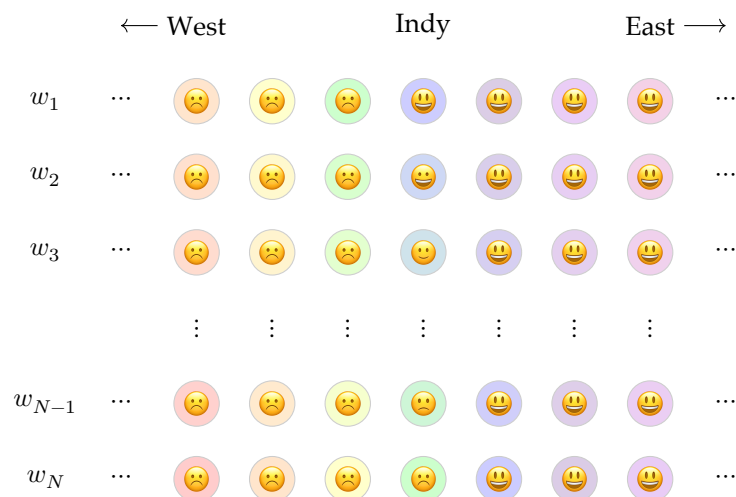


Figure 3.9: A morphing sequence.

which is in turn better than w_3 , and so on up to w_N . But w_N and w_1 are the very same possible world. So it seems that they are equally good:

$$w_1 > w_2 > \dots > w_N = w_1.$$

But such a cycle of betterness cannot be.

This argument shows that we cannot accept a certain principle about a betterness relation between *Lewisian worlds*:

Counterpart Pareto. Let v and w be Lewisian possible worlds. Suppose there is a counterpart function f from the individuals in v to the individuals in w such that for every individual i in v , $f(i)$ is at least as well off in w as i is in v , and for some individual i in v , $f(i)$ is strictly better off in w than i is in v . Then w is better than v .²⁵

Hare's argument demonstrates that Counterpart Pareto leads to contradiction.

One might naturally think that Counterpart Pareto just is the counterpart theorist's best version of the Pareto principle — but that is not so. In fact, the principle I have been calling Pareto is one the counterpart theorist can still accept at face value, without contradiction. As I have said, for the counterpart theorist, "possible worlds" come apart from *possibilities* — ways

²⁵Compare the relation Hare calls *Anonymous Pareto-Dominance* (2013, sec. 9.4).

things could have been. The Pareto principle I have been discussing is not about the counterpart theorist's "worlds", but rather about ways individuals could be. It says that it would be better (with respect to individual well-being) for everyone to be at least as well off as they are, while some people are even better off. The *betterness* involved is not a relation between Lewisian worlds (as in the Counterpart Pareto principle), but rather between ways for individuals to be. The counterpart theorist can make perfect sense of such ways for individuals to be, and of a betterness relation between them, and thus of the Pareto principle. This does not say the same thing as Counterpart Pareto at all. To apply Pareto, the counterpart theorist must consider not just a world, but the way things would be in that world, according to some choice of counterparts.

Consider how this goes in the case of the morphing sequence. Here the relevant counterpart relations have already been fully specified. This determines which possibilities *de re* we must be talking about. According to the counterpart theorist, a person's counterpart represents a way *that very person* could have been — not just a way some similar person could have been. That is the whole point.²⁶ So, here in w_1 we have Westley, Indy, Esther, and the rest. In w_2 we have their morphed counterparts. These counterparts represent the possibility that Indy is a little more like Westley is in w_1 (and a little sadder), that Esther is a little more like Indy is in w_1 , and so on for the rest. To distinguish it from the counterpart theorist's "possible world", let $\hat{w}_1$ be the *possibility* of these people being as they are in w_1 . Let $\hat{w}_2$ be the possibility of those same people being as their counterparts are in w_2 . Similarly, let $\hat{w}_3$ be the possibility of Westley, Indy, Esther, and the rest being as their counterparts' counterparts are in w_3 . And so on.

Each of these possibilities $\hat{w}_k$ is indeed a Pareto improvement over its successor $\hat{w}_{k+1}$. That is, if things were the $\hat{w}_k$ way, then everyone would be at least as well off as they would be if things were the $\hat{w}_{k+1}$ way, and one person

²⁶Recall Kripke's "Humphrey" objection to counterpart theory (1980, 45):

Probably, however, Humphrey could not care less whether someone *else*, no matter how much resembling him, would have been victorious in another possible world.

And recall Lewis's response (1986b, 196):

I think counterpart theorists [agree with Kripke] that there are other worlds ... according to which Humphrey — he himself! (stamp the foot, bang the table) — wins the election. ... Thanks to the victorious counterpart, Humphrey himself has the requisite modal property: we can truly say that *he* might have won.

would be strictly better off in $\hat{w}_k$ than in $\hat{w}_{k+1}$. So the Pareto principle tells us that it would be better for things to be as in $\hat{w}_k$ rather than as in $\hat{w}_{k+1}$. By transitivity, then, we conclude that $\hat{w}_1$ is better than $\hat{w}_N$.

But this is no problem. Even if w_1 and w_N are the very same possible world, $\hat{w}_1$ and $\hat{w}_N$ are distinct possibilities. The two possibilities are qualitatively indistinguishable, but they are different in how individuals fare. Moreover, they differ in that in $\hat{w}_1$ Indy is happy, while in $\hat{w}_N$ Indy is sad. This is precisely the sort of difference that matters morally, because Indy's well-being matters. We have already thrown out the moral telescope (section 3.3), and neither counterpart theory nor morphing gives us a reason to take it back up.²⁷

Now for Hare's second version of the morphing argument (2013, sec. 9.9), adapted to our present context. As I said above, Hare targets this version at non-counterpart-theorists; but really, it applies just as much to the counterpart theorist who has come with us this far. The basic idea is to replace our sequence of possibilities for the *same* people (represented by counterparts) with a sequence that explicitly involves possibilities for *different* people. We will build up our new morphing sequence with two kinds of steps. First, *morphing* steps just as before, in which we morph everyone a little, making one person a little sadder. Second, *swapping* steps, in which we replace everyone with duplicates.

Let's spell this out. This time we will consider two different sequences of

²⁷There is a separate, more minor difficulty with this argument — construed as an argument against Pareto — worth noting. Are $\hat{w}_1, \dots, \hat{w}_N$ all genuinely *possible* ways for Westley, Indy, Esther and the rest to be? This is only true if a quite radical anti-essentialism is true. Even though each change is very slight, cumulatively they amount to a dramatic changes in hair color, skin tone, gender, occupation. Indy could have been a *little* more like Westley than he is. And let us grant that *some* person a little more like Westley than Indy is could also have been even more similar to Westley. And so on for each step. But could *Indy himself* have been all these different ways — even up to being exactly the way Westley actually is, in every qualitative respect? We may not accept this. We may say, rather, that Indy has a non-trivial essence — he could have been happier, but he could not have been a qualitative duplicate of Taylor Swift, or a salamander, or a hard-boiled egg — or, perhaps, just the way Westley actually is. Accepting *haecceitism*, in the modest sense that not all facts supervene on qualitative facts, does not require us to go so far.

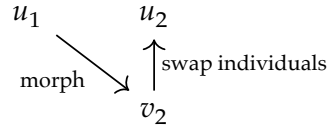
But this is only a minor difficulty for the argument. We might stipulate that the people in w_1 are not radically different from one another; rather, we can suppose that they are all quite similar, similar enough that any one of them represents a qualitative way that Indy might have been. That would limit some of the ambitions that Hare puts the argument toward, but it would not substantially affect the issue presently at hand: whether there is a problem for the Pareto principle.

possibilities (*de re*), $u_1, \dots, u_{N-1}$, and $v_2, \dots, v_N$.

Let u_1 be the same possibility as $\bar{w}_1$ from the first version of the argument. There are infinitely many happy people, including Indy and everyone to Indy's East, and infinitely many sad people to Indy's West.

Let v_2 be the same as the “morphed” possibility $\bar{w}_2$ from the first version of the argument. Everyone is morphed to be a little more similar to the way the person immediately to their West is in u_1 . By Pareto, u_1 is better than v_2 .

Now for the “swapping” step. Let u_2 be a possibility that is qualitatively indiscernible from v_2 , but which involves a completely different set of people. Everyone in u_2 is a perfect duplicate of someone in v_2 , but they are not the same people. Hare judges that v_2 and u_2 are equally good.²⁸ I return to this claim below.

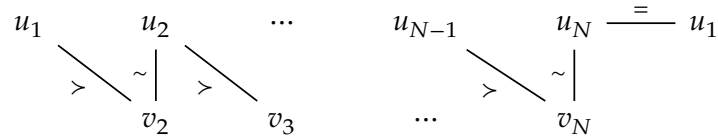


We can continue on in the same way. Let v_3 be what we get by “morphing” all of the people in u_2 a little, just as before. In particular, v_3 has the same population as u_2 , and u_2 Pareto-dominates v_3 . Then we “swap” populations again to produce a qualitatively indiscernible possibility u_3 . And so on: in each step k , the possibility u_{k-1} Pareto-dominates v_k , and v_k is qualitatively indiscernible to u_k , but with a disjoint population.

Finally, when we reach v_N we have a possibility which is qualitatively indistinguishable from our starting place u_1 . So in our last population swap, we can swap in the original population we began with, just as they were to begin with. So the final possibility u_N is not just *qualitatively* indiscernible from u_1 , but in fact it is precisely the same in *every* respect, both qualitative and “haecceitistic”. It is the very same possibility.

Now we are in trouble. We have claimed that u_1 is better than v_2 , that v_2 is equally good as u_2 , that u_2 is better than v_3 , and so on, until finally u_{N-1} is better than v_N , which in turn is equally good as u_N . But this means we again have a cycle of betterness, which cannot be.

²⁸To be more precise, Hare judges that a rational and minimally decent person will be indifferent between them — but I have translated his argument into my own terms, as throughout.



Once again, this argument has an anti-essentialist premise. The possibilities u_2 and v_2 are qualitatively indiscernible, and yet they have completely different populations. Even if we accept haecceitism (as I do), we need not go this far. The haecceitist says that *not all* facts supervene on the qualitative. But that is not to say that *no* facts about individuals supervene on the qualitative. But I will not pursue this. Let us assume that there are enough possibilities to run this morphing argument.

We must again reject the “telescope” step, where we say that qualitatively indiscernible possibilities are equally good. But I admit that this move is a bit weirder in this case. Hare considers (in the slightly different context of a simpler argument)²⁹ someone who prefers one of these duplicate worlds:

She is then in an odd position. I describe to her a maximal state of affairs in complete, exhaustive qualitative detail: down to the position and trajectory taken by every atom throughout world history. I describe to her another state of affairs in complete, exhaustive qualitative detail: down to the inner thoughts of every turtle and newt. I ask her which she prefers, and she replies “I don’t know. You have not told me enough about the states of affairs. You have told me, in glorious detail, all the qualitative facts about them, but I need to know some further, irreducibly haecceitistic facts about them in order to know which I prefer.” (Hare 2013, sec. 9.9)

In some cases there is something very natural to say in response to Hare’s challenge. Sometimes two scenarios are qualitatively just alike, and yet it is easy to find considerations that favor one of them over the other. In one scenario, this child drowns. In another, that same child is rescued. If this takes place in the context of a plenitudinous world full of qualitative

²⁹Specifically, in this passage Hare is considering a case of replacing a *single* individual out for a duplicate. This application seems much more defensible than the one I am considering, where we replace *infinitely many* individuals with duplicates. The arguments in this chapter against the claim that qualitatively isomorphic worlds are always equally good can leave intact the special case where the worlds are not only qualitatively alike, but also differ only for finitely many individuals.

duplicates of both ways for that child to be, then these two scenarios may be alike in every qualitative respect. But that does not make it at all mysterious that there is something that makes one of them better, even though that something is “irreducibly haecceitistic” — that is, it concerns how a particular individual fares. Particular individuals are important! Nor does saying this require any kind of partiality toward this one child over others. It isn’t as if this child’s rescue comes at the expense of some other child, who we are giving lesser consideration. We may say, impartially, that *every* similar rescue of a drowning child, whoever they may be, would make the world better to just the same extent.

But in the case at hand, we cannot say anything like this. The “swapped” possibilities v_2 and u_2 are qualitatively indiscernible. But also, besides that, neither of them is better than the other in any obvious respect. Unlike making an individual better off, merely replacing one individual with some alternative qualitative duplicate does not seem like it is making things better or worse, from an impartial perspective. And so, we might think, the same goes for replacing *infinitely many* someones.

Up to this point, I have steered clear of making comparisons between outcomes with different populations — the subject matter of *population ethics*. Before this morphing argument, whenever I have compared two possibilities I have stipulated that they both involve the very same people. Population ethics is notoriously difficult in many ways that are entirely separate from the notorious difficulties of *infinite* ethics; but to respond to the second morphing argument, we must at least take a few steps in that direction.

(This may require a change in subject: I have been discussing what is better *for certain individuals*. But does it still make any sense to ask whether one possibility is better than another for certain individuals if, in one of them, some of those individuals are never even born? This is a contentious question, sometimes called the question of *existence comparativism*: see Holtug (2001); Arrhenius (2004); and for overview Greaves and Cusbert (2022). If it does not make sense, this provides a straightforward answer to the immediate challenge: if we consider what is best for the population of u_1 , then none of the possibilities that do not include those people are any better or worse *for them*. But we might still ask a different question, central to population axiology, about what is better with respect to overall well-being. This is also a place where Hare’s focus on *preferences*, rather than *betterness*, might be important.)

Let’s start with a simpler puzzle. Forget about morphing for now; we

can do the whole thing in just two “swap” steps. Here once again are the worlds Happy* and Sad*. These have the same population, they are qualitatively indiscernible, and Happy* is a Pareto-improvement over Sad*. Now consider a third world Mezzo, which is qualitatively indiscernible from each of Happy* and Sad*, but which has a completely different population. (We are supposing that this kind of thing is possible.) Suppose we say that merely swapping out some people for the others makes no difference to goodness. Then we must say that Mezzo and Happy* are equally good, and that Mezzo and Sad* are equally good. Yet Happy* is better than Sad* — so this cannot be.

$$\text{Happy}^* > \text{Sad}^* \sim \text{Mezzo} \sim \text{Happy}^*.$$

What should we say about this short sequence?

Intuitively, we have already found that not all countably infinite sets of people are equally morally weighty. Let H be the infinite set of all happy people in Happy*, including Indy. Let H^- be the set of all the people in H except Indy. These two sets H and H^- each have the same infinite cardinality: there is a one-to-one correspondence between them. Nonetheless, there is a sense in which H is “bigger” than H^- — it contains one more individual. And this sense matters morally: it is better for all of the people in H to be happy than it is for just the people in the “smaller” set H^- to be happy.

Now, what about the set M of happy people in Mezzo? There is a one-to-one correspondence between M and H , and there is also a one-to-one correspondence between M and H^- . But we have seen in the case of H and H^- that one-to-one correspondence is not sufficient for having the same moral weight. So there is a question left open about Mezzo: does its set of happy people have the same moral weight as H , or as H^- ? It cannot be both. It need not be either. It might instead be the same weight as H with *two* elements removed, or the same weight as H with infinitely many elements added. Or M might not have the same weight as *any* set with elements in common with H , for all we’ve said.

Indeed, the question of whether Mezzo is as good as Happy* is underspecified. We simply described Mezzo as a world with the same qualitative pattern as Happy* and Sad*, but with a completely disjoint population. But there are many different possibilities that answer this description (supposing there are any). If Mezzo is one of them, then there is another qualitatively identical world Mezzo⁻ which has the very same population as Mezzo, but whose happy population M^- contains one fewer person than M . Mezzo⁻ is

worse than Mezzo. So if Mezzo is as good as Happy*, then Mezzo⁻ is not. But we did not say anything in the way we introduced Mezzo that would distinguish it from Mezzo⁻, or from infinitely many other possibilities. So we cannot say that Mezzo is better than Happy*, or that it is worse, or that it is equally good.

One natural thing to say is that each of these infinitely many Mezzo-like worlds is *incomparable* in value with both Happy* and Sad*. A different thing we might say (which is closely related) is that some of these Mezzo-like worlds are as good as Happy* and others are not, but it is an *indeterminate* or *vague* matter which ones they are. We might worry that such rampant incomparability or indeterminacy leaves us with too few value comparisons. (This worry is related to the “non-identity” problem Parfit 1984, ch. 16.) But the *morphing* problem presently under consideration only arises if we make *too many* value comparisons. It is a quite different problem if we instead make too few. I will take these issues up further in chapter 4.

In any case, we should not affirm that an infinite world is *equally good* as a qualitative duplicate world with a completely different population. Perhaps this sometimes holds, but certainly not always.

Now return to the morphing sequence. Before, we claimed that the world v_2 is just as good as the corresponding “swapped” world u_2 . But in fact, we cannot assert this, for the very same reason that cannot assert that Mezzo is equally good as Happy*. For any swapped world that is equally good as v_2 , there is another swapped world with fewer happy people which is *worse*, and thus not as good as v_2 . The way we introduced the swapped world did not specify enough to distinguish between pairs of worlds like these. So we cannot conclude that v_2 and u_2 are equally good.

There may be (for all I’ve said) *some* possibility answering u_2 ’s description that is equally good as v_2 . But no reason has been given to think that there is any sequence of possibilities that will satisfy *every* claim of equal goodness in the sequence. And we have a very good reason to suppose that there is no such sequence — simply because this would imply a cycle of betterness.

3.5 By the numbers

In section 3.2, I proposed that, when we evaluate whether one infinite world is better than another, we should set utility representations and cardinal arithmetic aside, and instead focus on our judgments about the relevant trade-offs.

But once we have made these judgments, there is nothing stopping us from returning to utility representations. Indeed, numerical representations — or anyway, something a lot *like* numerical representations — can be very useful. Instead of describing worlds in a lot of detail, we can represent them with abstract quantities, and we can reason about the value of worlds by doing arithmetic with these quantities. This will be especially useful when we wish to weigh up the value of different worlds against one another as we develop theories of risk. The catch is that that these quantities cannot always be *real numbers* — recall Basu and Mitra’s theorem (section 3.2).

Still, sometimes real numbers are enough. Let’s start small. As in section 3.2, let’s consider just *happy-sad* worlds, which can be represented by $\{1, 0\}$ -social outcomes. As a first step, we can simplify things even further: consider two-valued social outcomes in which only finitely many people are happy. Call these *finite outcomes*. Impartial Pareto orders are especially straightforward for outcomes like this — in fact, for finite outcomes, the counting order is the *unique* order that satisfies Pareto and Impartiality. Furthermore, this order can be straightforwardly represented by a finite quantity: the number of happy people. For any finite outcomes x and y ,

$$x \succeq^{\#} y \quad \text{iff} \quad \#H(x) \geq \#H(y)$$

where $H(x)$ is the set of happy people in x .³⁰

Finite outcomes can be quantitatively represented by their number of happy people — which is a *finite* quantity. But it is also possible to quantitatively represent arbitrary outcomes — using *infinite* quantities.

Some infinite number systems clearly will not do. Infinite *cardinal* numbers, of the sort discovered by Cantor, are no good. Two sets have the same cardinal number iff their elements can be put in one-to-one correspondence. In the case of a population of countably many individuals, *any* two outcomes

³⁰This can be generalized. For any $\{1, 0\}$ -outcome z , call an outcome x *finite relative to z* iff all but finitely many people are equally well off in x and z . If x is finite relative to z , the *net number of happy people* is a well-defined finite integer-valued function:

$$N_z(x) = \#(H(x) \setminus H(z)) - \#(H(z) \setminus H(x)).$$

Furthermore, for any impartial Pareto order, and any outcomes x, y which are finite relative to z , we have

$$x \succeq^{\#} y \quad \text{iff} \quad N_z(x) \geq N_z(y).$$

Furthermore, for any two choices of baseline z and z' which are relatively finite, the functions N_z and $N_{z'}$ only differ by adding a constant.

with infinitely many happy individuals have exactly the same cardinal number of happy individuals — there is just one countable infinite cardinal (standardly notated $\aleph_0$). Some have been tempted by this observation to Everything Bagel-ism — saying that these different outcomes are equally good — but we know better. Whatever abstract objects we use to represent the value of infinite outcomes, they had better not be Cantor’s cardinal numbers.

Fortunately, the mathematical universe abounds with infinite structures. One important kind of example that is often discussed in this context is known as the *hyperreal numbers* (Hewitt 1948; Robinson 1966; for overview, see Goldblatt 1998). These extend the standard real numbers with infinite numbers (as well as *infinitesimal* numbers, to which we will return in chapter 5). Furthermore, hyperreals have a key feature that distinguishes them from cardinal numbers, and which is crucial for our purposes: if ω is an infinite hyperreal, then $\omega + 1$ is a strictly *greater* hyperreal.

$$\omega + 1 > \omega.$$

Against Pascal and Zeno, we can say that unity joined to infinity *does* increase it. If we can assign a social outcome an infinite hyperreal number ω that (in some sense) measures its set of happy individuals, then making one *more* individual happy is an improvement — and this improvement can indeed be captured by a hyperreal representation.

We will have some use for hyperreals in what follows, but they also have some disadvantages. (I will take this up in chapter 4.) For our purposes, it is a different kind of structure — closely related to the hyperreals, but distinct from them — that will be most immediately useful.

Before we discuss this, let’s start with a conceptual point about numbers. Where do Cantorian cardinalities, like 6 or 15 or $\aleph_0$, come from? Frege ([1884] 1953) suggested that we can define such numbers by *abstraction*.³¹ He began by saying what it is for two sets to have the *same* cardinal number:

$\#A = \#B$ iff there is a one-to-one correspondence between A and B .

Furthermore (in the right context) this abstractive definition can be transformed into an explicit construction of cardinal numbers, using the method

³¹Frege ([1884] 1953, sec. 62):

If we are to use the symbol a to signify an object, we must have a criterion for deciding in all cases whether b is the same as a , even if it is not always in our power to apply this criterion.

of *equivalence classes* that Frege proposed. The idea is that we can identify the cardinal number $\#A$ with the equivalence class of all sets of individuals B that stand in one-to-one correspondence with A .³²

As we said, Cantorian cardinals are not the right mathematical objects for our purposes. But we can do something very similar for an alternative, non-Cantorian way of extending a notion of cardinality from finite sets to infinite sets. I will call this extended notion a set's **quasi-cardinality**, written $\#^*A$. Here's how it goes.³³

Say that a property holds *for all sufficiently large finite sets* iff there is *some* finite set of individuals J such that *every* finite set of individuals $K \supseteq J$ has the property. Then we can define quasi-cardinality by abstraction — along very similar lines to Frege's definition of Cantorian cardinality by abstraction. The idea is that to have the *same* quasi-cardinality is to share the same *finite* number of elements with any *sufficiently large* finite set.

$$\begin{aligned} \#^*A = \#^*B \quad \text{iff} \quad \#(A \cap K) = \#(B \cap K) \\ \text{for all sufficiently large finite sets } K. \end{aligned}$$

Again, we can transform this into an explicit definition using equivalence classes, if we like. (That is, explicitly, $\#^*A$ is the equivalence class of *all* sets of individuals B such that $\#(A \cap K) = \#(B \cap K)$ for all sufficiently large finite sets K).

Notice that if A and B are finite sets, this definition agrees with the ordinary notion of finite cardinality ($\#^*A = \#^*B$ iff $\#A = \#B$). But for infinite sets,

³²This explicit definition requires that we restrict our ambitions to defining cardinalities for subsets of a fixed set J , rather than for *arbitrary sets*. More generally, *restricted* abstractive definitions — those where the operator being defined, like $\#$, has some fixed set as its domain — can always be converted into constructive definitions in this way. (See also proposition B.20 in section B.2.) We cannot always do this with *unrestricted* abstractive definitions — which was Frege's original ambition. In fact, famously, some unrestricted abstractive definitions are inconsistent, like Frege's Basic Law V.

The set of F 's = the set of G 's iff every F is a G and every G is an F .

There is an interesting technical and philosophical program — *neo-Fregeanism* — exploring how far unrestricted abstractive definitions can be pushed, and what role they can play in the foundations of mathematics (see, for instance, Wright 1983; Heck 2011). But this is not important for our purposes.

³³For further details see section B.5; section B.6; section B.7. The approach I take is closely related to the *numerosity* theory of Benci and Di Nasso (2003; Benci, Di Nasso, and Forti 2006); for overview see Mancosu (2009). I also draw heavily on ideas from Pivato (2014). We will apply related technical ideas to probability in chapter 5.

quasi-cardinality comes apart from Cantorian cardinality. Say A is an infinite set, and B is A plus one more element b (where $b \notin A$). Then A and B are in one-to-one correspondence, but have different quasi-cardinalities.³⁴

Quasi-cardinalities can also be *ordered*.

$$\#^*A \geq \#^*B \quad \text{iff} \quad \#(A \cap K) \geq \#(B \cap K) \\ \text{for all sufficiently large finite sets } K.$$

Unlike with finite cardinalities, though, this is only a *partial* order, not a total order. For example, the set of all people assigned even numbers and the set of all people assigned odd numbers have different quasi-cardinalities, and neither is greater than the other. (There are arbitrarily large finite sets containing more odds than evens, and there are arbitrarily large finite sets containing more evens than odds.) In general, for any sets A and B such that the set-differences $A \setminus B$ and $B \setminus A$ are both infinite, their quasi-cardinalities $\#^*A$ and $\#^*B$ are incomparable.

Any social outcome x 's set of happy people has a *quasi-cardinality*, $\#^*H(x)$. Moreover, just as the ordinary finite number of happy people $\#H(x)$ represents the counting order for *finite* simple worlds, the *quasi-cardinality* of happy people $\#^*H(x)$ represents the counting order for *arbitrary* simple worlds.

$$x \succ^{\#} y \quad \text{iff} \quad \#^*H(x) \geq \#^*H(y).$$

Quasi-cardinality provides us with a “non-standard utility function” that represents an ordering of simple worlds that satisfies Pareto and Impartiality — something that we know cannot be done using standard reals, extended reals, or Cantorian cardinals.

In a way, this representation by abstraction is merely a redescription of the counting order.³⁵ But it becomes more powerful when we observe that quasi-cardinalities have a great deal of structure: they are *quantities*, much like integers or real numbers (see also section B.1). First, we have already observed that they are (partially) ordered. They also have a zero quantity (the quasi-cardinality of the empty set, $\#^*\emptyset$). We can naturally define addition, subtraction, and other operations as well.³⁶ Indeed, we can embed quasi-cardinalities in more general structures I will call the *quasi-rational numbers*

³⁴For any finite set J , however large, we can let $K = J \cup \{b\}$ and get $\#(A \cap K) < \#(B \cap K)$.

³⁵Each quasi-cardinal number (up to $\#^*\mathcal{I}$, the total quasi-number of individuals) corresponds directly to an equivalence class of social outcomes under the counting-order equivalence relation $\sim^{\#}$.

³⁶First, we can generalize our abstractive definition of quasi-cardinality by defining a

(notated $\mathbb{Q}^*$) or *quasi-real numbers* ($\mathbb{R}^*$), which have most of the mathematical structure that we are used to from their standard counterparts. I will not go into the details here, but they can be found in section B.5, and I will discuss this further in chapter 5.

As I remarked, the counting order is the *weakest* impartial Paretian order of $\{1, 0\}$ -social outcomes. Accordingly, the quantitative structure of quasi-cardinalities that we have invoked to represent this order is relatively weak, in that it includes a lot of incomparability. But we can also use very similar methods to define stronger orders than the counting order, and construct quantitative representations of those orders. The trick is to use alternative notions of what counts as “sufficiently large” finite sets.

For example, consider again the case from section 3.2 of Twos (each individual at an even numbered room is happy) and Sevens (each individual at a multiple of seven is happy). The counting order says that Twos and Sevens are incomparable: infinitely many multiples of seven are not multiples of two, and infinitely many multiples of two are not multiples of seven. But some people think that Twos is better than Sevens, on account of its higher “happiness density”. (I argued against this idea in section 3.2, but let’s take it up again just for the sake of illustration.) We can define a *strengthening* of the counting order that delivers this verdict. Let us modify our earlier definition, and instead say that a property holds “for all sufficiently large finite sets” iff there is some finite set I , such that all *intervals* $J \supseteq I$ have the property. For a large interval J of width n , the number of happy people in Twos is approximately $n/2$, and the number of happy people in Sevens is approximately $n/7$. Since the first is always larger, this *interval counting order* ranks Twos over Sevens. We can also similarly generalize our construction of quasi-real numbers using this alternative notion of “sufficiently large”, and once again we get a well-behaved quantity with familiar algebraic structure.

notion of a *quasi-limit*. Let f and g be any functions that take each finite set of individuals to a number. Then let

$$\lim^* f = \lim^* g \quad \text{iff} \quad f(K) = g(K) \\ \text{for all sufficiently large finite sets } K.$$

The *quasi-natural numbers* are precisely the quasi-limits of such natural-valued functions. Then we can straightforwardly define arithmetic operations, using limits. For example, we can define the sum of two quasi-natural numbers as the quasi-limit of the sum of their representing functions:

$$\lim^* f + \lim^* g \quad := \quad \lim^* (f + g).$$

See section B.5 for further details.

I will use the same label “quasi-real” for any structure that is generated in this way, from some appropriate notion of “sufficiently large finite sets”.³⁷ There are many other notions of “large enough finite sets” that can be useful for defining orders, and we will encounter further examples later on (see section 5.3).

We will also want to extend this project in a different direction, looking beyond simple worlds to cases where more is going on than just toggling “happy” and “sad”. As before, let $\mathcal{I}$ be a set of individuals, and let $\mathcal{X}$ be some set of *individual well-being levels*. Before we took this to just be $\{1, 0\}$, but now, more generally, we can let $\mathcal{X}$ be any partially ordered set. As before, we suppose that we can represent certain worlds with *social outcomes*, which are functions from $\mathcal{I}$ to $\mathcal{X}$. Here is the key assumption about this representation:

If social outcome x represents world v , and social outcome y represents world w , then for every individual $i \in \mathcal{I}$, $x(i) \geq y(i)$ iff i is (at least) as well off in v as they are in w .

Let an $(\mathcal{I}, \mathcal{X})$ -world be a world that is represented by some $\mathcal{X}$ -social outcome: that is, a world in which every individual $i \in \mathcal{I}$ has some well-being level in $\mathcal{X}$. As before, Pareto Equivalence implies that we can describe any order of $(\mathcal{I}, \mathcal{X})$ -worlds in terms of a corresponding order of $\mathcal{X}$ -social outcomes for $\mathcal{I}$.

If x is a social outcome representing world v , and y is a social outcome representing world w , then $x \succeq y$ iff $v \succeq w$.

That much is very general. As an important proof of concept, let’s now consider a way of making it very concrete, by describing an infinite generalization of *utilitarianism*. To be clear, right now I am not offering any argument for this as the *correct* way to order worlds. (Further details can be found in appendix B.) For this purpose, let’s assume that individual well-being levels can be represented by *real numbers*: we call these *individual utilities*. We’ll suppose that there is an *individual utility* function $u : \mathcal{X} \rightarrow \mathbb{R}$ such that $u(x) \geq u(y)$ iff $x \geq y$, for every $x, y \in \mathcal{X}$.

Utilitarians say that what is better is what has the greater sum of individual utilities. If we know what individual utilities are, then we know how to

³⁷In the same way, “hyperreal” is standardly used as a label for any one of infinitely many different structures that all arise as *ultrapowers* of the real numbers. Quasi-real numbers are more general *filter powers* of the real numbers: again, see section B.5 for the full lowdown. An appropriate notion of “sufficiently large” is represented by a *filter*. In fact, every hyperreal structure (in the ultrapower sense) is also a quasi-real structure, though not vice versa: hyperreals arise when the *filter* in question is in fact an *ultrafilter*.

make sense of this in finite cases. Let $K = \{i_1, i_2, \dots, i_n\}$ be any finite set of individuals. For a social outcome x , we can define the *social utility for K in x* as a sum of individual utilities:

$$U_K(x) = \sum_{i \in K} u(x(i)) = u(x(i_1)) + u(x(i_2)) + \dots + u(x(i_n)).$$

Now we want to extend the additive definition of social utility so that we can make sense of it for *infinite* sets. As we have already seen, the ordinary kind of infinite sum that one learns about in calculus classes won't work here. That only gives us an extended real number, which doesn't distinguish between Happy and Sad:

$$0 + 1 + 0 + 1 + 0 + \dots \stackrel{?}{=} \infty \stackrel{?}{=} 1 + 1 + 0 + 1 + 0 + \dots.$$

But we can define a different kind of infinite sum, which generalizes finite sums in a different way — just as we defined a different kind of “cardinality” that generalizes finite cardinality in a different way from Cantor's. We can do this in an exactly parallel way. We define by abstraction, for any functions $f, g : \mathcal{I} \rightarrow \mathbb{R}$,

$$\sum^* f = \sum^* g \quad \text{iff} \quad \sum_{i \in K} f(i) = \sum_{i \in K} g(i) \\ \text{for all sufficiently large finite sets } K.$$

Then $\sum^* f$ is another quasi-real number, which we will call the **quasi-finite sum** of f .³⁸ (We can also write quasi-finite sums using the alternative notation $\sum_i^* f(i)$, making the function argument explicit.)

In particular, we can define a social utility function for any infinite social outcome $x : \mathcal{I} \rightarrow \mathcal{X}$ as a quasi-sum:³⁹

$$U^*(x) = \sum_i^* u(x(i)).$$

We can define a corresponding order of social outcomes:

$$x \succeq y \quad \text{iff} \quad U^*(x) \geq U^*(y).$$

³⁸We can equivalently write this definition in terms of a quasi-limit:

$$\sum^* f := \lim^*_K \sum_{i \in K} f(i).$$

³⁹Equivalently: $U^*(x) = \lim^*_K U_K(x)$.

This defines an infinite generalization of utilitarianism — again, in a way, it is the *minimal* extension of utilitarianism to infinite cases.

Note in particular that this utilitarian order satisfies Pareto and Impartiality (where the latter is generalized in the obvious way from the happy-sad case to welfare levels in $\mathcal{X}$).

It follows that this order does *not* satisfy the principle of Strong Anonymity: in general, social outcomes that are related by a permutation can have different quasi-sums. For example, the outcome representing Happy has a strictly greater quasi-finite sum of utilities than the outcome representing Sad — as we should hope.

This quasi-utilitarian order is a variant of the *overtaking order* that is familiar from the literature. (Both say that an outcome x is as good as an outcome y iff the x 's total utility is “eventually” at least that of y .)⁴⁰ The main advantage that the quasi-finite sum formulation gives us is a direct *quantitative representation* for this kind of order. This will be especially useful when we return to the question of weighing risks. In particular, quasi-reals have the structure of a *vector space*: they can be added together and rescaled by real-valued probabilities. So they can play the role of vector utilities as we discussed in section 2.3. In particular, (as I alluded then) the quasi-reals are an *infinite-dimensional* vector space. I will return to the interplay of infinite value and risk in chapter 5.

Again, I am not saying that this quasi-utilitarian order is really the *correct* way to rank infinite worlds as better or worse. The basic point that the quasi-utilitarian order illustrates is that, even beyond the case of simple worlds, it is perfectly consistent to maintain Pareto and Impartiality. It also has many other attractive properties that we will consider in subsequent chapters. And even though we cannot represent such value orders using

⁴⁰The original overtaking criterion (Atsumi 1965; von Weizsäcker 1965; Koopmans 1967) was formulated in a way that depends on a privileged *order* of individuals:

$$x \succeq y \quad \text{iff} \quad \sum_{i=1}^n u(x(i)) \geq \sum_{i=1}^n u(y(i)) \text{ for all sufficiently large } n.$$

The ordering based on quasi-finite sums is in the same spirit, but need not privilege an ordering of individuals (as long as the underlying filter is suitably symmetric: see section B.8). Indeed, for the weakest filter — the one that corresponds to the counting order in the two-level case — the quasi-finite sum is equivalent to saying that the overtaking criterion holds with respect to *any* ordering of individuals. Compare also Asheim and Banerjee (2010); Asheim, d’Aspremont, and Banerjee (2010). See Pivato and Fleurbaey (2024) for discussion. See also our discussion of permutation invariance in section 4.2

standard real numbers, extended reals, or Cantorian cardinals, we do have quantitative representations using non-standard infinite “quasi-numbers”.

3.6 Conclusion

The Everything Bagel threatens the idea that finite changes can make a difference to the impartial value of individual well-being in an infinite world. We have found that this threat is resistible. To resist it, we should give up on using simple cardinal arithmetic to measure value. We should abandon “telescopic” thinking about value as inherent in a possible world’s overall qualitative pattern, and instead attend to the specific aspects of value that trade off against one another — such as individual well-being. When we have given this up, we can also resist Hare’s “morphing” argument. Moreover, none of this requires us to abandon quantitative mathematical representations of value. We just have to choose the right mathematical structures for the job, rather than letting our most familiar tools sway our moral judgments.

Even in an infinite world, one life can make a moral difference. Or, at least, the arguments we considered in this chapter should not persuade us otherwise. But we are far from done: many other puzzles and difficulties remain.

Chapter 4

Indeterminate Value

Surely, this commandment that I am commanding you today is not too hard for you, nor is it too far away. It is not in heaven, that you should say, 'Who will go up to heaven for us and get it for us so that we may hear it and observe it?' Neither is it beyond the sea, that you should say, 'Who will cross to the other side of the sea for us and get it for us so that we may hear it and observe it?'

Deuteronomy 30.12–13

4.1 Incomparability and small improvements

Althea has a sum of money that she is going to donate to a worthy charity. But she can't make up her mind. She could donate it to *Program A*, which has been shown to be highly effective at promoting education for girls in low-income countries. Or she could donate it to *Program B*, which has a strong track record at mitigating the horrible conditions for chickens on factory farms — for instance, by getting large restaurant chains to commit to going cage-free.

Althea can't make up her mind. The trouble isn't that Althea doesn't know how effective her donation would be in either case. Rather, it's that the donation to either organization would promote different values that she

cherishes. A donation for education would do more to promote autonomy and gender equality. A donation for animal welfare would do more to alleviate intense pain and discomfort. And Althea really has no idea how to trade off these different values against one another.

Cases like these¹ have led many to conclude that some options are simply *incomparable* with one another: neither is better than the other, but they are also not equally good.²

Neither program seems better than the other to Althea. But also, they don't seem equally good. Suppose Althea were to find that donating to Program A would also make her next-door neighbor slightly happier. (They'd be doing just fine either way, though.) This doesn't seem like the sort of thing that breaks the deadlock, to make one option clearly come out ahead.

This is one of the most common ways of arguing for incomparability with respect to what is better overall. Value is *multi-dimensional*. Even in cases where we can find a clear ranking of options with respect to each single dimension, imposing a total order of options from best to worst would require making trade-offs between very different dimensions, and in many cases we are at a loss about how to do that. These don't look like cases of ties, since making one option slightly better or worse doesn't tip the scales one way or the other. This is the *argument from small improvements*.

This is not just a feature of *value*, of course. Many different kinds of comparison are similarly multi-dimensional, in a way that may similarly leave us at a loss. Saul eats the same green salad for lunch every day, and has no desire for anything else; Sally has salad less frequently, but she tries a new salad whenever she has the opportunity — which of them likes salad more? Notre-Dame's flying buttresses are magnificent and inspiring; Sagrada Familia's organic columns are audacious and exuberant — which is more beautiful?

There are also powerful arguments for incomparability that are more specific

¹My case is modeled on one from Schoenfield (2014, 269).

²See Raz (1986); Chang (2002); for further references see Dorr, Nebel, and Zuehl (2023, footnote 11).

Chang (2002) prefers not to use the term "incomparability" for all such cases, since there is another sense in which they can be evaluatively compared. But I will simply use the term stipulatively to mean, specifically, that there is no comparison between the options with respect to the specific relation *as good*, leaving open the question of whether there are other value relations that hold between them.

to the project presently at hand: the impartial value of well-being in large or infinite worlds. Here is the key claim at issue.

(World) Comparability. Let $\mathcal{I}$ be a set of individuals. For any worlds v and w which are each good for the individuals in $\mathcal{I}$ to some degree, either v is at least as good as w for the individuals in $\mathcal{I}$, or w is at least as good as v for the individuals in $\mathcal{I}$ (or possibly both, in which case v and w are equally good for the individuals in $\mathcal{I}$).

The qualification, “which are each good for the individuals in $\mathcal{I}$ to some degree”, leaves open the possibility that a world w may not be suitable for social welfare comparisons at all — perhaps because some or all of the individuals in $\mathcal{I}$ don’t exist in w at all, or because they are not “welfare subjects” in w , the kind of entity that can be well off or badly off.³

As in sections 3.2 and 3.5, it can be useful to go up a level of abstraction. Let $\mathcal{I}$ be some infinite set of individuals, and let $\mathcal{X}$ be some ordered set of “well-being levels” — in the simplest case, these might just be “happy” or “sad”. An $(\mathcal{I}, \mathcal{X})$ -**world** w is one in which every individual $i \in \mathcal{I}$ has some well-being level $x(i) \in \mathcal{X}$. (I will assume that this suffices for w to be good for the individuals in $\mathcal{I}$ to some degree.) In that case, w can be represented by an $(\mathcal{I}, \mathcal{X})$ -**social outcome**, which is just a function from $\mathcal{I}$ to $\mathcal{X}$. We assume that $x(i) \geq y(j)$ whenever i is as well off in a world represented by x as j is in a world represented by y .⁴ As before, assuming Pareto Equivalence, we can describe orders of worlds by way of orders of social outcomes. We can stipulate that a social outcome x is **as good (for $\mathcal{I}$) as** a social outcome y (abbreviated $x \succeq y$) iff x and y represent worlds v and w , respectively, such that v is as good as w for the individuals in $\mathcal{I}$. Assuming Pareto Equivalence, this defines an order of social outcomes. Then we can restate Comparability in terms of social outcomes.

(Outcome) Comparability. For any $(\mathcal{I}, \mathcal{X})$ -social outcomes x and y , either x is as good for $\mathcal{I}$ as y or y is as good for $\mathcal{I}$ as x (or possibly both).

³That is, I am remaining neutral on the question of *existence comparativism*, briefly raised in section 3.4; see citations therein. Don’t take “degrees” too literally. Rather than saying “ w is good for the individuals in $\mathcal{I}$ to some degree”, we could say “ w is as good as w for the individuals in $\mathcal{I}$.” I am drawing on Dorr, Nebel, and Zuehl (2023) for this way of restricting Comparability; see further discussion in section 4.4 below.

⁴As I discussed in section 3.3, we may wish to be less committal than this about the extent to which *interpersonal* well-being comparisons are meaningful — in which case we might restrict the claim in the text to the case $i = j$. This would make a difference to the dialectic about symmetry principles in section 4.2. But I won’t keep track of this nuance in this chapter.

In general, I will omit the $\mathcal{I}$ and $\mathcal{X}$ qualifications. Given the standard logical relationships between *as good*, *better*, and *equally good*, this is an equivalent reformulation:

Trichotomy. For any social outcomes x and y , either x is better than y , y is better than x , or x and y are equally good.

In the next two sections, I'll discuss two powerful arguments against Comparability, specifically with respect to the impartial value of infinite worlds. After that (sections 4.4 and 4.5), I will consider arguments for Comparability. Finally (sections 4.6 to 4.8) I will propose a synthesis, appealing to *indeterminate* value.

4.2 Invariance

The first argument against Comparability is based on work by Askill (2018; see also Goodman 2025). First, recall the key principle from chapter 3: one life can make a difference.

Pareto. Let x and y be social outcomes for a set of individuals $\mathcal{I}$.

If each individual in $\mathcal{I}$ is as well off in x as in y , then x is as good as y for the individuals in $\mathcal{I}$, on balance.

If, furthermore, some individual in $\mathcal{I}$ is strictly better off in x than in y , then x is better than y for the individuals in $\mathcal{I}$, on balance.

Even in an infinite world, finite differences in well-being can make the world impartially better overall. In section 3.3 I discussed the inconsistency between Pareto and the *Strong Anonymity* principle. Let's restate that. Suppose π is a permutation of individuals. Then for each social outcome x , there is a *permuted* outcome πx such that every individual $i \in \mathcal{I}$ is equally well off in x as $\pi(i)$ is in $\pi(x)$.

$$(\pi x)(i) := x(\pi^{-1}(i)).$$

Strong Anonymity. For any social outcome x and any permutation of individuals π , $x \sim \pi x$.

I have argued against this principle: merely permuting which individuals are happy and which are sad is the sort of thing that can make a difference. Take an outcome x representing a *simple world* (as in section 3.2) where infinitely many people are happy and infinitely many people are sad. Then any outcome y representing another simple world that is a finite improvement

over x — making finitely many sad people happy — is related by a permutation to the original unimproved outcome x . So Strong Anonymity would effectively commit us to Everything Bagel nihilism: making a merely finite difference does not make individuals any better off. Moreover, I argued that *qualitatively indiscernible* possible worlds need not be equally good for their individuals (supposing there are any distinct qualitatively indiscernible possible worlds at all).

But Strong Anonymity should be distinguished from a different, weaker claim.

Permutation Invariance. For any social outcomes x and y and any permutation of individuals π , if $x \succeq y$, then $\pi x \succeq \pi y$.

Strong Anonymity implies Permutation Invariance.⁵ But the converse does not hold. For example, the *counting order* (section 3.2) satisfies Permutation Invariance, but not Strong Anonymity. This also shows us that, even though Pareto is inconsistent with Strong Anonymity, it is consistent with Permutation Invariance.

Permutation Invariance and Strong Anonymity both amount to claims about how much of the world's structure is relevant to value. Imagine we are about to approach the Chief Axiology Officer to inquire whether a certain world v is as good as another world w . The CAO is very good at her job, but also extremely busy, so we shouldn't impose on her patience by telling her *everything* about the two worlds — we need to prepare an executive summary. How much does she need to know? *Strong Anonymity* says that the executive summary only needs to include each world's *well-being profile*: how many people there are at each level of well-being. For big complicated worlds, this might still be a lot of information, but it also leaves out a lot. It doesn't tell us how people are arranged in space, for example, or what their names or genders or phone numbers are. The stuff we get to leave out is stuff we regard as *morally irrelevant*.

We have found that Strong Anonymity leaves out too much. The simple world Happy is better than the simple world Sad, but they have the very same well-being profile (section 3.3). (Each of them has a countable infinity of happy people, and a countable infinity of sad people.) If we just presented the CAO with this profile (twice over), she would be unable to tell us which is better. The question of whether someone is better off in one world than they are in another is precisely the sort of thing that is morally *relevant*. So

⁵Suppose $x \succeq y$. Then, by Strong Anonymity, $\pi x \sim x \succeq y \sim \pi y$.

this is the sort of thing that the executive summary should include: for instance, the CAO needs to be able to tell from the summary whether one of the worlds Pareto-dominates the other.

How much more do we need? We shouldn't have to go all the way to including arbitrary details like phone numbers. *Permutation Invariance* says that the executive summary can still be very austere. Say we prepare for the CAO a big table that lists each individual by name, together with their well-being in v and their well-being in w .

Individual	Well-being in v	Well-being in w
Indy	😬	😞
Westley	😬	😞
Esther	😬	😬
⋮	⋮	⋮

Then we accidentally delete all of the names. Disaster? Not if *Permutation Invariance* is true. We could arbitrarily reassign the rows of the table to *any* individuals, and it would make no difference to the relative ranking of v and w . Here's another way of putting this. *Permutation Invariance* is equivalent to the principle that whether v is as good as w is determined by the two worlds' *comparative well-being profile*. This says for any *pair* of well-being levels (x, y) how many people are at x in v and at y in w .

Like *Strong Anonymity*, *Permutation Invariance* has a very austere picture of what counts as morally relevant — it adds back in *just enough* information so that we can recover comparisons like Pareto dominance. This is attractive. It says that the overall value of individual well-being doesn't depend on arbitrary details like phone numbers, or on invidious distinctions like genders. Instead of arbitrariness, we have sweeping simplicity. Aiming for structural simplicity is a good guiding principle in all kinds of theorizing. Of course, *Permutation Invariance* might be *too* simple. Other things might matter that this summary leaves out. For example, in section 3.2 we discussed the idea that the spatial distribution of happiness might matter morally, after all. If so, we would have to put these things back into the executive summary. But it is at least a good working heuristic to see how far we can get with the simplest picture.⁶

⁶Pareto Equivalence is already imposing a quite strong assumption about what kind of

As I said, there is no direct conflict between Permutation Invariance and Pareto — this is evidenced by the counting order. But it is crucial that the counting order is not a *total* order of simple worlds. For example, according to the counting order, the world *Evens* where all and only the even-numbered people are happy is not better, worse, or equally good as the world *Odd*s where all and only the odd-numbered people are happy. More generally, let

feature of a world can make a difference to social welfare. As we discussed in the case of Twos and Sevens (section 3.2), if the very same people could be arbitrarily rearranged in space, then spatial arrangement cannot make a difference to what is best for those people. In general, Pareto Equivalence says that the only features of worlds that can make a difference to social welfare are those that supervene on individuals' well-being. But this could still include a lot more than just the information in a comparative well-being profile, if individuals have robust enough essences. (See also footnote 11 in section 3.2.)

Askell (2018) argues (in effect) for Permutation Invariance on the basis of two other claims in a more metaphysical key. (I will state them in a slightly modified form.) The first is that *as good* is a *necessary qualitative relation*. Let us take as given the notion of one pair of worlds $\langle v, w \rangle$ being a *qualitative duplicate* of another pair $\langle v', w' \rangle$: the idea

Qualitativeness. If a pair of worlds $\langle v', w' \rangle$ is a qualitative duplicate of a pair of worlds $\langle v, w \rangle$, then $v \succeq w$ iff $v' \succeq w'$.

The second is that for any worlds v and w , any permutation of individuals can be realized by a pair of worlds that is a qualitative duplicate of $\langle v, w \rangle$.

Permutation Principle. For any world pair $\langle v, w \rangle$ and any permutation of individuals π , there exists a world pair $\langle v', w' \rangle$ which is a qualitative duplicate of $\langle v, w \rangle$, such that v and v' are related by π , and w and w' are related by π .

I won't pursue this line of defense. In brief, the Permutation Principle is a very strong anti-essentialist, haecceitist claim: not only could individuals have been different while things are qualitatively just the same, but in fact they could have played *any* qualitative role that *any* individual in their world could play.

In reply, Askell (2018, 102) writes, "My response to this objection is that the world pair ... need not be a metaphysically possible world pair. We can appeal to a broader form of logical possibility in which it is possible for agents to have completely different qualitative properties while remaining the same agent." But right now the question before us is about what would really be better for certain individuals ("objectively", or *ex post*); this really is a question about *metaphysically possible* worlds. I take it that if it is impossible for individuals to be a certain way, then the question whether it would be *better* for them to be that way can only have an uninteresting answer. (It might be trivially *yes*, rather than *no* — but it would equally trivially be *worse* for them to be that way, and also $0 = 1$ would be true, etc.) In later chapters we will turn to questions about the *ex ante* value of prospects, and at that point we will sometimes need to consider "metaphysically impossible worlds" — for example, in cases of uncertain identity (see chapter 7, in particular).

The argument Askell gives for considering metaphysically impossible worlds is that the essential properties that make certain situations impossible (such as origins) seem *morally irrelevant* (p. 103). But it seems better to bring these considerations in more directly: rather than going via impossible *qualitative duplicate* world-pairs, we can consider world-pairs that are *not* duplicates, but which differ only in qualitative ways that seem morally irrelevant.

x and y be any social outcomes such that there are infinitely many people who are happy in x but not in y , and infinitely many other people who are happy in y but not in x . Then x and y are incomparable under the counting order. What's more, such incomparabilities are forced on us by Permutation Invariance.⁷

Theorem 4.1.

- (a) Pareto, Permutation Invariance, and Comparability are jointly inconsistent.
- (b) The counting order is the strongest order of $\{1, 0\}$ -social outcomes that satisfies Pareto and Permutation Invariance.
- (c) The counting order is the unique order of $\{1, 0\}$ -social outcomes that satisfies Pareto, Impartiality, and Permutation Invariance.

Here is the basic idea of the proof. (See figure 4.1.) Consider the two worlds Evens and Odds again. (I'll use the same names for the social outcomes as I did for the worlds they represent.) We will show first that Evens cannot be as good as Odds. Let X be the set of individuals at odd numbers $(1, 3, 5, \dots)$. We can split up the set of even numbered individuals into two infinite sets: let Y_1 contain the individuals at multiples of four $(4, 8, 12, \dots)$, and let Y_2 contain all the even numbers that are not divisible by four $(2, 6, 10, \dots)$. So just the X -people are happy in Odds, while just the people in Y_1 or Y_2 are happy in Evens.

Next, there is a permutation π that swaps each odd number with a multiple of four, and which takes each even number that is not a multiple of four to itself. That is, this permutation takes each individual in X to an individual in Y_1 , each individual in Y_1 back to its corresponding individual in X , and each individual in Y_2 to itself:

$$\pi(X) = Y_1 \quad \pi(Y_1) = X \quad \pi(Y_2) = Y_2.$$

First, $\pi(\text{Odds})$ is *Pareto-dominated* by Evens. The individuals in X are sad in both $\pi(\text{Odds})$ and in Evens. The individuals in Y_1 are happy in both $\pi(\text{Odds})$ and in Evens. The individuals in Y_2 are sad in $\pi(\text{Odds})$, and happy in Evens. So every individual who is happy in $\pi(\text{Odds})$ is also happy in

⁷The following theorem is based on Askill (2018, 128; see also Goodman 2025). See section A.6 for details.

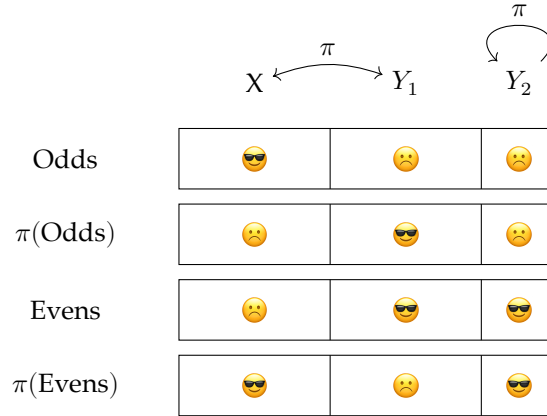


Figure 4.1: Permuting individuals in Odds and Evens leads to Pareto dominance.

Evens, and infinitely many individuals who are sad in $\pi(\text{Odds})$ are happy in Evens.

Second, Odds is Pareto-dominated by $\pi(\text{Evens})$, by similar reasoning: the individuals in X are happy in both Odds and $\pi(\text{Evens})$, the individuals in Y_1 are sad in both Odds and $\pi(\text{Evens})$, and the individuals in Y_2 are sad in Odds and happy in $\pi(\text{Evens})$.

Now let's put these two observations together. Suppose that Odds $\succeq$ Evens. Then, by Permutation Invariance and Pareto, we would have:

$$\begin{aligned}
 \text{Evens} &\succ \pi(\text{Odds}) && \text{by Pareto} \\
 &\succeq \pi(\text{Evens}) && \text{by Permutation Invariance} \\
 &\succ \text{Odds} && \text{by Pareto again.}
 \end{aligned}$$

So if Odds is as good as Evens, it follows that Evens is better than Odds — contradiction.

This much shows that Odds is not as good as Evens. But exactly parallel reasoning leads us to the conclusion that Evens is not as good as Odds, either. So Pareto and Permutation Invariance together imply that Evens and Odds must be incomparable.

Furthermore, we can use exactly the same reasoning to show that *any* two outcomes that are incomparable under the counting order must be incomparable under *any* permutation-invariant Pareto order. There isn't just a *little*

incomparability: the counting order is as far as you can go. For example, this shows that Pareto and Permutation Invariance together imply that Evens is not better than Sevens.

Does this mean we should give up on Permutation Invariance? Maybe we need to include more information in our executive summary after all. But what could this information be? It might have to be quite weird.

Indeed, Askill's argument against Comparability did not really rely on the full strength of Permutation Invariance. The argument that Evens and Odds are incomparable only required *two* carefully constructed permutations. This observation points toward a family of stronger arguments that turn on more natural-looking permutations.

Recall the *East–West worlds* from section 3.3: there are people arranged like the positive and negative integers, in a line from infinite West to infinite East. One very natural idea is that in worlds like these, there is no moral difference between East and West. For example, in the world Sad^* , Indy is Sad, everyone to Indy's East is happy, and everyone to Indy's West is sad. In the better world Happy^* , Indy is happy and everyone else is the same. Now imagine flipping both distributions across Indy, so instead everyone to Indy's *East* is sad, and everyone to Indy's *West* is happy. This makes no difference to the ordering: reflected Happy^* is also better than reflected Sad^* . And it would seem bizarre if this failed.

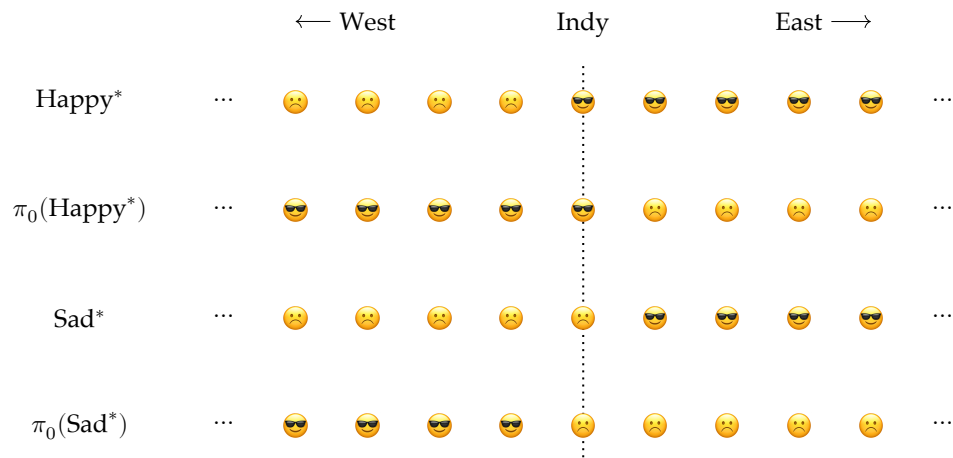


Figure 4.2: An example of reflection preserving betterness. Happy^* is better than Sad^* , and likewise the reflection of Happy^* is better than the reflection of Sad^* .

In general, if we choose any “North–South” axis (in the example, this passed through Indy), there is a corresponding *reflection* π , which takes each East–West world w to a corresponding world πw where the distribution of happiness is reflected from the people on the East side of the axis to the people on the West side, and vice versa. Here is a very natural symmetry principle:

Reflection Invariance. Let v and w be East–West worlds, and let π be any East–West reflection. If v is as good as w , then πv is as good as πw .

This is much, much weaker than full Permutation Invariance. But there is a corresponding result that works the same way as Askill’s. (For further details see section A.6; see also Pruss 2015 for a closely related result about reflection invariant *probabilities*.)

Proposition 4.2. Pareto, Reflection Invariance, and Comparability are jointly inconsistent.

The argument is straightforward — and it really only relies on invariance under *two* reflections. The first is the reflection around Indy that we just considered — call this π_0 . The second is the reflection around Esther, who is immediately to Indy’s East — call this π_1 . (See figure 4.3.)

Consider the world Happy^* again. Let *Mirror* be the reflection of Happy^* around Indy, $\pi_0(\text{Happy}^*)$. In *Mirror*, just Indy and everyone to Indy’s West is happy. Now reflect *Mirror* again, but this time around Esther. In this world $\pi_1\pi_0(\text{Happy}^*)$, all the people to Esther’s East are happy again — but not Indy, and not Esther. (Indy’s status gets swapped with the person immediately to Esther’s East, and in $\pi_0(\text{Happy}^*)$ this person is sad. Esther keeps her own status, which is sad.) In other words, this world $\pi_1\pi_0(\text{Happy}^*)$ is just the same as Happy^* , except that two happy people have been made sad. Call this world *Sadder*. By Pareto, *Sadder* is worse than Happy^* .

Since *Mirror* and Happy^* are reflections of one another, *Mirror* can’t be worse than Happy^* . If it were, then Reflection Invariance would imply that Happy^* is also worse than *Mirror*. Likewise, *Mirror* can’t be better than *Sadder*. If it were, then invariance under the reflection π_1 would imply that *Sadder* is also better than *Mirror*. So *Mirror* is not worse than Happy^* , and *Sadder* is not worse than *Mirror*.

$$\text{Sadder} \not\prec \text{Mirror} \not\prec \text{Happy}^*.$$

But if Comparability is true, this would imply that *Mirror* is *as good as*

Happy*, and Sadder is *as good* as Mirror:

$$\text{Sadder} \succeq \text{Mirror} \succeq \text{Happy}^*.$$

And we already know that Sadder is not as good as Happy* — contradiction.

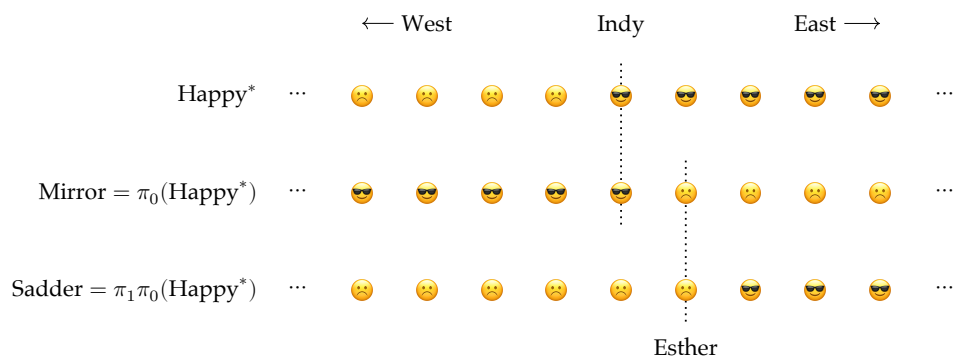


Figure 4.3: Reflecting a world twice, around two different axes, can take us to a strictly worse world.

Reflections are just one example of a kind of symmetry transformation that gives rise to this kind of conflict between Comparability and Pareto. In section A.6 I characterize the conflict more generally. The lesson is that there are many intuitively appealing symmetries of value that must be violated if Comparability and Pareto are both true. Any executive summary that is adequate for determining value comparisons between worlds will have to include lots of details that are intuitively morally irrelevant.

4.3 Non-constructiveness

The second argument against Comparability is based on a technical result independently due to Zame (2007) and Lauwers (2010). (This result establishes a conjecture originally from Fleurbaey and Michel 2003.) It requires a bit of mathematical background.

ZFC set theory (Zermelo-Fraenkel set theory with Choice) is the most widely accepted foundational framework for doing mathematics. This is a collection of axioms about the structure of sets, including the *Axiom of Choice*.

Choice. For any set of pairwise disjoint non-empty sets A , there exists a set C that contains exactly one element from each $A \in A$.

Choice is a workhorse in every part of orthodox mathematics that traffics in the infinite. Here are a few examples: we use it to prove that every consistent theory has a model, that vector space has a basis, that every cardinal number has a corresponding ordinal, that there are non-measurable sets of reals, and that there are non-standard hyperreal number systems. But even though Choice is widely deployed throughout modern mathematics without reservation, it is a bit spooky. Choice tells us that all of these various objects *exist*, but it gives us no clue about how to explicitly *construct* any of them. We can't explicitly give any examples. The Choice axiom selects elements from infinitely many sets for us, but we can't generally say *how* it selects those elements — the choices are arbitrary. Accordingly, proofs that essentially rely on Choice are called *non-constructive*. In the words attributed to Weyl, “A non-constructive existence proof informs the world that a treasure exists without disclosing its location” (Lauwers 2010).

Now let's go back to simple worlds, with infinitely many people who are either happy or sad. I have argued that the value order for such worlds should satisfy the *Pareto* principle: making a sad person happy makes things strictly better. And it should also satisfy *Impartiality*: if we just make one sad person happy (changing nothing else), this is just as good as just making a *different* sad person happy. It is not immediately obvious that these assumptions are consistent with Comparability. As we have discussed, the counting order satisfies Pareto and Impartiality, but it is very far from satisfying Comparability. And while there are various steps we might take to extend it (recall our example from section 3.5), these steps still leave us far away from a total order. It's not clear how we might get all the way to a total order. Fortunately, we can at least show that such orders *exist*.

Theorem 4.3 (Szpilrajn (1930); Arrow ([1951] 2012)). There exists an order of $\{0, 1\}$ -social outcomes that satisfies Pareto, Impartiality, and Comparability. (See theorem A.14 in section A.3)

That's the good news. The bad news is that the proof of this theorem is non-constructive: it relies on Choice. Furthermore, this is no accident: *any* proof of this theorem in standard ZFC set theory relies on Choice.

Theorem 4.4 (Zame (2007); Lauwers (2010)). It cannot be constructively proved that there exists an order of $\{0, 1\}$ -social outcomes that satisfies Pareto, Impartiality, and Comparability.^a

^aTo make that more precise, this means that the existence of such an order cannot be proved in ZF set theory without Choice. This theorem relies on the further auxiliary assumption that ZF set theory is consistent; because of Gödel's second incompleteness theorem, this further assumption is not itself provable in ZF set theory, nor indeed in ZFC.

Zame (2007) proves a further result, which helps accentuate the sense in which these orders are beyond our grasp: no *definable* order of $\{0, 1\}$ -social outcomes can be proved in ZFC to satisfy Pareto, Impartiality, and Comparability (in a precise and broad sense of definability).

(Note the difference between this claim and another that it might be confused with: "ZFC proves that no order that satisfies Pareto, Impartiality, and Comparability is definable." This claim is false. Indeed, ZFC is consistent with Gödel's constructibility axiom $V = L$, which implies that *every* relation is definable, in the relevant sense.)

Pruss (2014) proves a closely related theorem connecting such orders to the Banach–Tarski paradox: see theorem A.30 in section A.3 and theorem E.66 in section E.7.

Intuitively, coming up with any example of a *total* impartial Pareto order involves infinitely many arbitrary choices. It is hopeless to try to give a complete reckoning of which simple worlds are better than which. (And things do not get any easier when we go on to consider more complicated worlds.) If there is a complete moral ranking of worlds, it is too far away: it is in Platonic heaven, where we cannot ascend to get it, and where we cannot hear it or observe it.

4.4 The logic of comparatives

Those look like two powerful arguments against Comparability, which are specific to our setting of trying to describe an impartial social welfare order that respects the Pareto principle. Any order like this that satisfies Comparability must, first, violate Permutation Invariance, and second, be non-constructive in the sense discussed above. In short, the social welfare order for infinite worlds suffers either rampant *incomparability*, or else rampant *arbitrariness*.

But there are also some very powerful arguments *for* Comparability. We turn to those now.

Dorr, Nebel, and Zuehl (2023, henceforth DNZ) give two very general arguments for Comparability; they are not specific to value or betterness, but rather apply to any natural language comparatives at all. All natural language comparatives (*harder*, *better*, *faster*, *stronger*, etc.) satisfy certain logical principles. Some of these are relatively uncontroversial.

Irreflexivity. x is not more F than x .

Transitivity. If x is more F than y and y is more F than z , then x is more F than z .

More-As. x is more F than y iff x is (at least) as F as y and y is not as F as x .

DNZ argue that a generalization of Comparability has the same status.⁸ Even though their arguments are very general, I'll keep focusing on the case of *good*. (*Better* is analyzed as *more good*.)

Here is their first argument. The following two principles both sound intuitively valid.⁹

No Better. If x is no better than y , then y is at least as good as x .

Not As Good. If x is not better than y , then x is no better than y .

The first principle underwrites ordinary patterns of inference that sound unimpeachable, as in this example:

Taylor's paper is no better than Kyle's. So, Kyle's paper is at least as good as Taylor's. (Dorr, Nebel, and Zuehl 2023, 428)

The second principle sounds like a trivial change in the way negation is expressed. Together, these imply Comparability. (Let v and w be worlds. Suppose v is not at least as good as w . Then v is not better than w , so v is no better than w , and so w is at least as good as v .)¹⁰

DNZ's second argument turns on the relationship between comparatives and

⁸To be explicit, here is their generalization:

Restricted Comparability. If x is as F as x and y is as F as y , then x is at least as F as y or y is at least as F as x .

Note the restriction " x is as F as x and y is as F as y "; this plays the same role as the restriction in the statement of World Comparability in section 4.1, to worlds "which are each good for the individuals in $\mathcal{I}$ to some degree". The restriction rules out cases involving things that intuitively aren't F to any extent at all, or which are not both F in the same sense. (We do not argue that if the number 12 is not at least as good as a pesto sandwich, then a pesto sandwich is better than the number 12.) For short, DNZ say that x is *F-assessable* iff x is as F as x . I'll generally let the restriction to *good*-assessable worlds remain implicit in what follows.

⁹DNZ (2023, 427ff.) present their argument as an inference to the best explanation: they argue that Comparability makes better sense of certain intuitively valid patterns of inference than views that deny Comparability. Here I'll instead present direct arguments for Comparability, which are variants of theirs for which the dialectic is simpler.

¹⁰More generally: suppose x and y are each *F-assessable*, and it is not the case that x is at least as F as y . Then x is not more F than y . So x is no more F than y . And so x is at least as F as y .

“threshold” judgments.¹¹ Besides *ordinally* ranking things as better or worse than one another, sometimes we say that something is just *good*. Intuitively, this means that its goodness passes some contextually determined threshold. A sandwich that counts as “good” in a context where we are comparing it with other lunches we had this week may not count as “good” in a different context where we are comparing it with the best food in the world. But if we fix some particular context, however we do it, the following principle seems correct.

Strong Monotonocity. If x is good, and y is not good, then x is better than y .

Again, this seems to be a very general feature of comparatives.

- a. Kara is healthy, and Sam is not healthy. So Kara is healthier than Sam.
- b. Alex had fun at the fair and Cameron didn’t. So Alex had more fun at the fair than Cameron.
- c. Charlie likes broccoli and doesn’t like chocolate. So Charlie likes broccoli more than chocolate.

(Dorr, Nebel, and Zuehl 2023, 438)

This argument also depends on another structural claim. If there is incomparability, then it can appear in a certain pattern, which DNZ call *incomparability-connectedness*. Consider the argument from small improvements again (section 4.1). We had an education charity (Program A) and an animal welfare charity (Program B); neither seemed better than the other, and also a slight change in how good either one is does not seem to break the deadlock. Furthermore, in the example, both of these programs are *good*. But now consider Program A^- , which is just like Program A except a little bit less effective. Taking the judgments at face-value, Program A is not better than B , and B is not better than A^- , but A is better than A^- . We can also imagine an animal welfare charity just like Program B , except a little bit worse. So Program A^- is no better than Program B^- , while Program B is better than Program B^- . And we can go on this way. Call a set of things like Programs A , B , A^- , B^- , and so on, an *incomparability-chain*: in particular, no element of the sequence is better than the thing immediately after it.

¹¹Again we restrict to x and y that are *good-assessable* — each the right sort of thing to count as as good as itself. In our case, the relevant sort of thing is simple worlds.

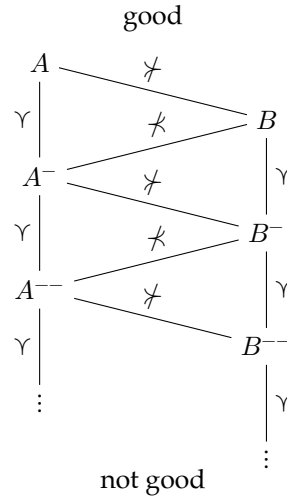


Figure 4.4: An incomparability chain built from small improvements.

But if we keep on going this way, successively making the education charity and the animal welfare charity worse and worse a little at a time, eventually we will reach programs that are pretty lousy at achieving their aims. In particular, even though we started with a pair of good charities, eventually we will reach charities that are not good. Here is the general principle:

Connectedness. If Comparability is false, then there exists an incomparability-chain $x_1, x_2, \dots, x_n$ such that x_1 is good, and x_n is not good.

But this contradicts Strong Montonicity. If we start the chain with something good, and end with something that is not good, then there must be an adjacent pair of x_k and x_{k+1} such that x_k is good and x_{k+1} is not good. Then Strong Monotonicity implies that x_k is *better* than x_{k+1} . This contradicts the assumption that x_k and x_{k+1} are part of an incomparability-chain. (If the resemblance between this argument and the sorites paradox concerns you — good! I’ll come back to this in section 4.7.)

It’s instructive to consider how this style of argument works in the specific case of the counting order. In this case, we can run the argument on a very short incomparability chain. Suppose we are in a simple world: either Happy or Sad (from section 3.2). The people in rooms 2, 4, 6, 8, ... are happy. The people in rooms 3, 5, 7, ... are sad. We don’t know how Indy is doing, in room one. Now, having fixed this context, I think we can fix the threshold for “good” in such a way that this is appropriate to say:

- (1) Happy (the world where Indy is happy) is good, and Sad (the world where Indy is sad) is not good.

Now consider the world Odds, where just the people in rooms 1, 3, 5, ... are happy. Is Odds good? It's hard to say, but whatever we say leads to trouble. According to the counting order, Happy is not better than Odds. Since Happy is good, Strong Monotonicity implies that Odds is also good. Also according to the counting order, Odds is not better than Sad. Since Sad is not good, Strong Monotonicity implies that Odds is not good. Contradiction.

As I said, these two arguments from Dorr, Nebel, and Zuehl (2023) are very general, applying across the board to any comparatives — not just *better*. There are also some powerful arguments for Comparability that are more specific to value theory. I'll turn to these next.

4.5 Opaque sweetening

Consider again the two charities from section 4.1: A , a highly effective education program; and B , a highly effective animal welfare program. It turns out, there are also two other programs seeking funding: A^+ , a marginally *more* effective education program than A ; and B^+ , a marginally more effective animal welfare program than B . If there can be incomparability between outcomes, then we can have cases structurally like this one: A^+ is better than A , B^+ is better than B , but each of A and A^+ is incomparable with each of B and B^+ :

$$\begin{aligned} A^+ &> A \\ B^+ &> B \\ A^+ &\not> B \\ B^+ &\not> A. \end{aligned}$$

Now let's consider how this interacts with *risk*. (This case is based on Hare 2010; Schoenfield 2014; see also Manzini and Mariotti 2008; Bader 2018; Bales, Cohen, and Handfield 2014; J. S. Russell 2024, sec. 3.2; Lederman forthcoming; Tarsney, Lederman, and Spears 2025.)

Opaque sweetening. Althea is still deciding on where to make her donation out of four options: A, A^+, B, B^+ . But then she discovers that her banking software has a very bad, very strange bug. (She is still going to have to use

it anyway, though, or else make no donations at all.) There are two buttons she can click: *Random*, and *Swapped*. Deep in the bank's server vault there is a coin being flipped. Random will send her donation to *A* if the coin comes up Heads, and to *B* if it comes up Tails. Swapped will send her donation to *B* if the coin comes up Heads, and to *A* if it comes up Tails. It doesn't seem like it makes much difference which she chooses.

But it turns out Althea also has a third option. If she clicks *Sweetened*, then her donation will instead go to one of the superior charities A^+ or B^+ — though this will still be chancy: A^+ on Heads, B^+ on Tails. (See table 4.2.)

Table 4.2: Opaque sweetening.

	Heads	Tails
Random	A	B
Swapped	B	A
Sweetened	A^+	B^+

If there are incomparable outcomes, then there are cases of opaque sweetening.¹² But these cases are paradoxical. For in fact, there are strong arguments for each of the following claims.

1. Sweetened is better than Random.
2. Random and Swapped are equally good.
3. Sweetened is not better than Swapped.

Take them one by one.

¹²In the situation before us, B is not better than A , and also a slightly *improved* program A^+ is not better than B ; but A^+ is better than B . This violates the following principle.

Negative Transitivity. If x is not better than y and y is not better than z , then x is not better than z .

Failures of Negative Transitivity suffice to raise opaque sweetening cases.

Negative Transitivity is weaker than Comparability. But arguments for Negative Transitivity lend much of their force to Comparability as well. First, if Negative Transitivity holds, there is no argument against Comparability from small improvements. Second, if Negative Transitivity holds, then the relation x is not better than y and y is not better than x is an *equivalence relation*. In that case, it is not clear what motivation we might have for saying that in *some* of these cases where neither is better than the other, the two are equally good, but in *other* cases they are not.

Sweetened is better than Random. This is a direct consequence of *Statewise Dominance* (section C.3): if X is certain to turn out better than Y , then X is strictly better than Y overall. If the coin comes up Heads, then Random will result in a donation to A , while Sweetened will result in a donation to the better charity A^+ . If the coin comes up Tails, then Random will result in a donation to B , while Sweetened will result in a donation to the better charity B^+ . In either case, Sweetened turns out better than Random. Since we know Sweetened will turn out better than Random, surely it is better overall!

Choosing Random rather than Sweetened would be leaving value on the table. We can make this more vivid by imagining that we can measure the relative effectiveness of A^+ and A in monetary terms: say, a \$900 donation to A^+ would promote education just as well as a \$1000 donation to A . Likewise, a \$900 donation to B^+ would promote animal welfare just as well as a \$1000 donation to B . Then choosing Sweetened is like choosing Random, and then additionally tacking on a free bonus \$100 donation that does some further good. It seems clearly worse to turn that down.

Random and Swapped are equally good. They certainly *sound* equally good. When I initially described the case, you might have wondered whether there was even any difference: Random gives a 50% chance of A and a 50% chance of B , and Swapped gives ... a 50% chance of B and a 50% chance of A . What's the difference? If her donation just goes to a random charity anyway, how could it matter which button Althea presses?

Or put it another way. Say we toss a coin, and ask Althea to call it in the air. If she wins the toss, the donation will go to A , and otherwise it will go to B . If Random and Swapped are not equally good, this means that calling Heads and calling Tails are not equally good. That isn't to say that one of them is better than the other — we can instead say that the two options are incomparable. But that is not that much comfort — they still differ in value in ways that can matter. For example, we will be pressured to say that calling Heads is worse than Sweetened, but calling Tails is not.¹³ So comparing the value of prospects would require us to inquire into matters that seem intuitively irrelevant.

We can put this point more generally. The claim that Random and Swapped are equally good directly follows from this principle (introduced in sec-

¹³Indeed, I have focused on the simple claim that Random and Swapped are equally good, but the same argumentative work can be done by the weaker claim that Random and Swapped are both worse than the same prospects.

tion 1.3.5):

Simple Stochasticism. For simple prospects X and Y , if X and Y have the same probability of resulting in x , for any possible outcome x , then X and Y are equally good overall.

(By *simple* prospects I mean those with only finitely many possible outcomes.) This is something we implicitly take for granted whenever we describe prospects like “probability 1/2 of saving three” or “probability 1/10 of saving a hundred” without specifying how those probabilities for different prospects line up with one another.

Sweetened is not better than Swapped. If the coin comes up Heads, Sweetened will result in A^+ , and Swapped will result in B . By hypothesis, these two outcomes are incomparable in value: A and B were incomparable, and a small improvement in A does not break the tie. So if the coin comes up Heads, Sweetened will not turn out better than Swapped. The same reasoning applies in the case of Tails: in this case, Sweetened will result in B^+ and Swapped will result in A . Again, the outcome of Sweetened is not better than the outcome of Swapped. So it is certain that Sweetened will not turn out better than Swapped. It seems that we should conclude that Sweetened is not better than Swapped overall.

This reasoning can be underwritten by a third principle:

Negative Statewise Dominance. For any prospects X and Y , if it is certain that X will not turn out better than Y , then X is not better than Y overall.

Why believe Negative Statewise Dominance? Here is an argument slightly adapted from Schoenfield (2014, 268).¹⁴

A theory of the value of prospects is designed to help us in our pursuit of value under conditions of uncertainty. Frequently, under conditions of uncertainty, we don’t know which prospect will produce the outcome with the highest value. But in cases in which there is certainty about the relative values of the available options, what maximizes the value of a prospect (*ex ante*) should line up with what maximizes the value of outcomes (*ex post*).

In the case at hand, we do have this kind of certainty: it is certain that

¹⁴She puts the point in terms of *expected value*, understood as a theory guiding the choices of rational agents with limited information. I have restated the point in terms of the value of prospects more generally

Swapped maximizes the value of outcomes (*ex post*), in the sense that no other option has a strictly better outcome. (Sweetened has this feature, too — more than one option can be maximal.) So “lining up” would mean that Swapped also maximizes the value of prospects (*ex ante*). If Sweetened is better than Swapped, this alignment is destroyed.

In general, if it is certain that *X* won’t have a better outcome than *Y*, and yet the prospect *X* is ranked as better than *Y*, then the value of prospects seems to have come loose from the thing this value is aiming at — namely, the value of the outcome that would result from the prospect. (See also section 6.4.)

To sum up: if Comparability is false, there are cases of opaque sweetening. These cases cannot satisfy all of Statewise Dominance, Negative Statewise Dominance, and Simple Stochastic Equivalence. One might reject one of these principles, but I think they are all very plausible, and have much to be said in their favor.¹⁵ An attractive way out of the difficulties is to accept them all — and Comparability as well.¹⁶

¹⁵Opaque sweetening cases have chiefly been discussed in the context of a theory of rational or moral decision-making, rather than in terms of the value of prospects as I have framed it. Hare (2010) raised this puzzle, and tentatively advocated giving up the analogue of Negative Statewise Dominance for rational choice. Bader (2018) has argued for this response more full-throatedly, defending analogues of Simple Stochastic Equivalence and Statewise Dominance. Schoenfield (2014) defends a version of Negative Statewise Dominance, and responds to the puzzle by rejecting Simple Stochastic Equivalence (in effect).

But note that we aren’t entirely out of the woods. Below, I will suggest that we should accept Comparability, but allow that it is sometimes vague which prospect is best. But this means that we still face questions of what one *ought to do* in such cases, and some similar considerations arise again in that setting. Nonetheless, I think we also have additional resources at our disposal for this reformulated question.

¹⁶Tarsney, Lederman, and Spears (2025; building on Lederman forthcoming) offer related arguments for Comparability, turning on similar idea. There are two key dialectical differences. First, their argument uses a weaker premise than Negative Statewise Dominance, which they call

Negative Dominance. If no possible outcome of *A* is better than any possible outcome of *B*, then *A* is not better than *B*.

The philosophical motivation for this premise is similar to that for Negative Statewise Dominance, but it is a bit more robust in that it avoids any question about the moral significance of *states*. (Though, for what it’s worth, I’m not especially worried about that.) The price for weakening this premise is adding another: either an *ex ante* Pareto premise (of the sort I discuss in chapter 7), or a *separability* premise (of the sort I discuss in chapter 8).

4.6 Classical indeterminacy

So far, we have surveyed arguments both for and against Comparability. I don't find any of the arguments on either side to be utterly decisive, but I do think they are powerful considerations. How shall we adjudicate between them?

I think the right thing to say here is that Comparability is true, but many comparisons of value are *indeterminate* or *vague*. (I'll use the two terms interchangeably.) (This is not an original idea: see especially Broome 1997, 2000; Dorr, Nebel, and Zuehl 2023, sec. 5, and references on p. 446.) I'll begin by saying some general things about the philosophy of vagueness, and then (in section 4.7) I'll discuss how this helps address the arguments against Comparability.

Let's forget about indeterminate *value* for a bit, and think about more hum-drum cases of vagueness. Take a series of a couple hundred color swatches (like you use to choose paint) forming a smooth gradient from imperial red to electric blue. In this series, adjacent swatches are so close in color that you can't really even tell them apart. Now, try to make a pile of all and only the red cards — no blues or pinks or purples allowed. It's a perplexing task. I wouldn't know how to do it, and I wouldn't know how to find out how to do it, or how to get significantly better at it. Of course, there are some tiles that clearly belong in the red pile, at the beginning of the series, and others that clearly don't belong, at the end. But many tiles in the middle of the spectrum are hard cases. We don't want to say they're red, and we don't want to say they aren't. They are borderline.

Many have been tempted to say that such borderline cases are counterexamples to the law of excluded middle, *A-or-not-A*: some of these swatches are neither red nor not red. But this is hard to make sense of. For

(2) Not (*A* or not *A*)

is logically equivalent, by de Morgan's law, to

(3) (Not *A*) and not (not *A*).

And this is a contradiction: not-*A* is at once affirmed and denied. Furthermore, the most familiar non-classical logics (such as those of Kleene and Łukasiewicz) also endorse this de Morgan equivalence: that is, "and" and "or" are dual (for overview see Priest 2008, chs. 7 and 11). Indeed, (2) and (3) both seem to be equally good formalizations of the English expression

“neither A nor not- A ”. There are some — *dialetheists* — who accept the contradictory consequence (3); they endorse contradictory descriptions of these these hard cases (Hyde 1997; see also Priest 1979). I won’t follow them.

Something similar goes for another natural thought, which is that there is a problem with *bivalence* — the principle that, for any particular tile, it is either true or false that it is red.¹⁷ If we understand “true” and “false” in their most familiar senses, then we should say:

- (4) (It’s true that A) iff A .
- (5) (It’s false that A) iff not- A .

But given these “deflationary” principles, saying

- (6) It’s not true that A , and it’s not false that A

is again equivalent to the contradiction

- (7) (Not A) and not (not A)

(see Williamson 1996, 188–89). So, if we are to consistently reject bivalence, we must understand “true” and “false” in some less familiar “thick” senses that don’t permit the equivalences (4) and (5). I’ll come back to this idea in a moment.

If we are to give up the principles of excluded middle or bivalence, we should do so in some subtler way than just asserting the negations of certain instances of those principles. Rather than *denying* A -or-not- A , by asserting its negation, one might instead *reject* A -or-not- A , refusing to assert it, and also refusing to assert its negation (for discussion of this distinction, see Restall 2005; Field 2008, sec. 4.6). Similarly, we might reject the claim that a certain tile is red, and also reject the claim that it is not red, without endorsing the contradictory claims that it is both non-red and not non-red.

It isn’t especially satisfactory to just remain silent, though. We’d like to say something to make it clear that it isn’t as if the question simply hasn’t occurred to us, or as if we have some secret opinion that would be too rude to say out loud, or any of the many other commonplace reasons for silence. Otherwise we would have no way to theorize about vagueness. We would like to express the problematic status of A , without denying A outright.

But we already have a way to do this: we can simply say that it is *vague* or

¹⁷A different thesis also called “bivalence” concerns a truth predicate for *sentences*, rather than the *it’s true that* operator I discuss, but that would take us too far off course.

indeterminate or *borderline* whether the tile is red. I'll say "it is *determinate* that *A*" to mean that it is true that *A*, and not indeterminate whether *A*. So, instead of saying "the tile is neither red nor not red", or "it's neither true nor false that the tile is red", we can say instead

- (8) It's not determinate that the tile is red, and it's not determinate that the tile is not red.

This is equivalent to just saying it's *indeterminate* whether it is red. Those who wish to deny bivalence in some sense can do so using the "thick" notions of *determinate truth* and *determinate falsity*, rather than the "thin" notions of truth and falsity that obey the deflationary principles (4) and (5). I prefer to reserve the simple words "true" and "false" for the "thin" notions that obey those principles: it's expressively useful to have *some* notions that behave that way, after all. But whatever words we use, we should always be careful to distinguish mere "thin" truth from determinate truth, and mere "thin" falsity from determinate falsity, or else we will run into all sorts of trouble.

I think this removes much of the pressure to reject the law of excluded middle. When we acknowledge that the tile is neither determinately red, nor determinately not red, then I feel no real pressure left over to *additionally* say that it is not determinately red-or-not-red. Our temptation to resist instances of excluded middle "*A-or-not-A*" can be explained more simply by our failure to distinguish ordinary truth from determinate truth (see Keefe 2000, 164).

A similar move helps defuse the sorites paradox. It's highly tempting to say of the color series, arranged in order along the red-to-blue gradient:

- (9) No red tile is next to a tile which is not red.

After all, how could two tiles we can't even tell apart be different colors? But (9) is untenable: it (classically) contradicts the evident fact that

- (10) There is a red tile which is next to a tile which is next to a tile which is ... which is next to a tile which is not red.

Again, this has led many to try to come up with alternative logics in which (9) and (10) are consistent with one another after all (or at least in which this contradiction does not explode into arbitrary absurd results). But instead, we can diagnose our temptation toward (9) as based on confusing it with an alternative principle:

- (11) No *determinately* red tile is next to a tile which is *determinately* not red.

Call a tile a *cut-off* iff it is red and next to a tile which is not red. Then we should clearly distinguish these two claims:

Narrow Scope. For some x , determinately x is a cut-off.

Wide Scope. Determinately, for some x , x is a cut-off.

Narrow Scope is false, and its falsehood accounts for our temptation to affirm the soritical principle (9). But Wide Scope can still be true. This goes a long way toward defusing the threat to classical logic raised by the sorites paradox.

In general, we must be careful. Indeterminacy is not an *alternative* to being true or false in the plain (“thin”) sense. Indeterminacy is compatible with truth, and it is compatible with falsity. Supposing there is any indeterminacy at all, excluded middle implies that there are true instances of each of these:

(12) A and it’s indeterminate whether A .

(13) Not- A and it’s indeterminate whether A .

So why do these sound so jarring? One reason is that they really do have a kind of defective status: they are *Moore-paradoxical*. Some philosophers once thought that

(14) It’s raining but I never know whether it’s raining

was inconsistent — taking this apparent observation as support for epistemic theories of truth. G.E. Moore (1952) noted that this is a mistake.¹⁸ “It’s raining but I never know whether it’s raining” can be true. The trouble with (14) is not that it cannot be true, but that it cannot be *known* (by me). For if I were to know (14), then I would know that it’s raining; but also, since knowing is factive, I would never know whether it’s raining, and this is a contradiction.¹⁹ Furthermore, since I can’t *know* (14), this would also be a bizarre thing for me to *assert*. But while ignorance is a barrier to assertion, and (trivially) to knowledge, this does not mean that ignorance is a barrier to *truth*.

The situation with vagueness is exactly analogous. The claim

(15) It’s raining and it’s indeterminate whether it’s raining

can be true. But it cannot be *determinately true*. For if (15) were determinately

¹⁸Moore actually discussed sentences involving *belief* rather than *knowledge*.

¹⁹This is making the slightly idealizing assumption that knowledge distributes over conjunctions; see Fitch (1963).

true, then it would follow that it's determinately raining, and also indeterminate whether it's raining. This is a contradiction. So while (15) can be true, it can be a vague truth at best. Furthermore, there is ordinarily something wrongheaded about *asserting* vague truths, and ordinarily we cannot *know* vague truths. So, while (15) can be true, typically it cannot be known or properly asserted.

In short, I do not think that vagueness requires us to depart from classical logic. Using the notions of determinacy and indeterminacy, we can theorize about borderline cases and sorites series perfectly well without sacrificing the law of excluded middle. This approach is what I call *classical indeterminacy*.

Classical indeterminacy is somewhat controversial, and I don't pretend that the brief story I've rehearsed puts the controversy to rest.²⁰ But it is not as controversial as you may think. Indeed, a classical logic of indeterminacy is a point of common ground between many philosophers who otherwise seem to disagree quite radically on the nature of vagueness — whether it is “epistemic” (Williamson 1996, ch. 7), or “linguistic” (K. Fine 1975; D. Lewis 1986a, 212–13), or “metaphysical” (Barnes 2010; Barnes and Williams 2011), or something else entirely.

Classical indeterminacy is not itself a theory of the *nature* of vagueness. It is neutral between these pictures, and many others. Indeed, it is a very subtle matter to articulate what genuinely distinguishes these different classical views from one another — but the points of disagreement are not to be found in anything I have said in this brief primer. (See Williamson 1996, 164; see also Akiba 2015.)²¹

For example, it is sometimes claimed that *epistemicists* distinctively say “there

²⁰See Bacon (2018, ch. 1) for further discussion of the logical difficulties facing non-classical theories.

²¹That is not to say that I have no opinions on the matter. In particular, I accept the widely held view that vagueness is “in representation”, rather than “in the world”. In particular, I hold that indeterminacy is *presentation-sensitive*, in the same sense I discuss in chapter 7, analogous to other “opaque” operators like “Amy believes” or “so-called”. For example, I said above that there is no tile x such that it is determinately a cut-off — the last red tile in the sequence. But there is a cut-off tile, so say we let “Cutty” be a proper name for the cut-off tile, whichever one it is. Then, I say, it is determinate that Cutty is the cut-off tile. And so there *is* a determinate cut-off tile — namely, Cutty! Contradiction? No — just equivocation. I take it that the claim “it is determinate that x is a cut-off” is *context-sensitive*, and specifically what it says is sensitive to a choice of *mode of presentation* for x . Under the mode of presentation *Cutty*, it is determinate that x is a cut-off. Under the mode of presentation *tile 38* (or whatever it may be), it is not determinate that x is a cut-off. Quantified claims are also presentation-sensitive. (This is exactly parallel to claims like “There is a tile that I believe to be a cut-off”.)

is a sharp cut-off in a sorites series". But if "sharp cut-off" means "*determinate* cut-off" — a tile such that *determinately* it is red and the next tile is not — then epistemicists say no such thing. If "sharp cut-off" is just a dramatic way of saying "cut-off", then all classical theorists say the same. (If a series contains both red tiles and tiles that are not red, then it includes a red tile next to a tile that is not red — as a matter of classical logic.) "Sharp cut-offs" are not a genuine point of disagreement between epistemicists and alternative classical theories of vagueness.

The usual paradigm of indeterminacy is the sorites series, but this is not the only place where it is to be found. I think it's helpful to keep a different paradigm in mind as well (see Breckenridge and Magidor 2012). Consider a bog standard bit of mathematical reasoning.

Theorem. The square of any even number is divisible by 4.

Proof. Let n be any even number. Then $n = 2k$ for some integer k . So $n^2 = (2k)^2 = 4k^2$. Since k^2 is an integer, n^2 is divisible by 4. $\square$

Consider the final statement " n^2 is divisible by 4". This is true, in the context where it is asserted. It follows from the fact that n is an even number. But which number is n ? Is it two? Twenty-six? Any answer to this question seems absurd.

One might think that this is because the question has no answer. But, since we know n is an even number, that seems to require a bizarre picture according to which n is an even number which is *distinct* from every one of 2, 4, 6, 8, ...²² (If you are tempted to reach for *non-standard* numbers, consider instead reasoning that begins, "Let n be an integer between 1 and 5.")

After we declare, "let n be any even number", we then know that n is an even number, and we do not know which number n is. We also don't have the slightest hope of ever discovering which number n is. And if it somehow turned out that, say, $n = 8$, this would be an entirely arbitrary and obscure matter. The question "Which number is n ?" exhibits the same hallmarks as

²²For sustained defense of this bizarre idea, see K. Fine (1985). Macnamara (1988) notes a parallel with Berkeley's criticism of Locke's doctrine of abstract ideas: "Does it not require some Pains and Skill to form the general Idea of a Triangle (which is yet none of the most abstract, comprehensive and difficult) for it must be neither Oblique nor Rectangle, neither Equilateral, Equicrural, nor Scalenen, but all and none of these at once?" ([1710] 2008, introduction, sec. XIII)

the question “Which is the last red tile?” I propose that they are the same phenomenon. It is *indeterminate* which even number n is.

We might want to say that n is not any *particular* even number. But I think that this is exactly like our pull to say that there is no *sharp* cut-off in the sorites series. We might mean that n is not any *determinate* even number: that is, it is not determinate that $n = 2$, nor that $n = 4$, and so on. This is perfectly correct. We might alternatively understand “particular even number” as just an emphatic way of saying “even number”. In that case, it is a blatant contradiction to say that n is an even number, but n is not any *particular* even number. And we know perfectly well that n is an even number — we rely on this fact in our proof!

We might ask another question. Suppose that in fact $n = 8$. (It may be, for all we know!) What *makes* it the case that $n = 8$, rather than some other number? What could explain this? It seems to me that the correct answer is *nothing at all*. After all — it is *arbitrary*. Arbitrariness is the opposite of reason. An arbitrary decision is one for which no adequate explanation can be provided. If n is an arbitrary number, then n is a number, and there is no reason at all for it to be one number rather than another. Is this spooky? Maybe, but that isn’t a very good reason to think it isn’t true. (Mind-controlling fungus and ball lightning are both pretty spooky.)²³

I suspect (very tentatively) that arbitrariness is the more basic variety of indeterminacy. Just as we can reason about arbitrary objects, we can also reason about arbitrary properties. Here is the color gradient again. Someone says, “Let *schmed* be an arbitrary property such that the first ten tiles are schmed, and the last ten tiles are not schmed.” Then this person knows the first tile is schmed, they know the last tile is not schmed, and there is a range of cases in the middle where they have no idea as to whether they are schmed. Of course, *red* is not the same as *schmed*. For one thing, unlike with

²³I am of the view that this is also what is going on with the principle of *counterfactual excluded middle*, which is of some relevance to understanding what the kind of counterfactual value we are exploring even is (see sections 1.3.3 and C.3). Here is an objectively chancy coin, which in fact we do not toss. We do not want to accept “If the coin were tossed, it would come up heads.” We also do not want to accept “If the coin were tossed, it would come up tails.” But we can still accept the disjunction of these two claims. This disjunction sometimes meets resistance, but I think this is well diagnosed by going on to say that it is *indeterminate* which of them is true. One kind of resistance to counterfactual excluded middle comes from the question: if it is an objectively chancy matter, then what could *make it the case* that the coin would have come heads, or that it would have come up tails? I think the right answer to this challenge might well be: *nothing* could make it the case. This is an entirely arbitrary matter.

red, there is a clear cut-off between the *clearly schmed* cases and the unclear cases. But we can imagine a more elaborate ceremony that wouldn't work this way. "Let *schmeveral* be a number between five and fifteen. Let *schmed*₂ be a property that applies to the first schmeveral tiles, and not to the last schmeveral tiles." We can imagine further complications that would bring things even closer to the real case. A just-so story is that *all* of our vague concepts have been introduced by some long chain of arbitrary introductions, lost in the mists of time.

4.7 Vague value

Now let's return to our original question about Comparability. Consider the example of the education program *A* and animal welfare program *B* again. Althea might be tempted to say,

A is not better than *B*, but *B* is not better than *A* either, and they are not equally good.

But as Dorr, Nebel, and Zuehl (2023, 444) point out, we are also sometimes tempted to say things like this:

George Costanza is not bald, but he is also not *not* bald.

DNZ offer the plausible diagnosis that this is a form of "metalinguistic negation": roughly, the speaker is distancing themselves from *asserting* that Costanza is bald, and likewise from asserting that Costanza is not bald — not because they are both false, but because they are both indeterminate. The same diagnosis makes sense for the judgment of comparative value. Program *A* is not *clearly* better than *B*. This is a good enough reason to distance oneself from asserting that *A* is better than *B*. The same goes for the claim that *B* is better than *A*, and likewise for the claim that they are equally good. But saying that none of these comparisons hold *clearly*, or *determinately*, is perfectly consistent with Comparability. Indeed, DNZ find that philosophers arguing against Comparability frequently hedge in just the ways you would expect in cases of vagueness.

For example, we have seen some say that 'there are no precise truths' (Parfit), that we 'may not ... be able to decide' (Sen), that there is 'no answer' (Keynes), or that it is 'perhaps not clear' (Kamp) whether one thing is more *F* than another. (p. 445)

So here is the key idea for resolving the conflict between the considerations

in favor of Comparability and against it: *value is vague*. That is, the relations of *better*, *worse*, *equally good*, and *at least as good* all have borderline cases, in much the same way as *red* or *bald*.

To be clear, the analogy here is between *as good* and *red*; the analogy is *not* between *as good* and *as red*. The color sorites series contains borderline cases of redness, but it need not contain any borderline cases of *as red*. It may well be that each tile is *clearly* at least as red as the next tile in the series. The color sorites can be a clear case of a color gradient, in which redness monotonically decreases while blueness monotonically increases. This can also happen with goodness. Consider the “value sorites” of cases $A, A^-, A^{--}, \dots$. This series begins with a clearly good charity, and ends with a clearly bad charity, while each charity in the series is *clearly better* than the next charity in the sequence, though not by much. This series can contain borderline cases of *good* without any borderline cases of *better* or *as good*.

The idea that the “threshold concept” *good* has borderline cases is not so surprising — it is typical to leave thresholds underspecified in a way that would lead to unclear cases. But the idea we are considering is the more dramatic one that the comparatives *as good*, *better*, and *worse* have borderline cases. These concepts also admit a sorites series, but it is slightly more complicated. Let C be a charity somewhere in the middle of the series $B, B^-, B^{--}, \dots$. We can choose C so that A is clearly at least as good as C , the much worse charity $A^{-- \dots -}$ (with many minus signs) is clearly not as good as C , and in the middle we have cases for which it is unclear how they compare with C . Then $A, A^-, A^{--}, \dots$ is not only a sorites series for *good*, but also — more importantly — it is a sorites series for *as good as C*.

In sections 4.2 and 4.3 I considered two main arguments against Comparability that were specific to the structure of infinite value. First, there was the argument from *invariance* principles, like permutation or reflection invariance: thus a total social welfare order relies on making distinctions beyond what is captured by the pattern of comparative well-being between worlds. Second, there was the argument from *non-constructiveness*: proving the existence of a total value order relies on making infinitely many arbitrary choices. Both of these objections are based, in different ways, on worries about *arbitrariness*. There does not seem to be any *principled* way of defining a value order on infinite worlds. Any way of doing it will make distinctions between pairs of worlds that seem alike in all important respects — and which no mortal could hope to describe or explain. But arbitrariness is exactly what we should expect in cases of indeterminacy.

There is another relation in the neighborhood of *at least as good*: namely, *determinately at least as good*. The analogue of Comparability does not hold for *this* relation.

Determinate Comparability. Either it is determinate that x is at least as good as y , or it is determinate that y is at least as good as x (or possibly both, in which case they are determinately equally good).

While we affirm Comparability, we should reject Determinate Comparability. But note that *determinately at least as good* is not a true comparative. By analogy, Comparability fails for other complex comparative expressions as well — along with other parts of the logic of comparatives. Take *better in some respects*. Program A is better in some respects than program B ; likewise, B is better in some respects than A . But A is not better in any respects than *itself*. So *better in some respects* is not transitive. Something similar is going on with *determinately better* and Comparability. (For further examples and discussion see Dorr, Nebel, and Zuehl 2023, 420.)

More directly to the point, the arguments we have marshalled for Comparability do not make good arguments for Determinate Comparability. Let's start with the first argument we discussed from Dorr, Nebel, and Zuehl (2023). Simplified a bit, it goes like this:

1. Suppose it is not the case that x is at least as good as y .
2. Then x is no better than y .
3. Then y is at least as good as x .

Now consider how an analogous argument for *Determinate Comparability* might go.

4. Suppose it is not the case that x is determinately at least as good as y .
5. Then it is determinate that x is no better than y .
6. Then it is determinate that y is at least as good as x .

But this time the step from 4 to 5 is a *non sequitur*. It gets the scope wrong: in general, from *it is not the case that determinately A* , we cannot conclude *it is determinately not the case that A* . (This reasoning would imply that there are no borderline cases at all!) Similar scope problems arise from other substitutes for step 5 that we might try.

Now consider the argument from Dorr, Nebel, and Zuehl appealing to incomparability-chains. Remember how this goes. We started from the idea that there are cases of incomparability with sweetening: A is better than A^- , but each of these is incomparable with B . Then we chained together a

sequence of such cases. By making programs A and B slightly worse, and then slightly worse again, and so on, we constructed a chain $x_1, x_2, \dots, x_n$ where (we suppose) no element is better than the next, but also x_1 is good, and x_n is not good. By classical logic, some good element x_k is followed by an element x_{k+1} that is not good. Then by *Strong Monotonicity*, x_k is better than x_{k+1} , contradicting our assumption.

Earlier I noted the similarity between this argument and the sorites paradox. Our classical diagnosis of the sorites series applies here as well. Given that we start with something good and end with something not good, it follows by classical logic that something good is immediately followed by something not good. I have argued that vagueness is not a good reason to reject classical logic and resist this kind of inference. Rather, we should observe that this is compatible with the claim that there is no *determinate* cut-off between good cases and those that are not good. Rather, the cut-off cases x_k and x_{k+1} are in the penumbra of borderline cases. No element of the sequence which is *determinately* good is immediately followed by an element that is *determinately* not good.

Now, instead of chains of *incomparability*, our diagnosis is that we have a chain of *borderline betterness*. No element of the chain is *determinately better* than the next element. Accordingly, let's relabel the case an *indeterminacy chain*. There are indeterminacy chains that start good and end not good. But these do not raise any conflict with Strong Monotonicity. We do have some good x_k followed by not-good x_{k+1} , and furthermore x_k is not *determinately better* than x_{k+1} . But this does not conflict with Strong Monotonicity, but rather with this stronger principle:

(16) If x is good and y is not good, then x is determinately better than y .

We should reject (16). If x and y are both borderline cases of being good, then x may also be merely a borderline case of being better than y . Instead of (16), we can accept this principle:

(17) If it is determinate that (x is good and y is not good), then it is determinate that x is better than y .

(Indeed, given the widely accepted principle that determinacy is preserved by logical consequence, (17) follows from the claim that the original Strong Monotonicity claim is determinately true.) But indeterminacy chains do not conflict with (17). A conflict would arise if the chain contained not merely a *cut-off* between good cases and those that are not good, but a *determinate* cut-off. This is exactly what we should deny.

In short, unlike incomparability chains, *indeterminacy* chains are compatible with the determinate truth of Strong Monotonicity. So we can endorse the argument for Comparability without being drawn into endorsing a superficially parallel argument for Determinate Comparability.

Next consider the argument from opaque sweetening. There we found a conflict between cases of incomparability and the principles of Statewise Dominance, Simple Stochastic Equivalence, and Negative Statewise Dominance. How does this argument look when we redescribe the case in terms of indeterminacy? Consider again the prospects Sweetened and Swapped.

	Heads	Tails
Swapped	B	A
Sweetened	A^+	B^+

The sweetened outcome A^+ is not determinately better than B , and B^+ is not determinately better than A . So, however the coin comes up, the outcome of Sweetened will not be determinately better than the outcome of Swapped. But we should not conclude that the prospect Sweetened is not better than Swapped. This analogue of Negative Statewise Dominance does not hold.

Negative Determinate Statewise Dominance. For any prospects X and Y , if it is certain that the outcome of X is not determinately better than the outcome of Y , then the prospect X is not better than Y overall.

Earlier, we argued (following Schoenfield) that the value of prospects is supposed to be a good guide to the value of outcomes, under conditions of uncertainty. If you know that one prospect won't turn out better than another, a good guide to the value of outcomes should reflect this, and say that the prospect is not better on balance, either. But a case where you know the outcome of X is not *determinately* better than the outcome of Y is not like this. Recall, indeterminacy is not an alternative to truth or falsity. If it is indeterminate whether A^+ is better than the outcome of B , then it still might be better! Likewise, B^+ might be better than A . So a guide that says that Sweetened is better overall than Swapped need not be leading us in the wrong direction, to end up with outcomes that are not better after all.

In fact, we can say more. Since A^+ is better than A and B^+ is better than B , Comparability implies that either A^+ is better than B , or else B^+ is better

than A (or perhaps both). (If $A^+ \not\asymp B$, then by Comparability $B \gtrsim A^+$. Since $B^+ > B$ and $A^+ > A$, it then follows that $B^+ > A$.) What's more, if A^+ is *determinately* better than A and B^+ is *determinately* better than B , then it is *determinate* that A^+ is better than B or B^+ is better than A . Even though each disjunct is indeterminate, the disjunction is determinately true. Thus there is no mystery here about why Sweetened is better than Swapped: in fact, we know that Sweetened has at least a 50% chance of turning out strictly better than Swapped. Either Sweetened will turn out better than Swapped on Heads, or it will turn out better on Tails, or perhaps even both. It is just indeterminate which of these cases holds.

4.8 Precisifications and hyperreals

Here is an important feature of (most) classical theories of vagueness: *vagueness admits of precisifications*.²⁴ Consider the sorites series of color tiles again, a spectrum from red to blue. Since blue tiles are not red, classical logic tells us that somewhere in this series there is a cut-off — a red tile followed by a tile that is not red. But it is indeterminate where that cut-off falls. There are some tiles that clearly are *not* at the cut-off. For example, the tile right after the first imperial red tile is certainly still bright red, and so the first tile is determinately *not* the cut-off. It is determinate that none of the first several tiles are the cut-off, and also that none of the last few are. But in between, there are borderline cases which *might* be where the cut-off lies. Intuitively, a *precisification* is a way of arbitrarily drawing a line, imposing a cut-off for *red*. The rule is just that the cut-off must not go in a place that is *determinately* wrong.

We can think of a *precisification* as a way the world does not determinately fail to be. We arbitrarily lay down cut-offs for every vague predicate in some way that is *consistent* with what is clearly true, even if it is not *determined* by any clear facts. It is common to think of precisifications as possible *interpretations*

²⁴To be precise, this holds as long as the *determinately* operator satisfies the logic of a *normal modal operator*. This amounts to saying that determinacy is closed under (multi-premise) logical consequence.

$$\text{if } A, B \vdash C \text{ then } \text{Det } A, \text{Det } B \vdash \text{Det } C.$$

When this holds, Kripke's completeness theorem implies that the theory has a *possible worlds* model. In this context, though, the interpretation of the modality is not *possibility*, but rather *indeterminacy*. Each "possible world" in the model is a *precisification*, a way that the world does not determinately fail to be.

of our language — though this presupposes a linguistic understanding of vagueness which I do not want to take for granted (*à la* K. Fine 1975).²⁵ If we are working within some other framework, we can instead think of precisifications not as precisifying vague linguistic expressions, but rather vague properties and propositions, or vague Fregean concepts. There is no need to arbitrate between these different pictures here.

The key thing is that precisifications are objects that play a certain useful logical role. First, each precisification tells a (classically) complete and consistent story about what the world is like. Second, no precisification's story is determinately false. Third, whatever is true according to every precisification is *determinately* true. In short, precisifications are to determinacy what possible worlds are to necessity.

Precisifications can be very useful for reasoning about vagueness. Essentially, for many argumentative purposes we can simply *ignore* vagueness. We can pick some arbitrary classical precisification, and reason about that. As long as we only make assumptions that are satisfied by every precisification (that is, that are determinately true), then our reasoning will lead to conclusions that also hold for every precisification.²⁶ So our conclusion will also be determinately true.

That much is completely general. Now we turn to our distinctive subject matter: the value of infinite worlds and Pascalian prospects. Here precisifications take a distinctive form. Recall the *counting order* on simple worlds. (I will focus here on the counting order as an example, but the same considerations will also apply to any explicitly definable strengthening of the counting order that we might put forward. Remember, though, that any stronger order will have to violate Permutation Invariance.) The counting order is a partial order: very many pairs of worlds are incomparable with respect to the counting order. So if what I have argued in this chapter is correct, the counting order is not the correct theory of the value of simple worlds — the correct theory satisfies Comparability. But the counting order has *completions*: these are *total* orders that faithfully extend the counting order (and thus still satisfy Pareto and Impartiality). On the other hand, we

²⁵*Supervaluationism* is a name commonly given to this linguistic theory of vagueness. But it is also a very good name for the much more ecumenical point that determinacy can be understood as truth-on-every-precisification.

²⁶This is just how reasoning by the rule of universal generalization works. We say, "Let x be an arbitrary F ." Then we draw the conclusion " x is G ". And finally we conclude "Every F is G ". In the application at hand, F is *precisification*, " x is G " is something of the form " A is true at x ", and then the conclusion is equivalent to " A is determinately true."

know from the Zame–Lauwers theorem that we cannot *prove* the existence of completions except by non-constructive means. In order to find a completion, we have to make infinitely many arbitrary choices. Intuitively, then, if the correct theory of value satisfies Pareto, Impartiality, and Comparability, then we can never know exactly what it is, or even hope to explicitly write down any candidates.

Here is a natural thing to say, then. The counting order is a candidate not for the correct theory of *value*, but rather for the correct theory of *determinate* value. More precisely, this may well be true:

For $\{1, 0\}$ -social outcomes x and y , x is determinately at least as good as y iff $x \succeq^\# y$.

If this is true, then whenever the counting order ranks x over y , we can conclude that x is better than y . But if the counting order is silent on how to compare x and y , we should conclude not that x and y are really incomparable in value, but rather that it is *indeterminate* how they compare.

Furthermore, every total extension of the counting order plays the role of a *precisification* of *at least as good*. In fact, the same proof that there exist total impartial Pareto orders in fact shows something stronger.

Theorem 4.5. Let $\succeq$ be any order of $\{1, 0\}$ -social outcomes that satisfies Pareto and Impartiality. Then for any outcomes x and y , $x \succeq y$ iff $x \succeq^+ y$ for every total order $\succeq^+$ which faithfully extends $\succeq$.

So here is our way forward. Given Comparability, a true complete theory of value is unattainable for us. We cannot ever hope to find it or write it down — it may as well be in heaven, or beyond the sea. But we can, perhaps, attain the more modest goal of discovering the correct theory of *determinate* value. (Though, of course, there is no guarantee of our success in this project, either.) For simple enough worlds, this might even just be the counting order. This theory will have infinitely many precisifications, each of which looks like a completely arbitrary choice within the bounds set by determinacy. Then we can say that the correct theory of value is described by one of these arbitrary precisifications, but it is indeterminate which one, and it is hopeless for us to try to choose among them. We then have two equally valid ways of reasoning about value. We can reason about the *partial* order of determinate betterness directly. Or we can reason about an *arbitrary total order* that precisifies it. Either way, we will come to determinately correct conclusions.

So far I have talked about precisifications of an order of goodness. Something similar applies to quantitative *representations* of such orders. For instance, recall that the counting order has a quantitative representation using what I called *quasi-real numbers*. These are an algebraic structure similar to the real numbers, but unlike the reals they are only partially ordered. We can do something very similar for the counting order's *completions*: each of these arbitrary and inscrutable total orders also admits a quantitative representation using a *different* quasi-real structure — one which is *totally* ordered. In fact, the totally ordered quasi-real structures that arise this way are a version of (or variation on) *hyperreal* structures known from non-standard analysis.

So (as I alluded in section 3.5) hyperreals do find a role in representing value. We can again think of the quasi-reals as representing *determinate* value — what is determinately as good as what. Each quasi-real structure can in turn be “precisified” by an infinite family of hyperreal structures. We can understand these as representing the possible *precisifications* of value — what might be better than what, for all the determinate facts tell us.²⁷

Nonetheless, even though hyperreals may be more familiar in a way (for a certain logically sophisticated audience), I think it is useful to linger on partially order quasi-reals, rather than passing immediately to their more exotic completions. Take the quasi-real structure representing the counting order. This is a number system (in a generalized sense) including ordinary finite numbers as well as infinite and infinitesimal numbers, with which we can do most ordinary arithmetic. But unlike the hyperreals, we can define this structure completely constructively, and examine its properties directly.

²⁷In section 3.5 and section B.5 I described how a *filter* can represent a “limiting concept” — a way of making sense of “all sufficiently large finite sets”. Using a filter, we can construct an order for infinite worlds, and also a corresponding quasi-real structure that represents this order. That quasi-real structure is a *filter power* of the real numbers. In more technical terms, the relationship I am describing here between quasi-reals and hyperreals is an application of the *ultrafilter lemma*: every filter can be extended to an *ultrafilter* — a maximal non-trivial filter. The order of worlds that arises from an ultrafilter is a total order, and the quasi-real structure that arises from an ultrafilter is a hyperreal structure — a so-called *ultrapower* of the reals. The subset relation between the filter and the ultrafilter gives rise to a relationship between the quasi-real structure and the hyperreal structure — a so-called *homomorphism*, intuitively telling us which hyperreal represents each quasi-real. Moreover, a version of Łoś’s theorem tells us that (in a sense to be made precise), for a large family of propositions, what is true for a quasi-real structure is just what is true for each of its hyperreal precisifications. (But not *all* propositions. For instance, each precisification is totally ordered, but the quasi-real structure is not. See section B.5 for details.)

Hyperreals gain us a little bit of convenience for doing arithmetic²⁸ at the price of being alien and inscrutable in their details.

But I think the biggest advantage in focusing on partially ordered structures is a change of mindset. When we invoke hyperreals, it is tempting to treat this as magic — out of a black box comes a wonderful mathematical structure, which we shouldn't inspect too closely. When we focus on constructible quasi-real structures, it is much easier to see the various choices involved in their construction — choices which are not at all arbitrary, and which potentially make a huge difference to important questions of value. (One context where this will become an issue is when we return to representing probability and value together in chapter 5.)

4.9 Conclusion

At the outset, we were pulled in different directions. There seemed to be strong arguments for incomparability between the values of different worlds, and also strong arguments against incomparability. I have defended a resolution of this tension that appeals to vagueness (following many others). In short, *as good* is a total order, while *determinately as good* is not. The upshot is that some of the value questions that we are investigating do not have any determinate answers. In the end, we must be satisfied with less than the complete truth of the matter. But indeterminacy does not stand in the way of exploring these questions using all the resources of classical logic, nor the representational resources of quantitative structures. We can aspire to precisely describe the relationship between those questions that have determinate answers, those that do not, and the infinitely many arbitrary and inscrutable ways of making those indeterminate matters precise.

²⁸Specifically, a total ordering and division.

Chapter 5

Infinitesimal Probabilities

“Where could you ever find a number so big?” scoffed the Humbug.

“In the same place they have the smallest number there is,” he answered helpfully.

Norton Juster, *The Phantom Tollbooth*

5.1 Probability and weighing value

An angel is going to flip a fair coin over and over, forever. It might come up heads on the first flip, and it might also come up heads on the second flip, and, indeed, it might keep coming up heads every time. How likely is this?

Orthodox probability theory says the probability is exactly *zero*. The argument is simple: the probability of the *first* flip coming up heads is $1/2$. The probability of the first *two* flips coming up heads is $1/4$. The probability of the first *three* flips coming up heads is $1/8$. And so on. For any finite number n , the probability of *every* coin coming up heads can't be greater than the probability of the first n flips coming up heads. So the probability must be less than $1/2$, and less than $1/4$, and less than $1/8$, and so on. Since every standard probability is a real number between zero and one, the only option left is zero. Call this view *Standardism*.

Standardism. All probabilities are standard real numbers. Thus the probability of all heads is zero.

That isn't to say that the angel *can't* get all heads. Indeed, the same argument applies to *any* infinite sequence of heads or tails. So each of those infinite sequences gets probability zero — and yet it is certain that some sequence or other will occur. The term of art is that all heads is *almost sure* not to happen — it has probability one. But that isn't the same thing as being genuinely *sure*. Indeed, we can be sure that *something* will happen that is almost sure not to happen.

Many people have found this to be absurd. If getting all heads *might* happen, how can it have the exact same probability as a contradiction? There is another possibility: perhaps the probability is something less than $1/2$, and less than $1/4$, and less than $1/8$, and so on — and yet still greater than zero. No standard real number fits the bill, so we must say that not all probabilities are standard real numbers. Instead, a probability can be an *infinitesimal* number, in some *non-standard* number system. (A popular choice is the *hyperreal* numbers, but these are not the only option.)

Non-Standardism. The probability of all heads is a positive infinitesimal.

Until recently, I was a contented Standardist. Non-Standardism seemed hard, and the arguments for it did not seem convincing enough to draw me out of the fold of statistical orthodoxy. The argument in this chapter changed my mind about this.

The argument for this conclusion is based on moral considerations. This is an unusual kind of argument, but it isn't shocking. We figure out what probabilities must be by examining the normative roles they play. There are standard arguments for various conclusions about the structure of probabilities based on the role they play in guiding our practical choices, as well as other arguments based on their role in aiming at the truth or responding to evidence. But besides these roles, probabilities have a key role to play in our theory of *value*. Recall this example.

A tsunami is approaching the two islands Sure and Unsure. You are driving a speedboat, and you only have time to rescue the people from one of them. There is one person on Sure. You don't know how many people there are on Unsure: the probability is $1/2$ that there are three people there, and the probability is $1/2$ that the island is empty.

There are two options here, and it is uncertain which of them will turn out better. Taking the different possibilities into account, which option is

better on balance? To answer this, it is extremely useful to be able to put quantitative weights — *probabilities* — on different epistemically possible ways the world might be. Probabilities give us the resources to weigh up this kind of trade-off. So our theory of probability should fit together with a theory of weighing up risky trade-offs of value.

The orthodox theory is to weigh the value of an outcome directly in proportion to its probability — in other words, the value of an uncertain outcome is its *expected value*. In this case, supposing the value of saving three lives is three times the value of saving one, that theory says that heading for Unsure Island is the better prospect. (This is the *risk-neutral* theory advocated by Pascal and the *Port-Royal Logic*, discussed in chapters 1 and 2.) Expected value is not the only way we might use probabilities to weigh uncertain value. We might instead use expected *utility*, where utility is some other increasing function of the number of lives. Or we might stray further from orthodoxy, and use risk-weighted expected utility (Buchak 2013), or discounted expected utility (Smith 2014; Monton 2019), or stochastic dominance (Tarsney 2025), or weighted-linear utility (Bottomley and Williamson 2024), or still other things. (I introduced some of these alternatives in section 1.2.) All of these different theories take a prospect's probabilities and values of outcomes as an input, apply some rule or formula, and spit out the value of the prospect. So they all share this common idea: *probabilities determine how values are weighed*. We can state this more officially.

Stochasticism. Prospects that have the same probability distribution over outcomes are equally good on balance.

In this way, probabilities are connected to the value of prospects. This kind of connection means that considerations about value can put constraints on our theory of probability. If one prospect is better than another, then the two prospects must have *different* probabilities of outcomes.

This idea is the basis for a familiar argument against Standardism: call it the *Simple Argument* (Skyrms 1980; Williamson 2007, 179). Let's say the angel offers you two options.

Nothing. With certainty, you get nothing.

All Heads. If the coin comes up heads every time, you save a million lives. Otherwise you get nothing.

According to Standardism, the probability of winning the bet on All Heads is just the same as the probability of saving lives by choosing Nothing: exactly zero. So the two options have the very same probability for every

outcome: saving no one has probability one, and getting any other result has probability zero.¹ Stochasticism, then, says that the two options are equally good.

But this conclusion seems wrong. There is no way that All Heads will turn out *worse* than Nothing. And if you take the All Heads bet, you *might* save a million lives, while if you take Nothing you certainly won't. All Heads is sure to turn out at least as well as Nothing, and it might turn out much better. So it seems like on balance, All Heads is strictly better than Nothing.

The term of art here is that All Heads *dominates* Nothing. Here is the general principle that underwrites this reasoning.

Strong Statewise Dominance. For any prospects X and Y , if X is certain to turn out at least as well as Y , and X might turn out strictly better than Y , then X is strictly better than Y on balance.

In this case we say that X **strongly statewise dominates** Y . Our example demonstrates the following fact:

Proposition 5.1. Standardism, Stochasticism, and Strong Statewise Dominance are jointly inconsistent.

As I said, this is an old argument. I am now inclined to think that it is sound. But in the past, like many other philosophers, I did not find it convincing. We can respond to the Simple Argument by giving up one of these two decision-theoretic principles (Stochasticism or Strong Statewise Dominance) *by the letter* while still keeping its spirit. There are two different ways to do this.

One response is to relax dominance. While it is *possible* for All Heads to turn out better than Nothing, there is not *much* of a chance. Even according to the non-standardist, the probability of this eventuality is infinitesimal. So while it is intuitive to say that All Heads is better than Nothing, this does not seem terribly important — the difference in value between the two is at most infinitesimal. So we may be willing to countenance this kind of *very small* violation of Dominance.

¹A technicality: two *distinct probability distributions* can assign the same probability to each outcome, individually. Classical continuous probability distributions can assign different probabilities to a *set* of outcomes, without assigning different probabilities to any particular outcome in that set. But in this case the full distributions over outcomes really are the same in each case: any set of outcomes that includes getting nothing has probability one, and any set of outcomes that does not include getting nothing has probability zero.

Indeed, some philosophers are willing to countenance even greater failures. As I discussed in section 1.2, there is a long line of people who have advocated *probabilistic discounting*, so as to treat sufficiently small *finite* probabilities as negligible. Consider another gamble.

Many Heads. If the coin comes up heads a hundred times in a row, you save a million lives. Otherwise, you save no one.

Some discounters are willing to say that, even though Many Heads *might* turn out better than Nothing, even so it is not better on balance. The event in which it turns out better than Nothing is “morally impossible”. But surely All Heads is no better than Many Heads — so this kind of discounter will also treat All Heads as no better than Nothing. (This result is not forced on the discounter, though. Some discounters instead go with the second familiar reply to the Simple Argument, discussed below: see Monton (2019), p. 20; Beckstead and Thomas (2024), p. 9.)

Retreating from Strong Statewise Dominance does not require retreating from dominance altogether. There are weaker forms of dominance that are still compatible with Standardism and Stochasticism. Here is one:

ϵ -Strong Statewise Dominance. For some real number $\epsilon \geq 0$, if X is certain to turn out at least as well as Y , and the probability that X turns out strictly better than Y is greater than ϵ , then X is better than Y on balance.

If we do not discount small *finite* probabilities, but only what the non-Standardist would consider to be *infinitesimal* probabilities, then we can even endorse the version where $\epsilon = 0$:

Non-Zero Strong Statewise Dominance. If X is certain to turn out at least as well as Y , and X has positive probability of turning out strictly better than Y , then X is better than Y on balance.

An even weaker dominance principle that the Standardist can endorse is this:

Strict Statewise Dominance. If A is certain to turn out strictly better than B , then A is better than B on balance.

This principle is harder to deny than Strong Statewise Dominance. As I have said before (sections 1.3.3 and 4.5), when we are comparing uncertain prospects, what we really care about is what will *in fact* turn out better. We don’t always know how to do this: as in the tsunami case, sometimes we are uncertain which prospect will turn out better. In these cases we must trade

off the possibilities of a prospect doing better or worse, in order to come up with an on-balance evaluation. Comparing the value of prospects can be a useful guide toward achieving the best outcome. But if it is *certain* that *A* will really turn out better than *B*, then it is already clear where this guide is supposed to point us. In this case, the trade-off should be easy.

The second familiar reply to the Simple Argument instead keeps Strong State-wise Dominance while relaxing Stochasticism. K. Easwaran (2014b) points out that probability theory using standard real numbers has more resources than just the numerical probability that is assigned to each proposition.

Probabilism uses a set of possibilities to represent propositions, and real numbers to represent credences. Because the real numbers are not fine grained enough to capture all the distinctions in these doxastic states, Skyrms and Lewis argue that we should use hyperreals instead. They focus on the numerical aspect of a probabilistic representation and seek to expand it so that it can represent all the relevant distinctions. But as I presented it above, probabilism uses a *set* together with some numbers, and a *conditional* credence function as well as an unconditional one. Both of these tools describe important features of credence that shouldn't be overlooked. (K. Easwaran 2014b, 3–4)

He goes on:

In cases where the expected value of a bet is exactly the same as the status quo, some non-numerical feature may serve as a tiebreaker. (2014b, 14)

In particular, *dominance* is one of these non-numerical features. (I will set conditional probabilities aside for now, returning to them in section 5.6.) Consider the bet All Heads: this has exactly the same (standard) expected value as the status quo option Nothing — namely zero. Even so, there is a possible state in which All Heads turns out better than Nothing. This fact is not captured by standard numerical probabilities, but that does not mean we must ignore it. It can still come in as a tie-breaker when standard expectations do not discriminate between prospects. (This idea is also endorsed by Hájek 2014; Monton 2019, 20.)

According to this picture, a prospect's standard probability distribution over outcomes is an imperfect guide to its value, but it is still a pretty good guide. Consider an implementation of this idea. Outcomes are assigned finite real numbers as utilities. A prospect's probability distribution over outcomes,

together with the utility function, determines a prospect's *expected utility* (when this is well-defined). Whenever two prospects X and Y have different expected utilities, we do not need to look any further: we can say that if X 's expected utility is *strictly greater* than Y 's, then X is strictly better than Y , end of story. And (we may think) this case is typical. It is only in the edge case where the expected utilities come out *exactly the same* that we need to consider non-numerical features like dominance.

Moreover, there is reason for optimism that we can extend this idea beyond the simple finite case. As we discussed (in chapters 2 and 3), even when we consider *infinite* outcomes, whose values cannot be represented by finite real numbers, we can still represent the value of outcomes using other mathematical systems, such as vector utilities. Then we can (in many cases) still calculate expected values for standard probability distributions over outcomes, and so we may hope that this Standardist approach can be taken even further. This approach combines standard probabilities with non-standard utilities. We may well hope that these tools are pretty much good enough for comparing prospects, except for the edge case where two prospects are *almost* exactly the same, differing on a set with probability zero. At this point tools like dominance can be brought in as a fallback.

If X strongly statewise dominates Y , it is still possible for the expected utility to come out tied, as in the example of Nothing and All Heads. But again, we can consider more demanding dominance concepts. If X *strictly* statewise dominates Y , then if both options have well-defined standard expected utilities, it follows that the expected utility of X is strictly greater than that of Y , and so X is strictly better than Y . Furthermore, this relationship is captured by their standard stochastic profiles: if some other prospect X' is stochastically equivalent to X , then it too is strictly better than Y . So even if we reject Stochasticism in edge cases, letting in dominance as a tie-breaker, we would still hope to be able to affirm this principle:²

Strict Stochastic Dominance. If X is stochastically equivalent to a prospect that strictly dominates Y , then X is strictly better than Y .

Strict Stochastic Dominance does not entail Stochasticism, so the defender of the tie-breaking strategy can also hope to maintain this principle. It is not threatened by cases like the choice between All Heads and Nothing. Moreover, the defender of the first reply to the Simple Argument — giving up

²There are subtleties about this formulation: see section E.4 and J. S. Russell (forthcoming) — as well as the rest of this chapter.

Strong Statewise Dominance while keeping Stochasticism — will also accept Strict Stochastic Dominance. (This is because they accept Strict *Statewise* Dominance, and Strict Statewise Dominance and Stochasticism together imply Strict Stochastic Dominance.)

We have surveyed two strategies for responding to the Simple Argument. In either case, we give up the letter of Strong Statewise Dominance or Stochasticism, while retaining the spirit. Accordingly, lots of people — including me — haven't been convinced by the Simple Argument against Standardism.

But there is another.

5.2 Return of the Everything Bagel

What I said in the previous section about weighing value might apply to any kind of value — what's good for you, or what's good for the philosophy department, what's good for the purpose of humiliating your enemies, or whatever. Indeed, in the decision-theoretic context these principles come from, people are often talking not about value but rather *preference*, and they often suppose (following Hume) that you can rationally prefer whatever outcomes you want — the destruction of the whole world rather than the pricking of your finger, say.

The argument that follows is not like that: it turns on substantive claims about the value of outcomes. It's not about any old kind of value: it's specifically about the kind of value we are discussing in this book, the value of individual well-being, considered impartially. As always, I'm not assuming that this is the only kind of value that exists, and I'm not making the *impartial consequentialist* assumption that we ought to do whatever best promotes this kind of impartial value. This argument, like the others in this book, just assumes that this is a genuine kind of value that can be weighed up in a context of risk.

Recall once again our toy model from section 3.2: we imagine *simple worlds* containing one person for each natural number, each of whom is either *happy* or *sad*. We suppose that these outcomes can be compared as better or worse overall than one another, with respect to how well-off these individuals are. Recall as well the two principles I have affirmed about these comparisons. First, making someone better off without any countervailing harm makes things better overall.

Pareto. For simple worlds u and v , if some individual is strictly better off in

u than in v , and every individual is at least as well off in u as in v , then u is strictly better than v overall.

Second, each person's happiness is just as valuable as any other person's happiness: making one sad person happy is equally good as making any other sad person happy.

Impartiality. For any individuals i and j and simple worlds v and w , if everyone besides i and j are equally well off in v and w , and in v , i is happy and j is sad, while in w , i is sad and j is happy, then v and w are equally good.

As I argued in chapter 3, we can consistently affirm these principles. In so doing, we escape from "Everything Bagel" nihilism: the idea that in infinite worlds, no finite differences make any difference to the overall value of the world.

Yet the Everything Bagel's threat still looms. In chapter 3, I discussed the value of *outcomes* — totally specific ways the world might be with respect to individuals' well-being. But if we are to strive to make the world a better place, we must do this without knowing exactly what the world is like. The world is not just vast, but also highly uncertain. So if we are going to escape the Everything Bagel, we don't just need a value ordering for specific outcomes, but also for uncertain *prospects*, which might turn out lots of different ways.

Accordingly, we need to extend our toy model to represent uncertainty. Let's consider a very simple choice between two uncertain prospects, which I'll call *Happy* and *Sad* (repurposing these labels). (See figure 5.1.) As before, there is one individual, Indy, whose happiness is up to us. In the prospect *Happy*, Indy is happy for sure; in *Sad*, he is sad for sure. This time, the fates of all of the other infinitely many people in the universe are determined at random. For each individual besides Indy, whether they are happy or sad is determined by a fair coin flip. All of these coin flips are independent of one another. They are the same coin flips for both of the two prospects: it is certain that *Happy* and *Sad* will turn out exactly the same way for everyone but Indy.³

³There's an interesting philosophical debate about chance and counterfactuals that bears on this, called the puzzle of "Morgenbesser's coin" (Slote 1978). There is a bunch of randomness in the world that is independent of our choice. Suppose you choose *Happy*. We might think that, if you had chosen *Sad* instead, all of that randomness would still have gone the *same* way as it actually did. Alternatively, we might think that in the counterfactual

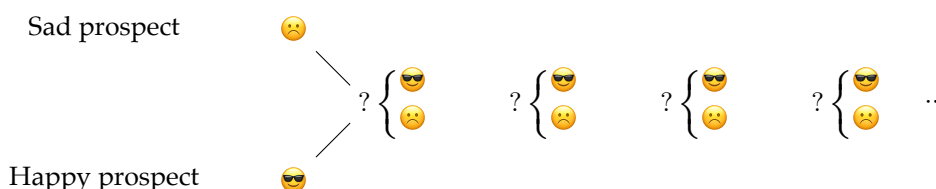


Figure 5.1: Making a difference in an infinite random world.

This is a toy model. It's not realistic. But it is a stylized version of a situation we may plausibly really be in. How well things will go in one part of the cosmos is pretty much independent of how well things will go in another super distant part of the cosmos. It is a not at all remote possibility that there are infinitely many such parts. And even though we can make a difference to more than one person in the world, it is plausible that our "sphere of influence" includes only finitely many people whose happiness is within our power to affect.

The choice between Happy and Sad seems easy — you can make Indy better off, while being sure that you won't make things worse in any respect. Surely this is a better prospect overall!

This conclusion is obvious, but we can also argue for it explicitly from principles we have put forward: Pareto and Strict Statewise Dominance. If s is any possible way the coins may have landed — for short, a *possible state* — let $\text{Happy}(s)$ and $\text{Sad}(s)$ be the simple worlds that would result from the prospects Happy and Sad, respectively. We know that for each possible state s , in $\text{Happy}(s)$ everyone is at least as well off as in $\text{Sad}(s)$, and Indy is strictly better off. So *Pareto* tells us that $\text{Happy}(s)$ is strictly better than $\text{Sad}(s)$. Since this holds whatever s may be, *Strict Statewise Dominance* tells us that the prospect Happy is strictly better than the prospect Sad.

Seems easy enough. But not so fast — we have new trouble. Given Standardism, we can also argue for an opposing conclusion, that Happy and Sad are *equally good*. Here is the key technical fact. (See proposition E.39 in

situation, all the randomness would get "rerolled" — so maybe how the coin flips would go if you choose Happy is *independent* of how they would go if you choose Sad. This is an interesting debate, but we don't have to get into it here. For our toy model, let's just stipulate that somehow you know for sure that whatever you do for Indy, the happiness coin flips would come up exactly the same for everyone else besides Indy. Of course, you still don't know which way that is. You can imagine that the coins have already been flipped far away, so their results are settled, but concealed from you.

section E.5, as well as proposition E.38 in section E.4.)

Proposition 5.2. Happy and Sad are **stochastically equivalent up to finite permutations**. That is to say, Happy has the same standard probability distribution over outcomes as another prospect X such that X and Sad are certain to have outcomes related by a finite permutation.

This is bad. We are once again in serious danger of Everything Bagelism. In chapter 3, we responded to arguments that changing one life makes no difference to the overall value of a world. A finite change in happiness might make no difference to a world's overall qualitative pattern of well-being; we concluded that these qualitative patterns do not contain enough information to discriminate between better and worse worlds. But now we find that changing one life does not change anything about the *standard probabilities* of how good the world might be overall. Analogously, we should conclude that these standard probabilities don't contain enough information to discriminate between better and worse prospects.

The proof of proposition 5.2 is a bit technical, and it is relegated to section E.5. But the main idea is not too hard to get across. For each of the prospects Happy and Sad, we can consider corresponding "partial" distributions, which tell us the probabilities for how things will go for just the first n people. For example, take the Sad probabilities (figure 5.2).

- Considering just the first person (Indy), it is certain that *zero* people will be happy.
- Out of the first two people, the probability is $1/2$ that zero people will be happy, and $1/2$ that one person will be happy.
- Out of the first three people, the probabilities are $1/4$, $2/4$, $1/4$.
- Out of the first four people, they are $1/8$, $3/8$, $3/8$, $1/8$.

And so on. In general, we get what is called a *Bernoulli distribution*.

Now consider how things go for the Happy prospect (figure 5.3). This just shifts everything up by one.

- It is certain that the first person will be happy.
- Out of the first two people, the probability is $1/2$ that one person will be happy, and $1/2$ that two people will be happy.
- Out of the first three people, the probabilities are $1/4$, $2/4$, $1/4$, but shifted up one.
- Out of the first four people, they are $1/8$, $3/8$, $3/8$, $1/8$.

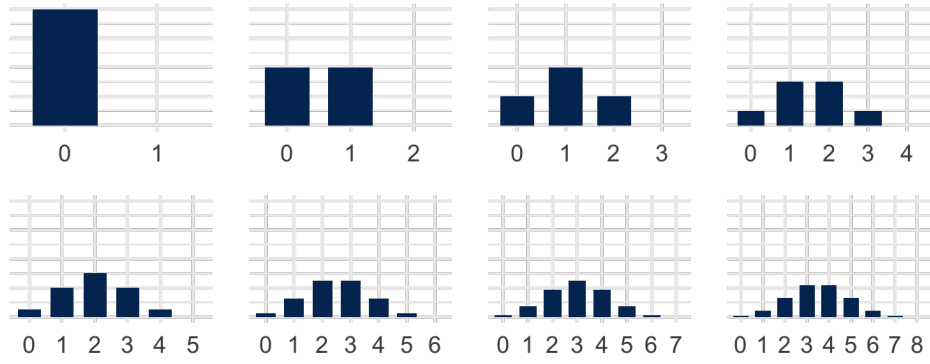


Figure 5.2: Partial distributions for Sad

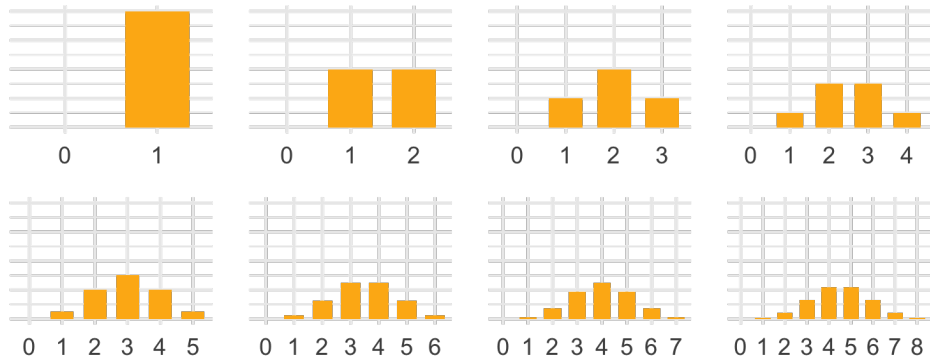


Figure 5.3: Partial distributions for Happy

Here's the key thing: as n gets larger, these probability distributions get wider and wider, and more and more *similar*. The difference you make to the first person matters less and less for the overall shape of the distribution. The precise notion of similarity at play here is called the **total variation distance** between measures. Intuitively, this measures the total amount that either probability distribution sticks up over the other (the total area of the blue and orange regions in figure 5.4). It can be shown that the partial distributions for Happy and Sad converge with respect to this total variation distance.

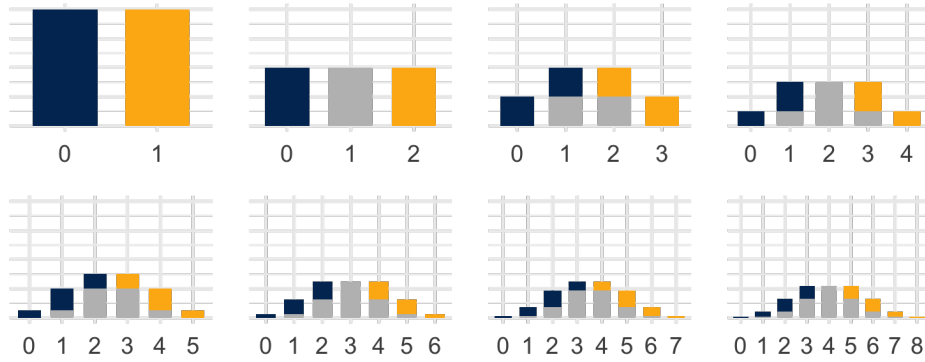


Figure 5.4: The two sequences of distributions get closer together as n increases.

Next, we can invoke a technical theorem: when partial distributions squish together like this, that guarantees stochastic equivalence up to finite permutations (theorem E.42 in section E.5). Intuitively, the overlapping part of two partial distributions represents a piece of Happy and a piece of Sad that are exactly alike, except that Indy's happiness has been exchanged for the happiness of some other person among the first n . As the overlapping parts get larger and larger, these pieces get closer and closer to Happy and Sad themselves. In the limit, there is nothing left — except a residue with probability zero. (We will return to this.) That proves proposition 5.2.

What does all of this mean? Proposition 5.2 tells us that there exists a prospect X that has precisely the same standard probability distribution over simple outcomes as Happy, and which is certain to have an outcome that is a finite permutation of the Sad outcome. By *Impartiality*, this means that X is certain to turn out just as well as Sad. But we already argued — by Pareto — that Sad is certain to turn out *strictly worse* than Happy. So the same goes for X .

In short, X is both *stochastically equivalent* to Happy and also *strictly dominated* by Happy. Then *Strict Stochastic Dominance* absurdly implies that Happy is strictly worse than Happy!

Proposition 5.3. Standardism is inconsistent with the combination of Impartiality, Pareto, and Strict Stochastic Dominance.

This fact is bad news for Standardism. Standardism and Strict Stochastic Dominance together conflict with the most basic things we can say about impartial moral value in an infinite world — which, again, may well be the kind of world we really live in.

There is a resemblance between this argument and the Simple Argument I discussed in section 5.1; but I think this new argument is much more powerful. Remember, the simple argument turned on applying *Strong Statewise Dominance* in a case where one option has a *zero or infinitesimal* chance of turning out better than another. We could escape the Simple Argument argument by relaxing either Stochasticism or Dominance just a *little*. We could say that for dominance to make an option better, it has to at least have a *significant probability* of turning out better. Or we could say that Stochasticism has to be amplified with a tie-breaking principle, while still providing a good guide in most cases.

Neither of these replies work against the new argument. Part of the appeal of those responses was that we could — apparently — maintain weaker principles like Strict Stochastic Dominance. But it turns out that Standardism conflicts with that principle, too! The Happy prospect doesn't just have a zero or infinitesimal chance of turning out better than Sad — it is *certain* to turn out better than Sad.

Furthermore, the argument that Happy and Sad are stochastically equivalent massively generalizes. Roughly, any prospect that incorporates infinite randomness strictly stochastically dominates any prospect that is only finitely different from it (see theorem E.46 in section E.5 for a more precise statement and proof). But this is plausibly a decent model for *any* realistic choices we face in an infinite world. For prospects like these, standard probability distributions aren't anywhere close to good enough: they give us no practical guidance at all. Basically *everything* comes out tied by the criterion of standard stochastic equivalence — and thus also by the criterion of standard expected value, whenever this is well-defined. The tie-breaking principle, whatever it is, will have to do *all* of the work.

One defeatist response is to finally surrender to the Everything Bagel, and give up on the idea that we can make a difference to the overall value of the world. The balance of cosmic value does not give us a reason to rescue a drowning child — though maybe we can still find reasons elsewhere. I have sometimes found this response tempting (and I will return to it in chapter 11). But there is a different response available: the problem is rather with standard probability theory. Let's explore an alternative.

5.3 Tiny but powerful

Let's look again at the equivalence we found between Happy and Sad — but this time, let's put on our non-standardist glasses.

Here's a simple question: how likely is it that *everyone* will be happy? If you choose Sad, then the answer is zero. In our toy model, it is completely certain that choosing Sad makes Indy sad. So at least one person will be sad no matter how all the coin flips go.

What about if you choose Happy? In this case, Indy is sure to be happy. If *all* of the coin flips come up heads, then everyone else will be happy as well. For the Standardist, that possibility also has probability zero — so the probability that everyone will be happy is the same either way, whether you choose Happy or Sad. But for the Non-Standardist, the probability of all heads is a *positive infinitesimal*. So choosing Happy makes it *more likely* that everyone will be happy. In the same way, it makes it infinitesimally *less* likely that everyone will be sad. In general, choosing Happy rather than Sad shifts an infinitesimal bit of probability up from each possible outcome to an outcome that is better for one person. All of those infinitely many infinitesimal changes end up making the overall distribution better by a finite amount — indeed, by the happiness of exactly one person.

That's the basic idea. To make it really convincing, though, we need to come up with a model. We don't just want an ad hoc answer to the particular question of Happy versus Sad. We want a general theory of non-standard probabilities and how these are used to represent uncertain prospects, and a general method for using them to make value comparisons — along the same lines as standard expected value theory for simple finite prospects. In the rest of this section I'll sketch the outlines of a model like this. (This combines and extends ideas from sections 2.3, 3.5 and 4.8.) (I am continuing to build on technical work by Pivato (2014). For further details see appendix F.)

We will use the same technical tools that we introduced for representing infinite value: what I have called *quasi-real numbers* (sections 3.5 and 4.8). These are in the same family as the more familiar hyperreal numbers, but more flexible in certain ways.

I'll continue to focus on the case of coin flips. In this context, a *state* is any infinite sequence of Heads or Tails. Let Ω be the set of all such states; I'll call this the *coin flip space*. There are some sets of states where we know how probability is supposed to go. For any number i , indexing a coin flip, let $H(i)$ be the set of states in which the i th coin comes up Heads. Then the probability of $H(i)$ should be exactly $1/2$. Similarly, for any finitely many distinct $i_1, \dots, i_n$, the probability of $H(i_1, \dots, i_n) := H(i_1) \cap \dots \cap H(i_n)$ should be $1/2^n$. This is how it works for standard real-valued probabilities, and it is also how it should work for any non-standard probabilities we come up with.

The first important point is that there exists a *regular uniform quasi-probability measure* $\mathbb{P}^*$ on Ω that satisfies these coin flip constraints. (See section F.4.) This is a function that assigns a quasi-real number to each set of states, and which satisfies the analogues of the usual probability axioms for quasi-real numbers.⁴ *Regularity* means that each individual state receives positive (infinitesimal) probability. *Uniformity* means that any two states receive the *same* probability.

Something to emphasize here is that the quasi-real structure that represents probability is *different* from the kind of quasi-real structure we introduced in section 3.5 to represent the value of infinite worlds. Here is the basic idea. A *filter* is a mathematical object that represents a way of understanding what it means for something to hold “in the limit”. For any filter $\mathcal{F}$, we can construct what is called a *filter power* of the real numbers, which I'll notate $\mathbb{R}^{\mathcal{F}}$. These filter powers are what I have been calling quasi-real structures. In section 3.5, we constructed a quasi-real structure that represents the value of infinite worlds, using a particular filter. That filter “lives on” the space of finite sets of individuals. That quasi-real structure is tailored specifically to the job that we made it for: representing the value of well-being for an infinite population.

Now, though, we have a different job to do: representing *probability*. We can use the same techniques, but in a different way — using a different filter. This

⁴One point worth noting, though, is that this function is *finitely* additive, but not *countably* additive in the standard sense.

time, it is a filter $\mathcal{G}$ that “lives on” the space of finite sets of *states*. Furthermore, the coin flip space has natural structure that the set of individuals does not. Consider the set of states $H(1)$ (heads on the first flip) and the set of states $H(2)$ (heads on the second flip). Each of these sets contains infinitely many states that the other does not. (There are infinitely many sequences that begin $H, T, \dots$, and there are infinitely many other sequences that begin $T, H, \dots$.) In the case of individuals, such cases raised hard questions, but in the case of probabilities, it is clear that we want to say that $H(1)$ and $H(2)$ are equally probable. We need to judiciously choose the filter $\mathcal{G}$ to ensure this. We can choose $\mathcal{G}$ so that it has certain *symmetry* properties — the *coin flip symmetries*. This will in turn ensure that the quasi-real probabilities that $\mathcal{G}$ generates also respect these symmetries: for example, $H(1)$, $H(2)$, and $\neg H(1)$ each have the same probability. (See section F.4 for details.) So we now have two different quasi-real structures: the old structure $\mathbb{R}^{\mathcal{F}}$ for representing the value of worlds, and a new structure $\mathbb{R}^{\mathcal{G}}$ for representing probabilities.

We aren’t done yet. The next thing we need is a generalization of the *expectation* operator. Let’s start by recalling how expectations work in humdrum finite cases. A *real random variable* is a function $X : \Omega \rightarrow \mathbb{R}$ from states to real numbers. Say that Ω is a *finite* set of states equipped with a probability distribution $\mathbb{P}$. In that case, the random variable’s *expectation* $\mathbb{E}X$ is a real number, which is a weighted average of the values that X takes:

$$\mathbb{E}X = \mathbb{P}(s_1) \cdot X(s_1) + \dots + \mathbb{P}(s_n) \cdot X(s_n) \quad \text{where } \Omega = \{s_1, s_2, \dots, s_n\}.$$

As we discussed in section 2.3, we can generalize expectations from *real-valued* random variables to *vector-valued* random variables. Recall that a *vector space* is a mathematical structure which has elements that can be added together and rescaled. If $X : \Omega \rightarrow \mathcal{U}$ is a function from states to elements of some vector space, we can define the expectation of X using exactly the same formula as we use for real-valued functions. Vectors are just the right general kind of things for this formula to make sense. The quasi-real structure $\mathbb{R}^{\mathcal{F}}$ that we introduced to represent the value of worlds is one example of a vector space. So we can straightforwardly make sense of expectations for finite probability distributions over infinite values.

Things get trickier when the state space is *infinite*, like the coin flip space. Here is one difficulty: we know from the St. Petersburg paradox that a random variable that only has finite values can, even so, have an infinite expectation. The generalized expectation of a real-valued random variable will not generally be a real number. Rather, it will be a *quasi-real* number, in some extended structure.

This generalizes to *infinite* values as well. Consider this version of the St. Petersburg lottery, on simple worlds: if the first heads is on the n th flip, then let the first 2^n people be happy, and everyone else sad. If we assume expected value theory for finite numbers of happy people, as well as Statewise Dominance, then this lottery must be better than any simple world that has only finitely many happy people. But — by dominance again — it is still not as good as *any* world with *infinitely* many happy people. We constructed a quasi-real space $\mathbb{R}^{\mathcal{F}}$ to represent the values of simple worlds, which includes both finite and infinite quantities. But the value of the St. Petersburg prospect is *greater* than any of these finite quantities in $\mathbb{R}^{\mathcal{F}}$, and *less* than any of the infinite quantities in $\mathbb{R}^{\mathcal{F}}$. Even though we have already constructed a space of infinite values representing simple worlds, it turns out that we need *new* infinite values to represent the value of *risky prospects* over simple worlds. (Compare J. S. Russell and Isaacs 2021, sec. 1.)

The upshot is that, if $\mathbb{R}^{\mathcal{F}}$ is the space of quantities that represent the value of infinite worlds, and $\mathbb{R}^{\mathcal{G}}$ is the space of quantities that represent probabilities, then the *expectation* of an $\mathbb{R}^{\mathcal{F}}$ -valued random variable with respect to a $\mathbb{R}^{\mathcal{G}}$ -valued probabilities is yet *another* kind of quasi-real quantity, in a new structure $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$ — it is a filter power twice over. More generally, for any vector space $\mathcal{U}$, the expectation of a $\mathcal{U}$ -valued random variable will be an element of an *extended* vector space $\mathcal{U}^{\mathcal{G}}$.

The headline result (finally) is that well-behaved expectation operators like this really do exist. We can even construct them explicitly (again, see appendix F). There is a certain filter $\mathcal{G}$ and an $\mathbb{R}^{\mathcal{G}}$ -valued *quasi-probability* operator $\mathbb{P}^*$ on the coin flip space which is *regular* and *uniform* in the sense that every state has the same positive (infinitesimal) probability, and which agrees with the standard coin flip probabilities for simple sets. (For instance, heads on the first three flips has probability exactly $1/8$.) For any ordered vector space $\mathcal{U}$, there is a corresponding *quasi-expectation* operator $\mathbb{E}^*$ that takes any $\mathcal{U}$ -valued random-variable to an element of $\mathcal{U}^{\mathcal{G}}$. Among other properties, this operator respects *Strong Statewise Dominance*:

For any random variables $X, Y : \Omega \rightarrow \mathcal{U}$, if $X(s) \geq Y(s)$ for every state $s \in \Omega$, and $X(s) > Y(s)$ for some state $s \in \Omega$, then $\mathbb{E}^*X > \mathbb{E}^*Y$.

Quasi-expectations also respect *stochastic equivalence*. To state this carefully, first we define the *distribution* of a random variable X . Given a quasi-real measure $\mathbb{P}^*$ on *states*, the distribution of X is a corresponding quasi-real

measure on *outcomes*. For $X : \Omega \rightarrow \mathcal{U}$, and $A \subseteq \mathcal{U}$, let

$$\mathbb{P}_X^*(A) = \mathbb{P}^*[X \in A] = \mathbb{P}^*\{s \in \Omega \mid X(s) \in A\}.$$

The sense in which quasi-expectations respect stochastic equivalence is that random variables with the same distribution have the same expectation.

For any random variables $X, Y : \Omega \rightarrow \mathcal{U}$, if $\mathbb{P}_X^* = \mathbb{P}_Y^*$, then $\mathbb{E}^*X = \mathbb{E}^*Y$.

(See theorem F.56.)

Quasi-expectations make sense for random variables whose values live in any ordered vector space. In particular, we can apply this to the quasi-real utility functions we discussed in section 3.5, which respect Pareto and Impartiality. This provides us with a working model for the value of infinite worlds on an infinite state space — a proof of concept. We can model possible outcomes as infinite utility streams, which represent each individual's level of well-being with a real number. In section 3.5 we defined a quasi-real utility function U^* on utility streams that generalizes the utilitarian theory for finite populations: it is the *quasi-sum* of individual utilities. So our basic *quasi-expectational* theory of value says that the value of a prospect over utility streams is represented by the *quasi-expectation* of the *quasi-sum* of individual utilities.⁵

$$X \succsim Y \quad \text{iff} \quad \mathbb{E}^*U^*X \geq \mathbb{E}^*U^*Y.$$

It is by no means clear that this theory is the *correct* theory of value, and I have done nothing to argue for it. But this demonstrates the *consistency* of various principles — principles that we know to be inconsistent if we require that utilities or probabilities can be represented by real numbers. As a start:

Theorem 5.4. Pareto, Impartiality, Strong Statewise Dominance, and Stochasticism (with respect to non-standard probability distributions) are jointly consistent.

We will see that this model has many other nice properties as well.

In section 5.2, we raised a problem, extending the various paradoxes of “infinite futility” from chapter 3: finite interventions seem to make no difference

⁵Note again that the asterisks stand in for different filters. We can use the more explicit notation $\mathbb{E}^{\mathcal{G}}$ for the quasi-expectation $\mathbb{E}^*$, and the more explicit notation $\sum_i^{\mathcal{F}} U_i$ for the quasi-sum of individual utilities U^* .

to the value of prospects in an infinite random world. We now have a satisfactory answer to this problem, as well. The key is to combine a theory of *infinite values* with a theory of *infinitesimal probabilities*. Just as reckoning with the very large requires us to go beyond the real numbers to represent value, reckoning with the very small requires us to go beyond the real numbers to represent probability.

5.4 Indeterminate probability

There is another analogy worth noting between probability and value. The natural quasi-real structure that we construct in the way just outlined to represent probabilities is only *partially ordered*. For example, consider the propositions *heads on every odd flip* $H(1, 3, 5, \dots)$ and *heads on every even flip* $H(2, 4, 6, \dots)$. According to the simplest and most natural quasi-probability measure that respects the coin flip symmetries, these two sets of states are incomparable. We might want to say instead that they are equally probable, though. We can use the same methods outlined in this section to cook up stronger probability orders (using filters that extend our chosen filter $\mathcal{G}$) — effectively, this imposes more structure on probabilities.

(It is not clear that we *should* say that $H(1, 3, 5, \dots)$ and $H(2, 4, 6, \dots)$ are equiprobable. After all, similar reasoning might lead us to say that $H(2, 4, 6, \dots)$ is equiprobable with $H(3, 5, 7, \dots)$. But $H(1, 3, 5, \dots)$ is strictly *less* probable than $H(3, 5, 7, \dots)$: indeed, it is exactly half as probable, since it requires one extra coin flip to come up heads. Moreover, we face a situation analogous to what we found with value orders for infinite worlds: stronger probability orders will violate a principle of *permutation invariance*, so merely rearranging coin flips will not preserve comparisons of probability. Still, this is something we can have if we want it.)

But there is a limit to how far we can go on in this way. The situation is, again, exactly analogous to what we said about value in chapter 4. In fact, another analogue of the Zame–Lauwers theorem holds (see section F.4):

Theorem 5.5. There is no constructive proof that there exists a regular uniform total comparative probability order on an infinite state space.

The quasi-probability measure P^* that I have been discussing is something we can explicitly construct. But there is no similarly constructive way of extending this to a measure that assigns *totally ordered* probabilities. We can

prove that such things exist in standard set theory, but only by appeal to the axiom of Choice. Assigning totally-ordered probabilities involves making infinitely many arbitrary choices to resolve incomparabilities between infinite sets.

Explicitly defining a total ordering of probability should not be our goal. I am inclined to take the same view here as I did in chapter 4. The weak probability order for coin flips I have discussed here — along with any particular stronger order that we might explicitly define — should not really be understood as a candidate for the *at least as probable* relation. Rather, it is a candidate for the relation of being *determinately* at least as probable. Like the value of infinite worlds, the probability of infinite sets of states is a vague matter. For many pairs of propositions, we are not in a position to know how their probabilities compare, and there is no hope of overcoming our ignorance on these matters.

The true order of probabilities may be any one of infinitely many different “precisifications” of the weak coin flip order (or whatever stronger order we eventually land on). Similarly to the case of value, a precisification of a *quasi-real* probability representation is a *hyperreal* probability representation. So hyperreals do come back into the story. But no particular hyperreal representation is determinately correct.

You do not have to take this perspective in order to use the formal apparatus I am describing. If you prefer straightforward incomparability rather than indeterminacy, you can simply take a partial order of probabilities as the final word. However we interpret them, we have two different kinds of representation. First, there are some nice constructively defined partially ordered quasi-real quantities — which I think of as representing determinate probability. Second, there are infinitely many totally ordered hyperreal quantities which we cannot construct explicitly — which I think of as “precisifications” of probability, different representations that we have no way of knowing to be correct, but which are at least not determinately incorrect.

In the approach I have outlined, we use different mathematical representations for value and for probability, but we find systematic analogies between them.

5.5 Williamson's paradox

There is an influential argument for Standardism from Williamson (2007). Consider once again the angel's infinite sequence of coin flips. Now consider two events (figure 5.5). First, the event we earlier called All Heads, which Williamson assigns the notation $H(1\dots)$. Second, the event that every coin *after the first* comes up heads (whether or not the first coin comes up heads), $H(2\dots)$.

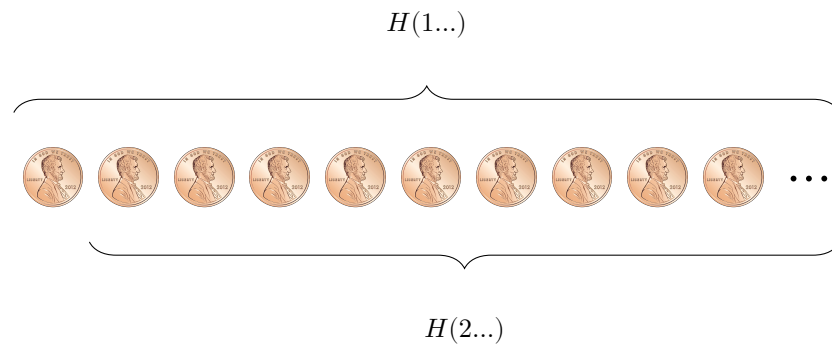


Figure 5.5: Two isomorphic events.

Note that every way for $H(1\dots)$ to obtain is also a way for $H(2\dots)$ to obtain. But the reverse is not true — if the first flip is tails, and every flip thereafter is heads, then $H(1\dots)$ fails while $H(2\dots)$ obtains. So regularity would imply that $H(1\dots)$ is strictly less probable than $H(2\dots)$.

But $H(1\dots)$ and $H(2\dots)$ are isomorphic events. More precisely, we can map the constituent single-toss events of $H(1\dots)$ one-to-one onto the constituent single-toss events of $H(2\dots)$ in a natural way that preserves the physical structure of the set-up just by mapping each toss to its successor. $H(1\dots)$ and $H(2\dots)$ are events of exactly the same qualitative type; they differ only in the inconsequential respect that $H(2\dots)$ starts one second after $H(1\dots)$ Thus $H(1\dots)$ and $H(2\dots)$ should have the same probability. (Williamson 2007, 175)

This observation is inconsistent with regularity, and with Non-Standardism. Williamson presses his point by introducing a second sequence of coin flips (figure 5.6).

To make the point vivid, suppose that another fair coin, qualita-

tively identical with the first, will also be tossed infinitely many times at one second intervals, starting at the same time as the second toss of the first coin, all tosses being independent. Let $H^*(1\dots)$ be the event that every toss of the second coin comes up heads, and $H^*(2\dots)$ the event that every toss after the first of the second coin comes up heads. Then $H(1\dots)$ and $H^*(1\dots)$ should be equiprobable, because the probability that a coin comes up heads on every toss does not depend on when one starts tossing, and there is no qualitative difference between the coins. But for the same reason $H^*(1\dots)$ and $H(2\dots)$ should also be equiprobable.

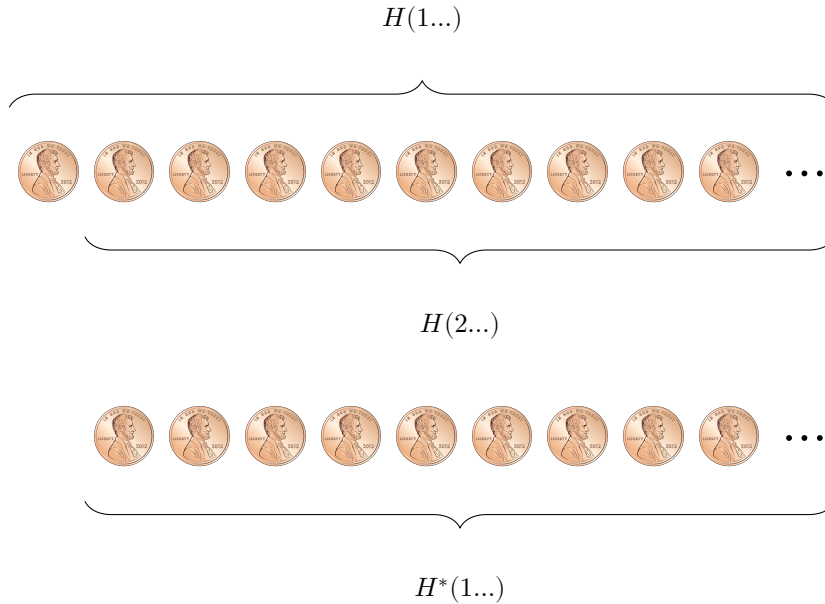


Figure 5.6: A second sequence of coin flips.

But if this much is right, then again we can conclude that $H(1\dots)$ and $H(2\dots)$ are equally probable, contradicting regularity.

Williamson is pointing us to something quite puzzling. How could just postponing the coin tosses change anything about their probabilities? I can't make this weirdness go away, but I have grown more more comfortable with it. As we have seen, there is an intimate connection between infinitesimal probabilities and infinite values. Furthermore, we have already encountered an exactly analogous puzzle for infinite values. Consider again the case from section 3.3: there are infinitely many individuals stretching from the

infinitely distant West to the infinitely distant East (figure 5.7). One of them is Indy. Everyone West of Indy is sad; everyone East of Indy is happy; only Indy's happiness is up to us. We might think similarly that the world where Indy is sad and the world where he is happy are *isomorphic* — they differ only in the “inconsequential aspect” that in one case the happiness starts one person later. And so, we may think, the two worlds have the same value.

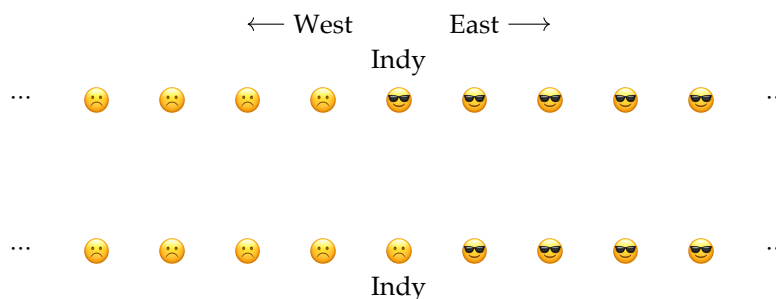


Figure 5.7: Some isomorphic pairs of worlds are not equally good.

My response to this puzzle was to throw away the “moral telescope”. We don’t need to try to read values off from the qualitative distribution of well-being. Rather, we can consider immediately the respects in which one world is better or worse than another — such as Indy’s happiness, which is a morally weighty thing. We shouldn’t think of aggregate moral value as some kind of cosmic scoreboard.

Williamson’s puzzle shows us we should also throw away our *probability* telescopes. We cannot read probabilities off from the “qualitative types” of events any more than we can read values off from the qualitative types of well-being distributions. But also, we don’t have to. We shouldn’t think of *probabilities* as cosmic scorecards, either. Rather, we should think of probabilities of infinitely complex events as involving trade-offs between different simpler *aspects* of probability — the “constituent events”, as Williamson puts it. When we compare $H(1\dots)$ and $H(2\dots)$ in this light, we find that $H(1\dots)$ is *less* probable in one respect — what it demands of the first coin — and *equally* probable in every other respect. This makes it less probable overall.

We must say something similar about the relationship between the first sequence $H(1\dots)$ and the second sequence $H^*(1\dots)$: qualitative similarity does not guarantee equal probability. But in this case we still face puzzling questions. How does $H^*(1\dots)$ compare to $H(1\dots)$ and $H(2\dots)$? It is very strange to say that one of the two simultaneous sequences of coin flips is

more likely than the other to come up all heads, and it is also very strange to say that when the sequence starts makes any difference.

K. Easwaran (2014b, 25) proposes that if we allow for incomparability, “the intuitions are preserved”:

Williamson claims that we have an intuition that $[H^*(1...)]$ is equiprobable with $[H(2...)]$, but I claim that our intuition is just that $[H^*(1...)]$ is neither more nor less probable than $[H(2...)]$ and that we can't reliably tell the difference between these types of intuition.

In line with my what I said in section 5.3, I would amend this to instead suggest that it may be *indeterminate* how $H^*(1...)$ and $H(2...)$ compare. This is the verdict delivered by the most straightforward way of extending the weak coin flip order to the case of two separate sequences of coin flips. (It works the same way as the example of $H(1, 3, 5, ...)$ and $H(2, 4, 6, ...)$ in section 5.3.) This might be the right thing to say.

Indeterminacy is not forced on us, though. There are stronger orders we could consider that would render $H^*(1...)$ equally probable with $H(2...)$ (and thus strictly less probable than $H(1...)$). There are also ways of extending the weak coin flip order that render $H^*(1...)$ just as probable as $H(1...)$ (and thus strictly more probable than $H(2...)$). None of these judgments seems obviously correct to me. But we should not be too embarrassed about not having all the answers. In discarding the probabilistic telescope, we are in a position to reject the trio of contradictory judgments

$$H(1...) \sim H^*(1...) \sim H(2...) < H(1...).$$

But this rejection leaves many questions open for further inquiry.

At a higher level of abstraction, Williamson's paradox is another instance of Zeno's paradox of addition, which we keep encountering at every turn. In general, you can add something to an infinite structure and get another infinite structure that is structurally identical to what we started with. It is tempting to conclude with Zeno that “the thing being added is nothing.” But we have learned to resist this. To lengthen an infinite ray by one foot is not to increase it by nothing. Even if improving a life makes no difference to the qualitative pattern of the world, that is not nothing, either. In the same way, when we add a state to $H(1...)$ to produce $H(2...)$, the resulting event is qualitatively identical to what we started with — but we should not conclude that what we have added is nothing.

In fact, we can make the analogy between probability and value even stronger. The Non-Standardist can uphold a tight connection between *probability* and *weighing value*. For any set of states A , we can consider a corresponding prospect 😊_A , which results in a good outcome 😊 if A is true or a worse outcome 😞 if A is false. The non-standardist can affirm this general principle:

$$\mathbb{P}(A) \geq \mathbb{P}(B) \quad \text{iff} \quad \text{😊}_A \succeq \text{😊}_B \quad \text{for any events } A \text{ and } B.$$

If 😊 and 😞 are the only possible outcomes, then the better prospect is whichever gives the good outcome greater probability. The Simple Argument that we discussed in section 5.1 shows that the *standardist* cannot say this, on pain of violating *Strong Statewise Dominance*. Consider two events: getting all heads from an infinite sequence of coin flips ($H(1\dots)$), or a contradiction ($\perp$). The standardist says that $H(1\dots)$ and $\perp$ have the same probability; Strong Statewise Dominance implies that $\text{😊}_\perp$ is not as good as $\text{😊}_{H(1\dots)}$.

We can define a *new* order of events: let $A \geq_{\text{😊}} B$ iff $\text{😊}_A \succeq \text{😊}_B$. Given modest assumptions, this relation $\geq_{\text{😊}}$ will satisfy the properties de Finetti proposed for a *comparative probability order* (Finetti 1937; see sections A.3 and F.1) (except that it may be only a *partial* order). The Standardist who accepts dominance will say that this relation $A \geq_{\text{😊}} B$ is *different* from the order $\mathbb{P}(A) \geq \mathbb{P}(B)$ — but it is still a relation they can theorize about. (As K. Easwaran (2014b) proposed, they might hope to define it somehow in terms of dominance or primitive conditional probabilities.) A more defensive version of the Non-Standardist picture I have been sketching is to say that *this* relation is what we have been talking about throughout this chapter, and this is what non-standard probabilities are intended to quantitatively represent. It is not clear that the Standardist should have any objection to this proposal, even if they think there is also a different theoretically fruitful kind of probability that is represented by standard real numbers.

Now return to the analogy. In chapter 3 I argued against the “moral telescope” basis of that *Pareto* principle: if improving lives at no cost to anyone makes the world better, then well-being isomorphism principles cannot be true. Analogously, we can argue directly against Williamson’s isomorphism principle — at least when it comes to the probability-like relation $\geq_{\text{😊}}$ — based on *Strong Statewise Dominance*: improving *states* at no cost in any other states makes a *prospect* better. If this is true, then since $\text{😊}_{H(2\dots)}$ strongly dominates $\text{😊}_{H(1\dots)}$, the qualitative isomorphism principle cannot be correct for the kind of probability that is most closely tied to value.

5.6 Conditional probabilities

Quasi-real numbers provide a very nice way of representing probability. But that is not to say they are the *only* way to represent probability. Many different kinds of representations are available. Standard real-valued probability distributions are also available, and they represent many important features — for example, that the probability of heads on the first flip is much closer to the probability of tails on the first flip than it is to the probability of all heads. But standard real representations leave out a lot of detail — for example, that the probability of all heads is greater than the probability of an impossibility. Some of this detail turns out to matter for the value of risky prospects. Other representations may impose too *much* structure — such as hyperreal-valued probability distributions. But representations that undershoot or overshoot the true structure of probabilities can be fine and useful, as long as we are aware of their limitations.

Let's consider a different way of representing probability, which does not give pride of place to infinitesimals: *primitive conditional probabilities*. (This also makes good on my promissory note to return to a response to the Simple Argument, endorsed by K. Easwaran (2014b), which I set aside in section 5.1.)⁶

In the orthodox framework of Kolmogorov (1950), conditional probabilities are defined as ratios of unconditional probabilities:

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This is fine so long as the denominator $\mathbb{P}(B)$ is non-zero. But what shall we say about the probability of All Heads ($H(1\dots)$) conditional on All But One Heads ($H(2\dots)$)? The question seems intelligible, and indeed it seems to have a clear answer: it's the same as the probability that the first flip comes up heads, which is $1/2$.

The non-standardist, using quasi-real probabilities, can affirm this consistently with the ratio formula: the probability of All Heads is a certain infinitesimal ϵ , and the probability of All But One Heads is another infinitesimal 2ϵ , and their ratio is indeed $1/2$.⁷ The Standardist cannot say this. But this does

⁶Another alternative kind of representation that I will not pursue are the *lexicographic probabilities* of Blume, Brandenburger, and Dekel (1991). See J. Y. Halpern (2010) for discussion of their relationship with both conditional probabilities and non-standard probabilities.

⁷More generally, they can say this as long as the denominator has a *reciprocal*. In a *field*,

not mean they must throw up their hands in defeat. Rather, they can expand on the orthodox framework by introducing conditional probabilities that are *not* defined as ratios of unconditional probabilities (Popper 1955; Rényi 1970; see also Hájek 2003b for discussion). (See section E.7 for technical details.)

Conditional probabilities provide resources for responding to the Simple Argument. Recall the bet All Heads (which pays off if the coin comes up heads every time) and the alternative bet Nothing (which never pays off). The standard unconditional probability of getting all heads is equal to the probability of something that certainly won't happen. So *Stochasticism* implies that All Heads and Nothing are equally good. But the Standardist might reject Stochasticism by the letter, while retaining something in the same spirit. As K. Easwaran (2014b, 16–17) argues, the Standardist need not regard events with the same *numerical unconditional probability* as *equally likely*. They also have a more discriminating notion of comparative probability available, understood in terms of conditional probabilities. The unconditional probability of getting all heads is zero, as is the unconditional probability of an impossibility. But they have different *conditional* probabilities: for instance, the probability of getting all heads conditional on getting all heads is one, while the probability of an impossibility conditional on getting all heads is zero. Thus even if getting all heads has unconditional probability zero, even the Standardist can acknowledge a sense in which getting all heads is *more probable* than an impossibility.

In general, consider the relation of *comparative conditional probability*, defined thus (Pruss 2014; K. Easwaran 2014b):

$$A \succsim^{\mathbb{P}} B \quad \text{iff} \quad \mathbb{P}(A \mid A \cup B) \geq \mathbb{P}(B \mid A \cup B).$$

Given the usual structural conditions on conditional probabilities, this does indeed define an order. Furthermore, this order discriminates between different propositions that each have (standard) *unconditional* probability zero: for instance, $H(1\dots) <^{\mathbb{P}} H(2\dots)$.

So conditional probabilities capture some aspects of the structure of probabilities that unconditional probabilities alone do not. But they still undershoot: there are probability comparisons that are still not captured. (This point is adapted from Thong forthcoming.) Consider these two propositions:

like the reals or hyperreals, every non-zero number has a reciprocal. In general, quasi-real structures do not have this property. But this is no serious problem. The regular quasi-probability measures discussed above are actually regular in a stronger sense: not only does each state have *non-zero* probability, but also each state's probability has a reciprocal.

- (A) The first coin flip comes up Heads.
- (B) The first coin flip comes up Tails, and at least one subsequent coin flip comes up Heads.

Intuitively, A is strictly more probable than B . After all, A is equally probable as a third proposition C :

- (C) The first coin flip comes up Tails.

By regularity, C is strictly more probable than B . However the coin flips go, if B is true, then so is C ; but also it is possible that the coin comes up Tails every time, in which case C is true but B is not. Similarly, a gamble that pays off only if A is true is strictly better than one that pays off the same prize only if B is true.

The comparative conditional probability relation $\succsim^{\mathbb{P}}$ does not capture this. In fact, both propositions A and B have probability $1/2$ conditional on $A \cup B$. Remember, we are still considering things as Standardists: primitive conditional probabilities are standard real numbers. Intuitively, while $\mathbb{P}(A \mid A \cup B) = 1/2$, the other conditional probability $\mathbb{P}(B \mid A \cup B)$ is only *infinitesimally less* than $1/2$. Standard real numbers do not discriminate between these two situations. (In contrast, the quasi-probabilities described in section 5.3 do deliver the verdict $\mathbb{P}^*(A) > \mathbb{P}^*(B)$. Likewise, the quasi-expectational order of prospects says that an A -gamble is strictly better than a B -gamble.)

A further result from Pruss (2025) strongly suggests that there is no good way to define regular comparative probabilities in terms of conditional probabilities and subset structure. A space equipped with just a conditional probability measure doesn't have enough structure to determine the true ordering of comparative probabilities. As in other cases we have discussed, we can think about "structure" in terms of *symmetries* (see sections 3.3, 4.2 and 9.3). Suppose there were a natural, "structure-respecting", recipe that took you from a conditional probability measure to a probability order. Then if we rearranged the space in a way that respected all of its inherent structure (via a *conditional probability space automorphism*), the result of applying the recipe should be a probability order that is rearranged in just the same way.

⁸More formally, let $(P, \Omega) \mapsto \succsim_{(P, \Omega)}$ be a function that takes a conditional probability measure to a regular probability order on the same space (see section A.3). If such a function is "natural", then any isomorphism of conditional probability spaces should also be an isomorphism of their corresponding comparative probability orders. For any (measurable)

Pruss proves two main results. One is that there is no such operation that takes every conditional probability measure to a *total* probability order. This much may be fine, and even desirable, as I will discuss below. But he also proves a second result about *partial* probability orders. In this case, the result is not that no such natural mappings exist.⁹ But the ones that do exist do not seem to be what is called for. His result involves probability measures on a circle, thought of as representing the outcome of a fair spinner. There are conditional probability measures on the circle that satisfy a condition he calls “strong invariance” with respect to arbitrary rotations.¹⁰ But any of these will necessarily be mapped to a very weak probability order, one that does not even make the two half-circles equally probable — they must be incomparable.¹¹

There is also another problem. Standard real-valued conditional probabilities do not just undershoot the structure of probabilities: in another respect, they *overshoot* it, including *too much* structure.

To appreciate this, let’s consider Easwaran’s (2014b) criticism of *hyperreal* probability representations — they have “too much structure to properly represent credences in ordinary propositions” (p. 4). His argument for this is based on two main considerations.

First (2014b, sec. 5.1), hyperreal numbers are *totally ordered*. Consequently,

function $f : \Omega \rightarrow \Omega'$ and any conditional probability measure P , we can define a conditional probability measure on Ω' such that for all measurable subsets $A, B \subseteq \Omega'$,

$$P^f(A \mid B) = P(f^{-1}(A) \mid f^{-1}(B)).$$

Likewise, given a probability order $\succsim$ on Ω , we can define a probability order on Ω' such that for all measurable subsets $A, B \subseteq \Omega'$,

$$A \succsim^f B \quad \text{iff} \quad f^{-1}(A) \succsim f^{-1}(B).$$

Then the naturality condition says that these two operations agree: for any measurable bijection $f : \Omega \rightarrow \Omega'$, we have

$$\succsim_{(P^f, \Omega')} = \succsim_{(P, \Omega)}^f.$$

(Pruss calls this condition “canonicity”.) As a special case, this says that any conditional probability space *automorphism* $f : \Omega \rightarrow \Omega$ must also be an automorphism of the corresponding probability order.

⁹There are at least trivial mappings, like the one that takes every conditional probability measure to the subset order.

¹⁰Namely, for any rotation f of the circle, and any (measurable) subsets of the circle A, B such that A and fA are both subsets of B , we have $P(A \mid B) = P(fA \mid B)$.

¹¹I do not know whether similar results hold for reasonable conditional probability measures on our central example, the coin flip space, but I suspect that they do.

hyperreal probability representations impose comparisons in cases where we may not want them. As I discussed in section 5.5, Easwaran claims that this is what is going on in the case of Williamson’s two sequences of coin flips.

Second (2014b, sec. 5.4), and relatedly, the existence of hyperreal probability representations can only be proved *non-constructively*. It is consistent with ZF set theory without the axiom of choice that there are no hyperreal structures at all. Easwaran argues that this shows that any relationship there might be between the credences of even idealized physical agents and hyperreal probabilities is “too complex to properly represent the physical manifestation of credences” (pp. 32–33).

In fact, though, conditional probabilities also have both of these very same features. Consider again the comparative probability relation $\succsim^{\mathbb{P}}$ defined in terms of primitive conditional probabilities. If conditional probabilities are standard real numbers, then these are also totally ordered. It follows that $\succsim^{\mathbb{P}}$ is a total order on propositions: for any propositions A and B , either $\mathbb{P}(A \mid A \cup B)$ is greater than $\mathbb{P}(B \mid A \cup B)$, or else it is less, or else they are equal. Consider, for instance, getting all heads in one sequence of coin flips, $H(1\dots)$, or getting all heads in a different sequence, $H^*(1\dots)$. Any standard real-valued conditional probability function will favor one of them or the other, or else say that they are equiprobable, conditional on their disjunction. In this sense, there is no probabilistic incomparability.

Conditional probabilities also have the second defect of the hyperreals: non-constructiveness. It turns out that the existence of an conditional probability measure on the coin flip space is *also* independent of ZF, just like the existence of hyperreals. (This is intimately connected to the fact that they impose a total order.)¹²

In short, conditional probabilities do not rescue us from the difficulties of indeterminacy or non-constructibility that arose for hyperreals. That is not to say we have no use for hyperreal probabilities or real-valued conditional probabilities. There is a natural correspondence between quasi-real representations and either one of these. But the correspondence is one-many. We can associate a *single* quasi-real representation with an infinite *family* of hyperreal representations — each of which “flattens out” the partially

¹²This follows from results in Pruss (2014); see section E.7 for discussion and an alternative proof. The fact is also closely related to the results I have discussed concerning the non-constructibility of total social welfare orders, and the theorems of Zame (2007) and Lauwers (2010).

ordered quasi-reals into some totally ordered hyperreal system. Likewise, we can associate a single quasi-real representation with an infinite family of real-valued *conditional* probability representations. These can be useful. But these representations add structure that is intuitively extraneous. On my preferred interpretation, no particular hyperreal representation is determinately correct. Likewise, no particular conditional probability function is determinately correct, either.

The hope was that conditional probability representations might hit the sweet spot, including just the right amount of structure to represent comparative probabilities. In fact, though, primitive conditional probabilities have *both* the defects of hyperreals and those of standard real unconditional probabilities. Like the hyperreals, conditional probabilities add extraneous structure; like the standard reals, conditional probabilities also leave out some important structure. In contrast, quasi-real representations have neither problem.

5.7 Conclusion

The key lesson we have learned in this chapter is that if we are to weigh up the value of risky prospects involving infinite worlds, we will need to be attentive to some *very unlikely* possibilities — so unlikely that their probabilities are not adequately represented by any finite numbers at all. When there are infinitely many such possibilities, their importance can add up dramatically. This means that the standard resources of orthodox probability theory are inadequate for our purposes. Prospects with even very large differences in value can be stochastically equivalent, as far as their standard real-valued probabilities of outcomes go. Instead, we must use finer-grained representations of probabilities. I have specifically advocated again for *quasi-real* representations, which are constructed directly in terms of the states whose probabilities we wish to measure.

The question of whether the probability of getting all heads in an infinite sequence of coin flips is *zero* or just *really really close to zero* might seem like the epitome of philosophical hairsplitting. But, surprisingly, issues of the greatest importance can turn on this question.

Chapter 6

Paradoxes of Dominance

As rain water falling on the mountain top runs down over the rocks on all sides; similarly, he who sees difference runs after them in various directions.

Katha Upanishad 2.1.14

6.1 Cain's paradox

Let's jump right into the next puzzle. (This is a slight variant of the "sphere of suffering" from Cain 1995.)

Cain's Paradox. Infinitely many immortal people are labeled by natural numbers. Consider two options.

Expanding Happiness.

- On day one, the first person is happy, and the rest are sad.
- On day two, the first two people are happy, and the rest are sad.
- And so on: on day n , the people labeled 1 to n are happy and the rest are sad.

Expanding Sadness.

- On day one, the first person is sad, and the rest are happy.
- On day two, the first two people are happy, and the rest are sad.

- And so on: on day n , the people labeled 1 to n are sad, and the rest are happy.

We can represent these options with two infinite matrices. Each row represents an individual, and each column represents a day.

$$\begin{array}{l}
 \text{Expanding Happiness:} \quad \begin{bmatrix} \text{😊} & \text{😊} & \text{😊} & \dots \\ & \text{😊} & \text{😊} & \\ & & \text{😊} & \\ & & & \ddots \end{bmatrix} \\
 \text{Expanding Sadness:} \quad \begin{bmatrix} \text{😞} & & & \\ \text{😞} & \text{😞} & & \\ \text{😞} & \text{😞} & \text{😞} & \\ \vdots & & & \ddots \end{bmatrix}
 \end{array}$$

Which of these is better? We can argue in two different ways to conflicting conclusions.

First, we reason person by person. In *Expanding Happiness*, each individual faces some finite duration of sadness followed by an eternity of happiness. In *Expanding Sadness*, each individual faces some finite duration of happiness followed by an eternity of sadness. The first of these options involves infinitely more happiness overall, so it is natural to say it is better on balance for each person. (We could argue for this from more basic principles, but that would be overkill for present purposes.) If it is better for everyone, then we can conclude that *Expanding Happiness* is better overall — by appeal to our now-familiar *Pareto* principle.

Next we reason *day by day*. In *Expanding Happiness*, every day there are finitely many happy people and infinitely many sad people. In *Expanding Sadness*, every day there are finitely many sad people and infinitely many happy people. It is natural to say that a day with finitely many happy people and infinitely many sad people is *worse* (with respect to individual well-being) than a day with infinitely many happy people and finitely many sad people. If that is right, then *Expanding Happiness* guarantees that *every day is worse* than it would be in *Expanding Sadness*. So it is natural to conclude that *Expanding Happiness* is worse than *Expanding Sadness* overall. We might justify this by appealing to a different principle:

Timewise Dominance. If in outcome x every day is better than it would be in outcome y , then x is better than y overall.

These two lines of reasoning cannot both be correct. *Personwise* dominance reasoning led us to the conclusion that Expanding Happiness is better. *Timewise* dominance reasoning led us to the conclusion that Expanding Sadness is better. Something has gone wrong.

Cain's Paradox is one example of a pattern that arises over and over. We represented social outcomes using two-dimensional matrices. One dimension represents individual lives, and the other dimension represents times. We can think of individual lives and times as orthogonal "dimensions of value". For each of these dimensions, there is a corresponding *dominance* principle. Timewise Dominance says that if one outcome is better than another on each day — for every "location" along the time dimension — then it is better overall. Similarly, Pareto says that if one outcome is better than another for every "location" along the "individual dimension", then it is better overall. These two dimensions cut across one another, and (in the presence of certain auxiliary assumptions) these two cross-cutting dominance principles lead to paradox. In subsequent sections, I will survey several more examples of this pattern: *paradoxes of dominance*.

These paradoxes raise obstacles to any kind of progress in understanding impartial value in the presence of infinity. We cannot have all of these dominance-style principles. This shakes our confidence in *each* of them — unless and until we can give some reasonably principled story about how to choose between them. This threat extends to two principles that have been our lodestones throughout our investigation of infinite value: the *Pareto Principle*, and *Statewise Dominance*. Without basic principles like these, the prospects for a theory of infinite value seem dire.

Call this the *bad company objection*.¹ We can't have all of the attractive dominance-style principles we might want. How are we to distinguish the true principles from the impostors? What principled defense can we give for them that will not overgeneralize?

Defeatism beckons once more. Perhaps the root problem behind all of these paradoxes is *Impartiality*, and the project of making sense of impartial value is somehow ill-conceived. (I defended this perspective in J. S. Russell 2022; for

¹I stole this name from neo-Fregean philosophy of mathematics (Dummett 1991; Boolos 1997). But just to be clear, the relationship between the bad company objection in infinite ethics and the objection to the neo-Fregean is only one of dialectical analogy.

discussion, see Nebel 2025.) I will return to this bleak thought in chapter 11.² For now, though, let's set this response aside; we will continue to pursue this project as far as we can.

In section 6.4 I will outline my strategy for responding to these paradoxes. Unfortunately, I have no master plan, no great insight that will defuse the paradoxes of dominance all at once. My response is more plodding: we will take up each principle in each paradox on its own, and search for principled reasons for distinguishing them, sorting out the good principles from among the bad. Fortunately, in many cases I think that we can find such reasons. But it will be no easy task.

But before we begin this task, we must get a clearer idea of the scope of the challenge. More paradoxes!

6.2 Subpopulation dominance

Cain's Paradox arose from carving up an outcome along the dimensions of *individuals* and *times*. But we can also carve up the *individual* dimension in more than one way.

Happy Pairs. There are infinitely many people. This time each person is labeled by a *pair* of natural numbers. Consider two possible worlds.

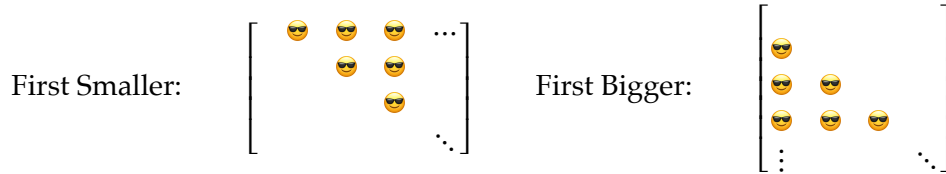
First Smaller. Each person who is labeled with a pair (m, n) such that $m < n$ is happy. Everyone else is sad.

First Bigger. Each person who is labeled with a pair (m, n) such that $m > n$ is happy. Everyone else is sad.

We can represent these options with our two familiar matrices. This time, the rows correspond to an individual's first number, and the columns correspond to their second number. There is no variation over time and no

²Note that such cross-cutting dominance principles are also deployed in important arguments for *expectational utilitarianism* — see section 7.1. So the paradoxes of dominance are an especially pressing challenge for those who are attracted to this kind of argument, though they do not face it entirely alone.

uncertainty to worry about in this case.



Consider the group of people whose first number is n . *First Smaller* guarantees that finitely many of these people are sad, and infinitely many of them are happy. *First Bigger* ensures the opposite. So *First Smaller* makes this group of people better off on balance than *First Bigger*. This is true for every n . So we can conclude that *First Bigger* is better overall, if we accept the following principle:

Subpopulation Dominance. Let $I_1, I_2, \dots$ be mutually exclusive and jointly exhaustive sets of individuals. If an outcome x is better than an outcome y for each subpopulation I_n , then x is better than y overall.

But the very same principle leads to the opposite conclusion, if we instead consider the group of people whose *second* number is n . Here we do not have a conflict between two different dominance principles; this single principle conflicts with itself, when applied two different ways.³

6.3 Paradoxes of risk

Cain's Paradox and Happy Pairs both concerned the value of *outcomes*. Another subfamily of paradoxes of dominance concerns the value of *prospects*, in the presence of risk.

The simplest version of the problem arises in a setting of non-standard probability (see discussion in chapter 5) — though we will see in a moment that the difficulty does not really depend on this. In particular, let's suppose for the moment that it is possible to conduct a fair lottery with a ticket for each natural number. (This kind of thing is impossible if we have standard countably additive probabilities. But it is possible in non-standard settings, including the framework I set out in chapter 5. I take this issue up in more detail in section 9.4.)

TODO. Fix off-by-one.

³TODO. Respond to Simpson's paradox / gerrymandering objection. Thanks to Hayden Wilkinson for pointing this out.

De Finetti's Social Lottery. There are infinitely many people labeled by the natural numbers. An angel chooses a natural number from a fair infinite lottery. There are two options.

Lucky Few. If the number n is drawn, then the first n people are happy, while everyone else is sad.

Unlucky Few. If the number n is drawn, then the first n people are sad, while everyone else is happy.

We can represent these prospects with *infinite* matrices. Again, each row represents an individual, and each column represents a possible state.

$$\text{Lucky Few: } \begin{bmatrix} \text{😊} & \text{😊} & \text{😊} & \dots \\ & \text{😊} & \text{😊} & \\ & & \text{😊} & \\ & & & \ddots \end{bmatrix} \quad \text{Unlucky Few: } \begin{bmatrix} \text{😊} & & & \\ \text{😊} & \text{😊} & & \\ \text{😊} & \text{😊} & \text{😊} & \\ \vdots & & & \ddots \end{bmatrix}$$

Let ϵ be the probability of drawing any particular number. This must be either infinitesimal or zero.

First, we reason person by person. In Lucky Few, the n th individual has probability $1 - n\epsilon$ of happiness, and probability $n\epsilon$ of sadness. This is very nearly as good as guaranteed happiness. In Unlucky Few, the n th individual has probability $n\epsilon$ of happiness, and probability $1 - n\epsilon$ of sadness. This is very nearly as bad as guaranteed sadness. So the risky prospect Lucky Few is, on balance, better for each individual than Unlucky Few. We might conclude that Lucky Few is better overall by appeal to this principle:

Ex Ante Pareto. For any social prospects X and Y , if X is better for each individual than Y , then X is better than Y overall.

Note that this is a principle about the value of *risky prospects* for individuals — this is what the qualification “*Ex Ante*” means. In contrast, the Pareto principle we discussed in chapter 3, said that if an *outcome* x is better than an outcome y for each individual, then x is better than y overall. In what follows I will call that version *Ex Post Pareto*, whenever I need to emphasize the contrast. I will take this up in detail in chapter 7.

In contrast, the Pareto Principle we have been discussing up to this point is about the value of *outcomes* rather than *prospects*.

But we can also reason state by state. In Lucky Few, whatever number is drawn, only finitely many people are happy, while infinitely many are sad. In Unlucky Few, whatever number is drawn, infinitely many people are happy, while only finitely many are sad. If it is better overall if infinitely more people are happy, then Unlucky Few is *certain* to turn out better overall than Lucky Few. So the following principle ensures that Unlucky Few is better overall.

(Strict) Statewise Dominance. If a prospect X is certain to turn out better overall than a prospect Y , then X is better than Y on balance.

As appealing as each line of reasoning is, they cannot both be correct. Either Ex Ante Pareto or Statewise Dominance must go.⁴

As I said, this straightforward version of the conflict relies on non-standard probability. But this is not essential. There is a modified version of this argument that works even in a standard setting with countable additivity, though it is a bit more complicated (Wilkinson 2022b; Kowalczyk, Manuscript; Goodsell 2021; Hong and Russell 2025). I will postpone this to chapter 7.

The paradox in De Finetti's Social Lottery arises from the cross-cutting dimensions of *individuals* and *states*. But we have a structurally analogous paradox that arises just from considerations about risk. For ease of exposition, let's help ourselves to non-standard probabilities again.⁵

Double De Finetti. Again we have a fair countable lottery. This time, each ticket is labeled with a pair of natural numbers. Only one person's happiness is at stake — Bruno's.

First Smaller. If the winning ticket is (m, n) for $m < n$, then Bruno is happy. Otherwise Bruno is sad.

First Bigger. If the winning ticket is (m, n) for $m > n$, then Bruno is happy. Otherwise Bruno is sad.

Suppose that the first number on the winning ticket is 1. In that case, *First*

⁴Indeed, a weaker principle of *Superdominance* would suffice for this argument.

Superdominance. If every possible outcome of X is better than every possible outcome of Y , then X is better than Y overall.

(See Tarsney, Lederman, and Spears 2025.)

⁵This case has many features in common with the *two envelope paradox* (see for example Broome 1995).

Smaller makes Bruno happy as long as the second number is not also 1. So it is all-but-assured that Bruno will be happy: the chance of being sad is either zero or an infinitesimal. *First Bigger* is sure to make Bruno sad. So First Smaller is better than First Bigger, conditional on the first number being 1. Similar reasoning holds if the first number on the winning ticket is 2 — and so on for *every* number. If the first number on the winning ticket is m , then *First Smaller* has probability $1 - m\epsilon$ of making Bruno happy, where ϵ is either zero or infinitesimal, while *First Bigger* has probability $(m - 1)\epsilon$ of making Bruno happy. So for every number m , conditional on the first number on the winning ticket being m , First Smaller is better for Bruno than First Bigger.

But of course exactly parallel reasoning applies when we consider the different possible cases for the *second* number. Thus paradox arises from this principle:

(Strict) Eventwise Dominance. Let $E_1, E_2, \dots$ be mutually exclusive and jointly exhaustive propositions. If a prospect X is better than a prospect Y conditional on each proposition E_n , then X is better than Y overall.

This problem also does not essentially rely on non-standard probabilities — though, again, I will defer discussion of these details to a later chapter (see chapter 9).

We are in hot water. We have considered a family of dominance-style principles: Ex Post Pareto, Statewise Dominance, Timewise Dominance, Subpopulation Dominance, Ex Ante Pareto, and Eventwise Dominance. Each of these principles seems quite plausible on its own. But, given Impartiality, we cannot have them all, and we cannot even have very many of them. This is a serious threat to the whole project of understanding the structure of impartial value.

6.4 Dimensions of value

We cannot trust dominance principles simply on the grounds that they are *dominance* principles, nor on the grounds of their immediate plausibility. But, while we cannot have them all, it would be dire to have none. We might hope for some kind of deep structural insight that unlocks all of these mysteries at once, a generally reliable standard for sorting the good from the bad. That would be lovely, but unfortunately I do not have any such insight to impart. Instead, I will take a more humble, piecemeal approach —

one paradox at a time.

There is a kind of metaphysical perspective according to which the different dimensions of value — individuals, states, times — are all fundamentally the same, just different abstract coordinates with respect to which we can locate value within a social prospect. This perspective makes it very hard to see how we could choose between (for example) dominance with respect to individuals and dominance with respect to times. If those are just like the arbitrary x and y coordinates on the plane, then it seems we cannot motivate one dominance principle over the other — and so we can accept neither.

But I do not think that all of these dimensions are alike. We can identify morally significant differences between them, and we can make some progress toward understanding how these differences give rise to dominance principles with different statuses. We can, at least to an extent, separate good dominance principles from their bad company.

In this section I will motivate this strategy and apply it to a few of the paradoxes of dominance. But much of the work lies ahead: the paradoxes involving *risk* will require much deeper discussion than would fit in one chapter. I will take them up one by one in the course of part II of this book.

Vallentyne (1995, 418, emphasis added) responds to Cain's paradox by arguing that a theory of value "must be appropriately sensitive to what the *basic bearers of value* are." A version of this idea has become fairly standard in the infinite ethics literature. Vallentyne's idea is that dominance principles hold with respect to the basic bearers of value, but not generally otherwise. He puts it this way (1995, 420, original emphasis):

And this principle ... is at the very heart of utilitarian thinking: if *every single bearer of value* bears more value under a given action than under another, then the given action is morally preferable.

If we dispense with the claims about what action is morally preferable, and instead talk about what is *better* overall, then the principle has more ecumenical appeal, not just for utilitarians.

In the case of Cain's paradox, where we found a conflict between times and individuals, it seems plausible to say that *people* are "bearers of value", while *times* are not. Thus Expanding Happiness (which gives each person a finite duration of sadness followed by an infinity of happiness) is better than Expanding Sadness (which gives each person a finite duration of happiness followed by infinite sadness).

Alternatively, one might think that *neither* of these are correct, and instead *temporal stages* of people are the basic bearers of value. Then, Vallentyne maintains, neither of these dominance principles apply, and we simply cannot tell which of these two worlds is best.

I think there is something to this idea. But we should be careful here.⁶ First, I don't think "bearers of value" is quite the right idea. What *bears value* is what is better or worse, in the first instance. This is a metaphysical question about the *better* relation: what are its relata? In our discussion this far, we haven't ever said that one *individual* is better or worse than another — as if we were giving individuals moral grades. Comparing the value of individual *lives* are a better idea. But again, this isn't actually how we have been putting things, and it does not seem mandatory. Rather, we have talked about whether one *outcome* or *prospect* is better than another. Even if outcomes are proper bearers of value, the relata of the *better* relation, this perspective does not require us to suppose that value is irreducibly "holistic" in any sense. We can also talk perfectly sensible about whether one outcome is better than another *for an individual*. I have argued (in chapter 3) that these individualized value comparisons are what ultimately matter; if one social outcome is better than another with respect to well-being, this is because it is better on balance *for individuals*. But the relation *better for individual i* may still have *outcomes* as its relata (or *possible worlds* if you prefer), rather than individual lives or something like that.

In subsequent discussion, "*locations of value*" has become the more standard term. This is an improvement over "bearers", and it is harmless enough if it is understood purely as a technical term of art. But it might suggest that we are thinking of individual people as mere "receptacles" (T. Regan 1985), and it may still be easily confused with the question of the basic *relata* of *better*.

I think what we are really talking about here are *respects of value* — *ways* in which one outcome (or prospect) can be better than another. An outcome can be better than another *for Indy*, or better *for the individuals Indy, Jay, and Kay on balance*, or better *on Tuesday*, or better *overall*. An outcome can also be better in respects that are far less tempting to call "locations": better *with respect to well-being*, or better *with respect to political equality*, or better *for the environment*. We can call these different respects of value *dimensions* of value — though this use of "dimension" can come apart from the kind of geometrical picture-thinking from section 6.1.

⁶Thanks to Teru Thomas, and indirectly to Ralf Bader, for discussion of the following line of thought.

The question, then, is how these different respects of value — or dimensions — are related to one another. Vallentyne asks what the *basic* bearers of value are; we might analogously ask what the *basic dimensions of value* are. Again, I think there is something right about this, but the language may still be misleading. It might suggest that we are concerned with what is most *metaphysically* basic or *atomic*. For example, we may take the view that *person-stages* are metaphysically more basic than *lives*, and then we may be tempted to conclude that *better for i at time t* must be more “basic” than *better for i on the whole*. But we need not accept this.

What should we say instead? I am not at all sure, but let’s give it a shot. I think this might be a better way of understanding the key idea, abstractly: some kinds of value matter only *for the sake* of other kinds of value; dominance principles track this relationship. This is a *moral* relation between different respects of value, rather than a metaphysical relation. It is not about *what they ultimately are*, but *why they ultimately matter*.

Say we have an option X that affects three people: Indy, Jay, and Kay. We can consider how good X is for Indy, and how good it is for Jay, and how good it is for Kay. We can also ask how good X is for Indy, Jay, and Kay *collectively, on balance*. Plausibly, this *collective* value “aims at” what is good for each of the three individuals. We don’t care about how things go for Indy because of how this might make a difference to how things go overall on balance; rather, we care about how things go overall, on balance, because this is a way of weighing up how things go for each individual. (Compare Bader 2022.) (You could take the opposite view, and hold that it is the group’s overall well-being that matters intrinsically, and how things go for each individual matters only instrumentally toward this end. But I don’t think that view is plausible.)⁷

Intuitively, the collective good of Indy, Jay, and Kay aims at each of their individual good; but it does *not* aim at other ways of arbitrarily carving up the social outcome into parts. We could split up how things go for Indy, Jay, and Kay collectively into just two components: how things go for Indy, and how things go for Jay and Kay collectively. But this is not what value for the three of them is aiming at.

We can divide up prospects into constituent parts in various ways. But not all of these ways of carving them up have the same moral import. When one

⁷We should contrast this with the idea that there are other “holistic” things that matter *besides* individual well-being, such as equality or diversity. See footnote 5 in section 3.2.

kind of value is a mere “balance sheet” for another — a kind of summary of how certain value trade-offs shake out — that is when we should expect a corresponding dominance principle to hold. Consider two options X and Y . Suppose we can compare them in various respects $\mathcal{I}$, and in some other more “global” respect j . The central idea is that when j -betterness matters *entirely for the sake of* i -betterness for each $i \in \mathcal{I}$, then we should expect a corresponding dominance principle to hold: if $X \succeq_i Y$ for each $i \in \mathcal{I}$, then $X \succeq_j Y$ as well. But if respects of i -betterness are just some motley mishmash, and not what j -betterness is ultimately *about*, then j -betterness need not respect their voices, even when they all agree.

This idea is far from entirely clear, but it is suggestive. I think it helps us see why Subpopulation Dominance can fail. In the case of Happy Pairs, we can carve up how things go overall into how things go for each collection of individuals who share a first number in their assigned pair. But how good things are for all of them collectively does not *aim* at this arbitrary partition of individuals. How well things go for everyone matters for the sake of how well things go for each individual. But the collective good does not matter, even in part, for the sake of how well things go for the people whose first number is 3.

The situation is analogous to gerrymandering.⁸ [TODO. I have to think this through.]

This also gives us a principled reason for rejecting Timewise Dominance. Carving up everyone’s lives into days is a motley mishmash partition, not one that reflects the directed structure of value. I think this is especially clear when we take relativity into account: in fact, if these individuals are spread out in space, then there is no frame-independent fact about what even counts as the *same day* for all of them. [TODO. Explain this better?]

In contrast, I think these two views are each quite plausible.

- The overall value of well-being for everyone in the world matters entirely for the sake of individual well-being.
- *Ex ante* (“subjective”) value of a prospect matters entirely for the sake of *ex post* (“objective”) value of outcomes.

The first idea corresponds to (*ex post*) Pareto. The second corresponds to Statewise Dominance. These ideas do not sound nearly as silly as the corresponding ideas about subpopulations or times. That is not to say they are

⁸Thanks to Hayden Wilkinson for suggesting this way of thinking about it.

obviously correct! Indeed, I don't think we have a firm enough grasp on what we are even talking about for *anything* to be obvious around here. But it's a start.

The perspective I have outlined remains quite impressionistic. I think it points us in a useful direction for resolving the paradoxes of dominance and weeding out bad company, even so, but it doesn't get us anywhere near all the way there. As I said, I don't have any general insight that unlocks every mystery.

In particular, it is plausible that *ex ante* value of a social prospect aims at *ex post* value of social outcomes. But we might also be tempted by the thought that *ex ante* value of a social prospect aims at *ex ante* value of *individual* prospects. (See, for instance, Broome 1991, ch. 8.) The first idea gives rise to Statewise Dominance; the second idea gives rise to Ex Ante Pareto. We know we cannot have both (in the presence of Impartiality). But, if we are to have either, it is still difficult to adjudicate between them. There is more work to be done.

6.5 The path ahead

Paradoxes of dominance threaten the coherence of the very project of infinite impartial axiology. The task before us is to give some principled reasons for rejecting certain principles that figure in the paradoxes, while retaining others — despite the formal similarity between the good principles and their bad company. In this chapter I have taken a small step in this direction.

So far, though, I have said very little to defuse the paradoxes involving *social risk*. Social risk will be the central topic of part II. There we will return to the questions raised in chapter 2 about the status of two principles about small probabilities: Risk Neutrality and Fanaticism. Paradoxes of dominance involving risk continue to loom over this exploration, and we will confront them in several places along on our way. In the end, I do think that we can find reasons for rejecting Ex Ante Pareto — reasons that do not extend to *ex post* Pareto, because they are specific to the value of risky prospects for individuals (chapter 7). We will also find reasons for rejecting Eventwise Dominance that do not extend to Statewise Dominance (chapter 9). In addition, we will explore certain *separability* principles, both for subpopulations (chapter 8) and for events (chapter 9), and disentangle them from their closely related dominance principles.

Throughout part I we have confronted paradoxes have threatened to sink the project of making sense of impartial value in an infinite world — not just the paradoxes of dominance, but also “infinite futility” (chapters 2 and 3), isomorphism (chapters 3 and 5), indeterminacy (chapter 4), and stochastic dominance (chapter 5). So far, that project has not entirely foundered. Along the way, we have pieced together a positive working theory of infinite value, a generalization of expectational utilitarianism using resources from non-standard analysis. (This is developed in much more detail in the appendices.) But there remains much work to do.

Part II

Weighing Small Probabilities

... such certain immediate wretchedness, such
uncertain future good.

Jane Austen, *Persuasion*

Introduction to Part II

In part I, we addressed a series of challenges to the coherence of infinite impartial value, and in so doing outlined how a positive theory of infinite value might go. To avoid “infinite futility” (or “infinitarian paralysis”), we should recognize that there are many different infinite values that can be quantitatively compared to one another. We should, correspondingly, reject a variety of qualitative isomorphism principles. We should recognize that value comparisons between infinite worlds can be deeply arbitrary (or else give out entirely). We should countenance infinitesimal probabilities. We also still carry the burden of providing better responses to certain *paradoxes of dominance*, concerning social risk. All of these ideas will have ramifications throughout the rest of the book.

Now that we have some idea of how to even begin to think coherently about infinite impartial value, it’s time for us to return to the question we began with: how shall we weigh tiny probabilities of vast value?

Let’s go back to Pascal’s argument for his principle that an arbitrarily small chance of an infinite good outweighs the certainty of any finite good (section 2.2). His argument turned on a claim about weighing chances of *finite* value:

Because there is equal chance of winning or losing, if you had to wager just two lives for one, you might still wager. But ... you would be imprudent ... not to stake your life to win three lives in a game where you have the same chance of losing and winning.

Pascal’s talk of “lives” concerned durations of happy life for one individual; but there are exactly analogous questions and arguments concerning happy lives for different individuals. This interpretation will be our focus. (See

section 2.5.)⁹

More generally, Pascal advocated *risk neutrality* with respect to happiness. In general, a chance at happy life should be valued directly in proportion to its quantity and to its probability.

Risk Neutrality. For any finite numbers n and k , chance $\frac{1}{n}$ of nk happy lives is equally good as k happy lives for sure. In symbols:

$$\frac{1}{n} * nk \text{ happy lives} \sim k \text{ happy lives}$$

More generally, the value of a risky prospect involving happy lives is the same as its *expected number of happy lives* — I call this the *Expected Number* view. As we discussed (section 2.6), the argument for Pascal’s Principle about infinite value really only needs a weaker principle than Risk Neutrality.

(Finite) Fanaticism. For any numbers n and k , there is some number K such that chance $\frac{1}{n}$ of K happy lives is at least as good as k happy lives for sure.

$$\exists K : \quad \frac{1}{n} * K \text{ happy lives} \succeq k \text{ happy lives}$$

Pascal did not give any argument for Risk Neutrality with respect to happy lives, nor for Fanaticism. But that is not because there are no arguments to be given. Indeed, there are some quite powerful arguments. In part II, I will assess several different lines of argument for Risk Neutrality or Fanaticism, with an eye toward their implications for Pascal’s principle.

- Chapter 7 takes up the *Ex Ante Pareto* principle, which says (roughly) that the value of *social prospects* is determined by the value of *individual prospects*.

⁹In particular, I will focus on *constant population* cases, where the alternative to a happy life is a *sad* life, rather than no life at all. We are focusing on “making people happy”, rather than “making happy people”. There are analogous considerations that apply to intrapersonal or variable population trade-offs. Of course these are also worth thinking through, but we can’t take on everything at once.

More concretely yet, most of the structural issues we will be investigating arise even in the setting of “simple worlds” introduced in chapter 3. There is a countable infinity of individuals, each assigned a natural number; in each simple world, each of those individuals is either happy or sad; nothing else of moral importance differentiates simple worlds from one another. In this simple setting, there is a straightforward way of interpreting Risk Neutrality: we can take “ n happy lives” to stand for a simple world in which exactly n individuals are happy. As we discussed in section 3.2, Impartiality implies that any two such worlds are equally good.

- Chapter 8 takes up a *Separability* principle, which says (roughly) that the value of a prospect is not sensitive to distant, unaffected, and apparently irrelevant goings-on.
- Chapter 9 takes up Savage's *Sure Thing Principle*, which says (roughly) that value is *dynamically consistent*, not susceptible to foreseeable reversals in the face of new evidence.
- Chapter 10 considers the challenge of consistently rejecting Fanaticism and Pascal's Principle even in contexts that involve apparently irrelevant "background" risks.
- Finally, in chapter 11 we take stock of where these arguments have led us, and what we should make of them when it comes to questions of what we ought to do.

Chapter 7

Individual Prospects

“Why do you wear a mask and hood?” Fezzik asked.

“I think everybody will in the near future,” was the man in black’s reply. “They’re terribly comfortable.”

William Goldman, *The Princess Bride*

7.1 The *Ex Ante* Pareto principle

In this chapter, I will consider an argument for Risk Neutrality that I think is ultimately unsuccessful. This will set the stage for a range of different, more successful arguments for Risk Neutrality, Fanaticism, and Pascal’s Principle, which I will take up in subsequent chapters.

Specifically, the argument in this chapter invokes the *Ex Ante Pareto* principle: a prospect that is better for each individual is better overall. As we discussed in chapter 6, this principle is implicated in certain *paradoxes of dominance*. So in this chapter we will also discuss my response to some of these paradoxes: *Ex Ante Pareto* must be clarified, and ultimately rejected.

But first, let’s see what this principle can do. Start with an example. (This is structurally the same as cases we discussed in section 1.2.)

A tsunami is coming. Two populated islands are in its path, and both of their early-warning systems are malfunctioning. You don't have cell service, but you do have your speed boat.

On **Sure Island**, there is one person. If you go there, you are sure that you can get that one person to safety.

On **Risky Island**, there are five people. If you can make it to Risky Island in time, you can get all five people to safety. But Risky Island is farther away than Sure Island, and weather conditions are bad. The probability that you will make it in time to save Risky Island is only $1/5$.

The islands are in opposite directions, so there's no way to save both of them. Everyone you save will be happy, overall. Everyone on the island you don't save will be sad, overall.

Table 7.1: Another tsunami.

Probability:	$1/5$	$4/5$
Sure	1 happy life	1 happy life
Risky	5 happy lives	0 happy lives

Sure certainly makes one person happy. Risky has probability one in five of making five people happy. According to Risk Neutrality, these two prospects are equally good.

Here is an argument that Risk Neutrality gets this case right.¹

It turns out there is a third option. One of the five people on Risky Island, chosen by a fair random lottery, can be evacuated to a nearer rendezvous point. If you choose this option, you will certainly be able save that one person. Call this option **Lottery**.

We can represent these three prospects as matrices (figure 7.1). Each column represents a possible state, and each row represents an individual. (The

¹Related arguments are given by Hare (2016); Horton (2020); Kowalczyk (2025); McCarthy, Mikkola, and Thomas (2020). See also footnote 6.

Sure islander is the first row, and the Risky islanders are the remaining five. Empty cells represent sadness.)²

We can reason as follows. First, *Lottery is just as good as Sure*. Lottery and Sure are each certain to save exactly one person. Impartiality tells us that it doesn't matter *which* person we save, so these two prospects are certain to have equally good outcomes. And two prospects that are certain to have equally good outcomes are equally good overall.

Second, *Risky is just as good as Lottery*. Risky and Lottery each give each person on Risky probability 1/5 of being saved, and each prospect is certain not to save the person on Sure. Since Risky and Lottery each give every person precisely the same chance of being rescued, they are equally good prospects for every individual. So they are equally good overall.

Putting these two steps together, we can conclude that Sure and Risky are also equally good (since *equally good* is a transitive relation).

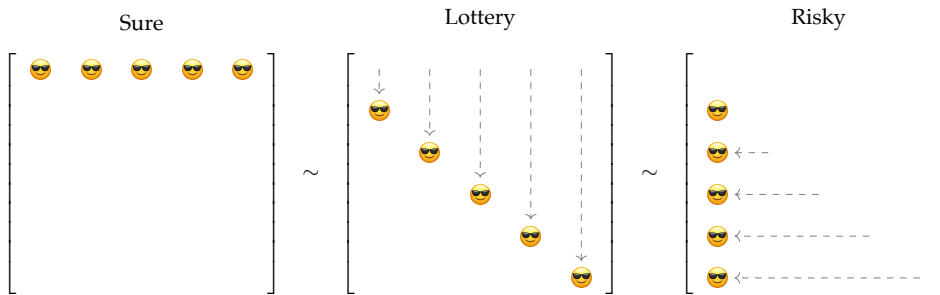


Figure 7.1: The argument for Risk Neutrality from Ex Ante Pareto and Statewise Equivalence. The arrows show how we are transforming the matrix at each step: we get from Sure to Lottery by rearranging happy lives within each column, and we get from Lottery to Risky by rearranging happy lives within each row.

Let's clarify a key point about this argument. In it, we said that the prospects Risky and Lottery are equally good for each individual. By this, we don't mean that the two prospects will *turn out* equally well for every individual.

²The matrices suggest that the outcome of the lottery is *correlated* with the outcome of the rescue attempt: successful rescue happens only in the first column of Risky, which seems to correspond to a specific winner of Lottery. This is not realistic, given the story. We could instead run this argument using a more artificial story — but in fact this is unnecessary. Nothing in this argument turns on the partition of states depicted in the Risky matrix being the *same* as the partition of states depicted in the Lottery matrix.

They won't: in cases where the Risky attempt fails, Lottery makes the lottery-winner better off than Risky. In the case where the attempt succeeds, Lottery makes the four lottery-losers worse off than Risky. The two options do not save the same individuals — but they give each individual the same *probability* of being saved. In the usual jargon, they are equally good for each individual *ex ante*, but they are not equally good *ex post*.

This argument straightforwardly generalizes to support Risk Neutrality (and even stronger conclusions). The general argument deploys four principles.

The first principle is our old friend *Impartiality* (section 3.2): making one person happy is equally good as making a different person happy. Second:

Statewise Equivalence. For social prospects X and Y , if X is certain to turn out just as well as Y , then X and Y are equally good overall.

This is a consequence of Statewise Dominance (as stated in sections 1.3.3 and 1.3.4).³ Impartiality and Statewise Equivalence together imply that the sure prospect Sure is equally good as the intermediate prospect Lottery. In terms of the matrices, this lets us take the happy person in Sure and swap them in each state with the happy person in Lottery — shifting the happiness down within each column.

The third principle:

Individual Stochastic Equivalence. If social prospects X and Y both give an individual i exactly the same probability distribution over outcomes, then X and Y are equally good for i .

Finally:

Ex Ante Pareto Equivalence. If social prospects X and Y are equally good for each individual, then they are equally good overall.

Individual Stochastic Equivalence and Ex Ante Pareto Equivalence together imply that Risky is equally good as the prospect Lottery, which rearranges each person's risk. Graphically, we can swap each Risky-islander's *anti-correlated* happy states in Lottery with their *correlated* happy state in Risky —

³For this argument, an even weaker principle would suffice.

Superequivalence. If there is some outcome x such that X and Y are both certain to have an outcome exactly as good as x , then X and Y are equally good.

(Compare *Superdominance*, in footnote 4 of chapter 6, as discussed by Tarsney, Lederman, and Spears (2025).) But Superequivalence does not suffice for the more general argument for Expected Number.

shifting the happiness over to the left within each row.

(As usual, by “equally good overall”, here we really mean equally good overall *with respect to the value of individual well-being*. In the cases we are discussing, we suppose that outcomes do not differ from each other in any other morally relevant respect. (On pattern goods like equality and diversity, see footnote 5 in section 3.2.))

Here is the general fact.⁴

Theorem 7.1. Impartiality, Statewise Equivalence, Individual Stochastic Equivalence, and Ex Ante Pareto Equivalence jointly entail Risk Neutrality.

The principle of Ex Ante Pareto Equivalence is related to the Pareto principle we discussed in chapter 3, but it is not the same. The key difference is in the *Ex Ante* part: this principle is about comparing the value of risky *prospects* for individuals. The principle we discussed in chapter 3 was an *ex post* principle about the value of certain *outcomes*.

A more superficial difference between Ex Ante Pareto Equivalence and the Pareto principle discussed in chapter 3 is in the *Equivalence* part: this principle is about what is *equally* good, rather than what is *better*. The Equiv-

⁴There is an alternative version of this result which is a bit more formally elegant (this is essentially a special case of McCarthy, Mikkola, and Thomas 2020, 82, theorem 2.2). It just appeals to two principles. For the “vertical” step:

Statewise Anonymity. If simple social prospects *X* and *Y* are certain to have outcomes that are related by a finite permutation of individual outcomes, then *X* and *Y* are equally good.

This can be immediately derived from Statewise Equivalence and Impartiality. But it is weaker than the combination of those two principles — in fact, it does not involve the notion of *equally good outcomes* at all.

For the “horizontal” step:

Reduction to Individual Prospects. If simple social prospects *X* and *Y* both give every individual *i* the same probability of each outcome, then *X* and *Y* are equally good overall.

Likewise, this can be immediately derived from Ex Ante Pareto Equivalence and Individual Stochastic Equivalence, but it is weaker than the combination of those two principles: it does not involve the notion of *equally good prospects for an individual* at all.

We can also strengthen the conclusion: not only does Risk Neutrality follow, but also the full *Expected Numbers* theory — for simple social prospects where all that is at stake is whether individuals are happy or sad, and where all of the probabilities involved are rational. We can also draw stronger conclusions yet: see section D.5 and McCarthy, Mikkola, and Thomas (2020).

alence principle is part of a more general family.⁵ Here is another important member of the family, which is more closely analogous to the *ex post* Pareto principle we discussed before.

(Strong) Ex Ante Pareto. Let X and Y be social prospects.

If X is as good as Y for every individual, then X is as good as Y overall.

If, furthermore, X is strictly better than Y for some individual, then X is strictly better than Y overall.

Ex Ante Pareto Equivalence is a consequence of Strong Ex Ante Pareto.

This result is part of a family of similar technical results. Broadly speaking, there are many reasonable-seeming *dominance* principles, like Ex Ante Pareto or Statewise Dominance. There is also a closely-related cluster of *separability* principles — more on these to come. Many different combinations of these principles point toward conclusions in the general neighborhood of *expectational utilitarianism*. (We have previously discussed this view in sections 3.5 and 5.3.) One of the most famous results in the family is Harsanyi's utilitarian aggregation theorem (1955). This has a somewhat different flavor from theorem 7.1, because it takes expected utility theory as a premise. But it similarly gives a central role to the *Ex Ante Pareto* principle: if a prospect X is as good as Y for every individual, then X is as good as Y overall.⁶

In the argument for Risk Neutrality, we represented social prospects using two-dimensional matrices. One dimension represented individual people, and the other dimension represented possible states. Combining dominance reasoning across both dimensions is very powerful, leading to strong conclusions like Risk Neutrality.⁷ Perhaps it is *too* powerful. Many have argued

⁵A very natural name for the first clause of Strong Ex Ante Pareto would be *Weak Ex Ante Pareto*, by analogy with dominance principles. Unfortunately, that terminology is standardly used for a different principle:

Strict Ex Ante Pareto. If X is strictly better than Y for each individual, then X is strictly better than Y overall.

(This is the version of the principle deployed in chapter 6.) Since I find the standard use of the term “Weak Pareto” quite confusing, I will simply avoid it. (Strict Pareto is indeed weaker than Strong Pareto — hence the usual name.)

⁶For discussion, see Broome (1991); other structurally related results include Fleming (1952); Gorman (1968); Deschamps and Gevers (1978); Blackorby, Donaldson, and Weymark (1999); Zhou (1997); Pivato (2013); McCarthy, Mikkola, and Thomas (2020); McCarthy, Mikkola, and Thomas (2019); for overview of the classics see Blackorby, Bossert, and Donaldson (2005).

⁷For parallel arguments invoking *time* as a dimension of value (as in chapter 6), see Broome (1991), ch. 11; Broome (2004); San (2026).

7.2 Ex Ante Pareto and infinity

This is one of the now-familiar *paradoxes of dominance* from chapter 6. There we introduced a version that turns on non-standard probability theory: *De Finetti's Social Lottery*. In particular, we suppose that we can randomly choose a natural number, where each number has an equal probability of being drawn. This probability must *minuscule*: either zero or infinitesimal. Let there be one person for each natural number. If we draw the number n , in *Lucky Few*, just the first n people are happy; in *Unlucky Few*, all but the first n people are happy.

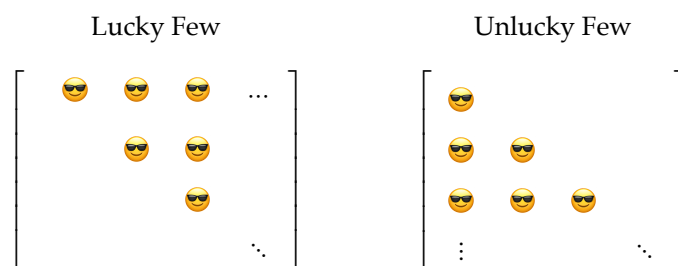


Figure 7.2: De Finetti's Social Lottery. In Lucky Few (left), if state n is drawn, the first n people are happy. In Unlucky Few (right), all but the first n people are happy.

Lucky Few gives each individual a minuscule probability of happiness, while

⁸These include critiques based on *egalitarianism* (such as Diamond 1967; Fleurbaey 2010; Otsuka and Voorhoeve 2009) or *risk aversion* (Epstein and Segal 1992; Grant et al. 2010; Blessenohl 2020; Nebel 2020). There is a separate line of attack based on the possibility of unanimous preferences arising from conflicting *beliefs* (Mongin 1995; Gilboa, Samet, and Schmeidler 2004). But this is not immediately relevant to the principle we are discussing, which is about what is *best* for individuals (with respect to some fixed body of information), rather than *preference* (for discussion see Broome 1991, ch. 7). (This point also affects some versions of the risk-aversion critiques.) Mahtani (2017); Mahtani (2020) raises a different important critique of Ex Ante Pareto, which I will discuss in section 7.3.

Unlucky Few gives each individual a minuscule probability of sadness. So it seems evident that Lucky Few is, on balance, better for each individual than Unlucky Few. (And this follows from Impartiality and Ex Post Pareto.) By Ex Ante Pareto, Lucky Few is better than Unlucky Few overall.

But also, for any possible outcome of the lottery, Lucky Few makes finitely many people happy, while Unlucky Few makes infinitely many people happy. Thus it seems evident that Unlucky Few will turn out *better* overall than Lucky Few. By Statewise Dominance, Unlucky Few is better than Lucky Few overall.

The two lines of reasoning lead to contradictory conclusions; at least one of them must be wrong.

As I said, this straightforward version of the conflict relies on non-standard probability. But this is not essential. There is a modified version of this argument that works even in a standard setting with countable additivity, though it is a bit more complicated.⁹

St. Petersburg Teams. Again, there are infinitely many people. This time, they are first divided up into two teams: Red and Blue. Each *team* is divided up into infinitely many *cohorts*. The first cohort of each team has two people, the second cohort has 4 people, the third has 8 people, and in general the n th cohort includes 2^n people.

An angel flips a fair coin until it first comes up heads. If this is on the n th flip, call n the *St. Petersburg number*. The angel also flips one extra coin. Call this the *team-choosing coin*. There are again two options.

One Team Wins. Suppose the St. Petersburg number is n . In this case, if the team-choosing coin comes up Heads, then everyone in the n th cohort of the Red team is happy, and everyone else is sad. If it comes up Tails, then just the people in the n th cohort of the Blue team is happy. (If there is no finite St. Petersburg number, then everyone is sad.)

Both Teams Win. If the St. Petersburg number is one, then no one is happy. Otherwise, if the St. Petersburg number is $n > 1$, then everyone in the $(n - 1)$ th cohort of *both* teams is happy. (The team-choosing coin is ignored.

⁹The following presentation is based on Hong and Russell (2025). Similar puzzles appear in Wilkinson (2022b); Kowalczyk (Manuscript); Goodsell (2021). The paradox of Seidenfeld, Schervish, and Kadane (2009) is also closely related; see discussion in section 8.2.

Again if there is no finite St. Petersburg number, then everyone is sad.)
Which is better?

We can repeat the same style of reasoning yet again. First, person by person. In *One Team Wins*, each person in the n th cohort of the Red team gets to be happy if they win the team-choosing toss, and the St. Petersburg number is n . This has probability $1/2^{n+1}$. The same goes for each person in the n th cohort of the Blue team. In *Both Teams Win*, a person in the n th cohort is happy if the St. Petersburg number is $n + 1$. Again, this has probability $1/2^{n+1}$. So everyone has exactly the same probability of happiness in either case; so the two prospects are equally good for everyone. By *Ex Ante Pareto Equivalence*, the two options are equally good.

Now state by state. If the first St. Petersburg flip comes up heads (state HH_1 or TH_1), then *One Team Wins* guarantees that exactly two people are happy; meanwhile, *Both Teams Win* makes no one happy. If the first flip comes up tails (all other states), then *One Team Wins* and *Both Teams Win* guarantee that *exactly the same* number of people will be happy: in HH_n or TH_n for $n > 1$, both prospects make exactly 2^n people happy. (For *One Team Wins*, these people comprise either the n th Red cohort or the n th Blue cohort. For *Both Teams Win*, these people comprise the $(n - 1)$ th cohort of *both* the Red and Blue teams, put together.) So, by *Statewise Dominance*, *One Team Wins* is strictly better than *Both Teams*.

Again, the two lines of reasoning cannot both be correct. To sum up:¹⁰

¹⁰This version of the argument used *Ex Ante Pareto Equivalence* and *Strong Statewise Dominance*. It is easy to change the numbers to raise problems for combinations of other related principles. By using a weighted coin to generate the St. Petersburg number, we can change the case so that *Both Teams Win* gives each person a strictly *greater* probability of happiness. In that case Pareto reasoning tells us that *Both Teams Win* is *better* — without changing the number of people who are benefitted in each state. This makes the case more closely analogous to the de Finetti lottery version. We can also change the cohort sizes so that *One Team Wins* is certain to make *strictly more* people happy than *Both Teams Win* — without changing the probabilities of happiness for any cohort. Then we can instead run the argument using *Strict Statewise Dominance*.

Alternatively, we can modify the argument to use *Statewise Equivalence*, if we correspondingly use either *Strong* or *Strict Ex Ante Pareto*.

Impartiality can also be substantially weakened. See Hong and Russell (2025) for discussion of these and other strengthenings of theorem 7.2.

One Team Wins											
Cohort	#	HH_1	TH_1	HH_2	TH_2	...	HH_n	TH_n	...		
Red 1	2	😎									
Blue 1	2		😎								
Red 2	4			😎							
Blue 2	4				😎						
⋮						⋮					
Red n	2^n						😎				
Blue n	2^n							😎			
⋮									⋮		
Both Teams Win											
Cohort	#	HH_1	TH_1	HH_2	TH_2	HH_3	TH_3	...	HH_{n+1}	TH_{n+1}	...
Red 1	2			😎	😎						
Blue 1	2			😎	😎						
Red 2	4					😎	😎				
Blue 2	4					😎	😎				
⋮								⋮			
Red n	2^n								😎	😎	
Blue n	2^n								😎	😎	
⋮											⋮

Figure 7.3: *St. Petersburg Teams*. The label HH_n denotes the state where the team-choosing coin is heads and the St. Petersburg number is n ; TH_n denotes tails.

Theorem 7.2. Impartiality, Statewise Dominance, Individual Stochastic Equivalence, and Ex Ante Pareto are jointly inconsistent.

The problem of St. Petersburg Teams only relies on standard, finite, countably additive probabilities. (It is worth noting that it also works perfectly well in the setting of non-standard probabilities discussed in chapter 5.) Furthermore, St. Petersburg Teams only involves comparing outcomes that differ from one another for finitely many people. This is not just a problem for *infinite* values, but for *large finite* values.

Theorem 7.2 substantially weakens the argument for Risk Neutrality presented in section 7.1 — especially in our context, where we are primarily interested in its consequences for prospects involving enormous numbers of people. That argument relied on almost the very same principles.¹¹ Combining personwise and statewise reasoning led to Risk Neutrality. But in cases involving tiny probabilities of enormous consequences, they don't stop there — they lead all the way to contradiction. What has gone wrong?

7.3 Modes of presentation

In order to diagnose this, let's step back from the infinite cases, and consider an important conceptual issue about the family of Ex Ante Pareto principles. Anna Mahtani (2017, 2020) points out that it is really not entirely clear what any of these principles even mean. Let's start with a simple example.

There are two islands: *Near* and *Far*. There is one person on each island: *Nero* on *Near*, and *Faro* on *Far*. As so often happens, both islands are threatened by a tsunami, and you can only try to save one of them. If you go for *Near*, you will certainly save *Nero*. *Far* Island is dicier: if you go for *Far*, you only have probability 1/2 of saving *Faro*.

Again we can represent these social prospects with matrices, with rows for individuals and columns for states (corresponding whether or not the

¹¹Technically, that argument relied on Statewise *Equivalence*, rather than (Strict) Statewise *Dominance*. As noted in footnote 10, a contradiction also arises from Statewise *Equivalence*, if we strengthen the Pareto premise to a *strict* version. Logically, we could hang onto both equivalence principles without either corresponding strict versions, but this seems pretty hopeless. Even though they are jointly *consistent*, together they trivialize comparisons between infinite prospects in a way that leads to a version of infinite futility.

attempt to save Far would succeed).

	Near			Far	
	$\frac{1}{2}$	$\frac{1}{2}$		$\frac{1}{2}$	$\frac{1}{2}$
Nero			Nero		
Faro			Faro		

Figure 7.4: *Near* and *Far*. Save Nero for sure, or save Faro with probability $1/2$?

It seems clear enough in this case that going for Near is the better prospect. But can we reach this conclusion just from Ex Ante Pareto? In other words, is Near *Pareto-superior* to Far — better for each individual? It seems not. If we save Near, then Nero is certainly saved and Faro is certainly lost. If we save Far, then Nero is certainly lost, while Faro has probability $1/2$ of being saved. So Near is better than Far for Nero, while Far is better than Near for Faro. Near is still plausibly *better on balance* than Far, but this judgment involves counterbalancing the conflicting claims of two individuals. Near is not Pareto-superior to Far.

Or is it?

Nero and Faro are actually the two individuals' *last* names. They also have *given* names: Al and Bea. But you've mixed up your files, and you're not sure whether the two individuals are named "Al Nero" and "Bea Faro", or else "Al Faro" and "Bea Nero". Both options are equally likely.

There are two individuals whose well-being is at stake in this situation: Al and Bea. What are their prospects? Now there are four possibilities to distinguish, corresponding to who is on each island, and whether the attempt on Far would succeed.

Now we see that saving Near gives Al probability $1/2$ of being saved (if he is the one on Near Island), while Far gives Al only probability $1/4$ of being saved (if he is on Far *and* the attempt there succeeds). Likewise, Near gives Bea probability $1/2$ of being saved, while Far only gives Bea probability $1/4$ of being saved. So Near is better than Far for *each* of them. Near is Pareto-superior to Far after all!

	Near					Far			
	Al = Nero Bea = Faro		Al = Faro Bea = Nero			Al = Nero Bea = Faro		Al = Faro Bea = Nero	
	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$		$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
Al					Al				
Bea					Bea				

Figure 7.5: The same choice between *Near* and *Far*, resdescribed. Save Al with probability 1/2 and Bea with probability 1/2, or save Al with probability 1/4 and Bea with probability 1/4?

So which is it, then? Does *Near* Pareto-dominate *Far*, or not? When we considered the individuals as *Nero* and *Faro*, the answer seemed to be no. When we considered the very same individuals as *Al* and *Bea*, the answer seemed to be yes. But *Nero* and *Faro* are the very same people as *Al* and *Bea*, and *Near* and *Far* are the very same prospects for those people! Which option is *better* does not depend on whether we call people by their first or last names. So it is unclear what the Ex Ante Pareto principle even says about this case.

Call this the *mode of presentation problem*.¹² Risky prospects involve epistemic probabilities: we don't know how things will go, and different possibilities are more or less probable given our evidence. The Ex Ante Pareto principle is specifically about the probabilities of different outcomes *for certain individuals*. It is not just about the probability of the world generally being a certain way; rather, it is about the probability that each *particular person* is happy or sad. As Frege (1892) observed, in general our epistemic situation with respect to an individual is, in a sense, sensitive to how we pick that individual out.¹³ This "how we pick them out" is standardly called a *mode of presentation*.¹⁴

¹²My discussion draws on Gustafsson and Kowalczyk (forthcoming).

¹³This general point does not just apply to *individuals*, but also to properties and entities of other types. But this is a complication we can safely ignore for now.

¹⁴One early point of departure from the following line of argument would be to turn our attention away from epistemic probabilities, and toward *objective chances*, the kind that may, for instance, figure in non-deterministic versions of quantum mechanics. The mode of presentation problem does not arise for objective chances: chance is a *metaphysical* modality, rather than an *epistemic* (or *representational*) modality. Thus a chancy construal of Ex Ante Pareto is unambiguous. This seems like an important idea to explore, but I will set it aside. Throughout this book, I am construing risk as epistemic uncertainty. Indeed, it might turn

Reid is reading *Middlemarch*, which says prominently on its cover that it is by George Eliot. In fact, George Eliot is the penname of Marian Evans — but Reid does not know this. Consider this one person, who goes by two different names: does Reid know that she wrote *Middlemarch*? In a sense, yes, and in another sense, no. Under one *mode of presentation*, naturally associated with the name “George Eliot”, Reid does know that she wrote *Middlemarch*. Under a different mode of presentation, naturally associated with the name “Marian Evans”, Reid does not know that she wrote *Middlemarch*.¹⁵ As for knowledge, so for epistemic probabilities. For example, it might be probable on Reid’s evidence that this person is a man, under the *George Eliot* mode of presentation, while the probability that of this same person being a man is much lower on the *Marian Evans* mode of presentation. And the same goes, in particular, for the epistemic probability that an individual is happy or sad.

Since epistemic properties of individuals — like their probability of being happy — are sensitive to a mode of presentation, unqualified statements about an individual’s epistemic properties — like “Al has probability 1/2 of being happy” — are generally *context-sensitive*. When we make such a statement, what we are saying is that Al has probability 1/2 *under M*, where *M* is some mode of presentation that can sensibly be inferred from context.

Consider the story of Near and Far again. Suppose, as it happens, that the two individuals in our story are called “Al Faro” and “Bea Nero”. (We can reason in just the same way if they are instead “Al Nero” and “Bea Faro”.) If we choose the option Near, what is the probability that Al Faro will be saved? This question does not have an unequivocal answer: it depends on a mode of presentation. Under the *Nero* mode of presentation, this person’s probability of being saved is zero. (Under this mode of presentation, he is certainly not on Near Island.) Under the *Al* mode of presentation, this very same person’s probability of being saved is 1/2. (Under this mode of presentation, he has probability 1/2 of being on Near Island.) This one person who goes by two

out that the world we live in is objectively deterministic, and there are no non-trivial chances at all. Even in that case, the questions in this book about the value of uncertain prospects remain pressing.

¹⁵What exactly *is* a mode of presentation? This is an old and hard question, which I will not try to answer. The key thing for us is just the role they play in determining epistemic probabilities about individuals. Mahtani (in both 2017, 2020) speaks of “designators”, and seems to think of them as linguistic expressions — like the *name* “George Eliot”; but modes of presentation need not be linguistic. Gustafsson and Kowalczyk (forthcoming), following others (see citations therein) favor an account in terms of *counterparts*. I don’t think that account is especially promising either, but the things I’ll say don’t turn on any of these details.

names, “Al” and “Faro”, has two different epistemic probabilities of being saved, corresponding to those two ways of presenting him.

Likewise, consider the question of which of the prospects Near or Far is *better* for Al Faro. The answer to this question also depends on which mode of presentation we select. Near is better for him under the *Al* mode of presentation: he has probability $1/2$ of being saved, rather than $1/4$. Far is better for him under the *Faro* mode of presentation: he will be saved with probability $1/2$, rather than probability zero.

(A point of clarification: we can make sense just fine of an ordering of *individual prospects* — probability distributions over individual outcomes — that does not depend at all on a mode of presentation. Rather, what depends on a mode of presentation is *which* individual prospect is *Al Faro’s*, for each option. What is *his* probability of being happy or sad?)

In contrast, how things *actually* go for an individual does not depend on a mode of presentation at all. We have supposed that Al is Faro, and Bea is Nero. In this case, if we save Near, then Bea gets saved, and Al does not.

Another more subtle contrast is between epistemic probabilities and *counterfactuals*. Suppose we save the Far island. Who *would* have been saved, had we chosen Near? This question is not sensitive to a mode of presentation either: counterfactuals are not an epistemic modality, but rather a *metaphysical* modality. If Al in fact is Faro, then Al *would have been* Faro, no matter what we chose, and no matter how the rescue attempt would have gone.

Evaluating social prospects involves the interaction between both kinds of modality at once: we are concerned with how *probable* it is that things *would* have gone a certain way, given a certain prospect. (For further discussion see appendix C.) Take the case of Near and Far again. It is epistemically possible that Al is Nero, and it is epistemically possible that Bea is Nero. So there are epistemically possible states of each kind. If Al is in fact Nero, then this is a matter of counterfactual necessity: Al would be Nero no matter what. If instead Al is Faro, then *this* would be so no matter what. We do not know whether *Al is Nero* or *Al is Faro* is metaphysically necessary, but one of them is.

We have often represented social prospects using two-dimensional matrices, with each row representing an individual and each column representing a possible state. Each cell (i, w) of the matrix represents how things would turn out (if the prospect were chosen) for individual i , if state w is actual. Whenever we have done this, we have implicitly relied on *presenting* each

individual in some particular way. Suppose that Al is Nero, though we do not know this. Under the *Nero* mode of presentation, we know that that Near would save him. Under the *Al* mode of presentation, we do not know whether Near would save him. That is, there is an epistemically possible (but metaphysically impossible) state where Al is Faro. Is this person Al Nero saved in this state? Under the *Nero* mode of presentation, yes; under the *Al* mode of presentation, no. So, should we put a happy face or a frowny face in the cell (i, w) of the matrix? This also depends on a choice of mode of presentation for i . In general, in order to use a matrix to represent a social prospect, we must make a choice of a mode of presentation for each individual.

For this representation to be sensible, these different modes of presentation for different individuals should be coordinated. A matrix with rows for *Al* and *Bea* is sensible; a matrix with rows for *Nero* and *Faro* is also sensible; but a matrix with a row for *Al* and a row for *Faro* is not sensible — even if, as it happens, these pick out the very same people as Al and Bea, respectively. The trouble is that, for all we know, Al and Faro are the same person, and in that case there is another person whose individual outcome will have been left off of the table. Let $M = (M_1, M_2, \dots)$ be a sequence of modes of presentation individuals $i_1, i_2, \dots$ (respectively). For M to be an appropriate basis for a matrix representation of a social prospect, it should be *certain* with respect to M that $i_1, i_2, \dots$ are all distinct individuals. In what follows, let a *sequential mode of presentation* be understood to satisfy this condition.

In short, for social prospects A and B and an individual i , each of the following questions is sensitive to a mode of presentation:

- How probable is it that i would be happy, if A were chosen?
- What *individual lottery* — what probability distribution over outcomes — does A give i ?
- For a possible state w , if w is actual, if A were chosen, would i be happy?
- What *individual prospect* — what function from states to outcomes — does A give i ?
- Is A better than B for i on balance?
- What two-dimensional matrix of individual outcomes represents A ?
- Is A *ex ante* Pareto-superior to B ?

In contrast, these questions are *not* presentation-sensitive:

- Would *A* actually make *i* happy?
- Would *A* turn out better for *i* than *B*?
- Would *A* turn out better for *B* overall?
- For a metaphysically possible world *w*, is *i* happy in *w*?¹⁶
- Is metaphysically possible world *w* better for *i* than *v*?
- Is world *w* Pareto-superior to world *v*?

Of course, we may be *uncertain* about the answers to any of these questions, and the *epistemic probabilities* we assign to different possible answers will be presentation-sensitive. But what the answer to the question actually *is* is not presentation-sensitive.

Now let's return to the Ex Ante Pareto principle. What we have found is that there is not really one such principle, but many. This is because there are many different relations of *ex ante* Pareto superiority. For *any* sequential mode of presentation *M* for a sequence of individuals, we can consider this relation:

For every individual *i*, *A* is better than *B* for *i* with respect to *M_i*.

Call this relation ***M*-Pareto superiority**. Each choice of *M* thus gives rise to a corresponding principle, which may or may not be true:

***M*-Pareto.** If *X* is *M*-Pareto superior to *Y*, then *X* is better than *Y* overall.

When we put forward the Ex Ante Pareto principle without qualification, it is unclear which of these many principles we meant, if any of them. Mahtani (2017) proposes that we might instead have meant something “supervaluational” — *M*-Pareto holds for *every* *M*; or else we might have meant something “subvaluational” — *M*-Pareto holds for *some* *M*. But we need not settle this interpretive question. The important issue is which, if any, of these many candidate Ex Ante Pareto principles is correct.

One idea is that *M*-superiority only matters for some very *special* kind of mode of presentation *M* — perhaps even just one for each individual. But

¹⁶It may be tempting to think that *counterpart theorists* will think that this question is sensitive to a choice of *counterpart relation*, and that a counterpart relation is at least very close in spirit to a mode of presentation. But this is incorrect — as long as *metaphysically possible world* is understood as a *complete way the world could have been*. See section 3.3 for discussion.

this seems hard to sustain. (Compare discussion in Mahtani 2017, 2020; Chalmers 2011; Hawthorne and Manley 2012.)

A tempting thought is that the important thing is to pick out an individual using a *rigid designator* — a representation that picks out the *same* individual in every possible world. There are two ways to take this, depending on what we mean by “possible world”.

If we are talking about *metaphysically* possible worlds, then this proposal is no help at all. *All* of the modes of presentation we have been considering are rigid designators in this sense. “Al” and “Bea” are rigid names for particular people, and so are “Nero” and “Faro”. Metaphysical rigidity does not narrow down the range of *epistemic* profiles that an individual may have under a mode of presentation at all.¹⁷

So suppose instead we are talking about *epistemically* possible worlds: we might restrict *M*-Pareto to modes of presentation that are *epistemically* rigid, in the sense that they are *certain* to pick out a certain individual. What does this mean, though? Say *M* is a mode of presentation for an individual *i*. We might say *M* is epistemically rigid if this holds:

For some individual *j*, it is certain under *M* that *i* is *j*.

Assume for simplicity that only truths are certain. Then the only possible witness for this condition is *i* itself. So this is equivalent:

It is certain under *M* that *i* is *i*.

But if *i* is picked out under *M* twice over, then this will hold for *any* mode of presentation *M*. Of course, we can be uncertain of identities: this holds when we pick out the same individual twice over under two different modes of presentation. So we might take another run at it:

For some individual *j* and mode of presentation *M'*, it is certain under (*M*, *M'*) that (*i*, *j*) is an identity pair.

This is not trivial. But we see that this kind of epistemic rigidity is not a property of a single mode of presentation, but rather a *relation* between modes of presentation *M* and *M'*. We might say a pair of modes of presentation *M* and *M'* is *relatively rigid* iff it is certain whether they pick out the *same* individual: that is, if *M* picks out *i* and *M'* picks out *j*, then it is certain

¹⁷Indeed, it is not totally clear to me what it even *means* to say a mode of presentation is metaphysically rigid.

under (M, M') whether (i, j) is an identity pair.¹⁸ But this relative notion of epistemic rigidity does nothing to pick out some modes of presentation as more suitable than others for the Ex Ante Pareto principle. All modes of presentation are rigid relative to themselves, and non-rigid relative to some others.

In order to choose special modes of presentation, a formal criterion like rigidity won't do; we would need some more *substantive* criterion, such as a special epistemic status. There is a tradition tracing to B. Russell (1910--1911) of singling out an intimate epistemic relation of *acquaintance* — so tight that it makes us immune to Frege puzzles of misidentification. If you are acquainted with an individuals i and j , then there are special mode of presentation for i and j with respect to which you cannot make a mistake about whether i and j are the same. But it is hard to see how we could stand in this kind of relation to *anyone*, except perhaps ourselves. So I don't see how acquaintance-based modes of presentation can help us recover a version of the Ex Ante Pareto principle that can be usefully applied to questions about social risk.

This is not to say that Ex Ante Pareto must be abandoned. Rather than trying to find some *special* modes of presentation to fill it in, we might say that it holds for *many* modes of presentation — perhaps even *all* of them.

Let's consider some class of *appropriate* (sequential) modes of presentation. This is a placeholder: we won't need to say exactly what these modes of presentation are, but we will later introduce assumptions about their structure — that the class is closed under certain natural ways of building new modes of presentation from old.

General Ex Ante Pareto. For every *appropriate* sequential mode of presentation M , if X is M -Pareto superior to Y , then X is better than Y overall.

This turns out to be a very powerful idea.

7.4 From modes of presentation to permutations

What we have found so far is that a social prospect can assign the very same individual different *individual prospect*, with respect to different ways of

¹⁸More generally, let M and M' be sequential modes of presentation for individuals $i_1, i_2, \dots$. Then we might say M and M' are *relatively rigid* iff for any numbers m and n , it is certain under (M, M') whether (i_m, i_n) is an identity pair.

presenting that individual. Thus the Ex Ante Pareto is ambiguous between many different *M-Pareto* principles, for different modes of presentation *M*. It is difficult to find a way of singling out very special modes of presentation. So, if we are to affirm any version of Ex Ante Pareto, plausibly we should affirm it for a range of *different* modes of presentation. This idea has important ramifications.

We saw in section 7.1 that the combination of Ex Ante Pareto, together with Impartiality and Statewise Equivalence, is powerful enough to imply Risk Neutrality (and more). But Ex Ante Pareto with respect to *the right modes of presentation* is powerful enough to secure Risk Neutrality even *without* the other two principles. This is because reasoning about modes of presentation can serve as a substitute for certain kinds of statewise reasoning.

Indeed, rearranging well-being between different individuals is closely related to representing the *same* distribution of well-being in a different way (Thomas 2019, sec. 3.2). Consider two matrices representing social prospects. In *X*, which person is happy is determined by a coin flip: Jay if Heads, and Kay if Tails. In the matrix *Y*, we have just rearranged well-being between different individuals in each column.

X	Heads	Tails
Jay	😄	😞
Kay	😞	😄

Y	Heads	Tails
Jay	😄	😄
Kay	😞	😞

There are two ways of interpreting this transformation. (Recall our discussion of *active* and *passive* transformations in section 3.3.) The most natural way is “active”, understanding *X* and *Y* as representing different prospects: while *X* gives Jay and Kay each probability 1/2 of being happy, *Y* makes Jay happy for sure and Kay sad for sure. But *probability* and *sure** are presentation-sensitive. To be more explicit, this is to say *X* gives Jay and Kay each probability 1/2 of being happy, *with respect to a certain mode of*

presentation *M*, while *Y* makes Jay happy for sure and Kay sad for sure, with respect to the *same* mode of presentation *M*.

There is also an alternative “passive” interpretation available: we can interpret *Y* as representing the very *same* prospect as *X* with respect to a *different* mode of presentation. Say *X* represents Jay and Kay’s well-being with respect modes of presentation *Jay* and *Kay*, associated with the names “Jay” and “Kay”. Then we could introduce two new names:

Let “Harper” be a name for whoever *X* makes happy, and let
“Sadie” be a name for whoever *X* makes sad.

Then the epistemic probability that Harper is happy and Sadie is sad is one. Furthermore, if the coin actually comes up Heads, then Jay is Harper, and Kay is Sadie. While Jay has probability 1/2 of being happy under the *Jay* mode of presentation, he has probability one of being happy under the *Harper* mode of presentation: with respect to *Harper*, Jay is happy in every possible state, no matter how the coin comes up. Likewise, while Kay has probability 1/2 of being happy under *Kay*, with respect to *Sadie*, is sad in every possible state. In short, the social prospect that *X* represents with respect to (*Jay*, *Kay*) is the *very same prospect*, for the very same people, that *Y* represents with respect to (*Harper*, *Sadie*).

Does this mean that the social prospects represented by *X* and *Y* are equally good? Here we must be careful: the question is ambiguous. If we interpret *X* and *Y* as representing the *same* prospect, in two different ways, then of course they are equally good: any prospect is just as good as itself. But if we interpret *X* and *Y* as representing *different* prospects, related to one another by rearranging well-being in each state, then *no* — it does not follow that they are equally good. Impartiality is a substantive thesis, and mode-of-presentation tricks do not trivialize it.

Thomas (2019, sec. 3.2) argues based on modes of presentation for (essentially) the following principle.¹⁹ (For a more precise statement see section C.2.)

Statewise Permutation Invariance. Suppose that a pair of social prospects *A* and *B* is related to another pair of prospects *A'* and *B'* simply by uniformly rearranging well-being between individuals in each state,

¹⁹Rather than betterness, Thomas’s argument concerns the *deontic* status of social prospects — whether it is permissible to choose an option out of a certain menu of options. He calls his principle Statewise Impartiality.

and A is as good as B , then A' is as good as B' .

The argument goes like this. Consider two matrices X and Y representing A and B . If (A', B') is related by a statewise permutation to (A, B) , then A' and B' are represented by a pair of statewise-permuted matrices X' and Y' (respectively). (That is, each column of X' is a permutation of the corresponding column of X , and each column of Y' related by the *same* permutation to the corresponding column of Y .) But then we can reinterpret X' and Y' as representing the *same* prospects as X and Y (respectively), with respect to alternative modes of presentation for the individuals involved. That is, X' represents A , and Y' represents B . If A is as good as B , then on this interpretation, of course the prospect represented by X' is as good as the prospect represented by Y' . So (we might conclude) A' is as good as B' .²⁰

That last step of the argument relies on a crucial assumption: if two pairs of options can both be represented by the same pairs of matrices, then their value comparisons must go the same way. In short, the matrices include all the information we need for value comparisons — without any further concerns about *how* those matrices represent individual prospects. This assumption seems plausible enough — and it is often made implicitly in discussions of welfare ethics — but it is substantive, and not obviously correct.

It might be instructive to compare this simpler argument for a stronger conclusion. Consider two possible worlds w and v whose distributions of well-being are related by a permutation of individuals. Must v and w be equally good? We should hope not — as we discussed in chapter 3, applying such an isomorphism principle to infinite worlds leads to the infinite futility problem. But we might be tempted by the following reasoning. We can represent the world w with a list of well-beings, with one item for each individual. We can also use the very *same* list to represent v , simply by reinterpreting the positions in the list as standing for different individuals. Since the same list represents both worlds, we might think they must be equally good.

But this argument is not persuasive. The very same representation can be

20

... [I]f cases X and Y stand in the relation supposed by statewise impartiality, then, by changing the way in which we designate people ('Adam' and 'Eve' or 'Alan' and 'Bea'?), we can transform the payoff tables in X into payoff tables isomorphic to those in Y . So X can be treated in the same way as Y . (Thomas 2019, 481)

used to represent very different things, which differ morally. If we used the same word ambiguously to mean either “happy” or “sad”, that wouldn’t make it just as good to be happy as it is to be sad. We could use the same position on a list to stand for a human or a fish, but that would not make fish equally important as humans.

In the same way, it is not at all clear that two *pairs of prospects* represented by the same pair of matrices (interpreted in different ways) must be treated the same. If not, then uninterpreted matrices do not include all the information we need for value comparisons, but so it may go. Indeed, this is a strengthening of the principle of Permutation Invariance discussed in section 4.2 — and there we found that there are strong reasons to reject this principle. (It is jointly inconsistent with Pareto and Comparability.) The principle is substantive, and not obviously correct. It cannot be established just on the basis of considerations about what alternative representations are possible for the same case.

So we cannot derive Statewise Permutation Invariance directly from considerations about modes of presentation. Nonetheless, we *can* derive some important conclusions. There may well be important facts about the relative value of prospects that are not unambiguously represented by their matrices. But *some* relationships certainly *are* captured in this way. In particular, relations of *ex ante* Pareto superiority are very closely tied to matrix representations.

Let $M = (M_1, M_2, \dots)$ be a sequential mode of presentation for individuals $i_1, i_2, \dots$. Let π be a *statewise permutation*: that is, for each state w , let π_w be some permutation of these individuals. A social prospect A assigns an individual outcome to each individual i , in each state w , with respect to M . Call this $A_M(i, w)$. So A_M is the matrix representation for A with respect to M . We can also apply the statewise permutation π to A_M to get a new collection of individual prospects,

$$(\pi A_M)(\pi_w(i), w) = A_M(i, w).$$

So πA_M is another matrix of individual outcomes. This can be interpreted in two different ways. We can interpret πA_M as representing a *new* prospect with respect to the *same* mode of presentation M . Or we can interpret πA_M as representing the *same* prospect A , with respect to a *new* mode of presentation M^π . To be concrete, we can introduce these modes of presentation like this:

For each number n and each state w , if w is actual, then let M_n^π

pick out the individual $\pi_w(i_n)$, where i_n is the individual picked out by M_n .

Call this a *conditional introduction* of a mode of presentation. We know that M^π picks out the same individuals as M , though we may not know in what order.

Now consider two social prospects A and B , a mode of presentation M , and a statewise permutation π . Suppose that the *permuted prospect* represented by πA_M (with respect to M) is M -Pareto superior to the permuted prospect represented by πB_M (also with respect to M). That is to say, each individual prospect $\pi A_M(i, -)$ is better than the corresponding individual prospect $\pi B_M(i, -)$. Then this relationship still holds on the *alternative* interpretation. For each individual i , the individual prospect that A gives i under the conditionally introduced mode of presentation M^π is the exactly the same as the individual prospect that πA_M gives i under M . The same goes for B . Since each of these individual A -prospects is better than the corresponding B -prospect, it follows that A is M^π -Pareto superior to B .

In short, if we can *permute* a pair of prospects to get them into a relationship of Pareto-superiority, then the *original* prospects *also* stand in a relationship of Pareto-superiority, with respect to an alternative mode of presentation.

For any social prospects A and B , any sequential mode of presentation M , and any statewise permutation π , if $\pi^M A$ is M -Pareto superior to $\pi^M B$, then A is M^π -Pareto superior to B .

Say that a class of modes of presentation $\mathcal{M}$ is **closed under conditional introductions** iff, whenever M is in $\mathcal{M}$, each statewise permutation of M is also in $\mathcal{M}$. Suppose that General Ex Ante Pareto, where the *appropriate* modes of presentation are some class that is closed under conditional introductions. This implies the following:

Permuted Pareto. For any social prospects A and B , any sequential mode of presentation M in the class $\mathcal{M}$, and any statewise permutation π , if $\pi^M A$ is M -Pareto superior to $\pi^M B$, then A is better than B overall.

Let's just assume that we are representing social prospects using *some* appropriate mode of presentation. If we hold fixed, then we don't need to keep repeating it over and over. Then we can restate the principle more simply:

Permuted Pareto. For any social prospects A and B , and any statewise permutation π , if πA is Pareto superior to πB , then A is better than B overall.

An exactly analogous principle of Permuted Pareto *Equivalence* can also be stated. Again, these each follow from the corresponding Ex Ante Pareto principle, together with the idea that any statewise permutations of our “base” mode of presentation is also appropriate — that is, in $\mathcal{M}$.

(We could also consider various ways of restricting the class of statewise permutations, thus weakening our assumptions about appropriate modes of presentation, which will not make much difference to the arguments that follow. We could restrict attention to statewise *finite* permutations. We could also require that the function from states to permutations is *measurable*.)

7.5 The power of Pareto

The key point from the preceding section is that, assuming the repertoire of appropriate modes of presentation is expressive enough, *General Ex Ante Pareto* effectively does much of the same work that our original Ex Ante Pareto (for fixed modes of presentation) can do in *combination* with Statewise Equivalence and Impartiality. But, as we saw in section 7.1, this combination is very powerful.

Consider once more the case of Near and Far: save Nero for sure, or save Faro with probability $1/2$? Initially it seemed that Ex Ante Pareto would not be enough to settle which of these is better, since Near does not Pareto-dominate Far — under the *(Nero, Faro)* mode of presentation. But suppose we also affirm the *(Al, Bea)*-Pareto principle. (And why shouldn’t we?) Under this mode of presentation, Near *does* Pareto dominate Far, and thus this principle implies that Near is better than Far after all.

This idea massively generalizes (compare Mahtani 2017, 315–16; 2020, 2090–91; McCarthy, Mikkola, and Thomas 2020; Thomas 2019).

Consider again the argument for Risk Neutrality from section 7.1. We showed that the prospects *Sure* (save one for sure) and *Risky* (probability $1/5$ of saving five) were equally good, by combining Ex Ante Pareto with Statewise Dominance and Impartiality. But if we have a free hand with modes of presentation, then we can drop the appeal to Statewise Equivalence and Impartiality in this argument. Here is the trick. Let s be the person on Sure Island, and let $r_1, \dots, r_5$ be the five people on Risky Island. Now throw a fair six-sided die, and don’t look at the result. We will introduce six new names $Randy_1, \dots, Randy_6$ according to this rule (table 7.4): if the die comes up 1, then $Randy_1, \dots, Randy_5$ are $r_1, \dots, r_5$, respectively, and $Randy_6$

is s . If it comes up 2, then Randy₁, ..., Randy₄ are $r_2, \dots, r_5$, respectively, Randy₅ is s , and Randy₆ is r_1 . If it comes up 3, then Randy₁, ..., Randy₆ are $r_3, r_4, r_5, s, r_1, r_2$, respectively. And so on, cycling all the way around.

Table 7.4: New modes of presentation for the people on Sure and Risky Islands.

Die	1	2	3	4	5	6
Randy ₁	r_1	r_2	r_3	r_4	r_5	s
Randy ₂	r_2	r_3	r_4	r_5	s	r_1
Randy ₃	r_3	r_4	r_5	s	r_1	r_2
Randy ₄	r_4	r_5	s	r_1	r_2	r_3
Randy ₅	r_5	s	r_1	r_2	r_3	r_4
Randy ₆	s	r_1	r_2	r_3	r_4	r_5

Now let's consider our options Sure and Risky, and the prospects they each offer the people Randy₁, ..., Randy₆ (under those modes of presentation). If we choose Sure, then Randy₁ is saved just if the die comes up 6, so the probability is $\frac{1}{6}$. If we choose Risky, then Randy₁ is saved just if the die comes up between 1 and 5, *and* the rescue is successful. The probability of this is $\frac{5}{6} \cdot \frac{1}{5} = \frac{1}{6}$. So Randy₁ has the same probability of being saved either way. And exactly symmetrical reasoning applies to Randy₂, Randy₃, ..., Randy₆. Each person has probability $\frac{1}{6}$ of being saved whichever option we choose. So it turns out that Sure and Risky are *ex ante* Pareto equivalent after all!

We can restate the point in terms of Permuted Pareto, slightly modifying the argument from section 7.1.²¹ In the original Sure and Risky choice, one individual (s) faces certain happiness under Sure, and certain sadness under Risky. The other five individuals ($r_1, \dots, r_5$) face certain sadness under Sure, and probability 1/5 of happiness under Risky. But now we can consider an alternative choice: we give each of the six individuals an *equal chance* at each of these six individuals' predicaments, based on the roll of a fair die. Call these two new prospects Lottery-Sure and Lottery-Risky. In this version,

²¹This argument is very close to the one given by McCarthy, Mikkola, and Thomas (2020, 82, theorem 2.2), except that it uses an even weaker assumption than Statewise Anonymity. It is also strongly reminiscent of Harsanyi's "veil of ignorance" style argument for utilitarianism (Harsanyi 1975).

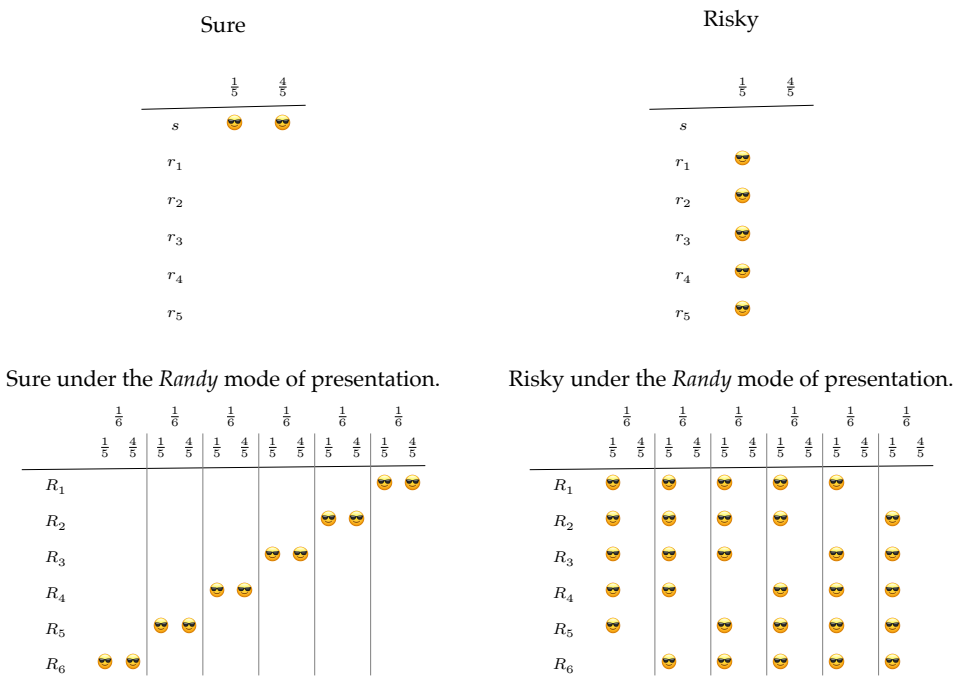


Figure 7.6: The same Sure and Risky prospects under alternative modes of presentation.

Sure gives each individual probability $1/6$ of being saved (in the case where they get to be the person on Sure Island), and Risky *also* gives each each individual probability $1/6$ of being saved (probability $5/6$ of being on Risky Island, times probability $1/5$ of the Risky rescue succeeding). These are *ex ante* equivalent. Furthermore, the pair of Lottery-Sure and Lottery-Risky are related to Sure and Risky by a *statewise permutation*. So *Permuted Pareto Equivalence* implies that Sure and Risky are equally good.

As we discussed in section 7.4, these two arguments — one based on randomized *modes of presentation*, and the other based on randomized *individual outcomes* — are structurally identical. Each of them turns on applying a statewise permutation uniformly to the two prospects; the difference is just in whether we interpret this permutation as representing a change in the *prospects* Sure and Risky, or a change in how we *represent* the very same prospects.

This argument is perfectly general: Permuted Pareto implies Risk Neutrality, and the Expected Number view more generally. Here is the idea. Let A and B be prospects that just involve making k people happy or sad. Then we can *randomize* each of these two prospects, constructing new prospects A^* and B^* , where A^* gives each individual chance $1/k$ of facing every possible individual predicament in A , and the same for B^* and B . We can do this in a way where (A^*, B^*) is related to (A, B) by a statewise permutation. Then each individual's probability of being happy under A^* is the same as the *average* of *every* individual's probability of being happy under A ; and the same goes for B^* and B . Moreover, a prospect's *expected number* of happy people (for a fixed population of size k) is exactly the same as k times its *average probability* of happiness. So if A has the same expected number of happy people as B , then A^* and B^* each give every individual exactly the same probability of happiness. So A^* and B^* are *ex ante* Pareto equivalent; by Permuted Pareto, A and B are equally good. Likewise, if A has a greater expected number of happy people than B , then A^* gives each individual a strictly *greater* probability of happiness than B^* — and by Permuted Pareto, A is *better* than B .

[TODO. What about vector EU? Check MMT on Independence.]

Theorem 7.3. Permuted Pareto implies Expected Number (and thus Risk Neutrality).

Thus, if the class of “appropriate” modes of presentation is closed under

conditional introductions, then General Ex Ante Pareto also implies the Expected Number view. This means that General Ex Ante Pareto is a very strong principle. Mahtani (2017) objects to it on the basis of this strength.²² Gustafsson and Kowalczyk (forthcoming, 10) regard it more approvingly.

But in a setting with *infinitely* many people, the strength of General Ex Ante Pareto becomes clearly unacceptable: it pulls us right past Expected Number all the way to *contradiction*.

7.6 From Pareto to paradox

Let's sum up the dialectic so far. In section 7.1, we considered an argument for Risk Neutrality based on Ex Ante Pareto, Statewise Dominance, and Impartiality. But in section 7.2, we found that the same three principles are *jointly inconsistent* when applied to infinite populations. What are we to make of this?

In section 7.3 I noted that Ex Ante Pareto is in fact presentation-sensitive, and I argued that it is hard to find a principled reason to affirm it with respect to *certain* modes of presentation, without also affirming it with respect to others. Thus we are pressed toward what I called *General Ex Ante Pareto*. In section 7.4, we found that this implies a kind of hybrid principle — *Permuted Pareto* — which combines the Ex Ante Pareto idea with a version of Impartiality and Statewise Dominance. Thus General Ex Ante Pareto *on its own* can do much of the same work that Ex Ante Pareto can do *in combination* with those other principles. In section 7.5 we saw that our original argument for Risk Neutrality can be run on the strength of General Ex Ante Pareto alone: Impartiality and Statewise Dominance drop out.

Now we will see that the same is true of the paradox from section 7.2. This too arises from General Ex Ante Pareto alone: again, Impartiality and Statewise Dominance drop out. This strongly suggests that the core problem here does

²²TODO. Fix this in relation to MMT. The principle is even stronger if we add the further assumption that value of an individual prospect is its *expected* value: in that case, general Ex Ante Pareto implies full-fledged expectational utilitarianism. Mahtani (2017, 316) says this consequence is “entirely opposed to the spirit of the concept” of *ex ante* pareto superiority, and an “uncharitable interpretation” Mahtani (2020, 2092). “Dialectically, nobody would accept the ex ante Pareto principle [so-construed] ... unless they were already committed to utilitarianism.” But one could just as well take this argument as applying dialectical pressure *toward* utilitarianism. Indeed, as Mahtani (2020, 2091) and Gustafsson and Kowalczyk (forthcoming, 10) each note, this argument amounts to a variant of Harsanyi’s argument for utilitarianism.

not have to do with either of those other two principles. The problem is Ex Ante Pareto.

Let's look again at *St. Petersburg Teams*. There is a Red Team and a Blue Team. Each team is divided into infinitely many cohorts. Cohort n (of each team) has 2^n members. There are two prospects: *One Team Wins* and *Both Teams Win* (table 7.5). *One Team Wins* is certain to make at least as many people happy as *Both Teams Win*, and it has probability $1/2$ of making strictly *more* people happy. But the two prospects give each individual the same probability of happiness — under certain modes of presentation, that is. We might pick out an individual as, say, “Red Team, Cohort 2, person 12”. Under this mode of presentation, that individual has probability $1/8$ of being happy under either prospect. As we showed in section 7.2, *One Team Wins* and *Both Teams Win* give the same probability of happiness to each individual under these modes of presentation. So Ex Ante Pareto implies that they are equally good.

Table 7.5: The happy cohort in each possible state of St. Petersburg Teams.

State	One Team Wins	Both Teams Win
HH_1	Red 1	No one
TH_1	Blue 1	No one
$HH_n, n > 1$	Red n	Red and Blue $(n - 1)$
$TH_n, n > 1$	Blue n	Red and Blue $(n - 1)$

One important thing about *One Team Wins* and *Both Teams Win* is that, in every state where the St. Petersburg number n is greater than 1, the *same number* of people are made happy: 2^n . (For *One Team Wins*, these are either the 2^n people in Red n , or else the 2^n people in Blue n . For *Both Teams Win*, these are the 2^{n-1} people in Red $(n - 1)$ and the 2^{n-1} people in Blue $(n - 1)$.) So in each of these states, there is a permutation that *swaps* these two blocs of 2^n people. This defines a statewise permutation π that transforms *One Team Wins* and *Both Teams Win* into two new prospects, *One Team $^\pi$* and *Both Teams $^\pi$* .

If the St. Petersburg number is greater than 1, then *One Team $^\pi$* has exactly the same effect as *Both Teams Win*, and *Both Teams $^\pi$* has exactly the same effect as *One Team Wins*. In the remaining case, where the St. Petersburg

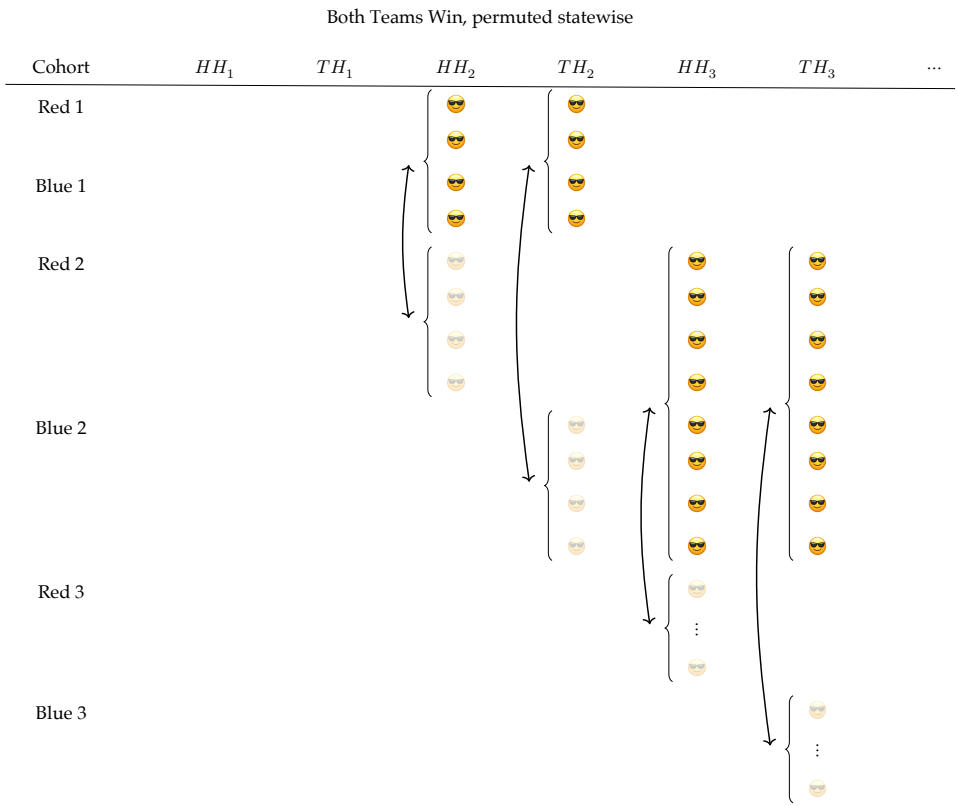


Figure 7.7: A statewise permutation of Both Teams Win.

number is 1, One Team $^\pi$ makes either the two Red 1 people happy, or the two Blue 1 people happy. In this case, Both Teams $^\pi$ makes no one happy. In other words, One Team $^\pi$ is exactly like Both Teams Win, except a little *better*, because it gives the four people in cohort 1 an extra way to be happy. And Both Teams $^\pi$ is exactly like One Team Wins, except a little *worse*, because it *takes away* cohort 1's chance at happiness. In particular, One Team $^\pi$ is *ex ante Pareto superior* to Both Teams $^\pi$.

Both Teams ^π										
Cohort	#	HH_1	TH_1	HH_2	TH_2	...	HH_n	TH_n	...	
Red 1	2									
Blue 1	2									
Red 2	4			😎						
Blue 2	4				😎					
⋮						⋮				
Red n	2^n						😎			
Blue n	2^n							😎		
⋮									⋮	

One Team ^π											
Cohort	#	HH_1	TH_1	HH_2	TH_2	HH_3	TH_3	...	HH_{n+1}	TH_{n+1}	...
Red 1	2	😎		😎	😎						
Blue 1	2		😎	😎	😎						
Red 2	4					😎	😎				
Blue 2	4					😎	😎				
⋮								⋮			
Red n	2^n								😎	😎	
Blue n	2^n								😎	😎	
⋮											⋮

Figure 7.8: The result of statewise-permuting each prospect.

Ex Ante Pareto told us that One Team Wins and Both Teams Win are equally good. But there is a statewise permutation that transforms One Team Wins

and Both Teams Win into One Team $^\pi$ and Both Teams $^\pi$, and One Team $^\pi$ is *Pareto superior* to Both Teams $^\pi$. Thus Permuted Pareto implies that One Team Wins is *strictly better* than Both Teams Win. This is a contradiction.

Moreover, as we have discussed, we can interpret this statewise permutation of St. Petersburg Teams as a mere change in how we *represent* the individuals involved. So this is a way of describing a direct clash between two applications of Ex Ante Pareto, with respect to two different ways of presenting the individuals involved. Ex Ante Pareto is not just inconsistent with Statewise Dominance and Impartiality: Ex Ante Pareto is inconsistent with *itself*.

The way I introduced the modes of presentation in this argument is rather baroque. This might seem suspicious. Maybe there is still room for making principled distinctions that would let us hold onto Ex Ante Pareto for *some* modes of presentation, but not *these* strange artificial ones. But in fact, the modes of presentation that give rise to the contradiction don't *have* to be any more strange or artificial than our usual ways of representing individual people.

Here's one way of filling in the story a bit. Here are infinitely many people, all standing in a long line. You go down the line and meet each of them. Look them in the eye, shake their hand. You get to know who each one of them is, under that mode of presentation. (You have a great memory for faces.)

You also have a very big telephone directory, listing the *same* infinitely many people by name. You go down the list of names, and you call up each person. You have a heart to heart with each of them. You get to know their voice, their life story, their hopes and fears. (You can do this very quickly, if you have to.) You know each of them as a unique and special person, under the *name-and-life-story* mode of presentation.

You are now acquainted with each person under two different modes of presentation. Call these two families of modes of presentation $M = (M_1, M_2, \dots)$ and $M^* = (M_1^*, M_2^*, \dots)$. You know each person under some M_i and some other M_j^* , but you do *not* know how these different families of modes of presentation are coordinated with one another: you don't know whether M_1 picks out the same person as M_1^* , or M_{532}^* , or whatever. Each of these families of modes of presentation is as normal and natural as mode of presentation for other people can be.

Now we let you in on something. As it turns out, the same coin flips we use to play St. Petersburg Teams were *also* used to arrange the line of people you met, as well as to assign people names for the telephone directory. This was carefully orchestrated so that the two families of modes of presentation are related to one another by precisely the same statewise permutation we used to transform One Team Wins and Both Teams Win into One Team* and Both Teams*. For example, we will tell you that, *if* the St. Petersburg number is 1, then $M_1, M_2, \dots$ and $M_1^*, M_2^*, \dots$ pick out the same people in the same order. If the St. Petersburg number is 2, and the team-choosing team came up Heads, then M_1 and M_2 correspond to M_5^* and M_6^* , while M_5 and M_6 correspond to M_1^* and M_2^* , and otherwise M_i and M_i^* both pick out the same person. And so on, for every other possible state. This is an entirely symmetric relationship between M and M^* : each is related to the other by the same statewise permutation.

And now, at last, we can describe the *very same situation* to you in two different ways: as One Team Wins vs. Both Teams Win (with respect to M), or as a choice that is isomorphic to One Team $^\pi$ vs. Both Teams $^\pi$ (with respect to M^*). Neither of these descriptions seems to have any priority over the other for you. If you think about each of the people you met, you are exactly as confident that *this person* will be happy in One Team Wins as you are that they will be happy in Both Teams Win. If you think about each of the people you spoke to on the phone, then you are *at least* as confident that this person will be happy in One Team Wins as you are that they will be happy in Both Teams Win — and for four of these people, your probability is 1/4 that One Team Wins will make them happy, while you are certain that Both Teams Win will make them sad.

With respect to one eminently reasonable mode of presentation (look them in the eye, shake their hand), Ex Ante Pareto implies that One Team Wins and Both Teams Win are *equally good*. With respect to another equally reasonable mode of presentation (call them up, learn their name and story), Ex Ante Pareto implies that One Team Wins is *strictly better* than Both Teams Win. Both conclusions cannot be true. But it is hard to see any reason to accept one version of Ex Ante Pareto but not the other. So we should not accept either version.

Our investigation in this chapter also gives us some insight into *why* Ex Ante Pareto might fail. In section 6.4, I proposed that some apparent dimensions of value are really just a motley mishmash, such as gerrymandering individuals into arbitrary subpopulations. Value for all individuals on balance “aims at”

value for *particular* individuals. The collective value of well-being is a mere “balance sheet” for weighing up what is better or worse for each individual. But, while this same value can in a sense be “decomposed” into values for *arbitrary groups* of individuals, it does not “aim at” such values, even in part. This (admittedly vague) idea gave us a principled reason for rejecting the principle of Subpopulation Dominance, while still maintaining the (Ex Post) Pareto principle. I proposed that a similar treatment can be given to *times*: there are many different ways of carving up space-time into “time-slices”, and any particular choice is arbitrary, not particularly natural, and not the sort of thing that overall value aims at.

We can say something similar about *individual prospects*. Individual prospects may well have seemed like a good candidate for what the value of a social prospect is aiming at. But in fact, an individual prospect is more of a motley mishmash than it initially seemed. We can “carve up” a social prospect into individual prospects in many different ways, corresponding to different individual modes of presentation. None of these ways of doing it seems particularly more natural than the others. Indeed, the value of a prospect for an individual is a bit like social value *at a time* for a group of spatially spread out individuals. We might say that epistemic probability for an individual, like simultaneity, is “frame-relative”, in that it depends on certain representational choices. But in that case, the dominance principle corresponding to this dimension of value — Ex Ante Pareto — is in no better standing than Subpopulation Dominance or Timewise Dominance. We should reject it.

Furthermore, this reason for rejecting Ex Ante Pareto is specific to *ex ante* evaluation. The unnaturalness of individual prospects does not arise from anything about *individuals*, but rather from something about *epistemic probabilities*. Alternative modes of presentation are relevant to epistemic probabilities concerning specific individuals, and they are thereby relevant to how good an uncertain prospect is for a specific individual. But modes of presentation are not relevant at all to the objective value of worlds. These considerations do not give us any reason to regard the question of whether an *outcome* is better or worse for an individual as somehow arbitrary or gerrymandered. We can still affirm the idea we began with in chapter 3: whether a particular individual is happy or sad is the sort of thing that matters, and the sort of thing that is to be weighed up when we consider the *overall* value of well-being for different individuals, on balance. Thus we also have a response to this particular version of the *bad company* objection to *ex post* Pareto principle.

7.7 What comes next

In section 7.1, I discussed a *two-dimensional* argument for Risk Neutrality, based on the combination of Ex Ante Pareto with Statewise Dominance. This argument seemed very tidy and powerful — but it has not fared so well. Its premises are jointly inconsistent: this is one member of the family of paradoxes of dominance from chapter 6.

But it turns out that Ex Ante Pareto is on shakier ground than it may have seemed. Even stating it clearly requires making some choice of a *mode of presentation* for each individual. There is no obvious default choice. And if Ex Ante Pareto holds on multiple disambiguations, with respect to a variety of different modes of presentation, then in fact it turns out to be internally inconsistent. This pushes us to the conclusion that Ex Ante Pareto is simply mistaken. This undermines the tidy and powerful argument for Risk Neutrality.

Can the argument be rehabilitated? Perhaps. There are restricted forms of Ex Ante Pareto that are strong enough to underwrite the argument for Risk Neutrality, but not so strong as to lead to contradiction. Perhaps we can affirm the principle when only *finitely* many individuals are implicated, but reject it in infinite cases. Of course we should not think that there are no differences between finite and infinite populations, and this might be one of them.

An argument for Risk Neutrality based on some finitary restriction of Ex Ante Pareto might well be sound. But it is on shaky ground, and hard to invest with much confidence. The only principled way I know of to motivate such a restricted principle is as a consequence of rather different ideas. Perhaps some restricted version of Ex Ante Pareto can be directly motivated, but I won't take up this challenge. (But see Kowalczyk 2025; Tarsney, Lederman, and Spears 2025.) Rather, I will leave Ex Ante Pareto and the argument of section 7.1 by the wayside.

In subsequent chapters, though, we will encounter a range of structurally related arguments that I find much more promising. In chapter 8, I will take up a different principle that is closely related to Ex Ante Pareto: *ex ante Separability* for populations. That is where we turn next.

Chapter 8

Distant Lives and Cosmic Value

The earth, that is sufficient,
I do not want the constellations any nearer,
I know they are very well where they are,
I know they suffice for those who belong to them.

Walt Whitman, "Song of the Open Road"

8.1 Separability

Consider a now-familiar story.

A tsunami is coming. Two populated islands are in its path, and both of their early-warning systems are malfunctioning. You don't have cell service, but you do have your speed boat.

The Small Island has a population of 100 happy people. If you go there, you are sure that you can get all hundred people to safety.

The Big Island has a population of 1,000,000 happy people. If you can make it to the Big Island in time, you can get all million people to safety. But the Big Island is further away than the Small Island, and weather conditions are bad. You almost certainly won't make it in time to save them from the tsunami. You *might* make it, though — the chances are one in a thousand.

The islands are in opposite directions, so there's no way to save both islands. Everyone on the island you *don't* go to will be hit by the tsunami — making their life sad, overall. Which way should you go?

Table 8.1: Another tsunami.

Probability:	10^{-3}	$1 - 10^{-3}$
Small	10^2 happy lives	10^2 happy lives
Big	10^6 happy lives	0 happy lives

Risk neutrality with respect to happy lives says that probability 1/1000 of saving a million lives is just as good as saving a thousand lives for sure — and thus attempting to save the Big Island is better than saving the Small Island, despite the small chance of success.

Perhaps, though, we should instead be *risk averse* with respect to happy lives. For example, instead of valuing prospects according to their expected number of happy lives, we might instead value them according to the expectation of some other quantity, which we call *utility* (see section 1.2.3). For concreteness, suppose that the utility of a number of happy lives is given by the square root function:

$$U(n) = \sqrt{n}.$$

The expected utility of Small is

$$U(100) = 10.$$

Meanwhile the expected utility of Big is

$$10^{-3} \cdot U(10^6) = 10^{-3} \cdot 10^3 = 1.$$

So even though Big has the greater expected *number of lives*, Big's expected *utility* is less than that of Small. This utility function, which bends downward as the numbers get bigger, recommends taking the safer bet. So, it seems, risk neutrality recommends going Big, while risk aversion recommends going Small.

But wait — is this really all that is going on? The world is bigger than just two islands. What about the rest of it? We have described the case in terms

of the *difference* you might make to the number of happy lives — you might save a hundred, or a million, or none. But that is not a complete description. For example, suppose that in addition to the Big and Small islands, there is a background population of a hundred million happy people living elsewhere, safe from tsunamis. Then the case is more properly described as in table 8.2 instead.

Table 8.2: Including a background population of a hundred million happy people.

Probability:	10^{-3}	$1 - 10^{-3}$
Small + Background	$10^8 + 10^2$ happy lives	$10^8 + 10^2$ happy lives
Big + Background	$10^8 + 10^6$ happy lives	$10^8 + 0$ happy lives

If the risk-neutral Expected Number view is correct, then adding in the background population makes no difference to the final verdict. The expected total number of happy lives if we save the Big Island is

$$\begin{aligned} &10^{-3} \cdot (10^8 + 10^6) + (1 - 10^{-3}) \cdot 10^8 \\ &= 10^8 + 10^3. \end{aligned}$$

Meanwhile, the expected total number of happy lives if we save the Small Island is

$$10^8 + 10^2.$$

So the comparison goes the same way as it did when we ignored everyone beyond the two islands. The *difference* in expectation between Big and Small is the same as their *expected difference*. So in order to figure out which expectation is greater, we really only need to pay attention to the *difference* in the number of happy lives between the two prospects.

But if the Expected Number view is *false*, then things are not so straightforward. We saw that when we considered the two islands in isolation, a square-root utility function gave us the verdict that saving Small is the better option. Now, though, consider how things go if we add in an additional hundred million happy lives, far away from these islands and unaffected by

the tsunami. In this case, the expected utility of Small is¹

$$U(10^8 + 10^2) \approx 10^4 + \frac{1}{2} \cdot 10^{-2}.$$

Meanwhile the expected utility of Big is

$$(1 - 10^{-3}) \cdot U(10^8) + 10^{-3} \cdot U(10^8 + 10^6) \approx 10^4 + \frac{1}{2} \cdot 10^{-1},$$

which is greater. So in this setting, the long shot attempt to save the Big Island is better on balance after all. In fact, the values assigned by the square root utility function approach risk neutrality as the background happy population increases.

The same goes for other theories that are not generally risk-neutral with respect to happy lives — such as risk-weighted expected utility or probability discounting (section 1.2). What they say about what is best overall will generally be highly sensitive to the big picture of what might be going on in the background.

When we posed the question of which islanders to rescue, we did not specify the background population of the Earth — or indeed the population of the distant past or future, or whether there are extra-terrestrials, or indeed *anything* about the distribution of well-being among other individuals who are unaffected by the choice between Big and Small. What's more, it seemed plausible that we didn't *need* to specify such extraneous background details in order to evaluate the two prospects. It seems like this choice can be evaluated on its own terms.

Of course, we are imagining a simple setting where all that is at stake in your choice is the amount of happy lives. In general, things may be more

¹We are using this approximation: for small ϵ ,

$$\sqrt{1 + \epsilon} \approx 1 + \frac{1}{2}\epsilon.$$

When 10^{2a} is more than an order of magnitude greater than 10^b , we have

$$\begin{aligned} \sqrt{10^{2a} + 10^b} &= \sqrt{10^{2a} \cdot (1 + 10^{b-2a})} \\ &\approx 10^a \cdot (1 + \frac{1}{2} \cdot 10^{b-2a}) \\ &= 10^a + \frac{1}{2} \cdot 10^{b-a}. \end{aligned}$$

complicated. Realistically, the total number of happy people also has knock-on effects — on resource consumption, the rate of innovation and economic growth, communities and relationships, and so on. These are all complications that we are setting aside.²

The core idea here is called *separability*. The rough idea is that what is going on elsewhere — perhaps in the distant past, or far away in space — should not matter for your choice. If there are certain lives that are “unaffected” in the sense that they are certain to go exactly the same way in each of two prospects *X* and *Y*, then for the purpose of comparing the value of *X* and *Y* with respect to overall well-being, we can simply set those unaffected lives aside. We can make the comparison between *X* and *Y* entirely on the basis of lives for which the choice between *X* and *Y* might make some difference.

This idea is the basis for so-called *Egyptology* arguments in the ethics of welfare aggregation, and especially in population ethics (McMahan 1981, 115). If we are deciding between saving Small or taking a chance at saving Big, it should not make a difference to our choice whether the ancient Egyptian civilization was large and prosperous or small and impoverished. As Parfit says (in a slightly different context), “research in Egyptology cannot be relevant to our decision” (1984, 420).

As we discussed, the Expected Number view implies separability in finite cases involving nothing but happy lives. (Again, this is because the *difference in expectation* is the same as the *expected difference* between two prospects.) It turns out that Expected Number is very nearly the *only* view with this property. Separability is the basis for a family of strong arguments for Risk Neutrality. Here is one.³

²An analogous point applies to the prudential value of happy life for an individual. In Pascal’s wager, he did not consider the question of how much happy life you might have already experienced. Happy life in the past is unaffected by your choice, and by hypothesis it also makes no difference to your chances of future happiness. So it seems that if we are evaluating a chance of gaining happy life, we can set the past aside, and simply focus on how the choice affects the future — insofar as your past does not make a difference to how the future goes. (This is to set certain complications aside. For example, if you have already lived a long time, then you might have made more promises that will need to be kept in the future. Or you might have had such a variety of experiences that the joy of novelty will be sapped from future experiences.)

³There is a family of Separability-based arguments for Risk Neutrality, and more generally for *expectational utilitarian* aggregation. See Thomas (2022) for overview. The argument I present here is based on J. S. Russell (2024); Kosonen (2022, sec. 3.3). (Hayden Wilkinson pointed out to me that Parfit 2003, fn. 16, gives a structurally-related argument for a different conclusion.) This argument is distinctive in that it uses especially spare background as-

Let's start by looking at how this goes in our concrete case of Big and Small. According to Big and Small, which prospect is better is the same no matter how many happy people there may be in far off realms. So let's consider a particular kind of information you might learn about those other separate lives.

In the distant past, a thousand-sided die was thrown. (See table 8.3.) This chance event determined two things.

First, this die roll determines the success of any attempt you may make to save the Big Island. If the die came up one, you can save the Big Island; otherwise, you will fail.

Second, the same die roll also determined the fate of a large background population, far away from Big and Small and unaffected by anything you might do. If the die came up one, then zero distant people have happy lives — everyone is sad. If it came up two, then there are two hundred distant happy lives. If it came up three, then there are four hundred distant happy lives. In general, if the die came up k , then exactly $100 \cdot (2k - 2)$ distant people are happy.

With this background context, now which prospect is better? Look at how they go.

Small + Random has chance $1/1000$ at resulting in 100 happy lives, chance $1/1000$ at 300, chance $1/1000$ at 500, and so on up to $100 \cdot (2,000 - 2)$ lives, and finally $100 \cdot (2,000 - 1)$ happy lives.

Big + Random has chance $1/1000$ at resulting in 200 happy lives, chance $1/1000$ at 400, chance $1/1000$ at 600 and so on up to $100 \cdot (2,000 - 2)$ lives.

sumptions — essentially just stochastic dominance. One branch of this family of arguments appeals to additional *topological* structure — the idea that value is *continuous* in a certain sense (for instance, Gorman 1968; see Broome 1991, ch. 4 for discussion). Beckstead and Thomas (2024) give a similar argument for Fanaticism that appeals to a *continuum* argument — a long sequence of prospects are strung together using the transitivity of betterness (see J. S. Russell 2024 for discussion). The argument I give here does not even rely on the assumption that betterness is transitive.

These separability-based arguments are closely related to arguments based on *Ex Ante Pareto*, discussed in section 7.1. But it is worth noting that the Separability principle stated below differs from *Ex Ante Pareto* in that it does not depend on any choice of mode of presentation for individuals. (But making precise the informal gloss in terms of a “near” and “far” part of the world *would* require choosing modes of presentation for the two corresponding subpopulations.)

That accounts for 999/1000 of the probability. In the remaining case, which also has probability 1/1000, Big + Random results in a million happy lives.

Every one of the outcomes of Big + Random is better than a corresponding equally-probable outcome of Small + Random, by at least a hundred lives! So surely Big + Random is better than Small + Random overall.

Here is another way of putting the point: we can *rearrange* the outcomes of Big + Random without affecting their probabilities. If we just switch around which die face corresponds to which outcome, this shouldn't make any difference to the prospect's value.

Table 8.3: Big and Small with a large uncertain background population. Here $n = 1000$. (So in particular $10^6 > 100n$.)

Probability:	$1/n$	$1/n$	$1/n$	...	$1/n$	$1/n$
Small	100	100	100	...	100	100
Big	10^6	0	0	...	0	0
Random	0	200	400	...	$100 \cdot (2n - 4)$	$100 \cdot (2n - 2)$
Small + Random	100	300	500	...	$100 \cdot (2n - 3)$	$100 \cdot (2n - 1)$
Big + Random	10^6	200	400	...	$100 \cdot (2n - 4)$	$100 \cdot (2n - 2)$
Big + Random, rearranged	200	400	600	...	$100 \cdot (2n - 2)$	10^6

If we do this, then we end up with Rearranged Big + Random, which is *certain* to turn out substantially better than Small + Random. We can sum this reasoning up with two familiar principles (see section 1.3.5).⁴ [stochastic-dominance] Call a prospect *simple* if it has only finitely many possible outcomes, each of which have rational probabilities.

Simple Stochasticism. Two simple prospects with the same probabilities for each outcome are equally good overall.

Strict Statewise Dominance. If X is certain to turn out better than Y , then X is better than Y overall.

By Simple Stochasticism, Big + Random is equally good as Rearranged Big + Random. By Strict Statewise Dominance, Rearranged Big + Random is better than Small + Random. So Big + Random is better than Small + Random. (Alternatively, we could appeal directly to the principle of (*Simple*) *Stochastic Dominance* — see section 1.3.5.)⁴

⁴In our present context, Stochastic Dominance is equivalent to the combination of Simple

This shows that there is *some* way of orchestrating our uncertainty about distant lives so that saving the Big Island is better than saving the Small Island, just by applying stochastic dominance reasoning. If *Separability* is correct, then our uncertainty about distant rearranged lives makes no difference to which is better — and so Big is better than Small *regardless* of the background context.

A point of clarification. As we said, the outcome of the background prospect Random does not affect the outcomes of Big or Small: we get the outcome of Big + Random just by adding up the happy lives from each component, rather than from some more complicated interaction between them. And we are adding precisely the same prospect Random to each of Big and Small. Even so, it is worth noting that the two prospects Random and Big are *correlated*, rather than probabilistically independent. This is consistent with the separability intuition; but if we are to dramatize that intuition in Parfit's way, saying "research into Egyptology cannot make a difference", then we have to be careful. Of course, Egyptological research *can* sometimes make a difference to our present choices: for example, learning about the fate of Egyptian projects might teach us how our own projects are likely to go. The sense in which Egyptology is plausibly irrelevant is that, if we take our choice (between Big and Small), it makes no difference if we simply *add on* some "Egyptian" prospect (like Random), in a way that leaves the original probabilities of the outcomes of either Big or Small intact. This is exactly what we did in the example. If we go on to learn how Random turns out, this would make a difference to our choice; what makes no difference is whether we add on Random as opposed to some *other* "Egyptian" prospect. (But this philosophical point is subtle; I won't try to defend it further here.)⁵

This argument generalizes. Consider some family of prospects where the

Stochasticism and Statewise Dominance. But that principle also makes sense in the von Neumann–Morgenstern framework, where we represent prospects as *lotteries*, rather than as functions from states to outcomes.

⁵Wilkinson (forthcoming) has argued that whether the "background" prospect is independent does make an important philosophical difference — partly in response to the impossibility result discussed in section 8.2. He offers a different argument for risk neutrality, which turns on a restricted form of Separability that requires the background prospect *B* to be probabilistically independent of each of the "foreground" prospects *X* and *Y*. The argument is based on Tarsney (2025) and Wilkinson (2022a), and it is closely related to the "cosmic background value" argument I discuss in section 8.4. This alternative argument faces some technical obstacles, for the reasons I discuss there.

What we will see in section 8.2 is that, instead of making a retreat to independent prospects, in fact even the unrestricted Separability principle can be consistently upheld.

only thing of value at stake is the number of happy lives.⁶ For any two such prospects X and Y , let $X + Y$ be the *statewise sum* of X and Y . That is, $X + Y$ is some prospect with the property that, with certainty, the number of happy lives resulting from $X + Y$ is the sum of the number of happy lives resulting from X and the number resulting from Y . (If the number of happy lives is all that matters, then we can arbitrarily choose *any* prospect with this property, and our arbitrary choice won't make any difference to how good the resulting prospect is.) Then we can write the Separability principle like this.

Separability. For prospects X, Y, Z ,

$$X \succeq Y \quad \text{iff} \quad X + Z \succeq Y + Z.$$

By generalizing the argument about Big + Random and Small + Random, we can prove the following.

Theorem 8.1. For simple prospects where only the number of happy lives is at stake, the Expected Number view is equivalent to the combination of Separability, Stochasticism, and Statewise Dominance.

In fact, we can generalize this further to *any* uncertain quantity Q . If prospects with the same probabilities of yielding any particular quantity of Q are equally good, and adding a fixed uncertain amount of Q makes no difference to value comparisons, then the value of a prospect is entirely determined by its expected Q -value. (This is made precise in section D.6.)

The Expected Number view implies Risk Neutrality, which in turn supports Pascal's principle. So Separability provides one way of filling the gap in Pascal's argument.

⁶Concretely, we can consider the simple worlds of section 3.2. Fix some "baseline" world z , and consider the equivalence class of all simple worlds that differ from z for only finitely many individuals. For any world w in this equivalence class, we can understand its "number of happy lives" $N(w)$ to be the *net* number of happy lives compared to z : that is, the number of people who are happy in w and sad in z , minus the number of people who are sad in w and happy in z . By Pareto and Impartiality, any two such worlds can be compared on the basis of their number of happy lives: w is as good as v iff $N(w) \geq N(v)$. Then we can let the relevant "family of prospects" be those whose possible outcomes are all in this equivalence class.

8.2 The SSK paradox

Separability is an attractive and powerful idea. Even so, in the past I have been inclined to reject it, due to a striking result originally proved by Seidenfeld, Schervish, and Kadane (2009; see also Goodsell 2021; my earlier reaction is in J. S. Russell 2024). I call it the *SSK paradox*. The paradox centers on a rather intricate case. (The case is closely related to the *St. Petersburg Teams* case from section 7.2.) We'll work up to it in four steps.

The first step is the famous St. Petersburg gamble (which I introduced in section 1.2).

Step 1. St. Petersburg. An angel flips a fair coin. If it comes up heads on the first flip, two individuals are happy. If the first heads is on the second flip, then four individuals are happy. If it's on the third flip, then eight, and so on. In the unlikely event that the coin *never* comes up heads, the game is called off and no one is made happy.

The original puzzle about this case (from N. Bernoulli 1713) had to do with the fact that its expected number of happy lives is apparently infinite or undefined. We won't let that bother us, though — we won't try to directly apply expectational reasoning to this prospect. We can use other tools to reason about its value.

Step 2. Delay. The angel flips the coin, but neither of you have seen how it landed yet. The angel says, "Hang on a sec, that one doesn't count. Let's start the game on the next flip. Is that a problem?"

So now we have two prospects: call them *Early* and *Late*.

Table 8.4: Delaying the St. Petersburg gamble.

	<i>HH</i>	<i>HTH</i>	<i>HTTH</i>	<i>...</i>	<i>TH</i>	<i>TTH</i>	<i>TTTH</i>	<i>...</i>
Early	2	2	2	...	4	8	16	...
Late	2	4	8	...	2	4	8	...

The outcome of Late is independent of the first flip in the sequence. If we get

heads on the *second* flip, there are two happy lives. If it's tails on the second flip and heads on the third, there are four happy lives. If the first heads after the first flip is on the fourth, then there are eight. And so on. (Again, if the coin never comes up heads, except perhaps on the first flip, then no one is happy.)

The natural response to the angel's question is, "No, it's not a problem." It seems that Early and Late are equally good. Each of them has probability $1/2$ of two happy lives, $1/4$ of four, $1/8$ of eight, and so on. It doesn't seem like the choice of *which* coins to play the St. Petersburg game with should make any difference to its value. (Indeed, I am personally on record succumbing to this temptation: see J. S. Russell 2024, and further discussion below.)

Let's continue to build up the case.

Step 3. Double up. All this so far has happened on the planet Alderaan, concerning happy Alderaanian lives. Meanwhile, another angel has appeared on the planet Bespin, and is offering an exactly similar choice between two St. Petersburg gambles, *Early B* and *Late B*. Neither planet's choice makes any difference to what happens on the other planet.

There are now four different prospects before us: Early A, Late A, Early B, and Late B. Each of the four prospects is a different isomorphic copy of the St. Petersburg gamble. So, on the face of it, all four prospects seem equally good.

Now the final step.

Step 4. Entangle. But there is a twist. The coin flips on Alderaan and the coin flips on Bespin are correlated with one another. (This is due to quantum entanglement.) In fact, every coin flip *after the first* is guaranteed to come up the same way on both Alderaan and Bespin. But the *first* coin flip is guaranteed to come up the *opposite* way on each planet.

The upshot is that Late A and Late B have *correlated* outcomes, while Early A and Early B are *anti-correlated*. This makes no difference to how the choice looks on Alderaan, and it also makes no difference for the choice on Bespin.

Table 8.5: Two copies of the choice between Early and Late St. Petersburg. (The columns are labeled with the sequence of coin flips on Alderaan.) The two Early options are anti-correlated, while the Late options are correlated. The sum of Early A and Early B strictly dominates the sum of Late A and Late B.

	<i>HH</i>	<i>HTH</i>	<i>HTTH</i>	<i>...</i>	<i>TH</i>	<i>TTH</i>	<i>TTTH</i>	<i>...</i>
Early A	2	2	2	...	4	8	16	...
Late A	2	4	8	...	2	4	8	...
Early B	4	8	16	...	2	2	2	...
Late B	2	4	8	...	2	4	8	...
Both Early	6	10	18	...	6	10	18	...
Both Late	4	8	16	...	4	8	16	...

Early A and Late A look equally good. So let's say that the Alderaanians choose Late — they might as well. And let's say that Bepin chooses Late as well. Both of their choices seem fine.

But now something weird happens. In fact, Both Late is guaranteed to result in two fewer happy lives than Both Early! (See table 8.5.) The happy lives will be differently arranged across the two planets: double-Late evenly distributes the happy lives across the two planets, while double-Early puts just two happy lives on one planet, and all the rest on the other. But we are supposing that only the number of happy lives is relevant here, not their distribution between planets.

If each planet chooses Late, that is just as good for that planet as choosing Early, and it is entirely irrelevant to what happens on the other planet. Yet somehow their *combined* choices make things *worse* overall. This is weird! How can we make things better overall for the *two* planets while making things no better at all for *either* planet?

One way of articulating the weirdness is by appealing to Separability. Suppose that Early A and Late A are equally good, and Early B and Late B are equally good. Then by applying Separability twice,

$$\text{Early A} + \text{Early B} \sim \text{Late A} + \text{Early B} \sim \text{Late A} + \text{Late B}.$$

But by Strict Statewise Dominance,

$$\text{Early A} + \text{Early B} \succ \text{Late A} + \text{Late B}.$$

Both conclusions cannot be correct. What gives?

In (2024, 573) I concluded, “This looks like very bad news for Separability.” But I now draw a different conclusion: these apparently equivalent prospects are not equally good. I previously expressed skepticism about this idea (2024, 573):

We would need to say that when it comes to gambles that only depend on the outcome of coin flips, it can make a moral difference *which* coin flips they are. ... This seems untenable. I am open to the idea that some kind of *isomorphism* between prospects which is stronger than stochastic equivalence is required to ensure that prospects are equally good. But my guess is that when we spell this notion out in any plausible way, the prospects [Early] and [Late] will still count as isomorphic even in the stronger sense.

I now think that this position that I judged “untenable” is the correct one: in fact, playing the St. Petersburg game with different sequences of coin flips *can* make a difference (compare Lauwers and Vallentyne 2016, 402–3). But that isn’t quite to say that we must give up stochastic equivalence — the idea that a prospect’s value is captured by its probability distribution over outcomes. In fact, the problem here is the very same one we encountered in chapter 5. Once again, we need to take *infinitesimal probabilities* into account. When we do, we see that these different St. Petersburg lotteries have *infinitesimally different* probability distributions. And once again, these infinitesimal differences in probability can result in a significant difference in value.

Recall Williamson’s (2007) argument, which I discussed in section 5.5. Consider an infinite sequence of coin flips. Let $H(1\dots)$ be the proposition that every flip comes up heads. Let $H(2\dots)$ be the proposition that every flip after the first comes up heads. Williamson argued thus:

$H(1\dots)$ and $H(2\dots)$ are events of exactly the same qualitative type; they differ only in the inconsequential respect that $H(2\dots)$ starts one second after $H(1\dots)$ Thus $H(1\dots)$ and $H(2\dots)$ should have the same probability.

But I argued against this verdict. Rather, the probability of $H(1\dots)$ is a certain positive infinitesimal ϵ , while the probability of $H(2\dots)$ is a *larger* infinitesimal 2ϵ . In general, qualitatively isomorphic propositions need not have the same probability — in just the same way that qualitatively isomorphic worlds need not have the same value (section 3.3).

This infinitesimal discrepancy in probability becomes significant when huge numbers of lives are potentially on the line. Early St. Petersburg and Late St. Petersburg are qualitatively isomorphic prospects, in a sense analogous to that in which $H(1\dots)$ and $H(2\dots)$ are qualitatively isomorphic events. They differ in the apparently “inconsequential respect” that one starts later. But they are not equally valuable — even though they are each *infinitely* valuable. (A recurring theme!) If you get a chance to play the St. Petersburg game, don’t delay!

The two gambles Early and Late certainly do look similar: each of them assigns probability exactly $1/2$ to making two people happy, exactly $1/4$ to making four people happy, and so on. But I left something out of table 8.4: each gamble *also* has a certain *infinitesimal* chance of making *zero* people happy. Moreover, these infinitesimal probabilities are not the same for Early and Late. The only case where the gamble Early makes no one happy is if the coin comes up tails on every flip. (This is the column labeled T^ω in table 8.6.) Call the infinitesimal probability of this event ϵ . But the Late makes no one happy if the coin comes up tails on every flip *after the first*—the first flip doesn’t count. This event has the *strictly greater* probability 2ϵ . (It is represented by the columns HT^ω and T^ω in the table.) The two gambles have *almost* the same probability distributions, but not quite.

Table 8.6: Delaying the St. Petersburg lottery. States are labeled by their initial sequence of coin flips. The notation T^n stands for a run of n tails in a row.

States:	HH	HTH	HT^2H	$\dots$	HT^ω	TH	T^2H	T^3H	$\dots$	T^ω
Early	2	2	2	$\dots$	2	4	8	16	$\dots$	0
Late	2	4	8	$\dots$	0	2	4	8	$\dots$	0

(You might try to fix this by changing the gambles. What if we alter Early so that in the event of heads followed by all tails, instead of helping two people, it helps no one? That gives both gambles the same infinitesimal probability 2ϵ of helping no one. But we just pushed the lump in the carpet over to the case of *two* people. The modified early gamble assigns this outcome probability $1/2 - \epsilon$, while the late gamble still assigns it exactly $1/2$. Other similar fudges also won’t work.)

This infinitesimal difference in probabilities doesn’t look very important, but

in fact, it matters quite a bit. Let's look at things another way. What is each gamble's probability of making *at least two* people happy? For Early, this is $1 - \epsilon$, and for Late it is the smaller probability $1 - 2\epsilon$. What about *at least four* people? For Early, this happens in every case where the first flip comes up tails and some flip eventually comes up heads. This has probability $1/2 - \epsilon$. For Late, this happens in every case where the *second* flip comes up tails, as long as some coin flip *after the first* eventually comes up heads. This has probability $1/2 - 2\epsilon$. And so on. In general, the delayed gamble not only has an infinitesimally *larger* probability of helping no one, it also has an infinitesimally *smaller* probability of reaching each value threshold. There are infinitely many infinitesimal gaps like this — and these can “add up” to a finite difference in value.

Table 8.7: The (non-standard) cumulative distributions for the two St. Petersburg gambles.

Value thresholds	2	4	8	...
Early	$1 - \epsilon$	$\frac{1}{2} - \epsilon$	$\frac{1}{4} - \epsilon$	...
Late	$1 - 2\epsilon$	$\frac{1}{2} - 2\epsilon$	$\frac{1}{4} - 2\epsilon$	...

(Notice the miracle of loaves and fishes that has been performed: we have taken away the infinitesimal probability ϵ from *one outcome*, zero, and bestowed the *same* infinitesimal probability ϵ on *infinitely many* different upward-closed sets of outcomes. Notice also: Early and Late assign precisely the same probability to helping two people, and also to helping four, and to eight, and so on ... and yet they assign *different* probabilities to the one remaining case, helping zero. Both of these points are related to an important fact about non-standard probability distributions: they do not satisfy *countable additivity*. See section 9.4 for further discussion.)

Just as qualitatively isomorphic worlds need not be equally good (section 3.3), and qualitatively isomorphic propositions need not be equally probable (section 5.5), we now similarly find that qualitatively isomorphic *prospects* need not be equally good. The intuition we began with — Early and Late are equally good — goes along with our intuition that $H(1\dots)$ and $H(2\dots)$ are equally probable, or that the qualitatively isomorphic worlds are equally good, or indeed Zeno's claim that adding a finite length to an infinite ray adds “nothing” (section 2.1). These ideas are natural, and giving them up is

strange — but, at this point, at least it is a familiar kind of strangeness.

These results are vindicated by the quasi-expectational model I introduced in section 5.3. The quasi-expected number of happy lives for Early and Late are both infinite. But — as we have found again and again — not all infinite quantities are created equal. It turns out that the infinite quasi-expectation for Early is *exactly one more* than the infinite quasi-expectation for Late. This fits perfectly with the result that led to paradox: the combined prospect Early A + Early B results in exactly two additional happy lives compared to Late A + Late B in every state.

What's more, the quasi-expectational model upholds Separability (see theorem F.69 in appendix F). Indeed, it does so for the same reason as the Expected Number view does in finite cases: the *difference in quasi-expectation* between two prospects is the same as their *quasi-expected difference*.⁷ When we are comparing the value of prospects just involving numbers of happy lives, we only need to consider the differences between them. We can subtract off anything that is held fixed between the prospects we are evaluating.

This means that the premises that led to the SSK paradox are consistent after all! We can have prospects over unbounded numbers of happy lives, where the number of happy lives is all that matters for value, and yet Separability, Statewise Dominance, and Stochasticism all hold. The way we avoid paradox is by broadening our understanding of a prospect's stochastic profile: different prospects with the same *standard* probability distributions — ignoring infinitesimal differences in probability — need not be equally valuable. When we consider the full *non-standard* probability distributions, Stochasticism is restored.

In (2024), I argued that the SSK paradox meant that we must give up Separability in general; moreover, I took this to undermine arguments for Risk Neutrality and Fanaticism that were based on Separability, like the one in section 8.1. But these qualms have now been laid to rest. In the setting of non-standard probability distributions, we *can* consistently maintain Separability and Stochastic Dominance.

⁷Indeed, like standard expectations, the quasi-expectation $\mathbb{E}^*$ is a *linear operator*: for any vector-valued random variables X and Y , and any scalar $a \in \mathbb{R}$,

$$\begin{aligned}\mathbb{E}^*(X + Y) &= \mathbb{E}^*X + \mathbb{E}^*Y \\ \mathbb{E}^*(a \cdot X) &= a \cdot (\mathbb{E}^*X)\end{aligned}$$

(proposition F.51(a)).

These are appealing principles. Together, they are elegant and theoretically powerful — and they are consistent. But perhaps they are still *too* powerful. For they lead us to Risk Neutrality, Fanaticism, and Pascal’s Principle: tiny probabilities of extreme values are of overwhelming importance.

8.3 A visit to Pasadena

A central lesson of section 8.2 is that not all St. Petersburg gambles are created equal. This is an important lesson for anyone who is interested in developing a decision theory for such infinite gambles. There has been a lively literature on this topic, kicked off by Nover and Hájek (2004). This literature has been situated almost entirely within the framework of standard probability theory. Furthermore, most of it has implicitly taken Stochasticism for granted, by representing the “input” to decision theory as a standard probability distribution over outcomes. (There are a few exceptions: we have already discussed Seidenfeld, Schervish, and Kadane 2009; see also Gwiazda 2014; Colyvan and Hájek 2016; and Hájek 2014.) But standard probability distributions collapse the distinction between infinitesimally different St. Petersburg gambles. Thus it elides important differences in value. The standard probabilistic representation of risk is simply inadequate for the project of evaluating infinite prospects.

Consider, for example, the *Alternating St. Petersburg gamble* (table 8.8). Suppose that in the “status quo” outcome, there will be infinitely many happy people and infinitely many sad people. The angel makes you a more dubious offer. If the first flip comes up heads, two people (who would otherwise be sad) will be made happy. If the first heads is on the second flip, four people (who would otherwise be happy) will be made sad. If it’s on the third flip, then eight people will be made happy. And so on, alternating between good and bad outcomes.

Table 8.8: Alternating St. Petersburg.

H	TH	T^2H	T^3H	T^4H	T^5H	...
+2	−4	+8	−16	+32	−64	...

The trick that Nover and Hájek (2004) pointed out (in the context of a different example, the *Pasadena game*, considered below) is that if you try to naïvely

calculate the expected value of the alternating gamble, you get something that looks like this:

$$1 - 1 + 1 - 1 + \dots$$

This series does not converge — it oscillates between 1 and 0 forever. But what if we rearrange the terms? We could do it like this:

$$(1 + 1 - 1) + (1 + 1 - 1) + (1 + 1 - 1) + \dots = 1 + 1 + 1 + \dots = \infty.$$

Looks great! But we could also do it like this:

$$(1 - 1 - 1) + (1 - 1 - 1) + (1 - 1 - 1) \dots = (-1) + (-1) + (-1) + \dots = -\infty.$$

Looks bad! But just rearranging the terms on paper like this doesn't correspond to any difference in the gamble itself. We are getting contradictory answers to the very same question.

The moral, of course, is that naïvely calculating limiting expectations in these ways does not accurately represent the value of a prospect. But we have better ways, using quasi-expectations.

But now we face some interesting new questions about *probabilities*. We know lots about the probabilities of sequences of coin flips: for example, the probability of getting at least two heads in the first five flips is exactly 13/16. In general, the standard symmetries of coin flips uniquely pin down the probabilities of arbitrary *cylinder sets*, which intuitively correspond to propositions that are about the outcomes of any finite set of coin flips. In the context of standard countably additive probabilities, this also uniquely fixes the standard real-valued probabilities for arbitrary *measurable* sets. (These form an algebra of propositions extending the cylinder sets that is also closed under countable conjunctions and disjunctions. This is a consequence of Carathéodory's extension theorem.) But from our current perspective, this still a quite coarse-grained representation of probability. For example, the standard real probability of getting all heads is zero — as is the probability of getting all heads after the first flip, as well as the probability of a contradiction. At best, these standard probabilities only get at the truth of the matter *up to infinitesimal differences*. When it comes to gambles like St. Petersburg or Pasadena these infinitesimal differences in probability can be very important. So in order to evaluate these gambles and compare them, we need to know more about probability than we currently know. There are a host of open questions here that the "blurry glasses" of standard real analysis have obscured from us. The "post-Pasadena" project

of evaluating infinite prospects is tightly bound up with this project of refining standard probabilities, and choosing the appropriate mathematical structure to represent them.

(There may not be a single correct answer to these questions: perhaps, unlike in the standard setting, there are no *uniquely* correct coin flip probabilities. There may be many different quasi-probability measures consistent with this canonical chancy set-up — and these in turn may be appropriately represented by different kinds of mathematical quantity.)

Furthermore, some of the different proposals people have made about how to extend the notion of expected value to infinite prospects actually correspond to different accounts of the *probabilities* in question. Here I'll very briefly survey a few options. (See section F.4 for further details.)

First, we can start with something *weaker* than the standard probabilities. The *weak coin flip* probabilities I discussed in section 5.3 are actually much less “determinate” than even the standard real-valued coin flip probabilities. They exactly agree on cylinder sets, but for many measurable sets the verdicts they deliver are just that the probabilities are incomparable. (Recall that quasi-probabilities are only *partially* ordered.) Likewise, while these weak quasi-expectations agree with the standard expectation up to infinitesimals for a certain class of functions,⁸ for many other functions the quasi-expectation is incomparable with any standard real value — even in some cases where the standard expectation is well-defined. We probably want more than this. The weak coin flip probabilities are a useful minimal starting point (in the spirit of offering a modular set of tools for theory-building), but probably not where we want to stop.

Something worth noting is that even when quasi-expectations “fall silent”, in a sense, they do so much more gracefully than standard expectations. The observation from Nover and Hájek (2004) is that in many cases the standard expectation is *undefined*. This leaves us without any clear idea of *anything* coherent to say about the relative values of these kinds of prospects — even basic properties like transitivity of *better* are hard to recover. In contrast, quasi-expectations are always defined, and they always satisfy the basic ordering and arithmetic properties we want from values. Sometimes these values are just incomparable with one another. What we get is a perfectly coherent theory of the value of prospects — just one that perhaps imposes a

⁸Namely, those that are *continuous* with respect to the product topology on the coin flip space.

weaker order on them than we would ultimately hope for. (But we should remember that we have no realistic hope of explicitly describing a *total* value order on prospects — recall the Zame–Lauwers theorem about non-constructiveness from section 4.3. See also Lauwers (2016) for a related and more immediately pertinent result.)

For example, Colyvan (2008) proposes extending standard expected utility theory by taking “relative expectations”. In some cases, even if two random variables X and Y do not have well-defined standard expectations, their *difference* does. For example, consider a uniform *improvement* of Alternating St. Petersburg, which adds one extra happy life in every state (table 8.9). Neither Alternating St. Petersburg nor its improved variant has a well-defined standard expectation. But their difference does: it is exactly one.

Table 8.9: Alternating St. Petersburg, and an improved version that adds one extra happy life.

	H	TH	T^2H	T^3H	T^4H	T^5H	...
Alternating St. P.	+2	−4	+8	−16	+32	−64	...
Improved Alt. St. P.	+3	−3	+9	−15	+33	−63	...

With quasi-expectations, there is no need to *extend* the theory to capture “relative expectations”. We get them automatically, since the difference in quasi-expectation just *is* the quasi-expected difference. This holds even in exotic cases like the one at hand. It turns out that Alternating St. Petersburg’s weak quasi-expectation is entirely “indeterminate”, in the sense that it is incomparable with any finite value. Likewise, the quasi-expectation of Improved Alternating St. Petersburg is *also* incomparable with any finite value. But they are comparable with *each other*: indeed, the difference between the two “indeterminate” expectations is exactly one. So even though the weak quasi-expectation is very weak in some ways — leaving indeterminate even in some cases where standard expected utility theory gives a definite answer — in other ways it already goes beyond the standard theory.

And there is no need to stop here. As a second step, we can construct a quasi-probability measure that agrees up to infinitesimal differences with standard probabilities on *all* measurable sets, and for which the corresponding quasi-expectation also agrees up to infinitesimal differences with the standard

expectation for all absolutely integrable functions.⁹ This recovers the power of standard expectations — but it also goes well beyond it. For instance, this automatically handles any case where the *difference* between two prospects is absolutely integrable, even if the prospects themselves are not. So this version of the quasi-expectational theory subsumes Colyvan and Hájek’s “relative expected utility theory” as well.

We can also construct an even stronger quasi-expectation operator (with a correspondingly stronger probability theory). For one more example, consider the *weak expectation* recommended by K. Easwaran (2008; see also 2014a). We need not go into the details; for present purposes we can just note that this is another expectation concept that extends the standard (“strong”) expectation, and which is well-defined for cases like the Pasadena gamble. Again, we can construct an even *stronger* expectation operator that agrees with the *weak* expectation whenever it is well-defined. Again, this doesn’t just refine our notion of the values of prospects, but also the probabilities of propositions. The quasi-probabilities corresponding to “weak expectations” further refine the quasi-probabilities corresponding to the standard “strong” expectation, introducing more comparability.

We could go on, exploring further generalized expectation concepts, but that is enough for now. The general upshot is that the non-standard approach to probabilities and expectations does not *settle* the questions of the post-Pasadena project about the values of prospects with no well-defined expectations. Rather, this approach opens even *more* questions — not just about value, but also about probability. It also provides a flexible and powerful set of tools for investigating these questions.

8.4 Cosmic background value

In section 8.2, we found Separability to be consistent with Stochastic Dominance. But in light of its radical consequences, we should still consider the possibility that it is false.

If we reject Separability, then insofar as we are concerned with impartial overall value, the project of understanding *cosmic background value* is of surprising importance. How many happy or sad lives *are* there out there,

⁹That is, functions f such that $\int |f|$ is finite; these are also called L^1 functions. These are the only functions for which the standard expectation is well-defined in a way that is independent of any choice of how to arrange states.

independently of what might be going on in our own local situation? What do our *overall* prospects really look like? What probabilities should we assign to the moral value of the whole universe we find ourselves in?

For example, take our toy model from section 8.1, where we evaluate a prospect according to the expected *square root* of its number of happy lives. As we discussed, this model says that if we know our universe to be otherwise empty, then strong risk-aversion is in order. It says that the safe option Small, which saves a hundred lives with certainty, is better than the risky option Big, which has a chance of one in a thousand of saving a million lives. But if instead there are a hundred million additional happy lives in the background, then this verdict is reversed: if we are guaranteed a lot of happiness anyway, then the risky option Big looks better than the safe option Small after all. Going further, if the universe is full of astronomical amounts of happy life apart from us, then when less-than-astronomical — but perhaps still gigantic — quantities are at stake we should be very nearly risk-neutral. (This toy model is silent on the situation where background value is huge and *negative*.)

This raises an interesting possibility. Even if we reject Separability and Risk Neutrality, certain theories of cosmic background value may end up basically washing this out. Even if cosmic value is not risk-neutral, even so, risk-neutrality may prevail for the kind of choices we ordinarily face, which have relatively small stakes compared to the potential magnitude of cosmic value.

Tarsney (2025) argues that this is precisely the situation we find ourselves in. His argument has two background presuppositions. First, we suppose that we are sure the total amount of well-being is a finite quantity — for simplicity, let's say it is represented by a real number, though the argument does not rely on anything quite this strong. Second, we suppose standard probability theory.¹⁰ So, for the moment, let us ignore both infinite values and infinitesimal probabilities.

Tarsney's striking argument has two steps.

Step 1. Heavy tails. What probabilities should we assign the cosmic background quantity of well-being? We might think that, whatever it is, it is somewhere between 10 billion terrible lives and 10 billion wonderful lives — since there are fewer than 10 billion people on Earth. But that number ignores past and future people. There have been somewhere around 100

¹⁰Including the principle of countable additivity.

billion individual humans so far, and it is highly uncertain how many there will be. If we broaden our perspective to include extraterrestrial well-being, things are even more uncertain. Even though there may not be any aliens out there, we shouldn't rule this possibility out — and if there are, there may be very many of them indeed.¹¹ So it seems presumptuous to be highly confident of any particular value, or even of any particular *bounds* on the value of cosmic well-being.

In the present setting of standard probability theory and finite quantities, the probability distribution over any quantity must trail off to zero as the quantity increases toward infinity, and likewise as it decreases toward negative infinity. But there are many different ways for a probability distribution to trail off. That is, there are many different possible shapes for the distribution's *tails*: its limiting behavior for very large positive or negative values. Consider two examples (figure 8.1): the *normal* distribution, and the *Laplacian* distribution.¹² Each of these has a central peak and tails that trail off to zero — but the shapes of these tails are very different. For the Laplacian distribution, the *relative rate of decrease* is constant: there is a constant factor α such that, for any $x > 0$, the ratio between the Laplacian probability density at x and the density at $x + 1$ is α . In contrast, the normal distribution falls off *faster* than exponentially. For this distribution, the relative rate of decrease is unbounded: as x gets bigger and bigger, its probability density decreases at a faster and faster relative rate.

For our purposes, we will say that distributions like the Laplacian, and unlike the normal distribution, have *heavy tails*. (Precise definitions are given in section E.6.) In this case, we can characterize how “spread out” the distribution's tails are with a number, which we will call its *relative k -concentration*. Fixing a number k , a distribution with higher relative k -concentration is more tightly concentrated around its peak, while a distribution with relative k -concentration close to zero is more spread out.

¹¹Of course, there are also non-human terrestrial animals, though I have generally been excluding them from discussion.

¹²A normal distribution is defined by

$$p(x) = k \cdot e^{-ax^2}$$

(where k is a normalization constant and a determines the “width”), while a Laplacian distribution is defined by

$$q(x) = k \cdot e^{-a|x|}.$$

The key difference is that x^2 increases faster than $|x|$ (in both the positive and negative directions), and accordingly e^{-ax^2} decreases faster than $e^{-a|x|}$.

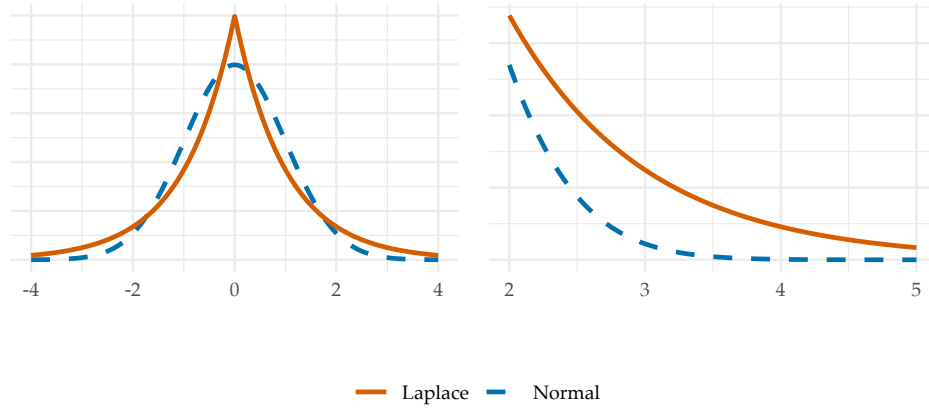


Figure 8.1: A normal distribution and a wider-tailed Laplace distribution. The second plot zooms in on the positive tails to show their shapes more clearly.

Tarsney argues that heavy tails are a good model for our actual state of uncertainty (2025, 1114–15). He defends this on cosmological grounds, as well as a more abstract reason: if our actual state of uncertainty is a *mixture* between several different models, some of which are heavy-tailed, then the overall mixture is heavy-tailed as well.

Step 2. Stochastic dominance. Let us suppose that the overall betterness ordering satisfies *Strict Stochastic Dominance*. If the cosmic background value distribution has heavy tails, then it follows that in many cases, risk-neutrality prevails for our local options.

Suppose we are considering two *local* options X and Y . Since we are not accepting Separability, the question of which is better overall depends on our background uncertainty. We can represent this with another random variable B : intuitively, this represents the total “far off” value of the rest of the world, unaffected by our choice. Then the question before us is whether $X + B$ is better overall than $Y + B$.

Suppose that the stakes of the local options X and Y are small, relative to our background uncertainty. Tarsney’s striking result is that, in this case, if X has greater *expectation* than Y — and the difference in *expectation* between X and Y is not tiny, relative to their stakes — then $X + B$ *stochastically dominates* $Y + B$.

The following theorem is a variant of Tarsney's (see section E.6; see also Pomatto, Strack, and Tamuz 2020).

Theorem 8.2. Let X, Y, B be real-valued random variables on a standard probability space, where B is independent of each of X and Y . Suppose that X and Y are supported by a finite interval $[a, a + k]$, and B has finite k -concentration α . Then, if

$$\mathbb{E}X - a \geq (1 + \alpha)^2(\mathbb{E}Y - a),$$

then $X + B$ strictly stochastically dominates $Y + B$.

Intuitively, the formula says that the expectation of X is greater than the expectation of Y , and the difference between them is not too small relative to the size of their stakes and the width of the background value distribution's tails.

The upshot is that if the quite weak relation of stochastic dominance is a good guide to the value of *cosmic* prospects, then these value comparisons are effectively *risk-neutral* when it comes to their *local* parts — at least in a wide range of plausibly realistic cases. So even if we *reject* expected value as a guide to the value of prospects in general, we can still ordinarily reason in exactly the same way as an expected value theorist in practice: subtract off cosmic background value, and then compare the resulting “local” prospects according to their expected values.

This is a very cool result. It seems to provide a justification for risk neutrality in *some* cases, but not all. In particular, where risk-neutrality gives out is just when things get either Pascalian — when the stakes are enormously high — or else when they get insignificant — when the differences in expectation are tiny. This would be a welcome result. In particular, it would accomplish the very neat trick of vindicating expected value as the right theory to apply in most ordinary cases, without pulling us all the way down the rabbit hole to Fanaticism and Pascal's Principle.

Unfortunately, I do not think the argument quite works. Once again, non-standard probability theory makes a crucial difference. If we bring infinitesimal probabilities back into the picture, then we find that the stochastic dominance result breaks down. The situation is similar to the one we found ourselves in in chapter 5. In that case, two distributions had the same standard probability distribution, but this equivalence disappeared once we paid attention to events that the standard distribution assigned probability

zero. The structure here is similar.

In fact, for the model of non-standard probabilities I proposed (in section 5.3 and appendix F) — what I have called *quasi-probability distributions* — we have this contrary fact: if the worst case outcome of X is worse than that of Y , then even if X has higher expectation than Y , there is no background prospect B such that $X + B$ stochastically dominates $Y + B$ (see proposition F.66). In the setting with infinitesimal probabilities, there is no magical way of producing dominance by adding on background uncertainty.

This is connected to a further point. In the case of standard probability distributions and real-valued random variables, there are two equivalent ways of stating stochastic dominance.¹³ As before, we say that a random variable X **stochastically dominates** a random variable Y iff there is some further random variable Z which has the same probability distribution over outcomes as X , and which statewise dominates Y . On the other hand, let us say that X **(weakly) cumulatively dominates** Y iff

$$\mathbb{P}[X \in U] \geq \mathbb{P}[Y \in U] \quad \text{for every upward-closed set of outcomes } U.$$

For a large class of standard probability measures (but not all!), these two conditions are equivalent. (See section E.4.) But not so when infinitesimal probabilities are taken into account. One direction of the equivalence still holds in the non-standard setting: for any random variables X and Y , if X *stochastically dominates* Y , then X also *cumulatively dominates* Y . But we can have random variables X and Y such that X *cumulatively dominates* Y , but X does not *stochastically dominate* Y . We will see an example of this below.

In fact, then, we *can* still get a result similar to Tarsney's even in the non-standard setting — where the conclusion is not “ $X + B$ stochastically dominates $Y + B$ ”, but rather “ $X + B$ *cumulatively dominates* $Y + B$ ”. This is still a striking result. So we can repair the argument, if we invoke this normative principle:

Cumulative Dominance. For prospects X and Y , if X weakly cumulatively dominates Y , then X is at least as good as Y .

I am not sure about the status of this principle, but it certainly at least has some intuitive appeal. If X is at least as likely as Y to turn out well, for any

¹³See J. S. Russell (forthcoming) for discussion. I there used the less descriptive term “setwise stochastic dominance” for what I am here calling cumulative dominance.

threshold we might set for what we mean by “well”, this certainly seems to speak in favor of X over Y .¹⁴

But if the Cumulative Dominance principle is true, it may show more than we bargained for. Tarsney’s argument deploys a long-tailed distribution over finite values. The longer those tails are, the broader the scope of Risk Neutrality, applying to prospects with higher and higher local stakes. But for any particular standard distribution, Risk Neutrality will give out somewhere. Consequently, we need not go all the way to Pascal’s principle about *infinite* value against the backdrop of any particular standard probability distribution.

But non-standard probabilities open new possibilities. We can consider the limiting case, where the tails get as long as can be: a perfectly *uniform* distribution over background value. Indeed, this seems like a more natural picture of the kind of uncertainty we might face than any particular long-tailed standard distribution. A distribution like the Laplacian singles out a privileged *zero* level of value around which it is centered. But it is not clear that there really is any such epistemically privileged place at the absolute center of the scale of moral value.

So how do things go if background cosmic value is represented by a uniform distribution over finite values? (We are still stopping short of considering *infinite* values, for now.) Consider two familiar *local* options, *Big* and *Small*. As usual, Big has chance 1/1000 of making one million people happy, and otherwise zero. Small is certain to make one hundred people happy. But in addition to whatever local option is chosen, there is also some background level of net happiness (compared to some baseline outcome) which is uniformly distributed over the integers; call this B . Each particular integer has the same positive infinitesimal probability ϵ of being the total cosmic background value.

Consider any integer n . Let p be the (non-standard) probability that $B \geq n$. Then the probability that the *total* value of Small + B will be at least n is the same as the probability that $B \geq n - 100$, which is $p + 100 \cdot \epsilon$. The probability

¹⁴J. S. Russell (forthcoming, sec. 3) proposed a counterexample to the normative principle Cumulative Dominance. But this counterexample turns on there being possible outcomes with probability zero. So we can reject the counterexample by insisting on *regularity* — which we may consistently do in the non-standard setting.

that the total value of $\text{Big} + B$ will be at least n is equal to

$$\begin{aligned} & \mathbb{P}[\text{Big} = 0] \cdot \mathbb{P}[B \geq n] + \mathbb{P}[\text{Big} = 10^6] \cdot \mathbb{P}[B \geq n - 10^6] \\ &= (1 - 10^{-3}) \cdot p + 10^{-3} \cdot (p + 10^6 \cdot \epsilon) \\ &= p + 1000 \cdot \epsilon. \end{aligned}$$

This is strictly greater than $p + 100 \cdot \epsilon$. So we have found that for *every* value threshold n , the probability that $\text{Big} + B$ exceeds n is strictly greater than the probability that $\text{Small} + B$ exceeds n (though only infinitesimally greater). In short, $\text{Big} + B$ *cumulatively dominates* $\text{Small} + B$.

(This example also illustrates the point made above, that we can have cumulative dominance without *stochastic* dominance. By proposition F.66, since Big does not stochastically dominate Small , likewise $\text{Big} + B$ does not stochastically dominate $\text{Small} + B$.)

This reasoning is entirely general.

Theorem 8.3. Let X, Y, B be random variables taking values in $\mathbb{Z}$ on a non-standard-valued probability space. Suppose that B is uniformly distributed, in the sense that for some infinitesimal probability $\epsilon > 0$, $\mathbb{P}[B = n] = \epsilon$ for each $n \in \mathbb{Z}$. Let X and Y each take only finitely many possible values, and let them each be independent of B . Then $\mathbb{E}X \geq \mathbb{E}Y$ iff $X + B$ cumulatively dominates $Y + B$.^a

^a

Proof. For $n \in \mathbb{Z}$,

$$\begin{aligned} \mathbb{P}[X \geq n] &= \sum_k \mathbb{P}[X = k] \cdot \mathbb{P}[B \geq n - k] \\ &= \sum_k \mathbb{P}[X = k] \cdot (\mathbb{P}[B \geq n] + k \cdot \epsilon) \\ &= \mathbb{P}[B \geq n] + \epsilon \cdot \mathbb{E}X. \end{aligned}$$

The same goes for Y . Thus $\mathbb{E}X \geq \mathbb{E}Y$ iff $\mathbb{P}[X \geq n] \geq \mathbb{P}[Y \geq n]$ for all $n \in \mathbb{Z}$. □

In short, in a context of uniform background uncertainty, we arrive at local risk neutrality once again. This time, though, it is *unrestricted* local risk neutrality, applying to arbitrarily high-stakes simple prospects. And *this* means that once again we are saddled with Pascal's principle, when it comes to our "local" prospects. Holding fixed uniform background uncertainty, if there is any chance, however small, at making an *infinite* difference to the world, this swamps any finite difference we might make.

There are many questions left to explore here. How far does this reasoning

generalize to other possible background distributions? What about when we take seriously the possibility of *infinite* background value? I don't know how far this will take us. But the glimmer of hope that Tarsney held out to us seems to be dimming. He offered a principled argument for risk neutrality across many ordinary cases, which would not take us all the way to Pascal's extreme conclusions. But once we have taken infinitesimal probabilities into account — as I have argued we must! — things look very different. First there is a technical matter: for Tarsney's argument to work, we need to appeal not to mere Stochastic Dominance, but rather to Cumulative Dominance, which is a stronger principle in this context. Second, though, in this non-standard setting, the stronger principle keeps on pushing us beyond where Tarsney's argument led — all the way to *unrestricted* risk neutrality and its troubling Pascalian consequences.

There is a short road from Separability to Risk Neutrality and Pascal's principle. But it is not clear that departing from Separability offers any real escape. A non-separable theory of impartial value is sensitive to cosmic uncertainty. When we seriously take this uncertainty into account, we may well be led to Risk Neutrality and Pascal anyway, when it comes to evaluating the "local" differences that our options might make in the world.

8.5 Conclusion

The central idea we have been exploring is that in impartially evaluating prospects, distant lives that go the same way in each of two prospects are irrelevant to which prospect is better. This idea is intuitive and powerful. Indeed, in the context of simple worlds where the net number of happy lives is the only valuable thing at stake, Separability and Stochastic Dominance together imply Risk Neutrality. This provides support for the key premise of Pascal's argument for his principle about infinite value.

In section 8.2, we considered a powerful challenge to Separability (and thus to this argument). We found that the challenge is answerable by appealing, once again, to infinitesimal probabilities. In section 8.4, we also explored what things are like without Separability. There we found pressure toward Risk Neutrality after all, given certain pictures of "cosmic background value".

There is some irony here. We might hope that when we are evaluating ordinary prospects, "the earth, that is sufficient." The prospects we might choose between don't normally have much to do with what is going on in

distant galaxies. We would like to be able to just ignore our options' cosmic background, and focus on the here and now.

There are two quite different ideas that might motivate such insouciance. One is the idea that what is going on in distant galaxies is just not that important, regardless of their scale. If we *discount* the value of distant lives, then they might be truly negligible for terrestrial decisions. But impartial axiology rules this idea out. If there might be very many distant lives, then Impartiality says that they might be extraordinarily important.

We have explored a second idea that can also motivate focusing our concern on terrestrial value: Separability. Even if what is going on in galaxies far, far away is intrinsically of immense importance, Separability tells us that insofar as the prospects we might choose between are exactly alike with respect to the probabilities of such distant goings-on, we can simply subtract that value off, however immense it might be, and evaluate prospects based solely on the differences that remain.

But Separability leads us to Pascal, instructing us to evaluate prospects according to probability-weighted sums of values — even when those values are immense or infinite. The irony is that this may foil our original mundane hopes. If cosmic background value is *exactly* equal for all of the prospects we might choose between, then we can subtract that immense value off and ignore it. But if Pascal is right, then this immense value is extremely sensitive to *tiny* probabilities. If there is even a very tiny, but finite and positive, chance of different prospects making a difference to the rest of the infinite cosmos, then this overwhelms terrestrial value. The probabilities of *infinite* differences between our options, however tiny (but not infinitesimal) they may be, is of overriding concern after all. The earth is not enough, and the constellations loom very near to our practical concerns.

Chapter 9

Consistency and Coordination

That's a problem for future Homer. Man, I don't envy that guy!

The Simpsons

9.1 The Sure Thing Principle

It's been a bad year for tsunamis. Once again, there is one approaching two islands, and as usual, you can only save one. (See table 9.1.)

The Small Island has a population of 100. If you go there, you are sure that you can get all hundred people to safety.

The Big Island is probably uninhabited. But there is a chance of one in a thousand that it has a population of 1,000,000.

This time, though, you think your boat might be broken. In fact, the probability is only one in a thousand that it is in good enough condition to get you to *either* island. If your boat breaks down, you won't save anyone at all. If it runs okay, then you will save whatever people are on whichever island you set out for.

Table 9.1: Illustrating the Sure Thing Principle

Probabilities:	$\frac{1}{1,000,000}$	$\frac{999}{1,000,000}$	$\frac{999,000}{1,000,000}$
Small	100	100	0
Big	1,000,000	0	0

Here is a natural line of reasoning. Overall, going for the Small Island has probability one in a thousand of saving 100 people. Going for the Big Island has probability one in a million of saving a million people. But in the case where your boat isn't working, the decision is moot. So for the purpose of deliberating about which island to go to, you may as well assume that your boat *does* work. You can set aside the cases where the boat is broken: nothing in your choice turns on those cases. In table 9.1, we can just cross out the last column, where nothing matters, and focus on the remaining columns. This reduces the question to one we considered before (in chapter 8): is it better to save 100 people for sure, or one million people with probability one in a thousand?

This reasoning relies on a principle which was influentially advocated by Savage (1954, sec. 2.7).¹

The Sure Thing Principle. Let X and Y be prospects and let E be a proposition. If X and Y are certain to have equally good outcomes conditional on not- E , then X is at least as good as Y unconditionally iff X is at least as good as Y conditional on E .

$$X \succeq Y \quad \text{iff} \quad X \succeq_E Y.$$

In our example, the prospects Small and Big are equivalent conditional on the boat not working. So the principle says that Small is as good as Big *unconditionally* if and only if it is as good as Big *conditional* on the boat working. This seems very plausible.

The Expected Number theory implies the Sure Thing Principle. If the value of a prospect is the probability-weighted average of the number of lives

¹There are a few differences from Savage's formulation. First, Savage was giving an account of rational preference, whereas I am talking about instead about what is better. Second, Savage does not wish to complicate his foundations with a basic notion of *conditional* preference, so he goes to some effort to offer a reduction of this to certain unconditional preferences. (This reduction requires a different complication, though: he makes strong structural assumptions about which prospects are in the domain of the preference relation.)

saved, then the *unconditional* value of X is a weighted average of the value of X conditional on E , and the value of X conditional on not- E .

$$\mathbb{E}X = \mathbb{P}(E) \cdot \mathbb{E}(X \mid E) + \mathbb{P}(\neg E) \cdot \mathbb{E}(X \mid \neg E).$$

If X and Y have the same value conditional on not- E , then when we are comparing their unconditional expectations we can just subtract that term off. But, as I discuss below, Expected Number is not the *only* theory compatible with the Sure Thing Principle.

The Sure Thing Principle is intimately connected to a principle we discussed in section 2.3: the *Independence* axiom deployed in von Neumann and Morgenstern's representation theorem. The Independence axiom is the natural analogue of the Sure Thing Principle for *lotteries*—probability distributions over outcomes. Indeed, if we assume *Stochasticism*—that the value of a prospect is determined by its distribution—then the Sure Thing Principle follows from the Independence axiom.² The converse also holds, if we assume the space of prospects and propositions is sufficiently rich.³

There is much to be said for the Sure Thing Principle. I will discuss one line of argument in section 9.4 and section 9.5 below. It is a widely accepted part of orthodox decision theory, and, in my view, there is good reason for this. But it is also far from uncontroversial.⁴

The Sure Thing Principle has powerful implications for Pascalian gambles. Let's start with an illustration (compare Kosonen forthcoming). As we introduced in section 1.2, some philosophers have proposed that we should avoid Pascalian conclusions by *discounting small probabilities*. If there is only a very tiny chance of a prospect achieving a certain outcome, then we can treat that small probability as equal to zero for practical purposes, even if the unlikely outcome would be enormously consequential.

Consider the tsunami case again. Suppose for illustration that probability one in a million is small enough to be negligible, but one in a thousand

²To be more exact, what we rely on in this case is the assumption that the *conditional* value of a prospect is determined by its *conditional* distribution.

³Stochasticism implies that there is a betterness ordering $\succsim$ on lotteries such that a prospect X is as good as a prospect Y iff X 's lottery is as good as Y 's lottery: $X \succsim Y$ iff $\mathbb{P}_X \succsim \mathbb{P}_Y$. Here is the statement of Independence again:

Independence. For any lotteries P, Q, R , and any probability $p > 0$,

$$P \succsim Q \text{ iff } p \cdot P + (1 - p) \cdot R \succsim p \cdot Q + (1 - p) \cdot R.$$

⁴Allais (1953); Quiggin (1993); Buchak (2013); [TODO More citations and discussion.]

is not. Then, considering Big and Small *unconditionally*, the probability of successfully saving anyone on the Big Island is negligible. So its probability-discounted expected value is negligibly small. The one in a thousand chance of saving a hundred people on the Small Island is not negligible, so Small is deemed better than Big overall.

But now consider how things go *conditional* on your boat starting. Given this, saving a million people on the Big Island has probability one in a thousand, which is non-negligible. So its value is its expectation: a thousand lives, which is more than a hundred lives. So Big is better than Small conditional on the boat working. The two prospects are equivalent conditional on the boat not working. Yet according to probability discounting, Big is worse than Small unconditionally. Probability discounting violates the Sure Thing Principle.⁵

Even though the Sure Thing Principle rules out probability discounting, it does not take us all the way to Risk Neutrality or Fanaticism on its own. Consider once again *expected utility theory* (section 1.2.3). Each number of happy lives n is assigned some number, its *utility* $u(n)$, where u is an increasing function. Then we compare prospects not according to their expected number of happy lives, but rather according to their *expected utility*. If the utility function is *linear*, then this agrees with risk-neutral Expected Number theory. Otherwise, though, the two come apart.⁶

In particular, if the utility function is *bounded*, then the resulting value comparisons are not Fanatical. For example, consider the utility function

$$u(n) = 1 - 2^{-n},$$

which takes values between 0 and 1. Then the certainty of one happy life for sure has expected utility $1/2$. Since any arbitrarily large number of happy lives has utility at most one, for any $p < 1/2$, there is no n such that chance p of n happy lives is better than one happy life for sure. This contradicts Fanaticism.

But also, expected utility theory satisfies the Sure Thing Principle—whether or not the utility function is linear, and whether or not it is bounded. This is

⁵In section 1.2.4 I discussed two versions of probability discounting: one where we discount small probabilities to *zero*, and another where we discount them to *infinitesimals*. The reasoning above goes the same way for either version.

⁶To be clear, expected utility with a non-linear utility function is not risk-neutral with respect to the *number of happy lives*. It is still risk-neutral with respect to *utility*, but that is not what we were talking about.

for essentially the same reason as the Expected Number theory: the expectation of any quantity is a weighted average of its expectation conditional on E and its expectation conditional on not- E . So, while the Sure Thing Principle rules out one way of avoiding Fanaticism—namely, probability discounting—it does not rule out all of them.

Even so, the distance from the Sure Thing Principle to Fanaticism is not vast. In the next two sections I will discuss two ways of bridging this distance, turning on two different additional premises. (The first argument supports the stronger conclusion of Risk Neutrality, while the second argument supports only a weaker conclusion, a version of Fanaticism.) After that, I will return to the question of whether the Sure Thing Principle is true.

9.2 Redistributing risks

This first argument turns on considerations of impartiality with respect to the allocation of risks. Structurally, it is very closely related to the argument from Separability for Risk Neutrality in section 8.1. (Basically, we turn the tables involved in that argument sideways.) Suppose we face a choice between one happy life for sure—*Safe*—or chance $1/n$ of saving n happy lives and otherwise none—*Risky*. Risk Neutrality with respect to happy lives says that Safe and Risky are equally good. Assuming the Sure Thing Principle, to show this, it is enough to show that Safe and Risky are equally good in the context of some wider prospects. This time, instead of additional *lives* off in some distant galaxy, what we want to introduce are additional *possible states*. We will construct two prospects X and Y . There are some possibilities in which the choice between X and Y boils down to the choice between Safe and Risky. In the rest of the possibilities, X and Y are sure to turn out exactly the same way. Then, if X and Y are equally good, the Sure Thing Principle will imply that Safe and Risky are equally good as well.

Here is how that goes.

There are n people, numbered $1, 2, \dots, n$. An n sided die is rolled and a fair coin is tossed. There are two prospects X and Y .

If the coin comes up heads, here is what happens.

- X : All n people are happy if the die comes up one; otherwise no one is happy.

- Y: Person 1 is happy no matter how the die comes up.

If the coin comes up tails, then X and Y both turn out exactly the same way, which is determined as follows. Person 1 gets to be happy if the die comes up one, and is otherwise sad. Person 2 gets to be happy if the die comes up 1 or 2. Person 3 gets to be happy if the die comes up 1, 2, or 3. And so on, up until the person n , who is guaranteed to be happy.

For concreteness we can illustrate this in the case where $n = 4$ (table 9.2 and table 9.3).

Table 9.2: The prospect X.

Person	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
1	😞				😞			
2	😞				😞	😞		
3	😞				😞	😞	😞	
4	😞				😞	😞	😞	😞

In the “risky” prospect X, person 1 has chance $2/2n$ at a happy life— $1/2n$ from the heads case and another $1/2n$ from the tails case. Person 2 has chance $3/2n$, person 3 has chance $4/2n$, and so on up to person n , who has chance $(n + 1)/2n$ at happiness.

Table 9.3: The prospect Y.

Person	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
1	😞	😞	😞	😞	😞			
2					😞	😞		
3					😞	😞	😞	
4					😞	😞	😞	😞

In the “safe” prospect Y, person 1 has chance $(n + 1)/2n$ of a happy life— $n/2n$ from the case of heads, and chance $1/2n$ from the case of tails. Person 2 just has chance $2/2n$ of a happy life, person 3 gets $3/2n$, and so on up to person

n , who has chance $n/2n$.

But these prospects are exactly the same, except that we have just switched around who gets which prospect. It shouldn't make any difference whether it is person 1 who gets the largest chance at happiness rather than person n . This is another version of the idea of *impartiality*: it doesn't matter morally who gets which prospect, if the same chances of happiness are being allocated either way. But this impartiality principle is different from the one we introduced in chapter 3. In particular, this is what is called an *ex ante* principle, concerning how we allocate *risks*, rather than an *ex post* principle about allocating *outcomes*.

Simple Ex Ante Impartiality. For any simple prospects X and Y , if X and Y are exactly alike except that in X person i has chance p of happiness and person j has chance q of happiness, while in Y those probabilities are switched, then X and Y are equally good.

Theorem 9.1. The Sure Thing Principle and Simple Ex Ante Impartiality together imply Risk Neutrality.

This is stated and proved more precisely in section C.6 (TODO).

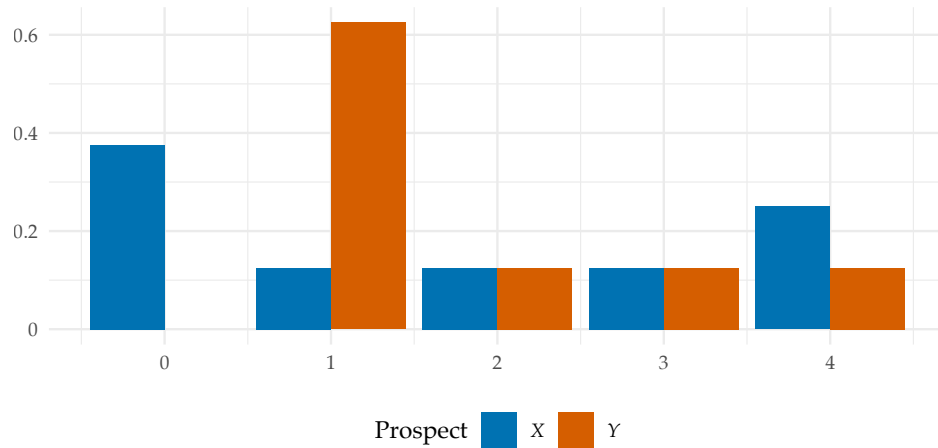
I think this is a strong argument, but we should unpack a move that may be less innocuous than it appears. The way we stated Ex Ante Impartiality implicitly assumed that what matters for the value of a prospect is the *probability* of happiness for each individual, and not which particular *states* they are happy in. This is a natural enough thought, especially when we consider the prospects from the perspective of individuals. But swapping an individual's happiness between different states, even if it makes no difference to the individual, can make a difference to the correlations between *different* individuals' happiness. This is the sort of thing that some people might think matters morally.

The prospects X and Y have the same profile of individual probabilities of happiness. But in the risky prospect X , these individual prospects are more strongly *correlated* with one another, while in the safe prospect Y they are more *decoupled*. This has implications for the shapes of the probability distributions for each prospect X and Y over different possible numbers of happy lives (see table 9.4 and section 9.2). The two distributions have the same mean, but X has higher *variance*—its distribution is more “spread out”, with more weight than Y on the extreme values of zero or four happy lives, and less weight than Y on the intermediate value of one happy life.

Risk aversion with respect to happy lives favors the prospect with lower variance.

Table 9.4: The distributions over number of happy lives for each prospect X and Y .

	0	1	2	3	4
X	$3/8$	$1/8$	$1/8$	$1/8$	$2/8$
Y	0	$5/8$	$1/8$	$1/8$	$1/8$



In a situation where individual happiness is the only value at stake, it doesn't matter to any particular person whether their happiness is correlated with other people's. In contrast, risk aversion with respect to a statistic about the whole population says that correlations between individuals *do* matter.

9.3 Symmetries of value

J. S. Russell (2024) gives a different style of argument also based on the Sure Thing Principle. The conclusion is weaker than Risk Neutrality—it is another form of Fanaticism, stated below.

Let's focus on the setting of *simple lotteries* (in the style of von Neumann and Morgenstern: section 1.3.5; section 2.3). As we noted, in this setting the Sure Thing Principle takes the form of the *Independence* axiom.

Our central example of a theory that satisfies Independence *without* Fanaticism is the one we discussed above: *bounded expected utility*. If we are considering just numbers of happy lives, we can use a utility function like this:

$$u_1(n) := 1 - 2^{-n}.$$

Even as the number of happy lives can increase without bound, its utility never exceeds one. So for small enough probabilities p , no matter how good a finite outcome x gets, chance p of x is still not that great.

But now let's also put very *bad* outcomes into the mix. Another form of Fanaticism looms in this direction. Can the certainty of a bad outcome be outweighed by a minuscule chance at a much worse outcome? There are two different forms of Fanaticism, one for the very good, and one for the very bad.

Positive Fanaticism. For any good finite outcome x and probability $p > 0$, there is some good finite outcome y such that chance p of y is better than the certainty of x .

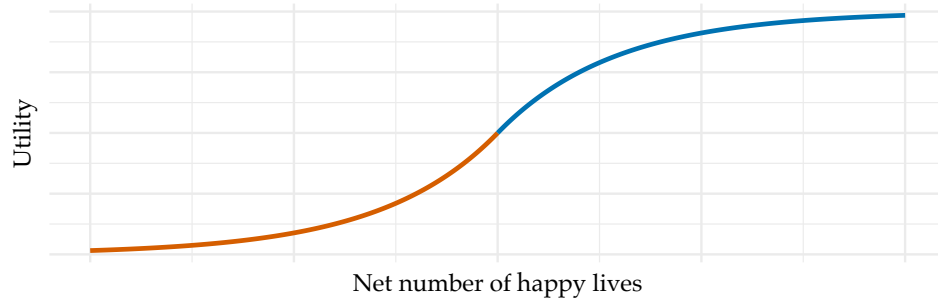
Negative Fanaticism. For any bad finite outcome x and probability $p > 0$, there is some bad finite outcome y such that chance p of y is worse than the certainty of x .

In the context of expected utility theory, if we are to avoid both forms of Fanaticism, then we need a utility function that is bounded both above and below.

Specifically, let's consider outcomes represented by their *net* number of happy lives, compared to some baseline. We might have massive amounts of additional happiness, or massive amounts of additional sadness. Then we might try a utility function like this one (section 9.3):

$$u_2(n) := \begin{cases} 1 - 2^{-n} & \text{for } n \geq 0; \\ 2^n - 1 & \text{for } n < 0. \end{cases}$$

This avoids both Positive and Negative Fanaticism.



But there is something strange about this utility function: it treats the *baseline* outcome—zero net happy lives—as if it were very special. For outcomes that are *better* than the baseline, this utility function displays risk-aversion; for outcomes that are *worse* than the baseline, it is instead risk-seeking. (Compare Beckstead and Thomas 2024, sec. 3.3.) But what is so special about *that* level of happiness, compared to any other?

This brings us to our key additional premise:⁷

Addition Invariance. Let X and Y be prospects for which the net number of happy lives is all that matters, and let d be an integer. If $X \lesssim Y$, then $X + d \lesssim Y + d$.

(We could generalize Addition Invariance to any *quantitative* prospects, as discussed in chapter 8.)

Before we discuss the philosophical case for Addition Invariance, let's state the mathematical result in which it figures. For a start, it is not too difficult to show that no real-valued bounded utility function satisfies Addition Invariance. Any such utility function will have to have a bend in it somewhere, like the example u_2 does at zero. Real-valued expected utility is not the only option: Independence guarantees the existence of a *vector* expected utility representation (section D.2). It turns out that something similar to the fact about real-valued bounded utility functions also holds for *vector* utilities: Addition Invariance is inconsistent with another *boundedness* property (which we can make precise).

This still leaves a window where we might find an escape from Fanaticism, by appealing to *incomparability* between lotteries. In the presence of incom-

⁷Wilkinson (2022a) argued for Fanaticism on the basis of a principle closely related to Addition Invariance. J. S. Russell (2024) showed that Wilkinson's argument does not quite work — for reasons connected to issues discussed in section 9.4 — and offered the following style of argument as a replacement.

parability, saying that the space of vector utilities is *not bounded* does not quite amount to affirming Positive or Negative Fanaticism, as we have stated them. Rather, what we get is one of these conclusions:

Weak Positive Fanaticism. For any good finite outcome x and probability $p > 0$, there is some good finite outcome y such that the certainty of x is not as good as chance p of y .

Weak Negative Fanaticism. For any bad finite outcome x and probability $p > 0$, there is some bad finite outcome y such that the certainty of x is not as bad as chance p of y .

Here we have modified the corresponding principles of Positive and Negative Fanaticism simply by replacing statements of the form “ Y is better than X ” with corresponding statements of the form “ X is not as good as Y ”.

At last we can state the main result (see section D.7 for details, and a generalization).

Proposition 9.2. Independence and Addition Invariance together imply that at least one of Weak Positive Fanaticism or Weak Negative Fanaticism is true.

Now we return to this question: why accept Addition Invariance? One reason is along the same lines as the *Separability* principle discussed in chapter 8. Recall the “Egyptology” argument. Suppose we are choosing between two risky policies, and then an archaeologist comes to us and reports that there was a huge and flourishing civilization in ancient Egypt. This does not seem like it should make any difference to our forward-looking policy proposals. Addition Invariance is weaker than full-fledged Separability, which concerned adding on an additional *risky prospect*—in particular, one that may have different correlations with the two policies under consideration. Addition Invariance only lets us tack on a *fixed* change in happiness, which is not only the same for both policies, but also the same in all possible states.⁸

But Addition Invariance can also be motivated in a quite different way from Separability. It is one principle in a family of what Goodsell (forthcoming) calls *symmetries of value*. It is common in the philosophy of science to think of symmetry principles as telling us that we do not have “excess structure”

⁸Addition Invariance follows from a restricted form of Separability: what Wilkinson (forthcoming) calls *Separability for Independent Options*. Wilkinson considers several reasons why we might accept this restricted principle without going all the way to unrestricted Separability.

(for overview, see Dasgupta 2021). For example, Aristotelian physics gives a special role to a special place in space—the center of the Earth, toward which the element earth naturally falls, and from which the element fire naturally ascends. A key part of the Copernican revolution was to give up this privileged place. The new physics was *translation invariant*: shifting everything uniformly in absolute space makes no difference at all to a system’s dynamical properties. No position in space is dynamically special. By analogy, a way of interpreting Addition Invariance is that there is no *privileged absolute zero* on the scale of value: no “position” on the scale from very bad to very good outcomes is *morally* special.

I think Addition Invariance is particularly attractive in an infinite setting. Consider once again our toy model of *simple worlds*: there are infinitely many individuals, each of whom is either happy or sad. Now consider some world w_0 in which infinitely many people are happy, and infinitely many are sad. Let E be the set of all simple worlds which are all only *finitely different* from w_0 . Impartiality tells us that among such worlds, the net number of happy people is all that matters—specifically, two worlds are equally good if they have the same net number of happy people with respect to w_0 . But w_0 is not special in this respect: *any* choice of a baseline world in E has the same property. No world in this infinite equivalence class is privileged over any other. Indeed, the ordering of E with respect to overall value is exactly the same as the ordering of the integers. But the order of the integers does not single out any particular element which is uniquely appropriate to serve as a zero; this ordering is invariant under shifts. Likewise, the *cardinal* values of these worlds is represented by *differences* between integers; but again, this cardinal structure is invariant under shifts as well. It would be strange if value facts broke this symmetry. No particular theory of the value of prospects over E will be *natural* with respect to the structure already present in those worlds—it will impose some “brute” additional structure on it.⁹

⁹It is worth distinguishing Addition Invariance from a related idea about *representations* (see Hájek 2014, 541; K. Easwaran 2014a, 523; Wilkinson 2022a, sec. 5; see Goodsell forthcoming, sec. 2 for discussion of the distinction). In general, if value can be represented by the expectation of a utility function, this representation is not unique. As K. Easwaran (2014a, 523) concisely puts the point, “utility is only defined up to a shift and a stretch — there is no well-defined 0 and no well-defined unit in which utility is measured.” That is to say, if a function u correctly represents the goodness ordering of lotteries, then other transformed utility functions also do this just as well, including the “shifted” function $u'(x) = u(x) + b$ for any number b , and the “stretched” function $u''(x) = a \cdot u(x)$ for any positive number a . (A similar point applies to *vector* utility representations. In both cases, the representational freedom directly follows from the fact that the expectation operator $\mathbb{E}$ is a *positive linear operator*.)

This is by no means a knock-down argument. In chapter 4 I discussed similar invariance principles — particularly Permutation Invariance and Reflection Invariance — which come into conflict with Pareto and Comparability. I argued there that we might draw the lesson that these invariance principles break down, but it is *indeterminate* exactly how they break down. A similar reaction is available here. One might say that Addition Invariance fails, that this does give rise to a certain amount of arbitrariness in what is better than what, and we try to put a good face on this by again appealing to indeterminacy. This might be the right thing to say, but I find it more disturbing in this case than it was in chapter 4. This time, arbitrariness and indeterminacy would arise not just for outcomes that differ for infinitely many people, nor for prospects involving infinitely many possibilities, but for *simple* prospects involving outcomes that only differ for *finitely* many people. Furthermore, even if appealing to indeterminate symmetry-breaking is right, we still face a fall-back argument. We can reinterpret the formal result underlying proposition 9.2: instead of applying it to the relation *as good*, we can consider the relation *determinately as good*. [TODO. More to say here, but I’m not sure how far to pursue this.]

9.4 Eventwise Dominance

(TODO. Need to talk about why the argument against Eventwise Dominance does not raise a problem for Statewise Dominance, and connect back to paradoxes of dominance from chapter 6.)

In section 9.2 and section 9.3, I presented two powerful arguments from the Sure Thing Principle (or the closely related Independence axiom) to Fanaticism. But there are reasons for concern here. I have previously argued that this kind of argument is dialectically unstable (J. S. Russell and Isaacs 2021; J. S. Russell 2024). The trouble is that the same grounds on which we

This point about representational degrees of freedom is not the same as Addition Invariance. Consider again the piecewise utility function u_2 from above. As we already observed, if the expectation of u_2 correctly represents the goodness order, then Addition Invariance fails. But of course this is not a counterexample to the general mathematical properties of expectations. A *shifted* utility function $u'_2(n) := u_2(n) + b$ (for any number b) represents the very same ordering of lotteries as u_2 does. But according to this theory of value, shifting the *utility* representation has a different effect from adding more *happy people*. For instance, there is no number b such that

$$u_2(n + 1) = u_2(n) + b \quad \text{for each } n.$$

The left-hand side represents the result of adding one happy person, while the right-hand side represents shifting the utility representation. These are different.

might accept the Sure Thing Principle also favor *stronger* principles, which are *inconsistent* with Fanaticism. It seems that something must be going wrong here.

Here is how Savage (1954, 52) originally motivated the Sure Thing Principle:

A businessman contemplates buying a certain piece of property. He considers the outcome of the next presidential election relevant to the attractiveness of the purchase. So, to clarify the matter for himself, he asks whether he would buy if he knew that the Republican candidate were going to win, and decides that he would do so. Similarly, he considers whether he would buy if he knew that the Democratic candidate were going to win, and again finds that he would do so. Seeing that he would buy in either event, he decides that he should buy, even though he does not know which event obtains ... It is all too seldom that a decision can be arrived at on the basis of the principle used by this businessman, but, except possibly for the assumption of simple ordering, I know of no other extralogical principle governing decisions that finds such ready acceptance.

Savage informally called this the sure-thing principle—but note that it is a bit different from the more official principle he eventually arrived at and gave that name. The businessman’s reasoning is more directly captured by the principle that unconditional value “reflects” conditional value.¹⁰

Finite Reflection. Let X and Y be prospects, and let E be a proposition.

(Weak) If X is at least as good as Y conditional on E , and X is at least as good as Y conditional on not- E , then X is at least as good as Y unconditionally.

(Strong) If, furthermore, X is strictly better than Y conditional on E , then X is strictly better than Y unconditionally.

Savage was not making some mistake in moving from one principle to the other. Given other assumptions that he was happy to accept, Finite Reflection and the Sure Thing Principle are equivalent. (These assumptions include Comparability. Without it, the relationship between the two principles is bit more delicate, but not in ways that I think substantially affect the discussion that follows.)¹¹

¹⁰TODO. Note with some context for reflection principles in philosophy more generally.

¹¹A further assumption is *statewise equivalence*: If X and Y are equivalent conditional on E , in the sense of being certain to have equally good outcomes, then $X \sim_E Y$. Another is that

As Savage said, Finite Reflection seems very plausible on its face. If one option is sure to be as good as another given some new information, whatever that information might be, then it is simply as good unconditionally. It is very strange to think that this could fail. Here is how Broome (1991, 129) puts the point (see also discussion in J. S. Russell 2024):

[P]robabilities derived from more information have a higher status than those derived from less. ... At the extreme, what would actually happen has the highest status of all. ... This at least is true: you ought not to found your judgements on lower-status probabilities when higher-status probabilities are available.

More metaphorically, think of betterness relations as people giving advice. Each advisor reports what is impartially best given their own evidence. Say Ada and Adam are two advisors; Ada has all of Adam's information, and additionally she knows the answer to a question Q . Then Ada is a "higher status" advisor than Adam. If Ada announces that X is as good as Y , then Adam ought to defer to her. If Adam does not defer to her and also say that X is as good as Y , then he is not doing his job, and we should ignore him.

But if this is our motivation for the Sure Thing Principle, why would we stop with *Finite* Reflection? Consider a variant on Savage's story.

The businessman considers next year's inflation rate relevant to the attractiveness of the purchase. So, to clarify the matter for himself, he asks whether he would buy if he knew that inflation would be less than 1%, and decides that he would do so. Similarly, he considers whether he would buy if he knew that inflation would be between 1% and 2%, and again finds that he would do so. Similarly, for any number n , he finds that if he knew inflation

the space of prospects is sufficiently rich.

The Sure Thing Principle implies Finite Reflection.

(Weak) Suppose $X \succeq_E Y$ and $X \succ_{\neg E} Y$. Let Z be a prospect equivalent to Y conditional on E and equivalent to X conditional on $\neg E$. Then also $X \succeq_E Z$ and X and Z are equivalent conditional on $\neg E$, so the Sure Thing Principle implies that $X \succeq Z$. Likewise, the Sure Thing Principle again implies $Z \succeq Y$. So $X \succeq Y$.

(Strong) Suppose furthermore that $X \succ_E Y$; so $X \not\succeq_E Y$. Then the Sure Thing Principle implies $X \not\succeq Z$, and so by Comparability $X \succ Z \succeq Y$.

Other direction. Let X and Y be equivalent conditional on $\neg E$. Then $X \sim_{\neg E} Y$, and so Weak Finite Reflection implies that if $X \succeq_E Y$, then $X \succeq Y$. Likewise, if $X \not\succeq_E Y$ then by Comparability $X \prec_E Y$, and Strong Finite Reflection implies $X \prec Y$, and thus $X \not\succeq Y$.

were between $n\%$ and $(n + 1)\%$, he would buy. Seeing that he would buy in any event, he decides that he should buy, even though he does not know which event obtains.

It seems we can generalize:

Reflection. Let X and Y be prospects, and let Q be a question.¹²

(Weak) If X is at least as good as Y conditional on every possible answer to Q , then X is at least as good as Y unconditionally.

(Strong) If, furthermore, X is strictly better than Y conditional on some possible answer to Q , then X is strictly better than Y unconditionally.

Finite Reflection is equivalent to the restriction of Reflection to *finite* partitions (as the name suggests). And the argument I gave above for Finite Reflection carries over, verbatim, to the general version. If Ada knows next year's inflation rate, this makes her a better-qualified expert just as well as if she knows the outcome of the next election.

Not everyone accepts this kind of argument for Reflection: for instance, Buchak (2013, 199–200) accepts that her favored theory has the “somewhat unpalatable upshot that more information is not always better for decision-making”. But she does not only reject the general Reflection principle for this reason, but also the finite case, and with it the Sure Thing Principle. J. S. Russell and Isaacs (2021) argue that this is the general situation:

We are certainly not claiming that [Reflection] is beyond question. We have already mentioned some reasons for doubt worth serious contemplation—decision theory is hard. But these reasons for doubt carry over to [Finite Reflection] as well. And despite these cogent doubts, [Finite Reflection] is a key part of orthodox decision theory. Those many decision theorists who are moved by the kinds of arguments we have discussed to accept the ortho-

¹²We model a question as a **partition**: a set of propositions—its *answers*—which are mutually exclusive and jointly exhaustive.

J. S. Russell and Isaacs (2021) call this principle the “Countable Sure Thing Principle”. (They restrict the principle to *regular* partitions, where each answer to the question has positive probability. According to some theories, conditional value only makes for antecedents with positive probability. In the case that “ X is at least as good as Y conditional on E ” is undefined for some E which is an answer to Q , then we should understand Reflection as vacuously true.) Lee (2013) calls it Dominance*, and Blackwell and Girshick (1954) and Hammond (1988) call the corresponding principle for lotteries D^* . It is closely related to the principle of *Conglomerability*, which concerns probabilities rather than values.

doxy in finite cases should also be moved toward its analogue in infinite cases.

I will return to their argument in section 9.5. If they are right, though, this is very *bad* news for the arguments for Fanaticism in section 9.2 and section 9.3. For in fact, Reflection is *inconsistent* with Fanaticism.

J. S. Russell (2024; drawing on Beckstead and Thomas 2024, sec. 4) gives the following kind of argument. (This version is slightly modified to avoid some technicalities involving stochastic dominance and infinitesimal probabilities.)

First, Fanaticism implies the existence of *generalized St. Petersburg prospects* ... We can find a sequence of outcomes that get better very fast, and use these to construct a prospect which is strictly better than any of its outcomes; J. S. Russell and Isaacs (2021) call such prospects *improper*. The existence of improper prospects is a particularly unsettling consequence of Fanaticism. (2024, 579)

Let's spell this step out. Start with any good finite outcome x_1 . We can repeatedly apply Fanaticism to find finite outcomes such that chance $1/4$ of x_2 is better than the certainty of x_1 , chance $1/8$ of x_3 is better than the certainty of x_2 , and in general chance $1/2^n$ of x_n is better than the certainty of x_{n-1} . Then consider this prospect X : we flip a coin until it first comes up heads. If this is on the first flip, we get x_1 ; if the second, then x_2 ; and in general outcome x_n has probability $1/2^n$. (In the infinitesimal-probability event that the coin never comes up heads, we get nothing.) Reflection (or indeed Statewise Dominance) tells us that, for each n , X is better than the prospect that has chance of $1/2^{n+1}$ of x_{n+1} , and otherwise nothing, and by construction this prospect is in turn better than the outcome x_n . So X is better than each of its possible outcomes.

For the second step, do this again. Let's flip another coin until *it* first comes up heads, and use the same pay-out schedule. So the prospect Y has the very same possible outcomes as X , and it too is improper (see tables 9.5 and 9.6).

Table 9.5: The outcomes for the first of two independent St. Petersburg gambles. The proposition A_n obtains when the first coin first comes up heads on the m th flip; B_n is the proposition that the second coin first comes up heads on the n th flip.

X	A_1	A_2	A_3	$\dots$	A_m	$\dots$
B_1	x_1	x_2	x_3		x_m	
B_2	x_1	x_2	x_3		x_m	
B_3	x_1	x_2	x_3		x_m	
$\vdots$					$\vdots$	
B_n	x_1	x_2	x_3	$\dots$	x_m	$\dots$
$\vdots$					$\vdots$	

Table 9.6: The outcomes for second of two independent St. Petersburg gambles.

Y	A_1	A_2	A_3	$\dots$	A_m	$\dots$
B_1	x_1	x_1	x_1		x_1	
B_2	x_2	x_2	x_2		x_2	
B_3	x_3	x_3	x_3		x_3	
$\vdots$					$\vdots$	
B_n	x_n	x_n	x_n	$\dots$	x_n	$\dots$
$\vdots$					$\vdots$	

How do X and Y compare? It is tempting to say they are equally good, since they are isomorphic, differing only in which coins are tossed—but we have learned to be suspicious of such reasoning (sections 5.5 and 8.2). Instead, we reason like this: suppose the first coin comes up heads on the n th flip. Then X has the outcome x_n , and Y has probability $1/2^m$ of outcome x_m , for each m . By construction, this generalized St. Petersburg gamble is better than x_n . So Y is strictly better than X conditional on the first coin coming up heads on the n th flip. This holds for every n . (Also, Y is better than X conditional on the exceedingly unlikely event of the first coin never getting heads.) So Reflection tells us that Y is strictly better than X unconditionally. But exactly parallel reasoning about the result of the *second* coin tells us that X is strictly better than Y ! These cannot both be correct. The conclusion is

that Fanaticism is inconsistent with Reflection.

If we combine this with the arguments in section 9.1, we get two immediate corollaries. First, Reflection is inconsistent with Ex Ante Impartiality. Second, Reflection is inconsistent with the combination of Addition Invariance and Comparability (Note that an exactly parallel argument shows that *Negative* Fanaticism is also inconsistent with Reflection. We invoke Comparability here to close the gap between Positive Fanaticism and *Weak* Positive Fanaticism, and likewise for the Negative versions. We don't really need the full strength of Comparability for this, as long as we have *enough* comparable prospects.)

In fact, though, this sort of problem is not specific to *Fanaticism*; indeed, it does not depend on any substantial commitments about the structure of value. Suppose there are just *two* possible outcomes, one better (😊) and one worse (😞). Even in this austere setting, Reflection has drastic consequences.

A *de Finetti lottery* is a probability distribution over a countable infinity of possible outcomes that assigns each individual outcome the exact same probability. This situation is ruled impossible in standard probability theory. Assigning each outcome the same *positive finite* probability would imply that the total probability must be infinite (by finite additivity). Assigning each outcome *zero* probability would imply that the total probability must be zero (by the standard principle of countable additivity). So, for the moment, let's assume that this feature of standard probability theory is incorrect, and that a de Finetti lottery is possible. (I have already argued against standard probability theory on independent grounds, in chapter 5.) After I have discussed the problem in that context, we can consider how it generalizes.

Suppose instead of St. Petersburg-style sequences of coin flips, we draw a natural number from a de Finetti lottery; then we draw a second number from a second, independent de Finetti lottery. (Compare the argument in Arntzenius, Elga, and Hawthorne 2004, 279–80.) Call the two results m and n .¹³ Now consider two prospects X and Y (tables 9.7 and 9.8). The prospect X results in a good outcome (😊) if $m > n$. The prospect Y results in the same good outcome if $m < n$.

¹³A technical point: assuming we have *two* de Finetti lotteries is not really any stronger than assuming *one*. We can put the set of numbers in one-to-one correspondence with the set of ordered pairs of numbers. Call this correspondence $p : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$. Then if we draw *one* number k from a de Finetti lottery, we can let $(m, n) = p(k)$. This distribution has the key property for the argument that the probability of drawing (m, n) is the same as that of (m', n) or (m, n') for any other numbers $m', n' \in \mathbb{N}$.

Table 9.7: A prospect based on two independent de Finetti lotteries. The proposition A_m obtains when the number m is drawn from the first lottery; B_n is the proposition that the the number n is drawn from the second lottery.

X	A_1	A_2	A_3	A_4	...
B_1		😬	😬	😬	
B_2			😬	😬	
B_3				😬	
B_4					
$\vdots$					

Table 9.8: Another prospect based on the same two independent de Finetti lotteries.

Y	A_1	A_2	A_3	A_4	...
B_1					
B_2	😬				
B_3	😬	😬			
B_4	😬	😬	😬		
$\vdots$					

As before, we can reason in two ways. First, suppose that the first number is 1. Then X is certain not to have the good result, while Y will have the good result as long as the second number drawn is not also 1—which has probability $1 - \epsilon$ for infinitesimal ϵ . Suppose instead that the first number is 2. Then X has probability ϵ of the good result, while Y has probability $1 - 2\epsilon$ of the good result, which is better. And so on for any number m : supposing the first number drawn is m , X 's probability of happiness is infinitesimal (namely, $(m - 1)\epsilon$), while Y 's probability of happiness is infinitesimally close to 1 (namely, $1 - m\epsilon$). So conditional on any possible answer to the question of what m is, Y is better than X . Reflection tells us that Y is better than X unconditionally. But once again, exactly parallel reasoning tells us that X is better than Y unconditionally. This is untenable.

In short, Reflection is inconsistent with the existence of a de Finetti lottery—a

uniform probability distribution over a countably infinite set Someone who likes standard probability theory, with its principle of countable additivity, may feel they have escaped.¹⁴

But even the “Standardist” faces difficulties here. As I argued in section 5.5, the Standardist should distinguish between two different orderings of events: the *standard probability* order $\mathbb{P}(A) \geq \mathbb{P}(B)$, and the *evaluative* order on events $\text{😊}_A \succeq \text{😊}_B$, where 😊_A is a prospect which results in 😊 if A is true and 😞 if A is false. We can write this relation $A \geq_{\text{😊}} B$. Strong Statewise Dominance implies that two events with the same standard probability can differ in the weight they give to their outcomes for the purpose of evaluating trade-offs. (For instance, getting infinitely many heads has standard probability zero, but Strong Statewise Dominance tells us that getting a good thing only on infinitely many heads is better than getting the same good thing only if $2 + 2 = 5$.)

The argument about de Finetti lotteries can be construed as an argument about this *evaluative* order. In this setting, here is the structural analogue of the de Finetti lottery:¹⁵

Infinite Indifference. There is a countably infinite set of pairwise disjoint possible events $E_1, E_2, \dots$ such that $E_m \sim_{\text{😊}} E_n$ for each m, n .

¹⁴TODO. This is not totally under control. Get a handle on the relationship between this and conglomerability, and its relationship to countable additivity (as well as full additivity), and the connection to regularity. K. Easwaran (2013); Stewart and Nielsen (2021); Seidenfeld et al. (2024); Schervish, Seidenfeld, and Kadane (2017); Schervish, Seidenfeld, and Kadane (1984).

¹⁵This version of the argument also turns on some modest structural conditions on value. Here is the principle appealed to most immediately:

- (1) Let $\mathcal{E}$ be a countably partition such that $E \sim_{\text{😊}} E'$ for all $E, E' \in \mathcal{E}$. Let A, B, F be events such that $A \cap F$ is a union of finitely many elements of $\mathcal{E}$, and $B \cap F$ is a union of infinitely many elements of $\mathcal{E}$, and $A \cap B = \emptyset$. Then $\text{😊}_A <_F \text{😊}_B$.

This can be derived from some more natural conditions (though these are jointly much stronger assumptions than the argument requires).

- (2) If $X(s) = Y(s)$ for all $s \in E$, then $X \sim_E Y$.
- (3) If $A <_{\text{😊}} B$, then $\text{😊}_A <_{A \cup B} \text{😊}_B$.
- (4) If $A \subsetneq B$, then $A <_{\text{😊}} B$. This follows from *Strong Statewise Dominance*.
- (5) If A and B are each disjoint from E , and $A \sim_{\text{😊}} B$, then $A \cup E \sim_{\text{😊}} B \cup E$.

Proof of (1). From (2), to show $\text{😊}_A <_F \text{😊}_B$ it suffices to show that $\text{😊}_{A \cap F} <_F \text{😊}_{B \cap F}$. From (3), it then suffices to show that $A \cap F <_{\text{😊}} B \cap F$. From (4), it then suffices to show that $A \cap F \sim_{\text{😊}} C$ for some $C \subsetneq B \cap F$. This in turn follows from (5). By assumption, $A \cap F = A_1 \cup \dots \cup A_n$, where these are n distinct elements of $\mathcal{E}$. There are also distinct $B_1, \dots, B_n \in \mathcal{E}$ such that $B_1 \cup \dots \cup B_n \subsetneq B \cap F$. By applying (5) n times, we find that $A_1 \cup \dots \cup A_n \sim B_1 \cup \dots \cup B_n$.

Should the Standardist accept that this is possible? It seems so. Here are two arguments.

First, the Standardist is normally happy to accept a uniform probability distribution over an *uncountable* set. Imagine throwing an ideally sharp dart at a continuous square dartboard, identified with the set of pairs of real numbers $[0, 1] \times [0, 1]$. The Standardist says that the probability of hitting any particular point is zero. It is natural for them also to say that these different events are *evaluatively* equivalent as well: for any points $x, y \in [0, 1] \times [0, 1]$, the prospect 🎯_x that pays off if x is hit and the prospect 🎯_y that pays off if y is hit are equally good. If that is so, then we can construct our “evaluative de Finetti lottery” by restricting attention to a countable subset of these points—say, those with rational coordinates. These are a countably infinite set of evaluatively equivalent non-empty events.¹⁶

The second argument is closely related, but may be more vivid. The Standardist is normally happy to accept infinite coin flip distributions: an infinite set of mutually independent events with probability $1/2$. For the standardist, this does not straightforwardly imply that getting heads on the n th flip and getting tails on the n th flip are *evaluatively* equivalent, but again it seems they should be willing to accept this. Indeed, consider any sequence of coin flips w , and let $\sigma_n w$ be the result of swapping the result of the n th flip from heads to tails or vice versa. Each of these events have standard probability zero; it is also natural to suppose that these two events are *evaluatively equivalent*. Call this principle *Coin Flip Indifference*.

Now consider the sequence of events of the form “heads on the first n flips, and tails thereafter”:

$$E_n = H(1, \dots, n) \setminus H(n+1, \dots).$$

Then E_2 is just like E_1 except that the second flip is switched from tails to heads. Likewise E_3 is just like E_2 except that the third flip is switched from tails to heads, and so on. Coin Flip Indifference implies $E_1 \sim \text{🎯} E_2 \sim \text{🎯} \dots$. This again yields an evaluative de Finetti lottery.

¹⁶This is related to a challenge raised by Skyrms (1995, sec. 2):

Suppose that your degrees of belief in our example correspond to Lebesgue measure on Borel propositions, as described above. Suppose that you are given the information that the outcome is a rational point. What should your new degrees of belief be?

But in the argument at hand we have merely *restricted our attention* to a countable set of states—we did not need to *update* on it.

In short, I think the possibility of Infinite Indifference should be highly appealing to Standardists and Non-Standardists alike. But this conflicts with Reflection, even *without* appealing to Fanaticism.

To sum up, the general Reflection principle conflicts with each of the following (individually):

1. Fanaticism
2. Ex Ante Impartiality
3. Addition Invariance and Comparability
4. Infinite Indifference

On the whole, this looks like bad news for Reflection!

But it also looks like at least somewhat bad news for *Finite* Reflection. As I argued in (2024, 584), to accept the finite principle without going all the way to the general version “seems unprincipled, when the arguments in support of each are so similar”. The finite principle seems like a mere ad hoc “truncation” of the general principle; so the fact that Reflection in general conflicts with other obvious-seeming principles seems

like [it] might be telling us that there is something wrong about the underlying ideas. Whatever motivations there are for the modified and truncated principles seem to equally well motivate the sweeping clean versions. (2024, 585)

In turn, these consequences of Reflection also seem to undermine the arguments for Fanaticism based on the Sure Thing Principle.

This is how I took things to stand until recently—but now I see things differently. While some motivations for the Sure Thing Principle do generalize to supporting full-fledged Reflection, I now think that one of the strongest arguments does not generalize this way, when properly understood. Let’s turn to it now.

9.5 Is Fanaticism dynamically consistent?

A standard and powerful argument for the Sure Thing Principle is based on the idea of *dynamic consistency*. In its original decision-theoretic setting, the idea is that agents who violate the Sure Thing Principle do not behave coherently across time as they gain new information. One way of dramatizing this incoherence is by way of a *money pump* argument (for discussion see Hammond 1988; Machina 1989; Buchak 2013, ch. 6; Briggs 2015; Gustafsson

2022, ch. 5). Adapting this to our present context, it becomes a *value pump* argument.

Suppose that the Sure Thing Principle is not true; concretely, let's specifically consider the probability discounting theory as in section 9.1. Recall the example we considered there of the Small Island and the Big Island, with an unreliable speedboat. According to probability discounting, if the boat is working, then it is better to save the Big Island. If the boat is not working, the two options are equally good. But unconditionally, it is better to save the Small Island.

$$\text{Big} \succ_E \text{Small}$$

$$\text{Big} \sim_{\neg E} \text{Small}$$

$$\text{Big} < \text{Small}$$

(where E is the proposition that the boat is working.)

Now let's consider a further variation on this case.

Moira is a moral agent who always chooses whichever available option is best overall in light of her evidence. She also knows this about herself. Moira is in the speedboat, facing the choice of whether to try to save the Big Island or the Small Island.

Moira's friend Esther is also aboard with her. Esther is an expert mechanic who has thoroughly inspected the engine, and she knows whether it is in working order, or whether it will stop working before she reaches either island. Esther says:

Do you want me to tell you whether your boat is working before you choose which way to go? If I don't, then instead I'll sing you a slightly annoying song. Which will it be?

Moira reasons as follows. Supposing that the boat is working, it is better to go for the Big Island. So that is what she knows she will do if gets that news. Supposing the boat is not working, she's just not going anywhere, so the result is again the same as that of trying to go to Big. So she knows in advance that hearing Esther out will have the same result as choosing Big in any case. But, in light of her current information, Big is *morally worse* than Small. Going for Big has only a negligible chance of saving anyone at all, while Small has a non-negligible chance of saving a hundred lives. Moreover, the harm of listening to a slightly annoying song does not outweigh the

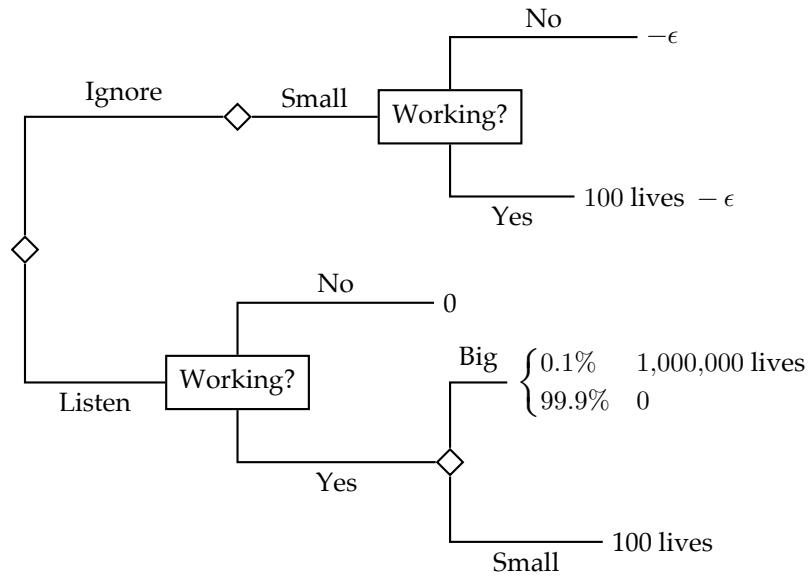


Figure 9.1: A decision tree representing Moira's situation. Diamonds are choice nodes.

chance of saving a hundred lives. So Moira, choosing what is best in light of her evidence, grits her teeth and says "No thanks, don't tell me. Just sing the song."

But this is a strange thing for Moira to say. Notice that there is another thing that she *could* do: she could hear her friend out, and then head for the Small Island anyway. In light of her current evidence, that course of action corresponds to a morally better prospect. In fact, that course of action is *certain* to be better than heading for the very same Small Island while listening to an annoying song. The strategy "Listen, Small" *statewise dominates* the strategy "Ignore, Small". Moira *could* take that alternative course of action, which by her own lights is certain to be better. But she also knows that she *wouldn't* do so. If only she wasn't such a conscientious moral agent, things would surely be morally better!

A similar thing goes for any violation of the Sure Thing Principle (or Finite Reflection): for any such violation, there is a possible situation of sequential choice in which an agent who knows themselves to always choose what is best given their evidence will choose a strategy that is certain to turn out morally worse than some other strategy they could have chosen instead.

That seems pretty bad!¹⁷

That is how the *dynamic consistency* argument for the Sure Thing Principle goes. It is a powerful and influential argument—though more recently it has fallen on hard times, for reasons I will discuss below. But why stop there? J. S. Russell and Isaacs (2021) argue that the same style of dynamic consistency argument does not just support *Finite* Reflection, but also the more general Reflection principle (see also Hammond 1988). If this is right, then since Fanaticism is inconsistent with Reflection, Fanatical theories are dynamically inconsistent!

Let's see how the argument goes with another concrete example.

A tsunami is approaching two islands, *Lesser St. Petersburg*, and *Greater St. Petersburg*. Petey is on a speedboat between the two islands, and he can only rescue the inhabitants of one of them.

Lesser St. Petersburg has probability $1/2$ of hosting two people, probability $1/4$ of hosting four people, probability $1/8$ of hosting eight, and so on.

Greater St. Petersburg has probability $1/2$ of hosting four people, probability $1/4$ of hosting eight, probability $1/8$ of hosting sixteen, and so on.

(The populations of the two islands are probabilistically independent of one another.)

Suddenly Petey gets a call from his friend Esther (more of an acquaintance, really):

I know how many people are on Greater St. Petersburg! Do you want me to tell you? If I don't, then instead I'll sing you a slightly annoying song. Which will it be?

Unconditionally, it is better overall to for Petey to save Greater St. Petersburg. (One way of arguing for this is by appealing to Stochastic Dominance.) But supposing that Greater St. Petersburg has only four people, it is better to save Lesser St. Petersburg. Likewise if Greater St. Petersburg has only eight people, or if it has only sixteen, and so on. So whatever Esther says, conditional on that new information, it will be better to save Lesser St. Petersburg.

Petey also knows that he will do whatever is best in light of his evidence. So

¹⁷TODO. Make this precise in appendix. section C.7

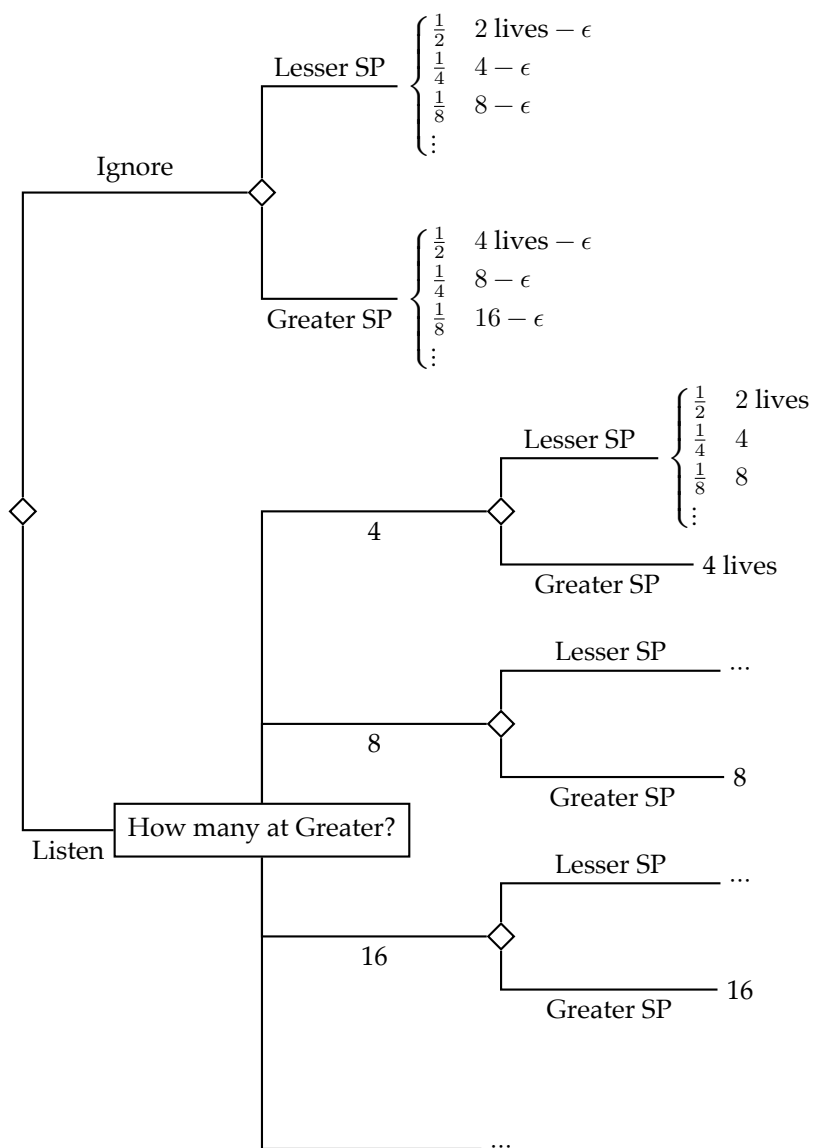


Figure 9.2: A decision tree for Petey's situation.

he reasons as follows.

If Esther tells me the news, whatever she tells me, I will save Lesser St. Petersburg. But, all things considered, it is best in light of my current evidence if I save Greater St. Petersburg—and it remains all-things-considered best even if I have to suffer the song. So that is what I should do.

Like Moira, Petey chooses the strategy (Ignore, Greater St. Petersburg) even though it is *dominated* by the alternative strategy (Listen, Greater St. Petersburg), which he could do instead. He knows he *wouldn't* do that, but the only thing stopping him is his own goshdarn morally optimal choices.

On the face of it, the value pump argument against Petey's values seems just as strong as the one against Moira's—and this is essentially what J. S. Russell and Isaacs (2021, sec. 2) argued. But I now believe that the two arguments are more different than they appear, and the difference is one with deep normative significance.

9.6 Equilibrium strategies

Before explaining that, first let's more carefully examine the normative upshot of the value pump argument in the simpler, finite case. What exactly is wrong with being value pumped, anyway? Many philosophers have said "nothing"—at least as far as *rationality* goes. Indeed, despite their popularity a few decades ago, more recently dynamic consistency arguments have met with so much resistance that I think it is fair to say that it has become unorthodox among philosophers to defend them.

The problem with being value-pumped, if there is one, seems to have to do with coordinating action across time: a single agent, acting at different times, ends up producing foreseeably bad results. A popular perspective on such issues is called "time-slice rationality": we can think about a single agent acting at different times in the same way that we think about a collection of distinct agents each acting separately (Moss 2012; Buchak 2013, 2015; Hedden 2015b, 2015a). But, as Hedden (2015b, 465) puts it, "It is uncontroversial that collections of distinct agents can act in a way that predictably produces a mutually disadvantageous outcome without there being any irrationality." In particular, he argues that value pumps "have the structure of a Prisoner's Dilemma". Consider the famous example (table 9.9).

Prisoner's Dilemma. Two prisoners, known as Row and Col, are being interrogated about a crime they committed together. Each of them can either *cooperate* with the other prisoner by keeping quiet, or *defect* by ratting out their accomplice. If both stay silent, they will both escape prosecution. If they both talk, they will both be punished. But the interrogators, being no fools, credibly offer each prisoner this deal: “If you tell us everything, and your partner stays silent, we will grant you immunity, buy you lunch, and throw the book at your partner.”

Table 9.9: The prisoner's dilemma. The lower left number in each cell represents the utility for Row, and the upper right number represents the utility for Col.

	Cooperate	Defect
Cooperate	+1 +1	-2 +2
Defect	+2 -2	-1 -1

Each prisoner can reason as follows.

My partner may *cooperate* with me, or they may *defect*. If they cooperate, I am better off if I defect by confessing: I get off scot-free, and they take the rap. If they defect, I am still better off if I also defect: I'll go to prison, but if we share the blame the punishment will be less severe. Defecting *dominates* cooperating, as far as my individual benefit goes: I am sure that things will go better for me if I defect than they will if I cooperate. So I had better defect.

If both prisoners reason this way, then they will both choose to defect. But this is worse for *both* of them than if they had both chosen to cooperate. In short: each does what is individually rational, yet they are collectively worse off than they could have been if they had only done things differently. This phenomenon is often called *collective self-defeat*: it is the interpersonal analogue of dynamic inconsistency (Parfit 1984, secs. 10–11, 21–24).¹⁸

¹⁸Buchak (2013); Buchak (2015) puts the point a bit differently. She says that from the “time-slice” perspective, “there is no agent to whom temporally extended acts are available: there is only a temporal extension of time-slices to whom acts-at-a-time are available.” (2015,

Here is how we can describe this using the standard terminology of game theory. A (*pure*) *joint strategy* is a choice of action for each agent in each situation they may find themselves in. (This is in contrast with *mixed strategies*, which we will not be discussing here.) A joint strategy is a *pure equilibrium* if it is optimal for each agent, on the supposition that every agent also plays by this strategy. One strategy *Pareto dominates* another if it is certain to turn out better for every agent. In the prisoner's dilemma, the *joint strategy* (Cooperate, Cooperate) Pareto dominates the strategy (Defect, Defect). Yet the strategy (Defect, Defect) is a pure equilibrium, while (Cooperate, Cooperate) is not.

I think Hedden's analogy provides some important insight. Let's press it further. In Moira's islands example, her reasoning is an example of what is called the "sophisticated choice" rule. In each situation she finds herself in, she reasons about what she will do in *other* situations, on the assumption that she will do something optimal in that situation. In each possible situation, she will choose an option that is best for *that* situation in light of her beliefs about what she will choose in *other* situations. In other words, a strategy selected by the sophisticated choice rule is an *equilibrium strategy*: it is optimal in each situation, on the supposition that she also plays by this strategy in each *other* situation. Moira's strategy (Ignore, Small) is an *equilibrium*, even though it is *dominated* by an alternative available strategy (Listen, Small).

From this game-theoretic perspective, the idea that the sophisticated chooser should not be "value pumped" seems to be underwritten by this principle:

Eq1. Pure equilibrium strategies are not dominated.

And this we know to be false, in general, as is shown by the prisoner's dilemma.

That said, our topic is a bit different from Hedden's (and Buchak's, and most of the others who have discussed dynamic consistency in depth). In order to engage with them, I have turned my attention toward *individual*

853). I don't think this is the best way of putting the point. The strategy is *available* in the straightforward sense that the agent *could* choose to act in accordance with it, simply by choosing each constituent act—in the same sense that the all-cooperate strategy is available to the prisoners. Just because they know (insofar as they are individually rational) that they *won't* choose that strategy does not mean that they *couldn't* have. The point is not that the strategy is not available—rather, the point is that failing to choose it does not display a rational deficiency for any particular agent at any particular time. From the time-slice perspective, this is the only kind of rational deficiency worth worrying about.

TODO. Something about "resolute choice" somewhere.

instrumental rationality, where what is at issue is whether each individual chooses the best means toward their peculiar ends, whatever those may be. But the topic of this book is not what is *rational* but rather what is *good*—and in particular, what is *impartially* good.¹⁹ Furthermore, it seems that in the prisoner’s dilemma, the prisoners find themselves in their predicament precisely because they are *not* motivated by what is impartially good. Each of them is pursuing their own narrow self-interest, rather than what is best for *both of them* together, on balance. Row chooses to defect because this is best for Row, even though it harms Col more than it benefits Row. Col chooses to defect because this is best for Col, even though it harms Row more than it benefits Col. The result is that even though each chooses the option that is in their own interest, both are harmed more than they are benefited.

This kind of dilemma only arises for those who are insufficiently benevolent. Indeed, plausibly part of the *point* of the idea of impartial moral value is precisely to solve coordination problems (see, for example, discussion in Parfit 1984, secs. 23–24). Suppose that the two prisoners are not self-interested, but rather pursue their common good. For concreteness, suppose this is the sum of the two individual values (table 9.10). In light of these impartial values, it is straightforwardly best for *each* prisoner to choose to cooperate. The strategy that is certain to turn out *impartially best* is also an *impartial equilibrium*: it is impartially better for each of them to choose their respective component act of the strategy, supposing that the other does as well.

Table 9.10: Impartial prisoners face no dilemma.

	Cooperate	Defect
Cooperate	+2	0
Defect	0	−2

¹⁹That said, we might understand “rationality” not merely as *instrumental* rationality, which might be directed at any end whatsoever, but rather as *prudential* rationality, concerned with achieving what is *in your own interest*—where *you* are a persisting agent with interest in how things go for you in the future as well as in the present. The points that follow about impartial value apply to this kind of rationality as well: the relationship between dynamic inconsistency and prudential rationality is analogous to that between collective self-defeat and impartial value.

It is tempting, then, to endorse this principle:

Eq2. Pure *impartial* equilibrium strategies are not dominated.

This principle also supports the Sure Thing Principle. Moira's case does not involve conflicting values; rather Moira is motivated throughout by impartial values. In general, violations of the Sure Thing Principle for impartial value give rise to dynamic inconsistency for choices that are guided entirely by impartial value. These are situations in which an *impartial* equilibrium is dominated.

But this replacement principle Eq2 is also false. We can see this by considering a variant of another famous example from game theory.²⁰ James (1896) offers an evocative description:

A social organism of any sort whatever, large or small, is what it is because each member proceeds to his own duty with a trust that the other members will simultaneously do theirs. Wherever a desired result is achieved by the co-operation of many independent persons, its existence as a fact is a pure consequence of the precursive faith in one another of those immediately concerned. A government, an army, a commercial system, a ship, a college, an athletic team, all exist on this condition, without which not only is nothing achieved, but nothing is even attempted. A whole train of passengers (individually brave enough) will be looted by a few highwaymen, simply because the latter can count on one another, while each passenger fears that if he makes a movement of resistance, he will be shot before any one else backs him up. If we believed that the whole car-full would rise at once with us, we should each severally rise, and train robbing would never even be attempted.

Train Robbery. Two passengers, Row and Col, are on a train being robbed by a highwayman (table 9.11). They know that if they both fight back together, they can successfully resist the robbery. If they both hang back meekly, they will be robbed, but at least keep their lives. The disastrous case is if one of them fights while the other hangs back.

²⁰This is a variant of the kind of case standardly called "stag hunts", after an earlier example from Rousseau ([1755] 1992). See also discussion in D. H. Regan (1980, ch. 2) and Parfit (1988, sec. II).

Table 9.11: James's train robbery case.

	Fight	Hang back
Fight	+2	−1
Hang back	−1	0

Crucially, we need not suppose that the two passengers are merely self-interested. Instead, let's suppose they are each motivated by their collective benefit. In this case, there are two different ways things might go. If a passenger believes the other will Fight, then it is best overall if they too choose to Fight. So (Fight, Fight) is an impartial equilibrium strategy: supposing the other goes with it, it is impartially best if you do too. But also, if a passenger believes the other will Hang Back, then it is best overall if they too choose to Hang Back—there is no virtue in pointlessly acting alone. So (Hang Back, Hang Back) is *also* an impartial equilibrium strategy: supposing the other goes with it, again it is impartially best if you do too. But note: (Hang Back, Hang Back) is *dominated* by (Fight, Fight). So this is a counterexample to the proposed principle Eq2: even a pure *impartial* equilibrium strategy can be dominated.

There is also a different style of counterexample to Eq2 that specifically has to do with infinity:

Satan's Apple. (Arntzenius, Elga, and Hawthorne 2004, 262) Satan has cut a delicious apple into infinitely many pieces, labelled by the natural numbers. Eve may take whichever pieces she chooses. If she takes merely finitely many of the pieces, then she suffers no penalty. But if she takes infinitely many of the pieces, then she is expelled from the Garden for her greed. Either way, she gets to eat whatever pieces she has taken.

Eve's first priority is to stay in the Garden. Her second priority is to eat as much apple as possible. She is deciding what to do when Satan speaks. 'Eve, you should make your decision one piece at a time. Consider just piece #1. No matter what other pieces you end up taking and rejecting, you do better to take piece #1 than to reject it. For if you take only finitely many other pieces, then taking piece #1 will get you more apple without incurring the greed penalty. On the other hand, if you take infinitely many other pieces,

then you will be ejected from the Garden whether or not you take piece #1. So in that case you might as well take it, so that you can console yourself with some additional apple as you are escorted out.'

Eve finds this reasonable, and decides provisionally to take piece #1. Satan continues, 'By the way, the same reasoning holds for piece #2.' Eve agrees. 'And piece #3, and piece #4 and ...' Eve takes every piece, and is ejected from the Garden. Where did she go wrong?

In this case, Eve finds herself in infinitely many different decision situations: in situation 1, the question is whether to take piece #1; in situation 2, whether to take piece #2, and so on. In this case, there is exactly one pure equilibrium strategy available to her: take every piece. Suppose that she knows this is what she will do. Then in each situation she finds herself, she knows that taking piece # n gains her a bit of apple, and makes no difference to her eventual ejection from the garden—so it is best for her. Moreover, it is not merely best for her with respect to some kind of narrow temporary self-interest—this is not a diachronic prisoner's dilemma, where one causes harm to one's past or future self for the sake of present benefit. Rather, from the perspective of temporally-neutral concern for Eve's well-being throughout her whole life, it is better if she takes this particular piece. So *take every piece* is an impartial equilibrium. Indeed, it is the only pure equilibrium for this game. For any pure strategy where she does not take piece # n , she can do strictly better by switching to taking # n , holding fixed what she does in every other situation. Yet even though *take every piece* is the unique pure equilibrium, it is still a dominated strategy. Indeed, it is dominated infinitely many times over, since any strategy that involves taking only finitely many pieces would be sure to turn out better. Again, then, an impartial equilibrium strategy can be dominated.²¹

So far this casts the value pump argument in a dim light. But I think there is still something strong to be said for it. There is something going wrong in Moira's case, and it is something different from what is happening in the prisoner's dilemma, the train robbery, or Satan's apple.

²¹TODO. The "deliberative dynamics" approach in Bartha, Barker, and Hájek (2014) is getting at essentially the same idea. Something worth noting is their "simplified setup" actually has totally different equilibrium structure from the original Satan's apple, and does not actually raise the same issues. That version has infinitely many (weak) equilibria, and the worst case of taking all pieces is *not* an equilibrium. TODO Also discuss McGee's (1999) "airtight Dutch book"?

Let's spell out some ideas a bit more. Consider a game set-up, which includes various possible states of nature, and various possible situations an agent might be in. (It doesn't matter whether we think of this as one agent choosing on different occasions, or different agents for different situations.) For each situation i , let $E(i)$ be the set of possible states consistent with being in i . For each situation i , there are also various *available acts*. A *strategy* is a choice of an available act for each situation. Each state, together with a choice of act for situations compatible with that state, determines an *outcome*. A theory of value then tells us whether a strategy A is as good as another strategy B , given any particular body of information E .²²

(I am dramatizing things in the usual fashion by describing things in terms of *choices* of particular acts. But, officially, this is not really what is at issue here. The question presently before us is not *what you ought to do*, but rather *what would be best on balance*. We can think of the "acts" available in each situation simply as certain ways things might be. If each act $A(i)$ is represented by a certain proposition, then "choosing" $A(i)$ in situation i amounts to making true the proposition *if $E(i)$ then $A(i)$* . A strategy amounts to a conjunction of such propositions. When we say one strategy is better than another, we just mean that it would be better for one strategy-proposition to be true than for another.)

Here are two different ways of selecting a strategy. First, one might "globally" choose the strategy that is best on balance, given only the information that one is in this game set-up.

Definition. A strategy A is a **a priori impartially optimal** iff for all strategies B , $A \succeq B$ unconditionally.

Second, one might go for a strategy that is consistent with "sophisticated choice": supposing that this strategy is chosen, it is optimal to choose it in each possible situation.

Definition. A strategy is a **(pure) impartial equilibrium** iff for each possible situation i , A is impartially at least as good conditional on $E(i)$ as any other strategy B such that $B(j) = A(j)$ for all situations $j \neq i$.

We have seen that these can come apart. In both the train robbery and Satan's apple cases, there is a strategy which is an impartial equilibrium, but not

²²I am presently excluding from consideration *unstable* decision situations, where the information that a certain strategy is chosen makes a difference to the probabilities strategies have of resulting in one outcome or another. [TODO. Come back to this.]

optimal—not only is it *worse on balance* than some other strategy, but in fact it is *statewise dominated* by another strategy.

But what about the reverse? Consider:

The Equilibrium Principle. Each a priori impartially stably²³ optimal pure strategy is a pure impartial equilibrium.

In *train robbery*, (Fight, Fight) is the optimal strategy—and lo, it is an equilibrium. In *Satan's apple*, there is no optimal strategy: every strategy that involves taking infinitely many pieces is surpassed by taking only finitely many, and every strategy that involves taking finitely many pieces is surpassed by any strategy that involves taking a larger finite number. So neither case is a counterexample to the Equilibrium Principle.

Moreover, the Equilibrium Principle makes good on the idea we introduced earlier: *impartial value is supposed to solve coordination problems*. Here we all are, about to be thrown into various idiosyncratic situations in a shared world. We are going to make a shared plan for how we will all behave in our various situations. There are two ways we might do this. We might spell out a specific rule in glorious detail: “For situation i , choose act $A(i)$. This law we have judged to be best on balance.” Or we might instead give a *principle*. “Here is an ordering of acts, given various possible bodies of information you may find yourself with. Adopt these values, then go and do what is best by your own lights.” The Equilibrium Principle says that these two ways of specifying what to do are consistent with one another: if a rule is acceptable from the original position, then it is also acceptable according to our principles about what will be best by our own lights.

For a different just-so story, imagine impartial value personified as an advisor, whispering to each of us what we ought to do given our evidence—including our evidence about what others will do. The advisor deems that a certain strategy is *best overall*. Happily, we all know that each of us will follow this strategy. And yet, perversely, this advisor tells some of us to defect from this optimal strategy that we know will be followed, and instead do something else that would be worse overall. A system of values that undermined effective coordination in this way would not be doing its proper job.

In contrast, in a case like *Satan's apple*, with *no* optimal strategy, it is far from clear what a system of values *should* be directing us toward—and so it

²³TODO: explain “stably” and what it’s doing

is far from clear whether it is getting things wrong. The unique equilibrium strategy, whispered by the voice of the Good in each of our ears, is worse than infinitely many alternatives—but *every* strategy is worse than infinitely many alternatives, so this does not seem like strong grounds for criticism.

The Equilibrium Principle has a similar flavor to Kant’s famous prescription: “act only in accordance with that maxim through which you can at the same time will that it become a universal law” (Kant [1785] 1997, 4:421). A straightforward reading of Kant requires a little bit of amendment. Suppose you are in *train robbery*, and as a matter of fact you know that the other passengers will Hang Back. In this case you too ought to Hang Back—even though, if you were in the position of supreme legislator, you ought to decree that everyone Fight. But consider this closely related principle:

For any maxim *A*, if, supposing *A* to become a universal law, one ought not conform to *A*, then *A* is not the true moral law.

This becomes the Equilibrium Principle if we apply the following translations. Understand “maxim” as “(pure) strategy”. Understand “*A* becomes a universal law” as “*A* is followed in every possible situation”. Understand “one ought not conform to *A*” as “*A*(*i*) is not the best available act in some situation *i*”. And understand “*A* is the true moral law” as “*A* is impartially best a priori”. Of course, this would be a very strained reading of Kant. But Kant at least points in the same general direction as the Equilibrium Principle. Indeed, I think it is not too implausible to imagine fitting the Equilibrium Principle into a larger Kantian ambition, deducing features of the moral law from the very idea of it being *universal, categorical, binding on all rational beings as such*. One such feature would be the Sure Thing Principle. But that ambitious rationalist project goes far beyond the scope of this book.

The Equilibrium Principle is inconsistent with the values described in Moira’s finite case. Indeed, the Equilibrium Principle straightforwardly implies one direction of the Sure Thing Principle [REF appendix.]

If prospects *X* and *Y* are equivalent supposing not-*E*, then if *X* is at least as good as *Y* unconditionally, *X* is also at least as good as *Y* conditional on *E*.

It also implies the other direction, given some further auxiliary assumptions [TODO].

But what about the infinite case? At long last, let’s return to Petey’s case. Petey chose the equilibrium strategy—refuse the news, suffer the song,

and save Greater St. Petersburg—even though this was dominated by an alternative available strategy—get the news, and save Greater St. Petersburg anyway. In contrast with Moira’s case, though, Petey’s does not violate the Equilibrium Principle!

In both Moira’s case and Petey’s, there is a unique pure equilibrium strategy: don’t listen to the news, and head to a certain island. In both cases, this equilibrium strategy is worse overall than the alternative of listening to the news and then going to that island anyhow. But in Moira’s case, (Listen, Small) is not just a *better* strategy than the equilibrium (Ignore, Small): it is an *optimal* strategy. Since it is optimal, and not an equilibrium, this violates the Equilibrium Principle. In Petey’s case, though, even though (Listen, Greater) is a *better* strategy than (Ignore, Greater), it is *not optimal*. Petey can do better yet.

Here is a different strategy: hear Esther out about the population of Greater St. Petersburg, and then save Greater St. Petersburg, *if* its population is at least eight people; otherwise go for Lesser St. Petersburg. Even from the perspective of unconditional value, while Greater is better than Lesser, “Save Greater St. Petersburg unless it turns out to be especially small” is better still. But also, here is an even better strategy: hear the news, and then save Greater St. Petersburg if its population is at least *sixteen* people. And an even better one: hold out for at least *thirty-two*.

In fact, Petey’s case is structurally analogous to Satan’s apple: there is *no* optimal strategy, only an ascending chain of better and better strategies. So the fact that Petey’s equilibrium strategy is *not* optimal is not a strike against it: *nothing* is optimal, so of course he can’t do what is best.

J. S. Russell and Isaacs (2021, 184) argued that the problem with Petey’s type of case is not merely that the equilibrium strategy is *dominated*, but that it is dominated by a strategy which is *not itself dominated*. This is in contrast with Satan’s apple: in that case, each of Eve’s alternative strategies is dominated by some better strategy. In Petey’s case, while the alternative strategies are better on balance a priori, they are not *certain* to turn out better.

But this difference is not as important as Russell and Isaacs make it out to be. Dominance is one way that a strategy can be better than another—an especially clear-cut way, but not the only way. Just because a strategy is *not dominated* does not mean it is best, or that it is what you should do. And just because a strategy is *dominated* by an undominated strategy does not mean you should do the other instead — nor does it imply that there is *anything*

you should do instead. In Petey's case, the pure equilibrium (Listen, Greater) is dominated by (Ignore, Greater), which itself is undominated. But even though (Ignore, Greater) is not *dominated* by other strategies, it is still *worse* than infinitely many other strategies. The same goes for any alternative to (Listen, Greater): no strategy is optimal.

So in fact there is a much deeper contrast than meets the eye between the value pump argument for the Sure Thing Principle and the value pump argument for Reflection. In fact, we have found what J. S. Russell and Isaacs (2021) challenged us to find: a cogent motivation for the Sure Thing Principle that does not extend to the infinitary version, Reflection. In particular, the value pump argument dramatizes a violation of the Equilibrium Principle; the Equilibrium Principle supports the Sure Thing Principle, but it does not support Reflection in general.

Moreover, Fanaticism is consistent with the Equilibrium Principle (section C.7). So arguments for Fanaticism on this basis—unlike appeals to Reflection—are dialectically stable after all.

TODO. Perhaps we also say something briefly here about the value of information and regret arguments, too. Not so clear how they work, if there is no optimal strategy...

That is not to say the Fanatic has given up nothing. Apart from dynamic consistency considerations, Reflection is plausible on its face. Life without it is strange. But, my earlier doubts notwithstanding, there really is a consistent, principled position here.

9.7 Conclusion

This chapter has taken us through some twists and turns. Its central topic all along has been Savage's *Sure Thing Principle*. We saw that this principle is inconsistent with discounting small probabilities, and (together with other auxiliary assumptions such as Ex Ante Impartiality or Addition Invariance) it presses us in the direction of Fanaticism. On the other hand, a natural generalization of the Sure Thing Principle — Reflection — is *inconsistent* with Fanaticism. Reflection conflicts with some other very natural ideas (such as Coin Flip Indifference, together with some auxiliary assumptions), so there are some strong reasons to resist it. Does this also undermine the case for

Fanaticism based on the Sure Thing Principle? Or is there a philosophically principled position that holds onto the Sure Thing Principle while giving up Reflection?

I have argued (against my past self) that there is. In particular, I have argued that the right kind of *dynamic consistency* argument is a strong consideration in favor of the Sure Thing Principle, without taking us all the way to Reflection (contrary to what I and others have said in the past). As others have argued, violations of dynamic consistency are best understood as analogous to interpersonal coordination problems — that is, collective self-defeat. But taking this analogy seriously does not undermine the dynamic consistency argument for the Sure Thing Principle, when this is understood as a principle about impartial value. Rather, it points us toward a clearer understanding of dynamic consistency's underlying principle: the *Equilibrium Principle*. A coherent system of impartial value does not publicly announce that a certain strategy is best overall, while privately inciting individuals to defect from that strategy even when they know others will follow it. Rather, there should be harmony between what the Good declares once for all and what it whispers in individual circumstances.

This Equilibrium Principle supports the Sure Thing Principle, but it does not support Reflection. We can then put this together with the (suggestive, but not decisive) arguments from the Sure Thing Principle for Fanaticism. The upshot is that escaping dynamic or collective self-defeat may well require Fanaticism.

Chapter 10

Small Changes in Probability

OPPENHEIMER: Chances are near zero.

GROVES: *Near zero.*

OPPENHEIMER: What do you want from theory alone?

GROVES: *Zero* would be nice.

Oppenheimer

10.1 Anti-Fanaticism

One of the central questions before us is whether a minuscule probability of attaining something of enormous value can itself be immensely valuable. *Fanatics* say yes: the certainty of any finite good can always be surpassed by a very small probability of something much better. *Anti-Fanatics* say no: a substantial enough *certain* benefit is better than an extremely tiny probability of a glorious utopia, no matter how wonderful it might be. Tiny enough probabilities are not weighty enough for that.

Here is how we have framed the principle, in its simplest form:

Fanaticism. For any finite probability $p > 0$ and any finite good x , there is some finite good y such that chance p of y is better than the certainty of x .

We have considered certain views that give up this form of Fanaticism,

including bounded expected utility theory and probability discounting. But there is a way in which these views do not really live up to their advertising. (This line of argument follows Cibinel 2023; Tarsney forthcoming; building on earlier observations by Beckstead and Thomas 2024; Kosonen 2022.)

For concreteness, consider *tail discounting*, which we introduced in section 1.2.4. For present purposes, we can focus on the especially simple case of prospects with only two possible outcomes, and where all that is at stake in an outcome is its number of happy lives. Extending the notation we introduced in chapter 2, we will write $p * (x, y)$ for a prospect with probability $1 - p$ of x and probability p of y . Fix some very small finite probability ϵ . (Let's not worry right now about where this number might come from.) Probabilities below the threshold ϵ are rounded down; probabilities above the threshold $1 - \epsilon$ are rounded up; and in between, we use ordinary expected value. Then we can write down a simple formula for such prospects' "tail-discounted expectation":

$$U_{\text{t.d.}}(p * (m \text{ lives}, n \text{ lives})) = \begin{cases} m & \text{if } p < \epsilon \\ n & \text{if } p > 1 - \epsilon \\ (1 - p) \cdot m + p \cdot n & \text{otherwise} \end{cases}$$

(As we noted in section 1.2.4, we may want to complicate this by instead treating small finite probabilities as if they were *infinitesimal* rather than exactly zero; this gives us a form of discounting that respects Simple Stochastic Dominance. But let's ignore this complication for now.)

Tail discounting does not satisfy Fanaticism, as we just formulated it. But even so, it does not quite live up to the Anti-Fanatic's ambition. Sometimes it still puts a lot of weight on tiny probabilities.

Surprise, it's another tsunami! Here is the situation this time (table 10.1).

The **Small Island** has ten inhabitants. The **Big Island** has a hundred million. But the Big Island is farther away, and you have only a small chance of making it to them in time.

This time, though, the Big Island's early warning system might be functioning just fine. In fact, they have a fifty percent chance of surviving whether or not you try to help them. You can increase their chances of surviving by one in a million, from 50% to 50.0001%.

Table 10.1: Save more lives for sure, or very slightly improve the probability of saving *many* more lives?

Probability:	50%	0.0001%	49.9999%
Status quo	10^8	0	0
Small	$10^8 + 10$	10	10
Big	10^8	10^8	0

Both of the options Small and Big are simple prospects, each with two possible outcomes, and all that is at stake is a finite number of happy lives. Small has 50% probability of its worse outcome (10 lives) and 50% probability of its better outcome (a hundred million and ten lives). Big has 49.9999% probability of its worse outcome (zero lives) and 50.0001% probability of its better outcome (a hundred million lives). None of these probabilities are especially close to zero or one — so the tail discounting formula does not actually tell us to discount anything in this case. It gives exactly the same verdict as the simple Expected Number theory: a one in a million chance at saving a hundred million lives outweighs the certainty of saving only ten. Big is better.

The probabilities of outcomes in this case are not close to zero or one—rather, they are very close to *each other*. Tail discounting doesn’t work as intended in this case. In this theory, the value of a small *change* in the probability of an outcome depends on what the *background* probability of that outcome is.¹

Tail discounting does not satisfy the principle of Fanaticism as it is stated above, but it still delivers intuitively fanatical verdicts: extremely unlikely possibilities of enormous effects can still be very weighty. What this reveals is that our formulation of Fanaticism was insufficiently general.

Consider the negation of our formulation of Fanaticism:

Narrow Anti-Fanaticism. For some finite probability $p > 0$ and some finite good x , there is no finite good y such that chance p of y is better than the certainty of x .

Intuitively, this says that if probability p is small enough, then there is a

¹Note that *Simple Separability* (chapter 8) implies that the baseline “status quo” prospect can be subtracted off from Big and Small without affecting their value order. But as we discussed in that chapter, Anti-Fanatics must reject Simple Separability.

limit to how good it can be to boost your chance at something good from zero to p . But this says nothing about the much more general case where your “status quo” chance at enormously good outcomes isn’t zero. What we would really like to uphold is a more general principle than Narrow Anti-Fanaticism that can be applied to other “status quo” prospects. The idea is that a substantial enough certain *improvement* from the status quo is better than an extremely tiny *increase* in the probability of a finite good outcome, no matter how extraordinarily good it might be. In the example above, we can consider a status quo prospect of saving neither island. Compared to this, Small is certain to save an additional ten people, while Big improves the probability of saving a hundred million people by one in a million. If sufficiently small probability events are not decisive, then Small should be better than Big. (We are supposing for the sake of illustration that one in a million counts as small enough, and saving ten lives counts as big enough. We can modify the case to make these numbers whatever we want.)

To state this general idea precisely, let’s again consider outcomes that can be represented as *finite quantities*, like the number of lives saved. In particular, these quantities have a sensible notion of addition. This way we can make sense of improving different possible outcomes by the *same amount*.² For a quantitative prospect X and value difference d , $X + d$ is the prospect that improves the outcome of X by d in every possible state. A *simple quantitative prospect* is a prospect with finitely many possible finite quantitative outcomes. Intuitively, the more general Anti-Fanatical principle says that a small enough *difference* in probabilities can never outweigh a large enough *difference* in value.

Weak Uniform Anti-Fanaticism. There is some finite probability $\epsilon > 0$ and some value difference d such that, for all simple finite quantitative prospects X and Y , if $\mathbb{P}[X \succeq Y] > 1 - \epsilon$, then Y is not as good as $X + d$.

The anti-fanatic plausibly will want to go a bit further yet, exchanging *not as good* for *strictly worse*.

Strong Uniform Anti-Fanaticism. There is some finite probability $\epsilon > 0$

²See section 3.5; section 8.1; section B.7. As we have discussed, very modest structural assumptions give rise to families of outcomes that can be represented by finite quantities. This simplifying assumption can be weakened even further: Tarsney (forthcoming) notes that, rather than assuming that outcomes have quantitative structure, we just need any general notion of an *improvement* for outcomes: a function I from outcomes to outcomes such that $I(x) \succ x$ for all outcomes x . But I think this additional abstraction would only be distracting in the present context.

and some value difference d such that, for all simple finite quantitative prospects X and Y , if $\mathbb{P}[X \succeq Y] > 1 - \epsilon$, then $X + d \succ Y$.

It is not just that small changes in probability don't outweigh big enough changes in value; rather, big enough changes in value *do* outweigh small changes in probability. If the Comparability principle is true (as I argued in chapter 4), then these two formulations are equivalent.

Another way of thinking of this principle is as an *approximate dominance* condition. If $\mathbb{P}[X \succeq Y] > 1 - \epsilon$, this means that X *nearly* statewise dominates Y : it is nearly certain to turn out as well as Y , except on a set of states with extremely tiny probability. If d is a large difference in value, then $X + d$ is nearly certain to turn out *much better* than Y . If very small differences in probability are never extraordinarily important, then this should be enough to guarantee that $X + d$ is better than Y .

Tail discounting does not uphold Uniform Anti-Fanaticism (whether Weak or Strong). As we already saw, the prospect Small has probability 99.9999% of saving ten more lives than Big; but as we discussed, tail discounting says that Big is better than Small. A similar construction shows the same thing for any finite probability $\epsilon > 0$ (perhaps much smaller than 0.0001%) and any value difference d (perhaps much greater than ten lives).

That seems to be a defect of *tail discounting*; but tail discounting is just one way of attempting to make probability discounting precise. Is there some other way of making good on the general idea that — as Nicolaus Bernoulli put it (as quoted by Monton 2019, 6) — “cases that have a very small probability must be neglected”?

10.2 Small probabilities matter

No, there is not. Or, slightly more cautiously: Strong Uniform Anti-Fanaticism is false. This is to construe Bernoulli's idea that “cases that have a very small probability must be neglected” as a claim about which prospects are better than which. In chapter 11 I will return to *non-axiological* construals of the idea of probability discounting, which may hold more promise.

Here is the basic problem: you can't *always* neglect small differences in probability, because small differences in probability can add up to arbitrarily large differences, which cannot be neglected. We can spell out this argument against (Strong) Uniform Anti-Fanaticism in two different ways.

The first is a continuum-style argument, due to Tarsney (forthcoming; building on earlier arguments by Beckstead and Thomas 2024; Cibinel 2023; Kosonen 2022). Recall the principle of *Simple Stochastic Equivalence* (section 1.3.5): in order to compare the value of two simple prospects, we do not need to know anything besides each of their probability distributions over outcomes. If we take this principle for granted, then we can restate Uniform Anti-Fanaticism as a principle about the value of probability distributions. We'll continue to use the notation $p * (x, y)$ for a simple prospect with probability $1 - p$ of outcome x and probability p of y . If y is better than x , then the prospect $p * (x, y)$ generally gets better as p increases. But Uniform Anti-Fanaticism implies that a very *small* increase in p is not good enough to outweigh a *big* enough improvement in both outcomes x and y . In short, Uniform Anti-Fanaticism and Simple Stochastic Equivalence together imply this principle:

AF. For some finite probability threshold $\epsilon > 0$ some value difference d , for all probabilities p and q such that $q < p + \epsilon$, and all finite quantitative outcomes x and y ,

$$p * (x + d, y + d) > q * (x, y).$$

(To see how this follows, let E be a set of states with probability p , let F be a disjoint set of states with probability $q - p$, let X be a prospect that results in y on E and x on $\neg E$, and let Y be a prospect that results in y on $E \cup F$ and x elsewhere, as in figure 10.1. Then $X \succeq Y$ everywhere except on F , so $\mathbb{P}[X \succeq Y] > 1 - \epsilon$. The left-hand side $p * (x + d, y + d)$ is the distribution of $X + d$, and the right-hand side $q * (x, y)$ is the distribution of Y .)

Suppose that AF is true. For concreteness, let's say in particular that one in a million is a small enough probability, and saving one life is a big enough difference in value. (Nothing turns on these numbers.) Then we have the following problem.

Tickets for Lives. You have a million tickets labeled one through a million. Here are the rules: some number from one to a million will be drawn from a fair lottery. If you hold the winning ticket, you save a million and one lives; otherwise you save no one. Right now you have all the tickets, so things look pretty good.

Now along comes Satan. "Listen," he says. "You have all the tickets. But *one* ticket only makes a minuscule difference to your probability of saving

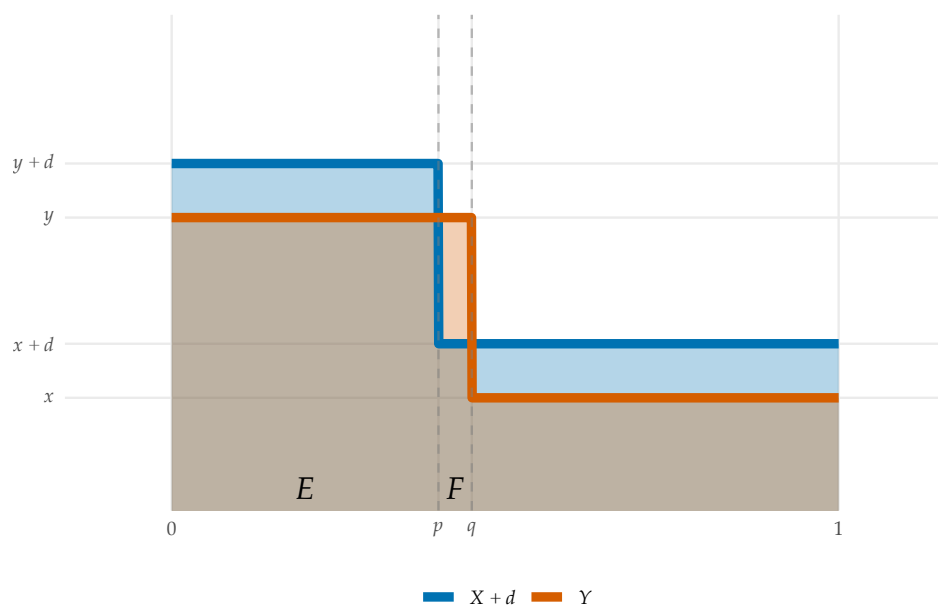


Figure 10.1: The distribution of $X + d$ is $p * (x + d, y + d)$, the distribution of Y is $q * (x, y)$, and $X + d$ has probability $1 - (q - p) < \epsilon$ of turning out d better than Y .

so many lives. I'll make you a trade: give me ticket #1, and I'll spare one additional life."

Principle AF tells you that this is a good trade: very slightly decreasing your probability of the best outcome is outweighed by the certainty of saving an additional life. So you make the trade: give up one ticket, and save one more life. Now you have chance 999,999 in a million of saving a million and *two* lives, and chance one in a million of only saving one life.

Satan goes on. "I'm feeling generous today: I'm happy to make the same trade again. Give me the *second* ticket, and I'll spare one more life. Whad-daya say?" Again, principle AF says that Satan is offering you a better prospect—so you trade again. Now you have chance 999,998 of saving a million and three lives, and chance 2 in a million of saving only two.

It's clear where this is going. Satan offers the same deal over and over again, until at last you find you have traded away your very last ticket. You now have no chance at all of winning the lottery. You didn't give it up for nothing, since at this point Satan has promised to spare a million lives. But you started out with the certainty of saving a million and *one* lives! In the end, you have sacrificed a life for no benefit at all. According to principle AF, every step along the way was an improvement—and yet you have only made things worse.

Let's spell out this argument precisely. Suppose principle AF is true, and let ϵ and d be as in the principle. Choose some n such that $1/n < \epsilon$. The story is slightly less dramatic, but the argument is slightly simplified, if we start with the certainty of saving n lives, rather than $n + 1$ lives. In that case, we can apply AF n times:

$$\begin{aligned}
 1 * (0, nd) &< \frac{n-1}{n} * (d, (n+1)d) \\
 &< \frac{n-2}{n} * (2d, (n+2)d) \\
 &< \dots \\
 &< \frac{1}{n} * ((n-1)d, (2n-1)d) \\
 &< 0 * (nd, 2nd).
 \end{aligned}$$

Each step corresponds to trading away one ticket (subtracting probability $1/n$) for one life (adding value difference d). But the first line $1 * (0, nd)$ and

the last line $0 * (nd, 2nd)$ represent the very same probability distribution over outcomes: getting the outcome nd with certainty. So AF implies that the betterness relation has *cycles*. This is impossible if betterness is really an *order*, as we have been implicitly assuming throughout.

Proposition 10.1. Uniform Anti-Fanaticism and Simple Stochastic Equivalence are inconsistent.

Philosophers call this style of argument a “continuum” or “spectrum” argument. (Of course, the sequence of prospects in the argument is *discrete*, rather than continuous in the mathematician’s sense.) Some philosophers treat this style of argument with suspicion. But we can also offer an alternative argument that does not involve any “continuum” of cases, and which does not turn on the structural assumption that *better* does not have cycles. (This is a variant of an argument we already discussed in section 8.1.)

Golden ticket. Once again, a number between one and a million is going to be drawn from a fair lottery. This time, whatever number comes up, that many lives will be saved. Call this prospect X . But now you are offered two options:

$X + d$. Save one additional life for sure.

Y . If ticket #1 is drawn (the *golden ticket*), save a million additional lives. Otherwise, get no benefit.

(See table 10.2.)

The Anti-Fanatic says that $X + d$ is better than Y : saving one extra life for sure is better than a minuscule chance at saving many lives. But $X + d$ and Y are essentially the same! Each of them has chance one in a million of saving two lives, chance one in a million of saving three lives, and so on up to a million and one. Preferring $X + d$ over Y amounts to having a preference regarding *which lottery tickets* are associated with which outcomes. This seems absurd. In particular, again we have a contradiction between Uniform Anti-Fanaticism and Simple Stochastic Equivalence.

Table 10.2: Strong Uniform Anti-Fanaticism implies that $X + d > Y$ (for large enough n and d); but $X + d$ and Y are stochastically equivalent.

Ticket:	1	2	3	...	$n - 1$	n
X	$1d$	$2d$	$3d$	...	$(n - 1)d$	nd
$X + d$	$2d$	$3d$	$4d$	...	nd	$(n + 1)d$
Y	$(n + 1)d$	$2d$	$3d$	...	$(n - 1)d$	nd

So there is a strong case against Uniform Anti-Fanaticism.

One reason one might reject Fanaticism is based on this intuitive idea: if one option is *nearly certain* to be *much better* than another, then it is better overall. If we give up Strong Uniform Anti-Fanaticism, then we must give up this idea. That intuition has led us astray. So the argument against Strong Uniform Anti-Fanaticism amounts to an indirect argument *for* Fanaticism, by undermining one reason for rejecting Fanaticism.

What the argument against Strong Uniform Anti-Fanaticism supports more *directly*, though, is not Fanaticism as we have officially formulated it, but rather this principle:

Weak Variable Fanaticism. For any finite probability $\epsilon > 0$ and finite value difference d , there are some simple finite quantitative prospects X and Y such that $\mathbb{P}[X \succeq Y] > 1 - \epsilon$ and $X + d$ is not as good as Y .

If we also assume the principle of Comparability (chapter 4), then we can strengthen this conclusion, replacing “ $X + d$ is not as good as Y ” with “ Y is better than $X + d$ ”.

Strong Variable Fanaticism. For any finite probability $\epsilon > 0$ and finite value difference d , there are some simple finite quantitative prospects X and Y such that $\mathbb{P}[X \succeq Y] > 1 - \epsilon$ and Y is better than $X + d$.

Either form of Variable Fanaticism is pretty fanatical. Arbitrarily small probabilities cannot be outweighed even by enormous certain goods, against the right status quo background conditions.

10.3 Variable Fanaticism and infinite value

But there is an important way in which Variable Fanaticism is much *less* fanatical than we might have feared (even in its Strong form). Even though it implies that small changes in probability can be very important, this variety of fanaticism does not have especially radical consequences for *infinite* value. Throughout this book we have discussed two important consequences of Fanaticism: the St. Petersburg paradox and Pascal's wager. Strong Variable Fanaticism does not immediately have either consequence. Let's consider each in turn.

First, the St. Petersburg paradox. Recall from section 9.4 the idea of a *generalized St. Petersburg prospect*. Fanaticism implies that there exists a prospect X such that X has only finite outcomes, yet X is better on balance than any finite outcome. Choose some good outcome x_1 . Then Fanaticism tells us there is an outcome x_2 which is better than chance $1/2$ of x_1 , an outcome x_3 which is better than chance $1/4$ of x_2 , and so on. We can choose these outcomes so that they increase in value without bound: for any finite good x , some outcome x_n is better than x . Then the prospect with chance $1/2$ of outcome x_1 , chance $1/4$ of x_2 , and in general chance $1/2^n$ of outcome x_n is a generalized St. Petersburg prospect.

This way of formulating the idea of the St. Petersburg paradox assumes a background context of certainty of getting some baseline "zero" outcome. But we can also ask: is there *any* background context in which we have generalized St. Petersburg prospects?

Definition. Quantitative prospects X and Y are a **St. Petersburg pair** iff it is certain that X and Y are only finitely different from one another, but there is no finite value difference d such that $X + d$ is as good as Y .

We might try to argue for the existence of St. Petersburg pairs in a similar way. Just as Fanaticism implies the existence of generalized St. Petersburg prospect, if Fanaticism holds in some *context* X , this implies the existence of a generalized St. Petersburg improvement of that context. But crucially, Strong Variable Fanaticism does *not* say that Fanaticism holds in any particular context. What Strong Variable Fanaticism says is that, for any finite improvement d and any positive probability p , we can find *some* context X where a probability p improvement of X outweighs the certainty of improving X by d . But for any *particular* context, there may still be a limit to how small p can get, or to how big d can get. If we reach such a limit, then we will not be able to continue our construction of the sequence of improvements.

To be explicit, we can contrast Strong Variable Fanaticism with this stronger claim, which rearranges the quantifiers.

Fixed Fanaticism. For any simple finite quantitative prospect X , any finite probability $\epsilon > 0$, and any finite value difference d , there is some simple finite quantitative prospect Y such that $\mathbb{P}[X \succeq Y] > 1 - \epsilon$ and Y is better than $X + d$.

Fixed Fanaticism intuitively captures the idea that our original formulation of Fanaticism is true in some fixed context X . (The prospect Y has at most a very tiny probability of providing any improvement over X , while the prospect $X + d$ is certain to provide an enormous improvement over X .) Accordingly, Fixed Fanaticism *does* imply the existence of a St. Petersburg pair, in just the way that was suggested above.³ But even Strong Variable Fanaticism is weaker than Fixed Fanaticism. Furthermore, the argument against Anti-Fanaticism in section 10.2 only directly supports Variable Fanaticism, rather than Fixed Fanaticism. So that argument does not take us all the way to St. Petersburg.

Next consider Pascal's wager. Pascal's central idea was that when infinite values are at stake, even tiny probabilities outweigh the certainty of any finite gain. In his own words once again:

This much is settled, wherever the infinite is and where there is not infinitely greater chance of loss than gain — there is no more weighing up to do: all must be given.

In chapter 2, we formalized this idea in the following way:

Pascal's Principle. For any finite probability $p > 0$, chance p of an infinite good is better than the certainty of any finite good.

We also discussed an argument for Pascal's Principle, generalizing Pascal's own argument (section 2.6). Consider any finite positive probability p and any finite good x . *Fanaticism* says that there is some *finite* good y such that chance p of y is better than x . If an *infinite* good w is better than any finite good, then dominance implies that chance p of w is also better than x .

Again, this statement of Pascal's principle is specific to cases where the *baseline* chance at an infinite good is zero. But we can also ask: is Pascal's

³This argument makes two structural assumptions about finite improvements. First, the *directed set* assumption: for any finite improvements d_1 and d_2 , some further improvement d is as good as either of them. Second, a *separability* assumption: there exists a countable set of finite improvements with no finite bound.

principle true in at least *some* contexts? Fix some choice of baseline prospect. Is it better to get a finite improvement for sure, or to increase the chance of an infinite good y by some arbitrarily tiny amount?

Definition. A simple quantitative prospect X is **positively Pascalian** iff there is some outcome y such that, for any prospect Y that everywhere either agrees with X or else has value y , if $\mathbb{P}[Y = y] > \mathbb{P}[X = y]$, then there is no finite value difference d such that $X + d$ is as good as Y .

Intuitively, X represents the status quo. If X is positively Pascalian, then not even an enormous gain in value d can outweigh an arbitrarily minuscule increase in the chance of getting the outcome y .

Positive Pascalianism. There exist positively Pascalian prospects.

We can generalize the Pascalian argument: if (Positive) Fanaticism holds in some context X , then Pascal's principle also holds in that context — that is, X is positively Pascalian. But for exactly the same reason as before, this generalized argument depends on finding a *fixed* context that upholds Fanaticism for *any* probability p and *any* finite improvement d . That is, this argument for Positive Pascalianism again relies on *Fixed* Fanaticism, rather than *Variable* Fanaticism. So the argument from section 10.2 does not take us all the way to Pascal, either.

Let's sum up what has happened so far. In section 10.2 we discussed a strong argument *against* (Strong) Uniform Anti-Fanaticism, and thus *for* (at least Weak) Variable Fanaticism. This conclusion is pretty fanatical, and it undermines various ways we might try to resist fanatical conclusions. Small probabilities of enormous outcomes *can* be so overwhelmingly important that even enormous gains don't outweigh them, at least in certain background contexts. But this conclusion is not quite as fully fanatical as we might have feared, particularly in its consequences for *infinite* value: it does not immediately give rise to either the St. Petersburg paradox or Pascal's wager.

10.4 The direct argument for Pascalianism

Not immediately. But the tyranny of the infinite is not to be escaped for long.

What might an anti-Pascalian theory actually look like? Let's back up a bit. Besides Fanaticism for finite value, probability discounting is also a natural way of trying to resist Pascal's principle. (Monton 2019, 21, proposes exactly

this.) Let's again begin by considering the same concrete implementation of this idea as we did in section 10.1: tail discounting. This time, we want to combine the theory with the possibility of *infinite* value; so the theory we will investigate is *tail discounted expected vector utility theory* (combining ideas from section 1.2.4 and section 2.3).

Consider Pascal's wager, in its original simple form (table 10.3).

Pascal's wager. There are two options: *wager for God*, or *wager against God*. There are two possible states: *God exists* or *God does not exist*. The existence of God has some probability p — let us suppose this to be very small. Let's also suppose wagering for God to come at some large finite cost — sacrificing a whole lifetime of happiness. If you wager for God and God exists, you gain an eternity of happiness. If you wager for God and God does not exist, you gain nothing.

We can represent the value of each possible outcome with a two-dimensional vector. One dimension represents earthly value. Let 1 on this dimension represent a lifetime of earthly joy, and let 0 represent its sacrifice. A second, lexicographically superior dimension represents eternal value. Let 1 on this dimension represent gaining eternal joy, and let 0 represent losing out on it.

Table 10.3: Pascal's wager, represented with vector utilities.

	p	$1 - p$
Wager against God	$(0, 1)$	$(0, 1)$
Wager for God	$(1, 0)$	$(0, 0)$

The expected vector utility of wagering against God is $(0, 1)$, while the expected vector utility of wagering for God is $(p, 0)$. So for any probability $p > 0$, the expected vector utility of wagering for God is lexicographically superior to that of wagering against God. This is Pascal's verdict.

Tail discounting gives a different verdict on these prospects. Right now we are only interested in two-outcome prospects, so we can use the same simple discounting formula that we introduced in section 10.1. Let $v(x)$ be the vector utility of an outcome x . Then for any probability p and outcomes x and y , the tail-discounted vector expected utility of the prospect with

probability p of y and probability $1 - p$ of x is given by this formula:

$$U_{\text{t.d.}}(p * (x, y)) = \begin{cases} v(x) & \text{if } p < \epsilon \\ v(y) & \text{if } p > 1 - \epsilon, \\ (1 - p) \cdot v(x) + p \cdot v(y) & \text{otherwise} \end{cases}$$

where ϵ is some small finite threshold. In the case at hand, if the probability of God's existence p is less than ϵ , we do not reach Pascal's verdict. The tail-discounted expected vector utility of wagering against God is still $(0, 1)$; but the tail-discounted expected vector utility of wagering for God is $(0, 0)$, rather than $(p, 0)$. The chance at eternal happiness is so slim that it is treated as negligible.

So far, so good. But now we recall the issue raised in section 10.1: what if your *background* chance of eternal happiness is not zero?

Pascal's new wager. The fate of your soul hangs on a coin flip. But *maybe, just maybe*, if you dedicate your life to propitiating the gods, you can *slightly* improve your odds.

To be more precise, you face this choice (table 10.4):

- If you don't propitiate the gods, you will have a long but finite lifetime of certain earthly joys, followed by chance $1/2$ of eternal happiness.
- If you propitiate the gods, you will have a long but finite lifetime of certain earthly sorrow, followed by chance $1/2 + p$ of eternal happiness, where p is a very small positive finite number—say, one in a billion billion.

Table 10.4: Pascal's wager with a substantial background chance at eternal happiness.

Probability:	$1/2$	p	$1/2 - p$
Don't propitiate the gods	$(1, 1)$	$(0, 1)$	$(0, 1)$
Propitiate the gods	$(1, 0)$	$(1, 0)$	$(0, 0)$

The possible outcomes in this case are the same as in the original wager, and

we can represent them with the same vector utilities. This time, the expected vector utility of failing to propitiate the gods is

$$\frac{1}{2} \cdot (1, 1) + \frac{1}{2} \cdot (0, 1) = \left(\frac{1}{2}, 1\right).$$

The expected vector utility of propitiating the gods is

$$\left(\frac{1}{2} + p\right) \cdot (1, 0) + \left(\frac{1}{2} - p\right) \cdot (0, 0) = \left(\frac{1}{2} + p, 0\right).$$

Just as in Pascal's original wager, the expected vector utility of propitiation is lexicographically superior to that of failing to propitiate.

This time, though, tail discounting provides no recourse. For prospects with two possible outcomes, tail discounting only kicks in when the probability of one of the outcomes is close to zero. That doesn't apply in this case: for both options, the probability of each outcome is middling. What is intuitively negligibly small here is not the *absolute* probability of gaining eternal happiness, but rather the *difference* between the probability assigned by each option. Tail discounting is no help with small differences between middling probabilities: it still delivers the Pascalian verdict that it is better to propitiate, no matter how small the probability p is.

Pascal's central idea was that when infinite values are at stake, even tiny probabilities outweigh the certainty of any finite gain. In chapter 2, we formalized this idea in the following way:

Pascal's Principle. For any finite probability $p > 0$, chance p of an infinite good is better than the certainty of any finite good.

But this formulation suffers from the same limitation that we discussed regarding our formulation of Fanaticism: it only concerns situations in which the *baseline* chance at an infinite good is *zero*. In the general situation, though, we are not deciding between the *certainty* of missing out on an infinite good and a very small probability of attaining it. Rather, we may face prospects that have very slightly *different* middling probabilities of attaining infinite goods.

Tail discounting escapes Pascal's principle, but it fails to uphold a more general anti-Pascalian idea: any sufficiently small *difference* in the probability of an infinite good can be outweighed by sufficiently large finite costs. Tail discounting still implies *Positive Pascalianism* as formulated in section 10.3: for many background prospects X , increasing the probability of an infinite good

by an arbitrarily small amount is better than any certain finite improvement to X . Indeed, tail discounting only delivers anti-Pascalian verdicts in cases where the baseline chance at an infinite gain is very close to *zero* or *one*.

Is this just because we have chosen the wrong kind of probability discounting? I will argue below that the answer is *no*: there is no plausible theory of value for prospects that is generally anti-Pascalian.

Before we take up the general argument, let's try out another approach. Instead of probability discounting, let's return to the idea of *bounded utility* (section 1.2.3).

Consider a simple model, where again outcomes can be represented by two dimensions of value: earthly and heavenly. So far we have only considered outcomes with two possible levels of earthly value and two possible levels of heavenly value. Let's generalize just a little a bit, but still keep things very simple: earthly value is represented by a non-negative number n , while heavenly value still comes at just two levels: 0 or 1. Which outcomes are better than which is still represented by the lexicographic order on these vectors, so heavenly value swamps even arbitrarily great earthly values. But even if these vectors determine the *ordinal* value of outcomes, they need not determine their *cardinal* value, which determines how probabilities of different outcomes are weighed. We have taken up this idea before in a context where only finite quantities were under consideration (sections 1.2.3 and 9.3). Instead of directly taking the expectation of these vector representations, we can apply a *bounded utility function* that respects their order. One way to do this is by applying our familiar exponential utility function *twice*. We can define

$$\begin{aligned} u(0, n) &= 1 - 2^{-n} \\ u(1, n) &= 2 - 2^{-n} \end{aligned}$$

(figure 10.2). This way, heavenly joy is worth more than any finite amount of earthly joy—but its *utility* is still only a finite quantity. In this case, gaining probability p at heavenly joy is worth a utility boost of at most p . Furthermore, if $p < 1$, then this can be outweighed by a sufficiently large difference in *earthly* joy.

This bounded expected utility model avoids Positive Pascalianism. In this model, the biggest utility difference you can get is still less than 2. For any particular choice of baseline prospect X , if the finite improvement d is big enough and the probability p is small enough, then multiplying p times a utility boost of less than 2 will be less than the finite difference

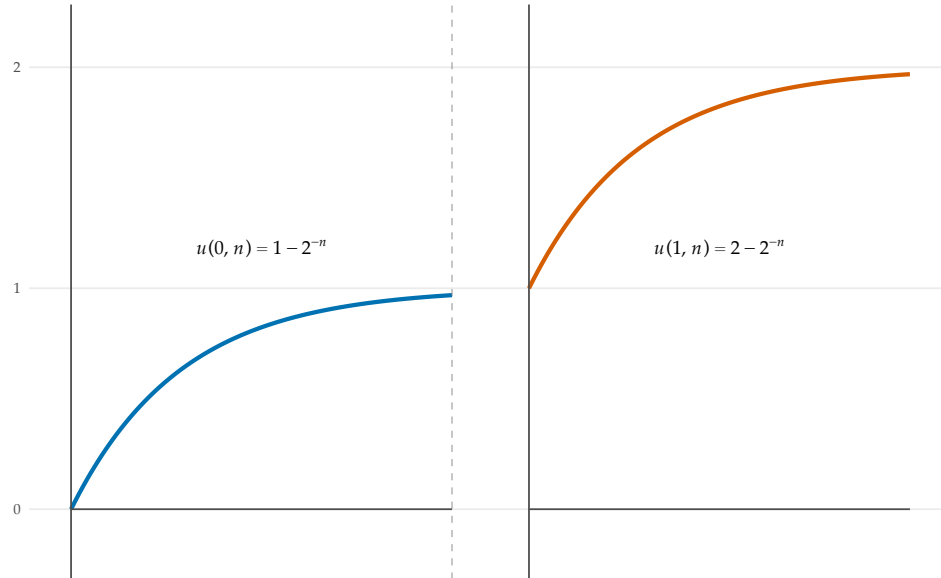


Figure 10.2: Bounded utilities for infinite goods.

$\mathbb{E}u(X + d) - \mathbb{E}u(X)$. Every baseline prospect has a finite “buffer” around it, where small enough probabilities of infinite goods are worth less than finite improvements.

Of course, we know from proposition 10.1 that the model *does* satisfy Variable Fanaticism. (Thus this expected utility model directly shows that Variable Fanaticism does not imply Positive Pascalianism, as was claimed in section 10.2.) For any particular finite improvement d , as n gets larger, the utility difference $u(0, n + d) - u(0, n)$ gets smaller and smaller, which means that it can be outweighed by improvements with smaller and smaller probabilities. If we let $p > 0$ be any probability and let d be any finite number, then we can choose as our baseline a prospect with probability p of the outcome $(0, 0)$, and probability $1 - p$ of the outcome $(0, n)$, where n is very large. The bounded expected utility model says that it is better to slightly improve the chance at the better earthly outcome from $1 - p$ to 1 than it is to

boost all outcomes by d with certainty.⁴ Even more so, then, it is better to get chance p at a *heavenly* outcome than it is to get d extra units of earthly joy with certainty. Smaller and smaller probabilities require us to look for bigger and bigger baseline values of n . But every baseline prospect has *some* finite buffer around it, within which small enough changes in probability aren't overwhelmingly important. It's just that how small counts as "small enough" depends on the background context.

Have we escaped Pascal, then? Not so fast. This model assumes that infinite value has a very simple structure. There are just two dimensions of value, and the "heavenly" dimension has only two possible levels. We can extend this model a little way, but we soon hit limits. Once we take into account the rich structure that infinite value plausibly has, we run into real trouble.

There is not just one possible level of infinite value. If there are infinite values at all, then it is plausible that there are *infinitely many* different regimes of infinite value. Recall our toy model from chapter 3 of *simple worlds*. There are rooms for each natural number, and each room contains one individual who is either happy or sad. Here is a very natural thing to say about the value of these worlds: it is better to make *infinitely many* people happy than it is to make only *finitely many* people happy.

∞ -Compensation. If v and w are simple worlds such that infinitely many people are sad in v and happy in w , and only finitely many people are happy in v and sad in w , then w is better overall than v .

The counting order (section 3.2) satisfies ∞ -Compensation. This tells us that ∞ -Compensation is a consequence of Pareto and Impartiality. But on its own it is substantially weaker than the combination of those two principles. If we can make sense of *impartial value* for simple worlds at all, then this should satisfy ∞ -Compensation. And it seems like any plausibly rich space of outcomes will have at least as much structure as the simple worlds do. In general, when an ordering of outcomes is at least this rich, let us say that it **embeds the ∞ -order**.⁵

⁴In particular, we can choose $n \geq d/p$. So $n \geq p \cdot d + (1 - p) \cdot (n + d)$. Then for any increasing, strictly concave down utility function $u : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$u(n) \geq u(p \cdot d + (1 - p) \cdot (n + d)) > p \cdot u(d) + (1 - p) \cdot u(n + d).$$

In particular, this holds where $u(n)$ is the exponential function $u(0, n)$ defined above.

⁵Let's make this more precise. Let $\mathcal{I}$ be an infinite set. The ∞ -**order** is the strict partial order $>^\infty$ on subsets of $\mathcal{I}$ such that, for $A, B \subseteq \mathcal{I}$, $A >^\infty B$ whenever $A \setminus B$ is infinite and $B \setminus A$ is finite. We say an order $\succsim$ on a set $\mathcal{X}$ **embeds the ∞ -order** if there is some function

Once we take this rich structure into account, the bounded utility strategy for escaping Pascal doesn't work so well. The space of value is not just two-dimensional, but *infinite dimensional*. There are infinite chains of outcomes such that each outcome is an infinite improvement over everything before it in the chain. Where can we put all these values, if we want to squeeze them all into the confines of real-valued utilities? Indeed, we already know from Basu and Mitra's theorem that this is impossible (theorem 3.1): no value ordering that embeds the ∞ -order has a *real-valued* utility representation. But this isn't just a problem for bounded real utilities. It turns out that it's a deep problem for *any* theory of the value of infinite prospects.

We are working up to a theorem that shows this. Before we get there, here is a familiar principle.

Weak Simple Stochastic Dominance. Let X and Y be simple prospects. If X has the same distribution over outcomes as a prospect that is certain to turn out at least as well as Y , then $X \succeq Y$.

As we have discussed, this can be derived from a combination of Simple Stochastic Equivalence with Weak Statewise Dominance (sections 1.3.4 and 1.3.5). Note that this principle shouldn't raise any objections from people who like discounting small probabilities. Some discounters are willing to give up *Strong* Statewise Dominance: X might be sure to be as good as Y , and have a very tiny probability of turning out better than Y , and yet not be better overall. But in this case, surely X is still *as good* as Y .

In fact, the result only really relies on applications of stochastic dominance with two especially simple forms.⁶

- (a) For any outcomes $x \preceq x'$ and $y \preceq y'$, and any probability p ,

$$p * (x, y) \preceq p * (x', y');$$

- (b) For any outcomes $x < y$ and probabilities $p \leq q$, then

$$p * (x, y) \preceq q * (x, y).$$

$f : 2^{\mathcal{I}} \rightarrow \mathcal{X}$ such that whenever $A \succ^{\infty} B$, $f(A) \succ f(B)$. Given such an embedding, we say $x \sim^{\infty} y$ for $x, y \in \mathcal{X}$ iff $f(A) \succ x, y \succ f(B)$ for some $A, B \subseteq \mathcal{I}$ such that $A \setminus B$ and $B \setminus A$ are both finite. (See sections A.3 and D.4.)

⁶[TODO. Double-check this. I think I also assume that if $x < y$ then $\delta x < \delta y$. Worth flagging that we also have a richness assumption? I've generally been cavalier about these.]

In section 10.3 we stated *Positive Pascalianism*: in some contexts, no increase in the chance of an infinite gain y (however tiny) can be compensated by any finite improvement (however vast). There is a corresponding principle about tiny chances of an infinite *loss*.

Definition. A simple quantitative prospect X is **negatively Pascalian** iff there is some outcome y such that, for any prospect Y that everywhere either agrees with X or else has value y , if $\mathbb{P}[Y = y] > \mathbb{P}[X = y]$, then there is no finite value difference d such that $X - d$ is as bad as Y .

Pascalianism. There exist either positively or negatively Pascalian prospects.

In short, a Pascalian prospect is a context X in which arbitrarily tiny changes to X with respect to the probability of *infinite* differences in value cannot be outweighed by arbitrarily great *finite* changes in value.⁷

Theorem 10.2. For any order of outcomes that embeds the ∞ -order, Weak Simple Stochastic Dominance implies that there exist uncountably many Pascalian prospects.

By my lights, the assumptions of this theorem are remarkably minimal constraints on a theory of impartial value under risk.

Given this, it is worth again highlighting another assumption of the theorem that I have normally kept in the background: betterness is a transitive relation. But even this assumption is not doing a lot of heavy lifting—there is no continuum-style argument buried in the proof of theorem 10.2, with some large number of applications of transitive improvements. Still, perhaps the sufficiently desperate may find some refuge by denying the transitivity of betterness—I won't take this up.

Another note: the theorem does not assume Comparability. But what it says is that no finite improvement is *as good* as a tiny improvement in the chance at infinite goods. Without Comparability, this is compatible with

⁷For simplicity, I have been stating things in the terminology of *quantitative prospects* with a corresponding notion of a *finite value difference*, which can be uniformly applied to different outcomes. This is a stronger structural assumption than the theorem really requires. Instead, we can just appeal to a more abstract notion of “finite” vs. “infinite” value differences. Outcomes x and y might be only *finitely different*, which we will write $x \sim^\infty y$. Similarly, for prospects X and Y , we will write $X \sim^\infty Y$ iff in every possible state, the outcome of X is only finitely different from the outcome of Y . Then instead of saying “there is no finite value difference d such that $X + d \succeq Y$,” we can say more simply “there is no $X^+ \sim^\infty X$ such that $X^+ \succeq Y$.” Likewise for the negative case. See section D.4.

also saying that a tiny improvement in the chance at infinite goods is not as good as some finite improvements. Perhaps accepting Pascal's new wager is not *better* than refusing it, but also not *worse*, nor equally good. Tarsney (forthcoming) defends a similar line; but this strikes me as not much of an escape. [TODO Should probably say more?]

The proof of theorem 10.2 is in section D.4. But there is a weaker theorem in the neighborhood whose proof is easier to intuitively understand. Suppose we assume the *Independence* axiom (section 2.3): this is tantamount to assuming that the value of prospects has a vector expected utility representation. But we also know, from Basu and Mitra's theorem (theorem A.23 in section A.3), that its value is as richly structured as the ∞ -order, then this representation cannot be one-dimensional. What's more, any *multi-dimensional* expected vector utility theory representation allows for Pascalian prospects. [TODO CHECK.]

As we have discussed, Independence is a very strong assumption (chapter 9). For instance, any reasonable implementation of probability-discounting will violate it. What theorem 10.2 shows us is that, at the core of the quick argument I just gave, there is a deeper argument that does not really turn on anything nearly as strong as Independence. Just Weak Simple Stochastic Dominance is enough.

This, then, is what I call the *direct argument* for Pascalianism. Pascal's own argument for his principle, which has loomed over the entire book, went by way of a premise about the value of small chances at enormous *finite* goods: Risk Neutrality, or in the generalization of Pascal's argument, Finite Fanaticism. The direct argument does not have any such premise: it goes straight for the infinite, without any detour through the finite. The conclusion is weaker than Pascal's Principle as we originally stated it in chapter 2. That principle said that if your baseline probability of an infinite good is zero, then any tiny increase in the probability of an infinite gain outweighs any finite gain. A closely related principle says that in this context, any tiny increase in the probability of an infinite *loss* outweighs any finite cost. What the direct argument shows is that, even if neither of these principles is true in a context where the baseline probability of infinite gains or losses happens to be *zero*, at least one of these principles *is* true in many other contexts.

Another dialectical upshot of this argument is that it provides further *indirect support* for Finite Fanaticism. This is because the direct argument defangs a strong objection to Fanaticism. The objection goes like this: Fanaticism implies Pascalianism; Pascalianism is false; so Fanaticism is false. What we

have seen here is that Pascalian prospects are not a problem specific to the finitary fanatic: they are a problem for *anyone* who takes seriously the idea of impartial value for prospects involving infinite worlds. Escaping Pascal would require much more radical steps than just giving up on Fanaticism.

10.5 Direct to St. Petersburg?

Let's recap. In section 10.2 we argued for a certain form of Finite Fanaticism. In section 10.3 we showed that this form does not have the radical consequences for infinite value we might fear — Pascal's wager and the St. Petersburg paradox. In section 10.4 we saw that, even so, there is a *direct* argument for Pascalian wagers that does not rely on any ideas related to Finite Fanaticism at all. So we might well wonder whether the same is true of the St. Petersburg paradox. Is this similarly difficult to escape, once the rich structure of infinite outcomes is taken into account?

Here we find some modest good news. It turns out that there are consistent theories of value that steer clear of St. Petersburg. Of course, we already know that these theories necessarily involve substantial sacrifices. From chapter 8 we know that they cannot satisfy *Separability*. From chapter 9 we know that they must give up either the Sure Thing Principle, minimal Comparability, or else certain natural symmetries of value. But we found that even making those sacrifices is not enough to escape from Pascal.

I'll briefly outline a model along these lines, and some of its key features. (Details are found in section D.8.) It is based on ideas from J. S. Russell and Isaacs (2021, sec. 5) and J. S. Russell (2020a): there is an interesting way to combine *vector utilities* and *bounded utilities* into a single structure.

Start with bounded real utilities. Say we have a set of outcomes that all get assigned utilities in the interval $(-1, 1)$. There is no St. Petersburg paradox to be found here. Any standard random variable X taking values in $(-1, 1)$ also has a well-defined *expectation* which lies in the same interval. Furthermore, this expectation $\mathbb{E}X$ is guaranteed not to be larger than every value of X :

$$\mathbb{E}X \leq \sup X.$$

There are no "improper" St. Petersburg-like prospects, which are more valuable than any outcome they could possibly have.

(By a *standard random variable* I mean one defined on standard probability space, where probabilities are represented by real values and are countably

additive. We will consider non-standard probabilities in a bit.)

Next, consider two-dimensional *vector* utilities, where *each coordinate* lies in the interval $[-1, 1]$. A very similar fact holds here as well. The expectation of a vector-valued random variable X is given by taking its expectation along each coordinate. So each coordinate of $\mathbb{E}X$ is again found in the interval $[-1, 1]$, and is bounded above by any bound on the first-coordinate of values of X . Again, no St. Petersburg prospects are possible for standard prospects that have these *bounded vector utilities*. The same continues to hold if we go on to three-dimensional vector spaces, or further.

Furthermore, we can even extend this idea to *infinite*-dimensional vector spaces — though we have to be careful about their structure. There is another kind of St. Petersburg paradox which we have to avoid. Suppose that we have outcomes

$$0 < x_1 < x_2 < \dots$$

where each outcome is *infinitely better* than the one before, in the sense that an arbitrarily small chance at x_{n+1} is better than the certainty of x_n . Now consider a prospect which has probability $1/2$ of getting the outcome x_1 , $1/4$ of getting the outcome x_2 , $1/8$ of x_3 , and so on. This prospect has a non-zero chance of getting x_2 , and is otherwise better than nothing, so it is better than x_1 . Similarly, it has a non-zero chance of getting x_3 , and is otherwise better than nothing; so it is better than x_2 . And so on. This is again an *improper prospect*, better than any outcome it could possibly have.

To avoid this, we have to be careful about the structure of our vector utilities. In particular, its lexicographic order shouldn't allow any *reverse ω -sequence* — an infinite sequence of lexicographically earlier and earlier dimensions. To put that another way, the lexicographic dimensions of value must be *well-ordered*. They must have the structure of *ordinal numbers*. These include the ordinary counting numbers, but they keep going from there.

$$1, 2, 3, \dots, \omega, \omega + 1, \omega + 2, \dots, 2\omega, 2\omega + 1, \dots, \omega^2, \dots, \omega^\omega, \dots$$

This brings us to the utility representation offered by J. S. Russell (2020a). Our space of utilities consists of *bounded transfinite sequences, lexicographically ordered*. A utility is a transfinite sequence of numbers between -1 and 1 . For any two distinct such sequences, there is an *earliest* component where they differ; this determines which of them is greater. Call this space of infinite-dimensional vectors $B(\alpha)$, where α is some ordinal number telling

us the length of the sequences.⁸

Furthermore, for any standard random variable X taking values in $B(\alpha)$, we can calculate its *expectation* componentwise, and this gives us another element of this space. And just as with ordinary bounded utilities, this expectation $\mathbb{E}X$ is bounded by any upper bound for X . There are no improper prospects to be found here, either.

These spaces of transfinite sequences can be truly vast, and contain massive amounts of structure. In particular — unlike the real numbers, or finite-dimensional vector spaces — they can be big enough and rich enough to assign values to infinite worlds. In fact, *any* order can be represented by bounded transfinite sequences (see theorem D.66). This applies not just to the ∞ -order, but also to orders with a lot more structure — for example, we might apply this to the *extended utilitarian order* described in sections B.7 and 3.5. A bounded lexicographic representation will respect the *ordinal* values of infinite worlds, while squeezing their *cardinal* utilities into infinitely many finite intervals.

There are no St. Petersburg prospects in this space of vector utilities — as long as prospects are represented by random variables on a standard probability space. We can also ensure that there are also no *St. Petersburg pairs* in the sense from section 10.3 — where one prospect is guaranteed to provide only a finite improvement over another, but even so is better than any particular finite improvement (theorem D.67).

There is one last complication to worry about. I argued in chapter 5 that *standard* probability spaces don't include enough structure to discriminate between the value of prospects over infinite worlds. These leave out information about *infinitesimal* probabilities, and these can make a decisive difference to the value of prospects. Fortunately, in the kind of models for non-standard probabilities I have put forward, we still do not get any St. Petersburg pairs.

10.6 Conclusion

Three ideas have come up many times throughout this book. *Finite Fanaticism* is about tiny probabilities of enormous finite values. *Pascal's wager* is about tiny probabilities of infinite values. The *St. Petersburg paradox* takes us from

⁸J. S. Russell (2020a)'s notation is $\mathbf{B}\ell^\infty(\alpha)$, the unit ball in the ordered Banach space $\ell^\infty(\alpha)$ of bounded transfinite sequences.

outcomes with enormous finite values to prospects of infinite value. These three ideas seemed to be tightly linked. Arguments led us from Finite Fanaticism to both Pascal and St. Petersburg, and we didn't really have any arguments for those conclusions that were independent of Finite Fanaticism.

But all of that discussion took place in a simplified setting where the "baseline" prospect was certain to be a status quo "zero" outcome. In this chapter we relaxed this assumption, and considered analogues to all three ideas that involve *changes* to an arbitrary status quo prospect. This shift of focus opened up a variety of new possibilities.

Finite Fanaticism has split apart into multiple different principles, with quite different properties. One form — *Weak Variable Fanaticism* — turns out to be basically inescapable. Even if one prospect is *nearly certain* to be *much better* than another, that cannot guarantee that it is better overall. But while this form of Fanaticism may be troubling enough, it does not have the same connections to infinity as the original principle did. Only a rather stronger form of Fanaticism leads us to Pascal and St. Petersburg.

But in this general context, a *new* argument for Pascalianism arose. Even if we can consistently resist Pascal's wager in some contexts of background risk, we cannot escape it everywhere all the time. The rich structure of infinite goods is enough to ensure that, at least in *some* circumstances, even the slightest change in the probability of an infinite good cannot be compensated for even by enormous finite benefits.

The St. Petersburg paradox went differently: we have a wider range of options for escaping it, even in the general setting of arbitrary background risks and richly structured outcomes. We have already learned that escaping St. Petersburg necessarily comes with some costs, but they are not as extreme as those required to escape Pascalianism.

In short, these three ideas — Finite Fanaticism, Pascal, St. Petersburg — seemed to be closely linked, but they have had different fortunes. Yet the idea out of the three that turns out to have the *strongest* support may well be the most troubling: the overwhelming importance of minuscule probabilities of infinite value. What escape can we find from the tyranny of the infinite?

Chapter 11

The Tyranny of the Infinite

“But by damn! I can’t add up figures like that. I don’t know if it’s right to count people like you count numbers. But then, what do you do?”

Ursula K. Le Guin, *The Dispossessed*

11.1 Where we find ourselves

Throughout this book so far, we have been exploring impartial value in a vast and uncertain world — supposing for the sake of argument that there is such a thing to be explored. The two central questions of part II have been, first, whether such value is *Fanatical* with respect to tiny probabilities of enormous finite values, and second, whether it is *Pascalian* with respect to tiny probabilities of *infinite* values. As I discussed in chapter 2, the two questions are related: Pascal gave a powerful argument that, in effect, Fanaticism about finite values implies Pascal’s Principle about infinite values. We considered several different arguments for finite Fanaticism.

In chapter 8 we considered *Separability*: the idea that distant lives that go exactly the same way in both of two prospects make no difference to the relative value of those prospects. Given very natural assumptions about the structure of impartial value, Separability implies Fanaticism. We considered the worry that Separability is actually *inconsistent* with basic principles about impartial value, and we found that this worry can be answered. So there is a dialectically stable argument from Separability to Fanaticism. But also, even

if Separability is *false* we are not in the clear. In that case, which prospects are best depends not just on the *local differences* between them, but also on what I called *cosmic background value* — and reasonable hypotheses about this question can again lead to fanatical conclusions.

In chapter 9 we considered *dynamic consistency*. It has been commonly held that such consistency implies some form of Savage's *Sure Thing Principle*. This principle is the basis for other arguments for Fanaticism. But if dynamic consistency indeed requires the Sure Thing Principle, then it seems that it requires even more — and these stronger requirements are again inconsistent with principles of impartial value, and indeed perhaps internally inconsistent as well. I argued that here again appearances are deceptive: dynamic consistency should be understood in terms of the principle about game-theoretic *equilibrium*. This principle says that when multiple agents are acting separately for a common purpose, or when one agent is acting individually at different times for a constant purpose, then the overall *best* joint strategy is also an *equilibrium* strategy. This *Equilibrium Principle* implies the Sure Thing Principle, without overgeneralizing. So here too we find dialectically stable ground on which to mount arguments for Fanaticism.

But again, even if Separability and the Sure Thing Principle both fail, we are not off the hook. In that case, when deciding which of two prospects is best, we can't just consider their *differences* in probabilities of outcomes; rather, we need to attend to the *absolute* level of risk. In chapter 10 we reconsidered Fanaticism and Pascal's Principle in a more general setting, taking into account the possibility that they are sensitive to the background probabilities of extreme outcomes. In this setting, it becomes much more difficult to consistently sustain anti-Fanatical ideas: at least in *some* background contexts, small probabilities of vast outcomes carry enormous weight. Moreover, we found that there is a separate case to be made for Pascal's Principle, which does not depend on Fanaticism about finite value at all. The richness of the *ordinal* structure of infinite value already implies that there are very many Pascalian prospects, given only very modest decision-theoretic principles.

In my opinion, the cumulative case for Fanaticism is quite strong, and for Pascal's Principle even stronger. Furthermore, in part I, I sketched a formal model of impartial value in infinite worlds, showing that this picture is at least internally consistent — along the way responding to a slew of objections to the very idea of infinite value. This model generalizes expectational utilitarianism for finite populations (and finite state spaces). It has certain counterintuitive features: notably, it violates a range of "qualitative

isomorphism" principles (chapter 3), as well as a range of dominance-style principles (chapter 6), and it requires pervasive incomparability or arbitrariness in facts about value (chapter 4). But I have argued that these are costs that must be borne by *any* tenable account of impartial value in infinite worlds.

In short, I think we have strong reasons to accept that Pascal was right about this much: if the idea of impartial value in infinite worlds is coherent at all, then this kind of value is *Pascalian*, in the sense that arbitrarily tiny increases in the probability of infinite benefits can outweigh the certainty of arbitrarily vast *finite* benefits.

In this chapter, let us suppose that this lesson is correct. What implications does it have for our moral lives? As I have emphasized, our focus so far has been entirely on questions of the *impartial value of well-being*. The time has come to look beyond these *axiological* questions about the value of prospects, and consider *deontic* questions about what we ought to do: at long last, we turn from the *good* to the *should*.

In preceding chapters I have often focused on simplistic situations, such as cases where all that is at stake is whether the individuals in some fixed infinite population are each happy or sad. The arguments in the preceding chapters support the idea that there are *some*, perhaps simplistic, outcomes that give rise to Fanaticism and Pascal's Principle. But, without more work, they don't tell us very much about *which* outcomes are Fanatical or Pascalian, beyond these very simplistic ones. And these simplistic outcomes are not really the ones we need to worry about bringing about. In order to draw any practical lessons, we need to carry over lessons we drew about simple worlds to much more complex worlds like ours. In this chapter I will do this freely, applying Pascal's Principle and Fanaticism to a variety of different infinite and enormously consequential outcomes. There is some danger that we will generalize in the wrong directions. But making the generalizations and applications completely rigorous would bog us down a lot, so let's not try to do that, and accept the danger. It will be worth revisiting the question later of whether our lessons are correctly applied.

I am deliberately lowering the standard of precision in this chapter. The goal is to take a quick look around at a bunch of the directions we might go from here, rather than carefully exploring any of them.

Spoiler: I'm not very happy with any of these directions. If Pascalian values should guide any of our moral choices, it is far from clear where they lead

us; if they shouldn't, then it is hard to see why not.

11.2 What should we do?

Let's begin by outlining a simple deontic theory. Here is our starting place:¹

Should-Best. You should do whichever of your **alternatives** would be **best** given the **evidence**.

There are three placeholder notions in this theory. Your **alternatives** comprise some set of mutually exclusive propositions A : intuitively, your menu of options. The **evidence** is some proposition E , a body of information. And **better given the evidence** is some order $\succ_E$ on these alternatives.

"Should" sentences are highly context-sensitive. We can equally well speak of what you should do to be as helpful as you can, or to be as harmful as you can. We can say what you should do in light of the facts, or in light of your evidence, or in light of someone else's evidence. We can consider whether you should eat your vegetables, which school you should choose, what character traits you should have, or whether there should be fewer hurricanes. We can understand each of these *should* claims by appropriately specifying a set of alternatives and a value order on those alternatives.²

We are presently especially interested in what we *morally* should do. That narrows things down quite a bit. The relevant set of alternatives should only

¹I take *should* and *ought* to be synonyms. But I generally find it more natural to theorize in terms of *should*, as a speaker of modern American English. *Ought*, while often favored by moral philosophers, is distinctly old-fashioned — and I think this sometimes interferes with judgments. (A Google Ngram query shows that "ought" was about 32 times less frequent than "should" in 2000 English, compared to only 4.8 times less frequent in 1800. "Should" is also more favored in American English than in British English, though the difference is much smaller. https://books.google.com/ngrams/graph?content=should%2Cought&year_start=1800&year_end=2022&corpus=en&smoothing=3&case_insensitive=false)

²This account is in the semantic tradition of Kratzer ([1981] 2008; drawing on Cariani 2013; Blumberg and Hawthorne 2021). Note that Should-Best as stated above is not entirely general. What about propositions that are *not* in the set of alternatives? And what if no proposition in the alternative set is uniquely best along the dimension in question? Following Blumberg and Hawthorne (2021), we say that *should A* presupposes that A is in the Boolean algebra of propositions generated by the set of alternatives A — that is, A is a disjunction of elements of A . Say that A' is *optimal* iff there is no $B \in A$ such that $B \succ A$. Note that if A is finite (and *better* is an order), it must contain at least one optimal element. Let $M(A)$ be the disjunction of all optimal elements of A . What the more general theory says is that, for any proposition A that satisfies the presupposition, *should A* is true iff A is entailed by $M(A)$. (This theory is silent on cases where A is infinite and has no optimal element.)

include things we *could do* (in some sense) — propositions like *I give to charity* rather than *there are fewer hurricanes*. The relevant value ordering should be a comparison of *moral* value (in some sense), rather than how good a way something is to win at chess or inflict suffering. And the relevant evidence should consist of information that is *accessible* to us (in some sense). But this still leaves many questions open.

We have spent this book exploring one particular dimension of value: the impartial value of well-being (supposing there is such a thing). So one thing it is natural to ask is what we should do toward *this* particular end. This is one of the things we might be asking about when we ask what we should do. The *impartial welfarist consequentialist* believes that this question coincides with the question of what we morally should do.

Impartial welfarist consequentialism has its defenders, but it is not an especially popular view. Very plausibly, sometimes what best impartially promotes well-being is not the morally best thing to do. Sometimes we should favor our own interests or the interests of those we love over the interests of strangers. Sometimes it is better to respect people's rights or autonomy even at the cost of making people worse off overall.

But even if the impartial value of well-being — the aim of *pure benevolence* — is not the *only* thing that matters morally, surely it is *among* the things that matter morally (again, supposing that there is any such thing at all). Furthermore, there may be certain decision contexts in which pure benevolence is the only important consideration. These would most plausibly be cases that do not involve trade-offs between strangers and our loved ones, and which do not run the risk of infringing anyone's rights, and which are otherwise free of messy moral entanglements. In such situations, our central goal is and ought to be doing as much good as we can for everyone ever overall, with certain resources at our disposal. (In recent years this project has come to be called *effective altruism*: for example, see MacAskill 2015).

Let an *altruistic decision* be one where what we morally should do coincides with what is best with respect to its consequences for impartial well-being. If the lessons of the preceding chapters are right, then the values that guide altruistic decisions are *Pascalian* — tiny probabilities of *infinite* value are decisive. Even if impartial consequentialism is false in general, if there are any settings in which the impartial value of well-being is of overriding concern, then in these settings we ought to pursue our best chances at infinite value, regardless of finite costs. Call this consequence of Pascal's Principle *Deontic Pascalianism*: in altruistic decisions, we should choose options with

arbitrarily small probabilities of infinite value over those which guarantee arbitrarily great finite values.

(One point of clarification: the simple deontic theory *Should-Best* is a theory of what you *should* do. We should distinguish this from what you *must* do — that is, what you *have* to do, what is *obligatory*. If one of your alternatives is better than all the others, then this is what you should do (in the sense of “should” that corresponds to the operative sense of “better”). But that is not to say you *have* to do that; that’s a separate question.)³

Consider a very simple example.

We face two options.

Utopia. With certainty, build a glorious utopia for everyone that will endure for a trillion trillion years.

Long Shot. *Almost* certainly, sink into depressing mediocrity for a trillion trillion years. But there is a one in a trillion trillion chance of a utopia that lasts literally *forever*.

Pascal’s Principle tells us that the Long Shot is overall better for people than Utopia. If this is an altruistic decision, where we ought to choose between these two options on the basis of impartial benevolence, then the ramifications are straightforward: we should take the Long Shot rather than Utopia. As Pascal put it, “wherever the infinite is ... all must be given” (or at least, all *should* be given). More than that, Pascal’s Principle tells us the numbers don’t even matter here. Multiply the finite cost by any number you want — divide that finite probability by any number you want — and you should *still* take the Long Shot gamble.

Could this conclusion be right? In part I, I responded to various unsuccessful objections to Pascal. First, it is often claimed that the idea of infinite value is simply incoherent — for instance, because it would require us to make sense of ill-defined quantities like $\infty - \frac{1}{2} \cdot \infty$. Second, many philosophers have been tempted to various forms of *infinite futility* (often called *infinitarian paralysis*): once infinite value is on the table, there is no room left for further

³For what it’s worth, my preferred view is that while *should* maximizes, *must* satisfies: while you *should* do what’s best; you *must* do something good enough. On this picture, *must* introduces a further dimension of context-sensitivity, beyond those already operative for *should*: a value threshold.

improvements. If every option leads to some (perhaps tiny) probability of infinite value, or if we might live in a world that already contains infinite value, then this seems to make it impossible for us to do any good at all. (More comfortingly, we can also do no harm, short of *infinite* harm.) The answer to both of these objections is that, insofar as we represent infinite values quantitatively at all, we must distinguish between *different* infinite quantities. If the value of eternal happiness is a certain infinite quantity, then the value of eternal happiness plus a pesto sandwich is a different infinite quantity, and the value of a fifty-fifty shot at eternal happiness is another infinite quantity again. In the course of part I, I outlined a mathematical framework that lets us systematize this idea (along similar lines to the more familiar system of *hyperreal* arithmetic). So I don't think the most popular challenges of incoherence or futility are successful.

Pascal's Principle implies that, in altruistic decisions, insofar as our actions make a difference only to finite things — our whole earthly lives, and all the mundane things that matter to us — they are utterly inconsequential, compared to even the most outlandish and quixotic reaches for eternity. Insofar as we are aiming at doing good impartially, our efforts should be completely dedicated to whatever last ditch efforts we can cobble together for truly *infinite* rewards — whether for religious causes like the salvation of immortal souls, or for science fiction causes like manufacturing eternally multiplying pocket universes (or whatever is in fashion right now). However far-fetched you may find these prospects, unless the probability that you can secure souls or pocket universes is literally *zero*, then they are effectively the *only* thing that should figure in your beneficent calculations.

Some effective altruists have been recently drawn to *longtermism*. If you really care about doing good for as many as possible, the distant future may well include a *lot* of people. Even very tiny per-person improvements or minuscule chances at averting disaster, when multiplied by the vast multitudes yet to come, can outweigh even very substantial more immediate benefits. Greaves and MacAskill (2021) summarize (see also MacAskill 2022; Bostrom 2003):

The idea, then, is that for the purposes of evaluating actions, we can in the first instance often *simply ignore* all the effects contained in the first 100 (or even 1000) years, focussing primarily on the further-future effects. Short-run effects act as little more than tie-breakers.

But (I have argued) the impartial values that lead them to this conclusion

are *Pascalian*. If tiny chances at arbitrarily large finite benefits in the distant future can swamp sure immediate benefits, then even tinier chances at *infinite* benefits swamp those in turn. So the implication is that we can in the first instance simply ignore all of the *further-future* effects of our actions, as well. Even if you are sure that an action will have enormous benefits for a trillion trillion years, you should just set that aside (except perhaps as a tie-breaker). What really matters is just your chance at *infinite* benefits. If you are guided by impartial benevolence, don't just forget about your local food bank, or anti-malarial bed nets — forget about securing an enduring, prosperous, and joyful future for trillions of generations, too. Infinite benefits or bust.

This is hard to accept. Consider again Wallace Stevens' "Sunday Morning" (1923): a poem about finding value and meaning in a secular life, rather than in religion, and in a mortal life, rather than in immortality.

Shall she not find in comforts of the sun,
In pungent fruit and bright, green wings, or else
In any balm or beauty of the earth,
Things to be cherished like the thought of heaven?

Deontic Pascalianism says, in essence, that the answer to Stevens' rhetorical question is *no*. The thought of heaven prevails — or at any rate the non-zero finite probability of heaven. This seems very hard to live with.⁴ We may respond with an incredulous stare. *Really?*

I will explore two possible escape routes. In this chapter we are setting aside *axiological* responses, which say that the structure of impartial value is *not* Pascalian. But other responses are still available. *Epistemic* responses appeal to features of our *evidence* that might restrain the tyranny of the infinite, by excluding infinite outcomes from consideration, or by counterbalancing them against one another. *Deontic* responses curtail the role of impartial values in our moral lives even further: we never (or hardly ever) face altruistic decisions in real life.

Before we get there, though, let's make the problem worse.

⁴Note, though, that the view expressed in "Sunday Morning" really seems to be rooted in the thought that immortality is *not as good* as a finite life. See section 2.5 for discussion.

11.3 Many gods and infinite cluelessness

We don't really face a stark choice between Utopia and Long Shot. Our real options are much messier. What does Pascal's Principle imply about which options before us are really best? It is very hard to tell.

Pascal, of course, used this idea to argue that you ought to try to live a life of religious faith, even if you have no evidence at all that religious doctrines are true. And indeed, some kind of religious activity might be a way of increasing the probability of some kind of infinitely good afterlife. Remember, it doesn't matter how *much* you increase that probability, in the contest with merely finite goods.

But things are a lot less clear cut than Pascal makes it seem. In section 2.1 I discussed the famous "Many Gods" objection to Pascal's wager. It isn't as if the *only* epistemically possible route to eternal life is through Catholic rites. There are many, many potential routes to infinite value. For a start, there are other religions. Protestants have different views of the means of grace than Catholics. Christians agree that salvation is attained through the second person of the Trinity, but Muslims say that denying the oneness of God is an unforgivable sin.⁵ In the soteriology of Hindus and Buddhists, *escaping* eternal life is infinitely better than the eternal cycle of suffering and rebirth (*samsāra*) that is our natural lot. Put together, adherents of those religions make up about three quarters of the world's population. Assigning the disjunction of all of their views probability *zero* seems strikingly dogmatic. But they each present very different views on what is infinitely better than what, and point us in very different directions to achieve it.

Then there is science fiction and speculative metaphysics. Inflationary cosmology might permit the creation of infinite chains of "bubble universes" (Guth 1981). The many-worlds interpretation of quantum mechanics might give each of us infinitely many futures, in some of which we are immortal (see D. Lewis 2004). Even if we live in a single universe rather than a multiverse, it might well be spatiotemporally infinite. Natural models predict that the eternal high-entropy chaos of heat death is populated by infinities of "Boltzmann bubbles" of local order (Dyson, Kleban, and Susskind 2002). Perhaps it is possible — with the right combination of science and the

⁵From the *Qur'ān* (5:72, with translator's interpolation):

Those who say, "Allah is the Messiah, son of Mary," have certainly fallen into disbelief. ... Whoever associates others with Allah [in worship] will surely be forbidden Paradise by Allah. Their home will be the Fire.

metaphysics of personal identity — to “Boltzmann hop” from one bubble to another.⁶ Issues are further complicated by plenitudinous metaphysical pictures that admit infinities of “personites” (Johnston 2016) — parts of people, temporal segments of people, things accidentally or temporarily coincident with people, and so on — which may also deserve our moral concern.

Furthermore, there are not just *Pascalian* hypotheses with *infinite* stakes to worry about, but also more mundane St. Petersburg style hypotheses with unbounded *finite* stakes. All it takes is for the distribution over possible finite benefits to be sufficiently long-tailed. For a toy model, imagine a happy human population where each generation has a fifty percent chance of extinction and a fifty percent chance of doubling in size. That has the same structure as the St. Petersburg prospect. If Finite Fanaticism is right (specifically, when applied to these variable population outcomes), then dominance reasoning implies an arbitrarily small chance at that kind of future is better than any certain finite good.⁷ Dominance also tells us that St. Petersburg style prospects are *worse* than an arbitrarily small chance at any properly *infinite* benefit; so this kind of infinity is only worth worrying about once we’ve managed to clear up the first category. But these possibilities still swamp any merely bounded benefit.

In short, the situation is a mess. There are many potential routes to infinity. We need to sensibly *compare* them, in order to decide which ones deserve our resources. This is a very hard problem.

The classic Many Gods objection says that the problem of comparing infinite goods leads to incoherence. As I have argued in chapter 2, this is incorrect. It is perfectly coherent to weigh up the probabilities of various infinite goods, in a way that is sensitive to their probabilities as well as their relative magnitudes. But while the infinite demands of Many Gods do not doom us to incoherence, they do give us serious difficulties. Which of our options is best hangs on questions about *how good* the various infinite goods are, and *precisely how probable* they are. These questions are deeply obscure. We do not know how to compare the values of different infinitely good or bad outcomes. Is it better to escape *samsāra*, or enjoy the beatific vision, or Boltzmann-hop from utopia to utopia, or please a continuum of personites?

⁶Greg Egan’s novel *Permutation City* offers a related picture (1994). See also Johnston (2010).

⁷In unpublished work, Hayden Wilkinson and Toby Ord each offer some more realistic hypotheses involving heavy-tailed distributions over total well-being.

Deontic Pascalianism implies that these obscure questions are the completely overriding concern for every altruistic decision we face. It seems that the *only* things that matter non-trivially are things of which we are deeply ignorant. We have no reason to think that weighing things according to the more scrutable probabilities of even incredibly momentous finite outcomes is even an anywhere-near-reasonable approximation for the correct ordering of options according to their overall value. We are just messing around with the least significant digits — only it's far worse than that.

We might think that many decisions can still be decided based on their finite stakes: decaf or regular, de-worming or bed nets — decisions where what is at stake is mundane, rather than soteriological or cosmic. But even the most ordinary decisions are infected with this problem. Any option may lead you down a path toward infinite gain or loss — putting you in the path of a persuasive evangelist or a scientific breakthrough. (“[E]very rational agent’s life is a constant series of such ‘‘gambles,’’ with wagering for God as one of the outcomes.” Hájek 2003a, 31.) The probabilities are small — but *how* small?

This makes apparently mundane altruistic decisions very hard. Say the difference in value between the best option and the worst option is finite. Now very slightly perturb these options: consider another decision situation, with options that are exactly the same with respect to their probabilities of outcomes for impartial well-being in all cases except for some event with minuscule probability: say, one in $10^{10^{10}}$, and in this event, the old worst option has an outcome infinitely better than the old best option. If Pascal’s Principle is correct (with sufficient generality — see chapter 10), finite value differences are swamped by tiny probabilities of infinite value differences — so the overall values of these options are reversed. Arbitrarily tiny perturbations of our decision situation can make all the difference to what we should do. Furthermore, even *infinite* value differences can be swamped by tiny probabilities of even *greater* infinite value differences. Thus, it seems that figuring out which of our options is best on balance requires us to figure out the probabilities of these infinite value differences *very precisely*: even heroically tight error bars will still include perturbed versions that reverse our decisions. How could we possibly hope to attain such precision?

This upshot of Deontic Pascalianism is related to the familiar “cluelessness” objection to consequentialism, which I will return to in section 11.4.⁸ But I

⁸See Lenman (2000), Greaves (2016); see also Mogensen (2021). Monton (2019)’s argument

think this distinctively Pascalian *infinite* cluelessness is especially deep and intractable.

Maybe we have options that circumvent this problem. If we have no idea what our best “first-order” option is, then perhaps what it is *really* best for us to do is investigate further, in the hopes of gaining a clearer picture of the infinite landscape. This might well be a very good thing to do. (If so — congratulations! Your time spent reading this book has not been in vain.)

Consider a version of the Value of Information Theorem (Hosiasson 1931; Blackwell 1951; Good 1966). Suppose we have a range of “first-order” available acts before us. We are deciding whether (a) to commit ourselves to the best of these options, or else (b) to commit ourselves to the best plan in which we *first* investigate matters further, and *then* perform some first-order act depending on the results of our investigation. In that situation, the inquisitive act can never be *worse* than the uninquisitive one. After all, the repertoire of inquire-act plans includes *constant* plans, which say to perform the same first-order act whatever information the investigation turns up. Effectively, then, the first-order acts are a subset of the inquisitive plans; so the best of the plans must be at least as good as the best of the first-order acts. Moreover, typically option (b) will be *strictly better* than option (a). If the Sure Thing Principle holds, then this is true whenever the inquiry has the potential to make any difference to which act is best. (That is, the best act conditional on some possible result *E* of the inquiry is not the same as the unconditionally best act.)⁹

The value of inquiry depends on the likelihood that we will learn something from it that improves our decisions. What is the outlook for this? Not very good, if it turns out that what is best turns on arbitrarily precise differences in probability for infinite effects that are basically chaotic. Our best hope is if we can make some progress in finding the greatest infinite values and ways we can systematically promote them. Sure, it’s not *easy* to survey the hierarchy of infinite values, our range of practical options, and the probabilities they each make to those values. But we’ve barely given it a serious try yet. If this

from “amorphousness” is closely related.

⁹Note that other versions of the Value of Information Theorem are subtly different. Suppose that instead of *committing* to a certain plan in advance, instead we *defer* the choice of the first-order act to our future selves, after the results are in, knowing that we will choose whatever is best in light of that new evidence. This deferential option is *not* automatically as good as any of our first-order options. Indeed, *this* value of information principle amounts to the Reflection principle discussed in chapter 9. As we discussed there, that principle is very difficult to consistently maintain.

project has a chance, then the upshot of Deontic Pascalianism may be that esoteric investigations into theology, speculative science, metaphysics, and (indeed) infinite ethics are among the most important human endeavors ever undertaken.

But even in that case, inquiry is not clearly what we should do. The Value of Information Theorem assumes that the exact *same* first-order options — with the same outcomes in every state — are available after the inquiry as before. This amounts to supposing that the investigation is *costless*. In real life this is never true. Monographs about infinite ethics can be quite expensive. But what is critical is *opportunity cost*. Time spent investigating these questions is time lost from winning souls, cultivating detachment from desire, or accelerating cosmological research. And how to weigh the cost of inquiry against its potential benefits for improving our chances at the greatest infinite goods seems like just as difficult a question as the choice between “first-order” options.

Inquiry into infinite value might be the best thing we could be doing, but I don’t think we are in any position to know this. Say we commit ourselves now to serious investigation, and we are able to figure out a lot about which options have the highest infinite value *ex ante*. If so, we might well discover after the fact that it would have been *better* for us to pursue some other very different route instead — that in our quest for knowledge we have squandered more than we could have dreamed, doomed countless people to infinite suffering. Our inquiry might turn out to have been very far from the best option available to us, even in light of the evidence we had prior to inquiry. At least we will then be in a position to choose better among the options that are still available *after* the inquiry. Better than nothing — but still, perhaps, overwhelmingly tragic.

I find infinite cluelessness deeply troubling and possibly tragic. But I do not think it is a *reductio* of Deontic Pascalianism. Perhaps it’s just *right* that we do not know what we morally should do when we face altruistic decisions, and maybe we never will know. Nobody said acting morally was easy. (Compare the “tragic outlook of the normative” of Srinivasan 2015.) Maybe in some sense we would be *blameless* for failing to do what we should in such cases, as long as we are trying our best with goodwill. Or maybe not.

This is an instance of the problem of *normative uncertainty*. What should we do when we don’t know what is best in light of our evidence? Different philosophers have very different reactions to this problem. Some say the answer is still that you should do *whatever is best* — and if you don’t know

what that is, too bad for you (and for the rest of us!) (Harman 2015) But some say that what we *morally* should do (in its deepest sense) must be *action-guiding*, in a sense that requires that we can know what is demanded of us (for example, Sepielli 2012; MacAskill and Ord 2020, 5). From this perspective, the upshot of infinite cluelessness about impartial value is that impartial value can't really be the right guide to what we should do.

(As a back-up plan, the altruist in search of action-guidance might go for “metanormativism” (see for example MacAskill, Bykvist, and Ord 2020). Say we don't know what act is best; instead we face a range of hypotheses about how valuable certain acts are in light of our evidence. Then perhaps we can assign those hypotheses probabilities, assign each act a “meta-expected value”, and try to maximize *that*. Perhaps — but that task doesn't sound any easier than the one we began with. The task is to articulate a range of hypotheses about the infinite value of different outcomes, and the probabilities of those outcomes for various available hypotheses, assign each of those hypotheses probabilities, and then calculate. But if we have got as far as this, then it seems that we have just about accomplished the task we faced initially. Metanormativism doesn't seem to make our hard choices any easier.)

In short, it seems that if impartial value is Pascalian — as it plausibly is! — then we don't have the slightest idea which of the things we might do are best with respect to impartial value. So, if we face any altruistic decisions, in which we ought to do what is impartially best, then we have no idea what we ought to do in those cases. Pure benevolence is intractable for us — at least for now, and perhaps forever.

11.4 Epistemic responses

Perhaps there is a way out of this bleak situation. I argued in section 11.3 that we have many potential routes to infinite benefits (or failing that, long-tailed finite benefits). But these hypotheses of religious afterlives, speculative cosmology, or metaphysical profusion may all sound a bit zany. Perhaps they are the sort of possibilities we can legitimately set aside in our moral decision making.

One way of understanding this proposal is as a kind of *probability discounting*, along the same lines as we have considered in section 1.2.4. The view we considered before was a view about which prospects are best: call this *axiological discounting*. I have given several arguments against this view

(sections 9.1 and 10.1). But in our present context another possibility opens up. The principle Should-Best says that you ought to do whichever of your alternatives is best given the *evidence*. Perhaps this evidence excludes sufficiently far-fetched hypotheses. If that is right, then even if Pascal's Principle is correct as far as the betterness ranking of prospects goes, and even if Deontic Pascalianism is right about what you should do when facing certain choices, this has no bite, because you never actually face such choices. The possible outcomes of your options are filtered by the evidence to exclude the infinite.

As I said, we can evaluate which prospects are better than which with respect to any supposition we like. One of the dimensions of context-sensitivity for *should* is a body of information; I have called this the "evidence", but any consistent proposition can play that role. What the present proposal amounts to is a claim about which contexts are relevant for the *should* claims of moral decision-making. Perhaps in such contexts the proposition that plays the evidence role is strong enough to take infinities off the table.

Here are two versions of this idea. One version stays true to the idea of discounting *small probabilities*. Your *evidence* rules out all hypotheses with sufficiently low probability. This proposal raises immediate questions. First, we are going to have to somehow narrow down which "hypotheses" we are talking about, because otherwise we will end up ruling out *every* possibility. (Any proposition A can be partitioned into 2^N propositions of the form " A , and if I were to flip a fair coin N times it would come up $b_1, b_2, \dots, b_N$ ", each of which has probability less than one in 2^N .) Second, what kind of probabilities are we talking about? Not what is probable *given your evidence*. That proposal amounts to saying just that our evidence rules out what it rules out. We need some other kind of background probabilities (perhaps what is probable given certain "proto-evidence" or "basic evidence"), which we can then use to eliminate certain improbable possibilities from the evidence that is operative for decision-making.

A second version appeals to *knowledge* (Hong 2025, 2024). The starting point is the anti-skeptical thought that some propositions that have (in one sense) a *chance* of being true, nonetheless we can know to be false. I know that if I drop this pen it will fall. But according to a collapse interpretation of quantum mechanics, there is an absurdly tiny, but non-zero objective chance that instead all of the pen's atoms will quantum tunnel through the ceiling. Of course that won't happen, though, and the tiny objective chance that it *could* happen does not destroy my knowledge of this fact.

Together with this anti-skeptical thought, the further thought is that *knowledge guides action* (Hawthorne and Stanley 2008). In our framework, the operative *evidence* for moral decision-making includes everything we know.

Consider how Hong applies this to the St. Petersburg paradox. We are about to flip a coin until it comes up heads. Let n be the number of flips it takes for this to happen. What do we know about n ? The chance of n being greater than, say, 10^{50} is comparable to the chance of my pen quantum tunneling. So it seems we are in a position to know this isn't going to happen. Plausibly we can know much more. (Of course, we don't know this in the extraordinarily unlikely scenario that the coin *does* come up tails so many times. But this isn't going to happen!) If we can know such things, then the value of playing the St. Petersburg lottery *given all we know* is not infinite after all.

I think the idea that our evidence (in a context of moral decision-making) includes what we know is appealing. But it only saves us from Pascal and St. Petersburg if we really know enough. Do we? I have no knock-down argument against such *finitist knowledge*, but here are a few observations.

First, suppose we do know that we're not in an infinite world, in the sense that we can take this for granted in our decision-making. This doesn't just ward us from taking exotic Pascalian wagers: it makes a difference to all sorts of practical decisions. For example, many research projects in physics and cosmology depend on an infinite universe being a live hypothesis for their success, or for their interest. Should we take for granted that this hypothesis is false for the purpose of scientific grant-making?

Second, for *finitist knowledge* it is not enough to know the world is finite. That still leaves us vulnerable to long-tail St. Petersburg prospects, which have only finite outcomes, but infinite values. We could block St. Petersburg if we know not just that the value of the world is *finite*, but that it is *bounded*: that is, there is some finite range of value such that we know things are within that range.¹⁰ But is there?

Third, finitist knowledge is incompatible with many versions of the idea that disagreement can defeat knowledge (see for example Christensen 2007; for discussion see Lasonen-Aarnio 2014; Hawthorne and Srinivasan 2013). For, as I have noted repeatedly, infinity's advocates are legion, and they include many epistemically upstanding individuals (at least, so it seems to

¹⁰This is not strictly required: we could know things that make it *sufficiently improbable* that the world is large in a way that rules out all of the heavy-tailed prospects.

me). Finitist knowledge would let us confidently rule out most people's worldviews.

Fourth, finitist knowledge conflicts with another influential idea in epistemology: *pragmatic encroachment*. Specifically, many philosophers have been taken by the idea that *long odds* threaten knowledge.¹¹

Here's an example. Homer lives at 123 Lovely Lane, and he has no plans to move. If he is ordering a package to be delivered next month, we would ordinarily say that he knows what his address will be then. But suppose instead Homer reasons, "I will live at 123 Lovely Lane next month; so my house won't burn down in the next month; so I need not pay for fire insurance this month." This seems bad.

What is going on? Some reject the *knowledge closure* step: Homer knows where he will live, but he doesn't know his house won't burn down. Others reject the idea that *knowledge guides action*: Homer knows his house won't burn down, but should not take this for granted in decision-making. But a third popular diagnosis turns on the point that the practical question of buying insurance involves *long odds*: the potential downside to having no insurance is much bigger than the potential upside. In especially long-odds practical situations, the standard for knowledge is higher than usual. So even though ordinarily we would say Homer knows where he will live, when he is making insurance decisions we should not.

Pascalian cases involve *very* long odds — indeed, *infinite* odds. So if there is anything to the pragmatic encroachment thought, these are cases where the standard for knowledge is *very* high. Indeed, a natural way of developing the pragmatic encroachment idea is to say that you can only know something in a practical situation where taking it for granted *makes no practical difference* (see Fantl and McGrath 2002). If this is right, then knowledge-first decision theory cannot, in principle, rescue us from the tyranny of the infinite.

Now let's turn to a different kind of epistemic response to Pascal. Rather than appealing to knowledge, we might appeal to ignorance.

Let us grant that there are many different potential routes to infinite benefits. But perhaps they all *cancel out*. This would directly address the problem of Pascalian cluelessness. Even though our actions may well have infinite

¹¹See Hawthorne (2004); Stanley (2005). People often instead say "high stakes" rather than "long odds"; for discussion see Weatherson (2012); Anderson and Hawthorne (2019). Alternatively one might apply a *contextualist* version of this idea: what "know" means is sensitive to stakes (DeRose 1992).

effects, or an unbounded range of finite effects, the different possibilities pull in equal and opposite directions — every chance of immense gain matched by an equal chance of equally immense loss.

By analogy, consider Lenman (2000)'s "cluelessness" argument. Here I am, deciding whether or not to help someone across the street. This has an immediate good effect. But it also has a massive number of unpredictable effects. Some people's trip home is slightly delayed, leading to them interacting with different people in different ways — some better, some worse than they would otherwise have been. If any of this makes a difference to the timing when some people conceive children, then we end up with some completely different people in the next generation. These differences propagate out over time, eventually chaotically perturbing the whole future. My simple act may have made the world overall massively better or massively worse, and I have no way of knowing.

Greaves (2016) offers a response. Perhaps I can still know that the *prospect* associated with helping this person across the street is better than the alternative. While there are various ways my action may make things better or worse, I haven't the slightest reason to think that helping will do this *more* than failing to help. The difference in value between the two acts is chaotic, and potentially massive — but it is equally likely to go either way. Greaves proposes that we apply an indifference principle, and conclude that all of these chaotic effects together make a difference of precisely zero to the *expected* value of my action.¹²

Can the same move save us from the tyranny of the infinite? Consider a simple model. I am choosing between two acts: make Indy happy or make Indy sad. But my act potentially makes a difference to infinitely many other individuals besides Indy. It might make all of them happy, it might make all of them sad, or it might have mixed effects. So I have no idea whether the act that makes Indy happy will be better overall than making Indy sad. For all I know, it will be infinitely worse. But suppose that for each person besides Indy I am utterly clueless about the effect of my acts. If I apply the

¹²Compare Ord, Hillerbrand, and Sandberg (2010, 203):

It is true that we cannot completely rule out any probability that apparently inconsequential actions might have disastrous effects, but there are a number of reasons why we do not need to worry about universal prohibition. A major reason is that for events like the dropping of a pencil which have no plausible mechanism for destroying the world, it seems just as likely that the world would be destroyed by not dropping the pencil. The expected losses would thus balance out.

natural indifference principle in this case, then I will conclude that the net expected effect on each of them is zero.¹³ This just leaves my immediate effect on Indy as the determining factor.

More generally, even if I am in an infinite world, I don't know that much about it. Furthermore, my ignorance deepens as we consider individuals further away from me in space and time. I might try to come up with some model to predict the effects of my actions, but there is always some possibility that the model doesn't apply at all, and this risk increases as I extrapolate further. Let's say that I am *in the dark* about an individual *i* if the way my actions affect them (if at all) is something I have no way of reckoning. If I am in the dark about *i*, then we might consider the expected net effect of any of my available actions on *i* to be zero, by a natural indifference principle. Otherwise, the probability of one act's effects on *i* would be different from the probability of another act's effects on *i*, even though I haven't the slightest reason to think one act rather than another would have any particular effect. Suppose furthermore that the probability that I am in the dark about an individual increases as we consider more and more individuals, approaching 1 in the limit. Then even if the total effects of my act are infinite or unbounded, the *predictable* expected net effects of my act may be finite, and entirely determined by the effects in cases where I am not in the dark.

This proposal amounts to a kind of *epistemic social discounting*. Different individuals matter more than others for the net value of my acts. But this is not because I prioritize some individuals over others. Rather, it is because the effects of my acts are more predictable for some individuals than for others. (Compare Askill and Neth 2025.)

I think this is a helpful proposal. More generally, I think a good tool in the face of the problem of cluelessness is to do our best to *explicitly model* our cluelessness: we should try to build into our models as clear a picture as we can of the places where our models give out. (See Tarsney 2020 for an instructive example of this exercise.) But this can only go so far.

Greaves (2016) distinguishes between *simple* and *complex* cluelessness. *Simple* cluelessness arises from the purely chaotic effects of our actions. In these cases, we don't know the chaotic long-run effects of an act will go, or which

¹³There is a technical issue about whether this reasoning is legitimate. The most straightforward way to run this argument appeals to Ex Ante Pareto — but in chapter 7 I argued that this principle fails. Fortunately, there are weaker, more tenable principles that suffice for this reasoning. The model I outline in section F.7 satisfies *Separability*, which suffices.

will be better, but this gives us no reason at all to think that the chaotic long-run effects of one act *A* would be better than those of another act *B*. In such cases, we can plausibly use indifference to cancel all those chaotic effects out. *Complex* cluelessness arises instead from *systematic* effects of our actions. In these cases, we *do* have reasons to believe that the long-run effects of *A* would be better than those of *B* — and we *also* have reasons to believe that the long-run effects of *B* would be better than those of *A* — and it is utterly unclear how to weigh up these competing reasons. In such cases, indifference reasoning is inappropriate.

Here is one argument for that last claim. Suppose we find some new evidence, in light of which the reasons for thinking *A* will turn out better than *B* are just a little stronger than they were. After taking this new evidence into account, we can *still* be in a position of complex cluelessness. But splitting our probability equally both before and after this change in our evidence would amount to disregarding relevant evidence entirely.

The trouble is that the way in which our actions touch the infinite are complex, rather than simple. Which soteriology is true — Christian, Muslim, Hindu, something else? It would be misguided to directly apply indifference reasoning to this question.

Suppose we are “in the dark” about all but finitely many people. Then we get a bit of evidence that a certain physics research program — with potential distant future applications to manipulating black holes and pocket universes — is slightly more promising than we initially thought. If we were exactly balanced on the value of increasing funding to this program, then it seems that this should now go up — and infinitely so. If we were “in the dark” before, and we become slightly more confident that one course of action will benefit infinitely many, then our perfectly indifferent ignorance has been disrupted. Moreover, we have all sorts of evidence like this, pushing one way or another for various infinitary hypotheses, and soon enough we will get more. Indifference about infinite effects does not seem like it can be stable.

In cases of complex cluelessness, it is not that the probabilities are exactly balanced. Rather, we just don’t have a clue what those probabilities are. This exactly the situation described in section 11.3 — not some way out of that situation.

11.5 Deontic responses

We have set aside *axiological* responses to Pascal: suppose that impartial value is Pascalian. Now let us set aside *epistemic* responses to Pascal: suppose that extreme consequences cannot be ruled out, and they do not stably counterbalance one another. We now turn to *deontic* responses. We have been considering what we should do in *altruistic decisions* — in contexts where what we morally should do is what best achieves the aim of pure benevolence. But perhaps there are no such contexts.

One reason might be because there are always other values or principles competing with and constraining benevolence. It is, of course, very plausible that other such values and principles are of pervasive importance in our moral lives. But *altruistic decisions* need not be cases where these other considerations are *absent*. They may merely be cases where they are *outweighed*. This could be because they aren't especially relevant to the decision at hand — none of the options involved breaking promises, violating rights, neglecting yourself or your family, etc. It also could be because individual well-being is especially relevant, because the consequences are drastic. Most deontologists accept that sometimes we morally ought to tell a lie to save a life, or to save the world (unlike Kant [1797] 1996). The case for altruism is especially strong when both of these conditions hold: the non-welfarist considerations are trifling, and the welfarist considerations are momentous.¹⁴

The trouble for restraining the tyranny of the infinite is that in Pascalian cases, the impartial welfarist considerations are extraordinarily high — higher than any finite value. So the only kind of deontic view that will prevent us from facing altruistic decisions (in the specified sense) in practice will be ones that say that other considerations can override impartial welfarist considerations *no matter how great those impartial welfarist considerations are*.

This is challenging, but not absurd. One promising move is to distinguish between different ways that the welfarist value of a prospect can be enormous. The value might be enormous because it is very likely to have enormous effects; or it might be enormous because it has a very *tiny* chance of vastly *more* enormous effects. The best bet here is to say that tiny probabilities play

¹⁴This line of reasoning follows Greaves and MacAskill (2021, sec. 9). They propose this principle:

When the axiological stakes are very high, there are no serious side-constraints, and the personal prerogatives are comparatively minor, one ought to choose a near-best option.

a special role in our deontic theory that *goes beyond* the way they contribute to the overall value of prospects. Such a view might take the form of *deontic discounting* — another alternative to the *axiological* and *epistemic* discounting views we have already considered. Non-consequentialists often reach for structurally similar ideas for other reasons. (For instance, perhaps killing someone to prevent a billion paper cuts is very wrong, while taking a one in a billion chance of killing someone to prevent one paper cut is fine.) It is a deep challenge to articulate and defend principles along these lines, and there is an active research program taking it up (for example, Frick 2015; Hawthorne 2026).

An even more sweeping response is that impartial values have *no* place in our moral lives. Perhaps there is no coherent system of impartial values at all. Throughout this book we have encountered a proliferation of paradoxes. I offered responses to many of them, with varying degrees of residual discomfort. Impartial values are inconsistent with structural isomorphism principles: for example, one possible world can be better than another world with a qualitatively identical distribution of well-being in space and time (chapter 3). Impartial values also require either pervasive incomparability or pervasive arbitrariness in what worlds are better than what (chapter 4). And they violate certain *dominance* style principles, such as *ex ante* Pareto or dominance over subpopulations (chapter 6). My arguments that these burdens were worth shouldering were all conditional — *supposing* that there are coherent impartial values for infinite populations, here is what they must be like. But in the end, the game may not be worth the candle. Maybe impartial benevolence in the face of infinity is simply incoherent. (Compare the views of Hare 2013; Nebel 2025.)

One could argue that it is infinity that has to go: infinite populations are conceptually incoherent. There is a long philosophical tradition (including Aristotle and Al-Kindi) of a priori arguments for a finite universe — though I don't know of any argument in this tradition that relies so essentially on premises from ethics. This would be a particularly dramatic intervention of armchair philosophy into empirical physics, and of ethics into metaphysics. (But that is not to call the move illegitimate! My own view is there is no first philosophy, and no last philosophy — anything might turn out to bear on anything.)

I have already argued that infinite populations are a live possibility that we do not know to be false. The same considerations also militate against the claim of conceptual incoherence. There are too many infinitarian pictures,

supported by too many different kinds of considerations — from physics, metaphysics, and theology — for me to dismiss them so quickly. Furthermore, even if we accept that *infinite* populations are conceptually incoherent, it seems absurd to say that *finite* populations beyond some fixed bound are not coherent possibilities. So we would escape Pascal only to land in St. Petersburg.

If we are stuck with the possibility of infinity, then it is impartiality that must go. There is no such thing as the impartial value of well-being. One upshot of this view would be that there is no such coherent view as impartial welfarist consequentialism, no such coherent project as effective altruism, and no altruistic decisions. There is no moral calculus that permits one to balance off the well-being of everyone who might ever live “from the point of view of the Universe”. So the question of whether we should ever orient our actions in accordance with such a calculus does not arise.

If pure impartial benevolence is impossible, what can replace it? Surely it is not acceptable to simply callously disregard the interests of others. Even if we are not impartially benevolent, we should be at least be morally decent. What does this demand? If we cannot be guided by impartial values, what *partial* values are acceptable — or even possible?

One classic model is *social discounting*. One of the ways that infinite ethics intrudes most directly into real-world policy making is when we consider decisions with an unbounded time horizon. The choices we make today affect our children, and our children’s children, and so on, we know not how long. If we take these unbounded possibilities seriously, then cost-benefit analysis becomes intractable, as long-run costs and benefits explode to infinity. So it is standard practice to introduce a *temporal discount factor*: for example, we may treat a harm next year as 1% less bad as the same harm today, and in two years 1% less bad again, and so on. This has the effect of keeping our calculations in the realm of ordinary finite arithmetic, even when an unbounded number of stakeholders are taken into account.

This practice has been widely criticized by philosophers (see section 1.2.3), but it is time to revisit it. Mogensen (2022) suggests that a version of temporal discounting is defensible as an approximate representation of our different strengths of relationships to different people. It is sometimes permissible for me to favor my children’s well-being over the well-being of strangers, at least to some degree. Similarly, it may be permissible for me to favor my children’s interests over my grandchildren’s (to some degree), and theirs over my great-grandchildren’s.

Of course, temporal discounting only directly addresses the infinities that arise from the potentially infinite future. There are also the potential infinities of the present to worry about — whether from extraterrestrials, or multiverses, or metaphysical plenitude. To handle all of these we will need to discount for more than just time.

Nebel (2025, sec. 5) proposes a similar idea.

We have only a finite amount of care and concern to spread around. This does not mean that we cannot care about each of infinitely many people. We can, and we should. But we cannot care equally about them, if we care about them at all. If we are to care about everyone, we must care about some people more than others.

We can have some inner moral circle of impartial concern. Outside of that, we must discount somehow. There may well be no best way to this. No coherent set of values embodies perfectly impartial benevolence — but some values are *more* impartial than others. Nebel (2025) elaborates:

Perhaps each of us should try to be *more* impartial, by expanding the set of people for whom we have maximal concern. And perhaps in certain contexts — especially when making collective decisions on behalf of a particular finite group of people — we should care equally about all members of that group in particular. But this does not mean that we must care equally about everyone. Sometimes, when dealing with the infinite, the best we can do is to try to do better, since every option may be worse than some alternative, and taking improvements to their limit can be worse than no improvement at all.

I think there is something very appealing about this picture, and it does spare us from many of the difficulties of infinite ethics.

But it is drastic. It is permissible for me to favor my children's interests over strangers', but there are limits. It wouldn't be all right for me to cause massive suffering to millions of strangers in order to spare my child a scraped knee.¹⁵ But even such modest limits on partiality are very hard to sustain in the face of infinity. It is not enough to be partial. "We can, and we should"

¹⁵Mogensen (2022) argues on this basis that, while temporal discounting may be defensible, a *constant discount rate* is not. Rather, he argues that the total discount factor for each generation must be bounded away from zero. Unfortunately, such modest discounting is of no help to save us from the explosion arising from unbounded time.

care about each of infinitely many people. But we can't care about them very *much*. As Hong and Russell (2025) note, many of the paradoxes of infinite ethics arise from assumptions far weaker than impartiality. The mere idea that a harm to one can be outweighed by *some* harm to *many more* is enough to get us into trouble. Anything less than very extreme partiality is of no help. This is also hard to accept.

11.6 Regrets

Our moral lives start small: don't bite people, use kind words, help clean up. Sometimes we step back further, taking a more expansive view, and we consider the effects of our choices on the coffee farmer, the sweatshop worker, the factory-farmed chicken. We also consider less foreseeable possibilities: risks of infection, injury, or catastrophe. But, having stepped back this far, from here we can see farther still. The small effects of our small lives ripple farther and farther out in ways fathomable and unfathomable, perhaps even cosmic or supernatural. Whatever we do touches the possibility of the infinite.

Maybe we can legitimately ignore such possibilities, but if so, that is also for moral reasons that call out for better understanding. Infinite ethics might seem like an exotic pastime for idiosyncratic puzzle-solvers. But all ethics must reckon with our place in a vast and uncertain world, one way or another.

This vertiginous perspective is frightening and confusing — even incomprehensible, we may worry. It is tempting to think that infinite ethics is beyond reckoning or reasoning, doomed to unresolvable paradox. It's instructive to remember that even infinite *mathematics* was once regarded this way. Reacting to this view, Cantor ([1883] 1976, 74) writes:

It is intended to contain the flight of mathematical speculation and conceptualization within its true limits, where it runs no danger of falling into the abyss of the "transcendent", the place where supposedly, as is said to inspire fear and wholesome terror, "all is possible".

Cantor urged us on to map this abyss, and mathematicians followed, with quite a bit of triumph.

Moral philosophers might hope for similar progress. Indeed, throughout this book, I have tried to tell an encouraging story. Yes, we are beset with

paradoxes, but they have not left us utterly dumbfounded. Many hopeless-seeming mathematical traps have turned out to have principled escapes. We have moved beyond Pascal's mystical aporia — "we know that the infinite exists, and we are ignorant of its nature" — and into a project of building and evaluating precise theories of unbounded and infinite value. This project is still in its early stages, and most questions are open. But it feels like we are getting somewhere.

And yet when it comes to practical action, we remain in the dark. If the theory of impartial value we have sketched is correct in outline, then it is hard to imagine how this kind of value could ever guide our action. Tiny fluctuations in probabilities of infinite or unbounded effects swamp any merely finite difference we could make in the world. Figuring out how to systematically promote the greatest infinite goods seems incredibly hard and obscure. And it just doesn't seem that plausible that throwing our resources into the abyss of the infinite is a great choice, compared to other ways we have of doing remarkable good in the world. If this is really where impartial benevolence leads us, this casts some doubt on its credentials as a moral guide. I have indicated some directions we might seek to retreat, but I am not very happy with any of them.

The arguments for Pascalian conclusions were not ironclad, and it is a good idea to challenge them further. I think it is clear, in any case, that avoiding Pascalian conclusions will come at other serious costs. But so will accepting them.

In the course of this book, I think we have learned some things and made some interesting progress; and yet it is unclear what practical upshot any of this has, or ever could have. But then again, we have only just begun.

Part III

Mathematical Appendices

"Would you like the detailed answer, or the simple answer?" asks Ibrahim.

"The simple answer, please, Ibrahim," says Elizabeth, without hesitation.

Ibrahim pauses. Perhaps he has phrased his question poorly? "But I have prepared a detailed answer, Elizabeth."

Richard Osman, *The Thursday Murder Club*

Introduction to Part III

TODO.

Appendix A

Ordering Social Outcomes

A.1 Basics

TODO Definitions.

- Cartesian product.

Definition A.1.

- (a) Let $\text{id} : X \rightarrow X$ be the identity function.
- (b) Let $\pi_1 : X \times Y \rightarrow X$ and $\pi_2 : X \times Y \rightarrow Y$ be the projection functions $(x, y) \mapsto x$ and $(x, y) \mapsto y$, respectively.
- (c) For functions $f : X \rightarrow Y$ and $g : X \rightarrow Z$, let $(f, g) : X \rightarrow Y \times Z$ be the product function $x \in X \mapsto (fx, gx) \in Y \times Z$.
- (d) Let $\text{diag} : X \rightarrow X \times X$ be the diagonal function $\text{id} \times \text{id} = (x \mapsto (x, x))$.
- (e) For functions $f : X \rightarrow Y$ and $g : X' \rightarrow Y'$, let $f \times g : X \times X' \rightarrow Y \times Y'$ be the product function $(x, x') \mapsto (fx, gx')$.

2^X is the power set of X . $\text{Fin } X$ is the set of all finite subsets of X . $\text{Fin}(X, Y)$ is the set of all functions from X to Y with finite support.

A subset A of a set X is **cofinite** iff its complement $X \setminus A$ is finite.

Disjoint union of sets $A \sqcup B$, and piecewise function $f \sqcup g$.

Characteristic function 1_A . Indicator function $\delta(x)$

TODO. Note about using 2^X ambiguously: subsets or characteristic functions.

Notation ω . Identification of sequences with functions.

Definition A.2. Let I be a set, and for each $i \in I$, let X_i be a set. The **Cartesian product** $\prod_{i \in I} X_i$ is the set of all functions x with domain I such that for each $i \in I$, $x(i) \in X_i$.

For $k \in I$, the **projection function** $\pi_k : \prod_{i \in I} X_i \rightarrow X_k$ is the function such that $\pi_k(x) = x(k)$.

Equivalence classes, quotient maps.

A.2 Orders

There are several different relevant concepts of an order. I will run through these quickly.

Definition A.3. Let $\succsim$ be a relation on a set X .

- (a) $\succsim$ is **reflexive** iff, for every $x \in X$, $x \succsim x$.
- (b) $\succsim$ is **transitive** iff, for every $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.
- (c) $\succsim$ is a **preorder** iff it is reflexive and transitive.
- (d) $\succsim$ is a **partial order** iff it is a preorder that is also **anti-symmetric**: for every $x, y \in X$, if $x \succsim y$ and $y \succsim x$, then $x = y$.
- (e) $\succsim$ is **total** (also sometimes called *complete*) iff for every $x, y \in X$, either $x \succsim y$ or $y \succsim x$.
- (f) A **total order** is a partial order that is also total.
- (g) A **total preorder** is a preorder that is also total.
- (h) A **well-order** is a total order such that every non-empty subset has a least element.

For any preorder $\succsim$, we define:

- (i) $x \sim y$ iff $x \succsim y$ and $y \succsim x$;
- (j) $x > y$ iff $x \succsim y$ and $y \not\succsim x$;
- (k) $x \lesssim y$ iff $y \succsim x$; likewise $x < y$.

For example, the natural numbers $\mathbb{N}$ and the real numbers $\mathbb{R}$ are each totally ordered by their usual $\geq$ orders. The natural numbers are well-ordered,

while the real numbers are not.

Example A.4. Let X be any set.

- (a) The **superset order** $\supseteq$ on 2^X is a partial order.
- (b) The **discrete order** on X , which is the identity relation $=$, is a partial order.
- (c) The **trivial order** on X is the relation such that $x \succeq y$ for all $x, y \in X$. This is a total preorder. (It is not a total order unless X has at most one element.)

Definition A.5.

- (a) An **ordered set** consists of a set X equipped with a (pre)order $\geq$ on X . Similarly **partially ordered set**, etc.

Let X and Y be preordered sets, and let $f : X \rightarrow Y$ be a function.

- (b) f is **(weakly) monotone** (or *order-preserving*) iff,

$$\text{if } x \geq y \text{ then } fx \geq fy \text{ for each } x, y \in X.$$

- (c) f is **strictly monotone** iff it is weakly monotone, and furthermore,

$$\text{if } x > y \text{ then } fx > fy \text{ for each } x, y \in X.$$

- (d) f is **order-reflecting** iff,

$$\text{if } fx \geq fy \text{ then } x \geq y \text{ for each } x, y \in X.$$

- (e) f is an **order embedding** iff it is both weakly monotone and order-reflecting. That is,

$$x \geq y \text{ iff } fx \geq fy \text{ for each } x, y \in X.$$

- (f) An **order isomorphism** is an order embedding which is also one-to-one and onto.

Despite the name, an order embedding $X \rightarrow Y$ between preordered sets need not be a one-to-one function. The implication does hold if X is *partially*

ordered.

Every order embedding $X \rightarrow Y$ is strictly monotone. The converse does not always hold; it does hold in the case where X is *totally* preordered.

We have often talked about “representations” throughout this book, but (as is usual throughout philosophy) this notion does not correspond to a single precise relation between ordered sets. There are various kinds of representation at issue in different places, from monotone functions all the way up to order isomorphisms. The literature contains many different definitions of “representation”, but this terminology is not standardized.¹ I will avoid it where precision is required.

Definition A.6. Let $\geq$ and $\geq^*$ be orders on a set X .

- (a) $\geq^*$ **(weakly) extends** $\geq$ iff the identity map from $(X, \geq)$ to $(X, \geq^*)$ is monotone. That is,

$$\text{if } x \geq y \text{ then } x \geq^* y \text{ for each } x, y \in X.$$

- (b) $\geq^*$ **faithfully extends** $\geq$ iff the identity map from $(X, \geq)$ to $(X, \geq^*)$ is strictly monotone. That is, $\geq^*$ weakly extends $\geq$, and also

$$\text{if } x > y \text{ then } x >^* y \text{ for each } x, y \in X.$$

Equivalently (given that $\geq^*$ weakly extends $\geq$),

$$\text{if } x \geq y \text{ and } y \geq^* x \text{ then } y \geq x \text{ for each } x, y \in X.$$

Definition A.7. Let X be a preordered set.

- (a) A subset $U \subseteq X$ is **upward-closed** (or an *upper set*) iff for all $x \in U$ and $y \in X$, if $y \geq x$ then $y \in U$.
- (b) For $x \in X$, the **principal upper set** $\uparrow x$ is the set $\{y \in X \mid y \geq x\}$.

¹For example, Debreu (1954)’s utility “representation” is an order embedding; Narens (1980, Def. 3) uses “probability representation” for strict monotonicity, and “weak probability representation” for weak monotonicity; meanwhile Aumann (1962, 446)’s “weaker sense” of representation is just the *strictness* clause of strict monotonicity. There is a similar proliferation of senses of “agrees”, qualified with “almost”, “partially”, “fully”, or “strongly” (see for example Konek 2019; Fishburn 1986; T. L. Fine 1973, [18]).

Proposition A.8. Any finite non-empty preordered set has maximal and minimal elements.

Here are a few useful constructions related to orders.

Proposition A.9. Let X be a set, let Y be a preordered set, and let $f : X \rightarrow Y$ be a function. The **pullback order** $\geq_f$ on X is defined such that

$$x \geq_f y \quad \text{iff} \quad fx \geq fy \quad \text{for each } x, y \in X.$$

Then $\geq_f$ is a preorder on X , and $f : (X, \geq_f) \rightarrow (Y, \geq)$ is an order embedding.

Equivalently, this is the *strongest* order on X that makes f monotone.

Proposition A.10. Let $\mathcal{O}$ be any set of preorders on X . Let $\geq_f$ be their intersection:

$$x \geq_f y \quad \text{iff} \quad x \geq y \text{ for every preorder } \geq \text{ in } \mathcal{O}.$$

Then $\geq_f$ is a preorder on X . If each order in $\mathcal{O}$ is a partial order, then $\geq_f$ is also a partial order.

Definition A.11. Let X be a preordered set, let Y be any set, and let $f : X \rightarrow Y$ be a function. The **pushforward order** $\geq^f$ on Y is the intersection of all preorders $\geq$ on Y such that f is monotone with respect to $\geq$.

Equivalently, this is the *weakest* order on Y that makes f monotone.

Proposition A.12. Let $(X, \succsim)$ be a preordered set. There is an order embedding of X in a partially ordered set $X/\sim$ with the following property.

For any partially ordered set Y and any monotone function $f : X \rightarrow Y$, there is a unique monotone function $h : X/\sim \rightarrow Y$ such that $f = h \circ q$.

$$\begin{array}{ccc} X & \xrightarrow{q} & X/\sim \\ & \searrow f & \downarrow h \\ & & Y \end{array}$$

Furthermore, $X/\sim$ has this property uniquely, up to order isomorphism.

We call $X/\sim$ the **quotient partial order** of X .

(It's important to remember here that an order embedding need not be one-to-one; in this case, not being one-to-one is the whole point!)

Proof. Let the domain of $X/\sim$ be the set of equivalence classes of X (under the equivalence relation $\sim$ arising from $\succeq$). Let $q : X \rightarrow X/\sim$ be the quotient map, which takes each element of X to its equivalence class. We equip $X/\sim$ with the pushforward order determined by q . $\square$

We will have a couple of occasions to appeal to a standard order-theoretic principle which is equivalent to the axiom of Choice.

If X is a preordered set, a **chain** in X is a subset $C \subseteq X$ which is totally preordered (by the order on X). An **upper bound** of a subset $Y \subseteq X$ is an element $x \in X$ such that $x \geq y$ for all $y \in Y$. A **maximal element** of a preordered set X is an element $x \in X$ such that, for any $y \in X$, if $y \geq x$ then $y \sim x$.

Lemma A.13 (Zorn's lemma). For any non-empty partially ordered set X , if every chain in X has an upper bound, then X has a maximal element.

Applications of Zorn's lemma often look very similar to proofs by induction; but they are much more powerful, applying even to uncountable sets. One standard way of using Zorn's lemma is to prove *extension theorems*. Here is a classic example, which is also quite important for infinite ethics.

Theorem A.14 (Szpilrajn's extension theorem). Every preorder on a set X has a faithful extension which is a total preorder on X .

Proof. The proof has two parts. The important part is to prove this key claim.

For any preorder $\geq$ on X , and any $x, y \in X$, if $y \not\geq x$, then $\geq$ has a faithful extension $\geq^*$ such that $x \geq^* y$.

The other part is to show why this suffices for the theorem — this is boilerplate, going essentially the same way in every Zorn argument (and so it can often be mostly omitted). Let $\geq$ be a preorder on X , and let O be the set of all preorders that faithfully extend $\geq$. This set O is partially ordered by the *faithful extension* relation. Clearly O is non-empty, and it can be checked

that the union of any chain of faithful extensions of $\geq$ is also a faithful extension of $\geq$, and this is an upper bound for the chain. So O satisfies the conditions of Zorn's lemma, and thus it has a maximal element $\geq^*$. The key claim implies that this *maximal* order is *total*: there would be some x, y which are $\geq^*$ -incomparable, and in that case the extension $\geq^{**}$ would be a *proper* extension of $\geq^*$, contradicting maximality.

Now we prove the key claim. Let $\geq$ be a preorder, let $x, y \in X$, and suppose $y \not\geq x$. Then we can define a preorder $\geq^*$ such that

$$x' \geq^* y' \quad \text{iff} \quad \{x' \geq y', \text{ or else } x' \geq x \text{ and } y \geq y'\} \quad \text{for each } x', y' \in X.$$

Then we have $x \geq^* y$ by construction, and it can be checked that $\geq^*$ is a preorder that faithfully extends $\geq$. $\square$

A.3 Finitely symmetric strictly monotone orders

Throughout the book, it was often convenient to imagine “simple worlds”: there are infinitely many individuals, perhaps each in rooms labeled by natural numbers, and each individual is either happy or sad. If $\mathcal{I}$ is the set of all individuals, then we can naturally represent each simple world with a subset of $\mathcal{I}$, corresponding to the set of happy individuals. Then the problem of deciding which simple worlds are as good as which formally becomes the problem of ordering subsets of $\mathcal{I}$. This is a much simplified version of the problem of infinite social welfare, but there are plenty of interesting things to say even in this simple case. In particular, there are lots of impossibility results, and these will of course raise trouble all the more for more complex cases. But also, even for positive theorizing it is helpful to start with these simple cases, and then work our way up to harder problems.

In particular, we will begin by exploring formal analogues of the *Pareto* and *Impartiality* conditions for simple worlds that played a major role in the book. In section A.4 we will explicitly draw the lessons for social welfare.

Let $\mathcal{I}$ be a set. A **permutation** of $\mathcal{I}$ is any bijection. We can apply a permutation $\pi : \mathcal{I} \rightarrow \mathcal{I}$ not only to individual elements of $\mathcal{I}$, but also to subsets of $\mathcal{I}$: let

$$\pi A := \{\pi(i) \mid i \in A\}.$$

The **support** of a permutation $\pi : \mathcal{I} \rightarrow \mathcal{I}$ is the set

$$\{i \in \mathcal{I} \mid \pi(i) \neq i\}.$$

Definition A.15. Let $\mathcal{I}$ be any set, and let $\succsim$ be a preorder on $2^{\mathcal{I}}$.

- (a) $\succsim$ is **monotone** iff it extends the superset order: that is, for all $A, B \subseteq \mathcal{I}$, if $A \supseteq B$ then $A \succsim B$. It is **strictly monotone** iff it *faithfully* extends the superset order. That is, it is monotone, and also if $A \supsetneq B$ then $A \succ B$.
- (b) $\succsim$ is **permutation-symmetric** iff, for every permutation $\pi : \mathcal{I} \rightarrow \mathcal{I}$,

$$\pi A \sim A \quad \text{for each } A \subseteq \mathcal{I}.$$

- (c) $\succsim$ is **finitely symmetric** iff for every permutation $\pi : \mathcal{I} \rightarrow \mathcal{I}$ with finite support,

$$\pi A \sim A \quad \text{for each } A \subseteq \mathcal{I}.$$

Here is one of the basic results that constrains infinite ethics (chapter 3).

Proposition A.16. Let $\mathcal{I}$ be an infinite set. No strictly monotone order on $2^{\mathcal{I}}$ is permutation-symmetric.

Proof. Since $\mathcal{I}$ is infinite, there is a one-to-one function from the integers to $\mathcal{I}$, and thus also a function $f : 2^{\mathbb{Z}} \rightarrow 2^{\mathcal{I}}$ which is an order embedding of the superset order on $2^{\mathbb{Z}}$. If $\succsim$ is a strictly monotone and permutation-symmetric preorder on $2^{\mathcal{I}}$, it follows that the pullback order $\succsim_f$ on $2^{\mathbb{Z}}$ is also strictly monotone and permutation-symmetric. So it suffices to prove the proposition in the case of $\mathbb{Z}$.

Now consider the function $\pi : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$\pi(z) = z + 1 \quad \text{for each } z \in \mathbb{Z},$$

which is a bijection. Let A be the set of positive integers. Thus $\pi A \subsetneq A$. Then strict monotonicity requires that $A \succ \pi A$, while permutation-symmetry requires that $A \sim \pi A$ — contradiction. $\square$

On the other hand, there do exist finitely symmetric strictly monotone orders for infinite sets, and we will be exploring many of them in detail throughout these appendices. We will start with the most basic example.

We will use ω as an alternative notation for the set of natural numbers. (This notation is used to emphasize the natural numbers' cardinality, and sometimes their order properties, rather than their algebraic properties.)

Definition A.17. The **counting order** on 2^ω is defined such that for all $A, B \subseteq \omega$,

$A \succsim^\# B$ iff the set difference $B \setminus A$ is finite, and $\#(A \setminus B) \geq \#(B \setminus A)$ (including the case when $A \setminus B$ is infinite).

This is also known as the *Suppes–Sen grading principle*, specialized to two-valued utility streams (Suppes 1966; Sen [1970] 2017; see Pivato and Fleurbaey 2024 for overview). Here is its central property.

Theorem A.18. The counting order is the weakest finitely symmetric strictly monotone preorder on 2^ω . That is, $\succsim^\#$ is finitely symmetric and strictly monotone, and every finitely symmetric strictly monotone preorder faithfully extends the counting order.

Proof. It is straightforward to check that the counting order is finitely symmetric and strictly monotone. For the second part, suppose $\succsim$ is any finitely symmetric strictly monotone preorder on 2^ω , and let $A, B \subseteq \omega$.

Suppose $A \succsim^\# B$: so $B \setminus A$ is finite, and there is some one-to-one function $f : (B \setminus A) \rightarrow (A \setminus B)$. Then f can be extended to a permutation $\pi : \omega \rightarrow \omega$ supported by the finite set

$$(B \setminus A) \cup f(B \setminus A)$$

such that

$$\pi(B \setminus A) \subseteq A \setminus B.$$

Since $f(B \setminus A) \subseteq A \setminus B$, it follows that

$$\pi B = \pi(B \setminus A) \cup \pi(A \cap B) \subseteq A.$$

Thus by symmetry and strict monotonicity,

$$A \succsim \pi B \sim B.$$

Similarly, if $A \succ^\# B$, then the function $f : (B \setminus A) \rightarrow (A \setminus B)$ in this argument is also not onto. It follows that

$$\pi B \subsetneq A.$$

So, in this case, $A \succ \pi B \sim B$. □

There is a subtlety worth noting about this definition. It is framed in the context of sets of natural numbers, which are guaranteed to be *countable*. In particular, any two sets of natural numbers are the same size. When we consider larger infinite sets, there is a distinction to make between two different generalizations of the counting order. One is the definition given above. This has the upshot that A and B are incomparable even in the case where $A \setminus B$ has *smaller infinite cardinality* than $B \setminus A$. There is an alternative definition, though.

Definition A.19. Let $\mathcal{I}$ be any set.

- (a) The **weak counting order** on $2^{\mathcal{I}} \gtrsim^{\#}$ is defined in the same way as the counting order on 2^{ω} : for all $A, B \subseteq \mathcal{I}$, $A \gtrsim^{\#} B$ iff $\#(B \setminus A)$ is finite and $\#(A \setminus B) \geq \#(B \setminus A)$.
- (b) The **strong counting order** is the order $\gtrsim$ on $2^{\mathcal{I}}$ such that for $A, B \subseteq \mathcal{I}$, $A \gtrsim B$ iff either $\#(A \setminus B) > \#(B \setminus A)$, or else $A \setminus B$ and $B \setminus A$ are of the same finite cardinality.

As the terminology suggests, the strong counting order faithfully extends the weak counting order. In the case where $\mathcal{I}$ is countable, these two definitions coincide, and so in that context we can unambiguously speak of the *counting order* without qualification. But in a more general setting we have to be careful about this distinction.

The generalization of theorem A.18 to uncountable sets holds for the *weak* counting order.

Theorem A.20. For any set $\mathcal{I}$, the weakest finitely symmetric strictly monotone preorder on $2^{\mathcal{I}}$ is the weak counting order.

The proof goes exactly the same way.

Definition A.21. Let $\mathcal{I}$ be any infinite set. For any $A, B \subseteq \mathcal{I}$, let

$$A \gtrsim^{\infty} B \quad \text{iff} \quad B \setminus A \text{ is finite.}$$

This is a preorder, which we call the ∞ -**order**.

We write $A \succ^{\infty} B$ for the strict part of $\gtrsim^{\infty}$. That is, $A \succ^{\infty} B$ iff $A \setminus B$ is infinite and $B \setminus A$ is finite. Call this relation the **strict ∞ -order**.

We write $A \sim^{\infty} B$ for the corresponding equivalence relation. That is, $A \sim^{\infty} B$

B iff $A \setminus B$ and $B \setminus A$ are both finite.

Lemma A.22 (Sierpiński (1965), as cited in Basu and Mitra (2003)). For any infinite set $\mathcal{I}$, there is an order embedding $A : \mathbb{R} \rightarrow 2^{\mathcal{I}}$ with respect to the ∞ -order. In particular, for all $r, r' \in \mathbb{R}$, if $r > r'$ then $A(r) \succ_{\infty} A(r')$.

Proof. Let J be a countably infinite subset of $\mathcal{I}$, and let $q : J \rightarrow \mathbb{Q}$ be a bijection with the rationals. Then for each real number $r \in \mathbb{R}$, let

$$A(r) = \{i \in J \mid q(i) < r\}.$$

If $r' > r$, then since there are infinitely many rationals between r and r' , $A(r') \succ_{\infty} A(r)$. (Clearly if $r' = r$, then $A(r') \sim^{\infty} A(r)$.) $\square$

Theorem A.23 (Basu and Mitra (2003)). For infinite $\mathcal{I}$, the counting order on $2^{\mathcal{I}}$ is not real-representable. Thus no preorder that embeds the counting order is real-representable.

Proof. Consider a function $A : \mathbb{R} \rightarrow 2^{\mathcal{I}}$ as in Sierpinski's lemma. For each $r \in \mathbb{R}$, let $B(r) = A(r) \cup \{i\}$ where i is an element of $\mathcal{I}$ not already contained in $A(r)$. So $B(r) \supsetneq A(r)$, and for any $r' > r$ we also have $A(r') \succ^{\infty} B(r)$.

Now suppose for contradiction that some function $u : 2^{\mathcal{I}} \rightarrow \mathbb{R}$ strongly represents the counting order. Then

$$u(B(r)) > u(A(r)) \quad \text{for each } r \in \mathbb{R}.$$

So there is some rational number $q(r)$ falling between $u(A(r))$ and $u(B(r))$. Also, if $r' > r$, then

$$q(r') > u(A(r')) > u(B(r)) > q(r).$$

So q is a one-to-one function from $\mathbb{R}$ to $\mathbb{Q}$. But this cannot be. $\square$

In chapter 4 we discussed at length the principle of Comparability: for any two outcomes, at least one of them is at least as good as the other. Is this true? There are some strong reasons for thinking so, but also some important obstacles. If Comparability is true, then the counting order falls very far short of the full theory of value, even for simple worlds. Can it be extended to a total order? Here are two central results bearing on this question.

The classic positive result is Szpilrajn's extension theorem (theorem A.14).

Corollary A.24 (Svensson (1980)). For any set $\mathcal{I}$, there exists a finitely symmetric strictly monotone total preorder on subsets of $\mathcal{I}$.

Proof. By Szpilrajn's extension theorem, there is a total preorder on $2^{\mathcal{I}}$ that faithfully extends the counting order, and which is therefore finitely symmetric and strictly monotone. $\square$

Second, an important negative result: it turns out that the appeal to Choice in the proof of Szpilrajn's extension theorem is indispensable.

Theorem A.25 (Zame (2007); Lauwers (2010)). There is no constructive proof that there exists a strictly monotone finitely symmetric total preorder on the subsets of an infinite set.

We motivated this section by interpreting ordering of subsets of $\mathcal{I}$ as a *value* order on simple worlds. But there is an alternative interpretation available of the same mathematical structure. Instead of thinking of $\mathcal{I}$ as a set of *individuals*, we can think of it as a set of *possible worlds*; instead of thinking of a subset of $\mathcal{I}$ as representing a *social outcome*, we can think of it as representing a *proposition*, a set of worlds. And instead of thinking of $\succsim$ as a *value* order, we can think of it as a *comparative probability* order: $A \succsim B$ means that A is at least as probable as B . This comparative notion of probability is much more general than the standard approach to probability theory, involving assignments of real-valued probability. We return to this in appendix F.

The following definition is based on de Finetti's axioms of comparative probability (Finetti 1937); but note that in contrast with de Finetti, we do not assume that the probability order is *total*: perhaps two propositions can be incomparable in probability.

Definition A.26. Let Ω be a set of **possible worlds** or **states**. A **(comparative) probability order** on Ω (called the **state space**) is a preorder $\succsim$ on 2^{Ω} such that for all $A, B, C \subseteq \Omega$,

- (a) *Non-triviality.* $\Omega \succ \emptyset$.
- (b) *Monotonicity.* If $A \supseteq B$, then $A \succsim B$.
- (c) *Quasi-additivity.* If $A \cap C = B \cap C = \emptyset$, then $A \succsim B$ iff $A \cup C \succsim B \cup C$.

The definitions of **finite symmetry** and **strict monotonicity** apply to probability orders without modification. A finitely symmetric probability order is called **uniform**. A probability order is **regular** iff $\{w\} > \emptyset$ for every $w \in \Omega$. In the presence of the other probability order conditions, this is equivalent to strict monotonicity. As we discussed in chapter 5, regularity is not captured by standard probability theory, but even so it is something we might naturally want to capture in our theory of *comparative* probability. (See also section E.1.)

Corollary A.27.

- (a) A probability order is regular and uniform iff it faithfully extends the weak counting order.
- (b) No regular uniform probability order on an infinite state space is real-representable.
- (c) There is no constructive proof that there exists a regular uniform total probability order on an infinite state space.

Here is one more interpretation of the formalism. Throughout the book, a running example has been the case of an infinite sequence of coin flips. In this application, ω represents the set of coin flips (indexed by numbers $1, 2, 3, \dots$), and subsets of ω represent sets of coin flips. We will think of the elements of 2^ω as *states* — ways the sequence of coin flips might turn out. But it is usually more natural in this setting to think of 2^ω not as a set of sets, but rather as the set of infinite *sequences* of coin flips — which we are identifying with *functions* $b : \omega \rightarrow \{0, 1\}$. Here $b(i) = 1$ represents Heads on the i th flip, and $b(i) = 0$ represents Tails. (As we discussed in section A.1, the ambiguous use of 2^ω is harmless, as anything we prove about subsets of ω can be straightforwardly translated into a fact about functions from ω to 2, and vice versa.)

Propositions about the outcomes of coin flips are represented by *subsets* of 2^ω — elements of 2^{2^ω} . A *probability order* on the coin flip space will thus be a preorder on such propositions. It might be a preorder on the whole power set 2^{2^ω} , the set of all propositions about coin flips. More generally, it might instead be a preorder on some *subalgebra* of this set; see section E.1.

Definition A.28. The **coin flip space** (also called the *Cantor space*) is the set 2^ω of all infinite sequences of ones and zeros, representing Heads and Tails.

(See also examples E.9 and A.44, definition E.8, and corollary F.25.)

For any set $A \subseteq \omega$, let $H(A)$ represent the proposition “Heads at least on the flips in A ”. That is,

$$H(A) := \{b \in 2^\omega \mid b(i) = 1 \text{ for each } i \in A\}.$$

For any sequence b in 2^ω and any position $i \in \omega$ in the sequence, we can define a “flipped” sequence $\sigma_i b$, which has the opposite value from b at position i , and the same value as b at every position besides i .

Here are two natural, fairly minimal conditions on a probability order for coin flips. First, no coin flip is sure to come up heads, even supposing that some other coin flips have come up heads: thus, if $i \notin A \subseteq \omega$, then heads on all of $A \cup \{i\}$ is less probable than heads on A . (This again encodes a regularity idea that is *not* captured by standard probability theory: in that setting, if A is infinite, then the probability of $H(A)$ and $H(A \cup \{i\})$ are both zero.) Second, any two coin flips are equally likely to come up heads, again even supposing that some other coin flips have come up heads: thus, if $i, j \notin A \subseteq \omega$, then heads on $A \cup \{i\}$ is equally probable as heads on $A \cup \{j\}$. Here are some negative results about any such order.

Corollary A.29. Let $\mathcal{A}$ be a set of subsets of 2^ω which (at least) includes each of the sets $H(A)$ for $A \subseteq \omega$, and let $\succsim$ be any (at least weakly) monotone order on $\mathcal{A}$.

Suppose that for each set $A \subseteq \omega$ and each $i, j \in \omega \setminus A$,

$$H(A \cup \{i\}) \sim H(A \cup \{j\}) < H(A).$$

- (a) $\succsim$ is not real-representable.
- (b) $\succsim$ is not symmetric under arbitrary permutations of coin flips: there are permutations $\pi : \omega \rightarrow \omega$ and sets $A \subseteq \omega$ such that $H(A) < H(\pi A)$.
- (c) There is no constructive proof that any such order $\succsim$ is total.

Proof. We can define a preorder on 2^ω :

$$A \sqsubseteq B \text{ iff } H(A) \succsim H(B) \text{ for each } A, B \subseteq \omega.$$

Note that the ordering of coin flips is *dual* to the probability order: a “bigger” set of coin flips has a *smaller* probability. It is straightforward to check

that the hypotheses imply that $\sqsubseteq$ is a strictly monotone finitely symmetric preorder on 2^ω . Thus part (a) follows from theorems A.18 and A.23, part (b) follows from proposition A.16, and part (c) follows from theorem A.25. $\square$

Independently, Pruss (2014) proves another striking result in the same vein. (The following is an immediate consequence of Pruss (2014, Theorem 1) together with Wagon (1994, Theorem 13.2). In fact, Pruss proves an even stronger result, using a more technical hypothesis that weakens strict monotonicity; see theorem E.66 in section E.7.)

Theorem A.30 (Pruss (2014)). There is no constructive proof that there exists a strictly monotone total preorder on the countable subsets of a set with the cardinality of the continuum (2^ω).

Strikingly, compared to the Zame–Lauwers theorem (and corollary A.29), Pruss’s theorem has no symmetry hypothesis at all — at the cost of an upgraded cardinality assumption about the underlying set.

Zame (2007), Lauwers (2010), and Pruss (2014) prove their respective results by quite different methods — and, accordingly, they are amenable to different strengthenings. Briefly, Zame shows that a strictly monotone finitely symmetric total preorder must be an *unmeasurable set* (see theorem E.5). Lauwers shows that such a preorder implies the existence of a *non-Ramsey set*. Pruss proves that a total strictly monotone preorder is sufficient for the *Banach–Tarski paradox*: a unit ball can be decomposed into finitely many disjoint pieces that can be rigidly rearranged to form *two* unit balls. Unmeasurable sets, non-Ramsey sets, and Banach–Tarski decompositions are all non-constructive; none of them have existence proofs in ZF alone. But they are not all equivalent: accordingly, the theorems of Zame, Lauwers, and Pruss have different implications for the place of different social welfare and probability orders in the realm of alternative set theories without Choice (see figure A.1).

[In particular, while the Banach–Tarski paradox implies the existence of non-measurable sets, non-Ramsey sets are known not to imply Banach–Tarski decompositions or non-measurable sets in ZF; the remaining implications are open questions. For discussion of the set-theoretic implications of Zame and Lauwers’ results, see Dubey and Laguzzi (2024). See also section B.5.]

One last point worth noting is that, in uncountable spaces, we can also relax the symmetry conditions for the analogue of Basu and Mitra’s theorem.

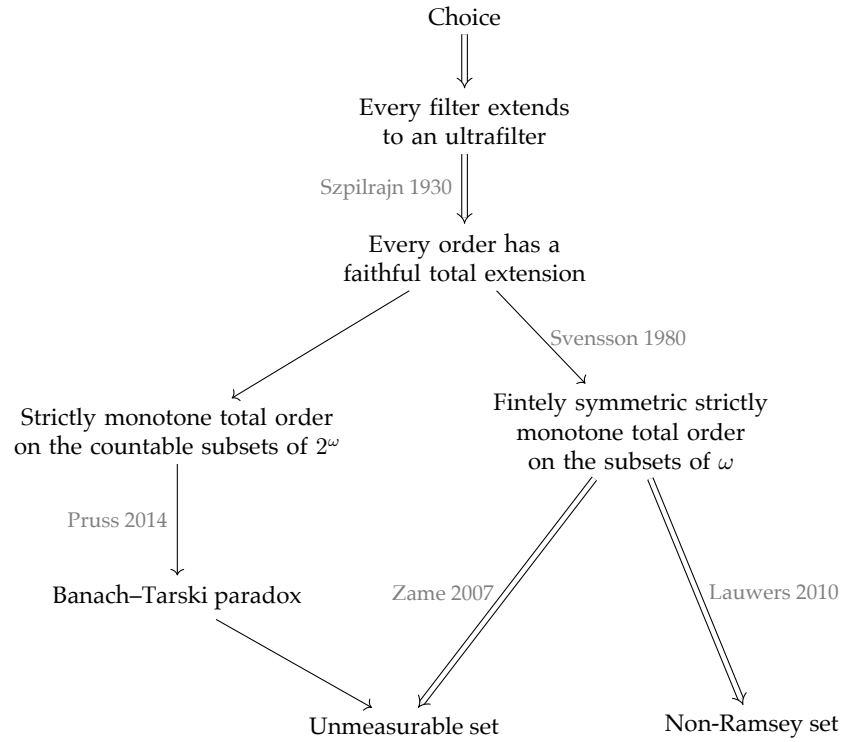


Figure A.1: The logical relationships between various “non-constructive” principles, in the context of ZF set theory. Single-line arrows represent known implications; double-line arrows represent known strict implications. (Ultrafilters do not imply Choice (J. D. Halpern and Lévy 1971); order extensions do not imply ultrafilters (Felgner and Truss 1999); non-measurable sets and non-Ramsey sets do not imply finitely symmetric strictly monotone total orders, even jointly (Dubey and Laguzzi 2024).)

(Compare corollary A.27 and corollary A.29(a).)

Theorem A.31. No strictly monotone preorder on (at least) the countable subsets of an uncountable set is real-representable.

Proof. Let ω_1 be the first uncountable ordinal; without loss of generality, let $\succsim$ be a strictly monotone preorder on countable sets of ordinals below ω_1 , and suppose for contradiction that u is a real-valued function that strongly represents $\succsim$.

For each $\alpha < \omega_1$, let $A(\alpha)$ be the set containing all ordinals below α . By strict monotonicity $A(\alpha) < A(\alpha + 1)$, and thus $u(A(\alpha)) < u(A(\alpha + 1))$. This non-empty interval contains a rational number. Thus for each $\alpha < \omega_1$, there is a rational number

$$q(\alpha) \in (u(A(\alpha)), u(A(\alpha + 1))).$$

This defines a strictly increasing — and thus one-to-one — function q from the uncountable ordinal ω_1 to $\mathbb{Q}$, which is impossible. $\square$

(The move to the ordinal ω_1 implicitly appeals to Choice again. A Choice-free proof can also be given, if the hypothesis “uncountable” is strengthened to “at least of continuum-cardinality”.)

A.4 Social welfare orders

For concreteness, we have focused so far on very simple toy worlds: a population indexed by natural numbers, each of whom is in one of just two states, “happy” or “sad”. But the negative results we just surveyed for orders on simple worlds apply much more generally to any domain that *embeds* these simple worlds. Let’s introduce a framework that will be generally useful throughout these appendices.

Definition A.32. A **social welfare structure** consists of the following components.

- (a) A preordered set $\mathcal{X}$ of **individual outcomes** (or *welfare levels*). We assume $\mathcal{X}$ has at least two elements $x \succ y$.
- (b) A set $\mathcal{I}$ of **individuals**. We will generally suppose that $\mathcal{I}$ is infinite.
- (c) The **social welfare order**, a preorder $\succsim$ on **social outcomes**. A social

outcome is any function from $\mathcal{I}$ to $\mathcal{X}$.

This framework is familiar, but it bears a bit of philosophical discussion.

The central abstraction is that we will think of a *social outcome* as a (possibly infinite) sequence of *individual outcomes*. An individual outcome is supposed to represent everything that matters for the well-being of a single person. The framework assumes that different individual outcomes are “recombinable”, so that we can freely modify one individual’s well-being independently of what happens to any other individual. This is not obviously true in general: my well-being is intimately — and perhaps constitutively — tied to the well-being of those I love, and of the community I am part of. But the assumption of recombability is a very useful simplification, and it is pervasive in formal ethics — usually passing without comment. I believe the assumption can be greatly relaxed without changing very much, but for now we will take it for granted, accepting the resulting limitation of scope for our results. (For insightful discussion of this issue, see San 2026.)

Another thing this framework assumes is that the *same* individual outcomes make sense for different individuals. This is required to make general sense of Anonymity-type principles, like Impartiality, which *switch* the well-beings of two different individuals. This requires us to make sense of certain *interpersonal comparisons of well-being*. This idea can be justified in terms of more basic assumptions.

[TODO Discuss Compensation interpretation.]

We can immediately apply the results from section A.3 to derive some negative conclusions about social orders for infinite populations.

For any permutation π of $\mathcal{I}$ and any social outcome $x : \mathcal{I} \rightarrow \mathcal{X}$, let $\pi x = x \circ \pi^{-1}$.

Definition A.33. Let $\succsim$ be a social order.

- (a) $\succsim$ is **Paretian** iff for all social outcomes x, y , if $x(i) \succsim y(i)$ for all $i \in \mathcal{I}$ and $x(i) \succ y(i)$ for some $i \in \mathcal{I}$, then $x \succ y$.
- (b) $\succsim$ is **strongly anonymous** iff for all social outcomes x and all permutations π of $\mathcal{I}$, $x \sim \pi x$.
- (c) $\succsim$ is **(finitely) anonymous** iff for all social outcomes x and all permutations π of $\mathcal{I}$ with finite support, $x \sim \pi x$.

As we discussed in chapter 3, for simple “happy-sad” social outcomes, the Impartiality principle that was central to much of the book (switching one person from sad to happy is equally good as switching another) is equivalent to finite anonymity.

Proposition A.34. Suppose the set of individuals $\mathcal{I}$ is infinite. For any finitely anonymous Paretian social order $\succsim$, there exists an order embedding of the counting order

$$f : (2^\omega, \succsim^\#) \rightarrow (\mathcal{X}^{\mathcal{I}}, \succsim).$$

Corollary A.35. Suppose the set of individuals $\mathcal{I}$ is infinite.

- (a) No Paretian social order is strongly anonymous.
- (b) No finitely anonymous Paretian social order is real-representable.
- (c) It is not provable in ZF that there exists a finitely anonymous Paretian total social order.

A.5 Symmetries

One running theme in this book has been the status of a variety of *symmetry* properties that value orders might have. Chief among these has been *Impartiality*, which says that swapping the well-being of two individuals leaves things equally good overall. We have also considered properties like *Strong Anonymity* and *Permutation Invariance*, as well as considering how values might be affected by shifting an assignment of well-being across space. Symmetry considerations were also crucial in our discussion of probability theory.

It will be helpful to have some general concepts and tools for reasoning about symmetries. This is a central part of the branch of mathematics called *group theory*. Consider, for example, a simple world in which individuals numbered by the positive and negative integers are arranged along an infinite one-dimensional line. We can transform assignments of well-being in such worlds in various ways. For example, we can shift any distribution over one step, producing a new distribution that assigns each individual z the well-being of their neighbor $z - 1$. We can apply this shift as many times as we like, displacing the distribution of happiness farther and farther. Or we can also undo this shift by moving everything back the other way.

Thus these well-being shifts, taken all together, comprise what is called a *group action*. We can combine shifts by *composition*, doing one and then the other. Each shift has an *inverse*, which shifts things equally the other direction. There is also a trivial *identity* shift — shifting everything by zero steps.

Definition A.36. A **group** is a set G (the group's *domain*) equipped with a binary operation $\cdot : G \times G \rightarrow G$ (the *group operation*) and an *identity* element $1 \in G$, where these have the following properties.

- (a) **Associativity.** $g \cdot (h \cdot k) = (g \cdot h) \cdot k$ for every $g, h, k \in G$.
- (b) **Identity.** $1 \cdot g = g \cdot 1 = g$ for each $g \in G$.
- (c) **Inverses.** For each $g \in G$, there is an element $g^{-1} \in G$ such that $g \cdot g^{-1} = g^{-1} \cdot g = 1$.

Definition A.37. A **group action** $G \curvearrowright X$ consists of a group G , a set X , and an operation $*$: $G \times X \rightarrow X$ with the following two properties:

$$\begin{aligned} 1 * x &= x & \text{for each } x \in X \\ g * (h * x) &= (g \cdot h) * x & \text{for each } g, h \in G, x \in X. \end{aligned}$$

(I used different symbols $\cdot$ and $*$ in this definition for clarity, but in practice we will often use the same symbol $\cdot$ for both the group operation and the group action.)

It is standard to use notation and terminology that evokes *multiplication* for groups. This is because composing symmetries and multiplying numbers have many important properties in common. But it is important not to be misled by this: the “multiplication” operation in a group can in some ways behave very differently from multiplying numbers. For one important example, multiplying *numbers* is *commutative*: $a \cdot b = b \cdot a$ for all numbers a and b . But this is not always true in a group.

Definition A.38. A group G is **abelian** iff $g \cdot h = h \cdot g$ for all g, h in G .

The group of shifts in space we just considered is indeed abelian. But, for example, the group of arbitrary rigid transformations of two-dimensional space is not abelian. Take an object at an “origin” point. First shift it a foot to the right, and then rotate it around origin. This has a different result from first (trivially) rotating it around itself, and *then* shifting it.

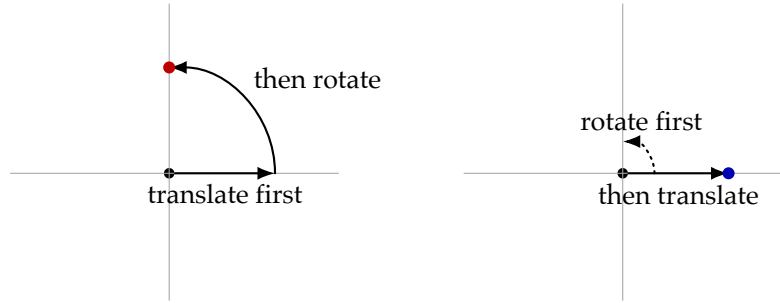


Figure A.2: Arbitrary transformations need not commute.

Sometimes we use alternative notation besides 1 and $\cdot$ for the features of a group, depending on what is most evocative. For example, sometimes we instead use 0 and $+$ for the identity and operation of a group — especially when the group is *abelian*. Here is another example.

Example A.39. For any set X , the **permutation group** $\Pi(X)$ is the group of all one-to-one and onto functions from X to X . Its group operation is function composition $\circ$, and its group identity element is the identity function id_X .

There is a natural **action** of $\Pi(X)$ on X , by function application:

$$\pi \cdot x := \pi(x) \quad \text{for each } \pi \in \Pi(X), x \in X.$$

The permutation group on X also acts in a natural way on other sets. For example, $\Pi(X)$ acts on 2^X by letting

$$\pi \cdot A := \{\pi(x) \mid x \in A\} \quad \text{for each } \pi \in \Pi(X), A \subseteq X.$$

The permutation group acts in the same way on $\text{Fin } X$, the set of all finite subsets of X (since if K is finite, then so is $\pi \cdot K$).

For another example, $\Pi(X)$ acts on the set of all functions $X \rightarrow Y$, by letting

$$\pi \cdot f := f \circ \pi^{-1} \quad \text{for each } \pi \in \Pi(X), f : X \rightarrow Y.$$

(If it isn't immediately clear, it is worth checking why the definition for the action on functions goes in the order it does, rather than using $f \circ \pi$. Notice also that if we think of a set $A \subseteq X$ in terms of its characteristic function $1_A : X \rightarrow \{0, 1\}$, then the two “lifted” actions on 2^X agree: $g \cdot 1_A = 1_{g \cdot A}$.)

Definition A.40. A **subgroup** H of a group G is a group such that

- (a) The domain of H is a subset of G ;
- (b) The group operation of H is the restriction of the group operation of G to H ; and
- (c) The identity element of H is the same as the identity element of G .

This requires that if $g, h \in H$, then $g \cdot h$ and g^{-1} are also contained in H .

If $G \curvearrowright X$ is a group action and H is a subgroup of G , then there is also a natural action $H \curvearrowright X$, by just restricting the action of G .

Example A.41. The group $\Pi_{\text{fin}} X$ is the group of all permutations of X with finite support. This is a subgroup of $\Pi(X)$. Thus it also has actions on X , 2^X , $\text{Fin } X$, and $X \rightarrow Y$, in by restricting the action of $\Pi(X)$.

Example A.42. The group of all shifts of the integers is a subgroup of $\Pi(\mathbb{Z})$, acting on $\mathbb{Z}$. In particular, it restricts $\Pi(\mathbb{Z})$ to those functions $f_a : \mathbb{Z} \rightarrow \mathbb{Z}$

$$f_a(z) = z + a \quad \text{for each } z \in \mathbb{Z},$$

where $a \in \mathbb{Z}$ is any fixed integer.

Example A.43. We can think of the interval $[0, 1)$ as representing a circle. We represent a rotation of the circle by adding a number $r \in [0, 1)$ and subtracting one if necessary (that is, taking the result modulo 1). This is an action of the additive group $[0, 1)$ on itself. This group is also labeled $\mathbb{R}/\mathbb{Z}$.

An important subgroup of the group of rotations is the group $\mathbb{Q}/\mathbb{Z}$ of *rational* rotations, which also acts on the interval $[0, 1)$ by addition modulo 1. Note that while the group of all rotations is uncountable, the group of rational rotations is countable.

Example A.44. Let 2^ω be the (Cantor) coin flip space, the set of infinite sequences of ones and zeros (Heads and Tails) (definition A.28).

For any sequence b in 2^ω and any position $i \in \omega$ in the sequence, we can define a “flipped” sequence $\sigma_i b$, which has the opposite value from b at position i , and the same value as b at every position besides i . By composing

these flip functions $\sigma_i : 2^\omega \rightarrow 2^\omega$, we can also flip any finite set of indices. We call these operations the **finitary flips**.

We can represent this as a group action $\text{Fin } \omega \curvearrowright 2^\omega$ of finite subsets of ω on 2^ω . The group operation on $\text{Fin } \omega$ is the “xor” symmetric difference, and the identity is the empty set. (Equivalently, we can think of the group operation as adding characteristic functions mod 2.)

More generally, for any set of indices $A \subseteq \omega$, whether finite or infinite, we can let $\sigma_A : 2^\omega \rightarrow 2^\omega$ be the “flip” function that takes each sequence $b \in 2^\omega$ to a sequence $\sigma_A b$ that has the opposite value from b at every position in A , and the same value as b at every position not in A . We can similarly equip 2^ω with the symmetric difference group structure. This is called the **Cantor group**. It also acts naturally on the coin flip space 2^ω by flipping: for $A \subseteq \omega$ and $b \in 2^\omega$, $A \cdot b = \sigma_A(b)$.

(We have distinguished subsets of ω from sequences in 2^ω to help motivate the picture conceptually, but it can also be useful to think of the Cantor as the action of the group 2^ω on *itself*.)

A third group of symmetries of the coin flip space involves switching the *order* of coin flips. The group of all permutations of ω , $\Pi(\omega)$, acts on 2^ω by permuting the indices of the sequence, so

$$(\pi \cdot w)(i) := w(\pi^{-1}(i)) \quad \text{for each } \pi \in \Pi(\omega) \text{ and } w \in 2^\omega.$$

We will call this the **coin-permutation action**. The group $\Pi(\omega)$ also has the countable subgroup $\Pi_{\text{fin}}(\omega)$, which also acts on 2^ω ; this is the **finitary coin-permutation action**.

Definition A.45. Let G be a group, and let X be any subset of G .

- (a) The **subgroup generated by** X , denoted $\langle X \rangle$, is the smallest subgroup of G that contains X . Equivalently, it is the intersection of all subgroups of G that contain X .
- (b) A group G is **finitely generated** iff it is generated by some finite subset: that is, $G = \langle F \rangle$ for some finite subset $F \subseteq G$.
- (c) A group G is **locally finite** iff every finitely generated subgroup of G is finite.

(Again, if it is not immediately clear, it is worth checking that the intersection

of any family of subgroups is itself a subgroup, and so the definition really guarantees that the subgroup generated by X exists.)

Example A.46.

- (a) For any set X , the group of all finite permutations $\Pi_{\text{fin}} X$ is locally finite. In contrast, if X is infinite, the group of all permutations $\Pi(X)$ is not locally finite.
- (b) The group of rational rotations $\mathbb{Q}/\mathbb{Z}$ (example A.43) is locally finite. In contrast, the group of all rotations $\mathbb{R}/\mathbb{Z}$ is not locally finite: for any irrational rotation $r \in \mathbb{R}/\mathbb{Z}$, the subgroup $\langle r \rangle$ is infinite.
- (c) The Cantor group 2^ω is locally finite, as is its subgroup $\text{Fin } \omega$ of finitary coin flip symmetries.
- (d) More generally, any subgroup of a locally finite group is also locally finite.

Definition A.47. Let $G \curvearrowright X$ be a group action, and let $x \in X$.

- (a) For any subset $F \subseteq G$, let

$$F \cdot x := \{f \cdot x \mid f \in F\}.$$

The **orbit** of x under the action of G is the set $G \cdot x$.

- (b) $G \curvearrowright X$ **has finite orbits** iff the orbit $G \cdot x$ is finite for every $x \in X$.
- (c) $G \curvearrowright X$ **is locally finite** iff the action of every finitely generated subgroup of G has finite orbits. That is to say, for every finite subset $F \subseteq G$ and every $x \in X$, $\langle F \rangle \cdot x$ is finite.

Proposition A.48. If G is a locally finite group, then any action $G \curvearrowright X$ is locally finite.

Example A.49. The natural actions of each of the locally finite groups in example A.46 are locally finite: finite permutations acting on any set, rational rotations acting on the circle, and both the finitary and the full Cantor coin flip actions on the coin flip space. In contrast, the action of the group of *all* rotations on the circle is not locally finite: again, an irrational rotation has an infinite orbit. Similarly, the coin permutation action of $\Pi(\omega)$ on the coin

flip space is not locally finite, since again there are coin permutations with infinite orbits.

We will also have some occasions to be interested in *orders* on groups.

Definition A.50. A **(left)-preordered group** is a group $(G, \cdot, 1)$ equipped with a preorder $\geq$ such that for all $g, h, k \in G$, if $g \geq h$ then $k \cdot g \geq k \cdot h$. In other words, $\geq$ is invariant under the action of G on itself.

(**Partially/totally ordered group** and **totally preordered group** are defined similarly.)

Here is a handy way to come up with orders on groups.

Definition A.51. Let G be a group, and let $P \subseteq G$ be a subset.

- (a) P is a **precone** iff
 - (i) $1 \in P$, and
 - (ii) If $g, h \in P$, then $g \cdot h \in P$.
- (b) P is a **partial cone** in G iff P is a precone such that $P \cap P^{-1} = \{1\}$ (where $P^{-1} = \{g^{-1} \mid g \in P\}$).
- (c) P is a **total precone** in G iff P is a precone such that $P \cup P^{-1} = G$.
- (d) If P and P^+ are precones in G , we say that P^+ **faithfully extends** P iff
 - (i) $P^+ \supseteq P$, and
 - (ii) $P \supseteq P^+ \cap P^{-1}$.

Proposition A.52. For any preordered group G , the set

$$P = \{g \in G \mid g \geq 1\}$$

is a precone in G .

Conversely, for any precone P in a group G , the relation $\geq$ defined by

$$g \geq h \quad \text{iff} \quad g^{-1} \cdot h \in P$$

is a preorder on G .

Similarly for partial/total (pre)-orders and their corresponding cone concepts.

Definition A.53. Let G and H be preordered groups. A **homomorphism** (of preordered groups) is a function $h : G \rightarrow H$ such that

- (a) $h(1) = 1$;
- (b) $h(g \cdot g') = h(g) \cdot h(g')$ for all $g, g' \in G$; and
- (c) if $g \geq g'$ in G , then $h(g) \geq h(g')$ in H .

A.6 Invariant orders

Now let us generalize the ideas of *finitely symmetric* and *permutation invariant* orders, to consider other symmetry groups, beyond just the group $\Pi(X)$ of *all* permutations, and the group $\Pi_{\text{fin}}X$ of all permutations with finite support.

Definition A.54. Let $G \curvearrowright X$ be a group action.

- (a) An order $\geq$ on X is **G -symmetric** iff

$$x \sim g \cdot x \quad \text{for each } x \in X, g \in G.$$

- (b) An order $\geq$ on X is **G -invariant** iff

$$\text{if } x \geq y \quad \text{then } g \cdot x \geq g \cdot y \quad \text{for each } x, y \in X, g \in G.$$

A caveat. It is nice to have a terminological distinction for these two conditions, but both words “symmetry” and “invariance” are very entrenched in mathematics, and they are often used interchangeably. In particular, if a structure (such as an order) is invariant with respect to a certain permutation, in ordinary practice this is called a “symmetry” of that structure.

Note that if G acts on $\mathcal{I}$, then we also automatically inherit an action $G \curvearrowright 2^{\mathcal{I}}$, by letting $g \cdot A = \{g \cdot x \mid x \in A\}$.

Any G -symmetric order is G -invariant. But the converse does not hold. For example, the superset order on $2^{\mathcal{I}}$ is *invariant* under all permutations of $\mathcal{I}$, but it is not *symmetric* with respect to any permutations of $\mathcal{I}$ besides the identity. The counting order is permutation-invariant in the same sense, but it is only symmetric with respect to permutations of $\mathcal{I}$ with finite support ($\Pi_{\text{fin}}\mathcal{I}$).

To be explicit: we call a preorder on $2^{\mathcal{I}}$ **permutation-invariant** iff it is in-

variant under the action of $\Pi(\mathcal{I})$. It is **finitely symmetric** iff it is symmetric under the action of $\Pi_{\text{fin}} \mathcal{I}$.²

Recall Szpilrajn’s extension theorem (theorem A.14): every preorder can be extended to a total preorder. As we discussed, this extension can involve infinitely many arbitrary choices. But also, there is no guarantee that these choices will preserve natural *invariances* of the original order. Typically they won’t. So passing from a preorder to a total preorder often means losing many nice properties along the way.

(Note that we do not lose *symmetries* in the narrow sense of equivalences. If $x \sim g \cdot x$, then of course we also have $x \sim^* g \cdot x$ for any preorder $\geq^*$ that extends $\geq$.)

In particular, while there exist strictly monotone permutation invariant orders on 2^ω , and these can be extended to total orders, the resulting total orders do not remain permutation invariant. This follows from the following fact.

Theorem A.55 (Goodman (2025)). Any permutation invariant total preorder on 2^ω is permutation-symmetric.

Proof. TODO □

We also know that no *strictly monotone* order on 2^ω can be permutation-symmetric.

Corollary A.56. There is no permutation-invariant strictly monotone total order on 2^ω .

This result was also proved by Askill (2018), in a somewhat different way. In fact, permutation invariant strictly monotone orders fall very far short of being total.

Theorem A.57 (Askill (2018)). The counting order is the strongest permutation-invariant strictly monotone order on 2^ω . Thus there is no

²In the social welfare setting these conditions go by a variety of names. *G*-symmetry is called *Q-anonymity* (for *Q* a group of permutations of ω). Permutation-invariance is called *relative anonymity* (Asheim and Banerjee 2010) or *isomorphism invariance* (Lauwers and Vallentyne 2004). Askill (2018)’s *qualitativeness* is closely related (see section 4.2). Again, see Pivato and Fleurbaey (2024) for overview.

permutation-invariant strictly monotone *total* order on 2^ω .

Proof. Let $\succsim$ be any permutation-invariant strictly monotone preorder of 2^ω . Let $A, B \subseteq \omega$, and suppose $A \succsim B$. We will show that $A \succsim^\# B$ as well. This uses the same idea as the proof sketch in section 4.2.

Recall that $A \succsim^\# B$ means that $B \setminus A$ is finite and $\#(A \setminus B) \geq \#(B \setminus A)$. If this fails, then either $B \setminus A$ is infinite, or else $B \setminus A$ is finite and $\#(A \setminus B) < \#(B \setminus A)$. In either case, there exists a function $f : (A \setminus B) \rightarrow (B \setminus A)$ which is one-to-one but not onto. Let $X = A \setminus B$, let Y_1 be the range of f , let Y_2 be $B \setminus A \setminus Y_1$. Let $Z = A \cap B$. So X, Y_1, Y_2, Z are pairwise disjoint subsets of ω such that

$$\begin{aligned} A &= X \cup Z \\ B &= Y_1 \cup Y_2 \cup Z. \end{aligned}$$

Moreover, the construction ensures that Y_2 is not empty.

We can extend f to a permutation of ω :

$$\pi(x) = \begin{cases} f(x) & x \in X \\ f^{-1}(x) & x \in Y_1 \\ x & \text{otherwise.} \end{cases}$$

By construction:

$$\begin{aligned} \pi(X) &= Y_1 \\ \pi(Y_1) &= X \\ \pi(Y_2) &= Y_2 \\ \pi(Z) &= Z. \end{aligned}$$

Now, by assumption, $A \succsim B$, and $\succsim$ is strictly monotone and permutation-invariant. Since Y_2 is not empty, $Y_1 \cup Z$ is a proper subset of $Y_1 \cup Y_2 \cup Z$.

Thus:

$$\begin{aligned}
 B &= Y_1 \cup Y_2 \cup Z \\
 &> Y_1 \cup Z && \text{(by strict monotonicity)} \\
 &= \pi(X \cup Z) \\
 &= \pi(A) \\
 &\succeq \pi(B) && \text{(by permutation-invariance)} \\
 &= \pi(Y_1 \cup Y_2 \cup Z) \\
 &= X \cup Y_2 \cup Z \\
 &> X \cup Z && \text{(by strict monotonicity)} \\
 &= A.
 \end{aligned}$$

Thus $B > A$, contradicting our assumption. $\square$

Corollary A.58. The counting order is the unique permutation invariant finitely symmetric strictly monotone order on 2^ω .

There is a subtlety when we generalize theorem A.57 beyond the case of the countable set ω . Recall from section A.3 the distinction between the *weak* and *strong* counting orders. The weak counting order is a permutation-invariant strictly monotone order, but it is not the *strongest* permutation-invariant strictly monotone order. That honor goes instead to the *strong* counting order.³

Corollary A.59. Let $\mathcal{I}$ be any infinite set. The strong counting order is the strongest permutation-invariant strictly monotone order on $2^{\mathcal{I}}$.

Proof. We can see this by inspecting the proof of theorem A.57. There we said that if $A \not\succeq^\# B$, then there is a function from $A \setminus B$ to $B \setminus A$ which is one-to-one but not onto. This does not generally follow for the *weak* counting order: it will fail in the case where $A \setminus B$ has greater infinite cardinality than $B \setminus A$. But it does follow for the *strong* counting order. $\square$

So in the uncountable case, we do not have a unique permutation-invariant finitely symmetric strictly monotone order. The *weakest* is the weak counting order, and the *strongest* is the strong counting order, and in general there are others which are intermediate between those two. (We could treat different

³The proof of corollary A.59 relies on the principle of cardinal comparability for subsets of $\mathcal{I}$. When $\mathcal{I}$ is uncountable, this is independent of ZF, but it can be proved using Choice.

infinite cardinalities in different ways.) The general situation with strictly monotone orders is interesting.

- For *finite* sets, the simple requirement of (finite) symmetry suffices to pin down a single strictly monotone order—the finite counting order.
- For *countably infinite* sets, this no longer suffices, but adding the requirement of permutation invariance restores uniqueness.
- For *uncountably infinite* sets, those two requirements together do not suffice for uniqueness.

It is an interesting question whether there are other natural symmetry conditions that restore uniqueness more generally.

As we discussed in section 4.2, the conflict of Permutation Invariance with Comparability and Pareto does not depend on anything close to Permutation Invariance's full strength. We discussed one stronger result there, involving reflections. Here we will generalize. First, we will consider constraints on *symmetry* principles (in the sense of equivalence) for arbitrary preorders. Next we will extend these to find constraints on *invariances* applying to *total* orders.

Recall one of our basic observations (section 3.3) about simple East–West worlds. Take the world where Indy and everyone to Indy's East is happy. Then *shift* the distribution of happiness one person to the East. Now things are just the same, except Indy is sad. So the Pareto principle is inconsistent with the principle of *shift* symmetry. This can be generalized (compare Mitra and Basu 2007).

Lemma A.60. Let $G \curvearrowright \mathcal{I}$ be a group action, and suppose there exists a strictly monotone G -symmetric order on $2^{\mathcal{I}}$. Then for each $g \in G$, the subgroup $\langle g \rangle$ with one generator has finite orbits.

Proof. Suppose that for some $i \in \mathcal{I}$, $\langle g \rangle \cdot i$ is infinite. Let

$$A = \{i, g \cdot i, g^2 \cdot i, \dots\}.$$

Then $g \cdot A \subsetneq A$, so by strict monotonicity $g \cdot A < A$; but G -symmetry implies $g \cdot A \sim A$. $\square$

As another application, we noted in example A.49 that an irrational rotation of the circle has an infinite orbit. Consequently, no strictly monotone pre-order on subsets of the circle is symmetric under all rotations. Similarly, no

strictly monotone preorder on subsets of the coin flip space is symmetric under arbitrary coin permutations (the action of $\Pi(\omega)$ on 2^ω), since again some such permutations have infinite orbits.

On the other hand, we will show in section B.8 that this finite orbit condition is not only necessary for the existence of a G -symmetric order, but also close to sufficient.

Theorem A.61. For any locally finite action $G \curvearrowright \mathcal{I}$, there exists a strictly monotone G -symmetric order on $2^{\mathcal{I}}$.

It seems plausible that local finiteness is also necessary.

Question A.62. If there exists a strictly monotone G -symmetric order on $2^{\mathcal{I}}$, does it follow that $G \curvearrowright \mathcal{I}$ is locally finite?

Now we turn to the generalization of the Reflection Invariance argument from section 4.2. The basic issue is that there the two reflections that each have *finite* orbits, but whose combination has an *infinite* orbit. (Compare Shinotsuka 1998, prop. 1.)

Lemma A.63. Let $G \curvearrowright X$ be a group action and let $\succeq$ be a G -invariant total preorder on X . Let $g \in G$, $x \in X$, and suppose the orbit $\langle g \rangle \cdot x$ is finite. Then $x \sim g \cdot x$.

Proof. We know there is some n such that $g^n \cdot x = x$. By totality, we either have $x \succeq g \cdot x$ or $g \cdot x \succeq x$. Suppose the first case. Then by repeatedly applying G -invariance, we have

$$x \succeq g \cdot x \succeq g^2 \cdot x \succeq \cdots \succeq g^n \cdot x = x.$$

Thus $x \sim g \cdot x$. Parallel reasoning applies to the case $g \cdot x \succeq x$. $\square$

Proposition A.64. Let $G \curvearrowright \mathcal{I}$ be a group action, and let $\succeq$ be a strictly monotone G -invariant total preorder on $2^{\mathcal{I}}$. For any set $F \subseteq G$, if each $g \in F$ has finite orbits, then each $g \in \langle F \rangle$ also has finite orbits.

Proof. Suppose $F \subseteq G$ is a set of group elements which each have finite orbits. Let $g \in \langle F \rangle$: that is to say, $g = g_1 g_2 \cdots g_n$ for some $g_1, g_2, \dots, g_n \in F$. Applying lemma A.63 n times, we find that $A \sim g \cdot A$ for every $A \subseteq \mathcal{I}$. That

is to say, the order $\succeq$ is symmetric under the action of $\langle F \rangle$. By lemma A.60, every element of $\langle F \rangle$ has finite orbits. $\square$

The Reflection Invariance result in section 4.2 immediately follows. The two reflections π_0 and π_1 each have order two, and thus have only finite orbits; but the combination of the two reflections, $\pi_1\pi_0$, has an infinite orbit.⁴

As we said, we can always extend a preorder to reach a total order, but this typically means losing many nice invariance properties along the way. But things are not hopeless: we can't keep *every* nice invariance, but we can at least keep *some* of them. Here is an important class of examples.⁵

Theorem A.65. Let G be any abelian group of permutations of a set X . Let $\succeq$ be any G -invariant preorder on X . Then $\succeq$ has a faithful extension which is a G -invariant total preorder.

Recall that preordered groups can be thought of in terms of *precones*: subsets of the group that are closed under multiplication and contain the identity. There is also a close relationship between precones in G and G -invariant preorders.

Lemma A.66. Let $G \curvearrowright X$ be a group action. The following are equivalent.

- (a) Every G -invariant preorder is faithfully extended by some G -invariant total preorder.
- (b) Every precone in G is faithfully extended by some total precone.

Proof. The (a) to (b) direction follows immediately by considering the action of G on itself by left multiplication.

The (b) to (a) direction is a bit more involved. We will use Zorn's lemma, in a proof analogous to that of Szpilrajn's extension theorem (theorem A.14). The key step is to show that for any G -invariant preorder $\succeq$, and any $a, b \in X$, $\succeq$ has a G -invariant faithful extension $\succeq^*$ such that either $a \succeq^* b$ or else $b \succeq^* a$.

We'll do this in three steps.

⁴Pruss (2015, Theorem 4) uses essentially the same method to prove a related result. Consider infinitely many coin flips ordered like the integers, from infinite West to infinite East. There is no *regular conditional probability measure* (see section E.7) on such coin flips which is invariant under reflections.

⁵These issues are closely related to the literature on *orderable groups* and *orderable group actions*, from which we can borrow some ideas. Pivato and Fleurbaey (2024, sec. 4)

Step 1. Choose any $a \in X$, and let

$$P := \{g \in G \mid g \cdot a \geq a\}.$$

Then P is a precone in G . By (b), there is a total precone P^* that faithfully extends P . Each precone P and P^* has a corresponding preorder on G :

$$\begin{aligned} g \geq h & \text{ iff } g^{-1} \cdot h \in P, \\ g \geq^* h & \text{ iff } g^{-1} \cdot h \in P^*, \end{aligned}$$

and $\geq^*$ is a total faithful extension of $\geq$.

Step 2. We will use the order on G to construct a preorder $\geq^*$ on X that faithfully extends $\geq$, with the property that $g \cdot a \geq^* a$ for every $g \in P^*$. For any $x, y \in X$, let

$$x \geq^* y \text{ iff } \begin{cases} x \geq y, & \text{or else} \\ x \geq g \cdot a \text{ and } h \cdot a \geq y & \text{for some } g \geq^* h \in G. \end{cases}$$

We can check that this is transitive: suppose $x \geq^* y \geq^* z$. The cases where $x \geq y$ or $y \geq z$ are easy; otherwise we have some $g \geq^* h$ and $i \geq^* j$ in G such that

$$x \geq g \cdot a, \quad h \cdot a \geq y \geq i \cdot a, \quad \text{and} \quad j \cdot a \geq z.$$

Since $h \cdot a \geq i \cdot a$, and P^* extends P , we also have $h \cdot a \geq^* i \cdot a$. Thus

$$g \geq^* h \geq^* i \geq^* j$$

and so $x \geq^* z$.

Next, we check that $\geq^*$ is G -invariant. Again, the case $x \geq y$ is easy, so let $g \in P^*$ and

$$x \geq i \cdot a \text{ and } j \cdot a \geq y \text{ for some } i \geq^* j \in G.$$

Then $g \cdot x \geq g \cdot i \cdot a$, $g \cdot j \cdot a \geq g \cdot y$, and $g \cdot i \geq^* g \cdot j$. So $g \cdot x \geq^* g \cdot y$.

Finally, we check that $\geq^*$ faithfully extends $\geq$. Let $x, y \in X$ and suppose $x \geq^* y$ and $y \geq x$; we will show $x \geq y$. Otherwise, we have

$$h \cdot a \geq y \geq x \geq g \cdot a \text{ for some } g \geq^* h \text{ in } G.$$

So $h \geq g$ in G . Since $\geq^*$ faithfully extends $\geq$, we must also have $g \geq h$ in G . Thus

$$x \geq g \cdot a \geq h \cdot a \geq y.$$

So $\geq^*$ is a faithful extension of $\geq$ such that $g \cdot a \geq^* a$ for every $g \in P^*$.

Step 3. Now let $b \in X$ be any other element, and suppose $b \not\geq^* a$. We can construct a faithful extension $\geq^{**}$ of $\geq^*$ such that $b \geq^{**} a$. Define

$$x \geq^{**} y \quad \text{iff} \quad \begin{cases} x \geq^* y, & \text{or else} \\ x \geq^* g \cdot a \text{ and } g \cdot b \geq^* y & \text{for some } g \in G. \end{cases}$$

Again, we check that this is transitive. Suppose

$$x \geq^* g \cdot a, \quad g \cdot b \geq^* y \geq^* h \cdot a, \quad \text{and} \quad h \cdot b \geq^* z \text{ for some } g, h \in G.$$

Since $\geq^*$ is total, either $h \geq^* g$ or $g \geq^* h$. In the first case, $h \cdot a \geq^* g \cdot a$, and so

$$g \cdot b \geq^* h \cdot a \geq^* g \cdot a.$$

But this contradicts our assumption that $b \not\geq^* a$. So we must instead have $g \geq^* h$. Then

$$x \geq^* g \cdot a \geq^* h \cdot a \quad \text{and} \quad h \cdot b \geq^* z,$$

which means that $x \geq^{**} z$.

We also have G -invariance for $\geq^{**}$. Let $g, h \in G$, and suppose

$$x \geq^* g \cdot a \text{ and } g \cdot b \geq^* y.$$

Then since $\geq^*$ is G -invariant,

$$h \cdot x \geq^* h \cdot g \cdot a \text{ and } h \cdot g \cdot b \geq^* h \cdot y.$$

So $h \cdot x \geq^{**} h \cdot y$ as well.

Finally we check that $\geq^{**}$ is faithful. As before, let $x, y \in X$ and suppose $x \geq^{**} y$ and $y \geq^* x$; we will show $x \geq^* y$. Otherwise, we have

$$g \cdot b \geq^* y \geq^* x \geq^* g \cdot a \text{ for some } g \in G.$$

But this implies $b \geq^* a$, contradicting our assumption. □

Thus theorem A.65 is equivalent to the following fact.

Lemma A.67. Every precone P in an abelian group G is faithfully extended by some total precone.

Proof. This is another Zorn's lemma argument.

First we will introduce a general construction. Let P be any preconce in G , and let $g \in G$ be any element. We can define a new preconce $P + g \supseteq P \cup \{g\}$ as follows:

$$P + g := \{g^n \cdot h \mid \text{for some } n \geq 0 \text{ and } h \in P\}.$$

For $g^m \cdot h$ and $g^n \cdot k$ in $P + g$, we have

$$(g^m \cdot h) \cdot (g^n \cdot k) = g^{m+n} \cdot (h \cdot k) \in P + g,$$

using the fact that G is abelian.

Furthermore, at least one of $P + g$ or $P + g^{-1}$ *faithfully* extends P . There are two cases to consider.

1. Suppose that there is some $n \geq 1$ such that g^n is in P . In this case, we show that $P + g$ faithfully extends P ; that is, $(P + g) \cap P^{-1} \subseteq P$.

Any element of $(P + g) \cap P^{-1}$ has the form $g^m \cdot h$ for some $m \geq 0$ and $h \in P$, where $(g^m \cdot h)^{-1}$ is also in P . Since $h \in P$, $g^{-m} = h \cdot (g^m \cdot h)^{-1}$ is also in P . Since g^n and g^{-m} are both in P , so is

$$(g^{-m})^{(n-1)} \cdot (g^n)^m = g^m.$$

Thus $g^m \cdot h$ is in P .

2. Otherwise, there is no $n \geq 1$ such that g^n is in P . In that case, we show $(P + g^{-1}) \cap P^{-1} \subseteq P$. Any element of $(P + g^{-1}) \cap P^{-1}$ has the form $(g^{-1})^n \cdot h = g^{-n} \cdot h$ for some $n \geq 0$ and $h \in P$. As before, since $(g^{-n} \cdot h)^{-1}$ is in P , so is $g^n = h \cdot (g^{-n} \cdot h)^{-1}$. But this implies $n = 0$. So $g^{-n} \cdot h = h$ is in P as well.

To complete the proof, note that the set $\mathcal{D}$ of all preconces that faithfully extend P satisfies the conditions of Zorn's lemma, and thus $\mathcal{D}$ has a maximal element P^+ . What we have just shown implies that P^+ is a total preconce. $\square$

Corollary A.68.

- (a) There is a translation-invariant strictly monotone finitely symmetric total preorder on subsets of $\mathbb{Z}$.
- (b) There is a rotation-invariant strictly monotone finitely symmetric total preorder on the subsets of the circle.

Proof. Apply theorem A.65 to the counting order on each space. $\square$

Question A.69. Abelianness is surely not a *necessary* condition for the existence of invariant total extensions. What is the appropriate generalization of theorem A.65?

Again, all of the results in this section can be interpreted in terms of *comparative probability orders* (section A.3). The definitions of G -symmetry and G -invariance apply to probability orders as a special case.

Corollary A.70.

- (a) The strongest permutation-invariant regular probability order on an infinite state space is the strong counting order.
- (b) Thus there is no permutation-invariant regular *total* probability order on an infinite state space.
- (c) More generally, if $G \curvearrowright \Omega$ is a group action, and there exists a regular G -symmetric probability order on the state space Ω , then for each $g \in G$, the subgroup $\langle g \rangle$ with one generator has finite orbits.

There are analogues to the positive results that apply to probability orders as well, but I will postpone this to section B.8 and appendix F.

Appendix B

Aggregation

B.1 Quantities

One central topic of part III is how we can *quantify* various things, including the values of outcomes and prospects and the probabilities of propositions. A running theme is that there are many different kinds of quantities, and we want to judiciously choose from among them whichever best suits the job at hand. These quantities include familiar number systems like the integers, rationals, or real numbers, somewhat less familiar systems like lexicographic vector spaces and hyperreals, and bespoke systems like what I have called “quasi-reals”. It will be helpful to impose some system on this proliferation of mathematical objects, and begin by considering certain abstract features that different kinds of quantity have in common. (This is another major topic, alongside groups and symmetries, within the field of *abstract algebra*.) There are many important facts we can prove about quantities at this very high level of abstraction.

But before things get terribly abstract, let’s start with a some familiar number systems.

Example B.1. The **natural numbers** $\mathbb{N}$ include the numbers $0, 1, 2, 3, \dots$. Natural numbers are totally ordered. They can also be added together, or multiplied. The number 0 plays a special role with respect to addition:

$$0 + a = a \quad \text{for each } a \in \mathbb{N}.$$

The number 1 plays a special role with respect to multiplication:

$$1 \cdot a = a \quad \text{for each } a \in \mathbb{N}.$$

Furthermore, addition and multiplication are *monotone* with respect to the ordering of natural numbers.

$$\begin{aligned} \text{if } a \geq b \quad \text{then } a + c \geq b + c & \quad \text{for each } a, b, c \in \mathbb{N} \\ \text{if } a \geq b \quad \text{then } c \cdot a \geq c \cdot b & \quad \text{for each } a, b, c \in \mathbb{N}. \end{aligned}$$

Example B.2. The **integers** $\mathbb{Z} \dots, -2, -1, 0, 1, 2, \dots$ are similar in many respects. Again, they are totally ordered, they have addition and multiplication, and there are special integers 0 and 1. The addition monotonicity property is exactly the same, but the one for multiplication needs to be slightly qualified:

$$\text{if } a \geq b \quad \text{and } c \geq 0 \quad \text{then } c \cdot a \geq c \cdot b \quad \text{for each } a, b, c \in \mathbb{Z}.$$

The integers also allow an operation that the natural numbers do not: *subtraction*. For any integer $a \in \mathbb{Z}$ there is an integer $-a$ with the property that $a + (-a) = 0$. This is called an **additive inverse**.

We can also extend the natural numbers in a different direction.

Example B.3. The **(weakly) positive rationals** $\mathbb{Q}^+$ include all of the natural numbers $0, 1, 2, \dots$, but also include positive *fractions* like $1/2$, $9/5$, or $127/2043$. (In general it will be convenient to count zero as a “positive” number, or “weakly positive” to be explicit. When we want to exclude zero, we will say “strictly positive”.) The positive rationals are also totally ordered, have addition and multiplication, and include the special numbers 0 and 1, and all of these have the same nice properties we already stated. They also introduce something new: *division*. For any positive rational number $q \neq 0$, there is a rational number q^{-1} with the property that $q \cdot q^{-1} = 1$. This called a **multiplicative inverse**.

Example B.4. The **rational numbers** $\mathbb{Q}$ combine the powers of the integers $\mathbb{Z}$ and the positive rationals $\mathbb{Q}^+$: they allow *both* subtraction *and* division.

Example B.5. The (weakly) positive real numbers $\mathbb{R}^+$ have all the properties we stated of the positive rational numbers $\mathbb{Q}^+$. In addition, they have a property called *order completeness*, but we will not bother spelling this out here.

Example B.6. The real numbers $\mathbb{R}$ have all the properties we stated for the rational numbers $\mathbb{Q}$, as well as order completeness.

These are six different number systems, each useful in different ways. They are related to each other by this pattern of embeddings:

$$\begin{array}{ccccc} \mathbb{N} & \longrightarrow & \mathbb{Q}^+ & \longrightarrow & \mathbb{R}^+ \\ \downarrow & & \downarrow & & \downarrow \\ \mathbb{Z} & \longrightarrow & \mathbb{Q} & \longrightarrow & \mathbb{R} \end{array}$$

Each of these kinds of quantities have their own distinguishing features, but they also have a great deal in common. It's useful to collect together some of the properties they share — paving the way for new applications to other distinctive kinds of quantities.

One thing these number systems have in common is that they are all *total orders*, as we discussed in section A.2. But now we will focus on their *algebraic* structure. Centrally, these quantities can all be *added*.

Definition B.7. A **commutative monoid** is a set R which is equipped with an operation of addition $+$ and an element 0 , with the following properties.

(a) *Associativity.*

$$a + (b + c) = (a + b) + c \quad \text{for each } a, b, c \in R.$$

(b) *Commutativity.*

$$a + b = b + a \quad \text{for each } a, b \in R.$$

(c) *Identity.*

$$0 + a = a.$$

A **preordered commutative monoid** is additionally equipped with a pre-order $\geq$ with this additional property:

(d) *Monotonicity*.

$$\text{if } a \geq b \text{ then } a + c \geq b + c \text{ for each } a, b, c \in R.$$

If the preorder $\geq$ is a partial order, then instead we say “partially ordered commutative monoid”, abbreviated p.o.c. monoid.

Despite its long name, a partially ordered commutative monoid is a very basic and general kind of quantity. It corresponds to the medieval notion of an *extensive quantity*, as received through modern measurement theory (see Krantz et al. 1971, ch. 3). All six of the structures $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}^+$, $\mathbb{Q}$, $\mathbb{R}^+$, and $\mathbb{R}$ are p.o.c. monoids. In general, we will call all such structures **quantities**.

In section A.5 we encountered a different kind of algebraic structure — a *group* — which was motivated by ideas about symmetry, rather than ideas about quantities. But these two ideas naturally converge. Recall that an *abelian group* is a group whose operation is commutative. If we write this operation as $+$, then we find that an abelian group is the same thing as a commutative monoid, but which also has *inverses*. This means that we can not only do addition, but also *subtraction*. The structures $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are all ordered abelian groups, while $\mathbb{N}$, $\mathbb{Q}^+$, and $\mathbb{R}^+$ are not.

Another important feature of these central examples is that they also all allow *multiplication*. This is also a useful feature to abstract on.

Definition B.8. A **commutative semiring** is a set R which is equipped with two operations of addition $+$ and multiplication $\cdot$, and elements 0 and 1, with the following properties.

(a) Addition and multiplication are each associative and commutative operations, with identities 0 and 1 respectively. This makes R a commutative monoid twice over, as $(R, +, 0)$ and $(R, \cdot, 1)$.

(b) Multiplication distributes over addition. For all $a, b, c \in R$:

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

If additionally each $a \in R$ has an additive inverse (making $(R, +, 0)$ an abelian group), then R is called a **commutative ring**.

If R is a commutative ring, and furthermore each $a \neq 0$ has a multiplicative inverse (making $(R \setminus \{0\}, \cdot, 1)$ an abelian group as well), then R is called a **field**.

A **preordered commutative semiring** is a commutative semiring R which is also equipped with a *preorder* $\geq$, with two further properties:

(c) Monotonicity for addition. For all $a, b, c \in R$,

$$\text{if } a \geq b \text{ then } a + c \geq b + c.$$

(This makes $(R, +, 0, \geq)$ a p.o.c. monoid.)

(d) Multiplication by positive quantities is also monotonic. That is, for all $a, b, c \in R$,

$$\text{if } a \geq b \text{ and } c \geq 0 \text{ then } c \cdot a \geq c \cdot b.$$

In the names of structures, “c” will abbreviate “commutative” and “p.o.” will abbreviate “partially ordered”. Accordingly, we will speak of *p.o. groups*, *p.o.c. semirings*, etc.

The structures $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}^+$, $\mathbb{Q}$, $\mathbb{R}^+$, and $\mathbb{R}$ are all *totally* ordered commutative semirings. Out of this list, only $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are commutative rings; only $\mathbb{Q}$ and $\mathbb{R}$ are fields.

There are also kinds of quantities that naturally come up which are a little different from these. For an example, consider the *lexicographic* quantities we introduced in chapter 2. Instead of representing values with numbers, we could represent them with pairs of numbers, ordered lexicographically. For example, we might represent the value of a sandwich as $(0, 2)$, and the value of eternal happiness as $(1, 0)$. We can still add these pairs together in a natural way: $(0, 2) + (1, 0) = (1, 2)$. We can also rescale them: $0.5 \cdot (1, 2) = (0.5, 1)$. These operations have natural meanings in value terms, since we can use them to calculate expectations. In contrast, while there are various ways of defining the result of multiplying $(1, 0)$ by $(0, 2)$, none of them have any obvious meaning in terms of value. These vector values have certain features in common with the number systems $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, but not all of them.

This helps motivate a general definition.

Definition B.9. Let R be a commutative semiring. An **R -semimodule** is a set $\mathcal{U}$ together with an operation of addition $+$, an additive identity element $0 \in \mathcal{U}$, and an operation of **scalar multiplication** by elements of R , with the following properties.

- (a) $\mathcal{U}$ is a commutative monoid under addition. That is, addition is associative, commutative, and has identity 0.
- (b) Scalar multiplication associates, and has identity 1. (This makes scalar multiplication an *action* of the multiplicative monoid $(R, \cdot, 1)$ on $\mathcal{U}$.) For $a, b \in \mathcal{U}$ and $q, r \in R$:

$$\begin{aligned}(q \cdot r) \cdot a &= q \cdot (r \cdot a) \\ 1 \cdot a &= a.\end{aligned}$$

- (c) Scalar multiplication distributes over addition. For $a, b \in \mathcal{U}$ and $r \in R$:

$$r \cdot (a + b) = r \cdot a + r \cdot b.$$

When R is a ring (rather than merely a semiring) we can say R -*module* rather than R -*semimodule*. When R is a field, we say R -*vector space*. Sometimes the parameter R will be left implicit.

If R is a *preordered* commutative semiring, then a **preordered R -semimodule** is an R -semimodule which is also equipped with a preorder $\geq$, with these additional properties:

- (d) Addition is monotone. For $a, b, c \in \mathcal{U}$,

$$\text{if } a \geq b \text{ then } a + c \geq b + c.$$

- (e) Multiplication by positive scalars is monotone. For $a, b \in \mathcal{U}$ and $r \in R$,

$$\text{if } r \geq 0 \text{ and } a \geq b \text{ then } r \cdot a \geq r \cdot b.$$

(If R is a ring, this makes $\mathcal{U}$ an **ordered R -module**; if R is a field, this makes $\mathcal{U}$ an **ordered R -vector space**.)

Example B.10. Every (ordered) semiring R is itself an (ordered) R -semimodule. Also, if an (ordered) semiring R is *embedded* in an (ordered) semiring R^+ (in a sense to be made precise in section B.2), then every R -semimodule is also an R^+ -semimodule. Thus:

- $\mathbb{N}$ is an ordered $\mathbb{N}$ -semimodule;

- $\mathbb{Z}$ is both a preordered $\mathbb{N}$ -semimodule and a preordered $\mathbb{Z}$ -module;
- $\mathbb{Q}^+$ is a preordered R -semimodule for each semiring $R \in \{\mathbb{N}, \mathbb{Z}, \mathbb{Q}^+\}$; and so on.

Example B.11. If U and $\mathcal{V}$ are each preordered R -semimodules, their **direct product** $U \times \mathcal{V}$ is also a preordered R -semimodule. The domain of $U \times \mathcal{V}$ contains all ordered pairs (u, v) where $u \in U$ and $v \in \mathcal{V}$. Addition, scalar multiplication, and the order are all defined componentwise:

- $(u, v) + (u', v') = (u + u', v + v')$
- $r \cdot (u, v) = (r \cdot u, r \cdot v)$
- $(u, v) \geq (u', v')$ iff $u \geq u'$ and $v \geq v'$

In section B.2 and section B.5 we will consider powerful generalizations of this idea.

Example B.12. If U and $\mathcal{V}$ are each totally ordered R -semimodules, their **lexicographic product** $U \times_{\text{lex}} \mathcal{V}$ is also a totally ordered R -semimodule. The domain of $U \times_{\text{lex}} \mathcal{V}$ is again the set of ordered pairs, with addition and scalar multiplication defined the same way as in the direct product. The order is defined lexicographically:

- $(u, v) \geq (u', v')$ iff $u > u'$ or $(u = u' \text{ and } v \geq v')$

The idea of a semimodule helpfully unifies several other algebraic concepts we have already encountered.

- An $\mathbb{N}$ -semimodule is the same thing as a commutative monoid. If $\mathcal{V}$ is a commutative monoid, then we can define scalar multiplication for $n \in \mathbb{N}$ and $a \in \mathcal{V}$ by

$$n \cdot a := \underbrace{a + a + \cdots + a}_{n \text{ times}}.$$

Likewise a *partially ordered* $\mathbb{N}$ -semimodule is just another way of describing a p.o.c. monoid — an *extensive quantity*.

- A $\mathbb{Z}$ -module is the same thing as an abelian group. Likewise a preordered $\mathbb{Z}$ -module is the same thing as a preordered abelian group.
- An $\mathbb{R}$ -module is a **(real) vector space**. A $\mathbb{Q}$ -module is a **rational vector space**.

Thus preordered R -semimodules (for various choices of R) unify many different kinds of quantities that are useful for representing values and probabilities.

We will have a lot to say about *partially ordered R -semimodules for an ordered commutative semiring R* . But that's quite a mouthful, so let's introduce a better name: call any such structure an **R -quantity**. All R -quantities are at least $\mathbb{N}$ -quantities — p.o.c. monoids — which, as before, we call simply **quantities**.

B.2 Structures

At this point we have a pretty large menagerie of different kinds of mathematical objects on our hands, such as, orders, groups, and number systems and more general quantities of various kinds. To keep things organized, it is useful to have some systematic ways of talking about these at a more abstract level. For this purpose, it is useful to introduce some concepts from model theory, and just a little bit of category theory. Some of this may be familiar from past experience with logic.

Definition B.13. A **signature** L is a pair of functions $(L_{\text{rel}}, L_{\text{fun}})$, each of which takes each natural number to some set, where all of these sets are pairwise disjoint. We call $L_{\text{rel}}(n)$ the set of **n -place relation symbols** in L , and we call L_{fun} the set of **n -place function symbols** in L . We also call 0-place function symbols **constants**.

Definition B.14. Given a signature L , an **L -structure** S consists of the following:

- A set $\text{dom } S$, the **domain** of S . (We typically also just use the same symbol S for the domain of S .)
- For each n -place relation $R \in L_{\text{rel}}(n)$, a set of n -tuples $R_S \subseteq (\text{dom } S)^n$.
- For each n -place function symbol $f \in L_{\text{fun}}(n)$, an n -place function $f_S : (\text{dom } S)^n \rightarrow \text{dom } S$.

We often leave the signature implicit.

Example B.15.

- The signature of *orders* includes a single two-place relation symbol (and no function symbols). Every preordered set is a structure with this signature (which additionally satisfies constraints like reflexivity and transitivity).
- The signature of *groups* includes a single two-place function symbol for the group operation, a constant for the identity element, and a one-place function symbol for inverses. Every group is a structure with this signature (satisfying the group axioms).
- For any p.o.c. semiring R , the signature of *R -quantities* includes a two-place relation symbol for the order, a two-place function symbol for addition, constants for 0 and 1, and many one-place function symbols for scalar multiplication: one for each scalar $r \in R$ in the corresponding semiring. Every *R -quantity* is a structure with this signature (again satisfying the appropriate axioms).
- The *empty* signature has no relation or function symbols at all. Every set is a structure with the empty signature.

Sets, orders, groups, and *R -quantities* are all examples of what we call **categories of structures**. While I won't formally define "category", I'll note that it is not just a synonym for "class": the term brings out that there are certain important *relationships* between structures of the same kind. (This notion of a category applies more broadly than model theory, but I won't pursue the generalization.)

Definition B.16. Let A and B be L -structures.

(a) An **(L -)homomorphism** $h : A \rightarrow B$ is a function from $\text{dom } A$ to $\text{dom } B$ with the following properties:

- For each n -place relation symbol R in L , and each set of n -tuples $(a_1, \dots, a_n) \in A^n$,

$$\text{if } (a_1, \dots, a_n) \in R_A \text{ then } (ha_1, \dots, ha_n) \in R_B.$$

- For each n -place function symbol f in L , and each set of n -tuples $(a_1, \dots, a_n) \in A^n$,

$$h(f_A(a_1, \dots, a_n)) = f_B(ha_1, \dots, ha_n).$$

(b) A homomorphism h is called an **embedding** iff h is a one-to-one function, and furthermore the converse of the relation condition also holds:

- For each n -place relation symbol R in L , and each set of n -tuples $(a_1, \dots, a_n) \in A^n$,

$$(a_1, \dots, a_n) \in R_A \quad \text{iff} \quad (ha_1, \dots, ha_n) \in R_B.$$

(c) A structure A is called a **substructure** of a structure B , written $A \leq B$, iff $\text{dom } A \subseteq \text{dom } B$ and the inclusion function that takes each element of A to itself is an embedding.

We will also sometimes use **morphism** as a synonym for “homomorphism”. Here are a couple important basic facts about homomorphisms.

Proposition B.17.

- (a) For any structure A , the identity function $\text{id}_A : A \rightarrow A$ is a homomorphism, and indeed an embedding.
- (b) For any structures A, B, C and any homomorphisms $g : A \rightarrow B$ and $h : B \rightarrow C$, the composition $h \circ g : A \rightarrow C$ is also a homomorphism. If g and h are each embeddings, so is $h \circ g$.

Definition B.18. A homomorphism $h : A \rightarrow B$ is called an **isomorphism** iff it has an **inverse**: some homomorphism $h^{-1} : B \rightarrow A$ such that

$$\begin{aligned} h^{-1} \circ h &= \text{id}_A \\ h \circ h^{-1} &= \text{id}_B. \end{aligned}$$

An **automorphism** of A is an isomorphism $h : A \rightarrow A$.

One important general tool is *abstraction*. Here is an example from Frege: we say two lines have the same *direction* iff they are parallel. In this way, the equivalence relation of being parallel gives rise to certain *object* which represent lines. We go from *being the same in a certain respect*, to *being represented by the same object*. Frege very ambitiously set out to make this idea the foundation for the whole of mathematics. This project requires an especially powerful version of the tool. But there is a humbler version of the same idea that has, after Frege, become a commonplace tool throughout mathematics. It is called a *quotient*.

Definition B.19. Let A be a structure. An equivalence relation $\sim$ on the domain of A is called a **congruence** iff it is compatible with the operations and relations in A , in the following sense:

- For each function symbol f (in the signature of A), for any n -tuples $(a_1, \dots, a_n)$ and $(a'_1, \dots, a'_n)$, if $a_i \sim a'_i$ for each i , then

$$f_A(a_1, \dots, a_n) \sim f_A(a'_1, \dots, a'_n).$$

- For each relation symbol R (in the signature of A), for any n -tuples $(a_1, \dots, a_n)$ and $(a'_1, \dots, a'_n)$, if $a_i \sim a'_i$ for each i , then

$$(a_1, \dots, a_n) \in R_A \quad \text{iff} \quad (a'_1, \dots, a'_n) \in R_A.$$

Proposition B.20. Let $\sim$ be a congruence on a structure A . Call a homomorphism $h : A \rightarrow B$ **compatible** with $\sim$ iff for all $a, a' \in A$,

$$\text{if } a \sim a' \text{ then } h(a) = h(a').$$

There exists a structure $A/\sim$, called the **quotient** of A by $\sim$, and a surjective homomorphism $q : A \rightarrow A/\sim$, called the **quotient map** which is compatible with $\sim$, and which has the following property:

For any homomorphism $h : A \rightarrow B$ that is compatible with $\sim$, there is a unique homomorphism $\hat{h} : A/\sim \rightarrow B$ such that $h = \hat{h} \circ q$.

$$\begin{array}{ccc} A & \xrightarrow{q} & A/\sim \\ & \searrow h & \downarrow \hat{h} \\ & & B \end{array}$$

Furthermore, the quotient structure $A/\sim$ is unique up to isomorphism.

Proof sketch. Let the domain of $A/\sim$ be the set of equivalence classes of $\sim$, and let $q : A \rightarrow A/\sim$ take each element of A to its equivalence class. The congruence conditions guarantee that we have well-defined functions and relations on $A/\sim$:

$$\begin{aligned} f_{A/\sim}(q(a_1), \dots, q(a_n)) &:= q(f_A(a_1, \dots, a_n)) \\ (q(a_1), \dots, q(a_n)) &\in R_{A/\sim} \quad \text{iff} \quad (a_1, \dots, a_n) \in R_A. \end{aligned}$$

It can be checked that these have the desired properties. $\square$

One important case is where the structure we start with has a *preorder* $\succeq$. In this case, quotienting by the corresponding equivalence relation $\sim$ gives us a *partial* order (proposition A.12). Furthermore, in the case of preordered monoids, groups, semirings, or semimodules, this equivalence relation is compatible with the structure, and the resulting quotient structures are partially ordered monoids, groups, semirings, or semimodules, respectively.

Sometimes it is useful to take a *logical* perspective on structures. We can use the basic symbols of a signature to build up a whole formal language, and then use that language to express various properties of structures. For instance, using the signature of monoids $(0, +)$, we can write a formal sentence that expresses the associative property:

$$\forall x \forall y \forall z ((x + y) + z = x + (y + z)).$$

Or using the signature of orders $\geq$, we can formalize the transitive property:

$$\forall x \forall y \forall z (x \geq y \rightarrow y \geq z \rightarrow x \geq z).$$

These are each formal sentences in the **language of first-order logic** build from the appropriate signature. Each of these formal sentences can be *interpreted* in an arbitrary structure.

We can define some important notions. Since these definitions are broadly familiar to philosophers, I won't go through the official details of any of these definitions (See Hodges 1997 for details.)

1. Given a signature L , we can define the notion of a (closed) **L -term** and a **first-order L -sentence**.
2. For any L -structure A and L -term α , we can define the **denotation** of α in A , which we will write $A(\alpha)$.
3. We can define a relation between L -structures and first-order L -sentences: **ϕ is true in A** , written $A \models \phi$.
4. A **(first-order) L -theory** is any set of first-order L -sentences.
5. A structure A is a **model** of a theory T iff every sentence in T is true in A .

Almost all of the classes of structures we have met so far can be redescribed as models of certain first-order theories.

Example B.21.

- A **commutative monoid** is a structure with signature $(0, +)$ that satisfies the following theory:

$$\begin{aligned} \forall x \forall y \forall z ((x + y) + z &= x + (y + z)) \\ \forall x \forall y (x + y &= y + x) \\ \forall x (0 + x &= x). \end{aligned}$$

Call this the **theory of commutative monoids**.

- A **preordered set** is a structure with signature $(\geq)$ that satisfies the following theory:

$$\begin{aligned} \forall x \forall y \forall z ((x \geq y \wedge y \geq z) &\rightarrow x \geq z) \\ \forall x x &\geq x. \end{aligned}$$

Call this the **theory of preorders**.

Notice that if the signature L^+ *expands* the signature L by adding more symbols, then every L -sentence is also an L^+ -sentence. So we can say, for example:

- A **preordered commutative monoid** is a structure with signature $(0, +, \geq)$ that satisfies the theory of commutative monoids *and* the theory of preorders, as well as this further sentence:

$$\forall x \forall y \forall z (x \geq y \rightarrow x + z \geq y + z).$$

We can similarly give first-order characterizations of partial orders, total orders, groups, abelian groups, (semi-)rings, rings, and (semi-)modules, as well as their ordered variants.

Not every important property of structures has a first-order characterization. One important counterexample is the **Archimedean property**.

A quantity $\mathcal{U}$ (that is, a p.o.c. monoid) is **Archimedean** iff for every $u, v \in \mathcal{U}$ such that $u, v > 0$, there is some $n \in \mathbb{N}$ such that $n \cdot u \geq v$.

For example, the natural numbers $\mathbb{N}$ themselves are an Archimedean quantity. But (as we will discuss in section B.5) there are non-Archimedean quantities $\mathbb{N}^*$ that satisfy *exactly the same* first-order sentences as $\mathbb{N}$.

The three examples of theories we gave above all have a particularly nice, straightforward form.

Definition B.22.

- (a) A **Horn clause** is one of the form

$$\forall x_1 \dots \forall x_k ((\phi_1 \wedge \dots \wedge \phi_n) \rightarrow \psi)$$

or else just

$$\forall x_1 \dots \forall x_k \psi$$

where $\phi_1, \dots, \phi_n$ are all atomic formulas, and ψ is either an atomic formula or else $\perp$. (In the latter case, sentence can equivalently be written as $\forall x_1 \dots \forall x_k \neg(\phi_1 \wedge \dots \wedge \phi_n)$.)

- (b) A **Horn theory** is a theory which is logically equivalent to a set of Horn clauses.

Horn clauses are a very common pattern for “nice” general properties of a structure. In fact, nearly all of the categories of structures we have surveyed are characterized by Horn theories: for example, preorders, partial orders, commutative monoids, (semi-)rings, (semi-)modules, or R -quantities (p.o. R -semimodules) for any particular preordered semiring R .

There are a few exceptional cases. One is the comparability condition on *total* orders:

$$\forall x \forall y (x \geq y \vee y \geq x).$$

Another is the division condition on *fields*:

$$\forall x (x \neq 0 \rightarrow \exists y (x \cdot y = 1)).$$

Neither of these is a Horn clause.

Definition B.23. Let A_i be a structure for each $i \in I$, where I is any set of indices. The **direct product** $\prod_{i \in I} A_i$ is the structure whose domain is the Cartesian product $\prod_i \text{dom } A_i$, and whose functions and relations are defined pointwise.

The **direct power** A^I is the direct product of the constant family $A_i = A$ for all $i \in I$.

Note that each of the projection functions $\pi_i : \prod_i A_i \rightarrow A_i$ from the direct product to any of its factors is a homomorphism. Likewise, the diagonal

function $A \rightarrow A^I$ that takes each element of A to its corresponding constant function is a homomorphism as well, and indeed an embedding (as long as I is nonempty).

Theorem B.24. Let T be a Horn theory.

- (a) If $(A_i)_{i \in I}$ is a family of models of T , the direct product $\prod_i A_i$ is also a model of T .
- (b) The direct power A^I satisfies precisely the same Horn theory as A itself.

Proof. See Hodges (1993, sec. 8.5). (Alternatively, this can be derived a more general fact about *filter products*; see section B.5) $\square$

Thus the direct power of p.o. abelian groups is another p.o. abelian group, the direct power of vector spaces is another vector space, and so on. In contrast, this does not hold for total orders or fields.

B.3 Embeddings and representations

In section B.1 we introduced a menagerie of quantitative structures, with different amounts of structure. How are these related to one another? In fact, in almost every case, the quantities with less structure can be represented by quantities with more structure, in some “most general” way. In some cases we have to impose some extra constraints: for example, a commutative monoid won’t necessarily be compatible with any reasonable notion of subtraction, so to get from a monoid to a group we have to add an extra constraint.

We saw that a great variety of important structures can be thought of as *partially ordered semimodules* corresponding to various number systems, $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{Q}^+, \mathbb{R}^+, \mathbb{R}$. These are related to one another by embeddings:

$$\begin{array}{ccccc} \mathbb{N} & \longrightarrow & \mathbb{Q}^+ & \longrightarrow & \mathbb{R}^+ \\ \downarrow & & \downarrow & & \downarrow \\ \mathbb{Z} & \longrightarrow & \mathbb{Q} & \longrightarrow & \mathbb{R} \end{array}$$

Recall that if R is any number system, we have a corresponding kind of quantity, **R -quantities**. (As in section B.1, a *number system* R is an ordered commutative semiring, and an *R -quantity* is a partially ordered R -semimodule.) As

it turns out, each of these embeddings of one number system in another has a corresponding *representation theorem*, which says under what conditions we can step up from one corresponding kind of *quantity* to the next.

In this section I'll state (mainly without proof) a collection of important representation theorems for different algebraic structures. These will give rise to a corresponding family of representation theorems for social welfare, in section B.4.

First, $\mathbb{N}$ embeds in $\mathbb{Z}$. There is a corresponding embedding theorem that takes $\mathbb{N}$ -quantities (p.o.c. monoids) to $\mathbb{Z}$ -quantities (p.o. abelian groups).¹

Definition B.25. A $\mathbb{N}$ -quantity (p.o.c. monoid) $\mathcal{U}$ is **cancellative** iff for any $x, y, z \in \mathcal{U}$, if $x + z \geq y + z$ then $x \geq y$.

Theorem B.26. An $\mathbb{N}$ -quantity (p.o.c. monoid) can be embedded in a $\mathbb{Z}$ -quantity (p.o. abelian group) iff it is cancellative.

(See Goodearl 1986.)

Next, as $\mathbb{Z}$ embeds in $\mathbb{R}$, there is a corresponding embedding theorem that takes $\mathbb{Z}$ -quantities to $\mathbb{Q}$ -quantities or $\mathbb{R}$ -quantities — p.o. vector spaces. (We have skipped over $\mathbb{Q}$ -quantities — rational vector spaces — in the chain of embeddings. This is because every rational vector space is also a real vector space, and it turns out that the conditions for embeddability in a real vector space are no more demanding than the conditions for rational vector spaces.)

Definition B.27. A $\mathbb{Z}$ -quantity (p.o. abelian group) G is **unperforated** iff for any $x \in G$ and any $n \in \mathbb{N}$, if $n \cdot x \geq 0$ then $x \geq 0$ as well.

(The intuition behind the term is that if $n \cdot x$ is in the group's positive cone but x is not, then x would be a “hole” in the cone.)

Theorem B.28. A $\mathbb{Z}$ -quantity (p.o. abelian group) can be embedded in an $\mathbb{R}$ -quantity (p.o. vector space) iff it is unperforated.

Note that any *totally* ordered abelian group is automatically unperforated, and thus embeds in a vector space. Furthermore, we can ensure that the vector space is also totally ordered.

¹The resulting $\mathbb{Z}$ -quantity is standardly called the *Grothendieck group*.

Theorem B.29. Any totally ordered abelian group can be embedded in a totally ordered vector space.

Moreover, we can represent totally ordered vector spaces concretely by *sequences of real numbers*, ordered lexicographically. Recall (chapter 2) that a natural way of coming up with quantitative representations for infinite goods uses ordered pairs of numbers, ordered lexicographically: so, for instance,

$$(2, 1) > (2, 0) > (0, 1) > (0, 0).$$

This can be generalized to sequences of real numbers of any length — including *infinitely* long sequences, though this requires a bit of care. First, clearly the components of the sequence must come in some order, so we can lexicographically prioritize “earlier” components over “later” components. But also, it is important for the definition of the lexicographic order that there is always an *earliest* place where two sequences disagree, if they ever do.

Recall that a set is *well-ordered* iff every nonempty subset has a least element. Note that the union of two well-ordered subsets of a totally ordered set is also well-ordered.

Definition B.30. Let I be a totally ordered set. The **lexicographic vector space** on I , $\mathbb{R}_{\text{lex}}^I$, is the vector space of all functions from I to $\mathbb{R}$ with *well-ordered support*, ordered lexicographically. That is, for any $f, g : I \rightarrow \mathbb{R}$ with well-ordered support, $f \geq g$ iff either $f = g$, or else $f(i) > g(i)$ for the least $i \in I$ such that $f(i) \neq g(i)$.

Theorem B.31 (Hausner and Wendel (1952)). Any totally ordered vector space can be embedded in a lexicographic vector space.

We can derive another classic theorem as a corollary.

Theorem B.32 (Hahn (1907)). Any totally ordered abelian group can be embedded in a lexicographic vector space. (See Glass 1999.)

We can also represent *partial* orders using *total* orders.

Theorem B.33. Any partially ordered vector space can be embedded in a direct product of totally ordered vector spaces.

Proof sketch. Let $\mathcal{U}$ be a partially ordered vector space. Let I be the set of all faithful extensions of $\mathcal{U}$'s partial order to a total order. The embedding $h : \mathcal{U} \rightarrow \mathcal{U}^I$ is the “diagonal” function that takes each $x \in \mathcal{U}$ to the constant function x_I . This is clearly a homomorphism. To check that it is order-reflecting, and thus an embedding, suppose that $x \not\geq 0$ in $\mathcal{U}$. It can be proved in essentially the same way as lemma A.67 that if $x \in \mathcal{U}$ and $x \not\geq 0$, then the partial order on $\mathcal{U}$ has a faithful extension $\geq^*$ such that $x <^* 0$. It follows that $h(x) \not\geq 0$ in the product order on $\mathcal{U}^I$. $\square$

We can combine the dominance and lexicographic constructions. This amounts to an order on (perhaps infinite) *matrices*. We can order two matrices by first comparing them *lexicographically* row by row, and then combining these results by *dominance*. For example, consider the two matrices

$$A = \begin{pmatrix} 2 & 1 & 3 & 0 \\ 1 & 4 & 2 & 1 \\ 0 & 3 & 3 & -5 \end{pmatrix} \quad B = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 1 & 4 & 2 & 0 \\ 0 & 3 & 2 & 1 \end{pmatrix}.$$

Comparing row by row, we find that each row of A is lexicographically superior to the corresponding row of B (treating entries furthest to the left as lexicographically most important). So by applying dominance to these lexicographic orders, we find that $A > B$.

This turns out to be a completely general way of representing partially ordered vector spaces. Thus we will also find that it is a very general way of representing *value*.

We can think of a matrix with I labeling its rows and J labeling its columns as a function $I \rightarrow (J \rightarrow \mathbb{R})$, which assigns each row a function that in turn assigns each column a value. So a matrix can be thought of as an element of $(\mathbb{R}^J)^I$.

Lemma B.34. Any family of lexicographic vector spaces can be embedded in a single lexicographic vector space.

Proof. Let $\mathbb{R}_{\text{lex}}^{J_i}$ be a lexicographic vector space for each $i \in I$. Let $J = \bigsqcup_i J_i$ be the disjoint union of the J_i 's. To define an order on J , first, by applying Szpilrajn's extension theorem (theorem A.14) to the discrete order on I , we can assume that I is totally ordered. Then, we can order J lexicographically: for any $i, i' \in I$, $j \in J_i$, and $j' \in J_{i'}$, let

$$j \geq j' \quad \text{iff} \quad \text{either } i > i', \text{ or else } i = i' \text{ and } j \geq j' \text{ in } J_i.$$

This makes I a totally ordered set that naturally embeds each ordered set J_i . These order embeddings also induce an embedding of $\mathbb{R}_{\text{lex}}^{J_i}$ in $\mathbb{R}_{\text{lex}}^{J_i}$ for each $i \in I$. $\square$

Corollary B.35. Any p.o. vector space can be embedded in a direct power of a lexicographic vector space, $(\mathbb{R}_{\text{lex}}^J)^I$ for some set I and some totally ordered set J .

Proof. Let $\mathcal{U}$ be a p.o. vector space. By theorem B.33, $\mathcal{U}$ has an embedding $f : \mathcal{U} \rightarrow \prod_i \mathcal{U}_i$ into a direct product of totally ordered vector spaces. By theorem B.32, each $\mathcal{U}_i$ has an embedding $g_i : \mathcal{U}_i \rightarrow \mathbb{R}_{\text{lex}}^{J_i}$ in a lexicographic vector space. By lemma B.34, these lexicographic vector spaces can be glued together in a single lexicographic vector space $\mathbb{R}_{\text{lex}}^J$ for a fixed totally ordered set J . Then we can compose these embeddings to produce an embedding $h : \mathcal{U} \rightarrow (\mathbb{R}_{\text{lex}}^J)^I$ into the direct power:

$$h(x)(i) := g_i(f(x)) \quad \text{for each } x \in \mathcal{U}, i \in I. \quad \square$$

The same follows for any unperforated p.o. abelian group, since this can always be embedded in a p.o. vector space. This gives us a very general, but quite concrete, representation of a large and natural class of ordered algebraic systems. Finally, let's consider very specific structures: those which can be represented specifically in the *real numbers*.

Definition B.36. A p.o. abelian group G is **Archimedean** iff for any $x, y \in G$ such that $x, y > 0$, there is some $n \in \mathbb{N}$ such that $n \cdot x \geq y$.

Intuitively, this says that no element of G is *infinitely greater* than any other element, or in other words, there are no relations of *lexicographic superiority* in G . A general group can be represented by a direct power of a lexicographic vector space $(\mathbb{R}_{\text{lex}}^J)^I$. Intuitively, if the order on the group is *total*, then we can collapse I down to a single dimension. If the order is *Archimedean*, then we can also collapse J down to a single dimension. Thus we get the following result.

Theorem B.37 (Hölder (1901)). Any Archimedean totally ordered abelian group can be embedded in the real numbers $\mathbb{R}$. (See Glass 1999.)

Every kind of quantity that is at least an $\mathbb{N}$ -quantity. In general, this just requires well-behaved notions of addition and a partial order. So we can say precisely when a general quantity can be represented by real numbers.

Corollary B.38. An $\mathbb{N}$ -quantity (p.o.c. monoid) can be embedded in the real numbers $\mathbb{R}$ iff it is totally ordered, cancellative, and Archimedean.

(The only thing that needs checking is that the total order and Archimedean condition are both preserved by the embedding of a monoid in a group.)

B.4 Finitary social aggregation

In this section we will state a few results characterizing *additive* aggregation for *unbounded finite* populations. This will be useful groundwork for the infinite case in section B.7. Here, “additivism” is a broad family of social welfare theories that includes utilitarianism, leximin, some forms of prioritarianism, and others, as we will discuss below.

Recall the framework from section A.4. We assume a set $\mathcal{X}$ of individual outcomes and an infinite set $\mathcal{I}$ of individuals. A social outcome is any function from $\mathcal{I}$ to $\mathcal{X}$, and a social order is a preorder of social outcomes.

Before we come to the task of characterizing a social order for these *infinite* social outcomes, we will begin by characterizing social orders for *finite parts* of these infinite social outcomes.

Definition B.39. A **finitary social welfare structure** consists of a non-empty set $\mathcal{X}$ of individual outcomes, an infinite set $\mathcal{I}$ of individuals, and, for each finite set of individuals $K \in \text{Fin } \mathcal{I}$, a preorder $\succeq_K$ on $\mathcal{X}^K$.

We also call this a **(finitary) social order**.

We can intuitively understand the relation $x \succeq_K y$ as saying that x is as good as y for the individuals in K , on balance. This leaves open whether there might be other individuals in the world besides those in K , or for that matter whether there are other matters of value besides individual well-being.

We will often omit the subscript on $\succeq_K$, since it can usually be inferred from context.

The classic example of a theory of social welfare is utilitarianism, which we can understand as the idea that social welfare is the *sum* of individ-

ual well-beings. This idea presupposes that well-being can be represented quantitatively, so that it makes any sense to talk about adding it up. (For discussion, see Nebel 2023.) We do not need to assume that these quantities are real numbers. Rather, what we assume is that well-being can be represented by *extensive quantities* in the sense discussed in section B.1. These will at least be elements of a *partially ordered commutative monoid*, what we have called an $\mathbb{N}$ -*quantity*. Recall, this gives us enough structure to make sense of quantities being greater or less than one another, and also a well-defined notion of a finite sum. This is close to the minimum amount of structure required for something like utilitarianism to make sense. Different conditions on the relationship between individual well-being and social orders will lead to different algebraic structure for this quantity — for example, whether it is a group, or a vector space, or the real numbers.

Definition B.40. Let $(\mathcal{I}, \mathcal{X}, \succeq_{(-)})$ be a finitary social welfare structure, and let $\mathcal{U}$ be a quantity. A function $v : \mathcal{X} \rightarrow \mathcal{U}$ is a **(weak) additive representation** iff for any $I \in \text{Fin } \mathcal{I}$ and any $x, y \in \mathcal{X}^I$,

$$\text{if } x \succeq y \text{ then } \sum_i v(x(i)) \geq \sum_i v(y(i)).$$

An additive representation is **strong** iff the converse also holds.

The simplest and most familiar paradigm of an additive theory of social welfare is utilitarianism. But the general notion of an additive representation is quite a bit more permissive than you might suspect, because the operative notion of *quantity* is very general (section B.1). For example, one example of a quantity is the set of non-negative real numbers with their usual order and their usual zero, but with the “addition” operation $x \oplus y$ defined to be *maximum* of x and y . (It is worth checking that this indeed satisfies all the conditions on a quantity.) So the “maximax” theory of social welfare is additively representable in this permissive sense.² For another example, take any preordered set of welfare levels $\mathcal{X}$, and consider the lexicographic vector space $\mathbb{R}_{\text{lex}}^{\mathcal{X}}$. Then the representation $v(x) = -\delta(x)$ corresponds to the *leximin* order on social outcomes. (For further examples, see McCarthy, Mikkola, and Thomas 2020; for philosophical discussion see San 2026.)

Also, note that the kind of representation in question here is only concerned with *same population* comparisons. So this kind of additive representation

²We can also get “maximin” by considering the *dual* order on this commutative monoid, so larger numbers represent *worse* outcomes, while holding fixed the operation $\oplus$.

does not rule out, for instance, the idea that populations of different sizes should be compared by taking *average* well-being rather than *total* well-being. The most natural interpretation of the formalism is silent on this question.

Each kind of quantity has a corresponding kind of additive representation, as well as its own representation theorem, characterizing which social welfare structures can be additively represented using that kind of quantity. For the sake of illustration, I will work through the case of $\mathbb{Z}$ -quantities (p.o. abelian groups). I will then state without a proof a variety of representation theorems for the gamut of other kinds of quantities.³

For any individual outcome x and set $K \in \text{Fin } \mathcal{I}$, let x_K be the constant social outcome that assigns x to each individual in K .

Definition B.41. Let $\mathcal{S} = (\mathcal{X}, \mathcal{I}, \succeq_{(-)})$ be a finitary social welfare structure.

- (a) $\mathcal{S}$ is **finitely anonymous** iff, for any social outcome $x \in \mathcal{X}^I$ for a certain set of individuals $I \in \text{Fin } \mathcal{I}$, any further individuals $i, j \in \mathcal{I} \setminus I$ such that $i \neq j$, and any individual outcomes $y, z \in \mathcal{X}$,

$$x \sqcup y_{\{i\}} \sqcup z_{\{j\}} \sim x \sqcup z_{\{i\}} \sqcup y_{\{j\}}.$$

Equivalently: for any finite set of individuals $I \in \text{Fin } \mathcal{I}$, any social outcome $x \in \mathcal{X}^I$, and any permutation $\pi : I \rightarrow I$,

$$x \sim \pi x.$$

- (b) $\mathcal{S}$ is **permutation-invariant** iff, for any social outcomes $x, y \in \mathcal{X}^I$ and any bijection $\pi : I \rightarrow J$, if $x \succeq_I y$ then $\pi \cdot x \succeq_J \pi \cdot y$ as well.
- (c) $\mathcal{S}$ is **monotone** iff for any disjoint $I, J \in \text{Fin } \mathcal{I}$, any social outcomes $x, x^+ \in \mathcal{X}^I$, and any social outcome $y \in \mathcal{X}^J$,

$$\text{if } x^+ \succeq x \text{ then } x^+ \sqcup y \succeq x \sqcup y.$$

Note that if $\mathcal{S}$ is anonymous, we can define a preorder on *individual* outcomes in terms of the order on singleton social outcomes:

$$x \succeq y \quad \text{iff} \quad x_{\{i\}} \succeq y_{\{i\}} \text{ for any } i \in \mathcal{I}.$$

³The representation theorem using abelian groups is a variant on Pivato (2014)'s theorem 1, which applies only to total social orders.

Without further assumptions, anonymity and permutation invariance are independent of one another.⁴

Before we prove the main representation theorem, let's start with some general observations. It can be useful to take a kind of census of a social outcome, counting up how many individuals get each individual outcome. The resulting summary statistic is a function from individual outcomes to numbers, which we will call the outcome's *distribution*. Finite anonymity amounts to the claim that the value of a social outcome is determined by its distribution.

Definition B.42. Let $x \in \mathcal{X}^I$ be a social outcome. The **social distribution** of x is the function in $\text{Fin}(\mathcal{X}, \mathbb{N})$ which takes each individual outcome y to the number of individuals whom x assigns outcome y :

$$D(x)(y) := \#\{i \in I : x(i) = y\}.$$

Equivalently,

$$D(x) = \sum_{i \in I} \delta x(i)$$

where $\delta x : \mathcal{X} \rightarrow \mathbb{N}$ is the indicator function that takes x to 1 and all other outcomes to 0. (Thus δ is a function $\mathcal{X} \rightarrow \text{Fin}(\mathcal{X}, \mathbb{N})$.)

The following can be straightforwardly checked.

Lemma B.43. Let $\mathcal{S}$ be a finitary social welfare structure.

- (a) For any social outcomes $x \in \mathcal{X}^I$ and $y \in \mathcal{X}^J$, $D(x) = D(y)$ iff there is a bijection $\pi : I \rightarrow J$ such that $x = \pi \cdot y$.
- (b) For any disjoint $I, J \in \text{Fin } \mathcal{I}$ and any social outcomes $x \in \mathcal{X}^I$ and $y \in \mathcal{X}^J$,

$$D(x \sqcup y) = D(x) + D(y).$$

- (c) For any $f : \mathcal{X} \rightarrow \text{Fin}(\mathcal{X}, \mathbb{N})$, and any finite set $I \in \text{Fin } \mathcal{I}$ such that

$$\#I = \sum f,$$

there is some social outcome $x \in \mathcal{X}^I$ such that $D(x) = f$.

⁴TODO. Explain why, with weird model. Some individuals are “pessimists”. For populations I with no pessimists, we compare by maximizing. If I includes any pessimists, we compare by minimizing.

Lemma B.44. If a social order is finitely anonymous and permutation invariant, then for any $I, J \in \text{Fin } \mathcal{I}$ and any social outcomes $x, y \in \mathcal{X}^I$ and $x', y' \in \mathcal{X}^J$, if

$$D(x) = D(x'), \quad D(y) = D(y'), \quad \text{and} \quad x \succeq y,$$

then $x' \succeq y'$.

Proof. By lemma B.43(a), there is a bijection $\pi : I \rightarrow J$ such that $\pi \cdot x = x'$. Since $D(\pi \cdot y) = D(y) = D(y')$, there is another bijection $\rho : J \rightarrow J$ such that $\rho \cdot \pi \cdot y = y'$. Thus, if $x \succeq y$, then by permutation invariance and anonymity,

$$x' = \pi \cdot x \succeq \pi \cdot y \sim \rho \pi \cdot y = y'. \quad \square$$

Theorem B.45. A finitary social welfare structure has a strong additive representation iff it is finitely anonymous, permutation-invariant, and monotone.

Proof. The left-to-right direction is straightforward. For the right-to-left direction, let $\mathcal{S} = (\mathcal{X}, \mathcal{I}, \succeq_{(-)})$ be a social welfare structure which is anonymous, permutation-invariant, and monotone; we will directly construct an additive representation.

We start with the set $\text{Fin}(\mathcal{X}, \mathbb{N})$ of all *social distributions*. This is a c. monoid (by pointwise addition). We will define a preorder on this monoid, which will then let us construct a *partially ordered* monoid by taking a quotient. That quotient monoid will be the quantity that represents social welfare.

Using lemma B.44, the following is a well-defined relation on $\text{Fin}(\mathcal{X}, \mathbb{N})$:

$$D(x) \succeq D(y) \quad \text{iff} \quad x \succeq y \quad \text{for any } x, y \in \mathcal{X}^I, I \in \text{Fin } \mathcal{I}.$$

We just need to check that this is a preorder compatible with addition of distributions.

Reflexivity follows from lemma B.43(c): every distribution arises from some social outcome, which is as good as itself.

For transitivity, suppose $f \succeq g$ and $g \succeq h$ in $\text{Fin}(\mathcal{X}, \mathbb{N})$. This means that there

are some I, J, x, y, y', z such that

$$\begin{aligned} f &= D(x) \\ g &= D(y) = D(y') \\ h &= D(z) \\ x &\succeq_I y \\ y' &\succeq_J z. \end{aligned}$$

Then there exists some bijection $\pi : J \rightarrow I$ such that $\pi \cdot y' = y$. So $D(\pi \cdot z) = D(z) = h$, and thus by lemma B.44, $y \succeq \pi \cdot z$. So, since $\succeq_I$ is transitive, $x \succeq_I \pi \cdot z$, and thus $f \succeq h$.

For compatibility with addition, let $f, g, h \in \text{Fin}(\mathcal{X}, \mathbb{N})$, and suppose $f \succeq g$; we need to show that $f + h \succeq g + h$. This means that there are some $x \succeq y \in \mathcal{X}^I$ such that $D(x) = f$, $D(y) = g$. By lemma B.43(c), there is some $z \in \mathcal{X}^J$ such that $D(z) = h$, where we can choose J to be disjoint from I . Thus by lemma B.43(b),

$$D(x \sqcup z) = f + h \quad \text{and} \quad D(y \sqcup z) = g + h.$$

And by monotonicity, $x \sqcup z \succeq y \sqcup z$. So $f + h \succeq g + h$.

Using this preorder, we can then construct a *quotient* partially ordered c. monoid, with its quotient map

$$q : \text{Fin}(\mathcal{X}, \mathbb{N}) \rightarrow \text{Fin}(\mathcal{X}, \mathbb{N})/\sim.$$

Here q is a homomorphism of preordered c. monoids, and it is also order-reflecting. Let

$$v = q \circ \delta : \mathcal{X} \rightarrow \text{Fin}(\mathcal{X}, \mathbb{N})/\sim.$$

We will prove that v is a strong additive representation of $\mathcal{S}$.

For any social outcome $x \in \mathcal{X}^I$, let

$$V(x) := \sum_{i \in I} v(x(i)) = \sum_{i \in I} q(\delta x(i)) = q\left(\sum_{i \in I} \delta x(i)\right) = q(D(x)).$$

What we need to show is just that $\mathcal{U} : \mathcal{X}^I \rightarrow \text{Fin}(\mathcal{X}, \mathbb{N})/\sim$ is both order-preserving and order-reflecting. This is clear by construction:

$$x \succeq y \quad \text{iff} \quad D(x) \succeq D(y) \quad \text{iff} \quad V(x) = q(D(x)) \geq q(D(y)) = V(y). \quad \square$$

Call the additive representation in this proof a social order's **initial additive representation**.⁵

This characterizes the most general kind of additive representability, where the *quantity* we are adding up is merely required to have a well-behaved notion of addition and order. In section B.1 we surveyed a gamut of different quantities; each of these has its own corresponding notion of an additive representation. In the remainder of this section I will state a few of these representation theorems, which follow pretty directly from the embedding theorems surveyed in section B.3.

Here is one property that has shown up frequently in this book.

Definition B.46. A social order is **separable** iff for any disjoint $I, J \in \text{Fin } \mathcal{I}$, any social outcomes $x, x^+ \in \mathcal{X}^I$, and any social outcome $y \in \mathcal{X}^J$,

$$x^+ \succeq x \quad \text{iff} \quad x^+ \sqcup y \succeq x \sqcup y.$$

Note that the monotonicity is one direction of separability. Separability also implies **strong monotonicity**, in the sense that if $x^+ \succ x$ then $x^+ \sqcup y \succ x \sqcup y$. For *total* social orders, separability and strong monotonicity are equivalent. In general, though, even strong monotonicity is a weaker condition than separability.

Separability simplifies the situation with permutation properties.

Lemma B.47. If a social order is finitely anonymous and separable, then it is also permutation invariant.

Proof. Let z be an arbitrary individual outcome. Recall that for any $K \in \text{Fin } \mathcal{I}$, z_K is the constant social outcome on K .

Let $x, y, I, \pi : I \rightarrow J$ be as in the statement of the lemma, where $J = \pi \cdot I$. There is a bijection $\rho : (J \setminus I) \rightarrow (I \setminus J)$, and $\pi \sqcup \rho$ is a permutation of $I \cup J$.

⁵This is because it is indeed an “initial object” in the sense from category theory. If $v : \mathcal{X} \rightarrow \mathcal{U}$ is the initial additive representation for $\mathcal{S}$, then for any additive representation $v' : \mathcal{X} \rightarrow \mathcal{U}'$, there is a unique quantity homomorphism $h : \mathcal{U} \rightarrow \mathcal{U}'$ such that $v' \circ h = v$.

Then

$$\begin{aligned}
 \pi \cdot x \sqcup z_{I \setminus J} &= \pi \cdot x \sqcup \rho \cdot z_{J \setminus I} \\
 &= (\pi \sqcup \rho) \cdot (x \sqcup z_{J \setminus I}) \\
 &\sim x \sqcup z_{J \setminus I} && \text{by anonymity} \\
 &\succeq y \sqcup z_{J \setminus I} && \text{by separability, since } x \succeq y \\
 &\sim \pi \cdot y \sqcup \rho \cdot z_{J \setminus I} && \text{similarly.}
 \end{aligned}$$

So by separability again, $\pi \cdot x \succeq \pi \cdot y$. $\square$

Theorem B.48. A finitary social welfare structure has a strong additive representation in a $\mathbb{Z}$ -quantity (a p.o. abelian group) iff it is finitely anonymous and separable.

This follows from theorem B.26. The key thing that needs to be checked is that if a social order is separable, then its initial additive representation is *cancellative*.

Next let's consider additive representations in vector spaces. Recall from theorem B.28 that a $\mathbb{Z}$ -quantity can be embedded in an $\mathbb{R}$ -quantity iff it is *unperforated*: if $n \cdot x$ is positive, then so is x . The corresponding condition on social welfare structures says that if many copies of one distribution x is as good as the same number of copies of another distribution y , then x is as good as y .

Definition B.49. A finitary social welfare structure is **unperforated** iff for any $x, y \in \mathcal{X}^I$ and any bijections $\pi_1 : I \rightarrow J_1, \dots, \pi_n : I \rightarrow J_n$, where $J_1, \dots, J_n$ are pairwise disjoint (and $n > 0$),

$$\text{if } \pi_1 x \sqcup \dots \sqcup \pi_n x \succeq_{J_1 \cup \dots \cup J_n} \pi_1 y \sqcup \dots \sqcup \pi_n y \text{ then } x \succeq_I y.$$

The converse of this principle — if x is as good as y then many duplicates of x are as good as many duplicates of y — follows from monotonicity and permutation invariance. But the forward direction is independent of anonymity and separability.

It can be checked that if a social welfare structure is anonymous, separable, and unperforated, then its group representation from theorem B.26 is also unperforated (in the sense appropriate for groups).

Corollary B.50. A finitary social welfare structure $\mathcal{S}$ is additively representable in a partially ordered vector space iff it is finitely anonymous, separable, and unperforated.

Corollary B.35 provided a general representation for partially ordered vector spaces, using the direct product of lexicographic vector spaces. This also applies to social orders.

Corollary B.51. Any finitely anonymous, separable, unperforated finitary social welfare structure can be additively represented in a direct product of lexicographic vector spaces $((\mathbb{R}_{\text{lex}}^J)^I$, for some set I and some totally ordered set J).

Now consider *total* social orders.

Definition B.52. A social order is **total** iff $\succsim_K$ is a total preorder for each $K \in \text{Fin } \mathcal{I}$.

In this case, we can drop the “unperforated” condition.

Proposition B.53. Any separable, finitely anonymous, total social order is unperforated.

Proof. We prove the contrapositive. Suppose $x \not\prec_I y$. Then by totality, $x \prec_I y$. We have permutation invariance, so $\pi_k x \prec_{J_k} \pi_k y$ for each k as well. Thus by applying separability (which implies strong monotonicity) n times,

$$\pi_1 x \sqcup \cdots \sqcup \pi_n x \prec_{J_1 \cup \cdots \cup J_n} \pi_1 y \sqcup \cdots \sqcup \pi_n y.$$

So

$$\pi_1 x \sqcup \cdots \sqcup \pi_n x \not\prec_{J_1 \cup \cdots \cup J_n} \pi_1 y \sqcup \cdots \sqcup \pi_n y. \quad \square$$

It is straightforward to check that in this case the initial additive representation is also totally ordered.

Theorem B.54. A finitary social welfare structure $\mathcal{S}$ is additively representable in a totally ordered vector space iff it is finitely anonymous, separable, and total.

And we can apply Hausner’s theorem (theorem B.31).

Corollary B.55. A finitary social welfare structure is additively representable in a lexicographic vector space iff it is finitely anonymous, separable, and total.

Finally, we can go all the way to our most concrete representations, using *real numbers* to represent individual outcomes. This is classic utilitarianism.

Definition B.56. A finitely anonymous social welfare structure is **Archimedean** iff for any individual outcomes $x > y \in \mathcal{X}$, and any social outcomes $a, b \in \mathcal{X}^I$, there is some finite set K such that

$$x_K \sqcup a \succeq y_K \sqcup b.$$

(As above, x_K represents a constant distribution which assigns x to each individual in K .)

It can be checked that if a social welfare structure is anonymous, separable, and Archimedean, then its initial additive representation is also Archimedean. So the following is a direct consequence of Hölder's theorem (theorem B.37).

Corollary B.57. A finitary social welfare structure is additively representable in $\mathbb{R}$ iff it is finitely anonymous, separable, total, and Archimedean.

B.5 Filter powers

Here is a recurring theme in what follows: we start with a certain family of *finite* objects, where we pretty much understand what is going on. Then we want to extend these to some *infinite* object, by taking some kind of a limit. Theorists have sometimes tried to apply the kind of limits one normally studies in a calculus class, but these are not well-suited to our purposes. If you start with a sequence of standard real numbers and try to take their limit, you either get another standard real number, an “extended” real number that can be ∞ or $-\infty$, or else nothing at all. But as we have seen over and over, these values are far too crude for comparing infinite worlds and infinite prospects. Fortunately, our mathematical resources are not so impoverished: there are many other kinds of limit concepts to draw on.

The key concept I will deploy in what follows is called a **filter limit**. Before

the formal definition, first the basic idea. Consider two infinite sequences

$$\begin{aligned}(x_1, x_2, \dots) &= (3, 0, 2, 1, 1, 2, 3, 4, \dots) \\ (y_1, y_2, \dots) &= (-1, 8, 3, 0, 2, 3, 4, 5, \dots).\end{aligned}$$

After a finite bit of bouncing around, the two sequences fall into a steady pattern: from the 5th element on, the x sequence is $1, 2, 3, 4, \dots$ and the y sequence is $2, 3, 4, 5, \dots$. We can say that *eventually* the y -sequence is always greater than the x -sequence. Or to put it another way, in the long run it is *determinate* that $y_k > x_k$. Contrast this with a pair of sequences like this:

$$\begin{aligned}(x_1, x_2, \dots) &= (3, 0, 2, 1, 1, 2, 3, 4, 5, 6, \dots) \\ (y_1, y_2, \dots) &= (-1, 8, 3, 0, 2, 1, 4, 3, 6, 5, \dots).\end{aligned}$$

In this case we might say that, in the long run, it is *indeterminate* which of the two sequences is greater. Sometimes it's x , and sometimes it's y . But there are other features that are more determinate: for example, it is determinate in the long run that $|x_k - y_k| \leq 1$.

Filters abstract from this idea. We can think of a filter as giving us a concept of "long-run determinacy". How frequently does a property have to hold to count as determinate? One simple and paradigmatic example of a determinacy concept is *holding for all numbers after a certain point*. If a property (such as $y_k = x_k + 1$) holds of all but finitely many numbers, then this is one sense in which it holds determinately in the long run. But there are many different reasonable concepts of "long-run determinacy". All of the senses we will be interested in have certain logical features in common. (Not coincidentally, these correspond to standard principles of modal logic.) Let ϕ and ψ be properties.

1. *Monotonicity*. If ϕ entails ψ , and ϕ holds determinately, then ψ holds determinately.
2. *Agglomeration*. If ϕ holds determinately, and ψ holds determinately, then ϕ -and- ψ also holds determinately.
3. *Non-triviality*. If $\phi(i)$ fails for every i , ϕ does not hold determinately.

For our applications, we will also typically want one more condition:

4. If ϕ holds for all but finitely many i , then determinately $\phi(i)$.

These motivate the following definition.

Definition B.58. Let I be a set. A **filter on I** is a set $\mathcal{F}$ of subsets of I with the following properties:

1. If $A \in \mathcal{F}$ and $A \subseteq B \subseteq I$, then $B \in \mathcal{F}$.
2. If $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.

If $\emptyset \in \mathcal{F}$ then $\mathcal{F}$ is called **trivial**.

A filter $\mathcal{F}$ is **free** if it is non-trivial and contains every co-finite subset of I .

In our motivating example, the index set I was the set of natural numbers. The simple concept of determinacy, “for all numbers after some finite n ” is modeled by the filter that consists of all *co-finite* sets of natural numbers. But we will find that filters on other sets are especially useful.

If $\mathcal{F}$ is a filter on an infinite set I , we say “for $\mathcal{F}$ -almost all i , $\phi(i)$ ” to mean that the set $\{i \in I \mid \phi(i)\}$ is an element of $\mathcal{F}$. We leave off the $\mathcal{F}$ when it is clear in context.

This gives us a concept of some property holding “in the limit”. But we also want certain limiting *objects*. For instance, we would like to have limits for the sequences x_k and y_k , and to be able to say that one of these limits is one less than the other. We can do this too.

Proposition B.59. Let $\mathcal{F}$ be a filter on I , and let A be a set. For any functions $f, g : I \rightarrow A$, define

$$f \sim^{\mathcal{F}} g \quad \text{iff} \quad f(i) = g(i) \text{ for } \mathcal{F}\text{-almost all } i.$$

Then $\sim^{\mathcal{F}}$ is an equivalence relation, and indeed a congruence relation on the direct power A^I . Therefore, there is a well-defined quotient structure $A^{\mathcal{F}} := A^I / \sim^{\mathcal{F}}$. This is called the $\mathcal{F}$ -**power** or **filter power** of A .^a We denote the quotient map

$$\lim^{\mathcal{F}} : A^I \rightarrow A^{\mathcal{F}}$$

and call it the $\mathcal{F}$ -**limit** or **filter limit**.

^aIn other texts, such as Hodges (1997), it is called a *reduced power* and notated $A^I / \mathcal{F}$.

The quotient map from A^I to $A^{\mathcal{F}}$ sends each function $f : I \rightarrow A$ to its equivalence class, the set of functions that are equal to f “ $\mathcal{F}$ -almost everywhere”. We commonly write $\lim^{\mathcal{F}} f$ using the indexed notation $\lim_i^{\mathcal{F}} f(i)$.

For any function $f : A \rightarrow A$, there is a corresponding “lifted” function $f^{\mathcal{F}} : A^{\mathcal{F}} \rightarrow A^{\mathcal{F}}$:

$$f^{\mathcal{F}}\left(\lim_i^{\mathcal{F}} a_i\right) = \lim_i^{\mathcal{F}} f(a_i).$$

In the quotient structure $A^{\mathcal{F}}$, each operation is interpreted as the lifted version of the corresponding operation in A .

We can think of $A^{\mathcal{F}}$ as a set of “infinite objects” extending A . There is a natural map from A to its power, “lifting” each object by taking a limit of a constant function:

$$a \in A \mapsto \lim_i^{\mathcal{F}} a.$$

As long as $\mathcal{F}$ is non-trivial, this is an *embedding* of A into $A^{\mathcal{F}}$. It is common to fudge things a little by identifying each object with its corresponding lifted object, and treat A as if it is a subset of $A^{\mathcal{F}}$.

Filter powers generalize the *direct powers* introduced in section B.2. The direct power A^I is just the filter power of A with respect to the weakest filter $\{I\}$ on I . (In this case, the equivalence relation $\sim^{\mathcal{F}}$ is identity.)

The case of a filter power that is most commonly discussed (especially among logicians) is when the filter is *maximal*.

Definition B.60. A non-trivial filter $\mathcal{F}$ on I is called an **ultrafilter** iff for every set $X \subseteq I$, exactly one of X or $I \setminus X$ is in $\mathcal{F}$. If $\mathcal{F}$ is a free ultrafilter, then $\mathcal{F}$ -powers are called **ultrapowers**.

Ultrapowers have especially nice logical properties.

Theorem B.61 (Łoś). Let A be a structure. Any ultrapower $A^{\mathcal{F}}$ satisfies the same first-order theory as A . (Hodges 1997, theorem 8.5.3)

Thus an ultrapower of a ring is a ring, an ultrapower of a totally ordered field is a totally ordered field, and so on. In particular, a **hyperreal** structure is an ultrapower of the real number structure $\mathbb{R}$. Since hyperreals satisfy the same first-order theory as the standard reals, this allows for very tidy arguments using what is called the “transfer principle”: whatever goes for the reals (in a suitably regimented language) also goes for the hyperreals.

This also shows something that was claimed in section B.2: the Archimedean property is not expressible by a first-order sentence: for $\mathbb{R}$ is Archimedean,

while an ultrapower $\mathbb{R}^{\mathcal{F}}$ is not. (To show this, consider a free ultrafilter $\mathcal{F}$ on $I = \mathbb{N}$. The element $\lim_n^{\mathcal{F}} n$ in $\mathbb{R}^{\mathcal{F}}$ is greater than $n \cdot 1$ in $\mathbb{R}^{\mathcal{F}}$ for every $n \in \mathbb{N}$.)

So ultrapowers are convenient for some purposes. But they are also unwieldy. Notably, appeals to ultrafilters are *non-constructive*, in the same sense that we discussed in section A.3. (See discussion in chapter 4; chapter 5.)

Theorem B.62. For any infinite set I , there is no constructive proof that there exists a free ultrafilter on I . That is, the statement “There exists a free ultrafilter on an infinite set” is independent of ZF set theory without choice.

Filter powers in general do not have this difficulty: filters are easy to come by, and straightforward to explicitly construct and describe. Furthermore, while their logical relationship to their underlying structures is not quite as neat as in the case of ultrapowers, it is still pretty nice.

Theorem B.63. For any structure A and any filter $\mathcal{F}$, A and its power $A^{\mathcal{F}}$ satisfy precisely the same Horn theory. (Hodges 1997, 248)

(This generalizes the property of *direct powers* stated theorem B.24.)

As we noted, many important categories of structures are characterized by Horn theories. Thus, for example, any filter power of a partial order is another partial order, any filter power of a ring is another ring, any filter power of an R -quantity is another R -quantity (for any preordered ring R), and so on.

But, as we noted in section B.2, some other important properties are not expressible by Horn clauses, and need not be inherited by filter powers. One especially important example is *comparability*: a filter power of a *totally* ordered set need not be totally ordered. In practice, this is desirable for our purposes. We don’t really want all of the infinite structures we’ll use to be totally ordered.

Likewise, the power of a *field* need not itself be a field. Specifically, the power may lack the *division* property, that every non-zero number has a multiplicative inverse. This is a minor inconvenience, but in practice we can ensure by other means that there is enough division to go around. For instance, one place where we want division is when we calculate conditional

probabilities. But we can have all the division we need for this if we require that *probabilities* (besides zero) have inverses.

One thing to bear in mind is that filters can be *trivial*. (Recall that the trivial filter on I contains *every* subset of I .) This does not break the filter power construction, exactly, but it does make it much less interesting. If $\mathcal{F}$ is a trivial filter, then for any structure A , the filter power $A^{\mathcal{F}}$ turns out to have just one element. With the right signature, the one-element structure will trivially be a ring (one in which $0 = 1$), a partial order, a vector space (containing only 0), or whatever.⁶ But this is not usually the case we are interested in. When $\mathcal{F}$ is non-trivial, then we don't get this collapse, and in particular the filter power preserves all of the original structure's diversity.

Proposition B.64. Let $\mathcal{F}$ be any non-trivial filter.

- (a) For any p.o.c. (semi-) ring R , the filter power $R^{\mathcal{F}}$ is a p.o.c. (semi-) ring that embeds R , and thus also an R -quantity.

In particular, $\mathbb{R}^{\mathcal{F}}$ is a p.o.c. ring that embeds $\mathbb{R}$, and thus also a p.o. vector space.

- (b) For any R -quantity $\mathcal{U}$, the filter power $\mathcal{U}^{\mathcal{F}}$ is an R -quantity that embeds $\mathcal{U}$.

In particular, for any (p.o.) vector space $\mathcal{U}$, the filter power $\mathcal{U}^{\mathcal{F}}$ is a (p.o.) vector space that embeds $\mathcal{U}$.

There is a tight relationship between general filters and ultrafilters.

Theorem B.65. Every non-trivial filter can be extended to an ultrafilter. That is, if $\mathcal{F}$ is a non-trivial filter on I , there is an ultrafilter $\mathcal{F}^* \supseteq \mathcal{F}$ on I .

Of course, the proof of this theorem again appeals to the axiom of choice.⁷

There is a corresponding relationship between general filter powers and ultrapowers.

⁶In general, the one-point structure satisfies every *positive* Horn clause — those where the conclusion is not $\perp$. All of the Horn theories we considered were in fact *positive* Horn theories.

⁷To be precise, the result is equivalent to the Boolean Prime Ideal Theorem, which is strictly weaker than choice, but also independent of ZF set theory (J. D. Halpern and Lévy 1971). See also section A.3, especially figure A.1.

Proposition B.66. Let $\mathcal{F}$ and $\mathcal{G}$ be filters on I such that $\mathcal{F} \subseteq \mathcal{G}$. Let A be a structure. Then there is a surjective homomorphism $h : A^{\mathcal{F}} \rightarrow A^{\mathcal{G}}$ defined by

$$h(\lim_i^{\mathcal{F}} a_i) = \lim_i^{\mathcal{G}} a_i \quad \text{for each } a \in A^I.$$

We can think of a filter power as “supervaluating” over all ultrapowers of a certain form. This corresponds to a two-step method we sometimes see in the philosophical literature (for example, in Arntzenius 2014). First, we say how to do things with respect to an arbitrary choice of ultrafilter. Second, we “supervalue” over all of the different choices we might have made. The filter power construction gives us the same results as this, in extension. But it has several advantages. First and most obviously, it is simpler, bypassing an intermediate mathematical step that we kick away. Second, we avoid any appeal to the axiom of choice. This may seem a merely technical advantage, but it has serious practical advantages. Since we are working with constructive objects that we can manipulate pretty directly, we are encouraged not to treat the choice of ultrafilter as a “black box”, a bit of inscrutable magic that gives us whatever it gives us. Instead, the choice of filter is something we can make intelligent decisions about. We will see in what follows that this is in fact a crucial choice point, which determines the answers to many interesting questions about value and probability. This choice point is obscured if we immediately leap up to completely specific — and completely obscure — ultrafilters.

B.6 Quasi-finite sums

In section B.5 we considered filters on an arbitrary set of “indices”. Now we will be a bit more specific, and consider filters on the set $\text{Fin } \mathcal{I}$ of finite subsets of some set $\mathcal{I}$. We can think of $\mathcal{I}$ as an infinite set of individuals; in appendix F we will alternatively think of it as an infinite set of states.

A filter $\mathcal{F}$ on $\text{Fin } \mathcal{I}$ is a set of sets of sets — but that makes it sound more confusing than it is in practice. Here is the general idea for how we will do things. First, we will say what is going on in finite cases: for example, if we want to order subsets of $\mathcal{I}$, we will start with the easy case of ordering finite subsets. A filter $\mathcal{F}$ on $\text{Fin } \mathcal{I}$ defines a limiting notion of what happens for “large enough” finite sets. Each element of the filter (a set of finite sets) represents one particular way of “getting larger” — one route along

finite sets toward the infinite destination $\mathcal{I}$. We can use this limiting notion to define a more general ordering for arbitrary sets, and also to construct representations for such orderings, using non-Archimedean quantities that include infinities and (later on) infinitesimals.⁸

Definition B.67. Let $\mathcal{F}$ be a filter on $\text{Fin } \mathcal{I}$. We call $\mathcal{F}$ **fine** (or a **fine $\mathcal{I}$ -filter**) iff it contains each set of the form

$$\{K \in \text{Fin } \mathcal{I} \mid i \in K\} \quad \text{for some } i \in \mathcal{I}.$$

In other words, for any particular $i \in \mathcal{I}$, $\mathcal{F}$ -almost all finite sets contain i .

Equivalently, a fine filter contains each set $\{K \in \text{Fin } \mathcal{I} \mid K \supseteq J\}$ for any $J \in \text{Fin } \mathcal{I}$. (This rules out, for example, the very weakest filter $\{\text{Fin } \mathcal{I}\}$.)

Note that a fine $\mathcal{I}$ -filter is a filter on $\text{Fin } \mathcal{I}$, and not on $\mathcal{I}$ itself. (The term *fine* does not make sense for filters on an arbitrary set.)

The weakest fine filter $\mathcal{F}_0$ is the one generated by just the principal sets. To put that another way, a property holds for $\mathcal{F}_0$ -almost all $K \in \text{Fin } \mathcal{I}$ iff there is some J such that every $K \supseteq J$ has the property. I will call $\mathcal{F}_0$ the **weak filter** on $\text{Fin } \mathcal{I}$.

Definition B.68. Let $\mathcal{I}$ be a set, and let $\mathcal{F}$ be any filter on $\text{Fin } \mathcal{I}$. For any function $f : \mathcal{I} \rightarrow \mathbb{R}$, define the **quasi-finite sum** as the filter limit of finite sums:

$$\sum_{\mathcal{F}} f := \lim_K^{\mathcal{F}} \sum_K f \in \mathbb{R}^{\mathcal{F}}.$$

For any subset $E \subseteq \mathcal{I}$, define

$$\sum_E^{\mathcal{F}} f := \sum_{\mathcal{F}} f \cdot 1_E = \lim_K^{\mathcal{F}} \sum_{E \cap K} f.$$

⁸This approach — especially the notion of *quasi-cardinality* below — is in the same spirit as *numerosity theory*, and uses similar technical ideas (Benci and Di Nasso 2003; Benci, Di Nasso, and Forti 2006; Benci, Bottazzi, and Di Nasso 2014). Two departures here: we do not assume that our quantitative representations are *totally ordered*, and (relatedly) we are able to give a constructive theory that does not rely on ultrafilters. For further related literature see footnote 1 in section F.1.

Theorem B.69. Let $\mathcal{I}$ be a set, and let $\mathcal{F}$ be a fine filter on $\text{Fin } \mathcal{I}$.

- (a) $\sum^{\mathcal{F}}$ is finitely symmetric, in the sense that for any permutation $\pi : \mathcal{I} \rightarrow \mathcal{I}$ with finite support and any function $f : \mathcal{I} \rightarrow \mathbb{R}$,

$$\sum^{\mathcal{F}} f \circ \pi = \sum^{\mathcal{F}} f.$$

- (b) $\sum^{\mathcal{F}}$ is a positive linear operator from $\mathbb{R}^{\mathcal{I}}$ to $\mathbb{R}^{\mathcal{F}}$ (considered as p.o. vector spaces). That is to say, for any $f, g : \mathcal{I} \rightarrow \mathbb{R}$ and any $a \in \mathbb{R}$,

$$\begin{aligned} \sum^{\mathcal{F}} (f + g) &= \sum^{\mathcal{F}} f + \sum^{\mathcal{F}} g \\ \sum^{\mathcal{F}} (a \cdot f) &= a \cdot \sum^{\mathcal{F}} f \\ \sum^{\mathcal{F}} f &\geq 0 \text{ if } f(i) \geq 0 \text{ for all } i \in \mathcal{I}. \end{aligned}$$

Suppose furthermore that $\mathcal{F}$ is non-trivial.

- (c) Then $\sum^{\mathcal{F}}$ is a *strictly* positive linear operator, in the sense that (in addition to positivity) if $f(i) \geq 0$ for all $i \in \mathcal{I}$ and $f(i) > 0$ for some $i \in \mathcal{I}$, then $\sum^{\mathcal{F}} f > 0$.
- (d) If $f : \mathcal{I} \rightarrow \mathbb{R}$ has finite support, then

$$\sum^{\mathcal{F}} f = \sum_{\mathcal{I}} f$$

where the right-hand side is the usual finite sum (lifted to $\mathbb{R}^{\mathcal{F}}$).

Definition B.68 can be straightforwardly generalized: instead of assuming a function takes *real* values, it can take values in any commutative monoid, and thus in any *quantity* in the sense of section B.1. If $f : \mathcal{I} \rightarrow \mathcal{U}$ is a function taking values in a quantity $\mathcal{U}$, then we can define the quasi-finite sum $\sum^{\mathcal{F}} f$ as the filter limit of finite sums, in exactly the same way as above. The value of $\sum^{\mathcal{F}} f$ will then be an element of the filter power $\mathcal{U}^{\mathcal{F}}$ — which is also a quantity, and generally a quantity of the same type as $\mathcal{U}$. (A commutative monoid, a p.o. vector space, etc. For any p.o.c. semiring R , if $\mathcal{U}$ is an R -quantity, then $\mathcal{U}^{\mathcal{F}}$ is also an R -quantity. This follows immediately from theorem B.63.)

All of the properties in theorem B.69 still hold in this more general setting, whenever they are well-defined.

We know that we cannot represent orders as rich as the counting order using real numbers (section A.3). But we can do this using filter powers.

Theorem B.70. Let $\mathcal{I}$ be any set, and let $\mathcal{F}$ be a filter on $\text{Fin } \mathcal{I}$. Define the **quasi-cardinality** function $\#^{\mathcal{F}} : 2^{\mathcal{I}} \rightarrow \mathbb{R}^{\mathcal{F}}$ such that

$$\#^{\mathcal{F}}(A) := \sum_{\mathcal{F}} 1_A \quad \text{for any } A \subseteq \mathcal{I}.$$

- (a) If $\mathcal{F}$ is non-trivial and fine, then $\#^{\mathcal{F}}$ represents a strictly monotone finitely symmetric order on $2^{\mathcal{I}}$.
- (b) If $\mathcal{F}$ is the weak filter $\mathcal{F}_0$, then $\#^{\mathcal{F}}$ represents the weak counting order on $2^{\mathcal{I}}$.
- (c) If $\mathcal{F}$ is a fine ultrafilter, then $\#^{\mathcal{F}}$ represents a strictly monotone finitely symmetric *total* order on $2^{\mathcal{I}}$.

Definition B.71. Let R be a quantity, let X be a set, and let $\mathcal{A}$ be a Boolean algebra of subsets of X . A **(finitely additive) R -valued measure on $(X, \mathcal{A})$** is a function $\mu : \mathcal{A} \rightarrow R$ such that

- (a) $\mu(\emptyset) = 0$, and
- (b) For any $A, B \in \mathcal{A}$, if $A \cap B = \emptyset$, then

$$\mu(A \cup B) = \mu(A) + \mu(B).$$

If $\mathcal{A}$ is not otherwise specified, we take it to be the power set algebra, which includes *every* subset of X .

Let μ be a measure on $(X, \mathcal{A})$.

- (c) μ is **regular** iff for any $A \in \mathcal{A}$,

$$\text{if } A \neq \emptyset, \text{ then } \mu(A) > 0.$$

- (d) μ is **uniform** iff for each $x \in X$, the singleton $\{x\}$ is in $\mathcal{A}$, and

$$\mu(\{x\}) = \mu(\{y\}) \quad \text{for any } x, y \in X.$$

Theorem B.72. For any set $\mathcal{I}$ and any non-trivial fine filter $\mathcal{F}$ on $\text{Fin } \mathcal{I}$, the quasi-cardinality operator $\#^{\mathcal{F}}$ is a regular uniform $\mathbb{R}^{\mathcal{F}}$ -valued measure on $\mathcal{I}$.

Note that, while we have considered the standard finite cardinality operator as taking *real* values, and thus the quasi-cardinality operator as taking *quasi-real* values in $\mathbb{R}^{\mathcal{F}}$, we can also think of the finite cardinality operator as taking natural-number values in $\mathbb{N}$. Thus the quasi-cardinality operator takes values in $\mathbb{N}^{\mathcal{F}}$, which we can consider as a substructure of $\mathbb{R}^{\mathcal{F}}$.

The ultrafilter lemma implies that there exist free ultrafilters. So it follows that there are total finitely symmetric strictly monotone orders on $2^{\mathcal{I}}$ (as we already knew, corollary A.24). Of course, as we discussed above, the ultrafilter lemma is non-constructive, so there is no contradiction with the Zame–Lauwers theorem (theorem A.25).

Here is one more property of quasi-finite sums that will be useful in the next section. There is a sense in which these sums are determined by their values on finite sets.

Proposition B.73. Let $\mathcal{F}$ be a fine $\mathcal{I}$ -filter. For $J \in \text{Fin } \mathcal{I}$, let $\uparrow J := \{K \in \text{Fin } \mathcal{I} \mid K \supseteq J\}$ be the principal filter generated by J . For any function $f : \mathcal{I} \rightarrow \mathbb{R}$, let

$$F(f) := \{K \in \text{Fin } \mathcal{I} \mid \sum_K f \geq 0\}.$$

For $f_1, f_2, \dots, f_n, g : \mathcal{I} \rightarrow \mathbb{R}$, if

$$\uparrow J \cap F(f_1) \cap \dots \cap F(f_n) \subseteq F(g)$$

and $\sum^{\mathcal{F}} f_k \geq 0$ for each k , then $\sum^{\mathcal{F}} g \geq 0$ as well.

Proof. In general, $\sum^{\mathcal{F}} f \geq 0$ iff $\sum_K f \geq 0$ for $\mathcal{F}$ -almost all $K \in \text{Fin } \mathcal{I}$, which is equivalent to saying that $F(f) \in \mathcal{F}$. Since $\mathcal{F}$ is fine, it also includes $\uparrow J$. So the result follows immediately from the properties of filters. $\square$

Note that, while we are presently focusing on interpretations involving social aggregation, all of these results also have direct applications to probability. We will return to this in appendix F.

B.7 Representing social welfare

Recall the framework from section A.4. We assume a set $\mathcal{X}$ of individual outcomes and an infinite set $\mathcal{I}$ of individuals. A social outcome is any function from $\mathcal{I}$ to $\mathcal{X}$, and a social order is a preorder of social outcomes.

We can now introduce our central example of an infinite social order.

Definition B.74. Let $\mathcal{U}$ be a quantity. Let $u : \mathcal{X} \rightarrow \mathcal{U}$ be a strictly monotone **individual utility** function, and let $\mathcal{F}$ be any non-trivial fine $\mathcal{I}$ -filter. Define a **social utility function** $U : \mathcal{X}^{\mathcal{I}} \rightarrow \mathcal{U}^{\mathcal{F}}$ by

$$U(x) = \sum_{\mathcal{F}} u \circ x.$$

The **additive social order** on social outcomes (given by u and $\mathcal{F}$) is defined such that

$$x \succeq y \quad \text{iff} \quad U(x) \geq U(y).$$

As in section B.4, “quantity” is very inclusive, and so additive orders form a pretty expansive family. In the case where $\mathcal{U} \subseteq \mathbb{R}$, we get the class of social welfare theories that I called *quasi-utilitarian* in section 3.5. But the family also includes interesting infinitary extensions of theories like prioritarianism or leximin, among others.

In what follows, fix $\mathcal{U}$, u , and $\mathcal{F}$, and their corresponding additive order.

Proposition B.75.

- (a) The additive social order is monotone and finitely anonymous. If $\mathcal{U}$ is cancellative, then the additive social order is Paretian.
- (b) For any action $G \curvearrowright \mathcal{I}$, if $\mathcal{F}$ is a G -invariant filter, the additive social order is also G -invariant. In particular, when $\mathcal{F}$ is the weak filter $\mathcal{F}_0$, the additive social order is invariant under arbitrary permutations of $\mathcal{I}$.
- (c) If $\mathcal{F}$ is an ultrafilter and $\mathcal{U}$ is totally ordered, the additive social order is total.

The additive social order agrees with the finitary additive order in finite cases.

Proposition B.76. Let x and y be social outcomes. If $u \circ x$ and $u \circ y$ each have finite support, then

$$x \succeq y \quad \text{iff} \quad \sum_{\mathcal{I}} u \circ x \geq \sum_{\mathcal{I}} u \circ y$$

where the right-hand side involves the usual finite sum.

More generally, if $\mathcal{U}$ is a $\mathbb{Z}$ -quantity (so subtraction is well-defined) and $u \circ x - u \circ y$ has finite support,

$$x \succeq y \quad \text{iff} \quad \sum_{\mathcal{I}} (u \circ x - u \circ y) \geq 0.$$

Another important feature is that, if individual utilities permit subtraction, then the additive social order is *separable*: we can always ignore what is going on with individuals who are unaffected by a choice between two social outcomes.

For $J \subseteq \mathcal{I}$, let $\pi_J : \mathcal{X}^{\mathcal{I}} \rightarrow \mathcal{X}^J$ be the projection map

$$\pi_J(x)(i) = x(i) \text{ for } i \in J$$

Write $x =_J y$ to mean that $\pi_J(x) = \pi_J(y)$; that is, x and y agree on the outcomes assigned to each individual in J .

The idea here is that if x and y differ only for J , then their order only depends on what is going on with individuals in J .

Definition B.77. A social order $\succeq$ is **separable** iff, for any social outcomes x, y, x', y' , if

$$x =_J x'$$

$$y =_J y'$$

$$x =_{\mathcal{I} \setminus J} y$$

$$x' =_{\mathcal{I} \setminus J} y',$$

then

$$x \succeq y \quad \text{iff} \quad x' \succeq y'.$$

The pattern looks like this:

$$\begin{array}{ccc}
 x & \xrightarrow{=_{\mathcal{I} \setminus J}} & y \\
 \downarrow =_J & & \downarrow =_J \\
 x' & \xrightarrow{=_{\mathcal{I} \setminus J}} & y'
 \end{array}$$

In this case, the comparison between x and y comes down to their respective parts on J , as does the comparison between x' and y' ; furthermore, since the corresponding J -parts of each pair are exactly the same, the two pairs are ordered the same way.

Proposition B.78. If $\mathcal{U}$ is cancellative, the additive social order is separable.

(For non-cancellative $\mathcal{U}$ quantities, we instead have a weaker *monotonicity* condition.)

An additive social order has a *finitary part*: namely, the finitary social order that is additively represented by u . Furthermore, the additive social order is, in a certain sense, *determined* by its finitary part.⁹ (We state the following property with slightly more generality than we really need here, so we can reuse it in the context of non-standard probability theory in appendix F. In our application, $X = \mathcal{X}^{\mathcal{I}}$ is the set of social outcomes.)

Definition B.79. Let X be any set, let $\succeq$ be a preorder on X , and let $(\succeq_K)_{K \in \text{Fin } \mathcal{I}}$ be a family of preorders on X . For any finite set $J \in \text{Fin } \mathcal{I}$, let $\uparrow J$ be the set of all finite sets $K \in \text{Fin } \mathcal{I}$ such that $K \supseteq J$. For any $x, y \in X$, define

$$F(x, y) = \{K \in \text{Fin } \mathcal{I} \mid x \succeq_K y\}.$$

We say that $\succeq$ is **finely filtered by** $(\succeq_K)$ iff for any finite set $J \in \text{Fin } \mathcal{I}$ and any $x_1, \dots, x_n, x^*$ and $y_1, \dots, y_n, y^*$ in X , if

$$\uparrow J \cap \bigcap_k F(x_k, y_k) \subseteq F(x^*, y^*)$$

and also $x_k \succeq y_k$ for each k , then $x^* \succeq y^*$ as well.

⁹The filtration property is closely related to Pivato's property of $\mathcal{F}$ -continuity. Being finely filtered amounts to being $\mathcal{F}$ -continuous with respect to *some* fine filter or other $\mathcal{F}$. (Note though that Pivato focused only on the case of ultrafilters, and filtration does not suffice for being $\mathcal{F}$ -continuous with respect to an *ultrafilter*.)

(One hiccup here: officially, a finitary social order $\succsim_{(-)}$ is a family of orders on the sets $\mathcal{X}^K$, rather than $\mathcal{X}^J$. We say that $\succsim$ is finitely filtered by a *finitary social order* iff it stands in this relation to the family of relations defined such that $x \succsim_K y$ iff $x|_K \succsim_K y|_K$, for $K \in \text{Fin } \mathcal{J}$, $x, y \in \mathcal{X}^J$.)

Lemma B.80. Let $\succsim$ be a preorder on a set X , and let $(\succsim_K)_{K \in \text{Fin } \mathcal{J}}$ be a family of preorders on X . The following are equivalent:

- (a) $\succsim$ is finely filtered by $(\succsim_K)$;
- (b) There is a fine filter $\mathcal{F}$ on $\text{Fin } \mathcal{J}$ such that $\succsim$ is the $\mathcal{F}$ -limit of $(\succsim_K)$, in the sense that for any $x, y \in X$,

$$x \succsim y \quad \text{iff} \quad x \succsim_K y \text{ for } \mathcal{F}\text{-almost all } K \in \text{Fin } \mathcal{J}.$$

Equivalently,

$$x \succsim y \quad \text{iff} \quad F(x, y) \in \mathcal{F}.$$

Proof. First, suppose that property (b) holds for a filter $\mathcal{F}$. Consider $J \in \text{Fin } \mathcal{J}$ and elements of X

$$x_1 \succsim y_1 \quad x_2 \succsim y_2, \quad \dots, \quad x_n \succsim y_n.$$

Then for each k , $F(x_k, y_k) \in \mathcal{F}$. Let x^* and y^* be elements of X such that

$$\uparrow J \cap \bigcap_k F(x_k, y_k) \subseteq F(x^*, y^*).$$

Since $\mathcal{F}$ is a fine filter, it also contains $\uparrow J$. Then by the properties of filters, $F(x^*, y^*) \in \mathcal{F}$, so $x^* \succsim y^*$.

For the converse, suppose that $\succsim$ is finely filtered by $(\succsim_K)$. Let $\mathcal{F}$ be the smallest fine filter containing all sets $F(x, y)$ for $x, y \in X$ such that $x \succsim y$.

If $x \succsim y$, then $F(x, y) \in \mathcal{F}$, so immediately $x \succsim_K y$ for $\mathcal{F}$ -almost all K .

For the other direction, suppose $F(x, y) \in \mathcal{F}$. By the construction of $\mathcal{F}$, this means that there are some

$$x_1 \succsim y_1, \quad x_2 \succsim y_2, \quad \dots, \quad x_n \succsim y_n$$

and some $J \in \text{Fin } \mathcal{J}$ such that

$$\uparrow J \cap \bigcap_k F(x_k, y_k) \subseteq F(x, y).$$

So the filtration property of $\succsim$ implies that $x \succsim y$ as well. □

Corollary B.81. The additive social order is finely filtered by its finitary part.

Lemma B.80 lets us turn representation theorems for finitary social orders into representation theorems for infinitary social orders based on filter limits. We can do this for all of the different additive representation theorems we proved in section B.4. Here is the most abstract version.¹⁰

Theorem B.82. Let $(\mathcal{X}, \mathcal{I}, \succeq)$ be an infinitary social welfare structure. The following are equivalent:

- (a) $\succeq$ is finely filtered by some finitely anonymous, permutation invariant, and monotone finitary social order.
- (b) There is some quantity $\mathcal{U}$, some monotone individual utility function $u : \mathcal{X} \rightarrow \mathcal{U}$, and some fine filter $\mathcal{F}$ on $\text{Fin } \mathcal{I}$ such that the additive social welfare function

$$U(x) = \sum_{\mathcal{F}} u \circ x \quad U : \mathcal{X}^{\mathcal{I}} \rightarrow \mathcal{U}^{\mathcal{F}}$$

strongly represents $\succeq$.

Proof. The direction from (b) to (a) is straightforward, using corollary B.81 and the properties of the finitary additive order.

So assume (a). Applying theorem B.45, the finitary social order $\succeq_{(-)}$ has a strong additive representation $u : \mathcal{X} \rightarrow \mathcal{U}$ for a quantity $\mathcal{U}$. For each finite set of individuals $K \in \text{Fin } \mathcal{I}$, let

$$U_K(x) = \sum_{i \in K} u(x(i)) \quad \text{for each social outcome } x : \mathcal{I} \rightarrow \mathcal{X}.$$

Then U_K strongly represents $\succeq_K$. By lemma B.80, there is a fine filter $\mathcal{F}$ such that $\succeq$ is the $\mathcal{F}$ -limit of $\succeq_{(-)}$, and

$$U(x) = \sum_{\mathcal{F}} u \circ x = \lim_K^{\mathcal{F}} U_K(x).$$

Thus

$$\begin{aligned} U(x) \geq U(y) &\iff U_K(x) \geq U_K(y) \text{ for } \mathcal{F}\text{-almost all } K \\ &\iff x \succeq_K y \text{ for } \mathcal{F}\text{-almost all } K \\ &\iff x \succeq y. \end{aligned}$$

¹⁰Theorem B.82 is very closely related to Pivato (2014)'s theorem 2; see the note on $\mathcal{F}$ -continuity above.

□

The same idea can be applied to any of the other additive representation theorems in section B.4. For example, we can similarly get an additive representation using *real* individual utilities by imposing a total order and the Archimedean property on finite social outcomes.

B.8 Symmetries and filter powers

In section A.6 we considered whether there are orders on simple worlds that are *invariant* or *symmetric* under various groups of symmetries. Filter powers bear on this question too, by providing us with resources for constructing additive social orders that satisfy various symmetry conditions. The key step is to use filters that are themselves symmetric in the right ways. This idea will be especially useful in the context of non-standard probability theory, but it makes sense to introduce some of the formal ideas here.¹¹

Consider a group action $G \curvearrowright X$. Recall that for any preorder $\succeq$ on X , we say $\succeq$ is ***G*-symmetric** iff for any $x \in X$ and $g \in G$, $x \sim g \cdot x$. We say $\succeq$ is ***G*-invariant** iff for any $x, y \in X$ and $g \in G$, if $x \succeq y$ then $g \cdot x \succeq g \cdot y$. We will make an analogous distinction for filters.

Recall that any group action $G \curvearrowright X$ also has “lifted” actions $G \curvearrowright 2^X$ on subsets of X , $G \curvearrowright \text{Fin } X$ on finite subsets of X , and $G \curvearrowright Y^X$ on functions from X to any set Y .

Definition B.83. Let $G \curvearrowright X$ be a group action, and let $\mathcal{F}$ be a filter on a set X .

(a) $\mathcal{F}$ is ***G*-symmetric** iff, for each $g \in G$,

$$g \cdot x = x \quad \text{for } \mathcal{F}\text{-almost every } x \in X$$

(b) $\mathcal{F}$ is ***G*-invariant** iff for all $g \in G$ and all $V \in \mathcal{F}$, also $g \cdot V \in \mathcal{F}$. In short, $G \cdot \mathcal{F} = \mathcal{F}$.

Here is a slightly different way of stating the definition of *G*-invariance. Suppose that some property ϕ applies to $\mathcal{F}$ -almost every $x \in X$. Then ϕ also applies to $g \cdot x$ for $\mathcal{F}$ -almost every $x \in X$.

¹¹The main ideas in this section are inspired by Pivato (2014)’s use of symmetric ultrafilters.

Proposition B.84. Every G -symmetric filter is G -invariant.

Proof. Let $g \in G$ and $V \in \mathcal{F}$. If $\mathcal{F}$ is G -symmetric, then it contains $\{x \in X \mid g \cdot x = x\}$. Straightforwardly if $x \in V \cap \{x \in X \mid g \cdot x = x\}$, then $g \cdot x = x \in V$. So by the properties of filters, $g \cdot V \in \mathcal{F}$. $\square$

I have stated these definitions very generally, but in our applications, the set X will always be $\text{Fin } \mathcal{I}$ for some set of “indices” $\mathcal{I}$. (So in the variable x from the definition will stand for a finite set of indices, and thus we will generally instead use a variable like K .) The group action G on $\text{Fin } \mathcal{I}$ will always be lifted from the action of some group of symmetries G on the underlying set $\mathcal{I}$.

In this setting, any fine filter $\mathcal{F}$ on $\text{Fin } \mathcal{I}$ gives rise to a quasi-finite sum

$$\sum^{\mathcal{F}} : (\mathcal{I} \rightarrow \mathbb{R}) \rightarrow \mathbb{R}^{\mathcal{F}}$$

as well as a quasi-cardinality function

$$\#^{\mathcal{F}} : 2^{\mathcal{I}} \rightarrow \mathbb{N}^{\mathcal{F}} \subseteq \mathbb{R}^{\mathcal{F}}.$$

Theorem B.85. Let $G \curvearrowright \mathcal{I}$ be a group action, and let $\mathcal{F}$ be a fine $\mathcal{I}$ -filter.

- (a) If $\mathcal{F}$ is G -symmetric, then $\sum^{\mathcal{F}}$ and $\#^{\mathcal{F}}$ are each G -symmetric, in the sense that for any $g \in G$,

$$\begin{aligned} \sum^{\mathcal{F}}(g \cdot f) &= \sum^{\mathcal{F}} f && \text{for each } f : \mathcal{I} \rightarrow \mathbb{R} \\ \#^{\mathcal{F}}(g \cdot A) &= \#^{\mathcal{F}}(A) && \text{for each } A \subseteq \mathcal{I}. \end{aligned}$$

- (b) If $\mathcal{F}$ is G -invariant, then $\sum^{\mathcal{F}}$ and $\#^{\mathcal{F}}$ are each G -invariant, in the sense that for any $g \in G$,

$$\begin{aligned} \text{if } \sum^{\mathcal{F}} f \geq 0 & \text{ then } \sum^{\mathcal{F}}(g \cdot f) \geq 0 && \text{for each } f : \mathcal{I} \rightarrow \mathbb{R} \\ \text{if } \#^{\mathcal{F}}(A) \geq \#^{\mathcal{F}}(B) & \text{ then } \#^{\mathcal{F}}(g \cdot A) \geq \#^{\mathcal{F}}(g \cdot B) && \text{for each } A, B \subseteq \mathcal{I}. \end{aligned}$$

Proof. Since $\#^{\mathcal{F}}(A) := \sum^{\mathcal{F}} 1_A$, it suffices to prove the facts about sums.

Part (a). Let $\mathcal{F}$ be G -symmetric, let $g \in G$, and let $f : \mathcal{I} \rightarrow \mathbb{R}$ be any function. Then for $\mathcal{F}$ -almost all $K \in \text{Fin } \mathcal{I}$, we have $g^{-1} \cdot K = K$, and thus

$$\sum_K (g \cdot f) = \sum_K f \circ g^{-1} = \sum_{g^{-1} \cdot K} f = \sum_K f.$$

Part (b). Let $\mathcal{F}$ be G -invariant, let $g \in G$, and let $f : \mathcal{I} \rightarrow \mathbb{R}$ be any function such that $\sum^{\mathcal{F}} f \geq 0$. So, for $\mathcal{F}$ -almost all $K \in \text{Fin } \mathcal{I}$,

$$0 \leq \sum_K f.$$

Thus, by the definition of G -invariance, for $\mathcal{F}$ -almost all $K \in \text{Fin } \mathcal{I}$

$$0 \leq \sum_{g^{-1} \cdot K} f = \sum_K f \circ g^{-1} = \sum_K (g \cdot f). \quad \square$$

This gives us a way of constructing symmetric or invariant orders on $2^{\mathcal{I}}$ and $\mathbb{R}^{\mathcal{I}}$, by using filters with the right symmetry properties. (Note that the conditions on filters are *sufficient* for the corresponding conditions on sums and orders, but not necessary. In general, the same G -symmetric order can arise from more than one filter, which need not all be G -symmetric in this sense.)

Note that the property we called *finite symmetry* (corresponding to finite anonymity for social orders) is the case of $\Pi_{\text{fin}} \mathcal{I}$ -symmetry, that is, symmetry under all permutations of $\mathcal{I}$ with finite support. We actually get this much symmetry more or less for free.

Proposition B.86. Every fine $\mathcal{I}$ -filter is $\Pi_{\text{fin}} \mathcal{I}$ -symmetric.

Proof. Let $\mathcal{F}$ be a fine $\mathcal{I}$ -filter, and let $\pi : \mathcal{I} \rightarrow \mathcal{I}$ be any permutation with finite support. Let J be the support of π . Since $\mathcal{F}$ is *fine*, for $\mathcal{F}$ -a.e. $K \in \text{Fin } \mathcal{I}$, we have $J \subseteq K$, and thus $\pi \cdot K = K$. $\square$

This is why quasi-sums were automatically symmetric under finite permutations, without any further restrictions on the filter. If we want more symmetries than this, we will have to choose our filter more carefully.

Definition B.87. For any action $G \curvearrowright \mathcal{I}$, let the **weak G -symmetric filter** $\mathcal{F}_G$ be the smallest G -symmetric fine filter on $\text{Fin } \mathcal{I}$.

Here is another way of characterizing the weak G -symmetric filter: it is the filter on $\text{Fin } \mathcal{I}$ generated by all sets of either of these two forms:

$$\begin{aligned} \{K \in \text{Fin } \mathcal{I} \mid i \in K\} & \quad \text{for some } i \in \mathcal{I} \\ \{K \in \text{Fin } \mathcal{I} \mid g \cdot K = K\} & \quad \text{for some } g \in G. \end{aligned}$$

So, for any group of symmetries G , we can always find a G -symmetric fine filter. But we also know that some symmetries are impossible for a strictly monotone order on $2^{\mathcal{I}}$ to satisfy — for example, shifts on the integers. The trouble is that the weak G -symmetric filter may be *trivial*, in which case the filter power $\mathbb{R}^{\mathcal{F}_G}$ is the trivial ring in which $0 = 1$. Then the order on $2^{\mathcal{I}}$ represented by $\#^{\mathcal{F}_G}$ is also trivial (and hence, in particular, not strictly monotone). This is not much use.

How can this happen? The problem arises when there is some finite set of permutations $H \subseteq G$, and some finite set of indices $K \in \text{Fin } \mathcal{I}$, such that *no* finite set $J \supseteq K$ is fixed by every permutation in H . In other words, the smallest H -closed set containing K is infinite. Or in other words again, the orbit of K under the finitely generated subgroup $\langle H \rangle$ is infinite. Equivalently, for some particular $i \in K$, the orbit $\langle H \rangle \cdot i$ is infinite.

Recall from section A.5 that a group action $G \curvearrowright \mathcal{I}$ is *locally finite* iff, for every finite subset of permutations $H \subseteq G$, the finitely generated subgroup $\langle H \rangle$ has finite orbits. The reasoning above proves the following result.

Proposition B.88. The following are equivalent:

- (a) $G \curvearrowright \mathcal{I}$ is locally finite;
- (b) The weak G -symmetric filter $\mathcal{F}_G$ is non-trivial.

Here are some important examples.

Example B.89.

- (a) Recall (examples A.43 and A.49) that the action of rational rotations on the circle is locally finite. Thus the weak rational-rotation symmetric filter is non-trivial, and gives rise to a regular uniform non-standard measure on the circle which is symmetric under each rational rotation

symmetry.

- (b) In contrast, the action of the group of *all* rotations on the circle is not locally finite, and thus the weak rotation-symmetric filter is trivial. Indeed, in this case we know that *no* regular uniform measure on the circle is symmetric under all rotations. Any such measure would represent a strictly monotone rotationally symmetric order, contradicting lemma A.60.
- (c) Recall (examples A.44 and A.49) the *coin flip space* 2^ω , the set of infinite sequences of zeros and ones. This is acted on by the *finitary coin flip group* $\text{Fin } \omega$, which flips between Heads and Tails at finitely many places. This action is locally finite, and thus its corresponding weak $(\text{Fin } \omega)$ -symmetric filter is non-trivial, and gives rise to a regular uniform non-standard measure on 2^ω which is symmetric under finitary flips.
- (d) Similarly, the action of the *Cantor group* 2^ω on 2^ω is also locally finite, and thus the weak Cantor-symmetric filter is non-trivial, and gives rise to a regular uniform non-standard measure on 2^ω which is symmetric under arbitrary flips.
- (e) In contrast, the *coin-permutation* action of $\Pi(\omega)$ on 2^ω is not locally finite, and again in this case we know as a consequence of lemma A.60 that *no* regular uniform measure on the coin flip space is symmetric under all coin permutations.

Corollary B.90. Suppose $G \curvearrowright \mathcal{I}$ is locally finite.

- (a) There exists a regular G -symmetric finitely additive non-standard measure on $2^{\mathcal{I}}$, namely $\#^{\mathcal{F}_G}$, taking values in $\mathbb{R}^{\mathcal{F}_G}$.
- (b) There exists a strictly monotone G -symmetric order on $2^{\mathcal{I}}$ (proving theorem A.61).
- (c) There exists a Paretian, finitely anonymous, G -symmetric social order on $\mathbb{R}^{\mathcal{I}}$.

What about *invariant* filters? As we discussed in section A.6, in principle we have much more liberty with invariance conditions than with the stronger notion of symmetry. The main conflicts we discussed involving invariance conditions had to do with *total* orders. We can immediately conclude:

Corollary B.91. No fine ultrafilter on $\text{Fin } \mathcal{I}$ is invariant under arbitrary permutations.

Proof. Any such ultrafilter would give rise to a permutation-invariant strictly monotone total preorder on $2^{\mathcal{I}}$ (by theorem B.70), which we know cannot exist by corollary A.56. $\square$

Proposition A.64 gives us further constraints on the invariances that can be satisfied by an ultrafilter. The condition is not as strong as local finiteness, but it is related. (Roughly, it says that we never get infinite orbits just by combining finite orbits.)

On the other hand, it is easy to find permutation-invariant fine filters that are not ultrafilters. For example, this is true of the weak filter $\mathcal{F}_0$ (the weakest fine filter on $\text{Fin } \mathcal{I}$). There is a generally useful way of identifying invariances of symmetric filters.

First, let's provide another characterization of the weak G -symmetric filter.

Lemma B.92. Let $G \curvearrowright \mathcal{I}$ be a group action. For any finite subsets $F \in \text{Fin } G$ and $K \in \text{Fin } \mathcal{I}$, let

$$[F, K] = \{J \in \text{Fin } \mathcal{I} \mid J \supseteq K \text{ and } F \cdot J \subseteq J\}.$$

Let $\mathcal{F}_G$ be the weak G -symmetric filter on $\text{Fin } \mathcal{I}$. For every set $V \subseteq \text{Fin } \mathcal{I}$, $V \in \mathcal{F}_G$ iff

$$[F, K] \subseteq V \quad \text{for some } F \in \text{Fin } G \text{ and } K \in \text{Fin } \mathcal{I}.$$

Definition B.93. We say that a group action $H \curvearrowright \mathcal{I}$ *normalizes* another group action $G \curvearrowright \mathcal{I}$ iff, for any $g \in G, h \in H$, there is some $g^h \in G$ such that

$$h \cdot (g \cdot i) = g^h \cdot (h \cdot i) \quad \text{for each } i \in \mathcal{I}.$$

This implies:

$$h \cdot G \cdot i = G \cdot (h \cdot i).$$

So H takes G -orbits to G -orbits.

This definition can be stated more simply in the case where G and H are both subgroups of the permutation group on $\mathcal{I}$. In that case, we say the group H

normalizes the group G iff G is fixed by conjugation by elements of H :

$$h^{-1} \cdot G \cdot h = G \quad \text{for each } h \in H.$$

Then the action $H \curvearrowright \mathcal{I}$ normalizes $G \curvearrowright \mathcal{I}$ iff the group H normalizes G . In the special case where $G \subseteq H$, we say G is a **normal subgroup** of H .

Proposition B.94. If $H \curvearrowright \mathcal{I}$ normalizes $G \curvearrowright \mathcal{I}$, then the weak G -symmetric filter $\mathcal{F}_G$ is H -invariant.

Proof. Let $V \in \mathcal{F}_G$, and let $h \in H$. We will show that $h \cdot V \in \mathcal{F}_G$ as well.

By lemma B.92, there exists $K \in \text{Fin } \mathcal{I}$ and $F \in \text{Fin } G$ such that $[F, K] \subseteq V$. For each $g \in F$, choose $g^h \in G$ such that

$$h \cdot (g \cdot i) = g^h \cdot (h \cdot i) \quad \text{for each } i \in \mathcal{I},$$

and let

$$F^h = \{g^h : g \in F\} \in \text{Fin } G.$$

We will show:

$$[F^h, h \cdot K] \subseteq h \cdot [F, K] \subseteq h \cdot V,$$

from which it follows that $h \cdot V \in \mathcal{F}_G$.

Let $J \in [F^h, h \cdot K]$: that is, $J \supseteq h \cdot K$ and $F^h \cdot J \subseteq J$. We will show that $J \in h \cdot [F, K]$: that is, $h^{-1} \cdot J \supseteq K$ and $F \cdot (h^{-1} \cdot J) \subseteq h^{-1} \cdot J$. The first condition is clear. For the second condition, let $g \in F$ and $i \in h^{-1} \cdot J$. Then $h \cdot i \in J$, and since $F^h \cdot J \subseteq J$, we have

$$h \cdot (g \cdot i) = g^h \cdot (h \cdot i) \in J.$$

So indeed $g \cdot i \in h^{-1} \cdot J$, completing the proof. $\square$

Corollary B.95. If $\mathcal{F}$ is the weak rational-rotation-symmetric filter on the circle, the measure $\#\mathcal{F}$ is invariant under all rotations.

Proof. It can be checked that the group of rational rotations is a normal subgroup of the group of all rotations. $\square$

Corollary B.96. If $\mathcal{F}$ is the weak finitary coin flip symmetric filter, the measure $\#\mathcal{F}$ is invariant under the coin-permutation action of $\Pi(\omega)$ on the coin

flip space 2^ω . That is, for any $A, B \subseteq 2^\omega$ and any permutation $\pi : \omega \rightarrow \omega$, if $\#^{\mathcal{F}}(A) \geq \#^{\mathcal{F}}(B)$, then $\#^{\mathcal{F}}(\pi \cdot A) \geq \#^{\mathcal{F}}(\pi \cdot B)$.

Proof. It can be checked that the group of coin permutations normalizes the finitary coin flip group in $\Pi(2^\omega)$. In particular, for $\pi : \omega \rightarrow \omega$ and $\sigma_i : 2^\omega \rightarrow 2^\omega$,

$$\pi^{-1} \cdot \sigma_i \cdot \pi = \sigma_{\pi^{-1}(i)}. \quad \square$$

Note the contrast: neither in the case of arbitrary rotations nor in the case of coin permutations can we have full measure *symmetry*; but in both cases, we can still have measure *invariance*. (This is directly analogous to the boundary we found in section A.6: in many cases, strictly monotone G -symmetric orders are impossible, which also raises obstacles for strictly monotone G -invariant *total* orders; but G -invariant *preorders* can still be attained.)

Appendix C

Frameworks and Foundations for Prospects

Throughout this book, I have freely appealed to three different frameworks for representing risky prospects.

- Most fundamentally, we have considered comparisons of *counterfactual value*. Prospects are understood to be *propositions*. These propositions are ordered by a relation $A \succeq B$, intuitively saying that it *would have been as good* if A were true as if B were true.
- The *Savage-style* framework represents prospects as *random variables*, functions from a *state space* to an *outcome space*.
- The *von Neumann-Morgenstern (VNM)-style* framework represents prospects as *probability distributions* over outcomes.

In subsequent appendices, I will generally focus on value orders on random variables (à la Savage), or on probability distributions (à la VNM). But I ultimately wish to interpret results proved in those contexts as telling us something about the counterfactual value of propositions. So it is worth being explicit about how each of these frameworks works, how they are related to one another, and under what assumptions results proved in one framework can be extended to the others. This will also help clear up some general philosophical issues about how we are understanding prospects and their values.

To be clear: I do not think we have to choose between these frameworks.

All three of them are legitimate ways to represent prospects. Each of them builds in its own assumptions about the structure of value; in each case, I think (contrary to what some have claimed) these assumptions are modest and defensible.

C.1 Possible worlds

The interplay between epistemic and metaphysical modalities is very important to the theory of value, though this does not seem to have been generally appreciated very systematically. Would it have been better to save one for sure or take a chance at saving three? In particular, which would be better *by the lights of incomplete evidence*, which does not settle how the rescue attempt for three would really turn out? What is relevant to this question is not just whether the rescue *would save three*, but also whether it *might* be true that the rescue would save three, and more generally how *probable* it is that the rescue would save three. This question embeds the metaphysical modality (what *would* ensue, were the rescue attempted) under an epistemic modality (what *might* actually be the case).

As is customary, we represent propositions by sets of *possible worlds*. To be clear, though, these are not generally *metaphysically* possible worlds (ways the world genuinely could have been), nor are they generally *epistemically* possible worlds (ways the world might actually be). Rather, we will call them *conceptually* possible worlds — intuitively, specific scenarios which are coherent, even if they could not obtain, or certainly don't. These include both metaphysical and epistemic possibilities, as well as certain iterations — such as worlds that (epistemically) *might* be *metaphysically* possible, even though in fact they are impossible.

I will understand “*A* is metaphysically possible” as equivalent to “*A* does not counterfactually lead to absurdity”. In symbols: $\Diamond A \equiv \neg(A \Box \rightarrow \perp)$.

Eliot is Evans; there is a certain sense of *proposition* according to which the proposition that Eliot wrote *Middlemarch* is the same as the proposition that Evans wrote *Middlemarch*. But this kind of proposition is ill-suited to play the standard role of propositions in probability theory. For certain bodies of evidence, it is natural to say that the epistemic probability that Eliot wrote *Middlemarch* is different from the probability that Evans wrote *Middlemarch*. Accordingly, we will distinguish two different propositions here. These propositions are like the ones Frege conceived of as the senses of sentences: they are the bearers of epistemic probability and *ex ante* value. Likewise, we

will also recognize conceptually possible worlds in which (so to speak) Eliot writes *Middlemarch* while Evans does not. We represent *propositions* as sets of conceptually possible worlds.

Fregean propositions and conceptually possible worlds raise some difficult philosophical issues — for instance, what are they? But we need not plumb these depths in order to make use of them. What matters for our purposes is just the role they will play in a framework for representing the value of prospects.

Among these conceptually possible worlds, we also have *epistemically* possible worlds — those that are consistent with our evidence — and *metaphysically* possible worlds — those that could have obtained. But epistemic possibilities are not a special restricted subset of the metaphysical possibilities, and likewise the metaphysical possibilities are not a special subset of the epistemic possibilities. Rather, epistemic and metaphysical possibility are orthogonal to one another.

Here is a way of showing this. For any (Fregean) proposition A , we can introduce a name 1_A by the following speech:¹

Let $1_A = 1$ if A is true, and otherwise let $1_A = 0$.

Then the Fregean proposition

$$1_A = 1 \text{ iff } A$$

is epistemically certain. Also, if the proposition A is actually true, then $1_A = 1$ is true, and — since true identities are metaphysically necessary — $1_A = 1$ is metaphysically necessary. But if it is not *certain* whether A is true, then it is likewise uncertain whether $1_A = 1$ is true. (If A is a contingent truth, then $(1_A = 1 \text{ iff } A)$ is an example of the *contingent a priori*. Likewise, if A is an empirical truth, then $1_A = 1$ is an example of the *necessary a posteriori*. See Kripke 1980)

For short, let *actually* A be understood to mean $1_A = 1$ (where 1_A is introduced as above). If A is true in the actual world, then *actually* A is a metaphysical necessity, and if A is false in the actual world, then *actually* A a metaphysical impossibility.² If A is uncertain, then *actually* A is also

¹The trick here is the same as in Davidson ([1969] 2001)'s "slingshot" argument, though he put it to rather different use.

²The first of these claims — if A is true, then $1_A = 1$ is metaphysically necessary — follows from the necessity of identity. The second claim — if A is false, then $1_A = 1$ is impossible — relies on the (plausible) claim that $0 \neq 1$ is metaphysically necessary. (It is certain that if

uncertain, but metaphysically necessary. If A is metaphysically contingent, then $(A \text{ iff actually } A)$ is also metaphysically contingent, but epistemically certain.

In fact, there is *exactly one* conceptually possible world which is both epistemically and metaphysically possible: the actual world. Let w^* be the actual world, satisfying all truths. Every epistemic possibility other than w^* is metaphysically impossible. For if w is an epistemic possibility distinct from w^* , then there is some proposition A which is true at w^* but false at w . Since A is true at w^* , $(\text{actually } A)$ is true, and thus metaphysically necessary. Since w is epistemically possible, $(A \text{ iff actually } A)$ is true at w . By hypothesis, A is false at w , so $(\text{actually } A)$ is also false at w . Thus a necessarily true proposition is false at w ; so w is metaphysically impossible. Equivalently, every metaphysically possible world w other than w^* is epistemically impossible.

The epistemic possibilities and metaphysical possibilities are also not *exhaustive*. There are also merely “possibly possible worlds”: specifically, there are conceptually possible worlds that are both epistemically and metaphysically impossible, where it is *epistemically possible* that they are *metaphysically possible*. For example, suppose that we do not know whether Al and Nero are the same person; also, we are certain that Al is unhappy, and we are uncertain whether Nero is happy. There is a conceptually possible world w according to which *Al is happy and Nero is not happy*. This world w is epistemically impossible, since we are certain Al is not happy. If in fact Al and Nero are the same person (which may well be true), then w is also metaphysically impossible, since it is metaphysically necessary that if Al is happy then so is Nero. So w is neither epistemically nor metaphysically possible. But it is epistemically possible that Al and Nero are distinct, and if this is true, then it is metaphysically possible that Al is happy while Nero is not. So it is epistemically possible that w is metaphysically possible.

There are other further-flung worlds yet — such as merely *possibly* possible possibilities — but we will have no need to appeal to them. All of the worlds that are relevant to evaluating prospects are at least (epistemically) possibly (metaphysically) possible: it might be true that they could have obtained. For short, call these *EM-worlds*.

For each world w , there are certain propositions A such that w satisfies *actually* A . There is exactly one world $\alpha(w)$ which satisfies *every* proposition A such that w satisfies *actually* A . This is the world that is *actual according*

$1_A \neq 1$ then $1_A = 0$.)

to w . For any proposition A (understood as a set of worlds), the set $\alpha^{-1}(A)$, which contains all worlds w such that $\alpha(w) \in A$, is the proposition *actually* A . (This is because, by construction, *actually* A is true at w iff A is true at $\alpha(w)$.) Every *epistemically possible* world w satisfies $(A \text{ iff actually } A)$, and so in this case $\alpha(w) = w$. For each world w , the worlds in $\alpha^{-1}\{\alpha(w)\}$ — those worlds v such that $\alpha(v) = \alpha(w)$ — are the *metaphysically possible worlds according to* w .

In particular, each EM-world is in $\alpha^{-1}\{w\}$ for exactly one epistemically possible w . If v and w are distinct epistemically possible worlds, then no world is metaphysically possible according to both v and w . Each epistemic possibility has its own conflicting vision not just of the truth, but of possibility and necessity.

One popular framework for understanding the interplay of epistemic and metaphysical modalities is *two-dimensionalism* (Davies and Humberstone 1980; Chalmers 1996; F. Jackson 1998; see Kment 2021 for overview and further citations). There is a sense (which I will explain further below) in which the framework I have described is “two-dimensionalist”. But “two-dimensionalism” is also used as a label for an approach that goes well beyond what I have described, with the aim of reducing the full space of conceptually possible worlds to *metaphysically possible* worlds. Here’s the basic idea. For short, call metaphysically possible worlds *M-worlds*. Think of a Fregean proposition as a *sentence*. Intuitively, a sentence like “Al is Nero” would express different things in different M-worlds. Say we are reading both names off of two papers. There are M-worlds where both papers originated from the same person, and “Al” and “Nero” each refer to that person — and thus “Al is Nero” expresses a necessary truth. There are other M-worlds where the two papers originated from two different people, and “Al” refers to one of them and “Nero” refers to the other. In that case “Al is Nero” expresses a necessary falsehood.

We can think of each of the “possibly possible” conceptually possible worlds as an *ordered pair* of M-worlds (v, w) . To evaluate whether a sentence is true at (v, w) , we consider what the sentence would have *expressed* if it were deployed in v , and then consider whether what it would have expressed would be *true* at the world w . If v is a single-source-of-paperwork world, then “Al is Nero” is true at *every* pair (v, w) , for any M-world w . If v' is a two-source-of-paperwork world, then “Al is Nero” is false at every pair (v', w) , for any M-world w . The *epistemically possible* worlds are the *diagonal* pairs (w, w) , for any M-world w : intuitively, a sentence is epistemically certain iff no matter how it was deployed, it would have expressed a truth.

The conceptually possible worlds that represent *metaphysical* possibilities are the “horizontal” pairs (w^*, w) where w^* is the actual M-world and w is any M-world. We see that these two classes of conceptual possibilities cross-cut one another, intersecting only in the pair (w^*, w^*) .

My own view is that the two-dimensional picture can sometimes be a useful heuristic, but it should not be pushed too far. (The following argument is similar to one in Soames 2005.) What kind of thing is a *sentence*? If it is just a string of uninterpreted symbols, then the proposal gives the wrong results. For example, we could have used the string of symbols “ $1 = 1$ ” to mean that water is not wet. So there is a possible world in which we deploy that string to express something that is not necessarily true. There are pairs (v, w) of M-worlds such that at v “ $1 = 1$ ” means that water is not wet, and at w water is wet. So this predicts that it is epistemically possible that $1 = 1$ is not metaphysically necessary. This is wrong, though: it is epistemically certain that $1 = 1$ is metaphysically necessary.

Instead, it seems better to think of sentences as individuated by their *meanings*. But it is crucial to the two-dimensionalist’s proposal that our sentences could have expressed *different* propositions than they in fact do. In order to make sense of the proposal, we need a second notion of meaning, which gets in between a sentence and its so-called “secondary intension” (the worlds in which what the sentence actually says could have been true). Each of these special meanings determines an intension at each M-world. But the problem of saying what these special meanings are, and how they determine their intensions, is at least as difficult as the problem we began with — saying what Fregean propositions are. It is not clear what we have gained from this two-dimensionalist “reduction”.

There really is something “two-dimensional” about conceptual possibilities — at least when we restrict attention to the “possibly possible” EM-worlds. If we arrange the epistemic possibilities vertically, then we can split up the EM-worlds into horizontal rows of metaphysical “co-possibilities” — the equivalence classes under the relation $\alpha(v) = \alpha(w)$. Each horizontal row crosses the vertical at a single point. Thus we can, if we like, represent an arbitrary set of EM-worlds as a function that takes each epistemically possible world w to a “subjunctive” or “metaphysical” intension — a set of worlds that are metaphysically possible according to w , a subset of $\alpha^{-1}\{w\}$.

But the two-dimensionalist (of the kind I just described) also postulates a correspondence between the vertical and horizontal dimensions, and also that

each of the horizontal rows really represents the *same* space of metaphysical possibilities, differently interpreted. (This is necessary, for instance, to make sense of a *diagonal* across this two-dimensional space.) I don't see any advantage in making these further postulates, and I think they are potentially misleading.³

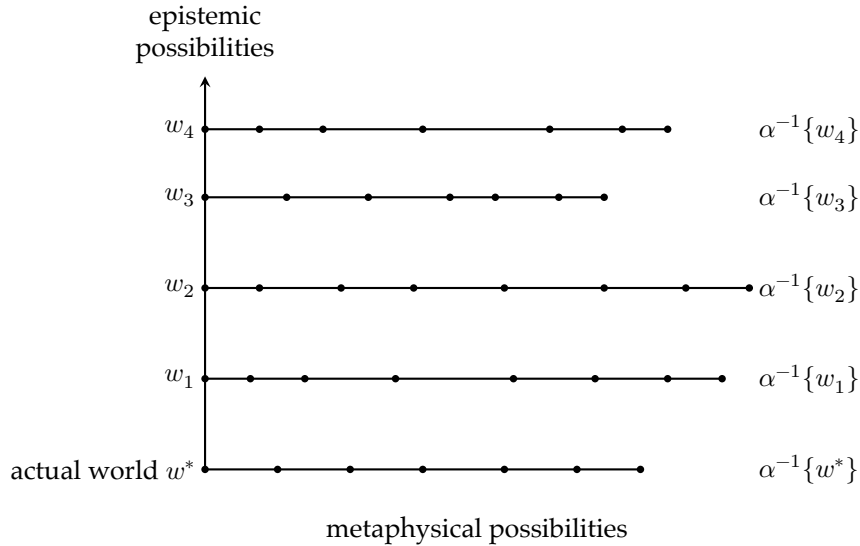


Figure C.1: The structure of EM-worlds: those conceptually possible worlds which might be metaphysically possible. For each epistemically possible world w , $\alpha^{-1}\{w\}$ is the set of worlds that are metaphysically possible according to w .

C.2 Modes of presentation

We have been particularly concerned with *social prospects*, which concern the well-being of individuals. In appendices A and B we represented a *social outcome* as a function from individuals to levels of well-being. We will want social prospects to present a range of different possible social outcomes, which can be associated with probabilities. But there is a complication: in section 7.3 we noted that what a prospect says about how a particular

³Chalmers (2002) characterizes two-dimensionalism in a more neutral way that is closer to the picture I have just described. In particular, that characterization is neutral on the relationship between epistemically possible “scenarios” and metaphysically possible worlds. But he still takes for granted that there is a *single* space of metaphysically possible worlds that can be combined with any epistemic possibility.

individual fares is sensitive to how that individual's "mode of presentation". Thus a social outcome is not really determined just by a conceptually possible world on its own, but rather in combination with a *mode of presentation* for each individual.

Very often we don't have to worry about this. We can simply assume that we have fixed *some* mode of presentation for each individual, and move on. But occasionally we have considered the same social prospect together with multiple different individual modes of presentation, and how these relate to one another. So it will be useful to have a basic formal framework for this. This will not be a full theory of modes of presentation adequate for every purpose, but just enough of one for our purposes.

Here is the basic framework. As before, we have some set $\mathcal{X}$ of *individual outcomes* or *well-being levels*.⁴ There is also some set $\mathcal{I}$ of individuals. The idea is that each conceptually possible world will assign an individual outcome to each individual — not on its own, but in conjunction with a *mode of presentation* for that individual. But also, while we can be uncertain about who is who, this is not to say that there is any *contingency* about who is who.⁵ Once we have figured out which individual we are talking about in a particular conceptually possible world w , this also determines what it is like in every every world that is metaphysically possible relative to w .

Here is a way of modeling this, within the framework of section C.1. Let Ω be the set of all conceptually possible worlds, and let $\Omega^* \subseteq \Omega$ be the set of epistemically possible worlds. As we discussed, there is a function $\alpha : \Omega \rightarrow \Omega^*$ that takes each world to an epistemically possible world; for $w \in \Omega^*$, $\alpha^{-1}\{w\}$ is the set of worlds that are metaphysically possible according to w . Let each epistemically possible world $w \in \Omega^*$ be associated with some set of *individual roles* $I(w)$. These are pairwise disjoint. (This is structurally analogous to the disjoint sets of metaphysically possible worlds for each epistemic possibility.)

⁴Individual outcomes can also be subject to epistemic confusion, and so in full generality we should also allow for them to have different modes of presentation. But we can ignore this complication.

⁵This may be oversimple. There are some good reasons to think that the necessity of *identity* and the necessity of *distinctness* come apart: while everything thing must have been itself, perhaps there could be two things which could have been one. (For discussion, see for example Bacon 2020.) This question is tied to the status of the modal principle B, which says that whatever is so is necessarily possible. Given B, the necessity of identity implies the necessity of distinctness. But, again, there are some good reasons to doubt B. For our purposes these are more complications we can safely ignore.

Each world $v \in \alpha^{-1}\{w\}$ determines a function $x_v : I(w) \rightarrow \mathcal{X}$. This is a kind of *proto-social outcome*: it says how well-being is distributed over individuals, in a way that does not yet settle which individuals those are. Call the function that takes each world $v \in \Omega$ to x_v a **social welfare profile**.

(I have generally set aside questions about population ethics, concerning either uncertainty or contingency about whether a given individual is anyone at all. We could accommodate this within the current framework in the usual way, but including a “null outcome” in $\mathcal{X}$ that represents not being anyone.)

Some world $w^* \in \Omega^*$ is *actual* — it represents the way things really are. Each actual individual $i \in \mathcal{I}$ corresponds to some role in $I(w^*)$. Formally, we might identify $\mathcal{I}$ with $I(w^*)$, but this is a bit philosophically misleading. (That suggests that the roles in *other* live worlds are some kind of *alternative* individuals. But they aren’t: they represent other ways individuals might actually be.) For clarity, we will posit a bijection $i^* : I(w^*) \rightarrow \mathcal{I}$ that takes each role in the actual world to the individual that actually occupies that role.

An **individual mode of presentation** is any function M that takes each epistemically possible world $w \in \Omega^*$ to some individual role $M(w) \in I(w)$. The individual **presented** by M is $i^*(M(w^*))$. Suppose we have a social welfare profile x , which assigns individual outcomes to individual roles in each world. Then an individual mode of presentation M determines a function $x^M : \Omega \rightarrow \mathcal{X}$ from worlds to individual outcomes:

$$x^M(w) = x_w(M(\alpha(w))).$$

Identities can be uncertain: it can be uncertain whether an individual under a mode of presentation M is the same as an individual under another mode of presentation M' . This corresponds to the case where $M(w) = M'(w)$ for some epistemic possibilities w , but not for all of them. But identities are not metaphysically contingent. If $M(w) = M'(w)$, by construction the individual picked out by M at any world v that is metaphysically possible according to w is $M(\alpha(v)) = M(w)$; and likewise $M'(\alpha(v)) = M'(w)$. So the framework correctly capture claims like this one:

Jay and Kay might be distinct, but if Jay is Kay, then, necessarily,
if Jay were happy Kay would also be happy.

A **sequential mode of presentation** (see section 7.3) is a sequence of individual modes of presentation $M = (M_1, M_2, \dots)$ that are certain to represent

distinct individuals: $M_k(w) \neq M_{k'}(w)$ for each $k \neq k'$ and each epistemic possibility $w \in \Omega^*$. Let M be a sequential mode of presentation. Let $\mathcal{I}$ be the set of individuals actually presented by M :

$$\mathcal{I} = \{i^*(M_k(w^*)) \mid k = 1, 2, \dots\}.$$

Each M_k determines an $\mathcal{X}$ -intension. We can put all of these together, so M determines a *social outcome intension*, a function $x^M : \Omega \rightarrow \mathcal{X}^{\mathcal{I}}$ from conceptually possible worlds to *social outcomes*.

$$x^M(w)(i) := x_w(M_k(\alpha(w))) \text{ where } i = i^*(M_k(w^*)).$$

That is, if i is the individual actually presented by M_k , then $x^M(w)(i)$ is the individual outcome for the M -individual in w , which is the same as the outcome in w for the individual in the role $M_k(\alpha(w))$.

Modes of presentation aren't just for individuals. We can alternatively think of a sequential mode of presentation not as a *sequence* of individual modes of presentation, but instead as a *mode of presentation* for a *sequence* of individuals: M takes each epistemically possible world w to a sequence of individual roles

$$M(w) = (M_1(w), M_2(w), \dots).$$

Similarly, we can think of M as determining a mode of presentation for a *social outcome*: a function from epistemically possible worlds to sequences of individual outcomes.

In section C.1, we used the trick of conditionally introducing a name: "If A is (in fact) true, let $1_A = 1$, and otherwise let $1_A = 0$." This had the effect of introducing a mode of presentation for a number (which is either 1 or 0), with special epistemic properties: we can be sure that A is true iff $1_A = 1$. This trick is very general. Let $E_1, E_2, \dots$ be a *partition* of the space of epistemically possible worlds, and let $a_1, a_2, \dots$ be names for individuals. Then we can introduce a new name by proclaiming:

Let a^* be a_k , where k is the unique index such that E_k is (actually) true.

This rigidly picks out some individual, though we do not know which one. We can now be certain that, if E_1 is true, then $a^* = a_1$ — and that this holds with metaphysical necessity — and likewise for 2, 3, These *names* correspond to certain modes of presentation. In general, let $M = (M_1, M_2, \dots)$ be a sequential mode of presentation, and let $E_1, E_2, \dots$ be a

partition of epistemically possible worlds. Then we can introduce a new individual mode of presentation M^* , where in each epistemically possible world w , if $w \in E_k$, then $M^*(w) = M_k(w)$.

In chapter 7 we introduced the idea of modes of presentation that are related to one another by *statewise permutations* of individuals. This is an extension of this same idea. For each $w \in \Omega^*$, let π_w be a permutation of $\mathcal{I}$, the set of actual individuals. For each $i \in \mathcal{I}$, let M_i be a mode of presentation for i (so $M_i(w^*) = i$). Then we can introduce a new family of modes of presentation M_i^π such that, for each $w \in \Omega^*$,

$$M_i^\pi(w) = M_{\pi_w(i)}(w).$$

Note that M_i^π may not be a mode of presentation for the individual i — rather, it is a mode of presentation for $\pi_{w^*}(i)$, which in general may be different. But collectively, the modes of presentation M_i^π present the same set of individuals as the modes of presentation M_i , just rearranged by π . Furthermore, we can think of M_i^π as arising from conditionally introducing a name. Let E_i be the set of epistemically possible worlds w such that $\pi_w(i) = i'$. Then M_i^π can be introduced thus:

Let M_i^π pick out the same individual as $M_{i'}$ if $E_{i'}$ is true.

Thus any family of modes of presentation M for the individuals $\mathcal{I}$ gives rise to a new family of modes of presentation M^π for the same individuals, related to M by statewise permutation. In section 7.4 we discussed how this trick gets us from the Ex Ante Pareto principle to a stronger principle, *Permuted Pareto*.

As we noted in section C.1, in addition to the Fregean “epistemic” notion of proposition we have foregrounded — the bearer of epistemic probability and *ex ante* value — there is also a “metaphysical” notion of a proposition. Since Evans is Eliot, for Eliot to write *Middlemarch* just is for Evans to write *Middlemarch*. To avoid ambiguity, I’ll call these “metaphysical propositions” *states of affairs*.⁶ There are two distinct Fregean propositions *Eliot wrote Middlemarch* and *Evans wrote Middlemarch*, which correspond to two different modes of presentation *Eliot* and *Evans* for the same individual. We can think of these two propositions as two different modes of presentation for the same *state of affairs* — Eliot writing *Middlemarch*, which is the very same entity as Evans writing *Middlemarch*.

⁶This is the notion of proposition that is central to higher-order metaphysics.

This can also be captured within our formal framework. We can model a state of affairs as a *metaphysical intension* — a set of *metaphysically possible worlds*.⁷ Just as we can be uncertain about which individuals are which, we can also be uncertain about which states of affairs are which. Just as an individual mode of presentation is a function from epistemically possible worlds to individual roles, a **propositional mode of presentation** is a function from epistemically possible worlds to “propositional roles”. In each epistemically possible world w , the propositional roles are the sets of metaphysically possible worlds *according to* w , subsets of $\alpha^{-1}\{w\}$. Again, these sets should be thought of as epistemically possible *roles* for the actual states of affairs, rather than as *alternative* states of affairs.

These propositional modes of presentation correspond straightforwardly to the things we have called Fregean propositions — sets of conceptually possible worlds. For any set of possible worlds A , there is a corresponding function

$$M(w) = A \cap \alpha^{-1}\{w\} \quad \text{for each } w \in \Omega^*.$$

This mapping from subsets of Ω to propositional modes of presentation is a one-to-one correspondence. The two objects are essentially different formal perspectives on the very same thing. The state of affairs *presented* by M is the set of metaphysically possible worlds $M(w^*)$.

I have treated Fregean propositions as the things involved in *ex ante* value comparisons. But it might be more perspicuous to instead think of *better* as a relation between *states of affairs*, which is *presentation-sensitive*. But this does not really make any formal difference: I am identifying Fregean propositions with propositional modes of presentation, and each of these presents a unique state of affairs. So we can easily translate back and forth between, on the one hand, saying $A \succsim B$, where A and B are Fregean propositions, and on the other hand saying $p^M \succsim^{M'} q$, where p and q are states of affairs and M and M' are modes of presentation for them. The first regimentation is technically more convenient, but the second I think is a bit more natural. (I don’t think we have any particularly convenient way of expressing or naming Fregean propositions in natural language, though.)

Now that we have clarified these issues, for the most part we can ignore them. Once we have fixed some sequential mode of presentation for the

⁷This, again, may be oversimple. The question of metaphysical propositions’ *fineness of grain* has been the subject of much recent philosophy, and nowadays there are many different accounts to choose from. (See Bacon 2023.) *Intensionalism* is one view among these, which is at least suitable as a working model for present purposes.

individuals involved in a social prospect, then each conceptually possible world straightforwardly determines a social outcome; we don't always have to be coming back to *how* it does this. So modes of presentation can generally remain in the background. The only time we really need to pay attention to them is when we are considering how things look under multiple *different* modes of presentation.

C.3 The counterfactual framework

As I said, our central value question about a prospect A is about how good it *would* have been if A , in light of certain incomplete evidence — a question that involves both metaphysical and epistemic modalities. Let's consider how to regiment such questions.

We will adopt a familiar approach to modeling counterfactuals using *selection functions* (Stalnaker 1968). (This incorporates some opinionated claims about the logic of counterfactuals.)

Definition C.1. A **counterfactual value structure** consists of the following components:

- (a) A set Ω of **(conceptually) possible worlds**. We call subsets of Ω **propositions** (or sometimes, traditionally, *events*).
- (b) A set $\mathcal{A}$ of non-empty subsets of Ω , whose elements are called **available prospects** (or sometimes, traditionally, *acts*).
- (c) A non-empty subset $\Omega^* \subseteq \Omega$, the set of **live epistemic possibilities**.
- (d) A **selection function** f , which takes each $A \in \mathcal{A}$ to a function $f_A : \Omega^* \rightarrow A$.
- (e) A set $\mathcal{E}$ of non-empty subsets of Ω^* , whose elements are called **consistent suppositions**. We suppose that $\Omega^* \in \mathcal{E}$, and $\{w\} \in \mathcal{E}$ for each $w \in \Omega^*$.
- (f) A four-place relation $(-|-) \succsim (-|-)$. For $A, B \in \mathcal{A}$ and $E, F \in \mathcal{E}$, $(A|E) \succsim (B|F)$, is read as

It would have been as good for A to be true, if in fact E , as it would have been for B to be true, if in fact F .

This is a preorder on $\mathcal{A} \times \mathcal{E}$. We write $A \succsim_E B$ for $(A|E) \succsim (B|E)$, and

we write $A \succsim B$ for $A \succsim_{\Omega^*} B$.

This is required to satisfy the following property:

For any $A, B \in \mathcal{A}$ and any $v, w \in \Omega^*$, if $f_A(v) = f_B(w)$ then $(A \mid \{v\}) \sim (B \mid \{w\})$.

Note that we only require the selection function to be defined for available prospects and epistemically live worlds. It could be extended to other propositions and worlds (and this would be necessary to evaluate nested counterfactuals). But for the sake of economy, we have restricted the function just to the cases we need the purposes of value theory.

The general four-place form of the value relation is used here for completeness, but we will generally restrict attention to the simpler unconditional evaluations $A \succsim B$.

Given a counterfactual value structure, we can define a two-place **counterfactual** propositional operator $\Box \rightarrow$: for $A \in \mathcal{A}$ and $B \subseteq \Omega$, we define

$$A \Box \rightarrow B := f_A^{-1}(B) = \{w \in \Omega^* \mid f_A(w) \in B\} \subseteq \Omega^*.$$

This model ensures that certain principles of counterfactual logic hold. Notably, for any $A \in \mathcal{A}$ and $B \subseteq \Omega$,

$$(A \Box \rightarrow B) \cup (A \Box \rightarrow \neg B) = \Omega^*.$$

That is, the principle of *counterfactual excluded middle* is taken to be epistemically necessary. I accept this principle, but it is rather controversial (see D. Lewis 1973, sec. 3.4; Stalnaker 1981). Those who wish to avoid it will have to complicate the framework somewhat, in ways that I will not pursue, but which I do not think affect anything very central to the project of this book. (In particular, they may still be able to avail themselves of the VNM framework, but they will have to justify this in a somewhat different way than the one I am sketching.)

We can define a set of **feasible worlds**: these are the worlds that *might* be what would result from some available prospect.

$$\Omega^\diamond := \{f_A(w) \mid w \in \Omega^*, A \in \mathcal{A}\} \subseteq \Omega.$$

The feasible worlds inherit a value order from the order of prospects: for $v, w \in \Omega^*$ and $A, B \in \mathcal{A}$,

$$f_A(v) \succsim f_B(w) \quad \text{iff} \quad (A \mid \{v\}) \succsim (B \mid \{w\}).$$

Let an **outcome proposition** be any equivalence class of worlds under the corresponding equivalence relation. For $w \in \Omega^\diamond$, let $O(w)$ be the unique outcome proposition containing w . Intuitively, this is the set of all conceptually possible worlds that would be just as good as w .

It is interesting to note that the Jeffrey–Bolker style framework for *evidential* value (“news value”) can be found within this counterfactual framework (Jeffrey 1983; Bolker 1966). Suppose Ω^* is included as a trivial “act” in $\mathcal{A}$ — “things are just as they are”. Then the news-value comparison “It is as good if (actually) E as if (actually) F ” can be understood as

$$(\Omega^* \mid E) \succeq (\Omega^* \mid F).$$

“It would be as good if things were just as they are, supposing in fact E , as if things were just as they are, supposing in fact F .” Thus this “counterfactual value” framework really encompasses both counterfactual and evidential value, and the interactions between them.

Which propositions count as “available prospects”? For some purposes it is convenient to be economical, including just a few prospects that are immediately relevant. For others — especially when it comes to representation theorems — it is instead convenient to be more extravagant, including some sufficiently *rich* space of propositions, perhaps even *all* the propositions which it makes any sense at all to evaluate. Which ones are these?

Crucially, to make sense of the question of how good it would be if A were true, it must at least be coherent to consider counterfactually how things would be if A were true. We do not want to include any hypotheses that counterfactually lead to absurdity. Equivalently, we require that A is *metaphysically possible*. (Recall that I am taking “ A is metaphysically possible” to be equivalent to $\neg(A \Box \rightarrow \perp)$.)

But we require a bit more than this. For every epistemically possible world w , we wish to make sense of how A would have turned out, supposing w actually obtains. For each available A and epistemically possible w , we are supposing that $f_A(w)$ is a conceptually possible world — and thus not absurd. This amounts to requiring that it is *epistemically certain* that A is metaphysically possible.

We might sometimes wish to consider possible acts that do not enjoy this status — which might, for all we are certain of, turn out to be metaphysically impossible. Someone might, say, consider whether how good it would be to visit Hesperus without visiting Phosphorus. But if Hesperus is Phosphorus,

this can't be done, and so there is nothing very useful to say about how good it would be to do it. I think what we should really be considering in these cases is some fallback proposition: we can consider how good it would be to do this *supposing it is possible*, and otherwise doing something trivial.⁸

So any proposition that is “available” for evaluation is certainly metaphysically possible. It will be useful to consider certain counterfactual value models which include many propositions like this. To capture this, it is useful to directly model metaphysical possibility via a feature introduced in section C.1 — the *actuality* function.

Definition C.2. A **counterfactual value structure with actuality** is a counterfactual value structure $\mathcal{S} = (\Omega, \mathcal{A}, \Omega^*, f, \mathcal{E}, \succeq)$ equipped with an **actuality function** $\alpha : \Omega \rightarrow \Omega$, with the following two properties.

- (a) For every $w \in \Omega^*$ and $A \in \mathcal{A}$,

$$\alpha(f_A(w)) = w;$$

- (b) For every $w \in \Omega^*$ and $A, B \in \mathcal{A}$,

$$\text{if } A \cap \alpha^{-1}\{w\} = B \cap \alpha^{-1}\{w\} \text{ then } f_A(w) = f_B(w).$$

Intuitively, condition (a) requires that (with certainty) counterfactually accessible worlds are metaphysically possible. Recall that for any epistemically possible world w , $\alpha^{-1}\{w\}$ is the set of conceptually possible worlds that w says are metaphysically possible. So a proposition $A \subseteq \Omega$ is metaphysically possible according to w iff $w \in \alpha(A)$ (the image of A under α). A proposition A is *certainly* metaphysically possible iff $\alpha(A) \supseteq \Omega^*$. Thus, the constraint on the actuality function ensures that each available proposition is certainly metaphysically possible: for each $w \in \Omega^*$, $\alpha(f_A(w)) = w$.

Condition (b) intuitively requires that (with certainty) the counterfactual is a *metaphysical* modality, which is not “hyperintensional”. If two propositions A and B have the same intension — they are true at the same metaphysically possible worlds — then they are also counterfactually equivalent — they would have the same counterfactual consequences.

⁸If we like, we can generalize our framework a bit. Instead of requiring each of our available acts to be possible at *every* epistemic possibility, we could instead restrict the value relation. More minimally, we could say that in general f_A can be a *partial* function on Ω^* , and require that, if $(A|E) \succeq (B|F)$, then f_A is defined on E and f_B is defined on F . But this adds complications that are irrelevant to our purposes.

We will sometimes want to consider *rich* structures which include lots of propositions, as long as they all have this property. In its strong form, we might require that *every* proposition that is certainly metaphysically possible is included as an available prospect. In its weaker form, we might just require that we include every such proposition out of some suitable class, which is closed under logical operators. We can articulate both requirements.

Definition C.3. Let $\mathcal{S} = (\Omega, \mathcal{A}, \Omega^*, f, \mathcal{E}, \succeq, \alpha)$ be a counterfactual value structure with actuality.

- (a) We say $\mathcal{S}$ is **full** iff $\mathcal{A}$ includes every proposition $A \subseteq \Omega$ such that $\alpha(A) \supseteq \Omega^*$.
- (b) Let $\mathcal{B}$ be a Boolean algebra of propositions containing $\mathcal{A}$. We say $\mathcal{S}$ is **$\mathcal{B}$ -rich** iff $\mathcal{A}$ includes every $A \in \mathcal{B}$ such that $\alpha(A) \supseteq \Omega^*$.

C.4 The Savage framework

We have often represented prospects using tables, like this one:

	Success	Failure
Safe	Save 1 life	Save 1 life
Risky	Save 3 lives	Save 0 lives

This amounts to representing each prospect as a function from certain “states” (like “Success”) to “outcomes” (like “Save 1 life”). This corresponds to the formal framework of Savage (1954) — though I interpret this framework in somewhat different terms from Savage.

When we use these tables, we implicitly assume that each available prospect can be represented by some function from states to outcomes, in a way that includes enough information to settle questions about which is better. Another assumption we have often implicitly made is that we have a fairly free hand in constructing these tables without losing contact with meaningful value comparisons. This is a “richness” assumption, about how we can mix and match states and outcomes.

What are states and outcomes? I will think of them as *features* of worlds. If possible worlds are fully specific conceptual possibilities, states and out-

comes are each somewhat less specific possibilities, each of which include certain relevant kinds of details and omit others. If you like, you can think of states and outcomes as *sets* of possible worlds. In practice, it is often more convenient to be more liberal about what sort of things they are: for example, it is often convenient to let outcomes be numbers, or distributions of well-being, or utility streams. For our purposes, it doesn't really matter what sort of thing they are; the important thing is what we use them to represent. Let's take outcomes and states in turn.

The purpose of outcomes is to represent *how good* a world is. This is not the *ex ante* value of a prospect, but rather the *ex post* value of how it might turn out. Accordingly, we can let the set of outcomes be any set $\mathcal{X}$, equipped with a function $O : \Omega \rightarrow \mathcal{X}$ that takes each world to its corresponding outcome. The key feature of this function is that if two worlds are equally good, then they have the same outcome.

The purpose of states is to represent those features the world might actually have which settle how each prospect would turn out. How fine-grained the states have to be depends on how rich the space of available prospects is, which we want to evaluate.

Definition C.4. A **Savage model** consists of the following:

- (a) A set Ω , called the **state space**;
- (b) A set $\mathcal{X}$ of **outcomes**;
- (c) A set $\mathcal{A}$ of functions from Ω to $\mathcal{X}$, the **Savage acts**;
- (d) A preorder $\succeq$ on $\mathcal{A}$.

It is also common to require that the state space Ω be equipped with a sigma algebra of *measurable* sets, and similarly for the outcome space $\mathcal{X}$. We would then require further that each Savage act is a *measurable* function (see appendix E). I will omit this, but it could be easily added back in.

Definition C.5. Let $\mathcal{S} = (\Omega, \mathcal{A}, \Omega^*, f, \mathcal{E}, \succeq)$ be a counterfactual value structure. A **Savage representation** of $\mathcal{S}$ consists of a Savage model $(\Omega', \mathcal{X}, \mathcal{A}', \succeq')$ together with functions

$$S : \Omega^* \rightarrow \Omega' \quad O : \Omega^\diamond \rightarrow \mathcal{X} \quad a : \mathcal{A} \rightarrow \mathcal{A}'$$

with the following properties:

$$\begin{aligned} a(A)(S(w)) &= O(f_A(w)) && \text{for every } A \in \mathcal{A} \text{ and } w \in \Omega^* \\ a(A) \succeq' a(B) &\text{ iff } A \succeq B && \text{for every } A, B \in \mathcal{A}. \end{aligned}$$

Theorem C.6. Let $\mathcal{S}$ be a counterfactual value structure satisfying the following condition:

Statewise Equivalence. Let $A, B \in \mathcal{A}$, and suppose $f_A(w) \sim f_B(w)$ for all $w \in \Omega^*$. Then $A \sim B$.

Then $\mathcal{S}$ has a Savage representation.

Proof. First, recall from section C.3 that for $w \in \Omega^\diamond$, $O(w)$ is the equivalence class of w under the relation of being equally good worlds. Let $\mathcal{X}$ be the set of all such equivalence classes.

Second, for $v, w \in \Omega^*$, say v and w are *counterfactually equivalent* iff

$$f_A(v) \sim f_A(w) \quad \text{for every } A \in \mathcal{A}.$$

Let $S : \Omega^* \rightarrow \Omega'$ be the quotient mapping arising from this equivalence relation.

Thus, for any $A \in \mathcal{A}$, and any $v, w \in \Omega^*$, if $S(v) = S(w)$, then $O(f_A(v)) = O(f_A(w))$. It follows that there exists a function $g : \Omega' \rightarrow \mathcal{X}$ such that

$$g(S(w)) := O(f_A(w)) \quad \text{for every } w \in \Omega^*.$$

We can let $a(A) = g$. Let

$$\mathcal{A}' := \{a(A) \mid A \in \mathcal{A}\}.$$

Finally, note that for any $A, B \in \mathcal{A}$, if $a(A) = a(B)$, then for every $w \in \Omega^*$,

$$O(f_A(w)) = O(f_B(w)).$$

Thus $A \sim B$ in the counterfactual value structure, by Statewise Equivalence. This means we can define a preorder on $\mathcal{A}'$ by

$$a(A) \succeq' a(B) \quad \text{iff} \quad A \succeq B \quad \text{for all } A, B \in \mathcal{A}.$$

Call this the **canonical Savage representation** of $\mathcal{S}$. □

It is also often useful to make *richness* assumptions about Savage acts (especially for the purpose of proving representation theorems). In its strongest and simplest form, we might want this:

Savage Richness. Every (measurable) function $f : \Omega \rightarrow \mathcal{X}$ from states to outcomes is in $\mathcal{A}$.

(This is sometimes called the *rectangular field* assumption.) Or we might impose these much more modest richness assumptions:

Constant Acts. For every outcome $x \in \mathcal{X}$, the function that is constantly x for every state is in $\mathcal{A}$.

Piecewise Acts. For any acts $f, g \in \mathcal{A}$ and any (measurable) $E \subseteq \Omega$, the function

$$(f \sqcup_E g)(w) := \begin{cases} f(w) & \text{for } w \in E \\ g(w) & \text{otherwise} \end{cases}$$

is in $\mathcal{A}$.

Even these more modest richness assumptions have been criticized. Joyce (1999, 107–8) writes:

To illustrate just how strong a requirement this is, imagine some exceptionally desirable future course of events O — world peace breaks out, cures for all known diseases are found, your children marry well, you win the Nobel prize, or whatever else suits your fancy. Now, let S be the state in which all living things on earth are destroyed at noon tomorrow by the impact of a huge asteroid. If [Constant Acts] is correct, then there must be some course of future events that is consistent with S and that would be just as good, from your perspective, as the wonderful outcome O . In fact, there has to be some contingent proposition A_O whose truth will guarantee you this wonderful outcome if S , or anything else, occurs. Savage calls A_O an act, of course, but ... it cannot be identified with something anyone might *do*. However, this is only part of the trouble.

As Joyce acknowledges, we can make sense of assigning values even to propositions that are not feasible courses of action.⁹ So we might consider

⁹Joyce calls requirements that *preferences* apply even to unfeasible propositions requirements of *coherent extendibility*. We are here focused on *value* rather than preference; it is plausible that the value relation *already* applies even to coherent but unfeasible propositions, without needing to be *extended* to them.

constant acts to be among these: not things anyone might *do*, but still ways things could be, which we consider better or worse than other ways things could be. But, Joyce argues, constant acts don't even make sense in this way. He goes on:

The real problem is that A_O might describe an *impossible* state of affairs. It would not at all be irrational for you to have desires that make it impossible for anything that happens before noon tomorrow to compensate you for the asteroid's devastating effects. If your desires are like this, then there will be nothing that can perform [a constant act's] function: there will be *no* proposition that is consistent with both the destruction of the world and all the wonderful things that O promises whose truth would make you indifferent between the two prospects.

I think this objection can be answered. Here is a place we must carefully distinguish between *epistemic* and *metaphysical* possibility (in a way that Joyce's reconstruction of Savage does not). In what sense must the act proposition A_O be *consistent* with the state proposition S ? These need not be *epistemically* consistent, in the sense that it is a live possibility that the two propositions are both true. We may be perfectly confident that the wonderful outcome O will *not* obtain, while still considering it a very good thing if it *were* to obtain. They also need not be *metaphysically* consistent, in the sense that both propositions *could have* been true together. Indeed, if I know that O will not obtain, then every epistemically possible state entails not- O , and thus O is not metaphysically compossible with any state. Again, this does not preclude considering how good it would be if O *were* to obtain.

(For a different example, suppose that Al and Faro are the same person, but Chloe does not know this. It is a live possibility for Chloe that Al and Faro are distinct, but this is in fact metaphysically impossible. In this case, we should countenance epistemically possible states S' according to which Al and Faro are distinct, and Chloe might consider what would result from her acts in such states. But no proposition Chloe could make true is metaphysically compossible with S' — nor is any other proposition, for that matter.)

The sense in which the state S and the "act" A_O really must be "consistent" is rather subtle. The requirement is that it is an *epistemically* live possibility that both (i) S is *actually true* and (ii) A_O is *metaphysically possible*. We need to be able to consider how things *would* be if A_O were true, on the supposition that S is *actually true*. And this is a perfectly coherent thing to do. Indeed, we can let the "act" A_O be the proposition O itself. Suppose that *in fact* the

world will be destroyed by an asteroid tomorrow. On this supposition, how good *would* things have been if all of the wonderful things promised by *O* occurred—instead? The straightforward answer is that it would be very good. We do not need to make the forbidding cognitive leap to imagining the asteroid impact and *O* both happening *together*. Rather, we imagine we are *actually* in a world of devastation, and then (as it were) imagine ourselves *imagining* how much better the world could have been instead.

(This relates to a different way in which my interpretation of the Savage framework differs from Joyce's. Joyce identifies acts with certain *counterfactual conditional* propositions, and considers the outcome of an act *A* in a state *S* to be what follows from *A* and *S* both being true together. I consider acts to be *arbitrary* metaphysically possible propositions, and the outcome of *A* is what *would* be true if *A* were true, on the supposition that *S* is *actually* true.)

As we discussed in section C.3, the constraint on an available prospect *A* is just that it is *epistemically certain* that *A* is *metaphysically possible*. This allows us, for each way the world might *actually* be, to make sense of how things *would* be if *A* were true. This is compatible with being certain that *A* is *actually* false, and thus that it is not in fact true in any possible state.

Constant Acts still imposes a substantive requirement here, but it is very modest and defensible. The requirement is that each feasible outcome is certainly metaphysically possible. In other words, for any way it *might* be that one of our available acts *would* turn out, it is *certain* that things *could have been* just as good as that. In terms of the model:

For any epistemically possible worlds *v* and *w* and any available proposition *A*, there is some world *v'* which is metaphysically possible according to *v* which is equally good as $f_A(w)$.

We can write the requirement more briefly in terms of structures with an actuality function:

$$\alpha(O(w)) \supseteq \Omega^* \quad \text{for each } w \in \Omega^\diamond.$$

This condition is straightforwardly satisfied in cases where there is no relevant uncertainty about which goods are metaphysically possible — something which is ordinarily taken for granted. I do not think this assumption really holds if we are taking into account *every* proposition that we might conceivably evaluate. But if we modestly restrict attention to certain prospects of the sort we would ordinarily be interested in comparing, then it is eminently reasonable to suppose that there is no relevant uncertainty about

what possible degrees of goodness are metaphysically possible. So Constant Acts has the status of a very modest simplifying assumption.

Proposition C.7. Call a counterfactual value structure δ **outcome-rich** iff, for each feasible world $w \in \Omega^\diamond$, the outcome proposition $O(w)$ is in the set A of available prospects. If δ is outcome-rich, then the canonical Savage representation of δ satisfies Constant Acts.

We can also defend Piecewise Acts. This also turns on carefully navigating the interplay between epistemic and metaphysical possibility. Recall, Piecewise Acts says that for any acts $f, g \in A$ and any proposition $E \subseteq \Omega$, there is also a piecewise act $f \sqcup_E g \in A$. Suppose that the acts f and g represent available propositions A and B , respectively. It is tempting to invoke this proposition for the piecewise act:

Either (A and E) or else (B and not- E).

But this doesn't quite work. Again, it might be that A and B are both known to be *false*. In that case, this complex proposition would be known to be metaphysically impossible, and so it cannot be an available proposition.

The trick is to invoke “actually” again. (Recall from section C.1 that *actually* A was defined as the proposition $1_A = 1$, where 1_A was introduced to be a name for 1 if A is true, and otherwise 0.) Let A and B be propositions which are known to be metaphysically possible, and let E be any proposition. Now consider the following proposition:

Either (A and actually E), or else (B and actually not- E).

Call this $A \sqcup_E B$. For each epistemically live world w , if E is true in w , then $A \sqcup_E B$ *would* be true at precisely those possible worlds where A is true. If E is false at w , then $A \sqcup_E B$ *would* be true at precisely those possible worlds where B is true. Thus $A \sqcup_E B$ is also known to be metaphysically possible, and it has the same outcome as A in each E world, and it has the same outcome as B in each not- E world.

Recall that in our structures with actuality, for any proposition A , the proposition *actually* A is represented by $\alpha^{-1}(A)$.

Proposition C.8. Let $\delta = (\Omega, A, \Omega^*, f, \mathcal{E}, \succsim, \alpha)$ be a counterfactual value structure with actuality. Let $\mathcal{B}$ be a Boolean algebra of propositions including A , which includes each proposition $\alpha^{-1}(E)$ for $E \subseteq \Omega^*$. If δ is $\mathcal{B}$ -rich, then

the canonical Savage representation for $\mathcal{S}$ satisfies Piecewise Acts.

Proof. Consider the canonical Savage representation of $\mathcal{S}$,

$$\begin{aligned} &(\Omega', \mathcal{X}, A', \succeq') \\ &S : \Omega^* \rightarrow \Omega' \\ &O : \Omega^\diamond \rightarrow \mathcal{X} \\ &a : A \rightarrow A'. \end{aligned}$$

Let $f, g \in A'$ be Savage acts. Then there are available prospects $A, B \in \mathcal{A}$ such that $a(A) = f$ and $a(B) = g$. So $\alpha(A) \supseteq \Omega^*$ and $\alpha(B) \supseteq \Omega^*$. Let $F \subseteq \Omega'$ be a set of states. Let $E = S^{-1}(F) \subseteq \Omega^*$. Thus $\alpha^{-1}(E)$ and $\alpha^{-1}(\Omega^* \setminus E)$ are in $\mathcal{B}$. Consider

$$C := (A \cap \alpha^{-1}(E)) \cup (B \cap \alpha^{-1}(\Omega^* \setminus E)).$$

Then C is in $\mathcal{B}$. Furthermore,

$$\alpha(C) = (\alpha(A) \cap E) \cup (\alpha(B) \cap (\Omega^* \setminus E)) \supseteq (\Omega^* \cap E) \cup (\Omega^* \cap (\Omega^* \setminus E)) = \Omega^*.$$

So by $\mathcal{B}$ -richness, C is in $\mathcal{A}$.

Finally, we check that $a(C) = f \sqcup_E g$. First, note that

$$\begin{aligned} C \cap \alpha^{-1}\{w\} &= A \cap \alpha^{-1}\{w\} && \text{for } w \in E \\ C \cap \alpha^{-1}\{w\} &= B \cap \alpha^{-1}\{w\} && \text{for } w \in \Omega^* \setminus E. \end{aligned}$$

By the intensionality condition on counterfactual value structures with actuality,

$$\begin{aligned} f_C(w) &= f_A(w) && \text{for } w \in E \\ f_C(w) &= f_B(w) && \text{for } w \in \Omega^* \setminus E. \end{aligned}$$

Thus, for any $s \in F$, we have some $w \in E$ such that $s = S(w)$, and

$$a(C)(s) = O(f_C(w)) = O(f_A(w)) = f(s).$$

Likewise, for any $s \in \Omega' \setminus F$, we have some $w \in \Omega^* \setminus E$ such that $s = S(w)$, and

$$a(C)(s) = O(f_C(w)) = O(f_B(w)) = g(s).$$

So $a(C) = f \sqcup_E g$. □

Indeed, we can even defend the strongest form of Savage Richness. For any function $f : \Omega \rightarrow \mathcal{X}$ from states to outcomes, we can consider the following proposition:

For some state s , the outcome is $f(s)$, while s is the state that actually obtains.

Call this A_f . Again, as long as every outcome is certain to be metaphysically possible, then A_f is also certain to be metaphysically possible. Furthermore, for every state s , if the actual state is s , then A_f would have the outcome $f(s)$. If we count as an “available prospect” *every* proposition which is certain to be metaphysically possible, then every function from states to outcomes will arise from some available prospect.

We can make sense of the value of any proposition that certainly could have been the case. So it is fine to include all of these propositions in a value structure, if it is useful to do so. (If, like Joyce, we wish to apply these models to *preference*, rather than value, we can still make sense of such propositions as part of the domain to which preferences must be coherently “extendible”.)

Proposition C.9. Let $\mathcal{S}$ be a counterfactual value structure. If $\mathcal{S}$ is outcome-rich and full, then the canonical Savage representation satisfies Savage Richness.

Proof. Take the components of the canonical Savage representation to be labeled as usual. Let $g : \Omega' \rightarrow \mathcal{X}$ be any function from states to outcomes. Then let

$$A = \bigcup_{w \in \Omega^*} g(S(w)) \cap \alpha^{-1}\{w\}.$$

First, we check that $\alpha(A) \supseteq \Omega^*$. Indeed, for each $w \in \Omega^*$, $g(S(w))$ is an outcome proposition, and thus by outcome richness there is some $v \in g(S(w))$ such that $\alpha(v) = w$. So $v \in A$, and thus $w \in \alpha(A)$.

Next, we check that $a(A) = g$. Indeed, for any state $s = S(w)$, for $w \in \Omega^*$, by the properties of selection function and α ,

$$f_A(w) \in A \cap \alpha^{-1}\{w\} \subseteq g(S(w)) = g(s).$$

Thus, by the construction of the representation function a ,

$$a(A)(s) = O(f_A(w)) = g(s) \quad \text{for every } s \in \Omega'.$$

So $g = a(A) \in \mathcal{A}'$. □

This vindicates much of Savage’s framework. Unfortunately, we cannot vindicate all of Savage’s axioms — at least, not if we are to use it to model *impartial* value. For Savage made a strong assumption about the *continuity* of value (1954, axiom P7, sec. 5.4), which amounts to ruling out many of the cases that interested us in this book, including the St. Petersburg paradox. It also amounts to ruling out some of the claims about *infinitesimal* probability that arose.

We cannot directly avail ourselves of Savage’s representation theorem, or standard expected utility theory. We will need to find new representations that are better suited to impartial value. But we can do this within the same general framework that Savage used, representing prospects as functions from states to outcomes.

C.5 The von Neumann–Morgenstern framework

The counterfactual and Savage frameworks did not say anything explicitly about probabilities. One approach we could take is to add further axioms constraining the value order on prospects in a way that would allow us to define a probability measure on the state space in terms of value. (This is the approach of Savage 1954; Jeffrey 1983; Bolker 1966.) For present purposes we don’t need to be quite so methodical about foundations. Instead, we can introduce a probability measure as an extra piece of data. For now, we will consider the simple case of standard *real-valued* probability; in appendix E we will discuss limitations of this approach, and in appendix F we will take up an alternative.

A **(real) probability space** is a set Ω equipped with an algebra of subsets $\mathcal{A}$ and a (real-valued) probability measure on $(\Omega, \mathcal{A})$. I will normally take the “built-in” probability measure of a probability space to be notated $\mathbb{P}$. I will use variables $P, Q, \dots$ for probability measures more generally. For a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and a set (or measurable space — see appendix E) $\mathcal{X}$, an **($\mathcal{X}$ -valued) random variable** is a (measurable) function from Ω to $\mathcal{X}$.

Let a **probabilistic Savage model** be a Savage model where the state space is equipped with the structure of a probability space. In this case, Savage acts are random variables. If Savage Richness is satisfied, then *every* random variable is a Savage act.

Similarly, let a **probabilistic counterfactual value structure** be a counterfactual value structure where the state space is equipped with the structure of a

probability space. If a probabilistic counterfactual value structure $\mathcal{S}$ satisfies Statewise Equivalence, then its canonical Savage representation also inherits a probability measure on its state space.

So prospects in such a model can be represented by random variables.

In this case, the problem of describing a value order on prospects is equivalent to the problem of describing a value order on a certain set of random variables on some probability space Ω , which take values in a some outcome space $\mathcal{X}$. If the counterfactual value structure also satisfies certain modest and plausible richness conditions, then this becomes the problem of describing a value order on *all* $\mathcal{X}$ -valued random variables on Ω .

Sometimes we have used an even more austere representation of prospects than Savage models. Often we have said things like “probability $1/n$ of n happy lives and otherwise none is just as good as saving one happy life for sure”. This does not specify a Savage function from states to outcomes: it just says the probability of each outcome, without saying anything further about the circumstances in which that outcome would come about. In particular, when we characterize prospects in terms of probabilities of outcomes, this leaves out any information about how those outcomes are *correlated* with one another. It is natural to think (and built into most standard decision theories) that this is fine: the probabilities of outcome are all the information we need to make value comparisons. This is the central idea behind the decision-theoretic framework of von Neumann and Morgenstern: prospects can be faithfully represented by *probability distributions*. Let’s spell this out explicitly.

Definition C.10. Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, and let $X : \Omega \rightarrow \mathcal{X}$ be a random variable. The **distribution** of X , written $\mathbb{P}_X$, is the probability measure on $\mathcal{X}$ defined by

$$\mathbb{P}_X(E) := \mathbb{P}(X^{-1}(E)) \quad \text{for every } E \subseteq \mathcal{X}.$$

Definition C.11. Let $\mathcal{S}$ be a probabilistic counterfactual value structure, and let A be a prospect in $\mathcal{S}$. Then the **distribution** of A is the distribution of the canonical Savage representation for A ; equivalently, this is the probability measure over outcomes

$$\mathbb{P}_A(E) := \mathbb{P}(\{w \in \Omega^* \mid O(f_A(w)) \in E\}) \quad \text{for every } E \subseteq \mathcal{X}.$$

The idea of the von Neumann–Morgenstern framework is that these probabil-

ity distributions over outcomes carry enough information for us to evaluate prospects. This is tantamount to the following assumption.

Stochastic Equivalence. For any prospects $A, B \in \mathcal{A}$, if A and B have the same probability distribution, then A and B are equally good. In short:

$$\text{if } \mathbb{P}_A = \mathbb{P}_B \text{ then } A \sim B.$$

Unlike Statewise Equivalence, this principle of Stochastic Equivalence is not plausibly true in complete generality. In general, not all of the information that is relevant to evaluating prospects can be captured by their *real*-valued probability distributions. Two prospects can have the very *same* real probability distribution, without having the same value: recall the bets All Heads or Nothing from (chapter 5). These two prospects have standard probability zero of turning out differently, and these probability zero differences get erased when we move to the VNM framework of probability distributions. To uphold Stochastic Equivalence in general we need to turn to non-standard probabilities: see appendix F.

Nonetheless, sometimes the information we lose isn't super important. Stochastic Equivalence can be plausibly upheld if scale back our ambitions a bit. A counterfactual value structure need not include *every* prospect that it makes sense to evaluate in principle. We don't always have to worry about prospects that are crowded as close together as All Heads and Nothing, with different values but only measure zero differences. For some purposes, we can simply restrict the space of prospects we are evaluating to exclude these cases. For example, if we restrict our attention to prospects whose outcomes turn on *finitely* many coin flips, then we don't have to worry about minuscule probability differences making a difference to their comparative values. The same goes if we only consider prospects that turn on the outcome of a certain n -ticket lottery. In general, even though Stochastic Equivalence does not hold in general, we can ensure by fiat that it holds in a particular value structure, by restricting the space of prospects to include at most one prospect for any given probability distribution. When we do this, we will call the prospects that remain **simple prospects**.

(I won't try to generically define what a simple prospect is — I'm not sure there is some sensible absolute notion here. The key feature is *relational*: what is important is that no two simple prospects differ from one another merely on a set that has real-probability zero.)

Then the problem of describing a value order on *simple* prospects is transformed again into the problem of describing a value order on *probability*

distributions over outcomes. But which ones? Perhaps not *every* probability distribution corresponds to any genuine prospect. But we can very naturally get many of them.

Consider any *finite-support* probability distribution with probabilities $p_1, \dots, p_n$ over outcomes $x_1, \dots, x_n$. We can imagine constructing a spinner divided into n corresponding sectors, with the right proportional sizes. Then the proposition E_i that if such a spinner were spun, it would land in sector i has probability p_i . Then we can use the trick from section C.4 to construct a metaphysically possible proposition that would result in outcome x_i if, actually, E_i is true.

Here is a natural condition on a probability space.

Definition C.12. A probability space $(\Omega, \mathcal{A}, \mathbb{P})$ is **non-atomic** if for every $A \subseteq \Omega$ and every real number $0 \leq p \leq \mathbb{P}(A)$, there exists $B \subseteq A$ such that $\mathbb{P}(B) = p$.

If a Savage space satisfies Constant Acts and Piecewise Acts, and it is equipped with a non-atomic probability measure, then in fact *every* probability distribution over outcomes with finite support arises from some prospect. (I argued in section C.4 that Constant Acts and Piecewise Acts are really quite modest assumptions).

Definition C.13. A **simple probability distribution** on a set $\mathcal{X}$ is a function $P : \mathcal{X} \rightarrow \mathbb{R}$ with finite support such that $\sum_{x \in \mathcal{X}} P(x) = 1$. Let $\Delta_0 \mathcal{X}$ be the set of all simple probability distributions on $\mathcal{X}$.

A **simple VNM model** consists of a preorder $\succeq$ on $\Delta_0 \mathcal{X}$, where $\mathcal{X}$ is some set of **outcomes**.

If a counterfactual value structure $\mathcal{J}$ has a non-atomic probability measure, and its space is not too impoverished — but also not so abundant that it violates Stochastic Equivalence — then it can be represented by a simple VNM model, where for any prospects A and B ,

$$A \succeq B \quad \text{iff} \quad \mathbb{P}_A \succeq \mathbb{P}_B.$$

This representation is more restricted than the Savage representation: we must restrict our attention to certain *simple* prospects, that do not have measure zero collisions. But for many purposes, we only need to occupy ourselves with a handful of prospects at a time. So often this restriction is no

serious obstacle. There are many issues that can be productively explored within the VNM framework; this will be the focus of appendix D; appendix E. But we should remember that ultimately this framework only captures a piece of the story about the value of prospects, and eventually it will be important to take a broader perspective, as we will in appendix F.

C.6 The Sure Thing Principle and the Independence axiom

TODO. State Sure Thing Principle and Independence axiom, and conditions under which they are equivalent. Possibly need to move some Eventwise Dominance details here, too.

C.7 Game theory

In chapter 9 we discussed the relationship between Independence, the Sure Thing Principle, and game-theoretic ideas about *equilibrium*. In particular, we argued that the Sure Thing Principle is equivalent to the principle that *every ex ante optimal strategy is an equilibrium* (in a certain sense). We will now spell this out more carefully. I'll point out in advance that the difficulty is entirely in *stating* the principle. Once we have stated it precisely, the proof is basically immediate.

Let's start with a couple of examples of the kinds of games we want to describe.

Train Robbery. Each of two impartial passengers chooses independently whether to fight the train robbers or hang back and get robbed.

Table C.2: Train Robbery.

	Fight	Hang back
Fight	+2	−1
Hang back	−1	0

Pay for Ignorance. Moira can either (a) pay a small price to save Small Island if her boat is working, or else (b) find out whether her boat is working, and, if it is, then choose whether to save Small Island or Big Island.

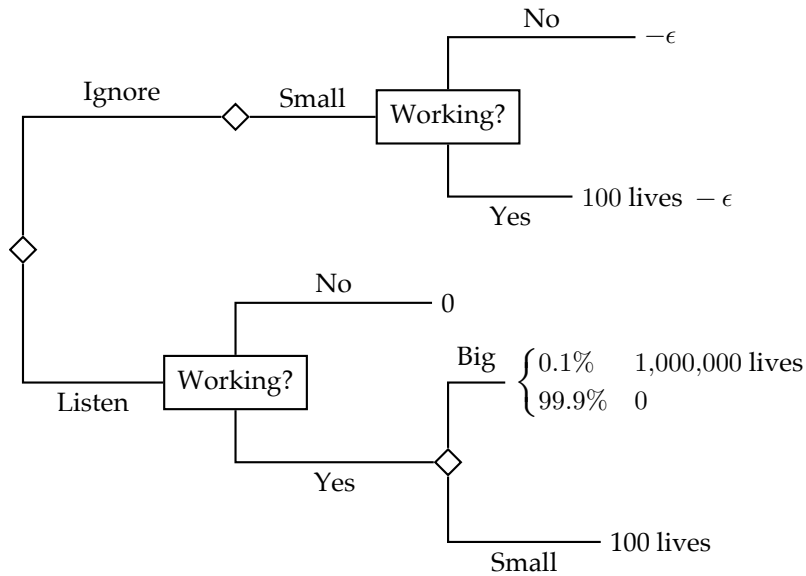


Figure C.2: Pay for Ignorance.

Satan's Apple. For each n , Eve chooses whether to eat the n th piece of the apple. Each piece she eats makes her better off, but if she eats infinitely many pieces she is ejected from Eden.

Let's consider an abstract way of representing all of these situations.¹⁰

¹⁰This kind of game representation does not exactly match any standard form I'm aware of. It sits somewhere in between extensive form, Bayesian strategic form, and agent form. It is an abstraction from extensive-form games, where we have set aside details about temporal precedence of actions, as well as bracketing whether different choices are made by the *same* player or *different* players. In this way it is similar to agent-form games, but that standardly also throws away the information about how choices depend on a state of nature (replacing pay-offs with *expected* pay-offs). The way we keep track of that information is very similar to Bayesian form games, with states of nature corresponding to "assignments of types"; but we are not explicitly representing individual players' beliefs about each other, and again we are ignoring whether different types correspond to a single player or different players. Also,

Definition C.14. A **game** consists of the following components:

- A set of **choice situations** $\mathcal{I}$.
- For each $i \in \mathcal{I}$, a set of **available acts** $A(i)$. Let $A = \prod_i A(i)$ be the set of **strategies**.
- A set of **states of nature** Ω .
- For each $a \in A$ and $i \in \mathcal{I}$, a set $E(a, i) \subseteq \Omega$, representing the event that the choice i would arise at all, if a were the chosen strategy.
- A set of **outcomes** $\mathcal{X}$, and an **outcome function** $X : A \times \Omega \rightarrow \mathcal{X}$. For $a \in A$ let $X_a = X(a, -)$.
- For each $i \in \mathcal{I}$ and $E \subseteq \Omega$, a **conditional value order** $\succsim_E^i$ on the set of functions from E to $\mathcal{X}$. (As usual, let the *unconditional* value order $\succsim$ be $\succsim_\Omega$, and for total functions $X, Y : \Omega \rightarrow \mathcal{X}$ let $X \succsim_E Y$ iff $X|_E \succsim_E Y|_E$.)

For any $w \in \Omega$, $a, b \in A$, and $i \in \mathcal{I}$, if $a \equiv^i b$, then:

- (i) $E(a, i) = E(b, i)$, and
- (ii) if $w \notin E(a, i) = E(b, i)$, then $X_a(w) = X_b(w)$.

In this case we can let $E = E(a, i) = E(b, i)$, and define $a \succsim_i b$ iff $X_a \succsim_E^i X_b$.

The motivation for condition (i) is that a choice cannot make a difference to whether that very choice arises at all. The motivation for condition (ii) is that if a choice does not arise, then it cannot make a difference to the outcome. The motivation for the definition of $\succsim_i$ is that in each choice situation, we evaluate strategies *conditional on that choice situation arising*.

We could generalize this definition to apply to games that are not pure coordination games by assigning *different* value orders to different choice situations.

we *are* including the possibility that whether a player with a given type arises can depend on other players' choices. Standardly, we might ignore this by just saying that the player's choice might make no difference to the outcome, if another player chooses a certain way. But treating this as equivalent amounts to assuming a form of the Sure Thing Principle, which is at issue here.

Let's apply this to a couple of our examples. In *Train Robbery*:

$$\begin{aligned}
 \mathcal{I} &= \{\text{Row}, \text{Col}\} \\
 A(\text{Row}) &= A(\text{Col}) = \{\text{Fight}, \text{Hang back}\} \\
 \Omega &= \{\bullet\} \quad (\text{a singleton set}) \\
 E(a, i) &= \Omega \quad \text{for each } a \in A, i \in \mathcal{I} \\
 \mathcal{X} &= \{-1, 0, +2\} \\
 X((\text{Fight}, \text{Fight}), \bullet) &= +2 \\
 X((\text{Fight}, \text{Hang back}), \bullet) &= -1 \\
 X((\text{Hang back}, \text{Fight}), \bullet) &= -1 \\
 X((\text{Hang back}, \text{Hang back}), \bullet) &= 0.
 \end{aligned}$$

In *Pay for Ignorance*:

$$\begin{aligned}
 \mathcal{I} &= \{1, 2, 3\} \\
 A(1) &= \{\text{Pay}, \text{Don't pay}\} \\
 A(2) &= \{\text{Small}, \text{Big}\} \\
 \Omega &= \{\text{Boat works}, \text{Boat doesn't work}\} \\
 E(a, 1) &= \Omega \\
 E(\text{Pay}, 2) &= \emptyset \\
 E(\text{Don't pay}, 2) &= \{\text{Boat works}\} \\
 \mathcal{X} &= \{100 \text{ lives} - \epsilon, 100 \text{ lives}, 1,000,000 \text{ lives}\} \\
 X((\text{Pay}, a), \text{Boat works}) &= 100 \text{ lives} - \epsilon \\
 X((\text{Pay}, a), \text{Boat doesn't work}) &= -\epsilon \\
 X((\text{Don't pay}, \text{Small}), \text{Boat works}) &= 100 \text{ lives} \\
 X((\text{Don't pay}, \text{Big}), \text{Boat works}) &= 1,000,000 \text{ lives} \\
 X((\text{Don't pay}, a), \text{Boat doesn't work}) &= 0.
 \end{aligned}$$

Notice that $E(-, 2)$ depends on the choice made in choice situation 1.

In *Satan's Apple*:

$$\begin{aligned}
 \mathcal{I} &= \mathbb{N} \\
 \mathcal{A}(i) &= \{\text{Eat}, \text{Don't eat}\} \\
 \Omega &= \{\bullet\} \\
 E(a, i) &= \Omega \quad \text{for each } a \in \mathcal{A}, i \in \mathcal{I} \\
 \mathcal{X} &= \mathbb{R} \\
 X(a, \bullet) &= \begin{cases} 1 - 2^{-n} & \text{if } n = \#\{i \in \mathbb{N} \mid a(i) = \text{Eat}\} \text{ is finite} \\ -100 - 2^{-n} & \text{otherwise} \end{cases}.
 \end{aligned}$$

Definition C.15. Let $\mathcal{G}$ be a game.

- (a) A strategy $a \in \mathcal{A}$ is a **(pure) weak equilibrium** iff for each $i \in \mathcal{I}$, $a \succsim_i b$ for each strategy $b \in \mathcal{A}$ such that $a \equiv^i b$.
- (b) A strategy $a \in \mathcal{A}$ is a **(pure) strict equilibrium** iff, for all $i \in \mathcal{I}$ such that $E(a, i)$ is non-null for $\succsim^i$, and all $b \equiv^i a$, if $a \neq b$, then $a \succ_i b$.
- (c) $\mathcal{G}$ is a **pure coordination game** iff there is a single conditional value order $(\succsim_E)_{E \subseteq \Omega}$ such that $\succsim_E^i = \succsim_E$ for each $i \in \mathcal{I}$ and $E \subseteq \Omega$.

Let $\mathcal{G}$ be a pure coordination game, and let $a \in \mathcal{A}$ be a strategy.

- (d) $a \in \mathcal{A}$ is **weakly ex ante optimal** iff $a \succsim b$ for all $b \in \mathcal{A}$. (That is, $X_a \succsim_\Omega X_b$ for all $b \in \mathcal{A}$.)
- (e) a is **strictly ex ante optimal** iff $a \succ b$ for all strategies $b \neq a$.

Definition C.16. Let $\succsim = (\succsim_E)_{E \subseteq \Omega}$ be a conditional value order on functions $\Omega \rightarrow \mathcal{X}$. Write $X \stackrel{E}{\equiv} Y$ iff $X(w) = Y(w)$ for all $w \in \Omega \setminus E$. As usual say E is **null** (for $\succsim$) iff $X \sim_E Y$ for all $X, Y : \Omega \rightarrow \mathcal{X}$.

- (a) $\succsim$ is **weakly reflective** iff for every $X, Y : \Omega \rightarrow \mathcal{X}$ and $E \subseteq \Omega$ such that $X \stackrel{E}{\equiv} Y$, if $X \succsim Y$, then $X \succsim_E Y$.
- (b) $\succsim$ is **strictly reflective** iff for every $X, Y : \Omega \rightarrow \mathcal{X}$ and non-null $E \subseteq \Omega$ such that $X \stackrel{E}{\equiv} Y$, if $X \succ Y$, then $X \succ_E Y$.

(Note that there is no need to exclude null events in the definition of weak reflectiveness, since if E is null then $X \succsim_E Y$ holds trivially.)

In the presence of Comparability, the conjunction of weak and strict reflectiveness is equivalent to the *Sure Thing Principle*.

Definition C.17. A conditional order $\succsim$ satisfies the **Sure Thing Principle** iff for any prospects X, Y and act E , if $X \stackrel{-E}{=} Y$, then

$$X \succsim Y \quad \text{iff} \quad X \succsim_E Y.$$

Theorem C.18. Let $\succsim$ be a conditional value order.

- (a) $\succsim$ is weakly reflective iff, for every pure coordination game $\mathcal{G}$ with common value order $\succsim$, every weakly *ex ante* optimal strategy in $\mathcal{G}$ is a pure equilibrium.
- (b) $\succsim$ is strictly reflective iff, for every pure coordination game $\mathcal{G}$ with common value order $\succsim$, every strictly *ex ante* optimal strategy in $\mathcal{G}$ is a strict equilibrium.

As I said, the proof is totally straightforward.

Proof. First, suppose that $\succsim$ is weakly reflective, and let $\mathcal{G}$ be pure coordination game with an ex ante optimal strategy a . Let $i \in \mathcal{I}$, and let $b \stackrel{-i}{=} a$. Then $a \succsim b$. By condition (i) of definition C.14, we can let $E = E(a, i) = E(b, i)$, and by condition (ii), $X_a \stackrel{-E}{=} X_b$. By weak reflectiveness, $X_a \succsim_E X_b$, and thus $a \succsim_i b$. Thus a is a weak equilibrium.

Similarly, if a is *strictly* ex ante optimal, then for any $i \in \mathcal{I}$ and $b \stackrel{-i}{=} a$ such that $a \neq b$, we have $a \succ b$. Thus, if $E(a, i)$ is non-null, strict reflectiveness implies that $a \succ_E b$, making a a strict equilibrium.

Next, suppose $\succsim$ is not weakly reflective: for some $E \subseteq \Omega$ and $X, Y : \Omega \rightarrow \mathcal{X}$, we have

$$\begin{aligned} X &\stackrel{-E}{=} Y \\ X &\succsim Y, \\ X &\not\succsim_E Y. \end{aligned}$$

Let $\mathcal{I} = \{i\}$, let $A(i) = \{a, b\}$, let $E(a, i) = E(b, i) = E$, and let $X_a = X$ and $X_b = Y$. Then by construction, a is ex ante optimal ($a \succsim b$), but not an equilibrium ($a \not\succsim_i b$).

Finally, if $\succsim$ is not strictly reflective, then we have non-null E and functions

X, Y such that

$$\begin{aligned} X &\stackrel{E}{=} Y \\ X &> Y, \\ X &\not\stackrel{E}{>} Y. \end{aligned}$$

Constructing the game exactly as before, we then have that a is strictly ex ante optimal ($X_a = X > Y = X_b$), but not a strict equilibrium ($a \not\stackrel{E}{>} b$ where $E(a, i) = E$ is non-null). $\square$

Note that reflectiveness actually follows if the equilibrium condition holds just for certain very simple games, with a single choice situation and two actions. So the equilibrium condition for *all* games is equivalent to its restriction to these very simple games.

Note that a different converse of theorem C.18 does not hold (see discussion in section 9.6). In general, equilibrium strategies need not be ex ante optimal. *Train Robbery* can give rise to *coordination failure* for impartial agents: (Hang back, Hang back) is a strict equilibrium: each passenger achieves a socially better outcome by hanging back, given that the other passenger hangs back as well; but (Fight, Fight) is better still. In this case, the optimal strategy (Fight, Fight) is *another* strict equilibrium. In *Satan's Apple*, there is *no* optimal strategy. But it does have a unique strict equilibrium: eat every piece.

Appendix D

Discrete Probability

D.1 Simple probability distributions

In appendix C we discussed how, at least sometimes, prospects can be represented by probability distributions over outcomes, and, in particular, probability distributions with *finite support*. This technical framework is limited in its generality, but it is still powerful enough to be worth exploring in some detail. Many of the technical results arising in this book can be stated and proved in this setting. In appendix E we will consider the standard approach to more general probability distributions, in which probabilities are standard real numbers (and countably additive). In appendix F we will venture further into the wilds, developing a non-standard theory of probability based on the filter power tools introduced in appendix B.

Definition D.1. A **simple probability distribution** on a set Ω is a function $P : \Omega \rightarrow \mathbb{R}$ with finite support such that $\sum P = 1$. Let $\Delta_0\Omega$ be the set of all simple probability distributions on Ω .

A set Ω equipped with a simple probability distribution $\mathbb{P}$ is called a **simple probability space**.

(I will normally use $\mathbb{P}$ as the name for the “built-in” probability distribution on a probability space, and $P, Q, \dots$ as variables for arbitrary probability distributions.)

This definition can be straightforwardly generalized beyond *real*-valued probabilities. In fact, it makes sense for any *partially ordered commutative*

semi-ring R (see section B.1). We will return to this higher level of generality in appendix F. In this appendix we will almost always be concerned with real probabilities. (Occasionally we will be a bit more restrictive, and focus on the case of *rational* probabilities.)

I defined a simple probability distribution in terms of its values on particular elements of Ω . We can alternatively think of this distribution as a *measure* on subsets of Ω .

Definition D.2. For $P \in \Delta_0 \Omega$ and $A \subseteq \Omega$, we define

$$P(A) := \sum_{w \in A} P(w).$$

This function $P : 2^\Omega \rightarrow \mathbb{R}$ is a probability measure. That is:

- (a) $P(\emptyset) = 0$.
- (b) $P(\Omega) = 1$.
- (c) For any disjoint $A, B \subseteq \Omega$, $P(A \cup B) = P(A) + P(B)$.

(In fact, this defines a *countably* additive probability measure; but this won't be important in this appendix.)

Definition D.3. If Ω is a simple probability space and $\mathcal{X}$ is some other set, a function $X : \Omega \rightarrow \mathcal{X}$ is called a **(simple, $\mathcal{X}$ -valued) random variable**.

Statisticians often think of a random variable X as representing an “unknown element” of $\mathcal{X}$, where each value $X(w)$ (for $w \in \Omega$) is one possibility for which element X really is. In language more familiar to philosophers, a random variable represents an element of $\mathcal{X}$ “in intension”. Statisticians commonly abuse notation a bit, with this interpretation in mind. If X is a real-valued random variable, for instance, we might write $P[X > 2]$ for the probability that the *value* of X is greater than two—that is, the probability of the set

$$\{w \in \Omega \mid X(w) > 2\}.$$

Similarly if X and Y are random variables we might write

$$P[X \geq Y] \quad \text{for} \quad P\{w \in \Omega \mid X(w) \geq Y(w)\}.$$

In general, any property of *elements* of $\mathcal{X}$ can be “lifted” to a property of $\mathcal{X}$ -valued *random variables* in a straightforward way, and we often use the very same notation for the “lifted” property as we do for the original property.

To spell this out, say $\phi(x, y, \dots)$ denotes some relation between elements of $\mathcal{X}$; then for $\mathcal{X}$ -valued random variables $X, Y, \dots$, let $[\phi(X, Y \dots)]$ denote the set

$$\{w \in \Omega \mid \phi(X(w), Y(w), \dots)\}.$$

Similarly, an *operation* on $\mathcal{X}$ can be lifted to an operation on random variables. If f is a function from $\mathcal{X}$ to some set $\mathcal{Y}$ and X is an $\mathcal{X}$ -valued random variable, then we sometimes use the notation $f(X)$ for the $\mathcal{Y}$ -valued random-variable $f \circ X$. These cheats are very convenient once you are used to them.

Another useful thing to do is push probabilities around from one set to another. We already discussed our central example in section C.5. Start with a probabilistic Savage space, where $\mathcal{X}$ is the set of outcomes, and the set of states Ω is equipped with a simple probability distribution $\mathbb{P}$. Each Savage act $X : \Omega \rightarrow \mathcal{X}$ is a random variable. Each Savage act also has a corresponding distribution over outcomes, which takes each outcome $x \in \mathcal{X}$ to the probability that X has this outcome, $\mathbb{P}[X = x]$. Once we have these probabilities of outcomes, for many purposes we can forget about the specific states and functions that they came from.

Here are two different perspectives on the same mathematical operation.

Definition D.4. Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a function, and let $P \in \Delta_0 \mathcal{X}$ be a simple probability distribution. The **pushforward distribution** of P along f is the distribution $f * P \in \Delta_0 \mathcal{Y}$ defined by

$$(f * P)(y) := \sum_{x \in \mathcal{X}: f(x)=y} P(x).$$

When $(\Omega, \mathbb{P})$ is a simple probability space and $X : \Omega \rightarrow \mathcal{X}$ is a random variable, we call this the **distribution of X** :

$$\mathbb{P}_X(y) := (X * \mathbb{P})(y) = \mathbb{P}[X = y].$$

The first perspective is more “functional”. We think of the function f as a transformation of that takes $\mathcal{X}$ -elements to $\mathcal{Y}$ -elements. The pushforward operation “lifts” this function $f : \mathcal{X} \rightarrow \mathcal{Y}$ to a function $(f * -) : \Delta_0 \mathcal{X} \rightarrow \Delta_0 \mathcal{Y}$ from *distributions* on $\mathcal{X}$ to *distributions* on $\mathcal{Y}$. The second perspective treats the function as a “random variable”, an element of $\mathcal{Y}$ in intension. In this case the pushforward measure is the distribution over this variable’s possible values.

The following can be checked.¹

Proposition D.5.

(a) Let $\text{id}_{\mathcal{X}} : \mathcal{X} \rightarrow \mathcal{X}$ be the identity function and let $P \in \Delta_0 \mathcal{X}$. Then

$$\text{id}_{\mathcal{X}} * P = P.$$

(b) Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ and $g : \mathcal{Y} \rightarrow \mathcal{Z}$ be functions, and let $P \in \Delta_0 \mathcal{X}$. Then

$$(g \circ f) * P = g * (f * P).$$

From the random variable perspective, let X is an $\mathcal{X}$ -valued random variable on a simple probability space, and let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be any function. Then the distribution of the random variable $f(X)$ (that is, $f \circ X$) is

$$P_{f(X)} = (P_X)_f = f * P_X.$$

Proposition D.6. For any $x \in \mathcal{X}$, let $\delta x \in \Delta_0 \mathcal{X}$ be the distribution that assigns probability 1 to x , and zero to every other element of $\mathcal{X}$. For any $f : \mathcal{X} \rightarrow \mathcal{Y}$ and $x \in \mathcal{X}$,^a

$$f * (\delta x) = \delta_{f(x)}.$$

^aWe can also take a categorical perspective on this. For each set $\mathcal{X}$, we have a corresponding function $\delta_{\mathcal{X}} : \mathcal{X} \rightarrow \Delta_0 \mathcal{X}$. For any function $f : \mathcal{X} \rightarrow \mathcal{Y}$, we have

$$\Delta_0 f \circ \delta_{\mathcal{X}} = \delta_{\mathcal{Y}} \circ f.$$

In category-ese, this is to say that δ is a *natural transformation* from the identity functor to the functor Δ_0 .

Sometimes it is useful to take random variables living on different probability spaces and stick them all together in a single space. Here is a useful construction for this purpose.

¹In the language of category theory, Δ_0 is a *functor* from the category of sets to itself. From this perspective, a third notation for the pushforward operation is convenient: define $\Delta_0 f P := f * P$. We can rewrite proposition D.5:

$$\begin{aligned} \Delta_0 \text{id}_{\mathcal{X}} &= \text{id}_{\Delta_0 \mathcal{X}} \\ \Delta_0 (f \circ X) &= \Delta_0 f \circ \Delta_0 X. \end{aligned}$$

Definition D.7. Let $(\Omega_1, \mathbb{P}_1)$ and $(\Omega_2, \mathbb{P}_2)$ be probability spaces and let $X : \Omega_1 \rightarrow \mathcal{X}$ and $Y : \Omega_2 \rightarrow \mathcal{X}$ be random variables with the same distribution:

$$X * \mathbb{P}_1 = Y * \mathbb{P}_2.$$

The **fiber product** $(\Omega_1 \times \Omega_2|_{X=Y}, \mathbb{P}_{X=Y})$ is the probability space whose state space is a subset of $\Omega_1 \times \Omega_2$

$$\Omega_1 \times \Omega_2|_{X=Y} := \{(w_1, w_2) \in \Omega_1 \times \Omega_2 \mid X(w_1) = Y(w_2)\}$$

and with the probabilities

$$\mathbb{P}_{X=Y}(w_1, w_2) := \mathbb{P}_1(w_1) \cdot \mathbb{P}_2(w_2 \mid [Y = X(w_1)]).$$

(Here $[Y = X(w_1)]$ is the set of all $w_2 \in \Omega_2$ such that $Y(w_2) = X(w_1)$, and $\mathbb{P}_2(w_2 \mid [Y = X(w_1)])$ is the conditional probability defined in the usual way.) Since X and Y have the same distribution, whenever $\mathbb{P}_1(w_1) \neq 0$ we have $\mathbb{P}_2([Y = X(w_1)]) \neq 0$, so the conditional probability is well-defined.

Let π_1 and π_2 be the projection functions from $\Omega_1 \times \Omega_2|_{X=Y}$ to Ω_1 and Ω_2 respectively. Then we can “lift” any random variable from Ω_1 to $\Omega_1 \times \Omega_2|_{X=Y}$ by composing it with π_1 , and similarly for variables on Ω_2 . In particular, we have

$$X \circ \pi_1 = Y \circ \pi_2.$$

If X and Y have the same distribution, then the pushforward distributions of $\mathbb{P}_{X=Y}$ along π_1 and π_2 are $\mathbb{P}_1$ and $\mathbb{P}_2$ respectively.

Note that $\Delta_0\Omega$ is a subset of $\text{Fin}(\Omega, \mathbb{R})$, the vector space of all functions from Ω to $\mathbb{R}$ with finite support. Thus all of the ordinary vector operations on functions also make sense for probability distributions: we can add two distributions together, or rescale them. In general the resulting function need not be another probability distribution — it may not be properly normalized. But certain operations do preserve normalization.

Definition D.8. Let $\mathcal{U}$ be a vector space. For any $x, y \in \mathcal{U}$ and any real number $p \in [0, 1]$, the **convex combination** of x and y is defined as

$$p \cdot x + (1 - p) \cdot y.$$

A subset $C \subseteq \mathcal{U}$ is **convex** iff it is closed under convex combinations: for any $x, y \in C$ and any $p \in [0, 1]$, we also have $p \cdot x + (1 - p) \cdot y \in C$.

Proposition D.9. The set of simple distributions $\Delta_0 \mathcal{X}$ is a convex subset of the vector space $\text{Fin}(\mathcal{X}, \mathbb{R})$ of functions from $\mathcal{X}$ to $\mathbb{R}$ with finite support.

In section 2.3 we introduced the idea of a *compound lottery*. We can first flip a weighted coin to decide which lottery you get, and then run the resulting lottery. This compound lottery, as a probability distribution, is a convex combination of its component lotteries.

D.2 Simple expectations

Consider a distribution with probability $2/3$ of making three people happy, and probability $1/3$ of making nine people happy. A simple and orthodox way of evaluating this kind of distribution, which appeared at many places in the main text, appeals to its *expected value*. This is the probability-weighted average of the values of its different possible outcomes.

$$\frac{2}{3} \cdot 3 + \frac{1}{3} \cdot 9 = 5.$$

Here we are taking the expected value of a distribution over *numbers*. But expectations make sense in a much more general setting — and this generality is very important when we take infinite values seriously. To make sense of expected value, we need to be able to *multiply* values by probabilities, and we need to be able to add the results together. If we want to use these expected values to make comparisons of betterness, then we also want these results to be ordered. If probabilities are a field — such as the real numbers, rationals, or hyperreals — then the kind of structure that fits the bill is a *vector space* (section B.1).²

Definition D.10. Let $X : \Omega \rightarrow \mathcal{U}$ be a vector-valued random variable on a simple probability space. The **expectation** of X is defined as

$$\mathbb{E}X := \sum_{w \in \Omega} \mathbb{P}(w) \cdot X(w).$$

(Since $\mathbb{P}$ has finite support, this is a finite sum even if Ω is an infinite set.)

²We noted in passing that simple probability distributions also make sense even when the “probabilities” are not real numbers, but rather elements of some p.o.c. semi-ring R . In this more general setting, the relevant kind of quantity for taking expectations is what we called an *R-quantity* in section B.1 — a (p.o.) *R*-semimodule. Almost everything in this section still holds when generalized to arbitrary *R*-probabilities and *R*-semimodules. But I will stick to the more familiar language of vector spaces.

Equivalently,

$$\mathbb{E}X = \sum_{v \in \mathcal{U}} \mathbb{P}[X = v] \cdot v.$$

(Again, since $\mathbb{P}[X = v] = 0$ for all but finitely many $v \in \mathcal{U}$, this is a finite sum.)

Note that this formulation makes it clear that the expectation of X only depends on the variable X 's *distribution*: if $\mathbb{P}_X = \mathbb{P}_Y$, then $\mathbb{E}X = \mathbb{E}Y$. This suggests an alternative definition, which applies directly to distributions rather than random variables.

Definition D.11. Let $\mathcal{U}$ be a vector space, and let $P \in \Delta_0 \mathcal{U}$ be a simple probability distribution on $\mathcal{U}$. The **expectation** of P is defined as

$$\mathbb{E}P := \sum_{v \in \mathcal{U}} P(v) \cdot v.$$

For $P \in \Delta_0 \mathcal{X}$ and $v : \mathcal{X} \rightarrow \mathcal{U}$, we also use the notation

$$\mathbb{E}_P v := \mathbb{E}(v * P).$$

These two definitions agree, in the sense that for any random variable $X : \Omega \rightarrow \mathcal{U}$,

$$\mathbb{E}X = \mathbb{E}(\mathbb{P}_X).$$

Proposition D.12. Let $\mathcal{U}$ be a p.o. vector space, and let $X, Y : \Omega \rightarrow \mathcal{U}$ be random variables on a simple probability space.

- (a) $\mathbb{E}(X + Y) = \mathbb{E}X + \mathbb{E}Y$
- (b) For any $a \in \mathbb{R}$, $\mathbb{E}(a \cdot X) = a \cdot \mathbb{E}X$
- (c) If $\mathbb{P}[X \geq Y] = 1$, then $\mathbb{E}X \geq \mathbb{E}Y$. If, furthermore, $\mathbb{P}[X > Y] \neq 0$, then $\mathbb{E}X > \mathbb{E}Y$.

Proposition D.13. Let $\mathcal{U}$ be a vector space.

- (a) For $v \in \mathcal{U}$, $\mathbb{E} \delta v = v$.
- (b) For $P, Q \in \Delta_0 \mathcal{U}$ and $p \in [0, 1]$,

$$\mathbb{E}(p \cdot P + (1 - p) \cdot Q) = p \cdot \mathbb{E}P + (1 - p) \cdot \mathbb{E}Q.$$

Definition D.14. Let $\mathcal{X}$ be a set, let $\mathcal{U}$ be a p.o. vector space, and let $v : \mathcal{X} \rightarrow \mathcal{U}$ be a vector-valued function. The **expectational order** on $\Delta_0 \mathcal{X}$ induced by v is defined such that, for $P, Q \in \Delta_0 \mathcal{X}$,

$$P \geq_v Q \quad \text{iff} \quad \mathbb{E}_P v \geq \mathbb{E}_Q v.$$

We call v an **expectational representation** of $\geq_v$.

There is no guarantee that $\geq_v$ is antisymmetric, and thus it may not be a partial order.

Definition D.15. We say a preorder $\succsim$ on $\Delta_0 \mathcal{X}$ **satisfies the Independence axiom** iff for all $P, P', Q \in \Delta_0 \mathcal{X}$, and all $p \in [0, 1]$ such that $p > 0$,

$$P \succsim P' \quad \text{iff} \quad p \cdot P + (1 - p) \cdot Q \succsim p \cdot P' + (1 - p) \cdot Q.$$

Recall that $\Delta_0 \mathcal{X}$ is a convex subset of the vector space $\text{Fin}(\mathcal{X}, \mathbb{R})$.

In section A.5 we introduced the notion of a *precone* in an abelian group, which is a convenient way of defining a preorder. This idea naturally extends to vector spaces.

Lemma D.16. A **cone** in a vector space $\mathcal{U}$ is a subset $C \subseteq \mathcal{U}$ such that

- (a) C is closed under addition: if $x, y \in C$, then $x + y \in C$.
- (b) C is closed under multiplication by positive scalars: if $x \in C$ and $a \geq 0$, then $a \cdot x \in C$.

If $\mathcal{U}$ is a preordered vector space, then

$$C = \{x \in \mathcal{U} \mid x \geq 0\}$$

is a cone. Conversely, for any cone C in a vector space $\mathcal{U}$, we can define

$$x \geq y \quad \text{iff} \quad x - y \in C$$

which makes $\mathcal{U}$ into a preordered vector space.

Lemma D.17. For any preorder $\succsim$ on $\Delta_0 \mathcal{X}$ that satisfies Independence, there exists a preorder $\geq$ on $\text{Fin}(\mathcal{X}, \mathbb{R})$ that faithfully extends $\succsim$ and makes $\text{Fin}(\mathcal{X}, \mathbb{R})$ a preordered vector space.

Proof. Define

$$C = \{a \cdot (P - Q) \mid P \succeq Q, a > 0\}.$$

This is clearly closed under multiplication by positive scalars. Independence implies that it is also closed under addition.

(To show this, let $P \succeq Q$, $P' \succeq Q'$, and $a, b > 0$. Then

$$a \cdot (P - Q) + b \cdot (P' - Q') = (a+b) \cdot \left(\frac{a}{a+b} \cdot P + \frac{b}{a+b} \cdot P' - \frac{a}{a+b} \cdot Q + \frac{b}{a+b} \cdot Q' \right).$$

Two applications of Independence imply that

$$\frac{a}{a+b} \cdot P + \frac{b}{a+b} \cdot P' \succeq \frac{a}{a+b} \cdot Q + \frac{b}{a+b} \cdot Q'.$$

So the sum $a \cdot (P - Q) + b \cdot (P' - Q')$ is indeed in C .)

So C is a cone, and thus makes $\text{Fin}(\mathcal{X}, \mathbb{R})$ a preordered vector space. Independence also implies that this order faithfully extends the original order $\succeq$.

(To show this it suffices to check that for $P, Q \in \Delta_0 \mathcal{X}$, if $P - Q \in C$, then $P \succeq Q$. Suppose that $P - Q = a \cdot (P' - Q')$ for some $P' \succeq Q'$ and $a > 0$. Then

$$\frac{1}{1+a} \cdot P + \frac{a}{1+a} \cdot Q' = \frac{1}{1+a} \cdot P' + \frac{a}{1+a} \cdot Q.$$

Since $P' \succeq Q'$, two more applications of Independence imply $P \succeq Q$.) $\square$

Theorem D.18 (vector expected utility theorem). A preorder on $\Delta_0 \mathcal{X}$ has an expectational representation iff it satisfies Independence.

Proof. Left-to-right. Straightforward, from the properties of expectations.

Right-to-left. Let $\text{Fin}(\mathcal{X}, \mathbb{R})$ be endowed with the order from lemma D.17, and consider the map $\delta : \mathcal{X} \rightarrow \text{Fin}(\mathcal{X}, \mathbb{R})$ that takes each $x \in \mathcal{X}$ to its indicator function $\delta x \in \Delta_0 \mathcal{X}$. We can check:

$$\mathbb{E}_P \delta = P \quad \text{for each } P \in \Delta_0 \mathcal{X}.$$

Thus

$$P \succeq Q \quad \text{iff} \quad \mathbb{E}_P \delta \geq \mathbb{E}_Q \delta \quad \text{for each } P, Q \in \Delta_0 \mathcal{X}.$$

$\square$

So far we have been talking about *preorders*. It is straightforward to pass from preordered vector spaces to *partially* ordered vector spaces, using a quotient construction (proposition B.20).

Lemma D.19. Let $\mathcal{V}$ be a preordered vector space. An **ideal** in $\mathcal{V}$ is a subspace $I \subseteq \mathcal{V}$ (meaning that it I is closed under addition and scalar multiplication) such that for any $x \in I$ and any $y \in \mathcal{V}$, if $x \geq y$, then $y \in I$.

Let I be an ideal in $\mathcal{V}$, and for any $x, y \in \mathcal{V}$, define

$$x \sim y \quad \text{iff} \quad x - y \in I.$$

Then $\sim$ is a congruence relation on $\mathcal{V}$, and the quotient $\mathcal{V}/\sim$ is a partially ordered vector space.

Proposition D.20. For any preordered vector space $\mathcal{V}$, there exists a strictly positive surjective linear map $h : \mathcal{V} \rightarrow \mathcal{V}'$, where $\mathcal{V}'$ is a partially ordered vector space.

Proof. Quotient out the subspace $I := \{v \in \mathcal{V} \mid v \sim 0\}$. □

So far we have considered a quite abstract notion of an expectational representation, in some arbitrary preordered vector space. But preordered vector spaces also admit a much more concrete *lexicographic* representation, via Hausner's representation theorem, which we introduced in section B.3. Recall:

- (a) Any p.o. vector space can be embedded in a direct power of a lexicographic vector space $(\mathbb{R}_{\text{lex}}^J)^I$, for some set I and some totally ordered set J .
- (b) Any totally ordered vector space can be embedded in a lexicographic vector space $\mathbb{R}_{\text{lex}}^J$, for some totally ordered set J .
- (c) Any Archimedean p.o. vector space can be embedded in a direct power of the real numbers $\mathbb{R}^I$, for some set I .
- (d) Any Archimedean and totally ordered vector space is isomorphic to the real numbers $\mathbb{R}$.

(In section B.3 we only stated (c) and (d) for embeddings of p.o. *abelian groups*, but essentially the same proofs carry over.)

We have already defined an Archimedean property for preordered groups; the same definition also applies to preordered vector spaces. But we need a slightly different definition for preorders on the convex set of distributions $\Delta_0\mathcal{X}$.

Definition D.21. A preorder $\succeq$ on $\Delta_0\mathcal{X}$ is **Archimedean** iff for any $P, Q, R \in \Delta_0\mathcal{X}$, if $p \cdot P + (1 - p) \cdot Q \succeq R$ for every finite probability $p > 0$, then $Q \succeq R$.

Theorem D.22. Let $\succeq$ be a preorder on $\Delta_0\mathcal{X}$ that satisfies Independence.

- (a) $\succeq$ has an expectational representation in some direct power of a lexicographic vector space $(\mathbb{R}_{\text{lex}}^I)^I$.
- (b) If $\succeq$ is a total preorder, then $\succeq$ has an expectational representation in some lexicographic vector space $\mathbb{R}_{\text{lex}}^I$.
- (c) If $\succeq$ is Archimedean, then $\succeq$ has an expectational representation in some direct power of the real numbers $\mathbb{R}^I$.
- (d) If $\succeq$ is an Archimedean total preorder, then $\succeq$ has an expectational representation in the real numbers $\mathbb{R}$.

Part (d) amounts to the standard von Neumann–Morgenstern representation theorem. (It is worth remembering that the other parts of this theorem rely on Choice.)

D.3 Simple stochastic dominance

Random variables can be ordered by *statewise dominance*.

Definition D.23. For random variables $X, Y : \Omega \rightarrow \mathcal{X}$, where $\mathcal{X}$ is a preordered set, say X **weakly statewise dominates** Y iff $X(w) \geq Y(w)$ for each $w \in \Omega$. This is written $X \geq Y$.

If we suppose that stochastically equivalent prospects are equally good, then we can represent a prospect by its probability distribution over outcomes without losing information relevant to how good it is. In this context, though, we do lose certain information about how different prospects are related to each other. In particular, we can't tell just from their probability distributions whether one prospect statewise dominates another. In this setting, we appeal to a more flexible dominance concept: *stochastic dominance*.

Definition D.24. Let $\mathcal{X}$ be a preordered set. For $P, Q \in \Delta_0 \mathcal{X}$, we say P **weakly stochastically dominates** Q , written $P \geq_s Q$, iff there are $\mathcal{X}$ -valued random variables X and Y on some simple probability space such that $P = \mathbb{P}_X$, $Q = \mathbb{P}_Y$, and $X \geq Y$.

In other words, P and Q are the respective distributions of random variables X and Y such that X statewise dominates Y .

Proposition D.25.

- (a) $\geq_s$ is a preorder on $\Delta_0 \mathcal{X}$.
- (b) The function $\delta : \mathcal{X} \rightarrow \Delta_0 \mathcal{X}$ is strictly monotone.

Proof. To see that $\geq_s$ is reflexive, let $\mathbb{P} = P$ and let $X = \text{id}_{\mathcal{X}}$. For transitivity, suppose we have probability spaces $(\Omega, \mathbb{P})$ and $(\Omega', \mathbb{P}')$, and random variables X, Y on Ω and Y', Z on Ω' , such that

$$\begin{aligned} 1 &= \mathbb{P}[X \geq Y] = \mathbb{P}'[Y' \geq Z] \\ P &= \mathbb{P}_X \\ Q &= \mathbb{P}_Y = \mathbb{P}'_{Y'} \\ R &= \mathbb{P}'_Z. \end{aligned}$$

The trick is to get all these variables onto a single probability space. We can do this using the fiber product construction from definition D.7: since Y and Y' have the same distribution, we can let $\Omega'' = \Omega \times \Omega'_{|Y=Y'}$, and we have $\mathbb{P} = \pi_1 * \mathbb{P}''$ and $\mathbb{P}' = \pi_2 * \mathbb{P}''$ (where π_1 and π_2 are the projection functions from Ω'' to Ω and Ω' , respectively). Thus $X \circ \pi_1$ has the distribution P , $Y \circ \pi_1 = Y' \circ \pi_2$ has the distribution Q , and $Z \circ \pi_2$ has the distribution R . Since $X \geq Y$ and $Y' \geq Z$ statewise, likewise

$$X \circ \pi_1 \geq Y \circ \pi_1 = Y' \circ \pi_2 \geq Z \circ \pi_2 \quad \text{statewise.}$$

For part (b), if $x \succ y$, then $\delta x \geq_s \delta y$ is witnessed by the one-element probability space and the random variables with constant values x and y , respectively. $\square$

There are a number of other alternative formulations of stochastic dominance. One important version uses *cumulative* distributions. In the special case where $\mathcal{X}$ is the set of real numbers, this can be stated as follows.

Proposition D.26. For $P, Q \in \Delta_0 \mathbb{R}$, $P \geq_s Q$ iff $P(\uparrow x) \geq Q(\uparrow x)$ for each $x \in \mathbb{R}$ (where $\uparrow x = \{y \in \mathbb{R} \mid y \geq x\}$).

When we look beyond the real numbers to other preordered sets of outcomes, though, this condition as stated is *not* equivalent to stochastic dominance; it needs to be generalized. Consider an example (based on J. S. Russell, forthcoming). Let $\mathcal{X}$ be the set of pairs $\{(0,0), (0,1), (1,0), (1,1)\}$, ordered by dominance. Let $P(0,0) = 1/3$ and $P(1,1) = 2/3$. Let Q be uniform on $\{(0,1), (1,0), (1,1)\}$. Then it can be checked that $P(\uparrow x) \geq Q(\uparrow x)$ for each $x \in \mathcal{X}$. But clearly P does not stochastically dominate Q : any random variable with distribution P must sometimes take the worst value $(0,0)$, while any random variable with distribution Q will have probability zero of taking the worst value.

The trick is to generalize our understanding of a cumulative distribution: instead of just paying attention to the *principal* upper sets of the form $\uparrow x$, we need to pay attention to *all* upward-closed sets of outcomes.

Definition D.27. Let $\mathcal{X}$ be a preordered set. We will use the notation $\text{Up } \mathcal{X}$ for the set of all upward-closed subsets of $\mathcal{X}$. For any simple distributions $P, Q \in \Delta_0 \mathcal{X}$, say that P **(weakly) cumulatively dominates** Q iff

$$P(U) \geq Q(U) \quad \text{for each } U \in \text{Up } \mathcal{X}.$$

I'll only prove the correspondence between stochastic and cumulative dominance for the special case of simple *rational* probability distributions. Nothing really turns on this restriction, but it substantially simplifies the proof, and this turns out to be the only case we will rely on (in appendix F; for a more general version see section E.4).

Proposition D.28. For any preordered set $\mathcal{X}$ and any simple rational probability distributions $P, Q \in \Delta_0^{\mathbb{Q}} \mathcal{X}$, P weakly stochastically dominates Q iff P weakly cumulatively dominates Q .

Proof. The left-to-right direction is straightforward. If $P = \mathbb{P}_X$, then $P(U) = \mathbb{P}[X \in U]$. Likewise if $Q = \mathbb{P}_Y$, then $Q(U) = \mathbb{P}[Y \in U]$. If U is upward-closed and $X \geq Y$ statewise, then $Y(w) \in U$ implies $X(w) \in U$ for any $w \in \Omega$. So $\mathbb{P}[X \in U] \geq \mathbb{P}[Y \in U]$.

The right-to-left direction takes more work.

In this case, there is some positive integer n such that $n \cdot P$ and $n \cdot Q$ are each natural-number valued functions with finite support. The result then follows from the following lemma. $\square$

Lemma D.29. Let $f, g \in \text{Fin}(\mathcal{X}, \mathbb{N})$, and let $F(U) = \sum_U (f - g)$. Suppose that $F(U) \geq 0$ for each $U \in \text{Up } \mathcal{X}$, and $F(\mathcal{X}) = 0$. Let $\Omega = \{(x, y) \in \mathcal{X} \times \mathcal{X} \mid x \succeq y\}$. Then there exists a function $h \in \text{Fin}(\Omega, \mathbb{N})$ such that for each $x, y \in \mathcal{X}$,

$$\begin{aligned} f(x) &= \sum_{y: x \succeq y} h(x, y) \\ g(y) &= \sum_{x: x \succeq y} h(x, y). \end{aligned}$$

Proof. Without loss of generality, we can suppose that $\mathcal{X}$ is finite.

We prove this by induction on the sum of values $\sum f$. If this sum is zero, then $f = g = 0$, and we can likewise let h be everywhere zero.

Otherwise, I claim:

(*) There exists $x \succeq y$ in $\mathcal{X}$ such that $f(x) > 0$, $g(y) > 0$, and $F(U) > 0$ for each upper set U that contains x but not y .

In this case, since f , g , and F are integer-valued, each of these positive values is at least one. Now we can consider the functions

$$\begin{aligned} f' &= f - \delta x \\ g' &= g - \delta y. \end{aligned}$$

We have $\sum f' = \sum f - 1$, and we can check that f' and g' still satisfy the cumulative dominance condition: $F'(U) = \sum_U (f' - g') \geq 0$ for each $U \in \text{Up } \mathcal{X}$.

(Let $U \in \text{Up } \mathcal{X}$. Then $F'(U)$ is the same as $F(U)$ *minus* one if x is in U , and *plus* one if y is in U . If U contains x but not y , then $F'(U) = F(U) - 1 \geq 0$. In all other cases, $F'(U) = F(U) \geq 0$.)

So by the inductive hypothesis, there exists a function $h' \in \text{Fin}(\Omega, \mathbb{N})$ such that

$$\begin{aligned} f'(x) &= \sum_{y: x \succeq y} h'(x, y) \\ g'(y) &= \sum_{x: x \succeq y} h'(x, y). \end{aligned}$$

Then we can let $h = h' + \delta(x, y)$.

It remains to prove (*). Suppose $f(x) > 0$. Since $\mathcal{X}$ is finite, and $F(\mathcal{X}) = 0$, it follows that there exists a minimal upward-closed subset $U \subseteq \mathcal{X}$ with the property that $x \in U$ and $F(U) = 0$.

There is some $x \succsim y$ in U such that $g(y) > 0$. Let $E = \{y \in U \mid x \succsim y\}$, and consider the upper set $V = U \setminus E$. We know that $x \in E$ and $f(x) > 0$, so if g is zero on E , it would follow that

$$F(V) = F(U) - \sum_E f + \sum_E g < F(U) = 0.$$

But by hypothesis $F(V) \geq 0$.

Finally we show that x and y have the required property for (*): for any upper set V such that $x \in V$ and $y \notin V$, $F(V) > 0$.

Let V be an upper set such that $x \in V$ and $F(V) = 0$. In that case, $U \cup V$ and $U \cap V$ are both upper sets as well, and so $F(U \cup V) \geq 0$ and $F(U \cap V) \geq 0$. But also,

$$F(U \cup V) + F(U \cap V) = F(U) + F(V) = 0.$$

So $F(U \cap V) = F(U \cup V) = 0$ as well. Furthermore, $x \in U \cap V$. Now the minimality condition on U tells us that $U \subseteq U \cap V$. Since we chose $y \in U$, we must also have $y \in V$. $\square$

Theorem D.30. Let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta_0 \mathcal{X}$. The following are equivalent:

- (a) **Realization.** $P \geq_s Q$: that is, there are $\mathcal{X}$ -valued random variables X, Y on some probability space, with distributions P and Q respectively, such that $X \geq Y$ pointwise.
- (b) **Realization 2.** There exist random variables $X \geq Y$ with distributions P and Q such that $\mathbb{P}[X \geq Y] = 1$.
- (c) **Coupling.** There exists a simple probability distribution $R \in \Delta_0(\mathcal{X} \times \mathcal{X})$, such that R is supported on the set $\{(x, y) : x \geq y\}$, and P and Q are the *marginals* of R :

$$P = \pi_1 * R \quad \text{and} \quad Q = \pi_2 * R.$$

- (d) **Minimality.** $P \geq Q$ with respect to the weakest order on $\Delta_0 \mathcal{X}$ that satisfies Independence and makes $\delta : \mathcal{X} \rightarrow \Delta_0 \mathcal{X}$ monotone.

- (e) **Expected vector utility.** For any monotone vector-valued function $u : \mathcal{X} \rightarrow \mathcal{U}$,

$$\mathbb{E}_P u \geq \mathbb{E}_Q u.$$

- (f) **Expected real utility.** For any monotone real-valued function $u : \mathcal{X} \rightarrow \mathbb{R}$,

$$\mathbb{E}_P u \geq \mathbb{E}_Q u.$$

- (g) **Cumulative dominance.** P weakly cumulatively dominates Q : for every upward-closed subset $U \subseteq \mathcal{X}$, $P(U) \geq Q(U)$.

Proof sketch. We'll first show that (a)–(c) are equivalent, then that (d)–(g) are equivalent, and finally bridge between these two groups of conditions.

For the first group:

- (a) $\Rightarrow$ (c). Given two $\mathcal{X}$ -valued random variables X, Y , their *Cartesian product* $X \times Y$ is an $\mathcal{X} \times \mathcal{X}$ -valued random variable. Let R be its distribution $\mathbb{P}_{X \times Y}$. Then $\pi_1 * R = \mathbb{P}_X$, $\pi_2 * R = \mathbb{P}_Y$, and R is supported on $\{(x, y) : x \geq y\}$.
- (b) $\Rightarrow$ (a). Restrict the probability space to the support of $\mathbb{P}$.
- (c) $\Rightarrow$ (b). Let $\Omega = \mathcal{X} \times \mathcal{X}$, let $\mathbb{P} = R$, and let X and Y be the projections π_1 and π_2 .

For the second group:

- (d) $\Rightarrow$ (e). If $u : \mathcal{X} \rightarrow \mathcal{U}$ is monotone, then the expectational order $\geq_u$ on $\Delta_0 \mathcal{X}$ satisfies Independence, and $\delta : \mathcal{X} \rightarrow \Delta_0 \mathcal{X}$ is monotone. So, if $P \geq Q$ with respect to the weakest such order, then $P \geq_u Q$ as well.
- (e) $\Rightarrow$ (f). Immediate, since $\mathbb{R}$ is a preordered vector space.
- (f) $\Rightarrow$ (g). For any upper set U , the function $1_U : \mathcal{X} \rightarrow \mathbb{R}$ is monotone, and $P(U) = \mathbb{E}_P 1_U$.
- (g) $\Rightarrow$ (d). It is straightforward to check that the cumulative dominance order satisfies Independence, and if $x \geq y$ in $\mathcal{X}$ then δx cumulatively dominates δy .

Finally, here are bridges between the two groups:

- (a) $\Rightarrow$ (e). By the properties of vector expectations (proposition D.12).

(b) $\Rightarrow$ (a). By proposition D.28.

□

Now we turn from *weak* stochastic dominance to *strong* stochastic dominance. This also has a variety of equivalent formulations.

Definition D.31. Let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta_0 \mathcal{X}$. We say P **strongly stochastically dominates** Q iff there exists some simple probability space $(\Omega, \mathbb{P})$ and random variables $X, Y : \Omega \rightarrow \mathcal{X}$ such that

$$\begin{aligned} P &= \mathbb{P}_X \\ Q &= \mathbb{P}_Y \\ \mathbb{P}[X \geq Y] &= 1 \\ \mathbb{P}[X > Y] &> 0. \end{aligned}$$

Theorem D.32 (Strong Stochastic Dominance). Let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta_0 \mathcal{X}$. The following are equivalent:

- (a) P strongly stochastically dominates Q .
- (b) $P \geq_s Q$ and $P \not\leq_s Q$.
- (c) **Expected vector utility.** For any strictly monotone vector-valued function $u : \mathcal{X} \rightarrow \mathcal{U}$,

$$\mathbb{E}_P u > \mathbb{E}_Q u.$$

- (d) **Expected real utility.** For any strictly monotone real-valued function $u : \mathcal{X} \rightarrow \mathbb{R}$,

$$\mathbb{E}_P u > \mathbb{E}_Q u.$$

- (e) P weakly cumulatively dominates Q , and Q does not weakly cumulatively dominate P . That is,

$$\begin{aligned} P(U) &\geq Q(U) \quad \text{for all } U \in \text{Up } \mathcal{X}, \text{ and} \\ P(U) &> Q(U) \quad \text{for some } U \in \text{Up } \mathcal{X}. \end{aligned}$$

The proof is pretty straightforward, using the corresponding equivalences for weak stochastic dominance, and the properties of expectations.

All of this is very nicely behaved—for *simple* probability distributions, which

have finite support. We will later see that many of these nice connections are broken in infinite cases (sections E.4 and F.6).

Here is one further point. In section 8.4 we discussed a striking result from Tarsney (2025): if a bounded prospect X has strictly higher expected value than another Y , then we can add on a “background value” prospect B such that resulting prospect $X + B$ *stochastically dominates* $Y + B$. This is true even if X is risky and Y is safe, so that X ’s worst case is worse than Y ’s. But Tarsney’s result is in a sense essentially infinitary: it requires an *unbounded* wide-tailed background distribution. The same phenomenon cannot arise for simple distributions with finite support. This also will turn out to be important when we move on to *non-standard* probabilities: in many ways, quasi-probabilities behave more like simple distributions than standard countably-additive probabilities do. (We will return to this in sections E.6 and F.6.)

Definition D.33. Random variables $X, Y : \Omega \rightarrow \mathcal{X}$ on a simple probability space $(\Omega, \mathbb{P})$ are **independent** iff

$$\mathbb{P}[X \in A \text{ and } Y \in B] = \mathbb{P}[X \in A] \cdot \mathbb{P}[Y \in B] \quad \text{for each } A, B \subseteq \mathcal{X}.$$

Definition D.34. Let $\mathcal{U}$ be any quantity, and let $P, Q \in \Delta_0 \mathcal{U}$. The **convolution** of P and Q is the distribution $P \star Q \in \Delta_0 \mathcal{U}$ defined by

$$(P \star Q)(z) = \sum_{x+y=z} P(x) \cdot Q(y) \quad \text{for each } z \in \mathcal{U}.$$

Proposition D.35. If X and Y are independent random variables on a simple probability space, then

$$\mathbb{P}_{X+Y} = \mathbb{P}_X \star \mathbb{P}_Y.$$

Proposition D.36. Let $\mathcal{U}$ be a $\mathbb{Z}$ -quantity, and let $P, Q, R \in \Delta_0 \mathcal{U}$.

- (a) If $P \geq_s Q$, then $P \star R \geq_s Q \star R$.
- (b) Suppose $\mathcal{U}$ is totally ordered. If $P \star R \geq_s Q \star R$, then

$$\begin{aligned} \max \text{supp } P &\geq \max \text{supp } Q \\ \min \text{supp } P &\geq \min \text{supp } Q. \end{aligned}$$

Proof. Part (a). Let $U \subseteq \mathcal{U}$ be an upward-closed set. Then for any $x \in \mathcal{U}$, the set $U - x$ is also upward-closed. Then $P \geq_s Q$ implies $P(U - x) \geq Q(U - x)$ for each $x \in \mathcal{U}$, and thus:

$$\begin{aligned} (P \star R)(U) &= \sum_{x \in \mathcal{U}} R(x) \cdot P(U - x) \\ &\geq \sum_{x \in \mathcal{U}} R(x) \cdot Q(U - x) \\ &= (Q \star R)(U). \end{aligned}$$

So $P \star R \geq_s Q \star R$.

Part (b). Suppose $P \star R \geq_s Q \star R$, and let $y = \max \text{supp } Q$. Also, let $z = \max \text{supp } R$. Then

$$0 < (Q \star R)(y + z) < (P \star R)(y + z).$$

This implies that there is some $x \in \text{supp } P$ and $z' \in \text{supp } R$ such that $x + z' = y + z$. So, since $z \geq z'$,

$$\max \text{supp } P \geq x = y + (z - z') \geq y.$$

The minimum case goes similarly. □

D.4 Pascalian theorems

Here we will prove generalizations of the main Pascalian theorems in chapter 2 and chapter 10. Throughout this section, let $\mathcal{X}$ be a preordered set of outcomes that embeds the ∞ -order on 2^ω (section A.3). That is, there is a function $f : 2^\omega \rightarrow \mathcal{X}$ such that whenever $A \succ^\infty B$, $f(A) \succ f(B)$. Also, let the set of simple distributions $\Delta_0 \mathcal{X}$ be equipped with a preorder $\succeq$ that (weakly) extends the weak stochastic dominance order.

Definition D.37.

- (a) For $A, B \in 2^\omega$, write $A \sim^\infty B$ iff they have all but finitely many elements in common: that is, the symmetric difference $(A \setminus B) \cup (B \setminus A)$ is finite.

Let $\mathcal{X}$ be a preordered set equipped with an embedding $f : 2^\omega \rightarrow \mathcal{X}$ of the ∞ -order. Let $x, y \in \mathcal{X}$.

(b) We say $x \sim^\infty y$ iff

$$f(A) \succeq x, y \succeq f(B) \quad \text{for some } A \sim^\infty B \in 2^\omega.$$

(c) We say $x \succ^\infty y$ iff

$$x \succeq f(A) \succ f(B) \succeq y \quad \text{for some } A \succ^\infty B \in 2^\omega.$$

Let $P, Q \in \Delta_0 \mathcal{X}$ be simple probability distributions.

- (d) We write $P \sim^\infty Q$ for the stochastic equivalence relation $P \sim^\infty_s Q$: in other words, P is stochastically equivalent to Q up to finite differences in outcomes.
- (e) We write $P \succ^\infty Q$ for the stochastic dominance relation $P \succ^\infty_s Q$: in other words, P and Q are the respective distributions of random variables X and Y such that $X \succ^\infty Y$ pointwise.
- (f) Write $p(x, y) \in \Delta_0 \mathcal{X}$ for the simple probability distribution $p \cdot \delta x + (1 - p) \cdot \delta y$. (That is, $p(x, y)$ is the distribution that assigns probability p to x and probability $1 - p$ to y .)

Lemma D.38.

- (a) $\sim^\infty$ is an equivalence relation.
- (b) If $x \sim^\infty x' \succ^\infty y' \sim^\infty y$ then $x \succ y$.
- (c) If $P \sim^\infty P' \succ^\infty Q' \sim^\infty Q$, then $P \succeq Q$.

We begin with a statement of Pascal's basic argument discussed in chapter 2.

Theorem D.39. Let $\mathcal{X}$ be a preordered set of outcomes equipped with an embedding of the ∞ -order, and let the set of simple distributions $\Delta_0 \mathcal{X}$ be equipped with a preorder $\succeq$ that weakly extends the weak stochastic dominance order. Suppose furthermore that $\succeq$ is **fanatical** in the following sense:

- For each $x \sim^\infty z \in \mathcal{X}$ and $p > 0$, there exists some $y \sim^\infty z$ such that $p(y, z) \succ x$.

Then the following form of **Pascal's principle** follows:

$$p(y, z) \succ x \quad \text{for each } y \succ^\infty x \sim^\infty z \in \mathcal{X} \quad \text{and} \quad p > 0.$$

Proof. Let $y \succ^\infty x \sim^\infty z$ and $p > 0$. By fanaticism, there is some $y' \sim^\infty z$ such that $p(y', z) > x$. Since $y' \sim^\infty x$ and $y \succ x$, lemma D.38(b) tells us that, $y \succ y'$. Then by stochastic dominance,

$$p(y, z) \succeq p(y', z) > x.$$

□

As we discussed in section 10.4, we get a version of Pascal's principle even without the fanatical assumption.

Definition D.40. Let $p \in [0, 1]$, and let $x \succ^\infty y \in \mathcal{X}$.

- (a) The probability distribution $p(x, y)$ is **positively Pascalian** iff $p < 1$, and there are no $p^+ > p$, $P, Q \in \Delta_0 \mathcal{X}$ such that

$$p(x, y) \sim^\infty P \succeq Q \sim^\infty p^+(x, y).$$

In other words, no increase in the probability of x , however small, can be outweighed by any finite differences in outcomes.

- (b) $p(x, y)$ is **negatively Pascalian** iff $p > 0$, and there are no $p^- < p$, $P, Q \in \Delta_0 \mathcal{X}$ such that

$$p^-(x, y) \sim^\infty P \succeq Q \sim^\infty p(x, y).$$

In other words, no decrease in the probability of x , however small, can be outweighed by any finite differences in outcomes.

- (c) A distribution is **Pascalian** iff it is either positively or negatively Pascalian.

Theorem D.41. Let $\mathcal{X}$ be a preordered set equipped with an embedding of the ∞ -order. Let $\succeq$ be any preorder on $\Delta_0 \mathcal{X}$ that (weakly) extends the weak stochastic dominance order. Then there exist uncountably many Pascalian prospects in $\Delta_0 \mathcal{X}$.

We will break the proof up into several lemmas.

Lemma D.42. For $z \in \mathcal{X}$ and $P \in \Delta_0 \mathcal{X}$, say z **splits** P iff $P' \succeq z$ for some but not all $P' \sim^\infty P$. If $P \succ^\infty Q$, then z splits at most one of P and Q .

Proof. If z splits Q , there is some $Q' \sim^\infty Q$ such that $Q' \gtrsim z$. Then for any $P' \sim^\infty P$, we know $P' \gtrsim Q'$ by lemma D.38(c), and so $P' \gtrsim z$. So z does not split P . $\square$

Lemma D.43. There exists $z \in \mathcal{X}$ and uncountably many pairs $x \succ^\infty y \in \mathcal{X}$ such that z does not split $p(x, y)$ for any rational $p \in [0, 1]$.

Proof. Recall Sierpinski's Lemma (lemma A.22): there is an order embedding of $\mathbb{R}$ into the ∞ -order on 2^ω . Let $z = f(A)$ where $A \subseteq \omega$ is an infinite and coinfinite set. Then we can apply Sierpinski's lemma twice, once to the ∞ -order on 2^A , and once to the ∞ -order on $2^{\omega \setminus A}$. This produces two order embeddings

$$B : \mathbb{R} \rightarrow 2^A \quad \text{and} \quad C : \mathbb{R} \rightarrow 2^{\omega \setminus A}.$$

For each $r \in \mathbb{R}$, let

$$\begin{aligned} x(r) &= f(A \cup C(r)) \\ y(r) &= f(B(r)) \\ P(p, r) &= p(x(r), y(r)) \quad \text{for each } p \in [0, 1]. \end{aligned}$$

By construction, if $r > r'$, then

$$A \cup C(r) \succ^\infty A \cup C(r') \succ^\infty A \succ^\infty B(r) \succ^\infty B(r').$$

So

$$x(r) \succ^\infty x(r') \succ^\infty z \succ^\infty y(r) \succ^\infty y(r').$$

So

$$P(p, r) \succ^\infty P(p, r').$$

Thus, by lemma D.42, for each p , there is at most one $r \in \mathbb{R}$ such that z splits $P(p, r)$. Call a number r *splittable* iff there is some rational p such that z splits $P(p, r)$. Since there are only countably many rational p , and for each of them there is at most one r such that z splits $P(p, r)$, there are only countably many splittable r . Thus there are uncountably many unsplittable r . For each unsplittable r , the pair $x(r) \succ^\infty y(r)$ satisfies the conditions of the lemma. $\square$

Lemma D.44. Let $p > q$ and $x \succ^\infty y$. Then for each $P \sim^\infty p(x, y)$ there exists some $Q \sim^\infty q(x, y)$ such that $P \gtrsim Q$. Likewise, for each $Q \sim^\infty q(x, y)$, there exists some $P \sim^\infty p(x, y)$ such that $P \gtrsim Q$.

Proof. If $P \sim^\infty p(x, y)$, then P assigns probability p to outcomes close to x , and probability $1 - p$ to outcomes close to y . Since $p > q$, we can further divide up the part of the distribution that is close to x into a part with probability q and a remaining part with probability $p - q$. That is, there are distributions $P_0 \sim^\infty P_1 \sim^\infty \delta x$ and $P_2 \sim^\infty \delta y$ such that

$$P = q \cdot P_0 + (p - q) \cdot P_1 + (1 - p) \cdot P_2.$$

Then let

$$Q = q \cdot P_0 + (p - q) \cdot (\delta y) + (1 - p) \cdot P_2.$$

Since $\delta y <^\infty P_1$, straightforwardly $P \succeq Q$ by stochastic dominance. Also, Q consists of a part with probability q close to x , and a part with probability $1 - q$ close to y . So $Q \sim^\infty q(x, y)$. The second part goes similarly. $\square$

Now we prove that each of the uncountably many pairs $x \succ^\infty y$ in lemma D.43 gives rise to a Pascalian prospect.

Proof of theorem D.41. Let $x \succ^\infty z \succ^\infty y$ be as in lemma D.43. Let

$$Q = \{q \in \mathbb{Q} \cap [0, 1] \mid q(x, y) \succeq z\}$$

and let $p = \inf Q$. There are three cases to consider.

- (a) $0 < p < 1$. Suppose for contradiction that $p(x, y)$ is neither positively nor negatively Pascalian. It follows that there are $p^- < p < p^+$ and P, Q such that

$$p^-(x, y) \sim^\infty P \succeq Q \sim^\infty p^+(x, y).$$

Then there exist rational $q^- \notin Q$ and $q^+ \in Q$ such that

$$p^- < q^- < p < q^+ < p^+.$$

By lemma D.44, we can push P up a bit and push Q down a bit, producing distributions P', Q' such that

$$\begin{aligned} P' &\succeq P \sim^\infty q^-(x, y) \\ Q' &\preceq Q \sim^\infty q^+(x, y). \end{aligned}$$

Furthermore, since $q^+ \in Q$, $q^+(x, y) \succeq z$, and z does not split $q^+(x, y)$, so

$$q^-(x, y) \sim^\infty P' \succeq P \succeq Q \succeq Q' \succeq z.$$

Since z does not separate $q^-(x, y)$ either, $q^-(x, y) \succeq z$. But this implies that $q^- \in Q$, which is a contradiction.

- (b) $p = 0$. Suppose for contradiction that p is not positively Pascalian: there exists $p^+ > 0$, $P \sim^\infty p(x, y) = \delta x$, and $Q \sim^\infty p^+(x, y)$, such that $P \succsim Q$. Then there exists rational $q \in Q$ such that $0 < q < p^+$. By lemma D.44, there is some $Q' \sim^\infty q(x, y)$ such that $P \succsim Q \succsim Q'$. Since $q \in Q$, $q(x, y) \succsim z$, and since z does not split $q(x, y)$, also

$$p(x, y) \sim^\infty P \succsim Q \succsim Q' \succsim z.$$

Since z does not split $p(x, y)$ (because zero is rational), it follows that $p(x, y) = x \succsim z$. But $z \succ^\infty x$, so $z \succ x$; this is a contradiction.

- (c) $p = 1$. This case goes similarly, for any $p^- < 1$.

□

D.5 Ex ante Pareto and statewise dominance

TODO. Change framing.

(McCarthy, Mikkola, and Thomas 2020)

Let $\mathcal{X}$ be a set of *individual outcomes*, and let I be a finite set of *individuals*. Let $\succsim$ be an ordering of $\Delta_0 \mathcal{X}^I$, the space of *social lotteries*.

Anteriority. For all $P, Q \in \Delta_0 \mathcal{X}^I$, if $\pi_i * P = \pi_i * Q$ for each $i \in I$, then $P \sim Q$.

Statewise Anonymity. Let Ω be any set, $P \in \Delta_0 \Omega$, and let $X, Y : \Omega \rightarrow \mathcal{X}^I$ be random variables. Suppose that for each $w \in \Omega$, there is some permutation $\pi : I \rightarrow I$ such that $X(w) = \pi(Y(w))$. Then $X * P \sim Y * P$.

We also define an ordering on *individual lotteries* in terms of the social ordering. Let $\kappa : \mathcal{X} \rightarrow \mathcal{X}^I$ be the function that takes each individual outcome $x \in \mathcal{X}$ to the constant social outcome that gives x to each individual.

$$(\kappa x)(i) = x \quad \text{for each } i \in I.$$

Let

$$P \succsim^\kappa Q \quad \text{iff} \quad \kappa * P \succsim \kappa * Q \quad \text{for each } P, Q \in \Delta_0 \mathcal{X}.$$

Idea: Harsanyi's "individualism". Reduce social choice to individual choice, by introducing uncertainty about *which individual you are*. If you face a social lottery P , and you randomize which individual prospect in P you get, then

your individual prospect is the average of each of the individual prospects that appear in P . Call this “randomized” prospect $\hat{P}$.

$$\hat{P} := \sum_{i \in I} \frac{1}{\#I} \pi_i * P.$$

Theorem D.45 (MMT aggregation theorem). Let $\succsim$ be a preorder of social lotteries $\Delta_0 \mathcal{X}^I$ that satisfies Anteriority and Statewise Anonymity. Then for all $P, Q \in \Delta_0 \mathcal{X}^I$,

$$P \succsim Q \quad \text{iff} \quad \hat{P} \succsim^\kappa \hat{Q}.$$

Proof. Let σ be a transitive permutation of I , so for any $i \in I$ we have

$$I = \{\sigma i, \sigma^2 i, \dots, \sigma^{n-1} i, \sigma^n i = i\}$$

(where n is the number of individuals).

Let $P \in \Delta_0 \mathcal{X}^I$ be a social lottery. Consider a two-stage random process: first we uniformly choose a number k from 1 to n , then we (independently) choose an outcome from P , and apply the permutation σ^k to its result. (This trick is essentially the same as in Mahtani (2017), as discussed in section 7.3.) This results in the following social lottery:

$$P' = \sum_{k=1}^n \frac{1}{n} \sigma^k * P.$$

By Statewise Anonymity, $P \sim P'$.

Furthermore, $P' \sim \kappa * \hat{P}$ by Anteriority. For in fact, for each $j \in I$,

$$\pi_j * P' = \sum_{k=1}^n \frac{1}{n} \pi_j * \sigma^k * P.$$

Since each $i \in I$ gets mapped to j by exactly one of the permutations $\sigma^1, \dots, \sigma^n$, this is the same as $\hat{P}$, the average of each of the individual lotteries $\pi_i * P$. We also have $\pi_i * \kappa * \hat{P} = \hat{P}$. So by Anteriority, $P' \sim \kappa * \hat{P}$.

Thus for any $P, Q \in \Delta_0 \mathcal{X}^I$, we have $P \sim \kappa * \hat{P}$ and $Q \sim \kappa * \hat{Q}$. So $P \succsim Q$ iff $\hat{P} \succsim^\kappa \hat{Q}$. $\square$

TODO. Discussion.

D.6 Separability and additivity

In chapter 8 we discussed how separability principles applied to prospects lead to risk neutrality. In this section we will show how this works in general terms — in particular, generalizing from “numbers of happy lives” to any quantity $\mathcal{U}$, in the sense of section B.1. (In particular, $\mathcal{U}$ is a partially ordered commutative monoid.) In REF we showed how social outcomes can be given quantitative representations, in a quite abstract sense, given certain principles like monotonicity and separability (*ex post*). Combining these results, we get a general representation theorem for simple social prospects. This is in the same spirit as many classic results in social welfare theory, deriving views in the neighborhood of expectational utilitarianism from separability ideas. (See footnote 3 for references.)

The argument there relied on Stochasticism: prospects with the same probability distribution over outcomes are equally good. Here we will bake this right into our framework, by working within the von Neumann–Morgenstern framework of orders on probability distributions. The principle we called Separability in chapter 8 turned on the idea of a *statewise sum* of two prospects. In the present context, we have to put this in a slightly roundabout way.

In this section, for any set X , let $\Delta X = \Delta_0^{\mathbb{Q}} X$ be the set of simple *rational* probability distributions over X .

Definition D.46. Let $\mathcal{U}$ be a quantity. A preorder $\succeq$ of $\Delta\mathcal{U}$ is **additively separable** iff, for any random variables $X, Y, Z : \Omega \rightarrow \mathcal{U}$ on any simple rational probability space $(\Omega, \mathbb{P})$,

$$\mathbb{P}_X \succeq \mathbb{P}_Y \quad \text{iff} \quad \mathbb{P}_{X+Z} \succeq \mathbb{P}_{Y+Z}$$

(where $X + Z$ is the statewise sum of X and Z , and likewise $Y + Z$).

If $\mathcal{U}$ is a p.o. rational vector space (a $\mathbb{Q}$ -quantity), then $\Delta\mathcal{U}$ can be endowed with the expectational order. This order is additive (by proposition D.12), and it faithfully extends the stochastic dominance order (by theorem D.30). In fact, it is characterized by these two properties.

Lemma D.47. Let $\mathcal{U}$ be any quantity, and let $\succeq$ be an additively separable preorder on $\Delta\mathcal{U}$. Then a uniform distribution over any outcomes $x_1, \dots, x_n \in \mathcal{U}$ is equivalent to a lottery with chance $1/n$ of getting the sum of those

outcomes, and otherwise nothing:

$$\sum_i \frac{1}{n} \cdot \delta x_i \sim \frac{1}{n} * \sum_i x_i.$$

(Here, as elsewhere, $p * x$ is the lottery $p \cdot \delta x + (1 - p) \cdot \delta 0$.)

Proof. Consider a state space $\Omega = \{1, \dots, n\}$ equipped with the uniform distribution $\mathbb{P}$, and define three random variables $X, Y : \Omega \rightarrow \mathcal{U}$:

$$\begin{aligned} X(i) &= x_i && \text{for } i \in \{1, \dots, n\} \\ Y(i) &= \begin{cases} \sum_{j=1}^n x_j & i = 1 \\ 0 & i > 1. \end{cases} \end{aligned}$$

Then by construction,

$$\begin{aligned} \mathbb{P}_X &= \frac{1}{n} \cdot \sum_i \delta x_i, \text{ and} \\ \mathbb{P}_Y &= \frac{1}{n} * \sum_i x_i, \end{aligned}$$

so it suffices to show that $\mathbb{P}_X \sim \mathbb{P}_Y$.

We will also define a third random variable. For each $i \in \{1, \dots, n\}$, define the partial sum

$$s_i = \sum_{j=1}^i x_j$$

and let $s_0 = 0$. Then define $Z : \Omega \rightarrow \mathcal{U}$ by

$$Z(i) = s_{i-1} \quad \text{for } i \in \{1, \dots, n\}.$$

Then it turns out that $X + Z$ and $Y + Z$ have the very same distribution over outcomes: both are uniform over the outcomes $s_1, \dots, s_n$. Thus

$$\mathbb{P}_{X+Z} = \mathbb{P}_{Y+Z}.$$

By additivity, $\mathbb{P}_X \sim \mathbb{P}_Y$. □

Theorem D.48. Let $\mathcal{U}$ be a $\mathbb{Q}$ -quantity (a p.o. rational vector space). The expectational order is the unique additively separable preorder on $\Delta\mathcal{U}$ that faithfully extends the stochastic dominance order.

Proof. Let $P, Q \in \Delta\mathcal{U}$ be rational probability distributions. Let n be the least common denominator of the probabilities appearing in the range of P and Q . Then we can write

$$P = \sum_i \frac{1}{n} \cdot \delta x_i$$

for some outcomes $x_1, \dots, x_n \in \mathcal{U}$ (possibly with repetitions), and likewise,

$$Q = \sum_i \frac{1}{n} \cdot \delta y_i.$$

Then

$$\begin{aligned} \mathbb{E}P &= \sum_i \frac{1}{n} \cdot x_i \\ \mathbb{E}Q &= \sum_i \frac{1}{n} \cdot y_i. \end{aligned}$$

By lemma D.47, we have

$$\begin{aligned} P &\sim \frac{1}{n} * \sum_i x_i \\ &= \frac{1}{n} * n \cdot \mathbb{E}P; \\ Q &\sim \frac{1}{n} * \sum_i y_i \\ &= \frac{1}{n} * n \cdot \mathbb{E}Q. \end{aligned}$$

Moreover,

$$\begin{aligned} \mathbb{E}P \geq \mathbb{E}Q &\text{ iff } n \cdot \mathbb{E}P \geq n \cdot \mathbb{E}Q \\ &\text{ iff } \frac{1}{n} * n \cdot \mathbb{E}P \geq_s \frac{1}{n} * n \cdot \mathbb{E}Q. \quad \text{ iff } P \succeq Q. \quad \square \end{aligned}$$

The expectational order is at home where $\mathcal{U}$ is (at least) a $\mathbb{Q}$ -quantity. We stated additivity in apparently more general terms, letting $\mathcal{U}$ be *any* quantity. But it turns out this generality is a bit illusory. If $\Delta\mathcal{U}$ admits an additively separable preorder, then $\mathcal{U}$ can be extended to a $\mathbb{Q}$ -quantity.

Lemma D.49. Let $\mathcal{U}$ be any quantity, and let $\succeq$ be an additively separable preorder on $\Delta\mathcal{U}$ that faithfully extends the stochastic dominance order. Then

for any $a, b \in \mathcal{U}$ and any $n \geq 1$,

$$\frac{1}{n} * a \gtrsim \frac{1}{n} * b \quad \text{iff} \quad a \geq b.$$

Proof. ($\Leftarrow$) If $a \geq b$, couple the two lotteries on $\Omega = \{1, \dots, n\}$ —placing a and b on the first state and 0 on the rest—to get $\frac{1}{n} * a \geq_s \frac{1}{n} * b$. Since $\gtrsim$ extends stochastic dominance, $\frac{1}{n} * a \gtrsim \frac{1}{n} * b$.

($\Rightarrow$) Suppose $\frac{1}{n} * a \gtrsim \frac{1}{n} * b$. For any $u \in \mathcal{U}$, realize these two lotteries on $\Omega = \{1, \dots, n\}$ by the random variables placing a (respectively b) on the first state and 0 elsewhere, and let Z place u on the first state and 0 elsewhere. Then $\mathbb{P}_{X+Z} = \frac{1}{n} * (a + u)$ and $\mathbb{P}_{Y+Z} = \frac{1}{n} * (b + u)$, so by additivity

$$\frac{1}{n} * (a + u) \gtrsim \frac{1}{n} * (b + u).$$

Chaining this to turn each copy of a into a copy of b in turn (taking $u = (n - 1 - j)a + jb$ at the j th step),

$$\frac{1}{n} * na \gtrsim \frac{1}{n} * ((n - 1)a + b) \gtrsim \dots \gtrsim \frac{1}{n} * nb.$$

But lemma D.47, applied to the constant sequences, gives $\delta a \sim \frac{1}{n} * na$ and $\delta b \sim \frac{1}{n} * nb$. So $\delta a \gtrsim \delta b$; since δ is faithful ($\delta a \gtrsim \delta b$ iff $a \geq b$), we conclude $a \geq b$. $\square$

Theorem D.50. Let $\mathcal{U}$ be any quantity (that is, a p.o. commutative monoid), and let $\gtrsim$ be an additively separable preorder on $\Delta\mathcal{U}$ that faithfully extends the stochastic dominance order. Then $\mathcal{U}$ has a representation $v : \mathcal{U} \rightarrow \mathcal{U}'$ in a $\mathbb{Q}$ -quantity $\mathcal{U}'$ (that is, a p.o. rational vector space). Furthermore, v is an expectational representation of $\gtrsim$.

Proof. Recall that a quantity can be embedded in a p.o. rational vector space iff it is *cancellative* ($x + z \geq y + z$ implies $x \geq y$) and *unperforated* ($nx \geq ny$ implies $x \geq y$, for any $n \geq 1$). (See theorems B.26 and B.28).

First some preliminaries. Recall that if $\gtrsim$ faithfully extends the stochastic dominance order, this implies in particular that $\delta : \mathcal{U} \rightarrow \Delta\mathcal{U}$ is faithful: $\delta x \gtrsim \delta y$ iff $x \geq y$.

Using additivity, we can also prove more generally:

$$\frac{1}{n} * x \gtrsim \frac{1}{n} * y \quad \text{iff} \quad x \geq y \quad \text{for each } x, y \in \mathcal{U}.$$

The right-to-left direction follows from stochastic dominance. Now suppose the left-hand-side holds. By applying additivity n times, we find

$$\begin{aligned} \delta x &\sim \frac{1}{n} * nx \\ &= \frac{1}{n} * (x + (n-1)x) \\ &\gtrsim \frac{1}{n} * (y + (n-1)x) \\ &= \frac{1}{n} * (x + y + (n-2)x) \\ &\gtrsim \frac{1}{n} * (y + y + (n-2)x) \\ &\gtrsim \dots \\ &\gtrsim \frac{1}{n} * ny \\ &\sim \delta y. \end{aligned}$$

And thus $x \geq y$.

Now we prove the main properties we need.

Cancellative. Suppose $x + z \geq y + z$; we show $x \geq y$. Consider the one-point probability space $\Omega = \{1\}$, with constant random variables $X(1) = x$, $Y(1) = y$, and $Z(1) = z$. Since $x + z \geq y + z$, we have

$$\mathbb{P}_{X+Z} = \delta(x+z) \gtrsim \delta(y+z) = \mathbb{P}_{Y+Z}.$$

By additivity,

$$\delta x = \mathbb{P}_X \gtrsim \mathbb{P}_Y = \delta y,$$

so $x \geq y$ by faithfulness.

Unperforated. Suppose $nx \geq ny$ for some integer $n \geq 1$; we show $x \geq y$. By lemma D.47 applied to the constant sequences $x, \dots, x$ and $y, \dots, y$, we have $\delta x \sim \frac{1}{n} * nx$ and $\delta y \sim \frac{1}{n} * ny$. Since $nx \geq ny$, we have $\frac{1}{n} * nx \gtrsim \frac{1}{n} * ny$, so $\delta x \gtrsim \delta y$, and hence $x \geq y$ by faithfulness.

Expectational representation. By (theorems B.26 and B.28), there is an embedding $v : \mathcal{U} \rightarrow \mathcal{U}'$ of $\mathcal{U}$ in a $\mathbb{Q}$ -quantity. To check that v is an expectational

representation, let $P, Q \in \Delta\mathcal{U}$. Then as in the proof of theorem D.48, there is a number n and outcomes $x_1, \dots, x_n, y_1, \dots, y_n \in \mathcal{U}$ such that

$$\begin{aligned} P &= \sum_i \frac{1}{n} \delta x_i \sim \frac{1}{n} * \sum_i x_i; \\ Q &= \sum_i \frac{1}{n} \delta y_i \sim \frac{1}{n} * \sum_i y_i; \\ \mathbb{E}_P v &= \sum_i \frac{1}{n} v(x_i); \\ \mathbb{E}_Q v &= \sum_i \frac{1}{n} v(y_i). \end{aligned}$$

Thus

$$\begin{aligned} P \succeq Q &\text{ iff } \frac{1}{n} * \sum_i x_i \succeq \frac{1}{n} * \sum_i y_i \\ &\text{ iff } \sum_i x_i \geq \sum_i y_i && \text{by the fact above} \\ &\text{ iff } \sum_i v(x_i) \geq \sum_i v(y_i) && \text{since } v \text{ is an embedding} \\ &\text{ iff } \sum_i \frac{1}{n} v(x_i) \geq \sum_i \frac{1}{n} v(y_i) \\ &\text{ iff } \mathbb{E}_P v \geq \mathbb{E}_Q v. \end{aligned}$$

□

Finally, let's spell out how this applies to risky social aggregation. As we saw in sections B.4 and B.7, finite anonymity and monotonicity suffice for the existence of the most general kind of additive representation for social welfare. Thus we can combine monotonicity directly with the additivity and stochastic dominance conditions above to get a kind of representation theorem for social aggregation under risk. But this way of doing it is a little awkward, since the additivity condition is stated in terms of a quantitative *representation* of social welfare, rather than directly in welfare terms. One more "direct" principle is *ex ante separability*.

(This approach also has disadvantage, though: unlike additivity, *ex ante separability* is *presentation-sensitive* in the same way as the *ex ante Pareto* principle discussed in chapter 7. What it says depends on how we pick out the *same* individual in different possible outcomes.)

We will consider here only *constant population* cases under uncertainty.

Definition D.51. A **(simple, risky, finitary) social welfare structure** consists of an infinite set of individuals $\mathcal{I}$, a set of individual welfares $\mathcal{X}$, and for each finite $I \in \text{Fin } \mathcal{I}$ a preorder $\succeq_I$ on $\Delta\mathcal{X}^I$.

Note that a risky finitary social welfare structure induces a finitary social welfare structure in the sense of section B.4, by considering certain outcomes as special cases of lotteries. (That is, we can define $x \succeq_I y$ iff $\delta x \succeq_I \delta y$ for $x \in \mathcal{X}^I$.) We say that a risky finitary social welfare structure has a property (such as separability or anonymity) *ex post* if its corresponding “riskless” structure has the property outright.

If $P \in \Delta\mathcal{X}^{I \cup J}$ is a prospect for a certain population $I \cup J$, then the prospect faced by just the *subpopulation* I is the pushforward $\pi_I * P$. (Here $\pi_I : \mathcal{X}^{I \cup J} \rightarrow \mathcal{X}^I$ is the projection map, $\pi_I(x)(i) := x(i)$ for $i \in I$.)

Definition D.52. A social welfare structure is **separable (ex ante)** iff, for any disjoint $I, J \in \text{Fin } \mathcal{I}$, and any lotteries $P, Q \in \Delta\mathcal{X}^{I \cup J}$, if $\pi_J * P = \pi_J * Q$, then

$$P \succeq Q \quad \text{iff} \quad \pi_I * P \succeq \pi_I * Q.$$

Definition D.53. A social welfare structure **respects stochastic dominance** iff for any finite population $I \in \text{Fin } \mathcal{I}$, the order $\succeq_I$ on $\Delta\mathcal{X}^I$ faithfully extends the stochastic dominance order (with respect to the induced “riskless” order on $\mathcal{X}^I$).

Notice that this *ex ante* separability condition entails separability *ex post*, again by considering certain outcomes as a special case of lotteries.

Definition D.54. An expectational additive representation of a risky finitary social welfare structure is a function $u : \mathcal{X} \rightarrow \mathcal{U}$ into a $\mathbb{Q}$ -quantity $\mathcal{U}$, such that for any finite population $I \in \text{Fin } \mathcal{I}$ and any lotteries $P, Q \in \Delta\mathcal{X}^I$,

$$P \succeq_I Q \quad \text{iff} \quad \mathbb{E}_P U \geq \mathbb{E}_Q U,$$

where

$$U(x) := \sum_{i \in I} u(x(i)) \quad \text{for each } x \in \mathcal{X}^I.$$

It is straightforward to get a representation of the *ex post* order in a quan-

tity $\mathcal{U}$, using the results of section B.4. We then want to take the various orders on *social lotteries* in $\Delta\mathcal{X}^I$ and “push them forward” into the space of *value lotteries* $\Delta\mathcal{U}$. One complication here is that we are not guaranteed that every value lottery can be realized by some constant population social lottery. Fortunately, we can always modify our value representation in a way that ensures this. The trick is to recalibrate values by picking some “zero” outcome; this lets us pad out a social outcome for a population I with additional “zero” individuals as needed, to get an “equivalent” outcome for a larger population. This is merely a formal trick: we are not making any assumptions about “real” cross-population value comparisons — we are just artificially introducing them into a value representation, for convenience.

[TODO. Fix clashing definition of “ideal”.]

Lemma D.55. Let $\mathcal{U}$ be an $\mathbb{N}$ -quantity. An **ideal** in $\mathcal{U}$ is a subset of $\mathcal{U}$ that contains zero, is closed under addition (for $x, y \in I, x + y \in I$), and is interval closed (for $x, y \in I$, if $x \leq z \leq y$, then $z \in I$).

For any ideal $I \subseteq \mathcal{U}$, the relation

$$x \sim_I y \text{ iff for some } i, j \in I, x + i \sim y + j.$$

is a congruence relation on $\mathcal{U}$ (section B.2). Thus it defines a quotient structure $\mathcal{U}/I$, which is also an $\mathbb{N}$ -quantity, and a quotient map $q : \mathcal{U} \rightarrow \mathcal{U}/I$, which is a surjective homomorphism of $\mathbb{N}$ -quantities. Furthermore, $q(i) = 0$ for all $i \in I$.

Proof. Straightforward to check. □

Theorem D.56. A social welfare structure admits an expectational additive representation iff it is finitely anonymous *ex post*, separable *ex ante*, and respects stochastic dominance.

Proof. Left-to-right. Straightforward, using the properties of expectations and quantities.

Right-to-left. Since separability *ex ante* implies separability *ex post*, and hence monotonicity, we can apply theorem B.45 and deduce that the *ex post* order has an additive representation $u : \mathcal{X} \rightarrow \mathcal{U}$ in an $\mathbb{N}$ -quantity $\mathcal{U}$. Moreover, we can choose $\mathcal{U}$ so that the corresponding function U is surjective: every element of $\mathcal{U}$ is the total value of some social outcome $x \in \mathcal{X}^I$, for some finite

population I . Also, considering the construction of $\mathcal{U}$, we can ensure that if $U(x) \geq U(y)$ in $\mathcal{U}$ then x and y both have the same population size.

Now, if we choose some “zero” element $o \in \mathcal{X}$, the set

$$\{n \cdot u(o) \mid n \in \mathbb{N}\} \subseteq \mathcal{U}$$

is an ideal. (Zero and addition are straightforward. Interval closure holds trivially, since for $m \neq n$, no element of $\mathcal{U}$ is comparable with both $m \cdot u(o)$ and $n \cdot u(o)$, by the “same population size” constraint.) Let $\tilde{\mathcal{U}}$ be the corresponding quotient $\mathbb{N}$ -quantity, with the quotient map $q : \mathcal{U} \rightarrow \tilde{\mathcal{U}}$. Let $\tilde{u} = q \circ u$, and likewise $\tilde{U} = q \circ U$. Then separability (*ex post*) ensures that $\tilde{u}$ is *also* an additive representation of the *ex post* order. In general, for any $I \in \text{Fin } \mathcal{I}$, let o_I be the social outcome which is constantly o on I . For any $x, y \in \mathcal{X}^I$, if $\tilde{U}(x) \geq \tilde{U}(y)$, then there exist $J, K \in \text{Fin } \mathcal{I}$ such that

$$U(x \sqcup o_J) = U(x) + (\#J) \cdot u(o) \geq U(y) + (\#K) \cdot u(o) = U(y \sqcup o_K) \quad \text{in } \mathcal{U}.$$

By the population size constraint, we must have $\#J = \#K$, and so $U(y \sqcup o_K) = U(y \sqcup o_J)$. Thus $x \sqcup o_J \succeq_{I \cup J} y \sqcup o_J$, and by separability, $x \succeq_I y$.

We will abuse notation: for $I \subseteq J \in \text{Fin } \mathcal{I}$, define the *padding function* $o_J : \mathcal{X}^I \rightarrow \mathcal{X}^J$ by

$$o_J(x) := x \sqcup o_{J \setminus I} \quad \text{for each } x \in \mathcal{X}^I.$$

Likewise, if $X : \Omega \rightarrow \mathcal{X}^I$ is a random variable, and $J : \Omega \rightarrow \text{Fin } \mathcal{I}$ is a population-valued random variable, where $J(w) \supseteq I$ for each $w \in \Omega$, then define the *padded random variable*

$$o_J(X)(w) := o_{J(w)}(X(w)) \quad \text{for each } w \in \Omega.$$

The same reasoning as above shows that for any $x \in \mathcal{X}^I, y \in \mathcal{X}^J, \tilde{U}(x) \geq \tilde{U}(y)$ iff

$$o_K(x) \succeq_K o_K(y) \quad \text{for each } K \supseteq I \cup J.$$

Likewise, for any simple random variables $X : \Omega \rightarrow \mathcal{X}^I$ and $Y : \Omega \rightarrow \mathcal{X}^J$, $\tilde{U} \circ X \geq \tilde{U} \circ Y$ statewise iff

$$o_K(X) \succeq_K o_K(Y) \text{ statewise} \quad \text{for each } K \supseteq I \cup J.$$

(Apply the previous fact to each state $w \in \Omega$.)

Similarly, for any $P \in \Delta \mathcal{X}^I, Q \in \Delta \mathcal{X}^J$, we have $\tilde{U} * P \geq_s \tilde{U} * Q$ iff

$$o_K * P \geq_s o_K * Q \quad \text{for each } K \supseteq I \cup J.$$

(We can realize P and Q by random variables X and Y such that $\tilde{U} \circ X \geq \tilde{U} \circ Y$ statewise, and apply the previous fact.) In particular, if $\tilde{U} * P = \tilde{U} * Q$, then $o_K * P \sim_K o_K * Q$ for all $K \supseteq I \cup J$ (since the order $\succeq$ on $\Delta\mathcal{X}^K$ respects stochastic dominance). The analogous fact also holds for strong stochastic dominance.

Likewise, the same padding tricks tell us that for any simple probability distribution $P \in \Delta\tilde{\mathcal{U}}$, there is some $P' \in \Delta\mathcal{X}^I$ such that $P = \tilde{U} * P'$. And for any simple random variable $X : \Omega \rightarrow \tilde{\mathcal{U}}$, there is a corresponding simple social outcome-valued random variable $X' : \Omega \rightarrow \mathcal{X}^I$ such that $X = \tilde{U} \circ X'$.

With all this established, we can now define a preorder on $\Delta\tilde{\mathcal{U}}$.

For any $P, Q \in \Delta\tilde{\mathcal{U}}$, $P \succeq Q$ iff for some (equivalently, for all) population $I \in \text{Fin } \mathcal{I}$ and social lotteries $P', Q' \in \Delta\mathcal{X}^I$,

$$\begin{aligned} P &= \tilde{U} * P', \\ Q &= \tilde{U} * Q', \text{ and} \\ P' &\succeq_I Q'. \end{aligned}$$

To check that the two formulations are equivalent, let $P', Q' \in \Delta\mathcal{X}^I$ and $P'', Q'' \in \Delta\mathcal{X}^J$, suppose $P = \tilde{U} * P' = \tilde{U} * P''$ and $Q = \tilde{U} * Q' = \tilde{U} * Q''$. Let $K \supseteq I \cup J$. Then by the facts above, $o_K * P' \sim o_K * P''$ and $o_K * Q' \sim o_K * Q''$. Then by separability *ex ante*,

$$\begin{aligned} P' \succeq_I Q' &\text{ iff } o_K * P' \succeq_K o_K * Q' \\ &\text{ iff } o_K * P'' \succeq_K o_K * Q'' \\ &\text{ iff } P'' \succeq_J Q''. \end{aligned}$$

It is straightforward to check that this defines a preorder. For reflexivity, we use the fact above that every $P \in \Delta\tilde{\mathcal{U}}$ is equal to $\tilde{U} * P'$ for some $P' \in \Delta\mathcal{X}^I$, and some I . For transitivity, we can suppose without loss of generality that $P \succeq Q$ and $Q \succeq R$ are each witnessed by the same population I (since we can again pad out to a larger common population as necessary). So suppose $P' \succeq_I Q', Q'' \succeq_I R''$, and $\tilde{U} * Q' = \tilde{U} * Q''$; it follows that $Q' \sim_I Q''$, and thus $P' \succeq_I R''$.

It is also straightforward to check that this order faithfully extends stochastic dominance. Suppose $P \geq_s Q$ in $\Delta\tilde{\mathcal{U}}$. Then there are $P', Q' \in \Delta\mathcal{X}^I$ such that $\tilde{U} * P' = P$ and $\tilde{U} * Q' = Q$ (again padding as necessary); then the above facts tell us that $P' \geq_{sI} Q'$, and thus $P' \succeq_I Q'$ (since $\succeq_I$ respects stochastic

dominance) and so $P \succeq Q$ by the definition. Similar reasoning shows that if P strongly stochastically dominates Q , then $P \succ Q$.

Next we check that the order is additive. Given simple random variables $X, Y, Z : \Omega \rightarrow \tilde{\mathcal{U}}$, we can find disjoint populations I and J and corresponding social outcome-valued random variables $X', Y' : \Omega \rightarrow \mathcal{X}^I$ and $Z' : \Omega \rightarrow \mathcal{X}^J$ such that $X = \tilde{U} \circ X', Y = \tilde{U} \circ Y'$, and $Z = \tilde{U} \circ Z'$. Then

$$\begin{aligned} X + Z &= \tilde{U} \circ (X' \sqcup Z'), \text{ and} \\ Y + Z &= \tilde{U} \circ (Y' \sqcup Z'). \end{aligned}$$

Thus

$$\begin{aligned} \mathbb{P}_X \succeq \mathbb{P}_Y &\text{ iff } \mathbb{P}_{\tilde{U} \circ X'} = \tilde{U} * \mathbb{P}_{X'} \succeq \tilde{U} * \mathbb{P}_{Y'} = \mathbb{P}_{\tilde{U} \circ Y'} \\ &\text{ iff } \mathbb{P}_{X'} \succeq_I \mathbb{P}_{Y'} \\ &\text{ iff } \mathbb{P}_{X' \sqcup Z'} \succeq_{I \cup J} \mathbb{P}_{Y' \sqcup Z'} && \text{by separability} \\ &\text{ iff } \mathbb{P}_{X+Z} \succeq \mathbb{P}_{Y+Z}. \end{aligned}$$

Thus we have defined an additively separable preorder on $\Delta\tilde{\mathcal{U}}$ that faithfully extends the stochastic dominance order; so we can apply theorem D.50, producing an expectational representation $v : \tilde{\mathcal{U}} \rightarrow \mathcal{U}'$ of the order on $\Delta\tilde{\mathcal{U}}$ in a $\mathbb{Q}$ -quantity $\mathcal{U}'$. Finally, we compose these two representations, and it can be checked that $v \circ \tilde{u}$ is an expectational additive representation of the original order. $\square$

D.7 Invariance and fanaticism

In section 9.3 we discussed an argument for Finite Fanaticism based on the Sure Thing Principle and a principle about symmetries of value. We will prove a general form of the underlying result here.

Definition D.57. Call a preorder $\succeq$ on $\Delta_0\mathcal{X}$ **bounded** iff there exists an element $0 \in \mathcal{X}$, elements $y \succ 0 \succ x$, and probabilities $p, q > 0$, such that

$$\begin{aligned} p * z &\succeq x && \text{for each } 0 \succ z \\ y &\succeq q * z && \text{for each } z \succ 0, \end{aligned}$$

where $p * z$ is the simple probability distribution $[z \mapsto p, 0 \mapsto 1 - p]$.

Let $G \curvearrowright \mathcal{X}$ be a group action. There is a corresponding action on simple probability distributions $G \curvearrowright \Delta_0 \mathcal{X}$:

$$(g \cdot P)(x) = P(g^{-1} \cdot x) \quad \text{for each } g \in G, P \in \Delta_0 \mathcal{X}, x \in \mathcal{X}.$$

Proposition D.58. Consider the action by addition of $\mathbb{Z}$ on $\mathbb{Z}$. No addition-invariant order of $\Delta_0 \mathbb{Z}$ that satisfies Independence is bounded.

Proof. By theorem D.18, there exists a vector expected utility representation $u : \mathbb{Z} \rightarrow \mathcal{U}$, where $\mathcal{U}$ is a partially ordered vector space, and $P \succeq Q$ iff $\mathbb{E}_P u \geq \mathbb{E}_Q u$ for all $P, Q \in \Delta_0 \mathbb{Z}$. Without loss of generality, we can suppose that $u(0) = 0$. If $\succeq$ is bounded, then there exist $x, y \in \mathbb{Z}$, $m, n \in \mathbb{N}$ such that for all $z \in \mathbb{Z}$,

$$m \cdot u(x) < u(z) < n \cdot u(y).$$

Suppose that $-x \geq y$. (The other case $y \geq -x$ goes similarly, but in the opposite direction, running from positive to negative values.) So $u(-y) \geq u(x)$. Then in particular $u(2y) \leq n \cdot u(y)$. By addition invariance,

$$u(y) + (n-1) \cdot u(-y) \leq 0.$$

So

$$u(y) \leq (n-1) \cdot (-u(-y)) \leq (n-1) \cdot (-u(-x)).$$

To simplify notation, let $v(k) = u(-kx)$ for each integer k . This is a strictly increasing function from $\mathbb{Z}$ to $\mathcal{U}$. Let $d = \frac{1}{m-1}(-v(-1))$. We will show next that

$$v(k) > kd \quad \text{for each positive integer } k.$$

To do this, we will prove by induction that

$$v(k) - v(k-1) > d$$

for each $k > 1$.

We know, by applying the lower bound on u to the particular case of mx ,

$$v(-m) > m \cdot v(-1).$$

So by addition invariance, for any integer k ,

$$v(k-m) + (m-1) \cdot v(k) > m \cdot v(k-1).$$

Rearranging this, we find

$$v(k) - v(k-1) > \frac{1}{m-1}(v(k-1) - v(k-m)).$$

First consider $1 \leq k < m$. Then since v is increasing, we have $v(k-1) \geq v(0) = 0$ and $v(k-m) \leq v(-1)$. Thus

$$v(k) - v(k-1) > \frac{1}{m-1}(-v(-1)) = d.$$

Next, let $k \geq m$, and suppose for induction that

$$v(j) - v(j-1) > \frac{1}{m-1}(v(j-1) - v(j-m))$$

for each integer j ranging from $k-m$ to $k-1$. Then

$$v(k-1) - v(k-m) = \sum_{j=k-m}^{k-1} (v(j) - v(j-1)) > (m-1)d.$$

Thus

$$v(k) - v(k-1) \geq \frac{1}{m-1}(v(k-1) - v(k-m)) > d$$

completing the inductive proof.

Thus

$$u(-kx) > \frac{k}{m-1}(-u(x)) \quad \text{for each } k > 0.$$

Putting this together with the first part, we conclude that

$$u(-n(n-1)(m-1)x) > n(n-1)(-u(x)) \geq n \cdot u(y).$$

This contradicts the claim that $n \cdot u(y)$ is an upper bound. □

TODO. We can generalize this, then. For any element $g \in G$ with an infinite orbit, there is a corresponding subgroup isomorphic to $\mathbb{Z}$. If also $g \cdot x > x$ for each $x \in \mathcal{X}$, then the orbit of any element under this subgroup is likewise order isomorphic to $\mathbb{Z}$.

D.8 Infinite prospects

In appendix E we will turn to standard measure theory, focusing on continuous measures. Before we get there, though, we have a few things to do that live in between simple finite-support prospects and the full resources of measure theory. These are issues related to prospects with countably infinite support — *discrete* prospects. Most of the things we have to say in this section are connected one way or another to the St. Petersburg paradox. The technical resources here are relatively simple compared to those of appendix E, but some of the harder philosophical issues about representing probability, which were not very pressing when we talked about finite-support distributions (especially those taking only rational probability values) arise here in full force. The results in this section take probabilities to be represented by real numbers, excluding infinitesimals; I argued in chapters 5 and 8 that this kind of representation actually omits information that is pertinent to the value of prospects.

In this section, for the most part I will collect some important results for reference, without recapitulating proofs.

Definition D.59. A **discrete probability distribution** on a set Ω is a function $P : \Omega \rightarrow \mathbb{R}$ with countable support such that $\sum P = 1$. Let $\Delta_\omega \Omega$ be the set of all discrete probability distributions on Ω .

A set Ω equipped with a discrete probability distribution $\mathbb{P}$ is called a **discrete probability space**.

A discrete probability distribution also induces a **discrete probability measure**,

$$P(A) := \sum_{w \in A} P(w) \quad \text{for each } A \subseteq \Omega.$$

This is countably additive.

Most of the concepts we introduced for finite support distributions carry over straightforwardly to discrete distributions: support, mixtures (now including *countable* mixtures), pushforwards, the VNM axioms, and stochastic dominance.

Not everything is straightforward, though: *expectations* need more care. Not every discrete distribution on the real numbers (or other quantities) has a well-defined expectation, if we define this by straightforwardly generalizing

definition D.11:

$$\mathbb{E}P = \sum_{x \in \mathcal{U}} P(x) \cdot x.$$

Before, since P had finite support, this was a well-defined finite sum. Now, though, it is an *infinite* sum, which may not converge (or even be meaningful for a general quantity $\mathcal{U}$). The most famous counterexample is the St. Petersburg lottery (letting $\mathcal{U} = \mathbb{R}$), which we have encountered many times in this book. This is related to the following facts.

Definition D.60. Let $\succeq$ be a preorder on $\Delta_\omega \mathcal{X}$, and let $Q \in \Delta_\omega \mathcal{X}$. A sequence of distributions $P_1, P_2, \dots \in \Delta_\omega \mathcal{X}$ is **supergeometric** with respect to Q iff there is some $p \in (0, 1)$ such that

$$p \cdot P_{n+1} + (1 - p) \cdot Q \succeq P_n \succ Q \quad \text{for each } n \in \mathbb{N}.$$

A preorder on $\Delta_\omega \mathcal{X}$ is **limited** iff it admits no supergeometric sequence, with respect to any $Q \in \Delta_\omega \mathcal{X}$. (J. S. Russell and Isaacs 2021; J. S. Russell 2020a)

Finite Fanaticism is one way for a preorder to be unlimited. A different way is for it to contain an ascending chain of lexicographically superior prospects — a chain of greater and greater regimes of infinite value.

Generalized St. Petersburg lotteries show us the following.

Theorem D.61. Let $\succeq$ be an unlimited preorder on $\Delta_\omega \mathcal{X}$, and suppose $\succeq$ respects stochastic dominance, in the sense that $P \succ Q$ whenever $P \succ_s Q$ (in the sense that P and Q can be realized by random variables X and Y such that $X > Y$ statewise).

- (a) $\succeq$ does not satisfy the Archimedean or Continuity axioms;
- (b) There exists an **improper prospect**, in the sense of a distribution $P \in \Delta_\omega \mathcal{X}$ such that $P \succ \delta x$ for every $x \in \text{supp } P$.

(section 9.4; Menger [1934] 1979; Beckstead and Thomas 2024; J. S. Russell 2020a)

Theorem D.62 (Seidenfeld, Schervish, and Kadane (2009)). Let $\mathcal{U}$ be a $\mathbb{Z}$ -quantity. No non-trivial preorder on $\Delta_\omega \mathcal{U}$ that faithfully extends strict stochastic dominance is additively separable, in the sense (from definition D.46) that for any $X, Y, Z : \Omega \rightarrow \mathcal{U}$ on any discrete rational probability

space $(\Omega, \mathbb{P})$,

$$\mathbb{P}_X \succeq \mathbb{P}_Y \quad \text{iff} \quad \mathbb{P}_{X+Z} \succeq \mathbb{P}_{Y+Z}.$$

On the other hand, we can define expectations in some cases. A *bounded* distribution on $\mathbb{R}$ still has a well-defined expectation, defined in the usual way as a (countable) sum of probability-weighted values. (Some unbounded distributions also have well-defined expectations, but we will put this off until section E.2.)

J. S. Russell (2020a) proves a representation theorem for limited total preorders, which specializes Hausner's lexicographic representation (discussed in section D.2). It involves a special kind of expected utility model.

Definition D.63. For any index set I , $\ell^\infty(I)$ is the set of all bounded functions from I to $\mathbb{R}$. If I is well-ordered, we can equip this space with the lexicographic order, which makes it a totally ordered vector space. (It is a subspace of the lexicographic space $\mathbb{R}_{\text{lex}}^I$: definition B.30.)

Let α be an ordinal, and let $B(\alpha) = [-1, 1]^\alpha \subset \ell^\infty(\alpha)$, the set of functions whose range is contained in the interval $[-1, 1]$. (This is the unit ball in $\ell^\infty(\alpha)$, as a Banach space.)

Every distribution on the interval $[-1, 1]$ has a well-defined expectation (also in the same interval). For any $P \in \Delta_\omega B(\alpha)$, a discrete distribution over transfinite sequences in $[-1, 1]$, there is a corresponding transfinite sequence of distributions on $[-1, 1]$, given by the pushforwards $\pi_i * P$ for each ordinal $i < \alpha$. So we can take the expectation of each of these component distributions, and this gives us a new sequence in $B(\alpha)$. This defines a well-behaved expectation operator $\mathbb{E} : \Delta_\omega B(\alpha) \rightarrow B(\alpha)$. And this in turn defines a preorder $\succeq$ on $\Delta_\omega B(\alpha)$, where $P \succeq Q$ if and only if $\mathbb{E}P \geq \mathbb{E}Q$ in $B(\alpha)$ (with respect to the lexicographic order), for any $P, Q \in \Delta_\omega B(\alpha)$. Call this the **expectational order** on $\Delta_\omega B(\alpha)$.

(Nothing here turns on *discrete* distributions; the construction carries over straightforwardly to standard countably additive probability measures on $B(\alpha)$.)

Here is the main representation theorem (which we will not reprove here).

Theorem D.64 (J. S. Russell (2020a)). Let $\succeq$ be a total preorder on $\Delta_\omega \mathcal{X}$. The following are equivalent:

- (a) For any probabilities $p_1, p_2, \dots \in [0, 1]$ such that $\sum_i p_i = 1$, and any distributions $P_1, P_2, \dots, Q_1, Q_2, \dots \in \Delta_\omega \mathcal{X}$:
- (i) If $P_i \succeq Q_i$ for all i , then $\sum_i p_i \cdot P_i \succeq \sum_i p_i \cdot Q_i$;
 - (ii) If, furthermore, $P_i \succ Q_i$ for some i such that $p_i > 0$, then $\sum_i p_i \cdot P_i \succ \sum_i p_i \cdot Q_i$.
- (b) For some ordinal α , there exists a function $u : \mathcal{X} \rightarrow B(\alpha)$ such that the function

$$P \in \Delta_\omega \mathcal{X} \mapsto \mathbb{E}(u * P) \in B(\alpha)$$

is an order embedding, with respect to the lexicographic order on $B(\alpha)$.

Property (a) is a discrete lottery version of Eventwise Dominance (sections 6.3 and 9.4). It implies as a special case that there are no improper prospects. (Let $x_1, x_2, \dots$ enumerate the support of P , let $p_i = P(x_i)$, $P_i = P$, and $Q_i = \delta x_i$.) As another special case, (a) implies that $\succeq$ respects stochastic dominance (supposing $\mathcal{X}$ to be equipped with the induced order $\delta x \succeq \delta y$). (For any discrete coupling $R \in \Delta_\omega(\mathcal{X} \times \mathcal{X})$, we can enumerate the support $(x_1, y_1), (x_2, y_2), \dots$ and let $p_i = R(x_i, y_i)$, $P_i = \delta x_i$, and $Q_i = \delta y_i$.) Consequently, (a) also implies limitedness, by theorem D.61(b). Furthermore, it implies the Independence axiom, by considering cases where $p_i = 0$ for all but two of the probabilities.

Note that the expectational order on $\Delta_\omega B(\alpha)$ is *itself* representable as in condition (b), letting u be the identity on $B(\alpha)$. Thus:

Proposition D.65. The expectational order on $\Delta_\omega B(\alpha)$ satisfies Independence and Comparability, faithfully extends the stochastic dominance order, has no improper prospects, and is therefore limited.

Now we prove this claim from section 10.5: the bounded lexicographic space shows that generalized St. Petersburg gambles are not as inevitable as generalized Pascalian gambles. Any order on outcomes at all can be given an expected vector utility representation that respects stochastic dominance and has no improper prospects. This is a skeletal version of theorem D.64; the hypotheses are dramatically weaker, but the conclusion only gives us a representation of *outcomes* rather than *lotteries*.

Theorem D.66. For any preordered set $\mathcal{X}$, there exists a strictly monotone function $f : \mathcal{X} \rightarrow B(\alpha)$ for some ordinal α . (If $\mathcal{X}$ is totally preordered, this is

an order embedding.)

Proof. We use Choice, via the well-ordering principle. For some ordinal α , there is an enumeration $(U_i)_{i < \alpha}$ of all upward-closed subsets of $\mathcal{X}$. Define $u : \mathcal{X} \rightarrow B(\alpha)$ by

$$u(x)(i) := \begin{cases} 1 & \text{if } x \in U_i, \\ -1 & \text{otherwise.} \end{cases}$$

If $x \geq y$ in $\mathcal{X}$, then whenever $y \in U_i$, we also have $x \in U_i$, so $u(x)(i) \geq u(y)(i)$ for any $i < \alpha$, and in particular $u(x) \geq u(y)$ lexicographically.

If $x > y$, then we already know $u(x) \geq u(y)$; it remains to show that $u(x) \neq u(y)$. For some i , $U_i = \uparrow x = \{z \in \mathcal{X} \mid z \geq x\}$, the principal upper set generated by x . If $u(x) = u(y)$, then since $x \in U_i$, we would also have $y \in U_i$, and thus $y \succeq x$. $\square$

This gives us a representation of an arbitrary order that avoids St. Petersburg-style improper prospects. On the other hand, it does not avoid Pascalian gambles. On the contrary, this model is quite extreme in that it makes *every* non-trivial gamble Pascalian! If $x > y$, then $u(x) > u(y)$ *lexicographically*. This particular model is not intended as a serious theory of value (though alternative embeddings in ℓ^∞ spaces may be more promising). Rather, it is an illustration of how sharply St. Petersburg and Pascal can come apart.

We also claimed in section 10.5 that there are bounded lexicographic utilities that do not allow *St. Petersburg pairs* — a generalization of improper prospects. Recall:

Quantitative prospects X and Y are a **St. Petersburg pair** iff it is certain that X and Y are only finitely different from one another, but there is no finite value difference d such that $X + d$ is as good as Y .

Making this precise requires a bit more structure on our space of outcomes: we need a notion of *finite value difference*. (Note that *value* differences should not be confused with differences between *utility* representations. We are considering generalizations of bounded utility models, which do not capture intuitive cardinal differences between outcomes — such as those measured by the net number of happy lives.)

Let $\mathcal{U}_0 \subseteq \mathcal{U}$ be quantities, where $\mathcal{U}_0$ is a substructure of $\mathcal{U}$, intuitively representing the *finite* quantities within a possibly larger space of quantities. We will suppose that $\mathcal{U}_0$ can be represented by finite numbers: there exists some strictly monotone function from $\mathcal{U}_0$ to $\mathbb{R}$. We can then compose this with a bounded strictly monotone function on $\mathbb{R}$ to get a strictly monotone function from $\mathcal{U}_0$ to $[0, 1]$.

We can capture the idea of a St. Petersburg pair with a pair of quantity-valued random variables X and Y , such that $Y - X$ is finite everywhere, while $Y \succ X + d$ for any finite quantity d .

Theorem D.67. Let $\mathcal{U}_0 \subseteq \mathcal{U}$ be $\mathbb{Z}$ -quantities, and let $v : \mathcal{U}_0 \rightarrow [0, 1]$ be strictly monotone. We call elements of $\mathcal{U}_0$ *small*.

Suppose $\mathcal{U}_0$ contains an increasing cofinal sequence. (That is, there are small $x_1 < x_2 < \dots$ such that for every $y \in \mathcal{U}_0$ is below some x_n .) Suppose furthermore that $\mathcal{U}_0$ is order convex in $\mathcal{U}$. (That is, if x is small and $0 \leq y \leq x$, then y small as well.)

There exists a strictly monotone utility function $u : \mathcal{U} \rightarrow B(\alpha)$, for some ordinal α such that, for any random variables $X : \Omega \rightarrow \mathcal{U}$ and $D : \Omega \rightarrow \mathcal{U}_0$ (on any discrete probability space), there is some small d such that $X + d$ has greater expected utility than $X + D$ (in $B(\alpha)$).

Proof. The idea is the same as the proof of theorem D.66, with a slight tweak. Call an upward-closed subset $U \subseteq \mathcal{U}$ *invariant* iff $U + \mathcal{U}_0 = U$: that is, U is closed under finite differences. Enumerate these as $(U_i)_{i < \alpha}$ for some ordinal α . This time we will define a function $u : \mathcal{U} \rightarrow B(\alpha + 1)$. For $i < \alpha$, the definition of u goes precisely the same way as in theorem D.66.

This leaves one coordinate to define. For $x, y \in \mathcal{U}$, let $x \sim^\infty y$ iff $x - y \in \mathcal{U}_0$. Choose one representative from each $\sim^\infty$ equivalence class, and let $z(x)$ be the representative of x 's equivalence class. Then $x - z(x)$ is in $\mathcal{U}_0$, so we can define

$$u(x)(\alpha) := v(x - z(x)).$$

Let $u_\alpha(x) := u(x)(\alpha)$: this is a bounded function from $\mathcal{U} \rightarrow [0, 1]$, which is strictly monotone on each $\sim^\infty$ equivalence class.

Note that $x \sim^\infty y$ iff $u(x)(i) = u(y)(i)$ for all $i < \alpha$.

Now we check that u is strictly monotone. As before, if $x \geq y$, then $f(x)(i) \geq f(y)(i)$ for each $i < \alpha$. If $f(x)$ and $f(y)$ differ before α , then $f(x) > f(y)$. If

they agree for all $i < \alpha$, then $x \sim^\infty y$, so $z(x) = z(y)$; so if $x \geq y$ then we have $v(x - z(x)) \geq v(y - z(y))$.

Now for strictness, suppose $x \geq y$ and $f(x) = f(y)$; we will show $x \sim y$. Consider $U = \uparrow x + \mathcal{U}_0$, which is upward closed and invariant. Since $x \in U$, we must also have $y \in U$; so there is some $d \in \mathcal{U}_0$ such that $y \geq x + d$. So $x - y$ lies between 0 and $-d$, and thus (by order convexity) is small, and so again $z(x) = z(y) = z$. Since $v(x - z) = v(y - z)$ and v is strictly monotone, $x \sim y$.

It is monotone for the same reason as before.

It remains to check that there are no St. Petersburg pairs. Let $X : \Omega \rightarrow \mathcal{U}$, $D : \Omega \rightarrow \mathcal{U}_0$. Then $X \sim^\infty X + D$ everywhere, so at every coordinate other than α , $u \circ X$ and $u \circ (X + D)$ agree statewise. Things go similarly for $X + d$. So the comparison between $\mathbb{E}u \circ (X + D)$ and $\mathbb{E}u \circ (X + d)$ comes down to their last coordinate. Let $Z = X - z(X)$, and for any $\mathcal{U}_0$ -valued random variable Y , let $V(Y) = \mathbb{E}(v \circ Y)$. What we want to prove is that, for some $d \in \mathcal{U}_0$,

$$V(Z + D) \leq V(Z + d).$$

Let $M = \sup_{\mathcal{U}_0} v$. For any $x \in \mathcal{U}_0$, there is some $y > x$, and thus $v(x) < v(y) \leq M$. Then by the properties of (real-valued) expectations

$$V(Z + D) < M$$

as well.

Now, let $d_1, d_2, \dots$ be a cofinal sequence in $\mathcal{U}_0$. For each n , let

$$T_n(w) = v(Z(w) + d_n) \quad \text{for each } w \in \Omega.$$

Then the functions $T_n \uparrow M$ in the sense of the monotone convergence theorem (theorem E.19). (First, T_n is pointwise increasing, since v is monotone, as is addition in $\mathcal{U}_0$. Second, for any $\epsilon > 0$ and $w \in \Omega$, there is some $x \in \mathcal{U}_0$ such that $v(x) > M - \epsilon$, and thus some d_n such that $d_n \geq x - Z(w)$, so $v(Z(w) + d_n) > M - \epsilon$.)

Thus, by monotone convergence, the expectations $\mathbb{E}T_n$ converge to M as well. Since $V(Z + D) < M$, this means there is some d_n such that

$$V(Z + d_n) > V(Z + D)$$

which was what we needed to show. □

One more thing to include in this section's round-up is a result we discussed in chapters 6 and 7, specific to infinite social welfare. As in (section A.4), let $\mathcal{X}$ be a preordered set of individual outcomes and let $\mathcal{I}$ be an infinite set of individuals. For any discrete prospect P on $\mathcal{X}^{\mathcal{I}}$, we can also define the *individual prospect* for each $i \in \mathcal{I}$, as the pushforward $\pi_i * P$ under the projection to the i th component. Here is one regimentation of one of the “paradoxes of dominance” involving individual prospects.

Theorem D.68. Let $\mathcal{X}$ be a non-trivial preordered set of individual outcomes, and let $\mathcal{I}$ be an infinite set of individuals. No preorder $\succsim^*$ on $\Delta_{\omega}\mathcal{X}^{\mathcal{I}}$ satisfies both of the following conditions:

- (a) *Ex Post Anonymity.* Let $\sim$ be the equivalence relation on $\mathcal{X}^{\mathcal{I}}$ of being related by a finite permutation of individuals. If $P \sim_s Q$, then $P \sim^* Q$.
- (b) *Ex Ante Pareto.* For any $P, Q \in \Delta_{\omega}\mathcal{X}^{\mathcal{I}}$, if $\pi_i * P \succ_s \pi_i * Q$ for every $i \in \mathcal{I}$, then $P \succ^* Q$.

The proof follows the sketch in section 7.2.

Appendix E

Standard Probability

In this appendix I will present orthodox foundations for decision theory using the standard tools of real-valued, countably-additive measure theory and integration. Many nice properties of the finite case do not carry over. In the next appendix I take up alternative tools of non-standard analysis, which have somewhat brighter prospects.

This appendix is somewhat less self-contained than the others, since the alternative would require making it extremely long.

E.1 Basic measure theory

The orthodox mathematical approach to probability theory models *propositions* as subsets of some state space Ω , and their *probabilities* as real numbers. In particular, probabilities are a *measure*. We have already encountered measures (definition B.71), in a setting of more general quantities: these are just finitely additive functions from sets to some quantity. In standard probability theory, we specialize to the real numbers, which have an additional important property:

Completeness. Any bounded set of real numbers $X \subseteq \mathbb{R}$ has a least upper bound (or *supremum*), written $\sup X$.

This allows us to strengthen the additivity property: probabilities standardly assumed to be *countably additive*: for any countable sequence of disjoint sets

$A_1, A_2, \dots$ that have probabilities,

$$\mathbb{P}(A_1 \cup A_2 \cup \dots) = \mathbb{P}(A_1) + \mathbb{P}(A_2) + \dots = \sum_i \mathbb{P}(A_i).$$

We have encountered a variety of infinite sum concepts throughout this book; this one really *is* the familiar one from calculus, making a serious appearance for the first time. That is, to be explicit:

$$\sum_i \mathbb{P}(A_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(A_i)$$

where this limit is the standard “epsilon-delta” limit, rather than anything fancy like a filter limit. Equivalently, this is the least upper bound of the partial sums:

$$\sum_i \mathbb{P}(A_i) = \sup_n \sum_{i=1}^n \mathbb{P}(A_i).$$

This is guaranteed to exist as long as the partial sums are bounded.

We can also split up countable additivity into two pieces: the familiar finite additivity property, and a new *continuity* property. Say $A_1, A_2, \dots$ are a sequence of sets; we write $A_i \uparrow A$ to mean that these sets monotonically approach A : that is, $A_1 \subseteq A_2 \subseteq \dots$ and $A = \bigcup_i A_i$. Then we can define countably additive measures as follows.

Definition E.1. Let $\mathcal{X}$ be a set, and let $\mathcal{A} \subseteq 2^{\mathcal{X}}$ be a family of subsets of $\mathcal{X}$. A **(countably additive) finite measure** on $(\mathcal{X}, \mathcal{A})$ is a function $\mu : \mathcal{A} \rightarrow \mathbb{R}^+$ such that

- (a) $\mu(\emptyset) = 0$; and
- (b) **Finite Additivity.** $\mu(A \cup B) = \mu(A) + \mu(B)$ for any disjoint $A, B \in \mathcal{A}$; and
- (c) **Continuity.** For any sequence of sets $A_i \uparrow A$,

$$\mu(A) = \sup_i \mu(A_i).$$

(We understand $\mu(A) = \dots$ to imply that $A \in \mathcal{A}$.)

The function μ is a **probability measure** if, furthermore, $\mu(\mathcal{X}) = 1$.

In a moment, we'll see why the domain of a measure is restricted to a family of subsets $\mathcal{A}$, rather than just letting it (more straightforwardly) be defined on *all* subsets of the space $\mathcal{X}$.

Countable additivity seems like a nice property, but it pretty quickly gets us into some technical difficulties. It turns out that there is no way to assign countably additive probabilities to arbitrary sets of states in a way that respects very natural symmetry conditions. Here's a start:

Proposition E.2. No countably additive probability measure P on a countably infinite set Ω is **uniform**, in the sense that $P\{\omega\} = P\{\omega'\}$ for every $\omega, \omega' \in \Omega$.

Proof. In this case, countable additivity tells us

$$P(\Omega) = \sum_{\omega \in \Omega} P\{\omega\}.$$

If $P\{\omega\} = c > 0$ for all $\omega \in \Omega$, then the sum on the right-hand side does not converge. If $c = 0$, then the sum is zero. In either case, $P(\Omega) \neq 1$. $\square$

Definition E.3. Let $G \curvearrowright \Omega$ be a group action. This also induces an action $G \curvearrowright 2^\Omega$ (example A.39). Say a probability measure P on $(\Omega, \mathcal{A})$ is *G-symmetric* iff

$$P(g \cdot A) = P(A) \quad \text{for every } A \in \mathcal{A} \text{ and } g \in G.$$

(Again, we understand this to imply that $g \cdot A \in \mathcal{A}$ as well.)

(Note that being *uniform* is equivalent to being symmetric under $\Pi_{\text{fin}}\Omega$, the group of finite-support permutations of Ω .)

Definition E.4. A group action $G \curvearrowright \mathcal{X}$ is **free** iff for any $x \in \mathcal{X}$ and $g \in G$, if $g \cdot x = x$ then g is the identity element of G . (Equivalently: G **acts freely** on $\mathcal{X}$.)

For example, the action of the finitary coin flip group is free (example A.44). Likewise, so are the actions of the group of translations of a line, and of the group of rotations of a circle (example A.43).

Theorem E.5. Suppose G has a countably infinite subgroup that acts freely on Ω . Then there is no countably additive G -symmetric probability measure

P on $(\Omega, 2^\Omega)$ — that is, where $P(A)$ is defined for every subset $A \subseteq \Omega$.

(This is a straightforward generalization of the standard *Vitali theorem*. The proof relies on the Axiom of Choice.)

Proof. If G is a subgroup of H , then any H -symmetric function is automatically G -symmetric. So it suffices to prove the case where G itself is a countably infinite subgroup that acts freely on Ω .

We can partition Ω into its *orbits* under G . These are disjoint non-empty sets; so by Choice, there is a set A that contains exactly one element from each orbit. Every element of Ω is in $g \cdot A$ for some $g \in G$. Also, for $g \neq g'$, the sets $g \cdot A$ and $g' \cdot A$ are disjoint. (Suppose $g \cdot a = g' \cdot a'$ for some $a, a' \in A$. Then $g^{-1}g' \cdot a' = a$, which means a and a' are in the same orbit; so our choice of A guarantees that $a = a'$. Then, since G acts freely, $g^{-1}g' = 1$, so $g = g'$ as well.)

Thus the sets $g \cdot A$ for $g \in G$ are a countably infinite partition of Ω . Furthermore, by G -symmetry,

$$P(g \cdot A) = P(A) \quad \text{for each } g \in G.$$

Moreover, if G is countable, then by countable additivity we have

$$P(\Omega) = P\left(\bigcup_{g \in G} g \cdot A\right) = \sum_{g \in G} P(g \cdot A) = \sum_{g \in G} P(A).$$

But this contradicts $P(\Omega) = 1$, for the same reason as in the proof of proposition E.2. $\square$

(Note that there is no difference between the G -symmetry and G -invariance for real-valued measures, because the real numbers are totally ordered. (See sections A.6 and B.8.))

Example E.6.

- (a) The group of finitary coin flip symmetries (example A.44) is countably infinite and acts freely on the coin flip space 2^ω . Thus no countably additive probability measure defined on all subsets of the coin flip space is symmetric under finitary coin-flip symmetries.
- (b) The group of rational rotations (example A.43) is countably infinite and acts freely on the circle (represented by the interval $[0, 1)$). Thus

there is no countably additive rotation-symmetric probability measure defined on all subsets of $[0, 1)$.

The orthodox response to impossibility results like these is to *restrict the domain* of the probability measures we study. Instead of trying to assign probabilities to all subsets of a state space, we only assign probabilities to certain *special* subsets, which we call *measurable*. We can do this for the coin flip space and for the unit interval, to recover suitably symmetric countably additive probability measures that are not defined on all subsets, but only certain nice ones. And indeed, the cases where the measure gives out seem pretty pathological. In particular, we can restrict the domain of the measure to a σ -algebra of subsets, which is closed under countable unions and complements.

Note, though, that this restriction strategy does not help at all with the case of uniform probabilities on a countably infinite set. In this case, the only σ -algebra small enough to avoid the result includes only $\emptyset$ and Ω .

Definition E.7.

- (a) Let $\mathcal{X}$ be a set. A **σ -algebra** on $\mathcal{X}$ is a collection of subsets A of $\mathcal{X}$ such that A contains $\emptyset$ and $\mathcal{X}$, and A is closed under complements and countable unions.
- (b) A **measurable space** is a set $\mathcal{X}$ equipped with a σ -algebra A of subsets, which are called **measurable subsets**. We will write $E \stackrel{\text{m}}{\subseteq} \mathcal{X}$ to say that E is a measurable subset of $\mathcal{X}$; that is, $E \in A$.
- (c) If $\mathcal{X}$ is a measurable space, by a **probability measure on $\mathcal{X}$** we understand a countably additive probability measure P defined on the measurable subsets of $\mathcal{X}$. For a measurable space $\mathcal{X}$, let $\Delta_\sigma \mathcal{X}$ be the set of all countably additive probability measures on $\mathcal{X}$.

Restricting our measures to special algebras of sets lets us recover nice symmetries.

Definition E.8. Let $\Omega = 2^\omega$ be the coin flip space. For $I \subseteq \omega$, let $\pi_I : \Omega \rightarrow 2^I$ be the projection function. A subset $E \subseteq \Omega$ is a **cylinder set** iff for some finite set I and some $A \subseteq 2^I$, $E = \pi_I^{-1}(A)$.

The **product σ -algebra** on Ω is the smallest σ -algebra of subsets of Ω that

contains all cylinder sets. When considered as a measurable space, the coin flip space is equipped with its product σ -algebra.

Example E.9. There exists a unique countably additive finitary coin flip symmetric probability measure defined on the coin flip space's product algebra. This is standardly called the **product measure**; I will also call it the **standard coin flip measure**.

Example E.10. The **Borel algebra** on $[0, 1)$ is the smallest σ -algebra containing all interval subsets of $[0, 1)$. There exists a unique countably additive rational-rotation-symmetric probability measure defined on the Borel algebra. (This is standardly called the **Lebesgue measure**.)^a

^aTo be precise, the standard domain of the Lebesgue measure — the *Lebesgue-measurable sets* — is a larger σ -algebra than the Borel algebra that I just described. But for our purposes the restriction of the Lebesgue measure to the Borel algebra will be all we need.

I will not prove this here. (They both follow from Carathéodory's extension theorem: see Klenke 2013, sec. 1.3. For the uniqueness part, see theorem F.34.)

Note that these measures also inherit some additional symmetries that we didn't have to bake into their characterization.

Proposition E.11.

- (a) The standard coin flip measure is symmetric under all coin flip symmetries (the action of the Cantor group 2^ω) and all coin permutations (the action of $\Pi(\omega)$).
- (b) The Lebesgue measure is symmetric under all rotations (the action of $\mathbb{R}/\mathbb{Z}$).

Something worth noting about both of the coin flip measure and the Lebesgue measure is that, in each case, each singleton set gets measure zero. This is a striking “something from nothing” fact: the whole space has measure one, but it is made up entirely of measure zero pieces. (This is closely related to another of Zeno's paradoxes: see Skyrms [1983] 2012 for discussion). Counterintuitive as it is, this feature is inescapable in the setting of standard measure theory.

Proposition E.12. A measure P on a measurable space $\mathcal{X}$ is **regular** iff, for every $E \subseteq \mathcal{X}$, if $P(E) = 0$ then $E = \emptyset$. If $\mathcal{X}$ can be partitioned into uncountably many disjoint measurable sets, then no real-valued probability measure on $\mathcal{X}$ is regular.

Proof. Given a partition $\mathcal{E}$ of $\mathcal{X}$ into measurable sets, there is an order $\succeq_P$ on countable subsets of $\mathcal{E}$ which is represented by $S \mapsto P(\bigcup S)$. If P is regular, this order is strictly monotone. But theorem A.31 tells us that no strictly monotone order on the countable subsets of an uncountable set is real-representable.¹ $\square$

Notably, proposition E.12 does not depend on countable additivity: not even a finitely additive measure on an uncountable space (with a rich enough σ -algebra) can be regular.

It will be useful to have a notion of a “probability one” set that we can apply even when the set in question is not really measurable. This is not just generality for its own sake; in fact, non-measurable sets are highly relevant in applications to infinite ethics. Consider the space $2^{\mathcal{I}}$ as “happy-sad” social outcomes for an infinite population $\mathcal{I}$. Zame (2007) proves that no *total* finitely anonymous Paretian order for an infinite population is a measurable subset of $2^{\mathcal{I}} \times 2^{\mathcal{I}}$. In the case where the population $\mathcal{I}$ is *uncountably* infinite things are even worse: non-measurable relations on $2^{\mathcal{I}} \times 2^{\mathcal{I}}$ include even the *identity* relation (that is, the diagonal of $2^{\mathcal{I}} \times 2^{\mathcal{I}}$), as well as the *Pareto dominance* relation (that is, $x \geq y$ coordinatewise). But in both cases it will still sometimes be an important question whether a probability measure on the product space $2^{\mathcal{I}} \times 2^{\mathcal{I}}$ is supported by such relations. (As we discuss in section E.4, this is directly tied to stochastic dominance and equivalence relations.)

Definition E.13. Let $P \in \Delta_{\sigma}(\mathcal{X})$, and let $E \subseteq \mathcal{X}$ (not necessarily measurable). We say E **supports** P iff for any measurable sets $A, B \subseteq \mathcal{X}$, if $A \cap E = B \cap E$ then $P(A) = P(B)$.

As with simple distributions, it is often useful to push probability measures around from one space to another.

¹Our proof of theorem A.31 relied on Choice. In this application, a direct constructive proof is also available. Consider $\mathcal{E}$ as a union of sets $\mathcal{E}_n$ that each contain only sets of measure at least $1/n$. By finite additivity, each of these sets is finite.

Definition E.14.

- (a) If $\mathcal{X}$ and $\mathcal{Y}$ are measurable spaces, a **measurable function** $f : \mathcal{X} \rightarrow \mathcal{Y}$ is one such that for each $E \subseteq \mathcal{Y}$, the preimage $f^{-1}(E) \subseteq \mathcal{X}$ is also measurable. Write $f : \mathcal{X} \xrightarrow{m} \mathcal{Y}$ in this case.
- (b) For $P \in \Delta_\sigma \mathcal{X}$ and $f : \mathcal{X} \xrightarrow{m} \mathcal{Y}$, the **pushforward** probability measure $f * P \in \Delta_\sigma \mathcal{Y}$ is defined such that

$$(f * P)(E) = P(f^{-1}(E)) \quad \text{for each } E \subseteq \mathcal{Y}.$$

Some basic facts from the simple case still hold here.

Proposition E.15. For $P \in \Delta_\sigma \mathcal{X}$, $f : \mathcal{X} \xrightarrow{m} \mathcal{Y}$, and $g : \mathcal{Y} \xrightarrow{m} \mathcal{Z}$, we have

$$(g \circ f) * P = g * (f * P).$$

As before, probability theorists often use alternative terminology and notation that can be usefully suggestive.

Definition E.16.

- (a) A **probability space** is a pair (Ω, P) , of a measurable space Ω and a probability measure P on Ω .
- (b) Let $(\Omega, \mathbb{P})$ be a probability space, and let $\mathcal{X}$ be a measurable space. An **$\mathcal{X}$ -valued random variable** on Ω is a measurable function $X : \Omega \xrightarrow{m} \mathcal{X}$.
- (c) The **distribution** of an $\mathcal{X}$ -valued random variable X (on a probability space $(\Omega, \mathbb{P})$) is the pushforward measure $X * \mathbb{P} \in \Delta_\sigma \mathcal{X}$. This is also written $\mathbb{P}_X$. (Probabilists also call this X 's *law*.)

Definition E.17. Let $\mathcal{X}$ and $\mathcal{Y}$ be measurable spaces. The **product space** $\mathcal{X} \times \mathcal{Y}$ is the measurable space whose underlying set is the Cartesian product of $\mathcal{X}$ and $\mathcal{Y}$, and whose σ -algebra is the smallest σ -algebra that makes each of the projection functions $\pi_1 : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{X}$ and $\pi_2 : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{Y}$ measurable.

For a probability measure $R \in \Delta_\sigma(\mathcal{X} \times \mathcal{Y})$, we call the measures $\pi_1 * R \in \Delta_\sigma \mathcal{X}$ and $\pi_2 * R \in \Delta_\sigma \mathcal{Y}$ the **marginals** of R .

For an $\mathcal{X}$ -valued random variable X and a $\mathcal{Y}$ -valued random variable Y , the **joint distribution** of X and Y is the pushforward measure $(X, Y) * \mathbb{P} \in \Delta_\sigma(\mathcal{X} \times \mathcal{Y})$.

$\Delta_\sigma(\mathcal{X} \times \mathcal{Y})$, where $(X, Y) : \Omega \xrightarrow{\text{m}} \mathcal{X} \times \mathcal{Y}$ is the measurable function defined by $(X, Y)(\omega) = (X(\omega), Y(\omega))$.

Note by the composition rule that the marginals of the joint distribution for X and Y are the distributions of X and Y , respectively.

In general, there are many different ways of “coupling” probability measures on two spaces to produce a joint probability measure on the product space: these represent different ways that the two measures might “interact”, with different kinds of correlation. One particularly important way to do this is the case where the two measures are *independent*. We state the following standard result without proof (see Klenke 2013, ch. 14).

Theorem E.18. For any measurable spaces $\mathcal{X}$ and $\mathcal{Y}$, and any $P \in \Delta_\sigma \mathcal{X}$ and $Q \in \Delta_\sigma \mathcal{Y}$, there exists a unique measure $P \otimes Q \in \Delta_\sigma(\mathcal{X} \times \mathcal{Y})$, called the **product measure**, with the property that for any $A \subseteq \mathcal{X}$ and $B \subseteq \mathcal{Y}$,

$$(P \otimes Q)(A \times B) = P(A) \cdot Q(B).$$

E.2 Expectations and integration

Standard probabilities also have a notion of an *expectation*, which is defined as an *integral*. But this notion is quite a bit more delicate than the simple case. For one thing, not every real-valued random variable has a well-defined expectation. For another, the generalization from the real case to other quantities (like vector spaces or semimodules) is not nearly as straightforward; there is more than one competing integral concept.² We will not go into this in any detail. But we will introduce some basic integration concepts and facts without definitions or proofs.³

²McCarthy, Mikkola, and Thomas (2020, 91 and pp. 104–105) propose the *Pettis (or weak) integral* as an appropriately general vector integral concept for applications in social welfare theory. But I will not go into these details.

³Besides the $\mathcal{L}^1$ space of functions, it is also common to define the quotient space $L^1(\mathcal{X}, P)$ of *equivalence classes* of functions in $\mathcal{L}^1(\mathcal{X}, P)$, up to equality P -almost everywhere. (This is the equivalence relation that holds between f and g when $P\{x \in \mathcal{X} \mid f(x) \neq g(x)\} = 0$.) This is convenient for some purposes (for example, it makes $L^1(\mathcal{X}, P)$ a Banach space) but not something we need to worry about here.

Theorem E.19. Let $\mathcal{X}$ be a measurable space, and let $P \in \Delta_\sigma \mathcal{X}$ be a probability measure on $\mathcal{X}$. There exists a unique set of measurable real-valued functions $\mathcal{L}^1(\mathcal{X}, P) \subset (\mathcal{X} \multimap \mathbb{R})$ — the **absolutely (P-)integrable functions** — and a unique operator

$$\int dP : \mathcal{L}^1(\mathcal{X}, P) \rightarrow \mathbb{R}$$

— the **(P-)integral** — with all of the following properties. (We write $\int f dP$ as equivalent to $\int dP f$, the value of the integral operator applied to f .)

- (a) **Subsets.** For any measurable set $E \subseteq \mathcal{X}$, the characteristic function 1_E is in $\mathcal{L}^1(\mathcal{X}, P)$, and

$$\int 1_E dP = P(E).$$

- (b) **Linearity and positivity.** $\mathcal{L}^1(\mathcal{X}, P)$ is a linear subspace of $\mathbb{R}^{\mathcal{X}}$, and the integral is a positive linear operator. That is to say, for any $f, g \in \mathcal{L}^1(\mathcal{X}, P)$ and $a, b \in \mathbb{R}$, the function $a \cdot f + b \cdot g$ is also in $\mathcal{L}^1(\mathcal{X}, P)$, and

$$\int (a \cdot f + b \cdot g) dP = a \int f dP + b \int g dP$$

and also,

$$\text{if } f \geq 0 \text{ pointwise, then } \int f dP \geq 0.$$

- (c) **Domination.** If $f \in \mathcal{L}^1(\mathcal{X}, P)$ and $g : \mathcal{X} \multimap \mathbb{R}$ is a measurable function such that $|g| \leq |f|$ pointwise, then g , $|g|$, and $|f|$ are in $\mathcal{L}^1(\mathcal{X}, P)$, and

$$\left| \int g dP \right| \leq \int |g| dP \leq \int |f| dP.$$

- (d) **Monotone convergence.** Let $0 \leq f_1 \leq f_2 \leq \dots$ be a pointwise non-decreasing sequence of functions in $\mathcal{L}^1(\mathcal{X}, P)$, such that the pointwise least upper bound $f := \sup_i f_i$ is well-defined. (That is, the sequence $f_1(x), f_2(x), \dots$ is bounded, for each $x \in \mathcal{X}$.) In this case we write $f_i \uparrow f$. If the sequence of integrals $\int f_i dP$ is bounded, we also have $f \in \mathcal{L}^1(\mathcal{X}, P)$, and

$$\int f dP = \sup_i \int f_i dP.$$

(See Klenke 2013, ch. 4; Folland 1984, ch. 2.)

This can be generalized beyond probability measures to arbitrary measures

with very little change; but we will focus on the probability case. In this case, we also call $\int f dP$ the **expectation** of f with respect to P , and use the notation $\mathbb{E}_P f$. For a probability space $(\Omega, \mathbb{P})$, we simply write $\mathbb{E}f$ for $\mathbb{E}_{\mathbb{P}} f$.

If $E \subseteq \mathcal{X}$, we define

$$\int_E f dP := \int f \cdot 1_E dP,$$

where as usual 1_E is the characteristic function of E . We also sometimes use the “pointed” notation

$$\int f(x) dP(x)$$

when it is convenient to spell out the function in terms of an argument.

Note that the properties in theorem E.19 guarantee that any *bounded measurable* function on a measurable space $\mathcal{X}$ is in $\mathcal{L}^1(\mathcal{X}, P)$, for any choice of a probability measure $P \in \Delta_{\sigma}\mathcal{X}$ at all. (If f is bounded, then for some $a \in \mathbb{R}$, $|f| \leq |a \cdot 1_{\mathcal{X}}|$.)

Integrals with respect to different measures, or measures on different spaces, are systematically related to one another.

Proposition E.20 (Convexity). Integrals are convex with respect to their measures. That is, for $p \in [0, 1]$, $P, Q \in \Delta_{\sigma}\mathcal{X}$, and $f \in \mathcal{L}^1(\mathcal{X}, P) \cap \mathcal{L}^1(\mathcal{X}, Q)$, we also have $f \in \mathcal{L}^1(\mathcal{X}, p \cdot P + (1 - p) \cdot Q)$, and

$$\int f d(p \cdot P + (1 - p) \cdot Q) = p \cdot \int f dP + (1 - p) \cdot \int f dQ.$$

Proposition E.21 (Change of variables). For any measurable function $f : \mathcal{X} \xrightarrow{m} \mathcal{Y}$ and any $g \in \mathcal{L}^1(\mathcal{Y}, f \ast P)$, we also have $g \circ f \in \mathcal{L}^1(\mathcal{X}, P)$, and

$$\int (g \circ f) dP = \int g d(f \ast P).$$

Proof sketch. It can be checked that the operator

$$g \mapsto \int (g \circ f) dP$$

satisfies the properties of theorem E.19, with respect to $\mathcal{L}^1(\mathcal{Y}, f \ast P)$. Thus by uniqueness, it must be the same as the integral with respect to $f \ast P$. $\square$

Here is another important standard fact. Say we have a “function of two variables” — that is, a function f defined on a product space $\mathcal{X} \times \mathcal{Y}$ — as well

as probability measures $P \in \Delta_\sigma \mathcal{X}$ and $Q \in \Delta_\sigma \mathcal{Y}$. There are three different strategies we might use for calculating its integral. First, we could apply the product measure integral $\int dP \otimes Q$ directly. Second, for any fixed value of y , we can apply the $\int dP$ integral to the resulting function $f(-, y)$ on $\mathcal{X}$. Doing this for every $y \in \mathcal{Y}$ produces a function from $\mathcal{Y}$ to $\mathbb{R}$, which we can then also integrate with respect to Q . Third, we can do the same thing in the opposite order: Q -integral first, followed by P -integral. Conveniently, it doesn't matter which way we do it.

Theorem E.22 (Fubini–Tonelli). Let $\mathcal{X}$ and $\mathcal{Y}$ be measurable spaces, and let $P \in \Delta_\sigma \mathcal{X}$ and $Q \in \Delta_\sigma \mathcal{Y}$. Then for any $f \in \mathcal{L}^1(\mathcal{X} \times \mathcal{Y}, P \otimes Q)$, we have

$$\begin{aligned} \int f d(P \otimes Q) &= \int \left(\int f(x, y) dQ(y) \right) dP(x) \\ &= \int \left(\int f(x, y) dP(x) \right) dQ(y). \end{aligned}$$

(As we will see in appendix F, though, this is a crucial point of difference between standard probability theory and the non-standard version.)

Here is one more useful fact (which again we state without proof). It is often convenient to represent probability measures using curves — like the famous “bell curve”, or *normal distribution* (figure E.1). (I’ll focus on a truncated distribution on a finite interval $[-1/2, 1/2]$, rather than all of $\mathbb{R}$, just to avoid some technical issues.) The relative heights of the curve at different points represents relative probabilities of those points — sort of. But not exactly! For a continuous distribution like the normal distribution, the probability of each particular point is zero. So the curve height doesn’t directly represent the probability of *points*. Rather, it represents the probability of *sets* of points — by way of its *integral*. The probability of getting a result in the set E is equal to the area under the curve, on that set. The curve itself — in this case, a function from $\mathbb{R}$ to $\mathbb{R}$ is not itself a probability measure, but it *induces* a probability measure on $\mathbb{R}$ via integration. The curve itself is called a *probability density function*.

Note that the integral also depends on some underlying measure — in this case, we implicitly appeal to the natural *uniform* measure on the interval $[-1/2, 1/2]$. In general, let $P \in \Delta_\sigma \mathcal{X}$ be a probability measure. For any non-negative function $f \in \mathcal{L}^1(\mathcal{X}, P)$, we can define a function

$$Q(E) := \int_E f dP \quad \text{for each } E \subseteq \mathcal{X}.$$

This defines another finite measure on $\mathcal{X}$. In this case, we call f the **Radon–Nikodym derivative** of Q with respect to P , notated $f = dQ/dP$. (This is because of the analogy with the fundamental theorem of calculus: integration “undoes” taking a derivative.) If $\int f dP = 1$, then Q is a probability measure. In this case, f represents a probability density function corresponding to the probability measure Q .

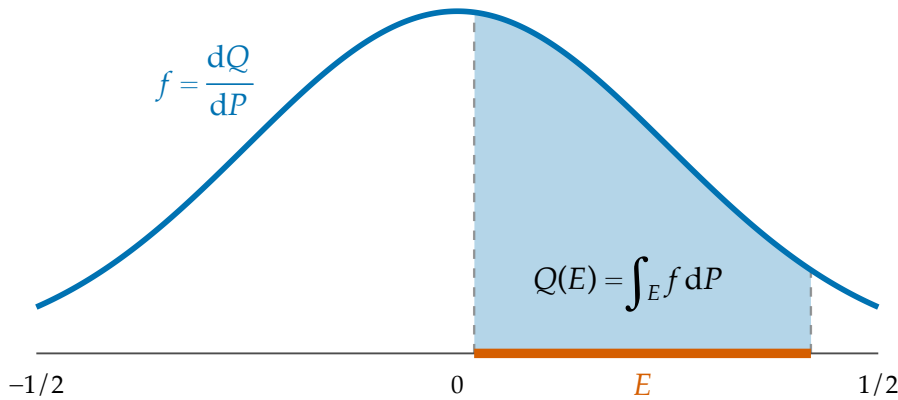


Figure E.1: A probability density function f determines a measure Q by integration: $Q(E) = \int_E f dP$, which is the area between the curve f and the set E . (Here P is the uniform measure.)

Which probability measures have density functions? Not all of them. For example, consider δ_0 , the indicator probability measure on $0 \in [-1/2, 1/2]$. This assigns probability one to a point. But any integrable function $f \in \mathcal{L}^1([-1/2, 1/2], P)$ must have

$$\int_{\{0\}} f dP = 0$$

because $P(\{0\}) = 0$. In general, the same problem arises if there is any set E that P assigns measure zero, while $Q(E) > 0$. In this case, intuitively, the density of Q on E would have to be infinite.

A central result in measure theory tells us that this is the *only* obstacle to having a density function.

Theorem E.23 (Radon–Nikodym). Let $\mathcal{X}$ be a measurable space and let $P, Q \in \Delta_\sigma \mathcal{X}$. We say Q is **absolutely continuous** with respect to P , written

$Q \ll P$, iff for all $E \subseteq \mathcal{X}$, if $P(E) = 0$ then $Q(E) = 0$. If this holds, then Q has a Radon–Nikodym derivative with respect to P : that is, there is a non-negative function

$$\frac{dQ}{dP} \in \mathcal{L}^1(\mathcal{X}, P)$$

such that for all $E \subseteq \mathcal{X}$,

$$Q(E) = \int_E \frac{dQ}{dP} dP.$$

E.3 Mixtures of probability measures

As I said, we can also make sense of integration for functions that don't have real values; we won't explore this in any detail, but one case is worth mentioning: the case of *measure-valued* functions. If $\mathcal{X}$ is a measurable space, we can also endow the space $\Delta_\sigma \mathcal{X}$ of probability measures with its own σ -algebra of measurable sets: in particular, we consider the smallest σ -algebra such that, for every measurable $E \subseteq \mathcal{X}$, the *evaluation* function

$$(P \in \Delta_\sigma \mathcal{X}) \mapsto P(E)$$

is measurable. Then we can make sense of measurable *probability-measure-valued* functions — and we can even integrate such functions.

Definition E.24. A **Markov kernel from $\mathcal{X}$ to $\mathcal{Y}$** is a measurable function $Q : \mathcal{X} \rightarrow \Delta_\sigma \mathcal{Y}$, which takes each $x \in \mathcal{X}$ to a probability measure Q_x on $\mathcal{Y}$.

Proposition E.25. Let Q be a Markov kernel from $\mathcal{X}$ to $\mathcal{Y}$, and let $P \in \Delta_\sigma \mathcal{X}$. Then we can also define a function

$$\hat{Q}(E) := \int Q_x(E) dP(x) \quad \text{for each } E \subseteq \mathcal{Y}.$$

(This is well-defined, because bounded functions are absolutely integrable). Then $\hat{Q}$ is a probability measure on $\mathcal{Y}$. We use the notation $\int Q dP$ for this probability measure.

Intuitively, this amounts to taking a probability-weighted *mixture* of a family of probability measures. In the case where P is a discrete measure, this reduces to the familiar notion of a convex combination of probability measures, like $p \cdot Q_1 + (1 - p) \cdot Q_2$.

Proof. We will prove that $\hat{Q}$ is countably additive. Let $A_1, A_2, \dots$ be a countable sequence of disjoint measurable subsets of $\mathcal{Y}$. For each $x \in \mathcal{X}$, the measures Q_x are countably additive. Note also that if we define a sequence of partial sum functions

$$g_n(x) = \sum_{i=1}^n Q_x(A_i)$$

$$g_\infty(x) = \sum_{i=1}^{\infty} Q_x(A_i),$$

then $g_n \uparrow g_\infty$, so we can apply monotone convergence. Thus:

$$\begin{aligned} \hat{Q}\left(\bigcup_i A_i\right) &= \int Q_x\left(\bigcup_i A_i\right) dP(x) \\ &= \int \sum_i Q_x(A_i) dP(x) \quad \text{by countable additivity for each } Q_x \\ &= \int g_\infty dP(x) \\ &= \sup_n \int g_n dP(x) \quad \text{by monotone convergence} \\ &= \sum_i \int Q_x(A_i) dP(x) \\ &= \sum_i \hat{Q}(A_i). \quad \square \end{aligned}$$

Proposition E.26. As in the simple case, for any $x \in \mathcal{X}$ we can define an **indicator probability measure** $\delta x \in \Delta_\sigma \mathcal{X}$, where

$$\delta x(E) = \begin{cases} 1 & \text{if } x \in E, \\ 0 & \text{otherwise.} \end{cases}$$

The function $\delta : \mathcal{X} \rightarrow \Delta_\sigma \mathcal{X}$ is a Markov kernel. For any $P \in \Delta_\sigma \mathcal{X}$ we have

$$\int \delta dP = P.$$

Finally, we introduce one more technical tool, which is a kind of generalization of conditionalization to standard probability measures, which gracefully accommodates probability zero events — known as **regular conditional probability**.⁴ Ordinary conditionalization takes a probability measure P and

⁴Due to Kolmogorov (1950). For philosophical discussion, see Rescorla (2015) and K.

a proposition $E \subseteq^m \mathcal{X}$ to a probability measure $P(\cdot|E)$. But for many purposes (for example, in applications of Bayes' theorem) we are really interested in the pair of conditional probabilities $P(\cdot|E)$ and $P(\cdot|\neg E)$ together. We can bundle these together into a function

$$\begin{cases} 1 \mapsto P(\cdot|E), \\ 0 \mapsto P(\cdot|\neg E). \end{cases}$$

Or equivalently,

$$(b \in \{0, 1\}) \mapsto P(\cdot|1_E = b).$$

This suggests a generalization: we can replace the “on-off” function $1_E : \mathcal{X} \rightarrow 2$ with *any* measurable function $\pi : \mathcal{X} \rightarrow \mathcal{Y}$. Intuitively, this tells us the probability conditional on the values of π . If π is a continuously varying function, often the specific values will have probability *zero*. But we can sort of make sense of conditional probabilities in these cases anyway.

This technical tool requires a bit more structure from our measurable spaces than we have needed for anything else so far: they must be *standard Borel spaces*. But before I explain this term, let's state the main theorem in which it is used.

Theorem E.27 (Disintegration, see Klenke (2013, sec. 8.3)). Let $\mathcal{X}$ and $\mathcal{Y}$ be standard Borel spaces, let $P \in \Delta_\sigma \mathcal{X}$, and let $\pi : \mathcal{X} \rightarrow \mathcal{Y}$ be measurable. There exists a Markov kernel Q from $\mathcal{Y}$ to $\mathcal{X}$ (unique up to $\pi * P$ -measure differences) such that $\pi^{-1}(y)$ supports Q_y for $\pi * P$ -almost every $y \in \mathcal{Y}$, and

$$P = \int_{\mathcal{Y}} Q_y \, d(\pi * P)(y).$$

The Markov kernel Q is called a **disintegration** of P along π .

We use the notation $P(\cdot | \pi)$ for Q , and we write $P(\cdot | \pi = y)$ for the probability measure Q_y for a specific $y \in \mathcal{Y}$.

Easwaran and Nielsen (2025, sec. 3), and further references therein. A terminological point: it is useful for our purposes to “conditionalize on” a *measurable function* $\pi : \mathcal{X} \rightarrow \mathcal{Y}$. The terminology of “regular conditional probability distributions” is associated with a different perspective, focusing on a *subalgebra* $\mathcal{B}$ of the algebra of measurable subsets of $\mathcal{X}$. Any measurable function $\pi : \mathcal{X} \rightarrow \mathcal{Y}$ straightforwardly generates an algebra $\mathcal{B} = \{\pi^{-1}(E) \mid E \subseteq \mathcal{Y}\}$. The *disintegration of P along π* , defined below, is the same as the **conditional probability distribution of P given $\mathcal{B}$** , where $\mathcal{B}$ is the subalgebra $\{\pi^{-1}(E) \mid E \subseteq \mathcal{Y}\}$ of the algebra of measurable subsets of $\mathcal{X}$.

To understand this, it is helpful to return to the special case where $\mathcal{Y} = 2$. In this case, the function $\pi : \mathcal{X} \xrightarrow{m} 2$ is the characteristic function of some subset $E \subseteq \mathcal{X}$, which we can think of as a proposition. The disintegration of P along π is a function that takes 1 to a probability measure $P(\cdot | \pi = 1) = P(\cdot | E)$ and 0 to a probability measure $P(\cdot | \pi = 0) = P(\cdot | \neg E)$. This has the property that, as long as $P[\pi = 1] > 0$, then the measure $P(\cdot | \pi = 1)$ is supported by the set $[\pi = 1]$: in other words, $P(\cdot | E)$ is supported by E . The same goes for $\pi = 0$; that is, if $P(\neg E) > 0$, then $P(\cdot | \neg E)$ is supported by $\neg E$. Finally, the integral property says

$$\begin{aligned} P &= \int_{\mathcal{Y}} P(\cdot | \pi = b) \, d(\pi * P)(b) \\ &= P(\cdot | \pi = 1) \cdot P[\pi = 1] + P(\cdot | \pi = 0) \cdot P[\pi = 0] \\ &= P(\cdot | E) \cdot P(E) + P(\cdot | \neg E) \cdot P(\neg E). \end{aligned}$$

This is the standard law of total probability. Together these support conditions and the law of total probability characterize standard conditional probabilities.

Note also that a disintegration $P(\cdot | \pi)$ is not generally *unique*, but only unique *up to $\pi * P$ -measure zero differences*. In the $\mathcal{Y} = 2$ case, this corresponds to the usual fact $P(\cdot | E)$ is standardly undefined in cases where $P(E) = 0$. In our case, when $P[\pi = y] = 0$, the value of $P(\cdot | \pi = y)$ is “not well-defined” just in the sense that it is *different* for different choices of the disintegration P . (It is still *defined* in the sense that any particular disintegration $P(\cdot | \pi)$ has some value $P(\cdot | \pi = y)$ — but this value is basically arbitrary.) The neat thing is that, in a case where π is continuously varying, even though no *particular* value $P(\cdot | \pi = y)$ is pinned down, even so the overall structure of the *function* $P(\cdot | \pi)$ can be quite determinate.

The key technical requirement for the disintegration theorem is that the measurable spaces in question are **standard Borel spaces**. Explaining this properly would require a long detour into topology, so I won’t.⁵ It is fine for our purposes just to think of standard Borel spaces as the “nice”, well-behaved ones. This includes most of our favorite spaces.

⁵Here is the official definition for those who will be able to make sense of it. A **Polish space** is a topological space which is *separable* and *completely metrizable*. The **Borel algebra** of a general topological space is the smallest σ -algebra that contains all open sets. A **standard Borel space** is a measurable space that is isomorphic to a Polish space equipped with its Borel algebra.

Proposition E.28. The following measurable spaces are standard Borel spaces:

- (a) The real line $\mathbb{R}$ (with its Borel algebra).
- (b) Any measurable subset $\mathcal{Y} \subseteq \mathcal{X}$ of a standard Borel space (with the subspace algebra, $\{E \cap \mathcal{Y} \mid E \subseteq \mathcal{X}\}$).
- (c) In particular, any finite or countably infinite set $\mathcal{X}$ (with its discrete algebra $2^{\mathcal{X}}$).
- (d) Any finite or countable product of standard Borel spaces (with the product algebra).
- (e) In particular, the coin flip space 2^{ω} .
- (f) Any finite or countable disjoint union of standard Borel spaces (with the disjoint union algebra).

In fact, a standard theorem tells us that standard Borel spaces are all very much alike, as far as their measurable structure is concerned: every standard Borel space is isomorphic (as a measurable space) to either a finite set, $\mathbb{N}$, or $\mathbb{R}$. (See Kechris 1995.)

A key example of a measurable space that is *not* standard Borel is the product space $2^{\mathcal{I}}$ where $\mathcal{I}$ is *uncountably* infinite. So using standard measure theory to study risky social welfare for uncountably infinite populations requires special care. But as we will see, plenty of pathologies already arise for merely countable populations.

E.4 Stochastic dominance

In section D.3 we defined weak and strong stochastic dominance relations for simple distributions, in several equivalent ways. For especially nice probability measures on preordered spaces the same equivalences still hold, and we have similarly well-behaved stochastic dominance concepts. Unfortunately, though, the preordered spaces that naturally arise in infinite ethics are not so nice; it turns out that stochastic dominance for standard probability measures on these spaces is not very well-behaved at all. We discussed some of these issues in chapter 5; here we will outline them a bit more systematically. In section F.6 we will see how things work better in a setting with infinitesimal probabilities.

Let's start with a relation that remains pretty well-behaved, even in the general case. We will present a rather abstract version that covers weak and strong versions of stochastic dominance as well as stochastic equivalence in one go.

Proposition E.29. Let $\mathcal{X}$ be any measurable space, let ρ be a relation on $\mathcal{X}$, and let $P, Q \in \Delta_\sigma \mathcal{X}$ be standard probability measures. The following are equivalent:

- (a) **Realization.** There are $\mathcal{X}$ -valued random variables X, Y on some probability space, with distributions $\mathbb{P}_X = P$ and $\mathbb{P}_Y = Q$, respectively, such that $X \rho Y$ pointwise.
- (b) **Realization 2.** There exist $\mathcal{X}$ -valued random variables X, Y with distributions $\mathbb{P}_X = P$ and $\mathbb{P}_Y = Q$ such that $\mathbb{P}[X \rho Y] = 1$.
- (c) **Coupling.** There exists a probability measure $R \in \Delta_\sigma(\mathcal{X} \times \mathcal{X})$, with marginals $\pi_1 * R = P$ and $\pi_2 * R = Q$, such that ρ supports R . (We call R a **coupling** of P and Q .)

We will call the relation ρ_s on $\Delta_\sigma \mathcal{X}$ characterized by these three equivalent formulations the **stochastic lift** of ρ . We will commonly say “ P stochastically ρ 's Q ” to mean that $P \rho_s Q$. (This relation ρ_s is also standardly called the **Barr extension** of ρ .)

Proof. TODO. (Straightforward.) □

Definition E.30.

- (a) For a preorder $\geq$ on $\mathcal{X}$, we call the relation $\geq_s$ on $\Delta_\sigma \mathcal{X}$ **(weak) stochastic dominance**.
- (b) In the case of an equivalence relation $\sim$, we call the relation $\sim_s$ **stochastic equivalence**.

When $\mathcal{X}$ is a preordered space, we understand these stochastic relations in terms of the weak order and equivalence relation that are defined in terms of the built-in preorder on $\mathcal{X}$.

Some nice properties of relations get lifted straightforwardly.

Proposition E.31. Let ρ be a relation on a measurable space $\mathcal{X}$.

- (a) If ρ is reflexive, then ρ_s is reflexive.
- (b) If ρ is symmetric, then ρ_s is symmetric.
- (c) If $\mathcal{X}$ is a standard Borel space and ρ is transitive, then ρ_s is transitive.
- (d) If $\rho \subseteq \rho'$, then $\rho_s \subseteq \rho'_s$.

Proof sketch. We use the coupling formulation. Let $P, Q, R \in \Delta_\sigma \mathcal{X}$.

- (a) The diagonal measure $\text{diag} * P \in \Delta_\sigma(\mathcal{X} \times \mathcal{X})$ is a coupling of P with itself supported by the diagonal of $\mathcal{X}$. If ρ is reflexive, the diagonal of $\mathcal{X}$ is a subset of ρ . So $P \rho_s P$.
- (b) Let R be a coupling of P and Q supported by ρ . Let $\sigma : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{X} \times \mathcal{X}$ be the function that swaps the two coordinates. Then $\sigma * R$ is a coupling of Q and P , supported by $\sigma(\rho)$. Since ρ is symmetric, $\sigma(\rho) = \rho$. So $Q \rho_s P$.
- (c) This part is the most technical. (It is known as the “gluing lemma”.) Let $S \in \Delta_\sigma(\mathcal{X} \times \mathcal{X})$ be a coupling of P and Q supported by ρ , and let $T \in \Delta_\sigma(\mathcal{X} \times \mathcal{X})$ be a coupling of Q and R also supported by ρ . Then we can “glue together” S and T (as in the simple case; see proposition D.25). Let S_x and T_x be disintegrations of S over π_2 and T over π_1 , respectively. Define

$$S \star_Q T := \int (\pi_1 * S_y) \otimes (\pi_2 * T_y) dQ(y).$$

It can be shown that $S \star_Q T \in \Delta_\sigma(\mathcal{X} \times \mathcal{X})$ is a coupling of P and R , supported by the “gluing” of ρ with itself:

$$\{(x, z) \mid \text{for some } y \in \mathcal{X}, x \rho y \text{ and } y \rho z\}.$$

If ρ is transitive, then this gluing is a subset of ρ ; so ρ supports $S \star_Q T$.

- (d) If $\rho \subseteq \rho'$, then any coupling of P and Q supported by ρ is also supported by ρ' .

□

Corollary E.32. On standard Borel spaces,

- (a) weak stochastic dominance is a preorder;

(b) stochastic equivalence is an equivalence relation.

As we will discuss below, though, other nice properties do not “lift” to stochastic relations so neatly. For example, the total order $\geq$ lifts to a preorder $\geq_s$ that is not ordinarily total. More strikingly, an *asymmetric* and *irreflexive* relation like a strict order $>$ may lift to a relation $>_s$ that is neither asymmetric nor irreflexive. Likewise, relations that stand in relations to each other like $\geq$, $>$, and $\sim$ lift to stochastic relations that do not generally stand in the same relations. For example, while $P \sim_s Q$ always entails $P \geq_s Q$, the conjunction of $P \geq_s Q$ and $Q \geq_s P$ does not always imply $P \sim_s Q$.

This can also lead to some notational clashes worth flagging. We have the convention that the *strict part* of a preorder $\geq$ is written $>$ (that is, $\geq \cap \not\leq$). The *stochastic* relation $\geq_s$ is a preorder, which has a strict part $\geq_s \setminus \leq_s$. But this is (in general) a different relation from $>_s$, the stochastic lift of the strict part of $\geq$. To put the point in slightly cryptic symbols:

$$(\geq_s \setminus \leq_s) \neq (\geq \setminus \leq)_s.$$

The left side is the strict part of the stochastic lift of $\geq$ (“ P dominates Q but Q does not dominate P ”), while the right side is the stochastic lift of the strict part of $\geq$ (“ P and Q have a coupling supported by the strict order $>$ ”). As we discuss below, for well-behaved orders the strict part of $\geq_s$ coincides with the relation we call *strong stochastic dominance*.

Similarly, the preorder $\geq_s$ has an associated equivalence relation $\geq_s \cap \leq_s$. But this is (in general) a different relation from $\sim_s$, which is the stochastic lift of the underlying equivalence relation $\sim$, that is, $(\geq \cap \leq)_s$.

When we discussed simple distributions, we also introduced another equivalent formulation of stochastic dominance: *cumulative dominance*.

Definition E.33. Let $P, Q \in \Delta_\sigma \mathcal{X}$, where $\mathcal{X}$ is a preordered set. We say that P **(weakly) cumulatively dominates** Q iff $P(U) \geq Q(U)$ for every measurable upward-closed subset $U \stackrel{\text{m}}{\subseteq} \mathcal{X}$.

Weak stochastic dominance straightforwardly implies weak cumulative dominance — but the converse can fail. Here is an example from J. S. Russell (forthcoming).

Example E.34. Let $\Omega = [0, 1] \times \{0, 1\}$ be equipped with the lexicographic order. Let P be the uniform measure on $I_0 = [0, 1] \times \{0\}$, and let Q be the uniform measure on $I_1 = [0, 1] \times \{1\}$. Every upward-closed subset of Ω takes one of the following forms:

$$\begin{aligned} &\{(x, b) \mid x \geq a, b \in \{0, 1\}\} \\ &\{(x, b) \mid x > a, b \in \{0, 1\}\} \\ &\{(x, b) \mid x > a, b \in \{0, 1\} \text{ or } x = a, b = 1\}. \end{aligned}$$

In each case, $U \cap I_0$ and $U \cap I_1$ differ by at most a single point; so $P(U) = Q(U)$. But it can be shown that there is no coupling of P and Q that is supported by the lexicographic order: that is, P does not weakly stochastically dominate Q .

Similar issues arise for the real expected utility formulation. [TODO CHECK]

For sufficiently *nice* orders, we do still have this equivalence in general. (See Fritz 2024 for discussion and further references.) I have been avoiding detours into topology, but here it is unavoidable to mention some topological concepts, though I will not explain them. We have already mentioned *standard Borel spaces*, in the context of the disintegration theorem (theorem E.27). These are more restrictive than general measurable spaces, but also include nearly all of our favorite spaces, such as the coin flip space 2^ω or the real line $\mathbb{R}$. These spaces arise from a certain kind of topological space, called a *Polish space*. A *nice* order on a Polish space is one that is *topologically closed*. For example, this is true of the standard order $\geq$ on $\mathbb{R}$. But, as we will discuss, many important orders are not closed.⁶

Theorem E.35 (Strassen (1965)). Let P and Q be probability measures on a Polish space $\mathcal{X}$, with its Borel algebra, equipped with a closed preorder $\geq$. The following are equivalent:

- (a) P weakly stochastically dominates Q ($P \geq_s Q$).

⁶What Strassen originally proved was slightly different (more general than this statement in some ways and less general in others). The generalization to Polish spaces is from Kamae, Krengel, and O'Brien (1977); the application to preorders is from Lindvall (1999). See Fritz (2024) for overview and further references.

Jaffe and Raban (2025) prove (independently of our work) that the equivalence can fail for non-closed orders, including orders that are countable unions of closed orders (called F_σ). The finitary part of the counting order on 2^ω is one example of such an order, though not one they directly discuss.

- (b) **Cumulative dominance.** For all upward closed measurable sets $U \subseteq^{\text{m}} \mathcal{X}$, $P(U) \geq Q(U)$.
- (c) **Expected utility.** For any bounded monotone functions $u : \mathcal{X} \rightarrow \mathbb{R}$,

$$\int u \, dP \geq \int u \, dQ.$$

Likewise, in this nice well-behaved case, our various *strong* stochastic dominance relations from section D.3 all agree (compare theorem D.32).

Theorem E.36. Let P and Q be probability measures on a Polish space $\mathcal{X}$, equipped with its Borel algebra and a closed preorder $\geq$. The following are equivalent:

- (a) **(Strict part)** $P \geq_s Q$ and $Q \not\geq_s P$;
- (b) **(Realization)** There exist $\mathcal{X}$ -valued random variables X, Y with distributions P, Q , respectively, such that

$$X \geq Y \text{ pointwise, and} \\ \mathbb{P}(X > Y) > 0;$$

- (c) **(Coupling)** There exists $R \in \Delta_\sigma(\mathcal{X} \times \mathcal{X})$ with marginals P and Q such that

$$R(\geq) = 1, \\ R(>) > 0;$$

- (d) **(Cumulative dominance)**

$$P(U) \geq Q(U) \text{ for every upward-closed } U \subseteq^{\text{m}} \mathcal{X}, \text{ and} \\ P(U) > Q(U) \text{ for some upward-closed } U \subseteq^{\text{m}} \mathcal{X};$$

- (e) **(Expected utility)** For every bounded strictly monotone $u : \mathcal{X} \rightarrow \mathbb{R}$,

$$\int u \, dP > \int u \, dQ.$$

This seems pretty good: for closed orders on Polish spaces, stochastic dominance is well-behaved and highly useful. Unfortunately, though, the orders of central importance in infinite ethics — finitely anonymous Paretian orders

— are not so well-behaved at all.

Let's focus on our central example of simple “happy-sad” worlds, represented by 2^ω . This naturally has the structure of a Polish space (specifically, endowing 2^ω with the product topology that arises from the discrete topology on $2 = \{0, 1\}$). With this topology, we have a notion of a (topological) *limit* of a sequence $x_1, x_2, \dots$ in 2^ω :

- $\lim_i x_i = x$ iff for each coordinate $k \in \omega$, there is some $N \in \mathbb{N}$ such that for all $i \geq N$, $x_i(k) = x(k)$.

What it means for a preorder $\succeq$ on 2^ω to be *topologically closed* is that it is preserved by limits: for any sequences $x_1, x_2, \dots$ and $y_1, y_2, \dots$ in 2^ω , if $x_i \succeq y_i$ for all i , then $\lim_i x_i \succeq \lim_i y_i$, if both limits exist.

Proposition E.37. The counting order on 2^ω is not topologically closed.

More generally, for any preordered Hausdorff topological space $\mathcal{X}$ and any infinite set $\mathcal{I}$, no finitely anonymous Paretian preorder on $\mathcal{X}^{\mathcal{I}}$ is topologically closed in the product topology.

Proof. Consider $x_1, x_2, \dots$ such that x_k consists of k zeros followed by all ones. Similarly let y_k consist of k ones followed by all zeros. Then for each k , $x_k \succeq^\# y_k$ with respect to the counting order. But

$$\lim_k x_k = (1, 1, 1, \dots) \quad \text{and} \quad \lim_k y_k = (0, 0, 0, \dots).$$

Clearly $(1, 1, 1, \dots) \not\succeq^\# (0, 0, 0, \dots)$. So the limits do not preserve the ordering. $\square$

Pivato and Fleurbaey (2024)

Thus we cannot apply theorem E.35 or theorem E.36 to anonymous Paretian orders. (You might wonder if we can weasel out of this by imposing a different topology on 2^ω ; we'll see that no such escape routes will work, because the theorems really do have counterexamples for the counting order, which arise independently of any topological assumptions.)

We can start most straightforwardly with the *expected utility* formulation of strong stochastic dominance:

For every bounded strictly monotone $u : \mathcal{X} \xrightarrow{m} \mathbb{R}$,

$$\int u \, dP > \int u \, dQ.$$

In fact, we know from Basu and Mitra's theorem (theorem 3.1) that *no* function from 2^ω to $\mathbb{R}$ is strictly monotone with respect to the counting order. So this relation holds vacuously for any P and Q in $\Delta_\sigma 2^\omega$ at all.

In contrast, the *strict part* and *cumulative dominance* formulations of strong stochastic dominance are clearly irreflexive: no probability measure can stand in either of these relations to itself. So the expected utility relation comes apart from them. This is not shocking, given what we already know about the deficiencies of real-valued utility functions in infinite ethics. We should simply set that formulation aside.

What about the *realization* and *coupling* formulations? We know these are equivalent to one another without any topological assumptions (proposition E.31). These are especially philosophically central: they correspond to the idea that *Strong Statewise Dominance* can be applied in a von Neumann–Morgenstern type setting where prospects are represented by probability measures. (And these two versions are indeed equivalent to each other in general.)

Here is a central fact that drives the problems.

Proposition E.38. Let $(2^\omega, \mathbb{P})$ be the coin flip space with the standard coin flip measure. There exists a probability-preserving function $\sigma : 2^\omega \xrightarrow{\mathbb{P}} 2^\omega$ such that $b \succ^\# \sigma b$ for almost every $b \in 2^\omega$.

Proof. For $n \in \mathbb{N}$, let $\sigma_n : 2^\omega \xrightarrow{\mathbb{P}} 2^\omega$ be the function that flips the first n bits of a sequence. Let $\sigma_\infty : 2^\omega \xrightarrow{\mathbb{P}} 2^\omega$ be the function that flips all bits of a sequence. Each of these functions preserves coin flip probabilities.

Let $f_n : 2^\omega \xrightarrow{\mathbb{P}} \mathbb{N}$ be the function that takes each sequence to the number of ones in the first n coordinates:

$$f_n(b) := \sum_{i=1}^n b(i).$$

For each n , let $E_n \subset 2^\omega$ be the set of sequences b such that n is the smallest number such that $f_n(b) > f_n(\sigma_n b)$. So for every $b \in E_n$, since b and $\sigma_n b$ agree on all but the first n coordinates, and b has more ones than $\sigma_n b$ in the first n coordinates, we have $b \succ^\# \sigma_n b$.

Furthermore, for $\mathbb{P}$ -almost every b , there is some n such that $f_n(b) > f_n(\sigma_n b)$. This follows from a standard fact about random walks (see Klenke 2013, sec. 17.5). The random sequence of numbers $f_n(b) - f_n(\sigma_n b)$ starts at 0, and goes

up by one with each Heads and down by one with each Tails. Almost surely, this running total will eventually be positive.

Thus $\bigcup_n E_n$ is a union of pairwise disjoint sets that differs from 2^ω by a measure zero set, which we can call E_∞ . By symmetry, the sets $\sigma_n(E_n)$ and $\sigma_\infty(E_\infty)$ are pairwise disjoint as well. (Each set $\sigma_n(E_n)$ is the set of sequences b such that n is the smallest number such that $f_n(b) < f_n(\sigma_n b)$ — just reversing the inequality. Similarly, $\sigma_\infty(E_\infty)$ is their remaining residue.)

Now define $\sigma : 2^\omega \rightarrow 2^\omega$ by

$$\sigma(b) := \begin{cases} \sigma_n(b) & \text{if } b \in E_n, \\ \sigma_\infty(b) & \text{if } b \in E_\infty. \end{cases}$$

Since this is a piecewise union of probability-preserving functions on a countable partition, it can be checked that σ also preserves probabilities, by countable additivity. Moreover, by construction,

$$b \succ^\# \sigma(b) \text{ for all } b \in 2^\omega \setminus E_\infty,$$

and thus for almost every $b \in 2^\omega$. □

In chapter 5 we discussed the case of an infinite random world, where infinitely many people have their happiness determined by independent coin flips. To model this, we'll consider the space 2^ω in two different roles: both as the state space Ω representing random coin flips, and as an outcome space $\mathcal{X}$ representing whether people are happy or sad. (To keep things conceptually clear, it is useful to use different notation for the same space of objects, depending on which role they are playing.)

Consider two random variables: the identity variable $X : \Omega \rightarrow \mathcal{X}$, and the permuted version $X \circ \sigma$, where σ is the probability-preserving function from proposition E.38. Then X and $X \circ \sigma$ have the same distribution: since $\sigma * \mathbb{P} = \mathbb{P}$,

$$\mathbb{P}_{X \circ \sigma} = (X \circ \sigma) * \mathbb{P} = X * (\sigma * \mathbb{P}) = X * \mathbb{P} = \mathbb{P}_X = \mathbb{P}.$$

But also, X is almost everywhere a counting-order improvement over $X \circ \sigma$. Indeed, we can modify X and $X \circ \sigma$ on a measure zero set to get new random variables Y and Z such that $Y \succ^\# Z$ everywhere (not just almost everywhere), and these variables still have the same distribution as X and $X \circ \sigma$ — namely, the coin flip measure $\mathbb{P}$. This shows that this measure $\mathbb{P}$ *strongly stochastically*

dominates itself, in both the *realization* and *coupling* senses. The very same probability measure can be realized by two different random variables, Y and Z , where one strictly dominates the other.

In fact, we can prove that this situation holds pervasively. Given enough background noise, *any* merely finite intervention falls through the measure-zero cracks. Roughly speaking, if two social outcome-valued random variables X and Y only disagree on how things go for finitely many people, and there is enough background randomness, then X and Y are stochastically equivalent. We will make this more precise in section E.5.

The upshot is that standard probability distributions lose a great deal of information that is relevant to evaluating prospects in infinite worlds. Two random variables can have the same distribution over outcomes, while one of them makes trillions of people happy, the other condemns those people to misery, and otherwise they are exactly the same. Accordingly, the stochastic dominance relations we can define between standard probability distributions are not only not very well-behaved technically, but also not very illuminating normatively.

E.5 Background randomness and infinite futility

Consider our “infinite random world” example (from chapter 5) once again. We considered two prospects, Happy and Sad. Happy makes the first person happy; Sad makes the first person sad; in both cases, everyone else’s happiness is determined by independent coin flips. In the chapter, we claimed proposition 5.2: Happy and Sad are *stochastically equivalent up to finite permutations*. Let’s state this precisely. Note that the relation $\sim^\#$ (the equivalence relation corresponding to the counting order) is the same as the relation on $2^{\mathcal{I}}$ of being equal up to finite permutations of individuals.

Proposition E.39. Let $B_1, B_2, \dots$ be 2-valued random variables which are independent and identically distributed (i.i.d.), and non-degenerate (that is, not concentrated on a single value). Let P be the joint distribution of $(1, B_1, B_2, \dots)$ and let Q be the joint distribution of $(0, B_1, B_2, \dots)$. Then $P \sim_s^\# Q$; that is, P and Q are stochastically equivalent up to finite permutations of individuals.

In the example of Happy and Sad, the B_n are independent fair coin flips.

The general moral is that a sufficiently random cosmic background wipes

out any finite differences between prospects — as far as standard probability distributions are concerned. The situation is analogous to trying to represent the value of infinite social outcomes using extended reals or infinite cardinals: standard real-probability representations of infinite prospects are too coarse-grained, to the point that “the finite is annihilated in the presence of the infinite”.

We will prove proposition E.39 by way of a general abstract characterization of stochastic equivalence relations, which will also extend the moral to much more general cases than happy-sad worlds with i.i.d. background variables.

To state the main abstract theorem, we need a standard metric for measuring the distance between two probability measures.

Definition E.40. The **total variation** distance between two probability measures $P, Q \in \Delta_\sigma \mathcal{X}$ is

$$\text{TV}(P, Q) := \sup_{A \subseteq \mathcal{X}} |P(A) - Q(A)|.$$

It can be checked that this has the usual properties of a *metric*.

Proposition E.41. For all $P, Q, R \in \Delta_\sigma \mathcal{X}$,

$$\text{TV}(P, P) = 0,$$

$$\text{TV}(P, Q) = \text{TV}(Q, P),$$

$$\text{TV}(P, R) \leq \text{TV}(P, Q) + \text{TV}(Q, R) \quad (\text{Triangle Inequality}).$$

Now we can state the core theorem, in abstract terms. The picture to have in mind is a sequence of increasingly coarse summaries of an outcome: we start with the full outcome $f_0(x) = x$, then $f_1(x)$ forgets some details about x , $f_2(x)$ forgets a bit more, and so on. In our applications, at each step we forget some of the details about the distribution for the first n people, just recording their relative *value*, while keeping track of the distribution for all the people further on.

For any function $f : \mathcal{X} \rightarrow \mathcal{Y}$, the **kernel** of f is the set

$$\ker f := \{(x, x') \in \mathcal{X} \times \mathcal{X} \mid f(x) = f(x')\}.$$

Theorem E.42. Let $\mathcal{X} = \mathcal{Y}_0, \mathcal{Y}_1, \mathcal{Y}_2, \dots$ be standard Borel spaces, and for each $n \geq 1$ let $g_n : \mathcal{Y}_{n-1} \xrightarrow{m} \mathcal{Y}_n$ be a measurable function. Let $f_n : \mathcal{X} \xrightarrow{m} \mathcal{Y}_n$ be the composite

$$f_n := g_n \circ \dots \circ g_1$$

(so that f_0 is the identity function on $\mathcal{X}$). In this case, $\ker f_1 \subseteq \ker f_2 \subseteq \dots$, and so

$$E := \bigcup_n \ker f_n$$

is again an equivalence relation. (In other words, E is the relation that holds between $x, y \in \mathcal{X}$ iff there is some n such that $f_n(x) = f_n(y)$.) For any probability measures $P, Q \in \Delta_\sigma \mathcal{X}$, the following are equivalent:

- (a) P and Q are stochastically equivalent with respect to E . (That is, $P E_s Q$.)
- (b) P and Q converge in total variation under f_n :

$$\lim_{n \rightarrow \infty} \text{TV}(f_n * P, f_n * Q) = 0.$$

It is worth flagging a technical point about this statement. The equivalence relation E is the union of an increasing sequence of equivalence relations $\ker f_n$. Any equivalence relation is the kernel of some function (recall the quotient map construction from proposition B.20) — but in general, this function may not go to a standard Borel space. The kernel of a measurable function between standard Borel spaces is called a *smooth* equivalence relation.

Here is a key example: if $\mathcal{X}$ is a standard Borel space and $\mathcal{I}$ is countable, then for any finite set $K \in \text{Fin } \mathcal{I}$, the relation $=_{\mathcal{I} \setminus K}$ is smooth. (Recall that $x =_{\mathcal{I} \setminus K} y$ iff $x(i) = y(i)$ for all $i \in \mathcal{I} \setminus K$.) But it turns out that the union of *all* of these relations — the relation of agreeing except at finitely many places — is not smooth (as long as $\mathcal{X}$ has at least two elements). Similarly, the counting equivalence $\sim^\#$ is not smooth. (This strengthens the more basic point from section E.4, that the counting order is *not closed*.) Since some of our machinery relies on standard Borel spaces, we will need to keep track of this technicality.

Before we prove this theorem, let's apply it. For concreteness, let's focus on cases where individual outcomes can be represented by real numbers, and individuals are identified with natural numbers. So our space of social outcomes is the space of **utility streams** $\mathbb{R}^\omega$. We will focus on class of social

welfare orders that generalize the counting order, the overtaking order, and additive social orders. (There are similar results that apply much more generally, but we won't pursue this here.)

Definition E.43. Let $s_n : \mathbb{R}^\omega \rightarrow \mathbb{R}$ be the **partial sum** function

$$s_n(x) := \sum_{i=1}^n x(i).$$

Let $t_n : \mathbb{R}^\omega \rightarrow \mathbb{R}^{\omega \setminus \{1, \dots, n\}}$ be the **tail** function that takes each $x \in \mathbb{R}^\omega$ to its restriction to $\omega \setminus \{1, \dots, n\}$.

A preorder $\succeq$ on $\mathbb{R}^\omega$ is **weakly additive** iff, for any $x, y \in \mathbb{R}^\omega$, if there is some n such that $s_n(x) \geq s_n(y)$ and $t_n(x) = t_n(y)$, then $x \succeq y$.

Here's one example of a weakly additive order.

Definition E.44. For $a \in \mathbb{R}$, let $a^+ = \max(a, 0)$. The **sum order** is the relation that holds between $x, y \in \mathbb{R}^\omega$ iff

$$x \succeq^\Sigma y \text{ iff } \sum (x - y)^+ \geq \sum (y - x)^+$$

and the right-hand side is finite (including the case where the left-hand side is infinite).

The sum order is a restriction of the *overtaking order*. (The restriction guarantees that the sum order does not depend on a privileged ordering of individuals — at the cost of some additional incompleteness.) The counting order is the restriction of the sum order to $\{0, 1\}^\omega$.

Note also that any extension of a weakly additive order is automatically weakly additive. So the results below also apply, for example, to any total order that extends the sum order. (We know from Zame (2007) that no such total order can be measurable — but the definition of stochastic dominance and the results below do not impose any measurability condition on the order.)

Let P and Q be the Happy and Sad distributions in the infinite random world example again. In this case, $s_n(x)$ is the number of people out of the first n that x makes happy. The *partial distributions* we discussed in chapter 5 are precisely the distributions $s_n * P$ and $s_n * Q$. Furthermore, in that example, later people's happiness is determined by independent coin flips, no matter

what happens to the first person; so s_n is *independent* of t_n for both P and Q , and P and Q have the *same* tails: $t_n * P = t_n * Q$ for $n \geq 1$. In this case the total variation term takes an especially simple form.

Corollary E.45. Let $\succeq$ be a weakly additive preorder on $\mathbb{R}^\omega$. Let $P, Q \in \Delta_\sigma \mathbb{R}^\omega$ be probability measures with *independent common tails*, in the sense that for all sufficiently large n , $t_n * P = t_n * Q$, and s_n is independent of t_n for both P and Q . If

$$\lim_n \text{TV}(s_n * P, s_n * Q) = 0,$$

then $P \sim_s Q$.

Thus, to show that P and Q are stochastically equivalent, it suffices to show that the total variation distance between the *distributions of partial sums* ($s_n * P$ and $s_n * Q$) converges to zero.

This total variation condition is less demanding than it may seem. The main thing it requires is that the tails are “sufficiently random”.

Abusing notation, we’ll write $\text{TV}(X, Y)$ for random variables X and Y as a shorthand for the total variation distance between their *distributions* $\mathbb{P}_X$ and $\mathbb{P}_Y$.

Theorem E.46. Let $\succeq$ be a weakly additive preorder on $\mathbb{R}^\omega$. Let $X_1, X_2, \dots, Y_1, Y_2, \dots$ be bounded real-valued random variables. Suppose there is some N such that, for all $n > N$, $X_n = Y_n = B_n$ and these are independent of $(X_1, \dots, X_{n-1}, Y_1, \dots, Y_{n-1})$. Let P and Q be the joint distributions of $(X_1, X_2, \dots)$ and $(Y_1, Y_2, \dots)$, respectively.

Suppose there exist $c > 0$ and $\epsilon > 0$ such that, for infinitely many n ,

$$\text{TV}(B_n, B_n + c) < 1 - \epsilon.$$

Then $P \succeq_s Q$ and $Q \succeq_s P$.

Furthermore, in the case where the difference of sums

$$\sum_{i=1}^N Y_i - \sum_{i=1}^N X_i$$

is almost surely an integer multiple of c , we have $P \sim_s Q$.

(Note that the second conclusion really is different from the first: as we noted

in section E.4, stochastic equivalence, $\sim_s$ is not the same as the equivalence relation arising from $\succeq_s$. That is,

$$(\succeq \cap \preceq)_s \neq (\succeq_s \cap \preceq_s),$$

in case that's at all illuminating.)

The total variation condition on the “background” variables B_n says that these variables’ distributions are each spread out in such a way that they overlap with a shift, at least a little.

In particular, if the B_n are also *identically distributed* (hence i.i.d.), then the condition on $\text{TV}(B_n, B_n + c)$ holds whenever the distribution of B_n is non-degenerate. (Suppose B_n has non-degenerate distribution β . If β has a continuous part, we can find some appropriate $\epsilon > 0$ for any sufficiently small $c > 0$. If β has no continuous part, it must have at least two atoms. Then we can take c to be the distance between them, and ϵ to be the minimum of their probabilities.)

Proposition E.39 (the infinite random world case) follows, in the special case where $N = 1$, $c = 1$, and the relation $\succeq$ is the sum order. (The restriction of $\sim^\Sigma$ to $\{0, 1\}^\omega$ coincides with the counting equivalence $\sim^\#$, which is the same as being related by a finite permutation.)

In summary, for any weakly additive preorder on utility streams, basically any non-trivial independent background noise washes out the difference between finite prospects — we face stochastic “infinite futility”. The trouble, again, is that standard probability distributions intuitively drop all finite value differences through their measure zero cracks.

In the rest of this section we will prove these claims: theorem E.42 (the abstract total variation characterization of stochastic equivalence), corollary E.45 (the case of common independent tails), theorem E.46 (the concrete case of independent background noise).

The proofs rely on some new technical ideas.

Definition E.47.

- (a) A **subprobability measure** P (on a measurable space $\mathcal{X}$) is a measure (on $\mathcal{X}$) such that $P(\mathcal{X}) \leq 1$.

Let P and Q be subprobability measures on a measurable space $\mathcal{X}$.

- (b) We write $|P|$ for $P(\mathcal{X})$, the **total mass** of P .

- (c) We write $P \geq Q$ iff $P(A) \geq Q(A)$ for all measurable sets $A \subseteq^m \mathcal{X}$.
- (d) A **subcoupling** of P and Q is a subprobability measure R on $\mathcal{X} \times \mathcal{X}$ such that $P \geq \pi_1 * R$ and $Q \geq \pi_2 * R$.

Here are some basic properties worth noting.

Lemma E.48. For subprobability measures P, Q ,

$$\begin{aligned} |P + Q| &= |P| + |Q|, \\ |P \otimes Q| &= |P| \cdot |Q|, \\ |f * P| &= |P| \quad \text{for } f : \mathcal{X} \xrightarrow{m} \mathcal{Y}, \\ \text{if } P &\geq Q \quad \text{then } |P| \geq |Q|. \end{aligned}$$

A subprobability measure P is a probability measure iff $|P| = 1$.

The following basic construction is central to the proof of theorem E.42: we can take a *monotone limit* of subprobability measures.

Lemma E.49. Any chain $P_1 \leq P_2 \leq \dots$ of subprobability measures on $\mathcal{X}$ has a least upper bound: that is, there is a subprobability measure P such that for all subprobability measures Q , $Q \geq P$ iff $Q \geq P_n$ for all n . In this case we write $P_n \uparrow P$. Furthermore,

$$|P| = \sup_n |P_n|.$$

If $E_1, E_2, \dots$ support $P_1, P_2, \dots$, respectively, then $\bigcup_n E_n$ supports P . Also, for any $f : \mathcal{X} \xrightarrow{m} \mathcal{Y}$,

$$f * P_n \uparrow f * P$$

as well.

Proof sketch. Let

$$P(A) := \sup_n P_n(A) \quad \text{for each } A \subseteq^m \mathcal{X}.$$

It follows (using the monotone convergence theorem) that P is countably additive, and hence a subprobability measure on $\mathcal{X}$. The remaining properties can be checked straightforwardly. $\square$

The following theorem is not trivial; we state it without proof (see Aliprantis and Border 2006, ch. 10).

Proposition E.50. For any subprobability measures P, Q on $\mathcal{X}$, there exists a (unique) subprobability measure $P \wedge Q$ on $\mathcal{X}$ which is a greatest lower bound of P and Q with respect to $\geq$: that is, for any subprobability measure R ,

$$P \wedge Q \geq R \quad \text{iff} \quad P \geq R \quad \text{and} \quad Q \geq R.$$

Furthermore:

(a) If $P \geq R$ and $Q \geq R$, then

$$(P \wedge Q) - R = (P - R) \wedge (Q - R).$$

(b) If $|P| = |Q|$, then

$$|P \wedge Q| + \text{TV}(P, Q) = |P| = |Q|.$$

(c) For any measurable $f : \mathcal{X} \rightarrow \mathcal{Y}$,

$$f * (P \wedge Q) \leq f * P \wedge f * Q.$$

The subprobability measure $P \wedge Q$ is intuitively the “common part” of P and Q . Accordingly, this is the part of each that can be matched up against the other: there exists a subcoupling R of P and Q supported by the identity relation such that $|R| = |P \wedge Q|$, and no other subcoupling can have greater mass. The following lemma generalizes this idea.

Lemma E.51. Let P, Q be subprobability measures on $\mathcal{X}$ and let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be measurable. If R is a subcoupling of P and Q supported by $\ker f$, then $|R| \leq |f * P \wedge f * Q|$. Furthermore, if $\mathcal{X}$ and $\mathcal{Y}$ are standard Borel spaces, this maximum is attained: there exists such a subcoupling R where $|R| = |f * P \wedge f * Q|$.

Proof sketch. The bound is relatively elementary, and holds for any measurable space. Let R be a subcoupling of P and Q . If R is supported by $\ker f$, then it can be checked that

$$f * \pi_1 * R = f * \pi_2 * R.$$

This measure is dominated by $f * P$ (because $\pi_1 * R \leq P$) and by $f * Q$ (because $\pi_2 * R \leq Q$), and so it is also dominated by their meet, $(f * P) \wedge (f * Q)$. Thus:

$$|R| = |f * \pi_1 * R| \leq |f * P \wedge f * Q|.$$

The existence proof is more technical, and it uses stronger assumptions on the spaces. It is straightforward to generalize the *integral* notion (theorem E.19) from probability measures to subprobability measures. This also applies to taking mixtures of probability measures. We can then reason as follows.

Let P_y and Q_y be disintegrations of P and Q , respectively, over f . Then let

$$R := \int P_y \otimes Q_y \, d(f * P \wedge f * Q)(y).$$

It can be shown that $\pi_1 * R \leq P$, $\pi_2 * R \leq Q$, and $|R| = |f * P \wedge f * Q|$. $\square$

Lemma E.52 (Extension). Let $g : \mathcal{X} \xrightarrow{m} \mathcal{Y}$ for standard Borel spaces $\mathcal{X}$ and $\mathcal{Y}$, and let R be a subcoupling of probability measures $P, Q \in \Delta_\sigma \mathcal{X}$ which is supported by $\ker g$. Then there exists a subcoupling $R' \geq R$ of P and Q also supported by $\ker g$, where

$$|R'| = |g * P \wedge g * Q| = 1 - \text{TV}(g * P, g * Q).$$

Proof. Since $P \geq \pi_1 * R$ and $Q \geq \pi_2 * R$, we can define subprobability measures

$$\hat{P} := P - \pi_1 * R,$$

$$\hat{Q} := Q - \pi_2 * R.$$

Then by lemma E.51, there exists a subcoupling S of $\hat{P}$ and $\hat{Q}$, also supported by $\ker g$, such that

$$|S| = |g * \hat{P} \wedge g * \hat{Q}|.$$

Let $R' = R + S$, which is also supported by $\ker g$. It can be checked that $\pi_1 * R' \leq P$ and $\pi_2 * R' \leq Q$.

The last thing to check is the total mass. Note that since R is supported by $\ker g$, it follows that

$$g * (\pi_1 * R) = g * (\pi_2 * R).$$

Call this measure M . Then $|M| = |R|$, and we have $g * \hat{P} = g * P - M$ and $g * \hat{Q} = g * Q - M$. Thus

$$\begin{aligned} |g * \hat{P} \wedge g * \hat{Q}| &= |(g * P - M) \wedge (g * Q - M)| \\ &= |(g * P \wedge g * Q) - M| \\ &= |g * P \wedge g * Q| - |R|. \end{aligned}$$

So, finally,

$$|R'| = |R| + |S| = |R| + |g * \hat{P} \wedge g * \hat{Q}| = |g * P \wedge g * Q|.$$

□

Proof of theorem E.42. Write $E_n := \ker f_n$, so that $E_1 \subseteq E_2 \subseteq \dots$ is a chain of equivalence relations on $\mathcal{X}$ with union E . Note that each E_n is measurable: it is the preimage of the diagonal of $\mathcal{Y}_n \times \mathcal{Y}_n$ under $f_n \times f_n$, and the diagonal of a standard Borel space is measurable.

For the first direction, suppose that R is a coupling of P and Q supported by E . Let R_n be the restriction of R to E_n . Then $R_n \uparrow R$, so

$$1 = |R| = \sup_n |R_n|.$$

Since each R_n is a subcoupling supported by E_n , by lemma E.51,

$$|R_n| \leq |f_n * P \wedge f_n * Q| = 1 - \text{TV}(f_n * P, f_n * Q).$$

Thus

$$\lim_n \text{TV}(f_n * P, f_n * Q) \leq \lim_n (1 - |R_n|) = 0.$$

For the other direction, suppose

$$\lim_{n \rightarrow \infty} \text{TV}(f_n * P, f_n * Q) = 0.$$

We will recursively construct a sequence of subcouplings R_n , each supported by the corresponding set E_n . For the base case, let R_0 be the trivial subcoupling of mass zero. Then for the recursive step, let $n \geq 1$, and suppose R_{n-1} is a subcoupling of P and Q supported by E_{n-1} . Then R_{n-1} is also supported by $E_n \supseteq E_{n-1}$, so we can apply lemma E.52: there exists a subcoupling $R_n \geq R_{n-1}$ of P and Q supported by E_n , where

$$|R_n| = |f_n * P \wedge f_n * Q| = 1 - \text{TV}(f_n * P, f_n * Q).$$

Next, we can apply lemma E.49 to take a limit of this sequence: there is a subcoupling R_∞ of P and Q supported by E , such that

$$|R_\infty| = \lim_{n \rightarrow \infty} |R_n| = 1 - \lim_{n \rightarrow \infty} \text{TV}(f_n * P, f_n * Q) = 1.$$

Since $\pi_1 * R_\infty \leq P$ and

$$|\pi_1 * R_\infty| = |R_\infty| = 1 = |P|,$$

it follows that $\pi_1 * R_\infty = P$, and similarly $\pi_2 * R_\infty = Q$. So R_∞ is in fact a coupling supported by E , exhibiting that P and Q are stochastically equivalent. $\square$

Next we apply theorem E.42 to the case of common independent tails. First, a useful fact about total variation distances.

Lemma E.53. For $P, Q \in \Delta_\sigma \mathcal{X}$ and $P', Q' \in \Delta_\sigma \mathcal{Y}$,

$$\text{TV}(P \otimes P', Q \otimes Q') \leq \text{TV}(P, Q) + \text{TV}(P', Q').$$

Proof. We have

$$\begin{aligned} \text{TV}(P \otimes P', Q \otimes P') &= \text{TV}(P, Q), \\ \text{TV}(Q \otimes P', Q \otimes Q') &= \text{TV}(P', Q'). \end{aligned}$$

The result follows from the triangle inequality. $\square$

Proof of corollary E.45. Let $P, Q \in \Delta_\sigma \mathbb{R}^\omega$ be probability measures with independent common tails. That means that for sufficiently large n , first,

$$t_n * P = t_n * Q$$

and second, s_n is independent of t_n :

$$(s_n, t_n) * P = (s_n * P) \otimes (t_n * P)$$

and similarly for Q . Thus (for sufficiently large n),

$$\begin{aligned} \text{TV}((s_n, t_n) * P, (s_n, t_n) * Q) &= \text{TV}((s_n * P) \otimes (t_n * P), (s_n * Q) \otimes (t_n * Q)) \\ &\leq \text{TV}(s_n * P, s_n * Q) + \text{TV}(t_n * P, t_n * Q) \\ &= \text{TV}(s_n * P, s_n * Q). \end{aligned}$$

By hypothesis, this converges to zero.

Each of the spaces $\mathbb{R}^\omega$ and $\mathbb{R} \times \mathbb{R}^{\omega \setminus \{1, \dots, n\}}$ is a standard Borel space, and each function $(s_n, t_n) : \mathbb{R}^\omega \rightarrow \mathbb{R} \times \mathbb{R}^{\omega \setminus \{1, \dots, n\}}$ is measurable.

Suppose that $s_n(x) = s_n(y)$ and $t_n(x) = t_n(y)$. Then clearly $t_{n+1}(x) = t_{n+1}(y)$, since these just further restrict the tail. Also, $x(n+1) = y(n+1)$, so

$$s_{n+1}(x) = s_n(x) + x(n+1) = s_n(y) + y(n+1) = s_{n+1}(y).$$

This shows that

$$\ker(s_n, t_n) \subseteq \ker(s_{n+1}, t_{n+1}).$$

Thus, letting

$$E = \bigcup_n \ker(s_n, t_n),$$

theorem E.42 tells us that $P E_s Q$.

Furthermore, weak additivity tells us that, if $s_n(x) = s_n(y)$ and $t_n(x) = t_n(y)$, then $x \sim y$. Thus $\sim$ contains E . By proposition E.31(d), stochastic lifts are monotonic. So $P \sim_s Q$ as well. $\square$

It is worth noting that this result could easily be generalized beyond weakly additive orders — very little depends on s_n being a *sum*. The same argument works for any family of functions s_n to any standard Borel space, as long as the kernels of (s_n, t_n) are nested and contained in the preorder $\succeq$.

We could also relax the assumption of common independent tails, at the cost of additional technicality. Instead of requiring that P and Q have identical tails, we can instead say that the tail distributions $t_n * P$ and $t_n * Q$ converge in total variation. Instead of requiring independent tails, we can instead impose a condition on the total variation of partial distributions *conditional* on the tails. But we will not go into these generalizations.

Now for theorem E.46, the case of independent background noise. For this, we will rely on an important standard theorem in probability theory.

Lemma E.54 (Mineka (1973), see Lindvall (2002), ch. II). Let $B_1, B_2, \dots$ be independent real-valued random variables, and let $S_n = \sum_{i=1}^n B_i$ be their partial sums. Let $c \in \mathbb{R}$. If the infinite sum

$$\sum_{n=1}^{\infty} (1 - \text{TV}(B_n, B_n + c))$$

diverges, then

$$\lim \text{TV}(S_n, S_n + c) = 0.$$

We also use two more basic properties of total variation distance.

Lemma E.55. For any real random variables X, Y and any $c \in \mathbb{R}$,

$$\text{TV}(X, Y) = \text{TV}(X + c, Y + c).$$

Lemma E.56. For any real random variables X, Y, S, T , if (X, Y) is jointly independent of (S, T) , then

$$\text{TV}(X + S, Y + T) \leq \int \text{TV}(S, T + z) dP(z)$$

where P is the distribution of $Y - X$.

Proof of theorem E.46. Let N be as in the theorem statement. Then we can let

$$\begin{aligned} X &= \sum_{n=1}^N X_n, \\ Y &= \sum_{n=1}^N Y_n, \\ B_n &= X_{N+n} = Y_{N+n}, \\ S_n &= \sum_{i=1}^n B_i. \end{aligned}$$

The theorem hypotheses tell us that X and Y are bounded, and $B_1, B_2, \dots$ are independent of each other and of X and Y . Furthermore, there is some $c > 0$ and $\epsilon > 0$ such that, for infinitely many n ,

$$\text{TV}(B_n, B_n + c) < 1 - \epsilon.$$

This implies that the sum

$$\sum_{n=1}^{\infty} (1 - \text{TV}(B_n, B_n + c))$$

diverges, so we can apply Mineka's theorem:

$$\lim_{n \rightarrow \infty} \text{TV}(S_n, S_n + c) = 0.$$

We will prove the equivalence case first. In this case we suppose that the value of $Y - X$ is almost surely an integer multiple of c . So, for some bounded

integer-valued random variable Z , $Y - X = c \cdot Z$ (almost surely). Say $|Z| \leq m$ almost surely. Our targets $s_{N+n} * P$ and $s_{N+n} * Q$ are the distributions of $X + S_n$ and $Y + S_n$, respectively. So the total variation distance between these is

$$\begin{aligned} \text{TV}(X + S_n, Y + S_n) &\leq \sum_{k=-m}^m \mathbb{P}[Z = k] \cdot \text{TV}(S_n, S_n + k \cdot c) \\ &= \sum_{k=-m}^m \mathbb{P}[Z = k] \cdot \text{TV}(S_n, S_n + |k| \cdot c). \end{aligned}$$

(The weighted sum here is the discrete case of the integral in lemma E.56.) By the triangle inequality we have

$$\begin{aligned} \text{TV}(S_n, S_n + |k| \cdot c) &\leq \text{TV}(S_n, S_n + c) + \text{TV}(S_n + c, S_n + 2c) + \cdots \\ &\quad + \text{TV}(S_n + (|k| - 1)c, S_n + |k| \cdot c). \end{aligned}$$

By translation invariance, this is equal to

$$|k| \cdot \text{TV}(S_n, S_n + c) \leq m \cdot \text{TV}(S_n, S_n + c).$$

Thus, putting it all together, we have

$$\text{TV}(s_{N+n} * P, s_{N+n} * Q) \leq m \cdot \text{TV}(S_n, S_n + c).$$

We know this converges to zero.

Now for the case of stochastic dominance (dropping the assumption that the difference is supported by integer multiples of c). In this case, since X and Y are bounded, there is some positive integer k such that $-kc \leq X, Y \leq kc$ almost surely. Let

$$X'_1 = -kc, \quad X'_2, \dots, X'_N = 0, \quad \text{and} \quad Y'_1 = kc, \quad Y'_2, \dots, Y'_N = 0.$$

Then let P' and Q' be the respective joint distributions of $(X'_1, \dots, X'_N, B_1, B_2, \dots)$ and $(Y'_1, \dots, Y'_N, B_1, B_2, \dots)$. By construction, the new X' -sequence has N th partial sum $-kc$, which is less than or equal to the N th partial sum of the original X -sequence, and they agree at all coordinates beyond N . So, since the order is weakly additive, the X -sequence weakly statewise dominates the X' -sequence in $\succeq$, and thus the distribution of X' weakly stochastically dominates the distribution of X : that is, $P \succeq_s P'$. Similarly, $Q' \succeq_s Q$.

Furthermore, by construction, the difference

$$\sum_{n=1}^N Y'_n - \sum_{n=1}^N X'_n = 2kc,$$

so it is certainly an integer multiple of c . So by the previous argument, $P' \sim_s Q'$. In particular (by monotonicity again) $P' \succeq_s Q'$.

In short,

$$P \succeq_s P' \succeq_s Q' \succeq_s Q.$$

Since $\succeq_s$ is transitive, $P \succeq_s Q$. Symmetric reasoning shows that $Q \succeq_s P$ as well. $\square$

E.6 The background value theorem

Here we precisely state and prove a variant of Tarsney's theorem (2025), discussed in chapter 8.

The result is about real-valued random variables, and it relies on representing these using *probability densities* on $\mathbb{R}$. This raises a minor snag for our presentation, because so far we have done everything in terms of *finite* measures. The background measure on $\mathbb{R}$ that we want to integrate against here — the standard (Lebesgue) measure on $\mathbb{R}$, which we will denote λ — is not a finite measure: $\lambda(\mathbb{R}) = \infty$. Instead, this is a σ -finite measure, meaning that it is a sum of countably many finite measures on $\mathbb{R}$. We have already officially introduced the Lebesgue measure on an *interval* $[0, 1)$; we can think of this as a measure on $\mathbb{R}$ supported by $[0, 1)$, and we can also translate it to any unit interval $[z, z + 1)$ for $z \in \mathbb{Z}$. The Lebesgue measure λ on $\mathbb{R}$ is the sum of all of these unit measures. (In general, the measure of a set lives in the extended non-negative reals, $\mathbb{R}^+ \cup \{\infty\}$). Extending the ideas we introduced in sections E.1 and E.2 from finite measures to σ -finite measures makes very little difference.

We will use this standard notation for integrals on real intervals:

$$\int_a^b f d\lambda := \int_{[a,b)} f d\lambda$$

(including the case where $a = -\infty$ or $b = \infty$).

First, we need to clarify the notion of “heavy tailed” probability distributions.

Definition E.57. For β a probability density function on $\mathbb{R}$ and $k > 0$, let the **relative k -concentration** of β be

$$\inf\{\alpha \mid \beta(x) \leq (1 + \alpha) \cdot \beta(y) \text{ for each } |x - y| \leq k\}.$$

Intuitively, the relative k -concentration tells us how quickly the tails of the distribution approach zero. If it is low, then the distribution at $x + k$ can't be much smaller than the distribution at x . For tails that decay faster than exponentially (for example, a normal distribution), the k -concentration is infinite. Lower k -concentration means the distribution has “thicker tails”, and is more disperse.

Definition E.58. Random variables X and Y are **independent** iff their joint distribution is the product measure:

$$\mathbb{P}_{(X,Y)} = \mathbb{P}_X \otimes \mathbb{P}_Y.$$

Theorem E.59. Let X, Y, B be real-valued random variables on a probability space, where B is independent of each of X and Y , and has an absolutely continuous distribution with probability density function β . Suppose that X and Y are supported by a finite interval $[a, a + k]$, and β has finite relative k -concentration α . If

$$\mathbb{E}X - a \geq (1 + \alpha)^2 \cdot (\mathbb{E}Y - a),$$

then the distribution of $X + B$ stochastically dominates the distribution of $Y + B$.

Intuitively, the hypothesis says that $\mathbb{E}X > \mathbb{E}Y$, and furthermore these expectations are *not too close together*, relative to how disperse the background distribution is. In this case, adding on background noise converts *greater expectation* into *stochastic dominance*.

The proof of theorem E.59 uses two standard facts, which we state without proof.

Lemma E.60 (Convolution). Let X and B be independent real random variables, where B has density β . Then for any $t \in \mathbb{R}$,

$$\mathbb{P}[X + B \geq t] = \int \beta(t - u) \cdot \mathbb{P}[X \geq u] \, d\lambda(u).$$

Lemma E.61. If $X \geq a$ almost surely, then

$$\mathbb{E}X - a = \int_a^\infty \mathbb{P}[X \geq u] \, d\lambda(u).$$

Proof of theorem E.59. We will show that $X+B$ cumulatively dominates $Y+B$: that is, for every $t \in \mathbb{R}$,

$$\mathbb{P}[X + B \geq t] \geq \mathbb{P}[Y + B \geq t].$$

Define:

$$F(x) := \mathbb{P}[X \geq x] \quad \text{for each } x \in \mathbb{R}.$$

Since X is supported by $[a, a+k]$, by lemma E.61 we have

$$\mathbb{E}X - a = \int_a^{a+k} F \, d\lambda.$$

Since B is independent of X , by convolution (lemma E.60),

$$\mathbb{P}[X + B \geq t] = \int \beta(t - u) \cdot F(u) \, d\lambda(u).$$

For $u \leq a$, $F(u) = 1$, and for $u \geq a+k$, $F(u) = 0$. So we can split this integral into pieces:

$$\mathbb{P}[X + B \geq t] = \int_{-\infty}^a \beta(t - u) \, d\lambda(u) + \int_a^{a+k} \beta(t - u) \cdot F(u) \, d\lambda(u).$$

Call the first term c . (In fact, $c = \mathbb{P}[B \geq t - a]$.)

Let $b = \beta(t - a) \geq 0$. For each $u \in [a, a+k]$ we have $|(t - u) - (t - a)| \leq k$, so by the definition of relative k -concentration,

$$\beta(t - u) \geq \frac{b}{1 + \alpha}.$$

Since $F \geq 0$, this gives a lower bound:

$$\begin{aligned} \mathbb{P}[X + B \geq t] &\geq c + \frac{b}{1 + \alpha} \int_a^{a+k} F(u) \, d\lambda(u) \\ &= c + \frac{b}{1 + \alpha} (\mathbb{E}X - a). \end{aligned}$$

Parallel reasoning, using the bound

$$\beta(t - u) \leq b(1 + \alpha) \quad \text{for each } u \in [a, a+k],$$

gives us an upper bound on Y 's cumulative probability:

$$\mathbb{P}[Y + B \geq t] \leq c + b(1 + \alpha)(\mathbb{E}Y - a).$$

So we have

$$\mathbb{P}[X + B \geq t] - \mathbb{P}[Y + B \geq t] \geq b \left(\frac{1}{1 + \alpha} (\mathbb{E}X - a) - (1 + \alpha)(\mathbb{E}Y - a) \right).$$

Since $b \geq 0$, our hypothesis

$$\mathbb{E}X - a \geq (1 + \alpha)^2 \cdot (\mathbb{E}Y - a),$$

implies that this is non-negative. In short,

$$\mathbb{P}[X + B \geq t] \geq \mathbb{P}[Y + B \geq t] \quad \text{for each } t \in \mathbb{R}.$$

That is, the distribution of $X + B$ cumulatively dominates that of $Y + B$. By Strassen's theorem (theorem E.35), this implies that the distribution of $X + B$ stochastically dominates that of $Y + B$.⁷ $\square$

Let's compare this result with the ideas from section E.5. They have a very similar flavor. In both of them, the idea is that as we add on more and more disperse background variables, we get closer and closer to stochastic dominance; and in both cases we can take a kind of limit. The key difference is that, for Tarsney's theorem, we are considering the case where "cosmic background value" is sure to be *finite* (though the range of possible values is unbounded). In this case, a sufficiently disperse distribution of background value can take a bounded variable X with *greater expectation* than Y to a case of stochastic dominance. In section E.5, we instead considered the case of *infinite* background value, represented by dispersion at the level of infinitely many individuals' welfare. In this case — with respect to a natural class of orders on these infinite outcomes — we get stochastic dominance even *without* any assumption of greater expectation. Even in cases where one prospect is certain to be massively (but only finitely) *worse* than another, it can still be transformed into a stochastically dominant prospect by adding on *infinitely* disperse cosmic background value.

⁷It is worth flagging that the appeal to Strassen's theorem is overkill here; for probability measures on $\mathbb{R}$ it can be proved much more straightforwardly that cumulative dominance implies stochastic dominance. If P and Q are probability measures on $\mathbb{R}$, then we can directly construct random variables on $[0, 1]$ given by

$$X(x) = \sup\{t \mid P[t, \infty) \geq x\},$$

and we can similarly define Y in terms of Q . If P cumulatively dominates Q , then $X \geq Y$ pointwise, exhibiting stochastic dominance in the realization sense.

I think what this shows, though, is that the stochastic dominance relations we get between standard probability measures out of disperse background noise are spurious, not representing real differences in the value of prospects. They show us that standard probability measures are losing information that is crucial for correctly assessing the value of prospects.

E.7 Conditional probability

Standard probability measures don't carry all the information we need for evaluating prospects. As I discussed in chapter 5, there are other simpler arguments for this conclusion, having to do with Strong Statewise Dominance and measure zero sets. Philosophers have chiefly taken two approaches to this problem. One is to appeal to non-standard probability measures that allow infinitesimal probabilities — I take this up in appendix F. The other, which I will briefly address in this section, is to appeal to primitive *conditional* probabilities (Popper 1955; Rényi 1970; for discussion see Hájek 2003b; K. Easwaran 2014b; see also section 5.6 in this book).

The main technical point I will make is that adequate conditional probabilities are *non-constructive*. In particular, it is not constructively provable that there exists a *coin-flip symmetric* conditional probability measure. Non-constructiveness is a feature conditional probabilities share with hyperreal probabilities, but (as we will see in appendix F) a more general system of “quasi-probabilities” can escape this limitation.

(It might be thought that these results are unnecessary, since there are results showing that conditional probability measures are “representationally equivalent”, in some sense, to hyperreal probability measures — which are known to be non-constructive. But in fact these correspondences are not that tight, especially when it comes to infinite spaces. See J. Y. Halpern (2010); Brickhill and Horsten (2018); Pruss (2021).)

Standardly, conditional probabilities are defined as ratios of unconditional probabilities: $P(A \mid E) = P(A \cap E)/P(E)$. But this definition cannot be applied in the case where $P(E) = 0$. This is a serious restriction in the context of standard probability theory, since many propositions that might well be true standardly get assigned probability zero (proposition E.12). In section E.3 we considered one strategy for getting around this — regular conditional probabilities, via the disintegration theorem — but while this approach gracefully avoids undefined ratios, it does not fully solve the problem, because regular conditional probabilities are only unique up to

measure zero differences. A different influential approach is to simply take conditional probabilities as primitive, rather than defined in terms of unconditional probabilities. (We can then turn things around and define *unconditional* probabilities in terms of *conditional* probabilities, via $P(A) := P(A \mid \Omega)$.)

Definition E.62. A **conditional probability measure** on a measurable space^a Ω assigns a real number $P(A \mid C)$ to each measurable $A \subseteq^m \Omega$ and each *non-empty* measurable $C \subseteq^m \Omega$. These satisfy the following axioms, for measurable $A, B \subseteq^m \Omega$ and non-empty measurable $C, D \subseteq^m \Omega$:

(a)

$$0 \leq P(A \mid C) \leq 1;$$

(b)

$$P(C \mid C) = 1;$$

(c) *Addition Rule.*

$$P(A \cup B \mid C) = P(A \mid C) + P(B \mid C) \text{ if } A \cap B = \emptyset;$$

(d) *Multiplication Rule.*

$$P(A \mid C) \cdot P(C \mid D) = P(A \mid D) \text{ if } A \subseteq C \subseteq D.$$

^aFor expository simplicity, we will continue to work with measurable spaces, even though this is overkill. A measurable space carries a σ -algebra of measurable sets, which is closed under complements and countable unions. But conditional probability measures are only assumed to be *finitely* additive, and they really only require that their algebra of measurable sets is *Boolean algebra* — closed under complements and *finite* unions. Sometimes one also allows the algebra of sets that can appear in the first argument place of $P(- \mid -)$ to be different from the collection of sets that can appear in the second argument place — the *conditioning sets* or “observables”. We could do this here, but it only adds unnecessary complexity.

For the rest of the section, let P be a conditional probability measure on a measurable space Ω .

The first key observation is that a conditional probability measure imposes a preorder on measurable sets. (Pruss 2025 calls it the *Union* ordering.)

Proposition E.63 (Pruss (2014, 3534)). Define, for $A, B \stackrel{\text{m}}{\subseteq} \Omega$:

$$A \succsim_P B \quad \text{iff} \quad P(A \mid A \cup B) \geq P(B \mid A \cup B) \text{ or } A = B = \emptyset.$$

Then $\succsim_P$ is a total preorder on the measurable subsets of Ω . Furthermore, $\succsim_P$ is monotone in the sense that $A \supseteq B$ implies $A \succsim_P B$.

An important note: while this defines an *order* that clearly has something to do with probability, $\succsim_P$ is *not* generally a *probability order* in the sense introduced in section A.3.

As Thong (forthcoming) and Pruss (2025) point out, it violates the *quasi-additivity* property (definition A.26; see also section 5.6). Suppose A and C are non-empty disjoint sets, $P(A \mid A \cup C) = 0$. (Example: A is “every coin flip comes up heads” and C is “the first coin comes up tails”.) Then we have $A \succ_P \emptyset$, but $\emptyset \cup C \sim_P A \cup C$, violating quasi-additivity (definition A.26). Likewise, even though we have $A \succ_P \emptyset$, it is not *strictly monotone* (definition A.15):

$$\text{if } A \supsetneq B \text{ then } A \succ_P B.$$

The same case is a counterexample: we have $A \cup C \supsetneq C$, but $A \cup C \sim_P C$. (Pruss 2025 calls strict monotonicity *strong regularity*; see also Pruss 2013b.)

In section A.3 we discussed *Pruss’s theorem* (theorem A.30): there is no constructive existence proof for a total strictly monotone preorder on the countable subsets of 2^ω . We cannot directly apply that form of the theorem to the order $\succsim_P$, since it is not strictly monotone. However, Pruss actually proved something even stronger than theorem A.30, which does apply to $\succsim_P$. In fact, the order $\succsim_P$ does satisfy the following *weakening* of strict monotonicity.

Definition E.64. Let $\mathcal{A}$ be a family of subsets of Ω . Call a preorder $\succsim$ on $\mathcal{A}$ **semi-strictly monotone** iff it is monotone, and for any disjoint sets $A, B \in \mathcal{A}$ if $A \lesssim B \neq \emptyset$, then $A < A \cup B$.

Note that in the counterexample above, we do not have $C < C \cup A$, but this relies on the fact that A is much *smaller* than C — the conditional probability of A given $A \cup C$ is zero. The definition of semi-strict monotonicity excludes this case. In fact, the following is straightforward to prove.

Lemma E.65 (Pruss (2014, 3534)). The relation $\succsim_P$ is semi-strictly monotone.

What Pruss proves is that this is good enough.

Theorem E.66 (Pruss (2014)). It is not constructively provable that there exists a semi-strictly monotone total preorder on the countable subsets of a set with the cardinality of the continuum (2^ω).

Corollary E.67. It is not constructively provable that there exists a conditional probability measure on the coin flip space.

Proof. Every countable subset of 2^ω is measurable. The conclusion follows from proposition E.63, lemma E.65, and theorem E.66. $\square$

I will also indulge in proving another theorem, which on its face is weaker than corollary E.67. This is worth doing anyway, for two reasons. The first reason is conceptual: the theorem is proved in a different way, which exhibits the close connection between these results about conditional probability measures and *social welfare orders* — specifically, the Zame–Lauwers theorem (theorem A.25). The second, technical reason is that theorem E.68 below can readily be strengthened in different ways than Pruss’s theorem, and, relatedly, it has different set-theoretic consequences for how the existence of conditional probability measures fits into the space of axioms weaker than Choice (again, see Dubey and Laguzzi 2024 — but I will not explore these implications here).

Theorem E.68. A conditional probability measure P on the coin flip space 2^ω is **coin flip symmetric** iff for all measurable $A, B \subseteq 2^\omega$ (B non-empty) and each coin flip symmetry σ ,

$$P(\sigma \cdot A \mid \sigma \cdot B) = P(A \mid B).$$

It is not constructively provable that there exists a coin flip symmetric conditional probability measure.

We will use this lemma:

Lemma E.69. For any non-empty $A, B \subseteq \Omega$, if $A \succeq_P B$, then $P(A \mid B) \geq P(B \mid A)$. If $P(A \mid B) > 0$ then the converse also holds.

Proof. By the multiplication rule,

$$\begin{aligned} P(A \mid B) \cdot P(B \mid A \cup B) &= P(A \cap B \mid A \cup B) \\ &= P(B \mid A) \cdot P(A \mid A \cup B). \end{aligned}$$

If $P(A \mid A \cup B) \geq P(B \mid A \cup B)$, then we must have $P(A \mid A \cup B) > 0$ — since otherwise both terms would be zero, and the addition rule would give $P(A \cup B \mid A \cup B) = 0$ rather than 1. So we can divide both sides by $P(A \mid A \cup B)$ and find that $P(A \mid B) \geq P(B \mid A)$. Likewise, if $P(A \mid B) > 0$ and $P(A \mid B) \geq P(B \mid A)$, then we can divide both sides by $P(A \mid B)$ to find that $P(B \mid A \cup B) \leq P(A \mid A \cup B)$. $\square$

Proof of theorem E.68. Recall from definition A.28 and section A.3 the definition of the sets $H(A)$ for a set $A \subseteq \omega$: “Heads on (at least) every flip in A ”. Each of these sets is measurable in the coin flip space. Let P be a coin flip-symmetric conditional probability measure on 2^ω . Then $\succeq_P$ is a monotone total preorder, and, as we will check below, for each set $A \subseteq \omega$ and each $i, j \in \omega \setminus A$,

$$H(A \cup \{i\}) \sim_P H(A \cup \{j\}) \prec_P H(A).$$

(And this can be proven constructively.) But corollary A.29 says that it is not constructively provable that such an order exists.

It just remains to check the order properties. Let $i \notin A$. If σ_i is the “flip the i th coin” symmetry, then we have

$$\sigma_i \cdot H(\{i\}) = \neg H(\{i\}) \quad \sigma_i \cdot H(A) = H(A).$$

Thus by symmetry under σ_i ,

$$P(H(\{i\}) \mid H(A)) = P(\neg H(\{i\}) \mid H(A)),$$

and so by finite additivity,

$$P(H(\{i\}) \mid H(A)) = \frac{1}{2} \quad \text{for each } i \notin A.$$

As a special case, if $i \notin A$, then

$$\begin{aligned}
 P(H(A \cup \{i\}) \mid H(A)) &= P(H(A) \cap H(\{i\}) \mid H(A)) \\
 &= P(H(\{i\}) \mid H(A)) \\
 &= \frac{1}{2} \\
 &< 1 \\
 &= P(H(A) \mid H(A) \cap H(\{i\})) \\
 &= P(H(A) \mid H(A \cup \{i\})).
 \end{aligned}$$

Thus, by lemma E.69,

$$H(A \cup \{i\}) <_P H(A).$$

As another special case, if $i \neq j$ and neither is in A , then

$$\begin{aligned}
 P(H(A \cup \{i\}) \mid H(A \cup \{j\})) &= P(H(A) \cap H(\{i\}) \mid H(A \cup \{j\})) \\
 &= P(H(\{i\}) \mid H(A \cup \{j\})) \\
 &= \frac{1}{2}.
 \end{aligned}$$

The same goes swapping the roles of i and j , so

$$P(H(A \cup \{i\}) \mid H(A \cup \{j\})) = P(H(A \cup \{j\}) \mid H(A \cup \{i\})) > 0$$

and so by lemma E.69,

$$H(A \cup \{i\}) \sim_P H(A \cup \{j\}).$$

□

Note that this argument only ever conditions on sets of the form $H(A)$ and (via the definition of $\succeq_P$) their finite unions. So it applies just as well to conditional probability measures defined on a smaller collection of sets than the product σ -algebra on 2^ω , as long as these sets are included. (Note also that this cross-cuts Pruss's requirement that the conditional probability measure be defined on all *countable* subsets of 2^ω .)

Appendix F

Non-Archimedean Probability

F.1 Probability orders and measures

The classical theory of real-valued, countably additive probability (appendix E) is the most familiar way of extending finite probability distributions to infinite sets—but it is not the only way. It is also not clearly the best. It has a variety of difficult features.

1. *Unmeasurable sets* (theorem E.5). Countable additivity implies that not all sets can be assigned measures, on pain of violating natural symmetries: hence the usual practice is to restrict attention to some sigma-algebra of *measurable* sets.
2. *Regularity* (proposition E.12). It is natural to think that the possible is more likely than the impossible. But this cannot be represented by real-valued probabilities on an uncountable space of possibilities, such as the space of infinite sequences of coin flips.
3. *Dominance* (proposition E.38). Stochastic dominance is badly behaved for standard probability distributions. A random variable can have the very same standard distribution as another variable that is better than it in every state.

In sections 5.6 and E.7 we discussed an alternative approach to representing probability, and two of its major drawbacks. First, there is (plausibly) no good way of defining a *comparative probability order* from a conditional probability measure. Second, conditional probability measures (on continuum-sized spaces like the coin flip space) are *non-constructive*: the existence of

such measures cannot be proved without appeal to Choice (or else some other tool beyond standard ZF set theory). This means that any such measure is deeply arbitrary. It also makes them cumbersome to work with, since we can't inspect any examples directly.

This second feature (non-constructiveness) is not idiosyncratic to conditional probability measures. In fact, we have seen that non-constructiveness is inescapable for any *total regular probability order* satisfying very minimal conditions (section A.3).

Here we explore an alternative approach that has none of these problems, though it does have certain drawbacks of its own—most prominently, the failure of countable additivity and related principles. It is a version of non-Archimedean probability, where probabilities can be infinitesimal. The most familiar theory like this uses *hyperreal* probabilities. That version imposes a total probability order, and accordingly is non-constructive. Here we take a more general approach, which subsumes certain versions of hyperreal probabilities, but also has implementations that impose only a perfectly constructive *partial* probability order, along with a quantitative representation for this order. For the most part this is a direct application of the ideas in sections B.6 and B.8.¹

These quantitative representations for probability are not real numbers. What mathematical structure should they have? Let's start by revisiting the standard Kolmogorov axioms — where Ω is a set of *worlds* (or *states*) and A and B are *propositions* (or *events*) understood as certain sets of worlds.

1. $0 \leq P(A) \leq 1$.
2. $P(\Omega) = 1$.
3. If $A \cap B = \emptyset$ then $P(A \cup B) = P(A) + P(B)$.

There is also a further axiom (sometimes treated as a definition) for *conditional* probability.

4. $P(A \cap B) = P(B \mid A) \cdot P(A)$ (at least for $P(A) > 0$)

By inspection we see that we require the following from mathematical objects that are to serve as probabilities:

¹We are building on ideas from Pivato (2014) and Benci, Horsten, and Wenmackers (2013, 2018). Those authors in turn are carrying on a tradition of non-standard measure theory from Bernstein and Wattenberg (1969) and Loeb (1975). All of that work generally relies heavily on ultrafilter constructions to construct measures taking their values in totally ordered fields. Here we extract a “constructive core” from that approach, focusing on partially ordered quantities.

1. There are some probabilities “zero” and “one”;
2. Probabilities are ordered;
3. Probabilities can be added;
4. Probabilities can be multiplied.

Looking through our menu of algebraic structures for various kinds of quantities (section B.1), we find a good fit in *partially ordered commutative rings*, or *p.o.c. rings* for short: these have well-behaved order, addition, subtraction, and multiplication. The real numbers are one important example of a p.o.c. ring; other examples include the rationals, the hyperreals — or “quasi-real” *filter powers* of the reals.

A ring includes many elements besides probabilities. This is convenient, since it is nice to be able to add or subtract probabilities, even though this sometimes takes us from probabilities to quantities that are outside the zero-to-one range of probabilities.

Definition F.1. Let $\mathcal{R}$ be a p.o.c. ring, and let Ω be a set. An $\mathcal{R}$ -valued **probability measure** on the **state space** Ω is a function $P : 2^\Omega \rightarrow \mathcal{R}$ such that, for all $A, B \subseteq \Omega$:

- (a) $0 \leq P(A) \leq 1$;
- (b) $P(\Omega) = 1$;
- (c) If $A \cap B = \emptyset$ then $P(A \cup B) = P(A) + P(B)$.

We will normally assume without comment $\mathcal{R}$ is **non-trivial**, in the sense that $1 > 0$.

In standard probability theory we restricted the measure to some *algebra* of subsets of Ω , rather than the full power set (definition E.7). But this adds additional complications, and it turns out to be unnecessary for non-standard probability theory, so we won’t bother with that.

In section B.6 we introduced the notion of an $\mathcal{R}$ -valued *measure*. The definition of a probability measure extends this just by adding the normalization condition $P(\Omega) = 1$. For this to mean anything, we have to know what 1 means: it is the multiplicative identity — so this condition brings along with it the idea that $\mathcal{R}$ has a sensible notion of multiplication, and not some less structured kind of quantity. Rings provide this. (In contrast, general measures make sense where $\mathcal{R}$ is any p.o.c. monoid.)

Let’s also repeat some definitions from section B.6.

Here and below, we continue to use $\mathcal{R}$ to denote the underlying p.o.c. ring.

Definition F.2. Let P be an $\mathcal{R}$ -valued probability measure on a state space Ω , and let $G \curvearrowright \Omega$ be a group action (which induces an action $G \curvearrowright 2^\Omega$).

(a) P is **regular** iff for any $A \subseteq \Omega$,

$$\text{if } A \neq \emptyset, \text{ then } P(A) > 0.$$

(b) P is **uniform** iff

$$P(\{v\}) = P(\{w\}) \quad \text{for any } v, w \in \Omega.$$

(c) P is **G-symmetric** iff $P(g \cdot A) = P(A)$, for each $A \subseteq \Omega$ and $g \in G$.

(d) P is **G-invariant** iff whenever $P(A) \geq P(B)$, also $P(g \cdot A) \geq P(g \cdot B)$, for each $A, B \subseteq \Omega$ and $g \in G$.

These extend notions we have already met for *orders* (sections A.3 and A.6), for standard measures (definition E.3), and for *filters* (section B.8). As at those levels, G -symmetry implies G -invariance, but not conversely. The condition that P is *uniform* is equivalent to P being symmetric under finite permutations of states (proposition B.86).

Recall from section A.3 that we defined a *comparative probability order* on Ω to be a preorder on 2^Ω that satisfies de Finetti's axioms of non-triviality, monotonicity, and quasi-additivity.

Proposition F.3. For any $\mathcal{R}$ -valued probability measure P on Ω , the relation

$$A \succsim_P B \quad \text{iff} \quad P(A) \geq P(B)$$

is a comparative probability order on Ω . We say $\succsim_P$ is **represented by** P . Moreover:

- (a) P is regular iff $\succsim_P$ is regular;
- (b) P is uniform iff $\succsim_P$ is uniform;
- (c) P is G -symmetric iff $\succsim_P$ is G -symmetric;
- (d) P is G -invariant iff $\succsim_P$ is G -invariant.

Note that we cannot always go the other direction: not every comparative probability order can be represented by an $\mathcal{R}$ -valued probability measure.

(This follows from a classic result from Krantz et al. 1971, originally stated for *real* probabilities.) To ensure the existence of an additive representation, we need to impose stronger conditions than de Finetti's axioms. We will return to this in theorem F.16.

We have some immediate corollaries from our negative results about orders.

Corollary F.4. Let Ω be an infinite state space, and let $\mathcal{R}$ be a p.o.c. ring.

- (a) No regular $\mathcal{R}$ -valued probability measure on Ω is symmetric under arbitrary permutations.
- (b) More generally, for any group action $G \curvearrowright \Omega$, if there exists a G -symmetric regular $\mathcal{R}$ -valued probability measure on Ω , then for every $g \in G$ and $w \in \Omega$, the orbit $\langle g \rangle \cdot w$ is finite.
- (c) There is no regular uniform real-valued probability measure on Ω .

Suppose furthermore that $\mathcal{R}$ is totally ordered. In that case:

- (d) No regular $\mathcal{R}$ -valued probability measure on Ω is invariant under arbitrary permutations.
- (e) More generally, for any group action $G \curvearrowright \Omega$, if there exists a G -invariant regular $\mathcal{R}$ -valued probability measure on Ω , then for every $F \subseteq G$, if every element of F has finite orbits, then so does every element of the generated subgroup $\langle F \rangle$.
- (f) There is no constructive proof that there exists a regular uniform $\mathcal{R}$ -valued probability measure on Ω .
- (g) There is no constructive proof that there exists any regular $\mathcal{R}$ -valued probability measure on $\Omega = 2^\omega$.

Proof. We prove each part by combining proposition F.3 with a corresponding result for probability orders.

- (a) Proposition A.16.
- (b) Corollary A.70(c).
- (c) Corollary A.27(b).
- (d) Corollary A.70(b).
- (e) Proposition A.64.

(f) Corollary A.27(c).

(g) Theorem A.30.

□

In addition to the notion of regularity above, it is also convenient to have a stronger notion. In a *field*, we know that it makes sense to *divide* by any non-zero number. In a general ring, this is not guaranteed. But division is often something we want, so we can make sense of *conditional* probabilities in the usual way. This motivates the following definition.

Definition F.5. For $p \in \mathcal{R}$, we call p **divisible** iff p has a positive multiplicative inverse in $\mathcal{R}$: that is, there is some $p^{-1} \in \mathcal{R}$ such that $p^{-1} > 0$ and $p \cdot p^{-1} = 1$.

An $\mathcal{R}$ -valued probability measure P on Ω is **divisible** iff for any $A \subseteq \Omega$, if $P(A)$ is positive, then $P(A)$ is divisible.

In the case where $\mathcal{R}$ is a totally ordered field, every $\mathcal{R}$ -valued measure is divisible. Note that the converse — if $P(A)$ is divisible, then it is positive — holds automatically for probabilities, since $P(A) \geq 0$.

Whenever $P(A)$ is divisible, we can define conditional probability in the usual way:

$$P(B | A) = P(A \cap B) / P(A).$$

If P is regular and divisible, then this is well-defined for every $A \subseteq \Omega$.

F.2 Limiting rings

Before we turn to some important constructions of non-Archimedean probability measures, let's draw some connections between *filter powers* (section B.5) and rings.²

The rationals are one important p.o.c. ring. For any set I , the set $\mathbb{Q}^I$ of all rational-valued functions on I can also be naturally equipped with the structure of a p.o.c. ring (as a Cartesian power of $\mathbb{Q}$, as discussed in section B.2). In section B.5 we introduced the *filter power* $\mathbb{Q}^{\mathcal{F}}$, for a filter $\mathcal{F}$ on a set I . This

²Compare the concept of a *non-Archimedean field equipped with an Ω -limit* of Benci, Horsten, and Wenmackers (2018).

is equipped with a *filter limit* operator $\lim^{\mathcal{F}}$, which is a surjective homomorphism from the ring $\mathbb{Q}^I$ to $\mathbb{Q}^{\mathcal{F}}$. We can also turn things round: rather than starting with the filter, we can start with this limit operation.

Definition F.6. Let I be a non-empty set. An **I -limiting ring** is a p.o.c. ring $\mathcal{R}$ equipped with a surjective homomorphism (of p.o.c. rings)

$$\lim^{\mathcal{R}} : \mathbb{Q}^I \rightarrow \mathcal{R}.$$

We will normally omit the superscript $\mathcal{R}$.

Even though we will use the notation $\lim$, this non-standard limit should not be confused with the standard limit concept from calculus.

We can think of I as a set of *indices*. For example, if $I = \mathbb{N}$, then the corresponding limit operator $\lim^{\mathcal{R}}$ takes an infinite sequence of numbers (a function $\mathbb{N} \rightarrow \mathbb{Q}$) to an element of $\mathcal{R}$. But we often won't use $\mathbb{N}$ as our index set.

Not much will turn on taking the *rational*s as our base: if we started with the reals (or some other ordered field), almost everything that follows would go the same way. But there are two reasons for preferring the rationals. First, this is the structure that arises most naturally in our constructions in the next section. Second, this seems philosophically correct. Rational probabilities have a clear probabilistic meaning, arising from equiprobable partitions. In a setting where we have countable additivity, real probabilities also acquire a clear meaning, as (calculus-style) limits of rational probabilities. But in what follows, we will not have countable additivity (or even a structure in which this makes precise sense), and the distinctive features of the real numbers (the least upper bound property) would have no work to do within the space of non-Archimedean probabilities. The limit operation $\lim^{\mathcal{R}}$ will do some analogous work instead. The real numbers are a kind of *completion* of the rationals. A limiting ring $\mathcal{R}$ is a different kind of completion; it would be technically otiose and conceptually distracting to be carrying around both of them at the same time.

It is clear from what we have said that any filter $\mathcal{F}$ on I gives rise to an I -limiting ring $\mathbb{Q}^{\mathcal{F}}$. The converse also holds.³

³This amounts to a standard correspondence between ideals in a product of fields (like $\mathbb{Q}^I$) and filters on I . (Gillman and Jerison 1960, ch. 2.)

Proposition F.7. Every I -limiting ring $\mathcal{R}$ is isomorphic to $\mathbb{Q}^{\mathcal{F}}$ for a unique filter $\mathcal{F}$ on I , with a unique limit-preserving isomorphism. We call this filter $\mathcal{F}(\mathcal{R})$.

Proof. Let $\mathcal{R}$ be an I -limiting ring. We can define a filter $\mathcal{F}$ on I by

$$V \in \mathcal{F} \quad \text{iff} \quad \lim 1_V = 1.$$

It is straightforward to check that $\mathcal{F}$ is a filter (using the homomorphism properties of $\lim$). Furthermore, the unique limit-preserving function from $\mathcal{R}$ to $\mathbb{Q}^{\mathcal{F}}$ is given by

$$h(\lim f) = \lim^{\mathcal{F}} f \quad \text{for each } f : I \rightarrow \mathbb{Q}.$$

It can be checked that h is well-defined (if $\lim f = \lim g$, then $f = g$ $\mathcal{F}$ -a.e.), and that h is an isomorphism (by the properties of limits). $\square$

Thus *filters on I* and *I -limiting rings* each capture exactly the same information.

We also have been considering filters with some special properties. These have counterparts for limiting rings. We call a p.o.c. ring $\mathcal{R}$ **non-trivial** iff $1 > 0$ in $\mathcal{R}$.

Proposition F.8. Let $\mathcal{R}$ be an I -limiting ring. The following are equivalent.

- (a) $\mathcal{R}$ is non-trivial;
- (b) The filter $\mathcal{F}(\mathcal{R})$ is non-trivial;
- (c) $\lim^{\mathcal{R}}$ is one-to-one on constant functions;
- (d) $\mathcal{R}$ contains an isomorphic copy of $\mathbb{Q}$.

Proof. The equivalence (a) $\Leftrightarrow$ (b) follows from the definition of $\mathcal{F}(\mathcal{R})$ in proposition F.7, together with the fact that $\lim$ is order-preserving, so that $1 \geq 0$ in $\mathcal{R}$ in any case. The implications (c) $\Rightarrow$ (d) $\Rightarrow$ (a) are straightforward. This just leaves (a) $\Rightarrow$ (c).

Write $c_a : I \rightarrow \mathbb{Q}$ for the constant function with value a . The map $a \mapsto \lim c_a$ is a ring-homomorphism from $\mathbb{Q}$ to $\mathcal{R}$, and so its kernel is an ideal of $\mathbb{Q}$. But $\mathbb{Q}$ is a field, and so (by a standard algebraic fact) its only ideals are $\{0\}$ and $\mathbb{Q}$ itself. In the latter case $1 = \lim c_1 = 0$, contradicting (a). So the kernel is $\{0\}$, which is to say that $a \mapsto \lim c_a$ is one-to-one. $\square$

Fine filters (definition B.67) also have a counterpart notion.

Definition F.9. Let Ω be a set. A **fine Ω -ring** is a $(\text{Fin } \Omega)$ -limiting ring $\mathcal{R}$ such that, for each $w \in \Omega$,

$$\lim_{K \in \text{Fin } \Omega} 1_K(w) = 1.$$

As with fine filters, note that a fine Ω -ring is indexed by $\text{Fin } \Omega$ rather than by Ω : the “finess” property only makes sense where the indices are finite sets. The following is straightforward to check (using the proof of proposition F.7).

Proposition F.10. Let $\mathcal{R}$ be a $(\text{Fin } \Omega)$ -limiting ring. Then $\mathcal{R}$ is a fine Ω -ring iff the corresponding filter $\mathcal{F}(\mathcal{R})$ is fine.

So, for example, we can equivalently characterize the quasi-finite sums introduced in section B.6 as follows:

Proposition F.11. For any fine Ω -ring $\mathcal{R}$, there is a corresponding **quasi-finite sum** operator $\sum^{\mathcal{R}} : \mathbb{Q}^\Omega \rightarrow \mathcal{R}$ defined by

$$\sum^{\mathcal{R}} f = \lim_K \sum_K f.$$

Likewise, there is a **quasi-cardinality** operator $\#^{\mathcal{R}} : 2^\Omega \rightarrow \mathcal{R}$ defined by

$$\#^{\mathcal{R}}(A) = \sum^{\mathcal{R}} 1_A = \lim_K \#(A \cap K).$$

These are identical (up to isomorphism) to the quasi-finite sum $\sum^{\mathcal{F}}$ and quasi-cardinality $\#^{\mathcal{F}}$, where $\mathcal{F} = \mathcal{F}(\mathcal{R})$ is the filter corresponding to $\mathcal{R}$.

Here are couple of important points.

Lemma F.12. If Ω is non-empty and $\mathcal{R}$ is a fine Ω -ring, then $\#^{\mathcal{R}}\Omega$ is divisible.^a

^aAn edge case is when $\mathcal{R}$ is trivial. In this case, $\#^{\mathcal{R}}\Omega = 0$; but also, the trivial ring is the only case where 0 is in fact *divisible*: 0 is its own multiplicative inverse, since $0 = 1$.

Definition F.13. A p.o.c. ring R is **Archimedean** iff for any $a, b \in R$ such that $a > 0$, there is some $n \in \mathbb{N}$ such that $n \cdot a \geq b$.

Proposition F.14. A non-trivial fine Ω -ring is non-Archimedean iff Ω is infinite.

Proof. Let $\mathcal{R}$ be a non-trivial fine Ω -ring.

If Ω is finite, then $\mathcal{R}$ is isomorphic to $\mathbb{Q}$, which is Archimedean. (In this case, since $\mathcal{I}$ has only finitely many elements, fineness and closure under finite intersections imply that $\mathcal{F}(\mathcal{R})$ must be the principal filter $\{V \subseteq \text{Fin } \Omega \mid \Omega \in V\}$. In this case $\mathbb{Q}^{\mathcal{F}(\mathcal{R})} \cong \mathbb{Q}$.)

Now suppose Ω is infinite. Let

$$N = \lim_K \#K = \#^{\mathcal{R}}\Omega.$$

In $\mathcal{R}$ we also have an isomorphic image of the standard number 1, and there is no standard natural number n such that $n \cdot 1 \geq N$. To see this, fix a subset $J \subset \mathcal{I}$ with $n + 1$ elements. Then for all finite K ,

$$\#K \geq \#(J \cap K) = \sum_{j \in J} 1_K(j).$$

Thus, by the properties of limits and fineness,

$$N = \lim_K \#K \geq \lim_K \sum_{j \in J} 1_K(j) = \sum_{j \in J} \lim_K 1_K(j) = \sum_{j \in J} 1 = n + 1.$$

Thus $N > n$ for all $n \in \mathbb{N}$, so $\mathcal{R}$ is non-Archimedean. $\square$

Putting these together, if Ω is infinite, then $\#^{\mathcal{R}}\Omega$ is an infinite divisible element of $\mathcal{R}$. So its inverse $1/\#^{\mathcal{R}}\Omega$ is a *positive infinitesimal*: it is smaller than $1/n$ for every standard natural number n .

F.3 Regular uniform probability measures and expectations

Back in section B.6 we used quasi-cardinalities to construct non-Archimedean measures. The exact same approach can also be used to construct *probability* measures: the only additional thing we need is to *normalize* the measure.

Definition F.15. Let Ω be a non-empty set, and let $\mathcal{R}$ be a fine Ω -ring. The **uniform $\mathcal{R}$ -valued probability measure** is the function $\mathbb{P}^{\mathcal{R}} : 2^{\Omega} \rightarrow \mathcal{R}$ defined by

$$\mathbb{P}^{\mathcal{R}}(A) := \frac{\#^{\mathcal{R}} A}{\#^{\mathcal{R}} \Omega} \quad \text{for each } A \subseteq \Omega.$$

Equivalently:

$$\mathbb{P}^{\mathcal{R}}(A) = \lim_K \frac{\#(A \cap K)}{\#K}.$$

One important case is the singletons: if Ω is infinite then for any $w \in \Omega$, the probability

$$\mathbb{P}^{\mathcal{R}}\{w\} = \frac{1}{\#^{\mathcal{R}} \Omega}$$

is infinitesimal.

We can now prove something in the neighborhood of a converse to proposition F.3 — a *representation theorem* for non-Archimedean probability measures. It relies on a further technical axiom. The intuitive idea is that the probability order is a “limit” of its behavior on sufficiently large finite sets — the same idea that we captured with the notion of a *finely filtered* order in definition B.79 in section B.7.

Theorem F.16. Let $\succsim$ be a comparative probability order on a state space Ω . The following are equivalent:

- (a) $\succsim$ is regular, uniform, and finely filtered by the family of preorders $(\succsim_K)_{K \in \text{Fin } \Omega}$ where

$$A \succsim_K B \quad \text{iff} \quad A \cap K \succsim B \cap K \quad \text{for each } A, B \subseteq \Omega;$$

- (b) $\succsim$ is represented by the uniform $\mathcal{R}$ -valued probability measure, for some non-trivial fine Ω -ring $\mathcal{R}$.

Proof. (b) $\Rightarrow$ (a). By theorem B.72, $\#^{\mathcal{R}}$ is a regular uniform measure, and thus it represents a regular uniform probability order $\succsim^{\mathcal{R}}$. Any such order faithfully extends the weak counting order (corollary A.27), so for any finite sets J and K , $J \succsim^{\mathcal{R}} K$ iff $\#J \geq \#K$. Thus for $K \in \text{Fin } \Omega$,

$$K \in F(A, B) \quad \text{iff} \quad \#(A \cap K) \geq \#(B \cap K) \quad \text{iff} \quad \sum_K (1_A - 1_B) \geq 0.$$

Proposition B.73 tells us, then, that $\succsim^{\mathcal{R}}$ is finely filtered. Furthermore, since $\mathcal{R}$ is non-trivial and fine, it follows that $\#^{\mathcal{R}}\Omega$ is divisible; so $\mathbb{P}^{\mathcal{R}}$ is a well-defined regular uniform probability measure, and it represents the same order as $\#^{\mathcal{R}}$.

(a) $\Rightarrow$ (b). Let $\mathcal{F}$ be the smallest fine filter containing the sets $F(A, B)$ for all $A \succsim B$. In this case, for all $A, B \subseteq \Omega$,

$$A \succsim B \quad \text{iff} \quad F(A, B) \in \mathcal{F}.$$

The left-to-right direction is immediate. For the right-to-left, if $F(A, B) \in \mathcal{F}$, then $\uparrow J \cap \bigcap_k F(A_k, B_k) \subseteq F(A, B)$ for some $J \in \text{Fin } \Omega$ and some family of pairs where $A_1 \succsim B_1, \dots, A_n \succsim B_n$; thus by fine filtration, $A \succsim B$.

It follows that $\mathcal{F}$ is non-trivial, since $\Omega \succ \emptyset$ and thus $F(\emptyset, \Omega) \notin \mathcal{F}$.

Now let $\mathcal{R} = \mathbb{Q}^{\mathcal{F}}$, a non-trivial fine Ω -ring. Then (using the definition of quasi-cardinality and corollary A.27 again), for any $A, B \subseteq \Omega$,

$$A \succsim B \quad \text{iff} \quad F(A, B) \in \mathcal{F} \quad \text{iff} \quad \#^{\mathcal{R}}A \geq \#^{\mathcal{R}}B \quad \text{iff} \quad \mathbb{P}^{\mathcal{R}}(A) \geq \mathbb{P}^{\mathcal{R}}(B).$$

So $\mathbb{P}^{\mathcal{R}}$ represents $\succsim$. □

Here is an alternative perspective on non-Archimedean probability (based on Benci, Horsten, and Wenmackers 2018). Standard probability measures satisfy a *continuity* condition, as a consequence of countable additivity:

Continuity. For any sequence of propositions $E_1 \subseteq E_2 \subseteq E_3 \subseteq \dots$ such that $\bigcup_k E_k = \Omega$, we have

$$P(A) = \lim_{n \rightarrow \infty} P(A \mid E_n) \quad \text{for each } A \subseteq \Omega.$$

(In particular, the conditional probability is well-defined for sufficiently large n .) This continuity condition is only non-vacuous when Ω is countable: in this case, we can approach Ω by an increasing sequence of finite sets. The limit in this condition is the standard limit of real numbers.

We can say something closely analogous for non-Archimedean probability measures, using non-standard limits, which give us a more general notion of “approaching” Ω via finite sets.

Definition F.17. Let $\mathcal{R}$ be a non-trivial fine Ω -ring, and let P be any $\mathcal{R}$ -valued probability measure on Ω . We say P is $\mathcal{R}$ -**continuous** iff, for sufficiently large K , the conditional probability $P(- \mid K)$ is well-defined and takes standard

values (in $\mathbb{Q}$, considered as a substructure of $\mathcal{R}$), and furthermore

$$P(A) = \lim_K^{\mathcal{R}} P(A \mid K) \quad \text{for each } A \subseteq \Omega.$$

Proposition F.18. The uniform $\mathcal{R}$ -valued probability measure $\mathbb{P}^{\mathcal{R}}$ is the unique uniform $\mathcal{R}$ -continuous $\mathcal{R}$ -valued probability measure on Ω .

Proof. Let P be any $\mathcal{R}$ -valued probability measure on Ω . Then P is uniform iff for all non-empty K , $P(- \mid K)$ is the uniform probability measure on K . Thus P is uniform and $\mathcal{R}$ -continuous iff, for all $A \subseteq \Omega$,

$$P(A) = \lim_K P(A \mid K) = \lim_K \frac{\#(A \cap K)}{\#K} = \mathbb{P}^{\mathcal{R}}(A). \quad \square$$

The main reason for this diversion into the foundations of probability is in order to develop a theory of *value* for uncertain prospects. Our working model for this is a generalization of the standard theory of *expected* value. So in addition to a notion of probability, we also want to define a notion of *expectation*.

Fortunately, once again we have just about already done this. In section B.6, we defined a notion of a *quasi-finite sum*. For any quantity $\mathcal{U}$, and any $\mathcal{U}$ -valued function f , the quasi-finite sum $\sum^{\mathcal{G}} f$ is an element of the filter power $\mathcal{U}^{\mathcal{G}}$. In order to avoid the detour back to filters, we will also write $\mathcal{U}^{\mathcal{R}}$ as a synonym for $\mathcal{U}^{\mathcal{F}(\mathcal{R})}$.

This is almost what we want: once again we just need to renormalize. For this purpose we should restrict attention to quantities which are at least *rational vector spaces* (or slightly more generally, $\mathbb{Q}^+$ -quantities). We will accordingly restrict attention to p.o. rational vector spaces — which include the real ones, on the occasions when we want them.

Definition F.19. Let $\mathcal{U}$ be a p.o. rational vector space, and let $f : \Omega \rightarrow \mathcal{U}$ be a $\mathcal{U}$ -valued function on a state space Ω . Let $\mathcal{R}$ be a non-trivial fine Ω -ring. Define the **uniform $\mathcal{R}$ -expectation** of f to be

$$\mathbb{E}^{\mathcal{R}} f := \frac{1}{\#\mathcal{R}\Omega} \sum^{\mathcal{R}} f \in \mathcal{U}^{\mathcal{R}}.$$

The following fact immediately follows from the corresponding properties of quasi-finite sums in section B.6.

Proposition F.20. Let $\mathcal{U}$ be a p.o. rational vector space, and let $\mathcal{R}$ be a non-trivial fine Ω -ring.

- (a) $\mathbb{E}^{\mathcal{R}}$ is a strictly positive linear operator from $\mathcal{U}^{\Omega}$ to $\mathcal{U}^{\mathcal{R}}$.
- (b) For any $A \subseteq \Omega$,

$$\mathbb{P}^{\mathcal{R}}(A) = \mathbb{E}^{\mathcal{R}} 1_A$$

(where as usual 1_A is the indicator function of A).

F.4 Symmetric measures

One of our central examples throughout the book is the case of an infinite sequence of coin flips (definition A.28 and example A.44). The (Cantor) *coin flip space* 2^{ω} is the set of infinite sequences of ones and zeros (corresponding to Heads and Tails). We have discussed several different kinds of natural symmetries of this space.

- The finitary coin flip action $\text{Fin } \omega \curvearrowright 2^{\omega}$ flips between Heads and Tails at finitely many positions in the sequence.
- The full *Cantor group* 2^{ω} acts on 2^{ω} by flipping the sequence at arbitrary sets of positions.
- The coin-permutation action $\Pi(\omega) \curvearrowright 2^{\omega}$ rearranges the sequence of coin flips.

All of these are symmetries of the standard coin flip probability measure (example E.9 and proposition E.11); but this comes at the cost of sacrificing regularity and measurable sets. In corollary A.29; theorem A.30; theorem A.31 we noted some general impossibility results about probability *orders* on the coin flip space. No regular probability order on the coin flip space is real-representable or symmetric under arbitrary coin permutations, and there is no constructive proof that a *total* regular probability order on the coin flip space exists at all.

We also met a family of related obstacles for probability orders on an interval, or a circle. In this case the standard (Lebesgue) probability measure is symmetric under arbitrary rotations. But no regular probability order is

symmetric under arbitrary rotations (lemma A.60), and again there is no constructive existence proof for a *total* regular probability order (theorem A.30).

Now we have some correlative positive results.

Our friends *symmetry* and *invariance* have accompanied us through our discussion of orders (section A.6), filters (section B.8), standard measures (definition E.3), and non-Archimedean measures (definition F.2). We can also apply them to rings.

Definition F.21. Let $G \curvearrowright \Omega$ be a group action. A fine Ω -ring $\mathcal{R}$ is **G -symmetric** iff the corresponding filter $\mathcal{F}(\mathcal{R})$ (proposition F.7) is G -symmetric. Likewise, $\mathcal{R}$ is **G -invariant** iff $\mathcal{F}(\mathcal{R})$ is G -invariant.

Accordingly, properties of filters are straightforwardly inherited by their corresponding rings: every G -symmetric ring is G -invariant (proposition B.84); every fine ring is symmetric under finite permutations (proposition B.86).

Proposition F.22. Let $G \curvearrowright \Omega$ be an action, let $\mathcal{R}$ be a non-trivial fine Ω -ring, and let $\mathcal{U}$ be a p.o. rational vector space. Then G also naturally acts on 2^Ω , as well as the space $\mathcal{U}^\Omega$ of $\mathcal{U}$ -valued functions on Ω .

- (a) If $\mathcal{R}$ is G -symmetric, then $\mathbb{P}^\mathcal{R}$ and $\mathbb{E}^\mathcal{R}$ are both G -symmetric, in the sense that for any $g \in G$,

$$\begin{aligned}\mathbb{P}^\mathcal{R}A &= \mathbb{P}^\mathcal{R}(g \cdot A) && \text{for each } A \subseteq \Omega; \\ \mathbb{E}^\mathcal{R}f &= \mathbb{E}^\mathcal{R}(g \cdot f) && \text{for each } f : \Omega \rightarrow \mathcal{U}.\end{aligned}$$

- (b) If $\mathcal{R}$ is G -invariant, then $\mathbb{P}^\mathcal{R}$ and $\mathbb{E}^\mathcal{R}$ are G -invariant, in the sense that for any $g \in G$,

$$\begin{aligned}\text{if } \mathbb{P}^\mathcal{R}A &\geq \mathbb{P}^\mathcal{R}B \\ \text{then } \mathbb{P}^\mathcal{R}(g \cdot A) &\geq \mathbb{P}^\mathcal{R}(g \cdot B) && \text{for each } A, B \subseteq \Omega; \\ \text{if } \mathbb{E}^\mathcal{R}f &\geq \mathbb{E}^\mathcal{R}h \\ \text{then } \mathbb{E}^\mathcal{R}(g \cdot f) &\geq \mathbb{E}^\mathcal{R}(g \cdot h) && \text{for each } f, h : \Omega \rightarrow \mathcal{U}.\end{aligned}$$

Proof. In theorem B.85 we proved the analogous facts about the quasi-cardinality $\#^{\mathcal{F}(\mathcal{R})}$ and the quasi-finite sum $\sum^{\mathcal{F}(\mathcal{R})}$. The probability measure $\mathbb{P}^\mathcal{R}$ is just $\#^{\mathcal{F}(\mathcal{R})}$, renormalized by dividing by $\#^{\mathcal{F}(\mathcal{R})}\Omega$ (which we know is

divisible in $\mathcal{R}$). Likewise the expectation $\mathbb{E}^{\mathcal{R}}$ is just the renormalized version of $\sum^{\mathcal{F}(\mathcal{R})}$. Both parts follow straightforwardly. $\square$

Furthermore, we can prove an important existence theorem, straightforwardly extending corollary B.90 from unnormalized measures to probability measures. Recall that in section B.8 we introduced the *weak G -symmetric filter* for a group action $G \curvearrowright \Omega$ (lemma B.92). Using the properties we showed there (especially propositions B.88 and B.94) it is straightforward to prove the following.

Proposition F.23. Let $G \curvearrowright \Omega$ be an action, and let $\mathcal{G}$ be the weak G -symmetric filter on $\text{Fin } \Omega$. We call the corresponding ring $\mathbb{Q}^{\mathcal{G}}$ the **weak G -symmetric ring**, and denote it $\mathcal{R}^G$.

If $G \curvearrowright \Omega$ is locally finite, then $\mathcal{R}^G$ is a non-trivial fine Ω -ring. We call the uniform $\mathcal{R}^G$ -valued probability measure the **weak G -symmetric probability measure** on Ω , and denote it $\mathbb{P}^G$; we write $\mathbb{E}^G$ for the corresponding uniform $\mathcal{R}^G$ -expectation.

- (a) $\mathbb{P}^G$ is uniform, regular, divisible, and G -symmetric.
- (b) Let $H \curvearrowright \Omega$ be another group action that *normalizes* $G \curvearrowright \Omega$. Then $\mathbb{P}^G$ is H -invariant.

(Properly speaking we should probably use the notations $\mathcal{R}^{G \curvearrowright \Omega}$ and $\mathbb{P}^{G \curvearrowright \Omega}$, since it is the *action* that matters, not just the group, but this is a bit unwieldy.)

For example, the group of *rational rotations* of the circle is locally finite, and it is normalized by the group of all rotations (example A.49 and corollary B.95).

Corollary F.24. Let $\mathbb{Q}/\mathbb{Z}$ be the group of rational rotations acting on the circle. The weak $(\mathbb{Q}/\mathbb{Z})$ -symmetric probability measure is symmetric under rational rotations, and invariant under arbitrary rotations.

For another example, which we will explore a bit more systematically, the group of coin flip symmetries is locally finite and normalized by the group of coin permutations (example A.49 and corollary B.96).

Corollary F.25. Let the **weak finitary coin flip symmetric measure** on the coin flip space 2^ω be the weak $(\text{Fin } \omega)$ -symmetric probability measure, where $\text{Fin } \omega \curvearrowright 2^\omega$ is the action of finitary flips. This is a uniform, regular non-

Archimedean probability measure on the coin flip space, which is symmetric under finitary flips, and invariant under arbitrary flips and coin permutations.

The **weak Cantor-symmetric measure** on 2^ω is the weak 2^ω -symmetric probability measure, where $2^\omega \curvearrowright 2^\omega$ is the Cantor coin flip action. This is a uniform, regular non-Archimedean probability on the coin flip space, which is symmetric under arbitrary flips, and invariant under coin permutations.

Furthermore, if $\mathcal{F}$ is a G -symmetric filter, then any filter $\mathcal{G} \supseteq \mathcal{F}$ is also G -symmetric. So even though we cannot constructively demonstrate totally ordered rings with certain symmetries, using Choice we can prove they exist. The same point does not apply to G -invariance, though.

Proposition F.26.

- (a) There exists a totally ordered fine 2^ω -ring $\mathcal{R}$ such that $\mathbb{P}^{\mathcal{R}}$ is symmetric under the full Cantor coin flip action. No regular measure is invariant under coin permutations.
- (b) There exists a totally ordered fine $[0, 1)$ -ring $\mathcal{R}$ such that $\mathbb{P}^{\mathcal{R}}$ is symmetric under rational rotations.

Proof. By theorem B.65, each weak G -symmetric filter can be extended to an ultrafilter, which inherits the properties of fineness and G -symmetry. This establishes both existence claims (via proposition F.22).

The negative part of (a) follows from corollary F.4. □

I do not know whether there exists a totally ordered fine $[0, 1)$ -ring that is invariant under arbitrary rotations. This is not settled by corollary F.4. Rather, we found in corollary A.68 that there *do* exist rotation-invariant strictly monotone orders, because the rotation group is abelian; it is not clear whether the same style of proof extends to fine filters. (We do know that no regular measure on $[0, 1)$ is *symmetric* under arbitrary rotations, by corollary F.4 (b), since an irrational rotation has infinite orbits.)

Next we'll examine how the non-Archimedean measures we have constructed on the coin flip space and the interval relate to their *standard* counterparts. These two spaces share some common structure. Essentially, we can think of their symmetries as "limits" of families of finite symmetry groups acting on finitely many "pieces" of the space.

(This is introducing yet another limit concept, not to be confused with filter limits or the limits from calculus; this one comes from category theory.)⁴

Definition F.27. A **directed set** is a partially ordered set I such that any two elements have an upper bound: for $i, j \in I$, there is some $k \in I$ such that $i \leq k$ and $j \leq k$.

Definition F.28. Recall that an action $G \curvearrowright \mathcal{X}$ is *free* (definition E.4) iff no element of $\mathcal{X}$ is fixed by any element of G besides the identity. The action is **transitive** iff it has exactly one orbit. When both these conditions hold, the action is **simply transitive**: that is, for any $x, y \in \mathcal{X}$, there is exactly one $g \in G$ such that $g \cdot x = y$.

Definition F.29. Let I be a directed set. A **family of finite actions** (indexed by I) consists of the following:

- For each $i \in I$, a simply transitive group action $H_i \curvearrowright \mathcal{X}_i$, where H_i is finite;
- For each $i \leq j$ in I , a function $f_{ji} : \mathcal{X}_j \rightarrow \mathcal{X}_i$,

which have the following properties, for any $i \leq j \leq k$ in I :

- (a) H_i is a subgroup of H_j ;
- (b) $f_{ki} = f_{ji} \circ f_{kj}$, and $f_{ii} = \text{id}_{\mathcal{X}_i}$;
- (c) f_{ji} is *equivariant*, in the sense that

$$f_{ji}(g \cdot x) = g \cdot f_{ji}x \quad \text{for each } g \in H_i \leq H_j \text{ and } x \in \mathcal{X}_j.$$

A **limit** of this family is a group action $G \curvearrowright \Omega$, together with a family of onto functions $h_i : \Omega \rightarrow \mathcal{X}_i$ for each $i \in I$, such that

- (d) $G = \bigcup_i H_i$;

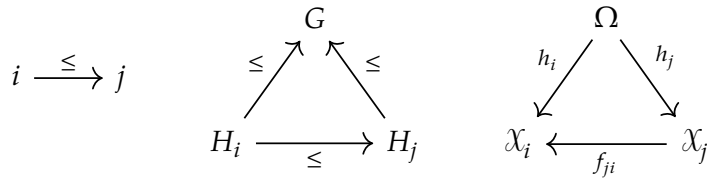
⁴In particular — just for those readers with the background to understand this — in definition F.29, the groups H_i form a *direct system* whose *direct limit* is G ; the sets $\mathcal{X}_i$ form an *inverse system*, and Ω can be embedded in its *inverse limit*. (We do not require that Ω be identically the inverse limit because this would exclude the interval.)

(e) For each $i \leq j$ in I ,

$$\begin{aligned} f_{ji} \circ h_j &= h_i; \\ h_i(g \cdot w) &= g \cdot h_i w \quad \text{for each } g \in H_i, w \in \Omega. \end{aligned}$$

(f) The functions $(h_i)_{i \in I}$ *separate points*, in the sense that for any $v, w \in \Omega$ there is some $i \in I$ such that $h_i v \neq h_i w$.

Let a **basic subset of Ω** be any set $h_i^{-1}(A)$ for $A \subseteq \mathcal{X}_i$.



Proposition F.30.

- (a) The action of finitary flips on the coin flip space is a limit of finite actions $H_K \curvearrowright 2^K$, indexed by the directed set $\text{Fin } \omega$. For each $K \in \text{Fin } \omega$, let H_K be the group of flips supported by K . For $J \subseteq K$, the function $f_{KJ} : 2^K \rightarrow 2^J$ is the projection function, and likewise $h_K = \pi_K : 2^\omega \rightarrow 2^K$ is also a projection, for each $K \in \text{Fin } \omega$. The basic subsets of 2^ω are precisely the cylinder sets.
- (b) The action of rational rotations on $[0, 1)$ is a limit of the family of finite actions $H_n \curvearrowright \mathcal{X}_n$ indexed by the positive integers, ordered by divisibility. In this case, H_n is the group of rotations by multiples of $1/n$, and $\mathcal{X}_n = \mathbb{Z}_n := \{0, \dots, n-1\}$. If n is divisible by k , so $n = ak$, the function $f_{nk} : \mathbb{Z}_n \rightarrow \mathbb{Z}_k$ takes each number b less than n to the integer part of b/a , which is less than k . The projection $h_n : [0, 1) \rightarrow \mathbb{Z}_n$ divides the interval into n equal segments, and takes each $r \in [0, 1)$ to the integer index corresponding to its segment (which is $\lfloor n \cdot r \rfloor$). The basic subsets of $[0, 1)$ are finite unions of half-open intervals with rational endpoints.

Lemma F.31. Any limit of finite actions is locally finite.

Proof. Straightforward (using the fact that I is directed, and thus any finitely many H_i are contained in some common H_j). $\square$

Note that the converse does not hold: for example the action of $\Pi_{\text{fin}} X$ on an infinite set X is locally finite, but not a limit of finite actions.

Theorem F.32. Let $G \sim \Omega$ be a limit of finite actions. Let $\mathcal{R}$ and $\mathcal{R}'$ be p.o.c. rings that each embed $\mathbb{Q}$. Let P be a (finitely additive) $\mathcal{R}$ -valued probability measure on Ω , and let P' be a (finitely additive) $\mathcal{R}'$ -valued probability measure on Ω . Suppose that both P and P' are G -symmetric. Then for every basic subset $A \subseteq \Omega$,

$$P(A) = P'(A) \in \mathbb{Q}.$$

Proof. Let $i \in I$ and $x \in \mathcal{X}_i$. Since H_i acts transitively on $\mathcal{X}_i$, it has a subset K with $\#\mathcal{X}_i$ distinct elements, each of which takes x to a different element of $\mathcal{X}_i$. Let $A = h_i^{-1}\{x\}$. The equivariance condition implies that $g \cdot A = h_i^{-1}\{g \cdot x\}$ for each $g \in K$. Thus the sets $g \cdot A$ for $g \in K$ are a finite partition of Ω . Since P is G -symmetric, we also have

$$P(A) = P(g \cdot A) \quad \text{for each } g \in K \subseteq H_i \leq G.$$

So by finite additivity we must have

$$P(A) = \frac{1}{\#K} \in \mathbb{Q}.$$

Since the same reasoning applies to P' , we have $P(A) = P'(A)$. Finally, every basic set is a finite union of sets of the form $h_i^{-1}\{x\}$. By finite additivity, P and P' agree on all basic sets. $\square$

Corollary F.33.

- (a) The weak finitary coin flip symmetric measure on 2^ω agrees exactly with the standard coin flip measure (example E.9) on all cylinder sets. The same holds for the weak Cantor-symmetric measure (since this is also symmetric under finitary flips).
- (b) The weak rational-rotation symmetric measure on $[0, 1)$ agrees exactly with the standard Lebesgue measure (example E.10) on half-open intervals with rational endpoints.

Incidentally, this shared structure also lets us prove the uniqueness of the standard measures on these two spaces.

Theorem F.34. Let $G \curvearrowright \Omega$ be a limit of finite actions, and let Ω be equipped with the smallest σ -algebra that contains every basic subset. There is at most one countably additive G -symmetric real-valued probability measure on Ω .

Proof sketch. By theorem F.32, any two such measures must agree on basic sets. Since I is directed, the basic sets are closed under finite intersections: for $i \leq k$ and $j \leq k$ we have $h_i^{-1}(A) \cap h_j^{-1}(B) = h_k^{-1}(f_{ki}^{-1}(A) \cap f_{kj}^{-1}(B))$. So by Dynkin's π - λ theorem, the two measures also agree on all sets in the generated σ -algebra (Klenke 2013, Lemma 1.42). $\square$

The uniqueness parts of examples E.9 and E.10 are immediate consequences. (Note that in both the case of the coin flip space and the interval, the standard σ -algebra in definition E.8 and example E.10 agrees with the one that arises from the limit of finite actions.)

We can also get *approximate* matching between these non-Archimedean measures and their standard counterparts, even in cases where we don't have *exact* matches. "Approximate" in this case means "up to infinitesimal differences".

Definition F.35. If $\mathcal{R}$ is a p.o.c. ring embedding $\mathbb{Q}$, for $r \in \mathcal{R}$ and $s \in \mathbb{R}$, we say $r \approx s$ iff every rational interval containing s also contains r .

As we will see, we do not get even approximate matching in this sense for *every* measurable set, but we do get them for an important general class.

Definition F.36. Let $G \curvearrowright \Omega$ be a limit of finite actions. For each basic set $A \subseteq \Omega$, we let the **content** of A , denoted $\text{con } A$, be the unique rational number q such that $P(A) = q$ for all G -symmetric measures.

(Since any limit of finite actions is locally finite, we know from proposition F.23 that such G -symmetric measures exist. In fact, if $A = h_i^{-1}(B)$ then $\text{con}(A) = \frac{\#B}{\#\mathcal{X}_i}$.)

Definition F.37. Let $G \curvearrowright \Omega$ be a limit of finite actions.

(a) Let $A \subseteq \Omega$. We define the **inner content** of A to be

$$\sup\{\text{con } B \mid B \text{ basic and } B \subseteq A\}$$

and the **outer content** of A to be

$$\inf\{\text{con } B \mid A \subseteq B \text{ basic}\}.$$

When these agree, we say A is **Jordan measurable**, and call this number the **Jordan content** of A , also denoted $\text{con } A$.

(b) A **basic function** is a finite sum

$$\sum_i a_i \cdot 1_{A_i}$$

where each $a_i \in \mathbb{R}$ and each A_i is a basic set. Define

$$\text{con} \left(\sum_i a_i \cdot 1_{A_i} \right) := \sum_i a_i \text{con } A_i.$$

For any bounded real-valued function $f : \Omega \rightarrow \mathbb{R}$ the **lower Riemann integral** is

$$\sup\{\text{con } g \mid g \text{ basic and } g \leq f\}$$

and the **upper Riemann integral** is

$$\inf\{\text{con } g \mid g \text{ basic and } g \geq f\}.$$

When these agree, we say f is **Riemann integrable**, and call this number the **Riemann integral** of f , which we also denote $\text{con } f$.

When Ω is the interval, these definitions agree with the standard notions of the same names.

Theorem F.38. Let $G \curvearrowright \Omega$ be a limit of finite actions, and let $\mathcal{R}$ be a non-trivial fine G -symmetric Ω -ring.

(a) For any Jordan measurable $A \subseteq \Omega$,

$$\mathbb{P}^{\mathcal{R}}(A) \approx \text{con } A.$$

(b) For any Riemann integrable $f : \Omega \rightarrow \mathbb{R}$,

$$\mathbb{E}^{\mathcal{R}} f \approx \text{con} f.$$

Proof. Since $\mathcal{R}$ is G -symmetric, $\mathbb{P}^{\mathcal{R}}$ is a G -symmetric probability measure (proposition F.22), and so by theorem F.32 it agrees with the content on every basic set.

Part (a). Let A be Jordan measurable, and let (p, q) be any rational interval containing $\text{con} A$. Since the inner and outer contents of A agree, there are basic sets $B \subseteq A \subseteq B'$ with

$$p < \text{con} B = \mathbb{P}^{\mathcal{R}}(B); q > \text{con} B' = \mathbb{P}^{\mathcal{R}}(B').$$

By monotonicity,

$$p < \mathbb{P}^{\mathcal{R}}(B) \leq \mathbb{P}^{\mathcal{R}}(A) \leq \mathbb{P}^{\mathcal{R}}(B') < q.$$

So every rational interval containing $\text{con} A$ contains $\mathbb{P}^{\mathcal{R}}(A)$; that is, $\mathbb{P}^{\mathcal{R}}(A) \approx \text{con} A$.

Part (b) goes similarly, noting first that for a *basic* function f , $\mathbb{E}^{\mathcal{R}} f = \text{con} f$ (exactly) by linearity and the fact that $\mathbb{E}^{\mathcal{R}} 1_A = \mathbb{P}^{\mathcal{R}}(A)$.

(Note that this time, since a basic function f can have irrational coefficients, $\mathbb{E}^{\mathcal{R}} f$ lives in $\mathbb{R}^{\mathcal{R}}$, rather than $\mathcal{R} = \mathbb{Q}^{\mathcal{R}}$. This raises no problem, since the definition of $\approx$ applies to $\mathbb{R}^{\mathcal{R}}$ just as well.) $\square$

Even so, these weak measures deserve the name. In some cases where standard measures give definite answers, they do not tell us much.

Proposition F.39. Let $G \curvearrowright \Omega$ be a limit of finite actions, let $A \subseteq \Omega$ be infinite with an infinite complement, and suppose $G \cdot A = A$. Then $\mathbb{P}^G(A)$ is incomparable with every rational strictly between zero and one.

Proof. Let $\mathcal{G}$ be the weak G -symmetric filter, and let $0 < p < 1$. To show that $\mathbb{P}^G(A)$ is incomparable with p , it is enough to show that for any $F \in \text{Fin } G$ and $K \in \text{Fin } \Omega$, the set $[F, K]$ (lemma B.92) contains some $J, J' \in \text{Fin } \Omega$ such that

$$\frac{\#(A \cap J)}{\#J} < p < \frac{\#(A \cap J')}{\#J'}.$$

There are infinitely many $\langle F \rangle$ -orbits contained in A , and infinitely many $\langle F \rangle$ -orbits disjoint from A . By including, besides $\langle F \rangle \cdot K$, sufficiently many $\langle F \rangle$ -orbits outside A in J , we can ensure that $\#(A \cap J)$ is much lower than $\#J$; likewise, by including many $\langle F \rangle$ -orbits inside A in J' , we can ensure that $\#(A \cap J')$ is very nearly $\#J'$. $\square$

Thus, for example the set of all rational points in $[0, 1)$ turns out to have completely *indeterminate* measure with respect to the weak rational-rotation symmetric measure, in the sense that it is incomparable with any rational number strictly between zero and one. Similarly, the set of all coin flip sequences containing finitely many Heads has indeterminate measure with respect to the weak finitary coin flip symmetric measure. (In contrast, the weak Cantor-symmetric measure escapes this kind of case, though not the general phenomenon; I believe it still assigns indeterminate measure to some standardly-measurable sets, but their construction is a more technical matter.)

So far we have focused on constructing non-Archimedean probability measures by way of symmetries. Next we will turn to a second strategy: we can also directly construct such measures as approximations of standard real-valued, countably additive measures. In the end, we will see that these two different approaches can be combined with very satisfying results.

Lemma F.40. Let $(\Omega, \mathbb{P})$ be an atomless (countably additive) real-valued probability space. Let $f_1, \dots, f_n : \Omega \rightarrow \mathbb{R}$ be finitely many $\mathcal{L}^1$ functions. Then for any $\epsilon > 0$, for any finite set $K \subseteq \Omega$, there exists a finite set $J \supseteq K$ such that the average value of each of $f_1, \dots, f_n$ on J is within ϵ of its expectation:

$$|\mathbb{E}_J f_i - \mathbb{E} f_i| < \epsilon \quad \text{for each } i \in \{1, \dots, n\}.$$

(Compare Klenke 2013, example 5.21.)

Proof. The strong law of large numbers (see Klenke 2013, sec. 5.3) tells us that, with respect to the product measure $\mathbb{P}^\omega$ on the infinite product space Ω^ω , for each $i \in \{1, \dots, n\}$, almost every sequence of points $w_1, w_2, \dots$ in Ω satisfies

$$\lim_{m \rightarrow \infty} \frac{1}{m} \sum_{j=1}^m f_i(w_j) = \mathbb{E} f_i.$$

Atomlessness ensures that almost every such sequence consists of distinct points, and likewise points distinct from those in K . So we can choose such

a sequence that satisfies each of these n properties, and let J_m be its first m points.

We choose m to be much larger than $\#K$ and so that the average value of each f_i on J_m is close to $\mathbb{E}f_i$. In particular, we can choose m so that, for each i ,

$$\begin{aligned} \frac{\#K}{\#K + m} \cdot |\mathbb{E}_K f_i - \mathbb{E}f_i| &< \epsilon/2; \\ |\mathbb{E}_{J_m} f_i - \mathbb{E}f_i| &< \epsilon/2. \end{aligned}$$

Let $J = K \cup J_m$. Then (for each i) the average value of f_i on J is a weighted average of the average values on K and J_m , and we have:

$$\begin{aligned} |\mathbb{E}f_i - \mathbb{E}f_i| &= \left| \frac{\#K}{\#K + m} \mathbb{E}_K f_i + \frac{m}{\#K + m} \mathbb{E}_{J_m} f_i - \mathbb{E}f_i \right| \\ &\leq \frac{\#K}{\#K + m} |\mathbb{E}_K f_i - \mathbb{E}f_i| + \frac{m}{\#K + m} |\mathbb{E}_{J_m} f_i - \mathbb{E}f_i| \\ &< \frac{\#K}{\#K + m} |\mathbb{E}_K f_i - \mathbb{E}f_i| + \frac{m}{\#K + m} \cdot \epsilon/2 \\ &< \epsilon. \end{aligned}$$

□

Theorem F.41. For any non-atomic standard probability space $(\Omega, \mathbb{P})$, there exists a filter $\mathcal{G}$ with the following properties:

- (a) The uniform $\mathbb{Q}^{\mathcal{G}}$ -valued probability measure $\mathbb{P}^{\mathcal{G}}$ on Ω agrees with $\mathbb{P}$ to within an infinitesimal for all measurable sets:

$$\mathbb{P}^{\mathcal{G}}(A) \approx \mathbb{P}(A) \quad \text{for each } A \stackrel{\text{m}}{\subseteq} \Omega.$$

- (b) The corresponding expectation operator $\mathbb{E}^{\mathcal{G}}$ agrees with the standard expectation $\mathbb{E}$ to within an infinitesimal for all absolutely integrable ($\mathcal{L}^1$) functions:

$$\mathbb{E}^{\mathcal{G}}f \approx \mathbb{E}f \quad \text{for each } f \in \mathcal{L}^1(\Omega, \mathbb{P}).$$

Proof. We first prove part (b). Let $\mathcal{G}$ be the smallest fine filter containing all sets of the form

$$[f, \epsilon] := \{K \in \text{Fin } \Omega \mid |\mathbb{E}_K f - \mathbb{E}f| < \epsilon\} \quad \text{for each } f \in \mathcal{L}^1(\Omega, \mathbb{P}) \text{ and } \epsilon > 0.$$

Since $\mathcal{G}$ is fine, it also contains all sets of the form

$$\{K \in \text{Fin } \Omega \mid w \in K\} \quad \text{for each } w \in \Omega.$$

To show that this filter is non-trivial, it suffices to show that each finite intersection of sets of these forms is non-empty. This immediately follows from lemma F.40. By construction, for any $f \in \mathcal{L}^1(\Omega, \mathbb{P})$ and any $\epsilon > 0$,

$$|\mathbb{E}_K f - \mathbb{E}f| < \epsilon \quad \text{for } \mathcal{G}\text{-almost every } K$$

and thus $\mathbb{E}^{\mathcal{G}}f \approx \mathbb{E}f$.

For part (a), we then have

$$\mathbb{P}^{\mathcal{G}}(A) = \mathbb{E}^{\mathcal{G}}1_A \approx \mathbb{E}1_A = \mathbb{P}(A) \quad \text{for each } A \stackrel{\text{m}}{\subseteq} \Omega. \quad \square$$

Thus it is possible to construct a non-Archimedean probability measure that approximates standard expectations to within an infinitesimal. But once we have done this, we have automatically achieved even more. For example, Colyvan (2008) proposes extending the standard notion of expectation using “relative expectations” (see also Colyvan and Hájek 2016) — even when two functions are not both integrable, they may have an integrable *difference*. In this case we can compare them based on this difference.

Corollary F.42. Let $f, g : \Omega \rightarrow \mathbb{R}$ be two measurable functions such that the difference $f - g$ is absolutely integrable. Then for the filter $\mathcal{G}$ constructed in theorem F.41, we have

$$\text{if } \mathbb{E}(f - g) > 0 \quad \text{then} \quad \mathbb{E}^{\mathcal{G}}f > \mathbb{E}^{\mathcal{G}}g.$$

Finally, we can combine these two different approaches, using *both* symmetries *and* countably additive measures in order to construct some quite powerful symmetric measures.

Definition F.43. Let $G \curvearrowright \Omega$ be a limit of finite actions, and suppose that there exists a G -symmetric countably additive probability measure $\mathbb{P}$ on Ω . (By theorem F.34, if such a measure exists, it is unique.) Let the **tight G -symmetric filter** be the smallest fine G -symmetric filter that, additionally, contains each set

$$[f, \epsilon] \text{ for } f \in \mathcal{L}^1(\Omega, \mathbb{P}) \text{ and } \epsilon > 0$$

(as defined in the proof of theorem F.41).

Recall that in this case G is locally finite. Also, if Ω is infinite, it can be checked that the measure $\mathbb{P}$ must be atomless.

Lemma F.44. The tight G -symmetric filter is non-trivial.

Proof. The proof goes very similarly to lemma F.40 and theorem F.41. The key thing that needs to be established is that, for $f_1, \dots, f_n \in \mathcal{L}^1(\Omega, \mathbb{P})$, $\epsilon > 0$, $K \in \text{Fin } \Omega$, and (the new part) $F \in \text{Fin } G$, there exists a finite set $J \supseteq K$ such that

$$|\mathbb{E}_J f_i - \mathbb{E} f_i| < \epsilon \quad \text{for each } i \in \{1, \dots, n\}$$

and (new again) $\langle F \rangle \cdot J = J$.

Let $H = \langle F \rangle$, which is finite. For any f , let

$$A_H f(w) := \frac{1}{\#H} \sum_{h \in H} f(h \cdot w)$$

be the average value of f on the orbit of w . (Note that since each of the actions in the limiting family acts freely, it follows that H acts freely on Ω as well; thus $\#(H \cdot w) = \#H$.) Because $\mathbb{P}$ is G -symmetric, we have $\mathbb{E} f = \mathbb{E}(f \circ h)$ for each $h \in H \leq G$, and so $\mathbb{E}(A_H f) = \mathbb{E} f$ as well.

Let K' contain exactly one element of K from each H -orbit. Now lemma F.40 tells us that there exists a finite set $J \supseteq K'$ such that

$$|\mathbb{E}_J A_H f_i - \mathbb{E} A_H f_i| < \epsilon \quad \text{for each } i.$$

Furthermore, by slightly tweaking the proof we can ensure that no two points in J are in the same H -orbit. (Since each H -orbit is finite and thus has $\mathbb{P}$ -measure zero, since $\mathbb{P}$ is atomless, almost every sequence never visits any H -orbit twice.) Thus $\mathbb{E}_{H \cdot J} f_i = \mathbb{E}_J A_H f_i$. Thus we have

$$|\mathbb{E}_{H \cdot J} f_i - \mathbb{E} f_i| < \epsilon \quad \text{for each } i.$$

So $H \cdot J \supseteq H \cdot K' \supseteq K$ is the set we sought. $\square$

Theorem F.45. Let $G \curvearrowright \Omega$ be a limit of finite actions that admits a G -symmetric countably additive real-valued probability measure $\mathbb{P}$. Let $\mathcal{G}$

be the tight G -symmetric filter, let $\mathcal{R} := \mathbb{Q}^G$, and let $\mathbb{P}^{\mathcal{R}}$ be the corresponding uniform $\mathcal{R}$ -measure.

- (a) $\mathbb{P}^{\mathcal{R}}$ is a regular uniform G -symmetric $\mathcal{R}$ -valued probability measure;
- (b) $\mathbb{P}^{\mathcal{R}}(A) = \mathbb{P}(A)$ exactly for all basic subsets of Ω ;
- (c) $\mathbb{P}^{\mathcal{R}}(A) \approx \mathbb{P}(A)$ to within an infinitesimal for all measurable subsets $A \subseteq \Omega$;
- (d) $\mathbb{E}^{\mathcal{R}} f \approx \int f d\mathbb{P}$ to within an infinitesimal for all absolutely integrable $f \in \mathcal{L}^1(\Omega, \mathbb{P})$;
- (e) For any $H \curvearrowright \Omega$ acting by measurable transformations, if $\mathbb{P}$ is H -invariant, then $\mathbb{P}^{\mathcal{R}}$ is H -invariant to within an infinitesimal, in the following sense: for measurable $A, B \subseteq \Omega$ and $g \in H$, if $\mathbb{P}^{\mathcal{R}}(A) \geq \mathbb{P}^{\mathcal{R}}(B)$, then $\mathbb{P}^{\mathcal{R}}(g \cdot A) + \epsilon \geq \mathbb{P}^{\mathcal{R}}(g \cdot B)$ for every rational $\epsilon > 0$.

Proof sketch. The proof is basically straightforward, after what we have already done. Parts (a) and (b) go the same way as with weak symmetric filters. Parts (c) and (d) go the same way as with the approximation filter.

Part (e) follows from (c). Suppose $\mathbb{P}^{\mathcal{R}}(A) \geq \mathbb{P}^{\mathcal{R}}(B)$. Let $p \geq \mathbb{P}(A)$ be a rational. Since $\mathbb{P}^{\mathcal{R}}(A) \approx \mathbb{P}(A)$, we also have $p \geq \mathbb{P}^{\mathcal{R}}(A) \geq \mathbb{P}^{\mathcal{R}}(B)$, and since $\mathbb{P}^{\mathcal{R}}(B) \approx \mathbb{P}(B)$, we then also have $p \geq \mathbb{P}(B)$. So $\mathbb{P}(A) \geq \mathbb{P}(B)$. Since $\mathbb{P}$ is H -invariant, for any $g \in H$, $\mathbb{P}(g \cdot A) \geq \mathbb{P}(g \cdot B)$.

Now let $\epsilon > 0$ be rational. We can choose rationals p and q such that

$$\mathbb{P}(g \cdot A) > p > \mathbb{P}(g \cdot A) - \epsilon/2 \quad \text{and} \quad \mathbb{P}(g \cdot B) + \epsilon/2 > q > \mathbb{P}(g \cdot B),$$

which implies that $p > q - \epsilon$. Since $\mathbb{P}^{\mathcal{R}}(g \cdot A) \approx \mathbb{P}(g \cdot A)$, we also have $\mathbb{P}^{\mathcal{R}}(g \cdot A) > p$, and likewise $q > \mathbb{P}^{\mathcal{R}}(g \cdot B)$. So $\mathbb{P}^{\mathcal{R}}(g \cdot A) + \epsilon > \mathbb{P}^{\mathcal{R}}(g \cdot B)$. $\square$

F.5 Distributions

In the context of classical measure theory, a random variable $X : \Omega \rightarrow \mathcal{X}$ on a probability space (Ω, P) has a *distribution* $X * P$, which is a probability measure on its space of values $\mathcal{X}$. This represents the probability of the variable's value landing in a particular part of the value space $\mathcal{X}$. This can be very useful, since typically a variable's distribution encodes all the information about it that is relevant to value, while leaving behind extraneous details

about the state space. (This corresponds to moving from the Savage framework of random variables to the von Neumann–Morgenstern framework of probability distributions.)

We can do something similar in the context of non-Archimedean probabilities, but there are some new subtleties. The non-standard measures we have constructed on a state space Ω take values in a fine Ω -ring $\mathcal{R}$. As the name suggests, this ring is specially tailored to the particular state space Ω . We can “push forward” a non-Archimedean probability measure from Ω to a space $\mathcal{X}$, but the resulting measure still takes values in the same ring $\mathcal{R}$, which is “attached” to the original space Ω . Thus certain properties — such as the non-Archimedean continuity condition (from Benci, Horsten, and Wenmackers 2018) — do not immediately make sense. Furthermore, the new measure does not directly encode all of the information we might want, such as how to calculate an expectation of a variable on the new space $\mathcal{X}$.

This motivates a more general idea. The key trick is that, besides just tracking the $\mathcal{R}$ -valued probabilities of each set, we will also keep track of some information about how we got there, in the form of a family of finite-support probability measures. The $\mathcal{R}$ -limits we have been focusing on tell us the limit of a family of *numbers* (indexed by finite sets). But more generally, we can also make sense of a limit of a family of *probability distributions* — using exactly the same filter limit machinery we have been using all along. If we have a filter $\mathcal{G}$ on $\text{Fin } \Omega$, and a family of finite-support probability distributions $(P_K)_{K \in \text{Fin } \Omega}$, then the limit

$$P = \lim_{\mathcal{G}} P_K$$

is (as usual) just the equivalence class of this family, under the equivalence relation of agreeing $\mathcal{G}$ -almost everywhere. This new object P is a kind of “non-standard distribution” on $\mathcal{X}$. It carries all the information we need to assign $\mathcal{R}$ -valued probabilities, but it also keeps track of how these different probabilities are coordinated with one another, in ways that turn out to be useful.

Once we have broken the connection between the ring $\mathcal{R}$ and the space $\mathcal{X}$ the $\mathcal{R}$ -measure lives on, we can also weaken our assumptions about the structure of $\mathcal{R}$: it just needs to be a ring equipped with an appropriate limit operator — that is, a *limiting ring* (definition F.6). (Recall that a fine Ω -ring

is just a $(\text{Fin } \Omega)$ -limiting ring that is fine.) We will state our definitions in this section at this appropriate level of generality.

Definition F.46. Let $\mathcal{R}$ be an I -limiting ring for some non-empty indexing set I . As before, let $\mathcal{F}(\mathcal{R})$ be the unique filter on I such that $\mathcal{R} \cong \mathbb{Q}^{\mathcal{F}(\mathcal{R})}$ (proposition F.7). Let

$$\Delta^{\mathcal{R}}\mathcal{X} := (\Delta_0^{\mathbb{Q}}\mathcal{X})^{\mathcal{F}(\mathcal{R})}$$

be the corresponding filter power of the set of all rational-valued distributions on $\mathcal{X}$ with finite support. The elements of $\Delta^{\mathcal{R}}\mathcal{X}$ are called **$\mathcal{R}$ -distributions**.

Each $\mathcal{R}$ -distribution $P \in \Delta^{\mathcal{R}}\mathcal{X}$ is the filter limit of some family of rational finite-support probability functions:

$$P = \lim_j^{\mathcal{F}(\mathcal{R})} P_j \quad \text{where } P_j \in \Delta_0^{\mathbb{Q}}\mathcal{X} \text{ for each } j \in I.$$

This family P_j is unique up to equality for $\mathcal{F}(\mathcal{R})$ -almost every j .

Furthermore, each $\mathcal{R}$ -distribution $P \in \Delta^{\mathcal{R}}\mathcal{X}$ gives rise to a corresponding $\mathcal{R}$ -valued probability measure $[P]$ on $\mathcal{X}$, defined by

$$[P](A) = \lim_j^{\mathcal{R}} P_j(A) \in \mathcal{R}.$$

(When the ambiguity is harmless, we will omit the brackets and write $P(A)$.)

As with real-valued functions, we will write $\lim^{\mathcal{R}}$ for the filter limit $\lim^{\mathcal{F}(\mathcal{R})}$ of finite-support distributions, and we will omit the superscript entirely when there is no risk of confusion.

In section F.3 we defined a notion of *uniform $\mathcal{R}$ -probability* on Ω , denoted $\mathbb{P}^{\mathcal{R}}$. This corresponds to a special case of $\mathcal{R}$ -distributions. When $\mathcal{R}$ is a fine Ω -ring, we can define the **uniform $\mathcal{R}$ -distribution** to be the filter limit of uniform finite distributions:

$$\mathbb{P}_{\mathcal{R}} := \lim_K \mathbb{P}_K$$

where $\mathbb{P}_K$ is the uniform distribution on K for each $K \in \text{Fin } \Omega$. Then the uniform $\mathcal{R}$ -valued measure $\mathbb{P}^{\mathcal{R}}$ is the measure induced by the distribution $[\mathbb{P}_{\mathcal{R}}]$. So we can see $\mathcal{R}$ -distributions as a generalization of the uniform $\mathcal{R}$ -measure.

One straightforward contrast between the general case of an $\mathcal{R}$ -distribution $P = \lim_j P_j$ and the special case of $\mathbb{P}_{\mathcal{R}}$ is that the finite-support measures P_j need not be uniform. A more important contrast is already visible in the notation. In the case of the uniform distribution, $\mathbb{P}_K$ is supported by the finite set K ; for a general distribution, the index j and the distribution P_j it labels have nothing to do with one another (even if j happens to be a finite set, as in the case where the indexing set is $\text{Fin } \Omega$). This separation of the two roles — the domain of indices generating $\mathcal{R}$'s algebraic structure, and the domain of states where the probability measure lives — is key to the power of $\mathcal{R}$ -distributions, allowing us to carry probabilities around with us as we travel from space to space.

One reason this generalization is useful is that it permits many basic operations and constructions involving probability measures that were not generally available for uniform $\mathcal{R}$ -measures.

Proposition F.47. For any set $\mathcal{X}$, the set $\Delta^{\mathcal{R}}\mathcal{X}$ of $\mathcal{R}$ -distributions is convex.

Proof. For $p \in [0, 1]$, $P = \lim_K P_K$, and $Q = \lim_K Q_K$ in $\Delta^{\mathcal{R}}\mathcal{X}$, we have

$$\begin{aligned} p \cdot P + (1 - p) \cdot Q &= p \cdot (\lim_K P_K) + (1 - p) \cdot (\lim_K Q_K) \\ &= \lim_K p \cdot P_K + (1 - p) \cdot Q_K. \quad \square \end{aligned}$$

Definition F.48. For any sets $\mathcal{X}$ and $\mathcal{Y}$, any function $f : \mathcal{X} \rightarrow \mathcal{Y}$, and any $P \in \Delta^{\mathcal{R}}\mathcal{X}$, define the **pushforward** $f_* : \Delta^{\mathcal{R}}\mathcal{X} \rightarrow \Delta^{\mathcal{R}}\mathcal{Y}$ by

$$f * (\lim_j P_j) = \lim_j f * P_j.$$

(We generally write $f * P$ rather than $f_* P$.)

Proposition F.49. This definition agrees with the standard definition of the pushforward *measure*: for any sets $\mathcal{X}$ and $\mathcal{Y}$, any function $f : \mathcal{X} \rightarrow \mathcal{Y}$, and any $P \in \Delta^{\mathcal{R}}\mathcal{X}$,

$$[f * P](A) = [P](f^{-1}(A)) \quad \text{for each } A \subseteq \mathcal{Y}.$$

It also has the standard property:

$$(f \circ g) * P = f * (g * P).$$

Proof. Straightforward using the properties of $\mathcal{R}$ -limits. $\square$

Recall that we call a probability measure *divisible* if whenever $P(A)$ is positive it has a positive multiplicative inverse. We likewise say an $\mathcal{R}$ -distribution P is divisible whenever its measure $[P]$ is. Note that the uniform distribution is divisible, and divisibility is preserved by pushforwards and real convex combinations. The upshot is that conditional probability will be well-behaved for the $\mathcal{R}$ -distributions of primary interest.

In section F.3 we defined a notion of expectation corresponding to a uniform $\mathcal{R}$ -measure. We can generalize this to arbitrary $\mathcal{R}$ -distributions.

Definition F.50. For $P \in \Delta^{\mathcal{R}}\mathcal{X}$ and $f : \mathcal{X} \rightarrow \mathcal{U}$, where $\mathcal{U}$ is a vector space, define

$$\mathbb{E}_P f := \lim_j \mathbb{E}_{P_j} f,$$

where $P = \lim_j P_j$. On the right-hand side, $\mathbb{E}_{P_j} f$ is the standard expectation of f with respect to the finite-support distribution P_j :

$$\mathbb{E}_{P_j} f = \sum_{x \in \mathcal{X}} P_j(x) \cdot f(x).$$

It is also convenient to use an alternative *integral* notation. We can rewrite the finitary expectation in integral notation:

$$\mathbb{E}_{P_j} f = \int f dP_j.$$

Likewise, it is convenient to define:

$$\int f dP := \mathbb{E}_P f = \lim_j \int f dP_j.$$

This notation is convenient both because it emphasizes certain analogies with standard measure theory, and because it emphasizes the function f and the distribution P more even-handedly.

As usual, we let $\int_E f dP := \int 1_E \cdot f dP$.

Here are some important properties of the integral.

Proposition F.51. Let $P \in \Delta^{\mathcal{R}}\mathcal{X}$, and let $\mathcal{U}$ be a preordered vector space. Define

$$f \geq_P g \quad \text{iff} \quad P[f \geq g] = 1 \quad \text{for each } f, g : \mathcal{X} \rightarrow \mathcal{U}.$$

This makes the space of functions $\mathcal{X} \rightarrow \mathcal{U}$ a preordered vector space as well.

- (a) $\int dP$ is a strictly positive linear operator from $\mathcal{X} \rightarrow \mathcal{U}$ to $\mathcal{U}^{\mathcal{R}}$.
- (b) $\int 1_E dP = P(E)$ for any $E \subseteq \mathcal{X}$.
- (c) If P has finite support and takes only standard rational probability values, we can identify P with an element of $\Delta_0^{\mathbb{Q}} \mathcal{X}$. In this case, $\int dP$ agrees with the standard finite vector expectation (definition D.11).

Proof. For each j , $\int dP_j$ is a linear operator which is strictly positive with respect to the order on functions $\geq_{P_j}$ (proposition D.12). Linearity and positivity are both straightforwardly inherited by $\int dP$ using the properties of $\lim^{\mathcal{R}}$.

For strict positivity, it is enough to show that if $P[f \geq 0] = 1$ and $\int f dP \leq 0$, then $P[f \sim 0] = 1$. And indeed, if $P[f \geq 0] = 1$ and $\int f dP \leq 0$, then for almost every j , $P_j[f \geq 0] = 1$ and $\int f dP_j \leq 0$, which together imply that $P_j[f \sim 0] = 1$ (using the fact that $\int dP_j$ is strictly positive). So $P[f \sim 0] = \lim_j P_j[f \sim 0] = 1$.

Part (b) is also inherited from the integrals $\int dP_j$, and then part (c) is straightforward using linearity. $\square$

Recall that $\delta : \mathcal{X} \rightarrow \mathbb{Q}^{\mathcal{X}}$ is the function that takes each $x \in \mathcal{X}$ to its indicator function $1_{\{x\}}$.

Lemma F.52. For any $P \in \Delta^{\mathcal{R}} \mathcal{X}$,

$$\int \delta dP = P \in (\Delta_0^{\mathbb{Q}}(\mathcal{X}))^{\mathcal{R}} \subset (\mathbb{Q}^{\mathcal{X}})^{\mathcal{R}}.$$

Proof.

$$\int \delta dP = \lim_j \int \delta dP_j = \lim_j P_j = P. \quad \square$$

Lemma F.53. For $f : \mathcal{X} \rightarrow \mathcal{Y}$, $g : \mathcal{Y} \rightarrow \mathcal{U}$, and $P \in \Delta^{\mathcal{R}} \mathcal{X}$,

$$\int g d(f * P) = \int (g \circ f) dP.$$

How tight is the relationship between $\mathcal{R}$ -distributions and non-Archimedean probability measures? Each distribution immediately gives rise to a corresponding $\mathcal{R}$ -valued probability measure. Does the converse hold? And do distributions carry any more information than their corresponding measures?

Question F.54. Consider the map $[\cdot]$ that takes each $\mathcal{R}$ -distribution P to its $\mathcal{R}$ -valued measure $[P]$.

- (a) Is $[\cdot]$ surjective? That is, given an $\mathcal{R}$ -valued probability measure Q on $\mathcal{X}$, does there exist a distribution $P \in \Delta^{\mathcal{R}}\mathcal{X}$ such that $[P] = Q$?
- (b) Is $[\cdot]$ injective? That is, if $P, Q \in \Delta^{\mathcal{R}}\mathcal{X}$ are distributions that give rise to the same measure $[P] = [Q]$, does it follow that $P = Q$?

I do not know the answer to either question in general, but I have a partial positive answer to part (b): it at least holds in certain important cases, where the distributions arise from random variables on especially well-behaved probability spaces.

Definition F.55. An $\mathcal{R}$ -distribution $P \in \Delta^{\mathcal{R}}\Omega$ is **nice** iff, for any set $\mathcal{X}$ and functions $f, g : \Omega \rightarrow \mathcal{X}$,

$$\text{if } [f * P] = [g * P] \text{ then } f * P = g * P.$$

That is, if two random variables are *stochastically equivalent* on P , then they have the same $\mathcal{R}$ -distribution with respect to P .

Theorem F.56. Let $G \curvearrowright \Omega$ be a locally finite action of a countable group. Let $\mathcal{R} = \mathcal{R}^G$ be the weak G -symmetric ring for this group action. The uniform $\mathcal{R}$ -distribution $\mathbb{P}_G$ is nice.

Note, in particular, that the group of finitary flips and the group of rational rotations are each countable; so theorem F.56 tells us that the weak finitary coin flip symmetric measure and the weak rational-rotation symmetric measure are each nice. It follows, in particular, that if f and g are stochastically equivalent random variables on either of these two spaces, then they have the same expectation. At least for these well-behaved measures, this vindicates the *Stochastic Equivalence* principle for $\mathcal{R}$ -expectational theories of value (see section 1.3.5 and chapter 5): the value of a prospect is determined by its probabilities of outcomes.

But the question of how far we can take this remains open. For instance, the full Cantor group 2^ω is uncountable, and it is an open question whether the weak Cantor-symmetric probability measure is nice. I believe that the proof below can be extended to the *tight* measures arising from countable symmetry groups (definition F.43), but I will not undertake this here.

Even for countable symmetry groups, the proof is surprisingly tricky! Proving theorem F.56 is the goal of the rest of this section.

Given $f, g : \Omega \rightarrow \mathcal{X}$, define for each $K \in \text{Fin } \Omega$:

$$\phi_K := \#K \cdot ((f * \mathbb{P}_K) - (g * \mathbb{P}_K))$$

where $\mathbb{P}_K$ is the uniform finite distribution on K . Equivalently,

$$\phi_K(x) = \#\{w \in K \mid f(w) = x\} - \#\{w \in K \mid g(w) = x\}.$$

Each ϕ_K is a finite-support signed measure on $\mathcal{X}$. Moreover, the family $(\phi_K)_{K \in \text{Fin } \Omega}$ is *additive* in the sense that

$$\phi_{I \cup J} = \phi_I + \phi_J \quad \text{for } I, J \in \text{Fin } \Omega \text{ such that } I \cap J = \emptyset.$$

If f and g are stochastically equivalent, this means that

$$[f * \mathbb{P}_G] - [g * \mathbb{P}_G] = 0.$$

In other words, for each $E \subseteq \mathcal{X}$,

$$[f * \mathbb{P}_G](E) - [g * \mathbb{P}_G](E) = \lim_K \frac{\phi_K(E)}{\#K} = 0.$$

Since $\#K > 0$ for almost every K , this is equivalent to $\lim_K \phi_K(E) = 0$. Thus theorem F.56 reduces to the following lemma.

Lemma F.57. Let $\mathcal{G}$ be the weak symmetric filter on $\text{Fin } \Omega$ for a locally finite action $G \curvearrowright \Omega$, where the group G is countable. Let $\phi_K : \mathcal{X} \rightarrow \mathbb{Z}$ for $K \in \text{Fin } \Omega$ be an additive family of finite-support signed measures. If, for each set $E \subseteq \mathcal{X}$,

$$\phi_K(E) = 0 \quad \text{for } \mathcal{G}\text{-almost every } K,$$

then $\phi_K = 0$ for $\mathcal{G}$ -almost all K .

Proof. We will reduce this to the simple case where there is a finite set F such that, for $\mathcal{G}$ -almost all K , ϕ_K is supported by F . Suppose otherwise for contradiction.

Since G is countable, we can choose an increasing sequence $F_1 \subseteq F_2 \subseteq \dots$ of finite subsets of G such that $\bigcup_n F_n = G$. We will begin by showing that there exists a sequence of pairwise disjoint finite sets $K_1, K_2, \dots \in \text{Fin } \Omega$, and a sequence $x_1, x_2, \dots \in \mathcal{X}$, with the following three properties, for each n :

$$\begin{aligned} F_n \cdot K_n &= K_n \\ \phi_{K_k}(x_n) &= 0 \quad \text{for each } k < n \\ \phi_{K_n}\{x_1, \dots, x_n\} &\neq 0. \end{aligned}$$

We prove this by induction. Let $K_1, \dots, K_{n-1} \in \text{Fin } \Omega$ and $x_1, \dots, x_{n-1} \in \mathcal{X}$. We will show that some K_n and x_n extends the sequence so that the three properties hold at n . Let $E = \{x_1, \dots, x_{n-1}\}$. By hypothesis, $\lim_K^{\mathcal{G}} \phi_K(E) = 0$. So, by the construction of the symmetric filter, there exists $I \in \text{Fin } \Omega$ and $F \in \text{Fin } G$ such that for all $J \supseteq I$, if $F \cdot J = J$, then $\phi_J(E) = 0$. We can suppose the following without loss of generality:

$$I \supseteq \bigcup_{k < n} K_k \quad F \supseteq F_n \quad F \cdot I = I.$$

Let

$$S = \text{supp } \phi_I \cup \bigcup_{k < n} \text{supp } \phi_{K_k}.$$

This is a finite set, so by hypothesis, it is not the case that for $\mathcal{G}$ -almost all J , ϕ_J is supported by S . In particular, then, there must be some $J \supseteq I$ such that $F \cdot J = J$ and $\phi_J(x) \neq 0$ for some $x \notin S$. Thus $\phi_I(x) = 0$ and $\phi_{K_k}(x) = 0$ for each $k < n$.

Let $K = J \setminus I$. Since $F \cdot J = J$ and $F \cdot I = I$, likewise $F \cdot K = K$, and so $F_n \cdot K = K$ as well. Finally, since $\phi_I(E) = \phi_J(E) = 0$, by additivity,

$$\phi_K(E \cup \{x\}) = \phi_J(E \cup \{x\}) - \phi_I(E \cup \{x\}) = \phi_J(x) - \phi_I(x) \neq 0.$$

Thus there exist $K_n = K$ and $x_n = x$ that allow us to extend both of our two sequences while preserving the three properties. This completes the induction.

Now that we have these sequences, let E be the infinite set $\{x_1, x_2, \dots\}$. Again, by hypothesis $\phi_K(E) = 0$ for $\mathcal{G}$ -almost all K . That is, for some $I \in \text{Fin } \Omega$

and $F \in \text{Fin } G$, for every $J \supseteq I$, if $F \cdot J = J$, then $\phi_J(E) = 0$. Again, without loss of generality suppose $F \cdot I = I$. So $\phi_I(E) = 0$. Since the sets $K_1, K_2, \dots$ are pairwise disjoint and I is finite, there must be some n such that $F_n \supseteq F$ and $I \cap K_n = \emptyset$. Since $F_n \cdot K_n = K_n$ by construction, the first fact implies $F \cdot K_n = K_n$. By construction, $\phi_{K_n}\{x_1, \dots, x_n\} \neq 0$ and $\phi_{K_n}(x_k) = 0$ for $k > n$. Thus

$$\phi_{K_n}(E) \neq 0.$$

Let $J = I \cup K_n$. Then $J \supseteq I$ and $F \cdot J = J$ as well. But by additivity,

$$\phi_J(E) = \phi_I(E) + \phi_{K_n}(E) \neq 0.$$

Contradiction.

That reduces the general problem to the finite case: there is some finite set $F \subseteq \mathcal{X}$ that supports ϕ_K for G -almost all K . By hypothesis, $\lim_K^G \phi_K(x) = 0$ for each $x \in F$. Thus, using finite additivity for limits,

$$\lim_K^G \sum_{x \in F} |\phi_K(x)| = \sum_{x \in F} \lim_K^G |\phi_K(x)| = 0.$$

But $\sum_{x \in F} |\phi_K(x)| = 0$ implies $\phi_K = 0$. So in fact $\phi_K = 0$ for almost all K . $\square$

F.6 Stochastic dominance revisited

We are now in a position to return to finish our story about *stochastic dominance* relations. We introduced weak and strong stochastic dominance relations in section D.3 via several different equivalent characterizations. In section E.4 we found that many of these equivalences break down in the setting of standard probability theory, when the ordering of the value space is sufficiently rich. Now, in the setting of general $\mathcal{R}$ -distributions, we can finally restore peace to the kingdom. As we will discuss, we don't get quite *all* of the equivalent characterizations that hold in the finite case, but we get many of them.

Theorem F.58. Let $\mathcal{R}$ be a non-trivial I -limiting ring, let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta^{\mathcal{R}}\mathcal{X}$. The following are equivalent:

- (a) **Realization.** There exist a set $\mathcal{Y}$, a distribution $R \in \Delta^{\mathcal{R}}\mathcal{Y}$, and functions $f, g : \mathcal{Y} \rightarrow \mathcal{X}$ such that $P = f * R$, $Q = g * R$, and $f \geq g$ pointwise.

- (b) **Realization 2.** There exist a set $\mathcal{Y}$, a distribution $R \in \Delta^{\mathcal{R}}\mathcal{Y}$, and functions $f, g : \mathcal{Y} \rightarrow \mathcal{X}$ such that $P = f * R$, $Q = g * R$, and $f \geq_R g$.
- (c) **Coupling.** There exists $R \in \Delta^{\mathcal{R}}(\mathcal{X} \times \mathcal{X})$ such that the marginals of R are $P = \pi_1 * R$ and $Q = \pi_2 * R$, and R is supported on the set $\{(x, y) : x \geq y\}$.
- (d) **Expected vector utility.** For any preordered vector space $\mathcal{U}$ and any monotone function $u : \mathcal{X} \rightarrow \mathcal{U}$,

$$\int u \, dP \geq \int u \, dQ.$$

- (e) **Lifted cumulative dominance.** If $\Delta_0^{\mathbb{Q}}\mathcal{X}$ is ordered by stochastic dominance (equivalently, cumulative dominance (theorem D.30)), then $P \geq Q$ with respect to the inherited order on $(\Delta_0\mathcal{X})^{\mathcal{R}}$. That is, if

$$P = \lim_j P_j \quad \text{and} \quad Q = \lim_j Q_j,$$

then for almost every j , P_j stochastically dominates Q_j .

In this case, we say P **weakly stochastically dominates** Q , written $P \geq_s Q$.

Proof. It suffices to prove:

$$(c) \Rightarrow (b) \Rightarrow (a) \Rightarrow (d) \Rightarrow (e) \Rightarrow (c).$$

(c) $\Rightarrow$ (b). Let $\mathcal{Y} = \mathcal{X} \times \mathcal{X}$, $f = \pi_1$, $g = \pi_2$.

(b) $\Rightarrow$ (a). Let $\mathcal{Y}' = \{y \in \mathcal{Y} \mid f(y) \geq g(y)\}$. Almost every R_j is supported by $\mathcal{Y}'$, so R restricts to a distribution on $\mathcal{Y}'$.

(a) $\Rightarrow$ (d). If u is monotone, then $u \circ f \geq u \circ g$ pointwise, and

$$\begin{aligned} \int u \, dP &= \int u \, d(f * R) \\ &= \int (u \circ f) \, dR \\ &\geq \int (u \circ g) \, dR \\ &= \int u \, d(g * R) = \int u \, dQ. \end{aligned}$$

(d) $\Rightarrow$ (e). Note that $\delta : \mathcal{X} \rightarrow \text{Fin}(\mathcal{X}, \mathbb{Q})$ is a vector-valued function which is monotone with respect to the cumulative dominance order (extended from

$\Delta_0^{\mathbb{Q}}\mathcal{X}$ to $\text{Fin}(\mathcal{X}, \mathbb{Q})$ in the obvious way). Thus

$$P = \int \delta \, dP \geq \int \delta \, dQ = Q.$$

(e) $\Rightarrow$ (c). Let $P = \lim_j P_j$ and $Q = \lim_j Q_j$. For almost every j , P_j stochastically dominates Q_j , and so by theorem D.30 there is a coupling R_j of P_j and Q_j such that

$$R_j\{(x, y) \in \mathcal{X} \times \mathcal{X} : x \geq y\} = 1.$$

Let $R = \lim_j R_j$. Then

$$\pi_1 * R = \lim_j \pi_1 * R_j = \lim_j P_j = P$$

and likewise $\pi_2 * R = Q$. So R is a coupling of P and Q , and

$$R\{(x, y) : x \geq y\} = \lim_j R_j\{(x, y) : x \geq y\} = 1.$$

□

Proposition F.59. Weak stochastic dominance is a preorder on $\Delta^{\mathcal{R}}\mathcal{X}$. If $\mathcal{X}$ is partially ordered, then $\geq_s$ is a partial order on $\Delta^{\mathcal{R}}\mathcal{X}$.

Proof. Immediate from the lifted cumulative dominance characterization, together with the corresponding facts about simple distributions (theorem D.30). □

Two items that are missing from theorem F.58, compared to the finitary version theorem D.30, are *cumulative dominance* and *expected real utility*. In their place, we have two new conditions: *expected vector utility*, and *lifted cumulative dominance*. Some of the implications are straightforward enough.

Proposition F.60. Let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta^{\mathcal{R}}\mathcal{X}$ be $\mathcal{R}$ -distributions. If $P \geq_s Q$, then

- (a) *Expected real utility.* $\int u \, dP \geq \int u \, dQ$ for every monotone real-valued function $u : \mathcal{X} \rightarrow \mathbb{R}$;
- (b) *Cumulative dominance.* $P(U) \geq Q(U)$ for all upward-closed subsets $U \subseteq \mathcal{X}$.

Proof. Condition (a) immediately follows from the expected vector utility condition (since $\mathbb{R}$ is an ordered vector space). Condition (b) immediately follows from condition (a), since if U is upward-closed, then $1_U : \mathcal{X} \rightarrow \mathbb{R}$ is monotone. $\square$

On the other hand, cumulative dominance does not imply either stochastic dominance or the expected real utility condition.

Proposition F.61. There exist a fine $\mathbb{N}$ -ring $\mathcal{R}$ and $\mathcal{R}$ -distributions $P, Q \in \Delta^{\mathcal{R}}\mathbb{N}$ such that

- (a) P cumulatively dominates Q , but
- (b) P does not stochastically dominate Q , and
- (c) For some monotone $u : \mathbb{N} \rightarrow \mathbb{R}$, $\int u dP \not\geq \int u dQ$.

Proof. Let $\mathcal{R}$ be the ring corresponding to the weakest fine $\mathbb{N}$ -filter $\mathcal{F}_0$ (section B.6). Define two functions $f, g : \mathbb{N} \rightarrow \mathbb{N}$:

$$f(n) = n$$

$$g(n) = \begin{cases} n + 1 & \text{if } n \text{ is even} \\ n - 1 & \text{if } n \text{ is odd.} \end{cases}$$

Let $\mathbb{P}_{\mathcal{R}}$ be the uniform $\mathcal{R}$ -distribution on $\mathbb{N}$, and let $P := f * \mathbb{P}_{\mathcal{R}}$ and $Q := g * \mathbb{P}_{\mathcal{R}}$. So $P = \lim_K P_K$ and $Q = \lim_K Q_K$, where $P_K := f * \mathbb{P}_K$ and $Q_K := g * \mathbb{P}_K$ are the pushforwards of the uniform distribution on the finite set K .

For part (a), let $U \subseteq \mathbb{N}$ be upward-closed: so $U = \{n : n \geq m\}$ for some m ; we may assume $m > 0$, since the case $U = \mathbb{N}$ is trivial. Note that g is a permutation of $\mathbb{N}$, and $g^{-1}(U)$ differs from U at most by exchanging m for $m - 1$. So for any finite $K \subset \mathbb{N}$ that contains both $m - 1$ and m , we have $\#(K \cap g^{-1}(U)) = \#(K \cap U)$, and so $P_K(U) = Q_K(U)$. So

$$P(U) = \lim_K P_K(U) = \lim_K Q_K(U) = Q(U).$$

So P weakly cumulatively dominates Q .

For part (b), we will show that *lifted cumulative dominance* fails (theorem F.58(e)): it is not the case that for $\mathcal{F}_0$ -almost every K , $P_K \geq_s Q_K$. That is to say, each finite subset of $\mathbb{N}$ is contained in some finite K such that P_K does not cumulatively dominate Q_K . In particular, we can consider the

finite set $K \subset \mathbb{N}$ that contains every natural number strictly less than m , for some sufficiently large odd m . Then $P_K(n) = 0$ for every $n \geq m$, while $Q_K(m) = 1/m$. Thus P_K does not cumulatively dominate Q_K .

For part (c), take $u = \text{id}_{\mathbb{N}}$. In this case,

$$\int u \, dP = \lim_K \int \text{id}_{\mathbb{N}} \, dP_K = \lim_K \int f \, d\mathbb{P}_K = \lim_K \frac{1}{\#K} \sum_K f$$

and similarly for Q and g . So $\int u \, dP_K \geq \int u \, dQ_K$ iff

$$\sum_K f \geq \sum_K g.$$

Since $g(n) = f(n) + 1$ for every even number, given any $K \in \text{Fin } \mathbb{N}$, we can choose $J \supseteq K$ that adds sufficiently many even numbers to ensure that the inequality fails. So it is not the case that $\int u \, dP_K \geq \int u \, dQ_K$ for $\mathcal{F}_0$ -almost every K . $\square$

This leaves one loose end.

Question F.62. Does the *expected real utility* condition imply stochastic dominance?

Now we turn to *strong* stochastic dominance, proving a counterpart of theorem D.32 and theorem E.36.

Theorem F.63. Let $\mathcal{R}$ be a non-trivial I -limiting ring, let $\mathcal{X}$ be a preordered set, and let $P, Q \in \Delta^{\mathcal{R}}\mathcal{X}$ be $\mathcal{R}$ -distributions. The following are equivalent:

- (a) $P \geq_s Q$ but not $Q \geq_s P$.
- (b) **Realization.** There exist a set $\mathcal{Y}$, a distribution $R \in \Delta^{\mathcal{R}}\mathcal{Y}$, and functions $f, g : \mathcal{Y} \rightarrow \mathcal{X}$ such that $P = f * R$, $Q = g * R$, and $f >_R g$.
- (c) **Coupling.** There exists $R \in \Delta^{\mathcal{R}}(\mathcal{X} \times \mathcal{X})$ such that the marginals of R are $P = \pi_1 * R$ and $Q = \pi_2 * R$, and R is supported on the set $\{(x, y) : x \geq y\}$, and $R\{(x, y) : x > y\} > 0$.
- (d) **Expected vector utility.** For any preordered vector space $\mathcal{U}$ and any strictly monotone function $u : \mathcal{X} \rightarrow \mathcal{U}$,

$$\int u \, dP > \int u \, dQ.$$

(e) **Lifted cumulative dominance.** $P > Q$ in $(\Delta_0^{\mathbb{Q}}\mathcal{X})^{\mathcal{R}}$, with respect to the lifted cumulative dominance order.

In this case say that P **strongly stochastically dominates** Q .

Proof. We will show:

$$(a) \Rightarrow (c) \Rightarrow (b) \Rightarrow (d) \Rightarrow (e) \Rightarrow (a).$$

$(a) \Rightarrow (c)$. By theorem F.58, if $P \geq_s Q$, then there is a coupling R of P and Q such that $R\{(x, y) : x \geq y\} = 1$. If $R\{(x, y) : x > y\} = 0$, then R is supported on the set $\{(x, y) : x \sim y\}$, demonstrating that $Q \geq_s P$ as well.

$(c) \Rightarrow (b)$. As in theorem F.58, let $\mathcal{Y} = \mathcal{X} \times \mathcal{X}$, $f = \pi_1$, $g = \pi_2$.

$(b) \Rightarrow (d)$. If u is strictly monotone, then wherever $f \geq g$ we also have $u \circ f \geq u \circ g$, and wherever $f > g$ we also have $u \circ f > u \circ g$. Since $f >_R g$, it follows that $u \circ f >_R u \circ g$ as well, and thus

$$\int u \, dP = \int (u \circ f) \, dR > \int (u \circ g) \, dR = \int u \, dQ.$$

$(d) \Rightarrow (e)$. Order $\text{Fin}(\mathcal{X}, \mathbb{Q})$ by cumulative dominance as in theorem F.58. Then δ is *strictly* monotone. Thus

$$P = \int \delta \, dP > \int \delta \, dQ = Q.$$

$(e) \Rightarrow (a)$. Let $P = \lim_j P_j$ and $Q = \lim_j Q_j$. If $P > Q$ in $(\Delta_0^{\mathbb{Q}}\mathcal{X})^{\mathcal{R}}$, then $P \geq Q$ and $Q \not\geq P$ in $(\Delta_0^{\mathbb{Q}}\mathcal{X})^{\mathcal{R}}$. By theorem F.58, the first clause implies $P \geq_s Q$. If $Q \geq_s P$ as well, then we would have $Q \geq P$ by theorem F.58 again. $\square$

Here is one further important application. Recall Tarsney's theorem once more, from sections D.3, E.6 and 8.4: if a bounded prospect X has strictly higher expected value than another Y , then we can add on a "cosmic background value" prospect B with wide enough tails such that $X + B$ stochastically dominates $Y + B$. We noted in section D.3 that this turns on an infinite space of possible background values. Indeed, if X has a chance of turning out worse than Y , then (roughly speaking) the way we get stochastic dominance is by spreading this chance out and pushing it off to infinity, so it ends up falling through minuscule measure-zero cracks. But those

minuscule events still matter, and can make a difference to the overall value of a prospect. In section 8.4 I claimed that the result analogous to Tarsney's theorem fails for quasi-probabilities. Here we justify this claim. The point is exactly analogous to the case of simple probability distributions, discussed in section D.3.

Definition F.64. Let $\mathcal{U}$ be any quantity, and let $P, Q \in \Delta^{\mathcal{R}}\mathcal{U}$ be $\mathcal{R}$ -distributions over $\mathcal{U}$. The **convolution** $P \star Q$ is the $\mathcal{R}$ -distribution over $\mathcal{U}$ defined by

$$P \star Q := \lim_j P_j \star Q_j$$

where $P = \lim_j P_j$ and $Q = \lim_j Q_j$.

(See definition D.34 for the definition of the convolution of finite-support distributions. The limit is computed index-wise on representatives, so the operation is well-defined.)

As in the simple case, the distribution of a sum of independent random variables X and Y is given by the convolution of their individual distributions. I will elide these details.

Definition F.65. Let $\mathcal{X}$ be a totally ordered set. For $P \in \Delta^{\mathcal{R}}\mathcal{X}$, define the **quasi-infimum** and **quasi-supremum** for P by

$$\begin{aligned} \inf^* P &:= \lim_j \min \text{supp } P_j \\ \sup^* P &:= \lim_j \max \text{supp } P_j \end{aligned}$$

(where $P = \lim_j P_j$).

In particular, in the case where P 's support attains a minimum value, $\inf^* P = \min \text{supp } P$; likewise, if it attains a maximum value, then $\sup^* P = \max \text{supp } P$.

Proposition F.66. Let $\mathcal{U}$ be a $\mathbb{Z}$ -quantity, and let $P, Q, R \in \Delta^{\mathcal{R}}\mathcal{U}$ be $\mathcal{R}$ -distributions over $\mathcal{U}$.

(a) If $P \geq_s Q$, then $P \star R \geq_s Q \star R$.

(b) Suppose $\mathcal{U}$ is totally ordered. If $P \star R \geq_s Q \star R$, then

$$\begin{aligned}\sup^* P &\geq \sup^* Q \\ \inf^* P &\geq \inf^* Q.\end{aligned}$$

Proof. Both parts follow straightforwardly from the corresponding results for finite-support distributions (proposition D.36), using the lifted cumulative dominance characterization (theorem F.58). $\square$

The upshot is that Tarsney's trick won't work in our context of non-standard $\mathcal{R}$ -probabilities. Suppose that a risky gamble P has higher expected value than a safe bet Q . If the worst-case outcome of P is worse than that of Q , then there is no way of conjuring stochastic dominance by adding on an independent background prospect.

F.7 Valuing social prospects

Now, at last, let's put all these technical pieces together to develop a working model of the main subject matter of the book: the value of infinite social prospects under uncertainty.

First, the set-up, combining the ideas from this appendix with elements from section B.7.

Definition F.67. Let Ω be a **state space**. Let $\mathcal{X}$ be a preordered set of **individual outcomes** (or *welfare levels*), and let $\mathcal{I}$ be an infinite set of **individuals**.

- A **social outcome** is any function from $\mathcal{I}$ to $\mathcal{X}$.
- A **social prospect** is any function from Ω to social outcomes.
- A **social order** is any preorder on the set of social prospects.

I state things in terms of a population of *individuals*, but note that all of these mathematical results apply just as powerfully to distributions over any other kind of "locations of value" (for philosophical discussion see footnote 11 and section 6.4).

(Some of the results below are stated in terms of *quantities*. These have direct applications to social welfare, even without assuming anything as strong as a utilitarian theory of value. Any finitely anonymous Paretian order embeds the counting order (proposition A.34), which has a faithful

quantitative representation (theorem B.70). In the general case this quantity is non-Archimedean, allowing infinite or infinitesimal values. But if we restrict attention to simple outcomes that are only finitely different from some baseline, then these can be faithfully represented by *finite* quantities — indeed, integers, the net number of happy lives for each outcome relative to some fixed baseline.)

Let's first round up the main negative results we have already established. I will group these according to six general lessons.

First, the real numbers are inadequate as faithful representations of value or probability.

- No finitely anonymous Paretian social order (chapter 3 and section A.3) is representable by a real-valued utility function (corollary A.35(b), via theorem A.23). For *uncountable* populations, no Paretian order is real-representable, even setting anonymity aside (theorem A.31).
- Analogous lessons apply to probability. No regular uniform probability order on an infinite state space can be faithfully represented by a real-valued probability measure (corollary A.27(b)). No regular probability order on a space with uncountably many disjoint events can be represented by a real-valued measure (proposition E.12).
- There are also familiar constraints on real-valued probability measures that specifically arise from *countable additivity*. No countably additive real-valued probability measure on a countably infinite state space is uniform (proposition E.2). Any appropriately *symmetric* countably additive measure must have unmeasurable sets (theorem E.5).
- No ordering of standard real-valued probability measures over social outcomes respects Stochastic Dominance (chapter 5 and section E.4). In fact, if we consider the case where individual outcomes are determined by the standard coin flip measure, there is a probability-preserving rearrangement of states that takes every outcome to a Pareto-inferior outcome, except for a set of outcomes with probability zero. In effect, probability zero events can make an arbitrarily large finite difference to the value of prospects. In general, finite differences in value can be washed out of standard probability distributions over outcomes, by adding on sufficiently disperse random background noise (theorem E.59).
- In general, stochastic dominance relations are badly behaved for contin-

uous real-valued distributions on infinite social outcomes (section E.4).

Second, *symmetry* is an important guide to theorizing about value and probability.

- No Paretian social order is strongly anonymous (corollary A.35(a)). More generally, for any group of transformations G acting on individuals, for there to exist any G -symmetric Paretian social order, every element of G must have finite orbits (lemma A.60). For example, Pareto is also inconsistent with symmetry under translations in space (the East–West argument in section 3.3).
- Even when *symmetry* under certain transformations is unavailable, we can still often have *invariance* (section A.6, definition A.54, and corollary A.68). For example, anonymous Paretian translation-invariant social orders do exist (corollary A.68). But there are strong limits on invariant orders as well. Indeed, no Paretian social order that is invariant under *arbitrary* permutations can be any stronger than the counting order (definition A.17, theorem A.57, and corollary A.59). In particular, no such order is total. More generally, G -invariant Paretian *total* social orders only exist when G satisfies a restrictive orbit condition (proposition A.64). For example, no total Paretian social order is invariant under arbitrary *reflections* of the line.
- Parallel positive and negative results apply to probability (corollary A.70). No regular probability order on the circle is symmetric under arbitrary rotations, though there are rotation-*invariant* regular probability orders (lemma A.60 and corollary F.24). No regular probability order on the coin flip space is symmetric under arbitrary coin permutations (lemma A.60), and no *total* regular probability order is even invariant under coin permutations (by proposition A.64).

Third, there is a tight relationship between *total orders* and *non-constructive mathematics*.

- There is no constructive proof that a *total* finitely anonymous Paretian social order exists (corollary A.35(c), via theorem A.25). For continuum-sized populations, we don't even need the hypothesis of anonymity (theorem A.30). Analogous results hold for probability orders (corollary A.29).

Fourth, hypersensitivity to tiny probabilities of extreme outcomes is hard to avoid: there are many roads to Finite Fanaticism.

- Ex Ante Pareto and Statewise Anonymity imply a form of *expectationalism*, and thereby Finite Fanaticism (theorem D.45).
- Any *separable* order on finite quantitative prospects that respects stochastic dominance is expectational, and thus fanatical (theorem D.50).
- Any addition-invariant order on simple quantitative prospects that satisfies Independence is fanatical (proposition D.58).

Fifth, some of these routes to fanaticism take us to even direr consequences when extended to discrete prospects that may have infinitely many possible outcomes.

- No order on discrete social prospects satisfies Ex Ante Pareto and Ex Post Anonymity (theorem D.68).
- No order on discrete prospects (with standard real-valued probabilities) that respects stochastic dominance is separable (theorem D.62).

Sixth, it is hard to escape St. Petersburg and Pascal.

- No fanatical order of discrete prospects respects the Archimedean or Continuity axioms, Eventwise Dominance, or outcome-prospect dominance (Savage's axiom P7) (theorems D.61(a) and D.61(b) and theorem D.64).
- Finite Fanaticism and stochastic dominance imply Pascal's Principle (theorem D.39).
- There is also a direct route to Pascal that does not assume Finite Fanaticism. In fact, any anonymous Paretian order on simple social prospects that respects weak stochastic dominance has uncountably many Pascalian prospects (theorem D.41).

Now (at last!) we turn to some positive results. Here is the general flavor. We can escape some of the constraints we just enumerated by turning away from real-valued representations of value and probability, and deploying non-Archimedean quantities in both cases. We can escape others by turning to quantities that are only *partially ordered* (or else embracing nonconstructiveness and arbitrariness). Some of the constraints we simply accept, rather than evading: for instance, we will not propose any strongly anonymous social orders, or rotation-symmetric regular probabilities, and our final model will not satisfy Ex Ante Pareto. Finally, and perhaps most starkly, the model I describe below does not escape Finite Fanaticism, St. Petersburg, or Pascal

in the slightest. Rather, it presents a novel coherent worldview where these principles can be *accepted* without contradiction.

Definition F.68. Let $\mathcal{U}$ be a p.o. rational vector space, and let $u : \mathcal{X} \rightarrow \mathcal{U}$ be a strictly monotone function, and let $\mathcal{F}$ be a non-trivial fine $\mathcal{I}$ -filter. Let $\mathcal{R}$ be a non-trivial fine Ω -ring, and let P be an $\mathcal{R}$ -distribution on Ω .

In definition B.74, for any social outcome $x \in \mathcal{X}^{\mathcal{I}}$, we defined

$$U(x) := \sum_{i \in \mathcal{I}}^{\mathcal{F}} u(x(i))$$

and

$$x \succeq y \quad \text{iff} \quad U(x) \geq U(y) \quad \text{for each } x, y \in \mathcal{X}^{\mathcal{I}}.$$

Now, for any social prospect $X : \Omega \rightarrow \mathcal{X}^{\mathcal{I}}$, define

$$\hat{U}(X) := \int U \circ X \, dP.$$

The **(infinitary ex ante) additive social order** (for P , u , and $\mathcal{F}$) is the preorder on social prospects defined by

$$X \succeq Y \quad \text{iff} \quad \hat{U}(X) \geq \hat{U}(Y) \quad \text{for each } X, Y : \Omega \rightarrow \mathcal{X}^{\mathcal{I}}.$$

(Note that since $\mathcal{U}$ is a vector space, it is automatically a cancellative quantity.)

Theorem F.69. The additive social order (for P , u , and $\mathcal{F}$) has the following properties:

Finite Anonymity (Ex Post). For any social outcome $x : \mathcal{I} \rightarrow \mathcal{X}$, and any permutation $\pi : \mathcal{I} \rightarrow \mathcal{I}$ with finite support,

$$x \sim x \circ \pi.$$

Strong Pareto (Ex Post). For any social outcomes x and y ,

- (i) If $x(i) \succeq y(i)$ for every $i \in \mathcal{I}$, then $x \succeq y$.
- (ii) If, furthermore, $x(i) \succ y(i)$ for some i , then $x \succ y$.

Statewise Dominance. For any social prospects X and Y ,

- (i) If $X(w) \succeq Y(w)$ for every $w \in \Omega$, then $X \succeq Y$.

- (ii) If, furthermore, $X(w) \succ Y(w)$ for some $w \in \Omega$ such that $P\{w\} > 0$, then $X \succ Y$.

Suppose that P is a divisible distribution: whenever $P(E) > 0$, $P(E)$ is a divisible element of $\mathcal{R}$ (definition F.5). For $E \subseteq \Omega$ such that $P(E) > 0$, we can define

$$\hat{U}(X|E) := \frac{1}{P(E)} \int_E U \circ X \, dP.$$

In this case, let

$$X \succeq_E Y \quad \text{iff} \quad \hat{U}(X|E) \geq \hat{U}(Y|E).$$

(When $P(E) = 0$, let $X \sim_E Y$ trivially for all prospects X and Y .)

The Sure Thing Principle. For any social prospects X, Y , and any event $E \subseteq \Omega$ such that $P(E) > 0$, if $X \sim_{\Omega \setminus E} Y$, then

$$X \succeq Y \quad \text{iff} \quad X \succeq_E Y.$$

For $J \subseteq \mathcal{I}$ and $x \in \mathcal{X}^J$, define

$$U_J(x) := \sum_{i \in J}^J u(x(i)).$$

For prospects $X, Y : \Omega \rightarrow \mathcal{X}^J$, define

$$\hat{U}_J(X) := \int U_J \circ X \, dP.$$

Define

$$X \succeq^J Y \quad \text{iff} \quad \hat{U}_J(X) \geq \hat{U}_J(Y).$$

Separability. For any social prospects X and Y , and any population $J \subseteq \mathcal{I}$, if $X \sim^{J \setminus J} Y$, then

$$X \succeq Y \quad \text{iff} \quad X \succeq^J Y.$$

Stochastic Dominance. For any social prospects X and Y ,

- if $X * P$ stochastically dominates $Y * P$ then $X \succeq Y$;
- if $X * P$ strongly stochastically dominates $Y * P$ then $X \succ Y$.

Proof. *Finite Anonymity* follows from the fact that the quasi-finite sum $\sum^J$ is finitely symmetric (theorem B.69(a)).

Strong Pareto follows from the fact that the quasi-finite sum is a strictly positive linear operator (theorems B.69(b) and B.69(c)): if x is a Pareto improvement over y , then $u \circ x - u \circ y \geq 0$ everywhere, with strictness holding somewhere. So $U(x) - U(y) = \sum^{\mathcal{F}} (u \circ x - u \circ y) > 0$.

Statewise Dominance follows from the fact that the integral $\int dP$ is a strictly positive linear operator (proposition F.51(a)).

The Sure Thing Principle follows from linearity of the integral, which implies

$$\hat{U}(X) = P(E) \cdot \hat{U}(X | E) + P(\Omega \setminus E) \cdot \hat{U}(X | \Omega \setminus E)$$

(unless $P(\Omega \setminus E) = 0$, in which case the Sure Thing Principle is trivially true). If $X \sim_{\Omega \setminus E} Y$, by cancellation,

$$\hat{U}(X) \geq \hat{U}(Y) \quad \text{iff} \quad P(E) \cdot \hat{U}(X | E) \geq P(E) \cdot \hat{U}(Y | E).$$

Multiplication by $P(E)$ preserves the order, since $P(E)$ is positive, and reflects it, since $P(E)$ is divisible.

Separability follows from linearity of both the quasi-finite sum and the integral; the first implies

$$U(x) = U_J(x) + U_{J \setminus J}(x) \text{ for each outcome } x,$$

and then the second implies

$$\hat{U}(X) = \hat{U}_J(X) + \hat{U}_{J \setminus J}(X).$$

Then if $\hat{U}_{J \setminus J}(X) = \hat{U}_{J \setminus J}(Y)$, we have $\hat{U}(X) \geq \hat{U}(Y)$ iff $\hat{U}_J(X) \geq \hat{U}_J(Y)$ by the properties of p.o. rational vector spaces.

Stochastic Dominance follows from the expected vector utility formulation of stochastic dominance theorems F.58 and F.63: by change of variables (lemma F.53),

$$\hat{U}(X) = \int U \circ X dP = \int U d(X * P),$$

and $U : \mathcal{X}^J \rightarrow \mathbb{R}^{\mathcal{F}}$ is a strictly monotone vector utility function. $\square$

The additive social order is a generalization of the finite expectational additive order.

Proposition F.70. Let X be a social prospect and let $J \in \text{Fin } \mathcal{I}$ be a finite subpopulation.

(a) We have

$$U_J(x) = \sum_{i \in J} u(x(i)) \text{ for } x \in \mathcal{X}^{\mathcal{I}},$$

where this is the standard finite (vector) sum.

(b) Suppose that the distribution $U_J * X * P$ has finite support and assigns each element of $\mathcal{U}$ a rational probability. In this case we can identify $U_J * X * P$ with a standard distribution in $\Delta_0^{\mathbb{Q}} \mathcal{U}$, and

$$\hat{U}_J(X) = \mathbb{E}(U_J * X * P),$$

where this is the standard (vector) expectation for discrete probability measures.

(c) Consequently, if X and Y agree at every state for all individuals not in J , then $\succeq$ ranks X and Y according to their standard expected total utility for the individuals in J .

Proof. Part (a) is because the quasi-finite sum agrees with the ordinary sum for functions with finite support (theorem B.69(d)).

Part (b) is because

$$\hat{U}_J(X) = \int U_J \circ X \, dP = \int \text{id} \, d(U_J * X * P) = \mathbb{E}(U_J * X * P),$$

since the integral agrees with the standard integral for simple distributions (proposition F.51(c)).

Part (c) follows from Separability. $\square$

When we look beyond these simple probability distributions, additive orders can still retain agreement with standard expected total utility to some extent, but things become a lot more delicate. Even when standard expected total utility is well-defined, it can miss infinitesimal differences in probability or value. As we discussed in chapter 5 and sections E.4 and E.5 and will return to below, even infinitesimal discrepancies can sometimes be quite consequential.

Finally, let's consider specifically what the additive social order has to say about extreme risks — tiny chances of enormous value. Assume the set of

individuals $\mathcal{I}$ and the set of states Ω are each infinite. We will make things a bit more concrete by assuming the space of individual utilities $\mathcal{U} = \mathbb{R}$. (This is just a simplifying assumption. There is no reason in principle why even individual utilities have to be either Archimedean or totally ordered.)

First, because this order extends the *standard* expectational additive order, it will satisfy Finite Fanaticism. Call a social outcome x *finite* if $U(x)$ is finite, and likewise call a social prospect X *finite* if $U(X(w))$ is finite for every $w \in \Omega$. For any positive finite social outcome x , and for any event $E \subseteq \Omega$ whose probability $P(E)$ is positive and finite (rather than infinitesimal), there is some finite social outcome y such that $y \sqcup_E 0 \succ x$. (Here $x, y, 0$ stand for their corresponding constant prospects.)

The additive properties of U ensure that something similar holds for finite *differences* between prospects. In particular, the additive social order satisfies the principle we called *Fixed Fanaticism* (section 10.3). Let X be any finite social prospect, let $E \subseteq \Omega$ have finite positive probability (intuitively very small), let d be any finite number (intuitively large), and let X^+ be a social prospect such that

$$U(X^+(w)) = U(X(w)) + d \quad \text{for every } w \in \Omega.$$

Then there exists a finite social prospect Y such that

$$Y \sqcup_E X \succ X^+.$$

Furthermore, Statewise Dominance ensures that these principles have their usual consequences. There are St. Petersburg-type finite social prospects that are better than any finite social outcome. And any *infinite* social outcome is Pascalian: an arbitrarily small finite probability of getting it is better than any finite social outcome.

In fact, though, the additive social order's "infinity obsession" extends even further. First, we need to get clear on a technical point. Each individual outcome is assigned an individual utility, which is a real number in $\mathbb{R}$. Each *social* outcome x is assigned a social utility $U(x)$, which is an infinite ("quasi"-) sum of individual utilities; so it takes its values in the quasi-real ring $\mathbb{R}^{\mathcal{F}}$. This structure is also a preordered vector space. This means that when we go on to calculate the utility for a social *prospect* X , the integral in the definition is not being applied to a *real*-valued function; rather it is being applied to a *vector*-valued function, taking values in $\mathbb{R}^{\mathcal{F}}$. The upshot is that the value of $\hat{U}(X)$ for social prospects is *even more* non-standard than the $U(x)$ for social

outcomes (so to speak). If $\mathcal{R} = \mathbb{Q}^{\mathcal{G}}$ is the ring where probabilities live, then the values of prospects live in the doubly-extended real numbers $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$.

In the standard utilitarian framework, we are used to using a single one-size-fits-all kind of quantity — the real numbers — to represent all sorts of things: probabilities, values of individual or social outcomes, and values of individual or social prospects. In the additive setting, these things are represented by *different* kinds of quantities. If $\mathcal{F}$ is our filter on individuals, and $\mathcal{G}$ is our filter on states, then where the standard framework just uses $\mathbb{R}$ all over the place, the additive theory deploys four different quantitative structures to represent four different kinds of thing:

- $\mathbb{R}$ represents the value of individual outcomes.⁵
- $\mathbb{R}^{\mathcal{F}}$ represents the value of social outcomes
- $\mathbb{Q}^{\mathcal{G}}$ represents probabilities.
- $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$ represents the value of social prospects

This is a feature, not a bug! One of the fundamental results in the field is that real numbers are inadequate for representing impartial social welfare (Basu and Mitra's theorem). Similarly, one of the lessons of the St. Petersburg paradox is that whatever space of quantities R we use to represent values of *outcomes* will be inadequate for representing the values of arbitrary *prospects* taking values in R . (This was discussed in chapter 2 and chapter 5.)

It is important that these four kinds of quantity are not arbitrary, but rather related to one another. There are embeddings of the smaller structures in the larger ones: $\mathbb{R}$ embeds in each of $\mathbb{R}^{\mathcal{G}}$ and $\mathbb{R}^{\mathcal{F}}$, and $\mathbb{R}^{\mathcal{G}}$ and $\mathbb{R}^{\mathcal{F}}$ each embed (in different ways) in $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$. Since these are all rings that embed $\mathbb{R}$, this also makes all four of these structures into preordered vector spaces. It also makes $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$ into an $\mathbb{R}^{\mathcal{G}}$ -module (as well as an $\mathbb{R}^{\mathcal{F}}$ -module). This is important, because probabilities and values have to play nicely together: we need to be able to multiply probabilities by values to get expectations (and probabilities live in $\mathbb{Q}^{\mathcal{G}} \subseteq \mathbb{R}^{\mathcal{G}}$, rather than $\mathbb{R}^{\mathcal{F}}$). Fortunately, all of these different quantities, while living their own separate lives in some respects, still cooperate harmoniously.

This also means there are some interesting things to investigate about how they interact. Crucially, the order in which we have taken these filter powers

⁵Again, we have supposed for simplicity that individual utilities are real numbers. If they are not, then we could start up this whole hierarchy not with *reals*, but with some other quantity $\mathcal{U}$.

matters. The two rings $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$ and $(\mathbb{R}^{\mathcal{G}})^{\mathcal{F}}$ are not the same — and this makes a big difference to arithmetic properties of “lifted” quasi-reals from $\mathbb{R}^{\mathcal{G}}$ and $\mathbb{R}^{\mathcal{F}}$. How do these two kinds of non-standard numbers compare to one another within $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$?

Proposition F.71. Let $r \in \mathbb{R}^{\mathcal{F}}$ be positive and infinite (that is, $r > s$ for every standard real number $s \in \mathbb{R}$). Let $s \in \mathbb{R}^{\mathcal{G}}$ be arbitrary (including the case where it is infinite). Then $r > s$ in $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$.

Proof. Suppose

$$r = \lim_K^{\mathcal{F}} r_K \quad \text{in } \mathbb{R}^{\mathcal{F}} \quad s = \lim_J^{\mathcal{G}} s_J \quad \text{in } \mathbb{R}^{\mathcal{G}}.$$

When we lift these values into $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$ to compare them, we get

$$\begin{aligned} r - s &= \lim_J^{\mathcal{G}} \lim_K^{\mathcal{F}} (r_K - s_J) \\ &= \lim_J^{\mathcal{G}} \left(\lim_K^{\mathcal{F}} r_K \right) - s_J \\ &= \lim_J^{\mathcal{G}} (r - s_J). \end{aligned}$$

Since r is infinite and s_J is a standard real number, $r - s_J > 0$. So this limit is positive, and thus $r > s$ in $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$. $\square$

So we have a hierarchical relationship, when all of these rings are considered as substructures of the largest structure $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$:

$$\mathbb{R} \ll \mathbb{R}^{\mathcal{G}} \ll \mathbb{R}^{\mathcal{F}}.$$

What this means is that, if we start with the puny finite real numbers, then $\mathbb{R}^{\mathcal{G}}$ contains infinite numbers that are greater than anything in $\mathbb{R}$; likewise, $\mathbb{R}^{\mathcal{F}}$ contains numbers that are even “more infinite”, greater than anything in $\mathbb{R}^{\mathcal{G}}$.⁶

⁶This question is harder than it looks: it cannot be proved in the same way that we showed $\mathbb{R} \ll \mathbb{R}^{\mathcal{G}}$. The question is equivalent to asking whether there exists a function from $\text{Fin } \Omega$ to $\mathbb{R}^{\mathcal{F}}$ which is unbounded in $\mathbb{R}^{\mathcal{F}}$. This is related to the *cofinality* of the non-standard order $\mathbb{R}^{\mathcal{F}}$, and how this relates to the order on $\text{Fin } \Omega$.

Question F.72. Does $\mathbb{R}^{\mathcal{F}} \ll (\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$ hold as well? That is, does there exist an element $r \in (\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$ which is larger than every element of $\mathbb{R}^{\mathcal{F}}$?

This has important implications for how the additive social order evaluates risk. Here is one important question for our purposes: what happens to prospects that have *infinitesimal* chances of *infinite* values? In general, multiplying an infinite value by an infinitesimal value within a single quasi-real structure (such as $\mathbb{R}^{\mathcal{G}}$) can be infinite, finite, or infinitesimal. But what happens when the infinitesimal value is a probability in $\mathbb{R}^{\mathcal{G}}$, and the infinite value is a social welfare in $\mathbb{R}^{\mathcal{F}}$? To multiply them together, we first embed each in $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$. Suppose $p \in \mathbb{R}^{\mathcal{G}}$ is a positive infinitesimal probability, $r \in \mathbb{R}^{\mathcal{F}}$ is an infinite value, and $s \in \mathbb{R}$ is a finite value. Then s/p is an infinite element of $\mathbb{R}^{\mathcal{G}}$, and we know by proposition F.71 that $r > s/p$. Thus $p \cdot r > s$. An infinitesimal probability times an infinite value is *infinite*.

It follows that the additive social order is Pascalian in a very strong sense. Even an *infinitesimal* chance of an infinite good is better than any certain finite good.

Corollary F.73. Let X be a social prospect such that $U(X(w)) \in \mathbb{R}^{\mathcal{F}}$ is infinite for some $w \in \Omega$ such that $\{w\}$ is non-null, and $U \circ X \geq 0$ everywhere. Let Y be a social prospect such that $U \circ Y \in \mathbb{R}^{\mathcal{F}}$ is finite everywhere. Then $\hat{U}(X) > \hat{U}(Y)$.

This is at least a bit surprising. Here is an example from Hayden Wilkinson (Wilkinson, n.d.). Imagine there are a billion happy people (each with utility 1); each generation faces an independent 50% chance of extinction, and otherwise is succeeded by another generation of a billion happy people. Standard utilitarianism apparently has no trouble with this kind of case: the prospect has a well-defined expected total utility of exactly 2 billion. But suppose, against the odds, *every* generation manages to thwart extinction, so all told there are *infinitely* many happy people. This event has standard probability zero, so it apparently makes no difference to the standard expected total utility. (Though this is not entirely clear, since we are stretching the standard theory slightly beyond its usual domain of application.) But in our current framework, surviving every toss of the coin has a positive *infinitesimal* probability. So the non-Archimedean expected quasi-finite sum of utilities for this prospect turns out to be *infinite*.

What's more, an infinitesimal chance at infinite paradise is also better than

any *St. Petersburg*-type prospect, which has finite but unbounded values. These prospects can have infinite values as well, but they are pretty small infinite numbers. If a random variable X takes only values in $\mathbb{R}$, then the integral $\int X dP^{\mathcal{G}}$ is an element of $\mathbb{R}^{\mathcal{G}}$ — the lowest level of infinite numbers. Thus *St. Petersburg*-type prospects can have values that are infinite with respect to *standard* real numbers, but not with respect to infinite numbers in $\mathbb{R}^{\mathcal{G}}$. And an *infinitesimal chance* at playing a *St. Petersburg* game can even have a value that is finite or infinitesimal.

Can we also go the other direction, and find prospects that are even *better* than simple Pascalian wagers? That is, can we *St. Petersburg* our way up from *infinite* values of social outcomes to prospects that are better than *any* particular social outcome? I do not know. (This is related to the question of whether $\mathbb{R}^{\mathcal{F}} \ll (\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$.)

Question F.74. Does there exist a random variable $X : \Omega \rightarrow \mathbb{R}^{\mathcal{F}}$ and a probability measure $P^{\mathcal{G}}$ such that $\int X dP^{\mathcal{G}}$ is larger than any element of $\mathbb{R}^{\mathcal{F}}$?

The feature of the additive social order that infinite values trump infinitesimal probabilities is not accidental. It is closely related to the *paradoxes of dominance* discussed in chapter 6. We have chosen a mathematical approach that keeps *statewise dominance* while sacrificing *ex ante Pareto*. This is reflected in the formalism by the fact that the filter for individuals $\mathcal{F}$ is applied *before* the filter for states $\mathcal{G}$. We could do things in the opposite order, taking the integral on individual prospects first, and the sum over individuals after:

$$V(X) := \sum_{i \in \mathcal{I}}^{\mathcal{F}} \int u(X(w)(i)) dP^{\mathcal{G}}(w).$$

This gives us a result in the ring $(\mathbb{R}^{\mathcal{G}})^{\mathcal{F}}$ rather than $(\mathbb{R}^{\mathcal{F}})^{\mathcal{G}}$. Furthermore, this version turns out to satisfy the *ex ante Pareto* principle, while giving up *Statewise Dominance* (as well as *Stochastic Dominance*).

Yet another alternative would be to treat individuals and states symmetrically, considering some fine $(\Omega \times \mathcal{I})$ -filter $\mathcal{H}$ based on the product space, and then calculating something like this:

$$W(X) := \frac{1}{N} \sum_{(w,i) \in \Omega \times \mathcal{I}}^{\mathcal{H}} p(w) \cdot u(X(w)(i))$$

where N is the normalization constant $N = \sum_{(w,i) \in \Omega \times \mathcal{I}} p(w)$. (Pivato (2014) proposes an approach along these lines.) In general, this symmetric approach will satisfy *neither* ex ante Pareto nor Statewise Dominance. It will only satisfy this very weak dominance principle:

If $X(w)(i) \succeq Y(w)(i)$ for every pair $(w,i) \in \Omega \times \mathcal{I}$, then $X \succeq Y$

(and similarly its $>$ counterpart).

I do not know how tight the connection really is between Statewise Dominance and the potentially enormous value of infinitesimal probabilities. This is how things work out for these particular models, using iterated filter powers. But, for all I know, there may be other formal approaches that work quite differently. This is another question that deserves further exploration.

Additive social orders arise very naturally, but I put them forward mainly as a proof of concept, showing that certain attractive properties can indeed be satisfied together. I do not put these models forward as a final theory of value for infinite social prospects. The truth may well be something even more alien, which no one has imagined.

References

- Adams, Robert Merrihew. 1979. "Primitive Thisness and Primitive Identity." *Journal of Philosophy* 76 (1): 5–26. <https://doi.org/10.2307/2025812>.
- Akiba, Ken. 2015. "How Barnes and Williams Have Failed to Present an Intelligible Ontic Theory of Vagueness." *Analysis* 75 (4): 565–73. <https://doi.org/10.1093/analys/anv074>.
- Aliprantis, Charalambos D., and Kim C. Border. 2006. *Infinite Dimensional Analysis: A Hitchhiker's Guide*. 3rd edition. Springer.
- Allais, M. 1953. "Le Comportement de l'Homme Rationnel Devant Le Risque: Critique Des Postulats Et Axiomes de l'ecole Americaine." *Econometrica* 21 (4): 503–46.
- Anderson, Charity, and John Hawthorne. 2019. "Knowledge, Practical Adequacy, and Stakes." *Oxford Studies in Epistemology* 6 (6): 234–57.
- Arnauld, Antoine, and Pierre Nicole. (1683) 1996. *Logic or the Art of Thinking*. Translated by Jill Vance Buroker. Cambridge University Press. <https://doi.org/10.1017/CBO9781139166768>.
- Arntzenius, Frank. 2014. "Utilitarianism, Decision Theory and Eternity." *Philosophical Perspectives* 28 (1): 31–58. <https://doi.org/10.1111/phpe.12036>.
- Arntzenius, Frank, Adam Elga, and John Hawthorne. 2004. "Bayesianism, Infinite Decisions, and Binding." *Mind* 113 (450): 251–83.
- Arrhenius, Gustaf. 2004. "The Person-Affecting Restriction, Comparativism, and the Moral Status of Potential People." In *The Repugnant Conclusion: Essays on Population Ethics*, edited by Jesper Ryberg and Torbjörn Tännsjö, 299–316. Dordrecht: Kluwer Academic Publishers.
- Arrow, Kenneth J. (1951) 2012. *Social Choice and Individual Values*. Vol. 12. Yale University Press.
- Asheim, Geir B., and Kuntal Banerjee. 2010. "Fixed-Step Anonymous Overtaking and Catching-up." *International Journal of Economic Theory* 6 (1): 149–65.

- Asheim, Geir B., Claude d'Aspremont, and Kuntal Banerjee. 2010. "Anonymous Overtaking and Catching-up." *Journal of Mathematical Economics* 46 (4): 519–33.
- Askell, Amanda. 2018. "Pareto Principles in Infinite Ethics." New York University. <https://askell.io/publication/pareto-principles-in-infinite-ethics>.
- Askell, Amanda, and Sven Neth. 2025. "Longtermist Myopia." In *Essays on Longtermism: Present Action for the Distant Future*, edited by Hilary Greaves, Jacob Barrett, and David Thorstad, 295–314. Oxford University Press.
- Atsumi, Hiroshi. 1965. "Neoclassical Growth and the Efficient Program of Capital Accumulation." *The Review of Economic Studies* 32 (2): 127–36.
- Aumann, Robert J. 1962. "Utility Theory Without the Completeness Axiom." *Econometrica* 30 (3): 445–62.
- Bacon, Andrew. 2018. *Vagueness and Thought*. Oxford: Oxford University Press.
- . 2020. "Logical Combinatorialism." *Philosophical Review* 129 (4): 537–89.
- . 2022. "Actual Value in Decision Theory." *Analysis* 82 (4): 617–29. <https://doi.org/10.1093/analys/anac014>.
- . 2023. *A Philosophical Introduction to Higher-Order Logics*. Routledge.
- Bader, Ralf M. 2018. "Stochastic Dominance and Opaque Sweetening." *Australasian Journal of Philosophy* 96 (3): 498–507.
- . 2022. "Person-Affecting Utilitarianism." In *The Oxford Handbook of Population Ethics*, edited by Gustaf Arrhenius, Krister Bykvist, Tim Campbell, and Elizabeth Finneron-Burns, 251–70. Oxford University Press.
- Bales, Adam, Daniel Cohen, and Toby Handfield. 2014. "Decision Theory for Agents with Incomplete Preferences." *Australasian Journal of Philosophy* 92 (3): 453–70. <https://doi.org/10.1080/00048402.2013.843576>.
- Barnes, Elizabeth. 2010. "Ontic Vagueness: A Guide for the Perplexed." *Noûs* 44 (4): 601–27. <https://doi.org/10.1111/j.1468-0068.2010.00762.x>.
- Barnes, Elizabeth, and J. Robert G. Williams. 2011. "A Theory of Metaphysical Indeterminacy." In *Oxford Studies in Metaphysics Volume 6*, edited by Karen Bennett and Dean W. Zimmerman, 103–48. Oxford University Press.
- Barnett, Zach. 2020. "Why You Should Vote to Change the Outcome." *Philosophy and Public Affairs* 48 (4): 422–46. <https://doi.org/10.1111/papa.12177>.
- Bartha, Paul. 2007. "Taking Stock of Infinite Value: Pascal's Wager and

- Relative Utilities." *Synthese* 154 (1): 5–52. <https://doi.org/10.1007/s11229-005-8006-z>.
- . 2016. "Making Do Without Expectations." *Mind* 125 (499): 799–827. <https://doi.org/10.1093/mind/fzv152>.
- Bartha, Paul, John Barker, and Alan Hájek. 2014. "Satan, Saint Peter and Saint Petersburg: Decision Theory and Discontinuity at Infinity." *Synthese* 191 (4): 629–60. <https://doi.org/10.1007/s11229-013-0379-9>.
- Basu, Kaushik, and Tapan Mitra. 2003. "Aggregating Infinite Utility Streams with Intergenerational Equity: The Impossibility of Being Paretian." *Econometrica* 71 (5): 1557–63.
- Beckstead, Nicholas, and Teruji Thomas. 2024. "A Paradox for Tiny Probabilities and Enormous Values." *Noûs* 58: 431–55. <https://doi.org/10.1111/nous.12462>.
- Benci, Vieri, Emanuele Bottazzi, and Mauro Di Nasso. 2014. "Elementary Numerosity and Measures." *Journal of Logic and Analysis* 6 (3): 1–14. <https://doi.org/10.4115/jla.2014.6.3>.
- Benci, Vieri, and Mauro Di Nasso. 2003. "Numerosities of Labelled Sets: A New Way of Counting." *Advances in Mathematics* 173 (1): 50–67. [https://doi.org/10.1016/S0001-8708\(02\)00012-9](https://doi.org/10.1016/S0001-8708(02)00012-9).
- Benci, Vieri, Mauro Di Nasso, and Marco Forti. 2006. "An Aristotelian Notion of Size." *Annals of Pure and Applied Logic* 143 (1–3): 43–53. <https://doi.org/10.1016/j.apal.2006.01.008>.
- Benci, Vieri, Leon Horsten, and Sylvia Wenmackers. 2013. "Non-Archimedean Probability." *Milan Journal of Mathematics* 81: 121–51. <https://doi.org/10.1007/s00032-012-0191-x>.
- . 2018. "Infinitesimal Probabilities." *British Journal for the Philosophy of Science* 69: 509–52.
- Berkeley, George. (1710) 2008. "A Treatise Concerning the Principles of Human Knowledge." In *Berkeley: Philosophical Writings*, edited by Desmond M. Clarke. Cambridge University Press.
- Bernoulli, Daniel. (1738) 1954. "Exposition of a New Theory on the Measurement of Risk." *Econometrica* 22 (1): 23–36. <https://doi.org/10.2307/1909829>.
- Bernoulli, Nicolaus. 1713. "Extrait d'une Lettre de M. N. Bernoulli à M. de M... Du 9. Septembre 1713." In *Essay d'Analyse Sur Les Jeux de Hazard: Revûe Et Augmentée de Plusieurs Lettres*, by Pierre Raymond de Montmort, Seconde edition, 401–2. Laurent le Conte.
- Bernstein, Allen R., and Frank Wattenberg. 1969. "Nonstandard Measure Theory." In *Applications of Model Theory to Algebra, Analysis and Probability*, edited by W. A. J. Luxemburg, 171–85. Holt, Rinehart; Winston.

- Bhikkhu, Ṭhānissaro, trans. 2012. *Udāna: Exclamations*.
- Blackorby, Charles, Walter Bossert, and David Donaldson. 2005. *Population Issues in Social Choice Theory, Welfare Economics, and Ethics*. Cambridge University Press.
- Blackorby, Charles, David Donaldson, and John A. Weymark. 1999. "Harsanyi's Social Aggregation Theorem for State-Contingent Alternatives." *Journal of Mathematical Economics* 32 (3): 365–87.
- Blackwell, David. 1951. "Comparison of Experiments." In *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*. The Regents of the University of California.
- Blackwell, David, and M. A. Girshick. 1954. *Theory of Games and Statistical Decisions*. New York: John Wiley & Sons, Inc. <https://onlinelibrary.wiley.com/doi/abs/10.1002/nav.3800010313>.
- Blessenohl, Simon. 2020. "Risk Attitudes and Social Choice." *Ethics* 130 (4): 485–513.
- Blumberg, Kyle, and John Hawthorne. 2021. "Inheritance: Professor Procrastinate and the Logic of Obligation1." *Philosophy and Phenomenological Research* 106 (1): 84–106. <https://doi.org/10.1111/phpr.12846>.
- Blume, Lawrence, Adam Brandenburger, and Eddie Dekel. 1991. "Lexicographic Probabilities and Choice Under Uncertainty." *Econometrica: Journal of the Econometric Society*, 61–79.
- Bolker, Ethan. 1966. "Functions Resembling Quotients of Measures." *Transactions of the American Mathematical Society* 124: 293–312.
- Boolos, George. 1997. "Is Hume's Principle Analytic?" *Richard G. Heck (Ed.), Language, Thought, and Logic: Essays in Honour of Michael Dummett*, 245–61.
- Bostrom, Nick. 2003. "Astronomical Waste: The Opportunity Cost of Delayed Technological Development." *Utilitas* 15 (3): 308–14.
- . 2009. "Pascal's Mugging." *Analysis* 69 (3): 443–45.
- . 2011. "Infinite Ethics." *Analysis and Metaphysics* 10: 9–59.
- Bottomley, Christopher, and Timothy Luke Williamson. 2024. "Rational Risk-Aversion: Good Things Come to Those Who Weight." *Philosophy and Phenomenological Research* 108 (3): 697–725. <https://doi.org/10.1111/phpr.13006>.
- Brahmegupta, and Bhascara. 1817. *Algebra with Arithmetic and Mensuration from the Sanskrit*. Translated by Henry Thomas Colebrooke. John Murray.
- Breckenridge, Wylie, and Ofra Magidor. 2012. "Arbitrary Reference." *Philosophical Studies* 158 (3): 377–400. <https://doi.org/10.1007/s11098-010-9676-z>.
- Brickhill, Hazel, and Leon Horsten. 2018. "Triangulating Non-Archimedean

- Probability." *Review of Symbolic Logic* 11 (3): 519–46. <https://doi.org/10.1017/S1755020318000060>.
- Briggs, R. A. 2015. "Costs of Abandoning the Sure-Thing Principle." *Canadian Journal of Philosophy* 45 (5): 827–40.
- Broome, John. 1991. *Weighing Goods: Equality, Uncertainty and Time*. Wiley-Blackwell.
- . 1994. "Discounting the Future." *Philosophy & Public Affairs* 23 (2): 128–56.
- . 1995. "The Two-Envelope Paradox." *Analysis* 55 (1): 6–11.
- . 1997. "Is Incommensurability Vagueness?" In *Incommensurability, Incomparability, and Practical Reason*, edited by Ruth Chang. Harvard.
- . 2000. "Incommensurable Values." In *Well-Being and Morality: Essays in Honour of James Griffin*, edited by Roger Crisp and Brad Hooker, 21–38. Clarendon Press.
- . 2004. *Weighing Lives*. Oxford University Press.
- Buchak, Lara. 2013. *Risk and Rationality*. Oxford University Press.
- . 2015. "Revisiting Risk and Rationality: A Reply to Pettigrew and Briggs." *Canadian Journal of Philosophy* 45 (5): 841–62.
- Buffon, Comte de (Georges-Louis Leclerc). (1777) 2009. "Essai d'arithmétique Morale." Translated by Richard J. Pulskamp. *Histoire Naturelle, Générale Et Particulière*.
- Cain, James. 1995. "Infinite Utility." *Australasian Journal of Philosophy* 73 (3): 401–4. <https://doi.org/10.1080/00048409512346731>.
- Cantor, Georg. (1883) 1976. "Foundations of a General Theory of Manifolds." Translated by Uwe Parpart. *The Campaigner: Journal of the National Caucus of Labor Committees* 9 (1-2): 69–97.
- Cariani, Fabrizio. 2013. "'Ought' and Resolution Semantics." *Noûs* 47 (3): 534–58. <https://doi.org/10.1111/nous.12006>.
- Chalmers, David J. 1996. *The Conscious Mind*. Oxford University Press.
- . 2002. "On Sense and Intension." *Philosophical Perspectives* 16: 135–82.
- . 2011. "The Nature of Epistemic Space." In *Epistemic Modality*, edited by Andy Egan and Brian Weatherson, 60–107. Oxford University Press.
- Chang, Ruth. 2002. "The Possibility of Parity." *Ethics* 112 (4): 659–88.
- Chen, Eddy Keming, and Daniel Rubio. 2018. "Surreal Decisions." *Philosophy and Phenomenological Research*.
- Chipman, John S. 1960. "The Foundations of Utility." *Econometrica* 28 (2): 193–224.
- Christensen, David. 2007. "Epistemology of Disagreement: The Good News." *Philosophical Review* 116 (2): 187–217.
- Cibinel, Pietro. 2023. "A Dilemma for Nicolausian Discounting." *Analysis*

- 83 (4): 662–72. <https://doi.org/10.1093/analys/anac095>.
- . Unpublished. “Independent Aspects of Well-Being.” University of Southern California.
- Clarke, Samuel, and Gottfried Wilhelm Freiherr von Leibniz. (1717) 1956. *The Leibniz-Clarke Correspondence: Together With Extracts from Newton’s Principia and Opticks*. Edited by H. G. Alexander. Manchester University Press.
- Colyvan, Mark. 2008. “Relative Expectation Theory.” *Journal of Philosophy* 105 (1): 37–44. <https://doi.org/10.5840/jphil200810519>.
- Colyvan, Mark, and Alan Hájek. 2016. “Making Ado Without Expectations.” *Mind* 125 (499): 829–57. <https://doi.org/10.1093/mind/fzv160>.
- Cowling, Sam. 2015. “Non-Qualitative Properties.” *Erkenntnis* 80 (2): 275–301. <https://doi.org/10.1007/s10670-014-9626-9>.
- D’Aspremont, Claude, and Louis Gevers. 1977. “Equity and the Informational Basis of Collective Choice.” *The Review of Economic Studies* 44 (2): 199–209.
- Dasgupta, Shamik. 2021. “Symmetry and Superfluous Structure: A Metaphysical Overview.” In *Routledge Companion to the Philosophy of Physics*, edited by Eleanor Knox and Alistair Wilson, 551–62. Routledge.
- Dattani, Saloni, Lucas Rodés-Guirao, Hannah Ritchie, Esteban Ortiz-Ospina, and Max Roser. 2023. “Life Expectancy.” *Our World in Data*. <https://ourworldindata.org/life-expectancy>.
- Davidson, Donald. (1969) 2001. “True to the Facts.” In *Inquiries into Truth and Interpretation*, 37–54. Clarendon Press.
- Davies, Martin, and Lloyd Humberstone. 1980. “Two Notions of Necessity.” *Philosophical Studies* 38 (3): 1–30.
- Debreu, Gerard. 1954. “Representation of a Preference Ordering by a Numerical Function.” In *Decision Processes*, edited by Robert M. Thrall, Clyde H. Coombs, and Robert L. Davis, 159–65. New York: Wiley.
- DeRose, Keith. 1992. “Contextualism and Knowledge Attributions.” *Philosophy and Phenomenological Research* 52 (4): 913–29.
- Deschamps, Robert, and Louis Gevers. 1978. “Leximin and Utilitarian Rules: A Joint Characterization.” *Journal of Economic Theory* 17 (2): 143–63.
- Diamond, Peter A. 1965. “The Evaluation of Infinite Utility Streams.” *Econometrica* 33 (1): 170–77.
- . 1967. “Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparison of Utility: Comment.” *Journal of Political Economy* 75 (5): 765–66.
- Diderot, Denis. (1746) 1950. *Pensées Philosophiques*. Edited by Robert Niklaus. Geneva: E. Droz.

- Dorr, Cian, Jacob M. Nebel, and Jake Zuehl. 2023. "The Case for Comparability." *Noûs* 57 (2): 414–53. <https://doi.org/10.1111/nous.12407>.
- Dubey, Ram Sewak, and Giorgio Laguzzi. 2024. "Social Welfare Relations and Irregular Sets." *Annals of Pure and Applied Logic* 174 (9): 103302.
- Duff, Antony. 1986. "Pascal's Wager and Infinite Utilities." *Analysis* 46 (2): 107–9. <https://doi.org/10.1093/analys/46.1.107>.
- Dummett, Michael. 1991. *Frege: Philosophy of Mathematics*. Cambridge, MA: Harvard University Press.
- Duncan, Craig. 2007. "The Persecutor's Wager." *Philosophical Review* 116 (1): 1–50. <https://doi.org/10.1215/00318108-2006-021>.
- Dyson, Lisa, Matthew Kleban, and Leonard Susskind. 2002. "Disturbing Implications of a Cosmological Constant." *Journal of High Energy Physics* 2002 (10): 011.
- Earman, John. 1989. *World Enough and Spacetime: Absolute Versus Relational Theories of Space and Time*. MIT Press.
- Easwaran, Eknath, trans. 2007. *The Bhagavad Gita*. Second edition. Nilgiri Press.
- Easwaran, Kenny. 2008. "Strong and Weak Expectations." *Mind* 117 (467): 633–41. <https://doi.org/10.1093/mind/fzn053>.
- . 2013. "Why Countable Additivity?" *Thought: A Journal of Philosophy* 2 (1): 53–61. <https://doi.org/10.1002/tht3.60>.
- . 2014a. "Principal Values and Weak Expectations." *Mind* 123 (490): 517–31. <https://www.jstor.org/stable/24489477>.
- . 2014b. "Regularity and Hyperreal Credences." *Philosophical Review* 123 (1): 1–41. <https://doi.org/10.1215/00318108-2366479>.
- Easwaran, Kenny, and Michael Nielsen. 2025. "Updating by Maximizing Expected Accuracy in Infinite Non-Partitional Settings." *Journal of Philosophical Logic* 54 (5): 1095–1134. <https://doi.org/10.1007/s10992-025-09814-6>.
- Egan, Greg. 1994. *Permutation City*. Millennium Orion Publishing Group.
- Epstein, Larry G., and Uzi Segal. 1992. "Quadratic Social Welfare Functions." *Journal of Political Economy* 100 (4): 691–712.
- Fantl, Jeremy, and Matthew McGrath. 2002. "Evidence, Pragmatics, and Justification." *Philosophical Review* 111 (1): 67–94.
- Felgner, Ulrich, and John K. Truss. 1999. "The Independence of the Prime Ideal Theorem from the Order-Extension Principle." *The Journal of Symbolic Logic* 64 (1): 199–215.
- Field, Hartry. 2008. *Saving Truth from Paradox*. New York: Oxford University Press.
- Fine, Kit. 1975. "Vagueness, Truth and Logic." *Synthese* 30 (3–4): 265–300. <https://doi.org/10.1007/bf00485047>.

- . 1985. *Reasoning with Arbitrary Objects*. New York, NY, USA: Blackwell.
- Fine, Terrence L. 1973. *Theories of Probability: An Examination of Foundations*. New York: Academic Press.
- . 2008. "Evaluating the Pasadena, Altadena, and St Petersburg Gambles." *Mind* 117 (467): 613–32. <https://doi.org/10.1093/mind/fzn037>.
- Finetti, Bruno de. 1937. "La Prévision : Ses Lois Logiques, Ses Sources Subjectives." *Annales de l'Institut Henri Poincaré* 7 (1): 1–68.
- Fishburn, Peter C. 1971. "A Study of Lexicographic Expected Utility." *Management Science* 17 (11): 672–78. <https://doi.org/10.1287/mnsc.17.11.672>.
- . 1986. "The Axioms of Subjective Probability." *Statistical Science* 1 (3): 335–45. <https://doi.org/10.1214/ss/1177013611>.
- Fitch, Frederic B. 1963. "A Logical Analysis of Some Value Concepts." *Journal of Symbolic Logic* 28 (2): 135–42. <http://www.jstor.org/stable/2271594>.
- Fleming, Marcus. 1952. "A Cardinal Concept of Welfare." *The Quarterly Journal of Economics* 66 (3): 366–84.
- Fleurbaey, Marc. 2010. "Assessing Risky Social Situations." *Journal of Political Economy* 118 (4): 649–80.
- Fleurbaey, Marc, and Philippe Michel. 2003. "Intertemporal Equity and the Extension of the Ramsey Criterion." *Journal of Mathematical Economics* 39 (7): 777–802.
- Folland, Gerald B. 1984. *Real Analysis: Modern Techniques and Their Applications*. New York: John Wiley & Sons.
- Frege, Gottlob. 1892. "Über Sinn Und Bedeutung." *Zeitschrift Für Philosophie Und Philosophische Kritik* 100 (1): 25–50.
- . (1884) 1953. *The Foundations of Arithmetic: A Logico-Mathematical Enquiry into the Concept of Number*. Translated by J. L. Austin. 2d rev. Oxford: Blackwell.
- Frick, Johann. 2015. "Contractualism and Social Risk." *Philosophy & Public Affairs* 43 (3): 175–223.
- Fritz, Tobias. 2024. "Characterizing the Asymptotic and Catalytic Stochastic Orders on Topological Abelian Groups." *Bernoulli* 30 (4): 2932–54. <https://doi.org/10.3150/23-BEJ1700>.
- Gaertner, Wulf. 2009. *A Primer in Social Choice Theory*. Revised edition. Oxford University Press.
- Gibbard, Allan, and William L. Harper. 1978. "Counterfactuals and Two Kinds of Expected Utility." In *Foundations and Applications of Decision Theory, Volume i: Theoretical Foundations*, edited by Clifford Alan Hooker, James J. Leach, and Edward F. McClennen, 125–62. Dordrecht: D. Reidel.
- Gilboa, Itzhak, Dov Samet, and David Schmeidler. 2004. "Utilitarian Aggre-

- gation of Beliefs and Tastes." *Journal of Political Economy* 112 (4): 932–38.
- Gillman, Leonard, and Meyer Jerison. 1960. *Rings of Continuous Functions*. Princeton: Van Nostrand.
- Glass, A. M. W. 1999. *Partially Ordered Groups*. World Scientific.
- Goldblatt, Robert. 1998. *Lectures on the Hyperreals: An Introduction to Non-standard Analysis*. New York: Springer.
- Good, I. J. 1966. "On the Principle of Total Evidence." *British Journal for the Philosophy of Science* 17 (4): 319–21.
- Goodearl, K. R. 1986. *Partially Ordered Abelian Groups with Interpolation*. Mathematical Surveys and Monographs 20. American Mathematical Society.
- Goodman, Jeremy. 2025. "Permutation-Invariant Social Welfare Orders Are Anonymous." *Journal of Mathematical Economics* 120: 103–53. <https://doi.org/https://doi.org/10.1016/j.jmateco.2025.103153>.
- Goodsell, Zachary. forthcoming. "Symmetries of Value." *Noûs*, forthcoming. <https://doi.org/10.1111/nous.12549>.
- . 2021. "A St Petersburg Paradox for Risky Welfare Aggregation." *Analysis* 81 (3): 420–26. <https://doi.org/10.1093/analys/anaa079>.
- Gorman, W. M. 1968. "The Structure of Utility Functions." *Review of Economic Studies* 35 (4): 367–90.
- Grant, Simon, Atsushi Kajii, Ben Polak, and Zvi Safra. 2010. "Generalized Utilitarianism and Harsanyi's Impartial Observer Theorem." *Econometrica* 78 (6): 1939–71.
- Greaves, Hilary. 2016. "Cluelessness." *Proceedings of the Aristotelian Society* 116 (3): 311–39. <https://doi.org/10.1093/arisoc/aow018>.
- Greaves, Hilary, and John Cusbert. 2022. "Comparing Existence and Non-Existence." In *Ethics and Existence: The Legacy of Derek Parfit*, edited by Jeff McMahan, Tim Campbell, Ketan Ramakrishnan, and Jimmy Goodrich, 61–92. Oxford: Oxford University Press.
- Greaves, Hilary, and William MacAskill. 2021. "The Case for Strong Longtermism." *Global Priorities Institute Working Papers Series*, no. 5.
- Gustafsson, Johan E. 2022. *Money-Pump Arguments*. Cambridge: Cambridge University Press.
- Gustafsson, Johan E., and Kacper Kowalczyk. forthcoming. "Ex-Ante Pareto and the Opaque-Identity Puzzle." *Journal of Philosophy*, forthcoming. <https://doi.org/10.5840/jphil20251177>.
- Guth, Alan H. 1981. "Inflationary Universe: A Possible Solution to the Horizon and Flatness Problems." *Physical Review D* 23 (2): 347–56.
- Gwiazda, Jeremy. 2014. "Orderly Expectations." *Mind* 123 (490): 503–16. <https://doi.org/10.1093/mind/fzu059>.

- Hahn, Hans. 1907. "Über Die Nichtarchimedischen Größensysteme." *Sitzungsberichte Der Kaiserlichen Akademie Der Wissenschaften, Wien* 116: 601–55.
- Hájek, Alan. 2003a. "Waging War on Pascal's Wager." *Philosophical Review* 112 (1): 27–56.
- . 2003b. "What Conditional Probability Could Not Be." *Synthese* 137 (3): 273–323.
- . 2008. "Pascal's Wager." In *Stanford Encyclopedia of Philosophy*.
- . 2014. "Unexpected Expectations." *Mind* 123 (490): 533–67. <https://doi.org/10.1093/mind/fzu076>.
- Hájek, Alan, and Harris Nover. 2008. "Complex Expectations." *Mind* 117 (467): 643–64.
- Halpern, James D., and Azriel Lévy. 1971. "The Boolean Prime Ideal Theorem Does Not Imply the Axiom of Choice." In *Axiomatic Set Theory*, 13, part 1:83–134. Proceedings of Symposia in Pure Mathematics. Providence: American Mathematical Society.
- Halpern, Joseph Y. 2010. "Lexicographic Probability, Conditional Probability, and Nonstandard Probability." *Games and Economic Behavior* 68 (1): 155–79.
- Hamkins, Joel David, and Barbara Montero. 2000. "With Infinite Utility, More Needn't Be Better." *Australasian Journal of Philosophy* 78 (2): 231–40.
- Hammond, Peter J. 1988. "Consequentialist Foundations for Expected Utility." *Theory and Decision* 25 (1): 25–78.
- Hare, Caspar. 2009. "The Ethics of Morphing." *Philosophical Studies* 145 (1): 111–30.
- . 2010. "Take the Sugar." *Analysis* 70 (2): 237–47. <https://doi.org/10.1093/analys/anp174>.
- . 2013. *The Limits of Kindness*. Oxford: Oxford University Press.
- . 2016. "Should We Wish Well to All?" *Philosophical Review* 125 (4): 451–72. <https://doi.org/10.1215/00318108-3624764>.
- Harman, Elizabeth. 2015. "The Irrelevance of Moral Uncertainty." *Oxford Studies in Metaethics* 10.
- Harsanyi, John C. 1955. "Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparisons of Utility." *Journal of Political Economy* 63 (4): 309–21.
- . 1975. "Can the Maximin Principle Serve as a Basis for Morality? A Critique of John Rawls's Theory." *American Political Science Review* 69 (2): 594–606.
- Hausner, M. 1952. "Multidimensional Utilities." *Rand Corp*, August.
- Hausner, M., and J. G. Wendel. 1952. "Ordered Vector Spaces." *Proceedings*

- of the *American Mathematical Society* 3 (6): 977–82.
- Hawthorne, John. 2004. *Knowledge and Lotteries*. Oxford: Oxford University Press.
- . 2026. *Killing and Other Dastardly Deeds*. Princeton: Princeton University Press.
- Hawthorne, John, and David Manley. 2012. *The Reference Book*. Oxford University Press.
- Hawthorne, John, and Amia Srinivasan. 2013. “Disagreement Without Transparency: Some Bleak Thoughts.” In *The Epistemology of Disagreement: New Essays*, edited by David Christensen and Jennifer Lackey, 9–30. Oxford: Oxford University Press.
- Hawthorne, John, and Jason Stanley. 2008. “Knowledge and Action.” *Journal of Philosophy* 105 (10): 571–90.
- Hazen, Allen. 1979. “Counterpart-Theoretic Semantics for Modal Logic.” *The Journal of Philosophy* 76 (6): 319–38. <http://links.jstor.org/sici?sici=0022-362X%28197906%2976%3A6%3C319%3ACSFML%3E2.0.CO%3B2-H>.
- Heck, Richard G. 2011. *Frege’s Theorem*. Oxford: Oxford University Press.
- Hedden, Brian. 2015a. *Reasons Without Persons: Rationality, Identity, and Time*. Oxford, United Kingdom: Oxford University Press UK.
- . 2015b. “Time-Slice Rationality.” *Mind* 124 (494): 449–91. <https://doi.org/10.1093/mind/fzu181>.
- Hewitt, Edwin. 1948. “Rings of Real-Valued Continuous Functions.” *Transactions of the American Mathematical Society* 64 (1): 45–99. <https://doi.org/10.2307/1990738>.
- Hodges, Wilfrid. 1993. *Model Theory*. Cambridge University Press.
- . 1997. *A Shorter Model Theory*. Cambridge University Press.
- Hölder, Otto. 1901. “Die Axiome Der Quantität Und Die Lehre Vom Maß.” *Berichte Über Die Verhandlungen Der Königlich Sächsischen Gesellschaft Der Wissenschaften Zu Leipzig* 53: 1–64.
- Holtug, Nils. 2001. “On the Value of Coming into Existence.” *The Journal of Ethics* 5 (4): 361–84. <https://doi.org/10.1023/A:1013957425591>.
- Hong, Frank. 2024. “Know Your Way Out of St. Petersburg: An Exploration of ‘Knowledge-First’ Decision Theory.” *Erkenntnis* 89 (6): 2473–92. <https://doi.org/10.1007/s10670-022-00678-3>.
- . 2025. “Fanaticism and Knowledge.” *Synthese* 206 (1): 1–30. <https://doi.org/10.1007/s11229-025-05114-3>.
- Hong, Frank, and Jeffrey Sanford Russell. 2025. “Paradoxes of Infinite Aggregation.” *Noûs* 59 (3): 809–27. <https://doi.org/10.1111/nous.12535>.
- Horton, Joe. 2020. “Aggregation, Risk, and Reductio.” *Ethics* 130 (4): 514–29. <https://doi.org/10.1086/708534>.

- Hosiasson, Janina. 1931. "Why Do We Prefer Probabilities Relative to Many Data?" *Mind* 40: 23–36.
- Huggett, Nick. 2019. "Zeno's Paradoxes." Edited by Edward N. Zalta. <https://plato.stanford.edu/archives/win2019/entries/paradox-zeno/>.
- Hume, David. (1739) 2000. *A Treatise of Human Nature*. Edited by David Fate Norton and Mary J. Norton. Oxford University Press.
- Hyde, Dominic. 1997. "From Heaps and Gaps to Heaps of Gluts." *Mind* 106 (424): 641–60. <https://doi.org/10.1093/mind/106.424.641>.
- Isaacs, Yoaav. 2014. "Duty and Knowledge." *Philosophical Perspectives* 28 (1): 95–110.
- Isaacs, Yoaav, Adam Lerner, and Jeffrey Sanford Russell. forthcoming. "Counting Your Chickens." *Australasian Journal of Philosophy*, forthcoming.
- Jackson, Elizabeth, and Andrew Rogers. 2019. "Salvaging Pascal's Wager." *Philosophia Christi* 21 (1): 59–84.
- Jackson, Frank. 1991. "Decision-Theoretic Consequentialism and the Nearest and Dearest Objection." *Ethics* 101 (3): 461–82. <https://doi.org/10.1086/293312>.
- . 1998. *From Metaphysics to Ethics: A Defence of Conceptual Analysis*. Oxford University Press.
- Jaffe, Adam Quinn, and Daniel Raban. 2025. "Coupling Theory, Optimal Transport, and Strassen's Theorem Beyond Regular Orders." <https://arxiv.org/abs/2509.21616>.
- James, William. 1896. "The Will to Believe." *The New World* 5: 327–47.
- Jeffrey, Richard. 1983. *The Logic of Decision*. University of Chicago Press.
- Johnston, Mark. 2010. *Surviving Death*. Princeton University Press.
- . 2016. "The Personite Problem: Should Practical Reason Be Tabled?" *Noûs* 50 (4): 617–44. <https://doi.org/10.1111/nous.12159>.
- Joyce, James M. 1999. *The Foundations of Causal Decision Theory*. Cambridge University Press.
- Kamae, T., U. Krengel, and G. L. O'Brien. 1977. "Stochastic Inequalities on Partially Ordered Spaces." *The Annals of Probability* 5 (6): 899–912.
- Kant, Immanuel. (1797) 1996. "On a Supposed Right to Lie from Philanthropy." In *Practical Philosophy*, edited by Mary J. Gregor, 611–15. Cambridge: Cambridge University Press.
- . (1785) 1997. *Groundwork of the Metaphysics of Morals*. Edited by Mary J. Gregor. Cambridge Texts in the History of Philosophy. Cambridge University Press.
- Kechris, Alexander S. 1995. *Classical Descriptive Set Theory*. New York: Springer-Verlag.

- Keefe, Rosanna. 2000. *Theories of Vagueness*. New York.
- Klenke, Achim. 2013. *Probability Theory: A Comprehensive Course*. Springer.
- Kment, Boris. 2012. "Haecceitism, Chance, and Counterfactuals." *Philosophical Review* 121 (4): 573–609.
- . 2021. "Varieties of Modality." In *The Stanford Encyclopedia of Philosophy*, edited by Edward N. Zalta, Spring 2021. <https://plato.stanford.edu/archives/spr2021/entries/modality-varieties/>; Metaphysics Research Lab, Stanford University.
- Kolmogorov, A. N. 1950. *Foundations of the Theory of Probability*. Chelsea.
- Konek, Jason. 2019. "Comparative Probabilities." In *The Open Handbook of Formal Epistemology*, edited by Richard Pettigrew and Jonathan Weisberg. PhilPapers Foundation.
- Koopmans, Tjalling C. 1967. "Objectives, Constraints, and Outcomes in Optimal Growth Models." *Econometrica* 35 (1): 1–15.
- Kosonen, Petra. forthcoming. "Probability Discounting and Money Pumps." *Philosophy and Phenomenological Research*, forthcoming. <https://doi.org/10.1111/phpr.13053>.
- . 2022. "Tiny Probabilities of Vast Value." PhD thesis, Oxford University.
- Kowalczyk, Kacper. 2025. "Saving Fanaticism." *Australasian Journal of Philosophy*. <https://doi.org/10.1080/00048402.2025.2489120>.
- . Manuscript. "Risk Times Infinity."
- Krantz, David H., R. Duncan Luce, Patrick Suppes, and Amos Tversky. 1971. *Foundations of Measurement, Volume i: Additive and Polynomial Representations*. Academic Press.
- Kratzer, Angelika. (1981) 2008. "The Notional Category of Modality." In *Formal Semantics: The Essential Readings*, edited by Paul H. Portner and Barbara H. Partee, 35. John Wiley & Sons, Incorporated.
- Kripke, Saul A. 1980. *Naming and Necessity: Lectures Given to the Princeton University Philosophy Colloquium*. Edited by Darragh Byrne and Max Kölbel. Cambridge, MA: Harvard University Press.
- Lasonen-Aarnio, Maria. 2014. "Higher-Order Evidence and the Limits of Defeat." *Philosophy and Phenomenological Research* 88 (2): 314–45.
- Lauwers, Luc. 1997. "Continuity and Equity with Infinite Horizons." *Social Choice and Welfare* 14 (2): 345–56. <https://doi.org/10.1007/s003550050070>.
- . 2010. "Ordering Infinite Utility Streams Comes at the Cost of a Non-Ramsey Set." *Journal of Mathematical Economics* 46 (1): 32–37.
- . 2016. "Why Decision Theory Remains Constructively Incomplete." *Mind* 125 (500): 1033–43.

- Lauwers, Luc, and Peter Vallentyne. 2004. "Infinite Utilitarianism: More Is Always Better." *Economics and Philosophy* 20 (2): 307–30.
- . 2016. "Decision Theory Without Finite Standard Expected Value." *Economics and Philosophy* 32 (3): 383–407. <https://doi.org/10.1017/S0266267115000334>.
- Lederman, Harvey. forthcoming. "Of Marbles and Matchsticks (1st Edition)." In *Oxford Studies in Epistemology, Vol. 8*, edited by Tamar Szabó Gendler, John Hawthorne, Julianne Chung, and Alex Worsnip.
- Lee, Chunghyoung. 2013. "The Two-Envelope Paradox: Asymmetrical Cases." *Mind* 122 (485): 1–26.
- Lenman, James. 2000. "Consequentialism and Cluelessness." *Philosophy and Public Affairs* 29 (4): 342–70. <https://doi.org/10.1111/j.1088-4963.2000.00342.x>.
- Lewis, David. 1968. "Counterpart Theory and Quantified Modal Logic." *Journal of Philosophy* 65 (5): 113–26. <https://doi.org/10.2307/2024555>.
- . 1973. *Counterfactuals*. Harvard University Press.
- . 1979. "Counterfactual Dependence and Time's Arrow." *Noûs* 13 (4): 455–76.
- . 1986a. *On the Plurality of Worlds*. Oxford Blackwell.
- . 1986b. *Philosophical Papers: Volume II*. Oxford: Oxford University Press.
- . 2004. "How Many Lives Has Schrodinger's Cat?" *Australasian Journal of Philosophy* 82 (1): 3–22. <https://doi.org/10.1080/713659799>.
- Lewis, Gregory. n.d. "Reducing Global Catastrophic Biological Risks." 80,000 Hours. <https://80000hours.org/problem-profiles/preventing-catastrophic-pandemics/full-report/>.
- Lindenstrauss, Elias. 2001. "Pointwise Theorems for Amenable Groups." *Inventiones Mathematicae* 146: 259–95. <https://doi.org/10.1007/s002220100162>.
- Lindvall, Torgny. 1999. "On Strassen's Theorem on Stochastic Domination." *Electronic Communications in Probability* 4: 51–59.
- . 2002. *Lectures on the Coupling Method*. Courier Corporation.
- Liu, An, John S. Major, Sarah A. Queen, Andrew Seth Meyer, and Harold D. Roth, eds. 2010. *The Huainanzi*. Columbia University Press.
- Loeb, Peter A. 1975. "Conversion from Nonstandard to Standard Measure Spaces and Applications in Probability Theory." *Transactions of the American Mathematical Society* 211: 113–22.
- Luce, R. Duncan, and Howard Raiffa. 1989. *Games and Decisions: Introduction and Critical Survey*. Revised edition. New York: Dover Publications.
- MacAskill, William. 2015. *Doing Good Better: Effective Altruism and How You*

- Can Make a Difference*. New York: Gotham Books.
- . 2022. *What We Owe the Future*. Basic books.
- MacAskill, William, Krister Bykvist, and Toby Ord. 2020. *Moral Uncertainty*. Oxford: Oxford University Press.
- MacAskill, William, and Toby Ord. 2020. "Why Maximize Expected Choice-Worthiness?" *Nous* 54 (2): 327–53. <https://doi.org/10.1111/nous.12264>.
- Machina, Mark J. 1989. "Dynamic Consistency and Non-Expected Utility Models of Choice Under Uncertainty." *Journal of Economic Literature* 27 (4): 1622–68.
- Macnamara, John. 1988. *Journal of Symbolic Logic* 53 (1): 305–6. <https://doi.org/10.2307/2274449>.
- Mahtani, Anna. 2017. "The Ex Ante Pareto Principle." *The Journal of Philosophy* 114 (6): 303–23.
- . 2020. "Frege's Puzzle and the Ex Ante Pareto Principle." *Philosophical Studies* 178 (6): 2077–2100. <https://doi.org/10.1007/s11098-020-01524-w>.
- Mancosu, Paolo. 2009. "Measuring the Size of Infinite Collections of Natural Numbers: Was Cantor's Theory of Infinite Number Inevitable?" *Review of Symbolic Logic* 2 (4): 612–46. <https://doi.org/10.1017/S1755020309990128>.
- Manzini, Paola, and Marco Mariotti. 2008. "On the Representation of Incomplete Preferences over Risky Alternatives." *Theory and Decision* 65 (4): 303–23. <https://doi.org/10.1007/s11238-007-9086-7>.
- Mathieu, Edouard, Hannah Ritchie, Lucas Rod  s-Guirao, Cameron Appel, Daniel Gavrilov, Charlie Giattino, Joe Hasell, et al. 2020. "Excess Mortality During the Coronavirus Pandemic (COVID-19)." *Our World in Data*. <https://ourworldindata.org/excess-mortality-covid>.
- Maudlin, Tim. 1988. "The Essence of Space-Time." In *PSA: Proceedings of the Biennial Meeting of the Philosophy of Science Association, 1988*:82–91. 2. Cambridge University Press.
- May, Todd. 2009. *Death*. Routledge.
- McCarthy, David, Kalle M. Mikkola, and J. Teruji Thomas. 2019. "Aggregation for Potentially Infinite Populations Without Continuity or Completeness." In *arXiv:1911.00872 [Econ.TH]*, 1–27.
- McCarthy, David, Kalle Mikkola, and Joaquin Teruji Thomas. 2020. "Utilitarianism with and Without Expected Utility." *Journal of Mathematical Economics* 87: 77–113.
- McClennen, Edward. 1994. "Pascal's Wager and Finite Decision Theory." In *Gambling on God: Essays on Pascal's Wager*, edited by Jeff Jordan, 115–37.

- Lanham, MD: Rowman & Littlefield.
- McGee, Vann. 1999. "An Airtight Dutch Book." *Analysis* 59 (4): 257–65. <https://doi.org/10.1093/analys/59.4.257>.
- McGinnis, Jon. 2007. "Avicennan Infinity: A Select History of the Infinite Through Avicenna." *Philosophy* 32: 233–50.
- McMahan, Jefferson. 1981. "Problems of Population Theory." *Ethics* 92 (1): 96–96.
- Meacham, Christopher J. G. 2020. "Too Much of a Good Thing: Decision-Making in Cases with Infinitely Many Utility Contributions." *Synthese* 198 (8): 7309–49. <https://doi.org/10.1007/s11229-019-02522-0>.
- Menger, Karl. (1934) 1979. "The Role of Uncertainty in Economics." In *Selected Papers in Logic and Foundations, Didactics, Economics*, 259–78. Springer.
- Milford, James. 2022. "An Examination of Certain Counterpart Accounts." University of Oxford.
- Mineka, J. 1973. "A Criterion for Tail Events for Sums of Independent Random Variables." *Zeitschrift für Wahrscheinlichkeitstheorie Und Verwandte Gebiete* 25 (3): 163–70.
- Mitra, Tapan, and Kaushik Basu. 2007. "On the Existence of Paretian Social Welfare Quasi-Orderings for Infinite Utility Streams with Extended Anonymity." In *Intergenerational Equity and Sustainability*, edited by John Roemer and Kotaro Suzumura, 85–99. London: Palgrave Macmillan.
- Mogensen, Andreas L. 2021. "Maximal Cluelessness." *Philosophical Quarterly* 71 (1): 141–62.
- . 2022. "The Only Ethical Argument for Positive Delta?" *Philosophical Studies* 179: 2731–50.
- Mongin, Philippe. 1995. "Consistent Bayesian Aggregation." *Journal of Economic Theory* 66 (2): 313–51.
- Monton, Bradley. 2019. "How to Avoid Maximizing Expected Utility." *Philosophers' Imprint* 19.
- Moore, George Edward. 1952. "A Reply to My Critics." In *The Philosophy of g. E. Moore*, edited by Paul Arthur Schilpp. Tudor Pub. Co.
- Moss, Sarah. 2012. "Updating as Communication." *Philosophy and Phenomenological Research* 85 (2): 225–48. <https://doi.org/10.1111/j.1933-1592.2011.00572.x>.
- Muñoz, Daniel. 2025. "Impartiality, Anonymity, and Caring Who." *Free and Equal* 1 (2).
- Narens, Louis. 1980. "On Qualitative Axiomatizations for Probability Theory." *Journal of Philosophical Logic* 9 (2): 143–51.
- Nebel, Jacob M. 2020. "Rank-Weighted Utilitarianism and the Veil of Ignorance."

- rance." *Ethics* 131 (1): 87–106.
- . 2023. "The Sum of Well-Being." *Mind* 132 (528): 1074–104. <https://doi.org/10.1093/mind/fzad019>.
- . 2024. "Ethics Without Numbers." *Philosophy and Phenomenological Research* 108 (2): 289–319. <https://doi.org/10.1111/phpr.12966>.
- . 2025. "Infinite Ethics and the Limits of Impartiality." *Noûs*. <https://doi.org/10.1111/nous.70010>.
- Nelson, Cassidy. 2020. "Dr Cassidy Nelson on the Twelve Best Ways to Stop the Next Pandemic (and Limit COVID-19)." Interview. Edited by Robert Wiblin. 80,000 Hours Podcast. <https://80000hours.org/podcast/episodes/cassidy-nelson-12-ways-to-stop-pandemics/>.
- Nelson, Mark T. 1991. "Utilitarian Eschatology." *American Philosophical Quarterly* 28 (4): 339–47.
- Nover, Harris, and Alan Hájek. 2004. "Vexing Expectations." *Mind* 113 (450): 237–49. <https://doi.org/10.1093/mind/113.450.237>.
- Ord, Toby, Rafaela Hillerbrand, and Anders Sandberg. 2010. "Probing the Improbable: Methodological Challenges for Risks with Low Probabilities and High Stakes." *Journal of Risk Research* 13: 191–205.
- Ortiz-Ospina, Esteban, and Max Roser. 2016. "Global Health." *Our World in Data*. <https://ourworldindata.org/health-meta>.
- Otsuka, Michael, and Alex Voorhoeve. 2009. "Why It Matters That Some Are Worse Off Than Others: An Argument Against the Priority View." *Philosophy & Public Affairs* 37 (2): 171–99.
- Parfit, Derek. 1984. *Reasons and Persons*. Oxford University Press.
- . 1988. "What We Together Do."
- . 2003. "Justifiability to Each Person." *Ratio* 16 (4): 368–90. <https://doi.org/10.1046/j.1467-9329.2003.00229.x>.
- Pascal, Blaise. (1670) 1962. *Pensées*. Edited by Louis Lafuma. Editions du Seuil.
- . (1670) 2021. *Pensées*. Edited by Pierre Zoberman and David Wetsel. Project Muse. Washington, DC: The Catholic University of America Press. <https://muse.jhu.edu/book/89919>.
- Pivato, Marcus. 2013. "Risky Social Choice with Incomplete or Noisy Interpersonal Comparisons of Well-Being." *Social Choice and Welfare* 40: 123–39.
- . 2014. "Additive Representation of Separable Preferences over Infinite Products." *Theory and Decision* 77: 31–83.
- . 2023. "Population Ethics in an Infinite Universe." *Philosophical Studies* 180 (12): 3383–3414.
- Pivato, Marcus, and Marc Fleurbaey. 2024. "Intergenerational Equity and

- Infinite-Population Ethics: A Survey." *Journal of Mathematical Economics* 113: 1–29.
- Plate, Jan. 2022. "Qualitative Properties and Relations." *Philosophical Studies* 179 (4): 1297–1322. <https://doi.org/10.1007/s11098-021-01708-y>.
- Pomatto, Luciano, Philipp Strack, and Omer Tamuz. 2020. "Stochastic Dominance Under Independent Noise." *Journal of Political Economy* 128: 1877–1900. <https://doi.org/10.1086/705555>.
- Popper, Karl. 1955. "Two Autonomous Axiom Systems for the Calculus of Probabilities." *British Journal for the Philosophy of Science* 6 (1): 51–57.
- Priest, Graham. 1979. "The Logic of Paradox." *Journal of Philosophical Logic* 8 (1): 219–41. <https://doi.org/10.1007/bf00258428>.
- . 2008. *An Introduction to Non-Classical Logic: From If to Is*. New York.
- Pruss, Alexander R. 2013a. "Linear Extensions of Orders Invariant Under Abelian Group Actions."
- . 2013b. "Null Probability, Dominance and Rotation." *Analysis* 73 (4): 682–85.
- . 2014. "Regular Probability Comparisons Imply the Banach–Tarski Paradox." *Synthese* 191: 3525–40. <https://doi.org/10.1007/s11229-014-0458-6>.
- . 2015. "Popper Functions, Uniform Distributions and Infinite Sequences of Heads." *Journal of Philosophical Logic* 44: 259–71. <https://doi.org/10.1007/s10992-014-9317-7>.
- . 2021. "Underdetermination of Infinitesimal Probabilities." *Synthese* 198: 777–99. <https://doi.org/10.1007/s11229-018-02064-x>.
- . 2025. "The Impossibility of Generating Comparative Probabilities from Primitive Conditional Probabilities." *Synthese* 206 (103). <https://doi.org/10.1007/s11229-025-05177-2>.
- Quiggin, John. 1993. *Generalized Expected Utility Theory: The Rank-Dependent Model*. Springer Netherlands.
- Ramsey, Frank Plumpton. 1928. "A Mathematical Theory of Saving." *Economic Journal* 38 (4): 543–59.
- Raz, Joseph. 1986. "Value Incommensurability: Some Preliminaries." *Proceedings of the Aristotelian Society* 86 (1): 117–34. <https://doi.org/10.1093/aristotelian/86.1.117>.
- Regan, Donald H. 1980. *Utilitarianism and Co-Operation*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780198246091.001.0001>.
- Regan, Tom. 1985. *The Case for Animal Rights*. University of California Press.
- Rényi, Alfréd. 1970. *Foundations of Probability*. Holden-Day.
- Rescher, Nicholas, and Haig Khatchadourian. 1965. "Al-Kindī's Epistle on the Finitude of the Universe." *Isis* 56 (4): 426–33. <https://www.jstor.org>

- g/stable/228770.
- Rescorla, Michael. 2015. "Some Epistemological Ramifications of the Borel-Kolmogorov Paradox." *Synthese* 192 (3): 735–67.
- Restall, Greg. 2005. "Multiple Conclusions." In *Logic, Methodology, and Philosophy of Science*, edited by Petr Hájek, Luis Valdés-Villanueva, and Dag Westersth. College Publications.
- Robinson, Abraham. 1966. *Non-Standard Analysis*. Princeton University Press.
- Rota, Michael. 2016. "A Better Version of Pascal's Wager." *American Catholic Philosophical Quarterly* 90 (3): 415–39. <https://doi.org/10.5840/acpq20166288>.
- Rousseau, Jean-Jacques. (1755) 1992. *Discourse on the Origin of Inequality*. Oxford University Press.
- Russell, Bertrand. 1910--1911. "Knowledge by Acquaintance and Knowledge by Description." *Proceedings of the Aristotelian Society (New Series)* 11 (1910--1911): 108–28.
- Russell, Jeffrey Sanford. 2013. "Possible Worlds and the Objective World." *Philosophy and Phenomenological Research* 87 (1). <https://doi.org/10.1111/phpr.12052>.
- . 2020a. "Non-Archimedean Preferences over Countable Lotteries." *Journal of Mathematical Economics* 88 (May 2020): 180–86.
- . 2020b. "On the Probability of Plenitude." *Journal of Philosophy* 117 (5): 267–92. <https://doi.org/10.5840/jphil2020117517>.
- . 2022. "Problems for Impartiality." Global Priorities Institute, University of Oxford: Derek Parfit Memorial Lecture. <https://globalprioritiesinstitute.org/parfit-memorial-lecture-jeffrey-sanford-russell-problems-for-impartiality/>.
- . 2024. "On Two Arguments for Fanaticism." *Noûs* 58: 565–95. <https://doi.org/10.1111/nous.12461>.
- . Forthcoming. "Fixing Stochastic Dominance." *The British Journal for the Philosophy of Science*.
- Russell, Jeffrey Sanford, and John Hawthorne. 2018. "Possible Patterns." *Oxford Studies in Metaphysics* 11.
- Russell, Jeffrey Sanford, and Yoaav Isaacs. 2021. "Infinite Prospects." *Philosophy and Phenomenological Research* 103 (1): 178–98.
- San, Weng Kin. 2026. "Aggregation and the Structure of Value." *Noûs*.
- Sathaye, Avinash. n.d. "Infinity in Classical Indian Mathematics." https://www.ms.uky.edu/~sohum/ma330/files/Infinity_in_india.pdf.
- Savage, Leonard J. 1954. *The Foundations of Statistics*. Wiley Publications in Statistics. New York: John Wiley & Sons.

- Scanlon, T. M. 1998. *What We Owe to Each Other*. Belknap, Harvard University Press.
- Schervish, Mark J., Teddy Seidenfeld, and Joseph B. Kadane. 1984. "The Extent of Non-Conglomerability of Finitely Additive Probabilities." *Zeitschrift Für Wahrscheinlichkeitstheorie Und Verwandte Gebiete* 66 (2): 205–26. <https://doi.org/10.1007/bf01025890>.
- . 2017. "Nonconglomerability for Countably Additive Measures That Are Not κ -Additive." *Review of Symbolic Logic* 10 (2): 284–300. <https://doi.org/10.1017/s1755020316000221>.
- Schlesinger, George. 1994. "A Central Theistic Argument." In *Gambling on God: Essays on Pascal's Wager*, edited by Jeff Jordan, 1–16. Lanham, Md.: Rowman & Littlefield.
- Schoenfield, Miriam. 2014. "Decision Making in the Face of Parity." *Philosophical Perspectives* 28 (1): 263–77. <https://doi.org/10.1111/phpe.12044>.
- Seidenfeld, Teddy, Joseph B. Kadane, Mark J. Schervish, and Rafael B. Stern. 2024. "Finite Additivity, Complete Additivity, and the Comparative Principle." *Erkenntnis* 90 (5): 1945–68. <https://doi.org/10.1007/s10670-023-00757-5>.
- Seidenfeld, Teddy, Mark J. Schervish, and Joseph B. Kadane. 2009. "Preference for Equivalent Random Variables: A Price for Unbounded Utilities." *Journal of Mathematical Economics* 45 (5-6): 329–40.
- Sen, Amartya. (1970) 2017. *Collective Choice and Social Welfare: An Expanded Edition*. Harvard University Press.
- Sepielli, Andrew. 2012. "Subjective Normativity and Action Guidance." *Philosophical Studies* 161 (1): 17–35.
- Shinotsuka, Tomoichi. 1998. "Equity, Continuity, and Myopia: A Generalization of Diamond's Impossibility Theorem." *Social Choice and Welfare* 15 (1): 21–30.
- Sierpiński, Waław. 1965. *Cardinal and Ordinal Numbers*. Polish Scientific.
- Skow, Bradford. 2007. "Haecceitism, Anti-Haecceitism and Possible Worlds." *The Philosophical Quarterly* 58 (230): 98–107.
- Skyrms, Brian. 1980. *Causal Necessity*. New Haven: Yale University Press.
- . 1995. "Strict Coherence, Sigma Coherence and the Metaphysics of Quantity." *Philosophical Studies* 77 (1): 39–55. <https://doi.org/10.1007/bf00996310>.
- . (1983) 2012. "Zeno's Paradox of Measure." In *From Zeno to Arbitrage: Essays on Quantity, Coherence, and Induction*, 5–37. Oxford: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199652808.003.0002>.
- Slote, Michael A. 1978. "Time in Counterfactuals." *The Philosophical Review* 87 (1): 3–27.

- Smith, Nicholas J. J. 2014. "Is Evaluative Compositionality a Requirement of Rationality?" *Mind* 123 (490): 457–502. <https://doi.org/10.1093/mind/fzu072>.
- Soames, Scott. 2005. *Reference and Description: The Case Against Two-Dimensionalism*. Princeton University Press.
- Srinivasan, Amia. 2015. "Normativity Without Cartesian Privilege." *Philosophical Issues* 25 (1): 273–99. <https://doi.org/10.1111/phis.12059>.
- Stalnaker, Robert. 1968. "A Theory of Conditionals." In *Studies in Logical Theory*, edited by Nicholas Rescher, 98–112. Blackwell.
- . 1981. *Letter to David Lewis*. Edited by William Leonard Harper, Robert Stalnaker, and Glenn Pearce. Dordrecht: D. Reidel.
- Stanley, Jason. 2005. *Knowledge and Practical Interests*. Oxford: Oxford University Press.
- Stevens, Wallace. 1923. "Sunday Morning." In *Harmonium*, 100–104. Alfred A. Knopf.
- Stewart, Rush T., and Michael Nielsen. 2021. "Conglomerability, Disintegrability and the Comparative Principle." *Analysis* 81 (3): 479–88. <https://doi.org/10.1093/analys/anab012>.
- Strassen, V. 1965. "The Existence of Probability Measures with Given Marginals." *Annals of Mathematical Statistics* 36: 423–39. <https://doi.org/10.1214/aoms/1177700153>.
- Suppes, Patrick. 1966. "Some Formal Models of Grading Principles." *Synthese* 16 (3–4): 284–306. <https://doi.org/10.1007/BF00485084>.
- Svensson, Lars-Gunnar. 1980. "Equity Among Generations." *Econometrica* 48 (5): 1251–56.
- Szpilrajn, E. 1930. "Sur l'extension de l'ordre Partiel." *Fundamenta Mathematicae* 16: 386–89.
- Tarsney, Christian. forthcoming. "Against Anti-Fanaticism." *Philosophy and Phenomenological Research*, forthcoming. <https://doi.org/10.1111/phpr.13120>.
- . 2020. "The Epistemic Challenge to Longtermism." Working Paper 10-2020. Global Priorities Institute.
- . 2025. "Expected Value, to a Point: Moral Decision-Making Under Background Uncertainty." *Noûs* 59 (4): 1093–1125. <https://doi.org/10.1111/nous.12544>.
- Tarsney, Christian, Harvey Lederman, and Dean Spears. 2025. "A Dominance Argument Against Incompleteness." *Philosophical Review* 134 (4): 455–90. <https://doi.org/10.1215/00318108-11964820>.
- The Holy Bible: New Revised Standard Version*. 1996. Oxford University Press.
- Thomas, Teruji. 2019. "The Asymmetry, Uncertainty, and the Long Term."

- Philosophy and Phenomenological Research*, no. 2: 470–500. <https://doi.org/10.1111/phpr.12927>.
- . 2022. “Separability in Population Ethics.” In *The Oxford Handbook of Population Ethics*, edited by Gustaf Arrhenius, Krister Bykvist, Tim Campbell, and Elizabeth Finneron-Burns, 271–95. Oxford University Press.
- Thong, Joshua. forthcoming. “Primitive Conditional Probabilities, Subset Relations and Comparative Regularity.” *Analysis*, forthcoming.
- Vallentyne, Peter. 1993. “Utilitarianism and Infinite Utility.” *Australasian Journal of Philosophy* 71 (2): 212–17. <https://doi.org/10.1080/00048409312345222>.
- . 1995. “Infinite Utility: Anonymity and Person-Centredness.” *Australasian Journal of Philosophy* 73 (3): 413–20.
- Vallentyne, Peter, and Shelly Kagan. 1997. “Infinite Value and Finitely Additive Value Theory.” *Journal of Philosophy* 94 (1): 5–26.
- Venkatesananda, Swami, trans. 1993 [c. 500 CE–1500 CE?]. *Vāsiṣṭha’s Yoga*. State University of New York Press.
- von Neumann, John, and Oskar Morgenstern. (1944) 2007. *Theory of Games and Economic Behavior: 60th Anniversary Commemorative Edition*. Princeton University Press.
- von Weizsäcker, Carl Christian. 1965. “Existence of Optimal Programs of Accumulation for an Infinite Time Horizon.” *The Review of Economic Studies* 32 (2): 85–104.
- Wagon, Stan. 1994. *The Banach-Tarski Paradox*. Cambridge: Cambridge University Press.
- Weatherston, Brian. 2012. “Knowledge, Bets and Interests.” In *Knowledge Ascriptions*, edited by Jessica Brown and Mikkel Gerken, 75–103. Oxford University Press.
- Wilkinson, Hayden. forthcoming. “Egyptology and Fanaticism.” *Philosophical Studies*, forthcoming. <https://doi.org/10.1007/s11098-024-02180-0>.
- . 2022a. “In Defense of Fanaticism.” *Ethics* 132 (2): 445–77.
- . 2022b. “Infinite Aggregation and Risk.” *Australasian Journal of Philosophy*, 1–20.
- . n.d. “Comments at the Global Priorities Workshop.” Personal communication.
- Williams, Bernard. 1973. “The Makropulos Case: Reflections on the Tedium of Immortality.” In *Problems of the Self*. Cambridge University Press.
- Williamson, Timothy. 1996. *Vagueness*. New York: Routledge.
- . 2007. “How Probable Is an Infinite Sequence of Heads?” *Analysis* 67 (3): 173–80.

- Wright, Crispin. 1983. *Frege's Conception of Numbers as Objects*. Aberdeen: Aberdeen University Press.
- Zame, William. 2007. "Can Intergenerational Equity Be Operationalized?" *Theoretical Economics* 2 (2): 177–98.
- Zhou, Lin. 1997. "Harsanyi's Utilitarianism Theorems: General Societies." *Journal of Economic Theory* 72 (1): 198–207.
- Zhuangzi. 1968. *The Complete Works of Chuang Tzu*. Translated by Burton Watson. Columbia University Press.

